S. M. NIKOLSKY

A Course of Mathematical Analysis

Volume

1

MIR PUBLISHERS MOSCOW

Academician S.M NIKOLSKY a State Prize winner, the author of more than 130 scientific papers and several monographs including *Quadrature Formulas*, *Approximation of Functions of Several Variables and Embedding Theorems* and *Integral Representation of Functions and Embedding Theorems* (co-author).

He contributed many fundamental results to the theory of approximation of functions, variational methods for solving boundary-value problems and functional analysis. For his monograph *Approximation of Functions of Several Variables and Embedding Theorems* he was awarded the Chebyshev Prize of the USSR Academy of Sciences.

С. М. НИКОЛЬСКИЙ

КУРС МАТЕМАТИЧЕСКОГО АНАЛИЗА

ТОМ 1

ИЗДАТЕЛЬСТВО «НАУКА»

МОСКВА

S. M. NIKOLSKY

Member, USSR Academy of Sciences

A Course of Mathematical Analysis

Volume 1

Translated from the Russian
by
V. M. VOLOSOV, D. Sc.

MIR PUBLISHERS
MOSCOW

ИЗДАТЕЛЬСТВО · МИР ·

First published 1977
Revised from the 1975 Russian edition

На английском языке

Preface to the English Edition

The major part of this two-volume textbook, with the exception of some supplementary material, stems from the course in mathematical analysis given by the author for many years at the Moscow Physico-technical Institute.

The first chapter is an introduction. It treats of fundamental notions of mathematical analysis using an intuitive concept of a limit. Ultimately, it even establishes, with the aid of visual interpretation and some considerations of a physical character, the relationship between the derivative and the integral and gives some elements of differentiation and integration techniques necessary to those readers who are simultaneously studying physics.

The second chapter is devoted to the notion of a real number which is interpreted on the basis of its representation as an infinite decimal. The part of this chapter given in small print may be omitted by the reader in his first acquaintance with the subject-matter.

It is my belief, which incidentally coincides with the traditional point of view, that the fundamentals of mathematical analysis should be first presented for functions of one independent variable and only after that extended to functions of several variables although this leads to some unavoidable recapitulations. On the other hand, in a comprehensive course intended for students in the fields of mathematics and physics it is quite possible to pass from functions of one variable not to the cases of two and three variables but directly to n variables. Here the whole question rests merely on a suitable choice of notation. However, such notation has already been elaborated in scientific journals and monographs and has proved expedient, and now it only remains for the authors of textbooks to accept it. Such an approach provides the necessary prerequisites for the second half of the course where in the study of such topics as the Fourier series and the Fourier integral the reader should acquire a good understanding of the concept of an infinite-dimensional function space.

The notions of an n-dimensional Euclidean space, of an arbitrary space with scalar product and of a Banach space are introduced in the course at a rather early stage. They are then widely used but only to an extent required for the realization of the plan of the book.

The presentation of the subject-matter of the book is based mainly on the notion of the Riemann integral. To save the reader time and effort I have always tried, when possible, to give similar proofs of the analogous theorems in the one-dimensional and many-dimensional cases.

There arises the subtle question of the completeness of the spaces L and L_2. In this connection I do not introduce abstract elements which substitute for the Lebesgue integrable functions simply confining myself in the general discussion of the question to an explanation of the corresponding fact in terms of the Lebesgue integral.

Incidentally, the textbook includes a supplementary section (Chapter 19, Vol. 2) devoted to the Lebesgue integral. I hope that many of the readers will be interested enough to look through it. The concept of the Lebesgue integral is essential for modern mathematical physics. Without using the Lebesgue integral it is quite impossible to study the direct variational methods of mathematical physics.

Chapters 17 and 18 (Vol. 2) also contain supplementary material. Chapter 18 deals with such important notions of modern calculus as the Sobolev regularization of functions and the partition of unity.

Chapter 17 is devoted to differentiable manifolds and differential forms. It culminates in the proof of the Stokes theorem for the n-dimensional space. This chapter may serve as a test of the reader's grasp of the material of the book.

My intention was to help the reader to proceed more easily to the study of mathematical physics. A number of supplementary topics included in the course have been chosen particularly in consideration of their use in mathematical physics. There still remain a number of pedagogical problems to be solved in teaching the theory of functions of several variables. It is hoped that this book will contribute to the purpose.

I owe a great deal to two books that have helped me greatly.

One is *Course d'analyse infinitesimal* by Ch. J. de la Valleé Poissin, a book which I studied assiduously in my student days.

The other is *An Introduction to the Theory of Functions of a Real Variable* by P. S. Alexandrov and A. N. Kolmogorov whose many lectures I have also attended. I wish to express my deepest gratitude to these distinguished scientists and especially to A. N. Kolmogorov to whom I am greatly indebted for scientific inspiration.

In the prefaces to the first and second Russian editions of the book I mentioned a number of persons and organizations from whom I received valuable advice in the preparation of the present course. These were the Chairs of Mathematics of the Moscow Physico-technical Institute and of the Moscow Institute of Electronic Engineering and also O. V. Besov, A. A. Vasharin, I. N. Vekua, E. A. Volkov, R. V. Gamkrelidze, A. A. Dezin, L. D. Kudryavtsev, P. I. Lizorkin and Yu. S. Nikolsky.

To all of them I once again express my warmest gratitude.

Finally, I wish to express my appreciation to the Mir Publishers English Department and to the translator of the book Professor V. M. Volosov for their most efficient handling of the publication of this English translation.*

S. M. Nikolsky

---

* The present translation incorporates suggestions made by the author. — *Tr.*

Contents

CHAPTER 1

Introduction

§ 1.1. Preliminary Remarks

The name "Mathematical Analysis" is a modification of the old name "Infinitesimal Analysis". The latter provides some more information but is itself an abbreviation. It would be more precise to call our course "The Analysis by Means of Infinitesimals".

It would be still better if the title of the book characterized the objects subjected to analysis (investigation). In classical mathematical analysis these objects are mainly functions, that is variable quantities dependent on other variable quantities.

We say "mainly" since further development of mathematical analysis showed that its methods can be applied to the study of some more complex objects than functions (functionals, operators, etc.). But at the present stage we shall not speak of them. Our immediate task is the investigation of some sufficiently general functions, used in practice, with the aid of infinitesimal methods or, which is the same, with the aid of limits. The essence of these methods will be gradually presented to the reader in the course of the study of the book. Now we only confine ourselves to saying that, in particular, these methods lead to the extremely important operations of differentiating and integrating functions.

Sections 1.2 and 1.3 deal with the concepts of a set and of a function.

The following three sections (1.4-1.6) are of a purely introductory character. They will give the reader a general idea of the fundamental notions of mathematical analysis which will be studied in detail later on: the continuity, the derivative, the indefinite and definite integrals.

In this chapter we of course use the notion of a limit but do not define it or explain it relying completely on the intuition of the reader. There can also be suggested an alternative way of studying the book: the reader can first omit Secs. 1.4-1.6 and then come back to them whenever they are referred to.

§ 1.2. Set. Open and Closed Intervals

In mathematics any collection or system of objects is called a *set*. For instance, we can speak of the set of all trees on a given glade, of a number of geese feeding on it or of the set of all integers. If A denotes a given set of objects and x is one of these objects we say that x is an *element* of the set A (synonymously, x *belongs to* A) and write $x \in A$.

If y is not an element of A we write $y \bar{\in} A$ or $y \notin A$.

If one and the same set is denoted by two different letters A and B we write $A = B$; it should be stressed, when necessary, that this is a set-theoretic equality which must not be confused with an equality of numbers.

If, for any x, the relation $x \in A$ always implies $x \in B$ the set A is said to be *contained* in B or is called a *subset* of B. In this case we write $A \subset B$. It should be taken into account that, according to this definition, the relation $A = B$ is a special case of the relation $A \subset B$. For if not only $A \subset B$ but also $B \subset A$ then $A = B$ and vice versa.

If a set consists of a single element x it is preferable to denote this set by another letter, say A, because, logically, we should distinguish between this single-element set and the element itself. It is also necessary to introduce the notion of the *empty* (*void*) *set* containing no elements at all and to denote it by a special letter, for instance, O. By definition, $O \subset A$ for any set A.

As is known from elementary mathematics, it is possible to establish a one-to-one correspondence* between the real numbers and the points of a line by applying the following rule. With the number 0 there is associated an arbitrary point O, the *origin*. The length of a certain line segment is taken as unit measure. Then with every real number $\pm a$ $(a > 0)$ there is associated the point of the line lying at a distance of a from the origin to the right or to the left of O depending on whether there is the sign "+" or "−" in front of a. Conversely, if A is a point of the line lying at a distance of a from O we associate with it the number $+a$ or $-a$ depending on whether A lies to the right or to the left of O.

The straight line for whose all points a one-to-one correspondence with all real numbers has been established, as described above, is called the *number line* or the *number axis* or the *real line* (*axis*). Its points are called (i.e. are identified with) the numbers which they represent. Thus, we can speak of points 0, 1, 1.2, $\sqrt{2}$ and the like. Bearing in mind what has been said we shall interpret the numbers as *points* (*of the number line*) and, conversely, the points of the line as numbers.

Let some numbers (points) a and b satisfy the inequality $a < b$.

The set of all numbers x satisfying the inequalities $a \leqslant x \leqslant b$ is referred to as a *closed interval* (with end points a and b) and is denoted as $[a, b]$.

The set of the numbers x satisfying the inequalities $a < x < b$ is spoken of as an *open interval* (with end points a and b) and denoted (a, b).

The set of the numbers x satisfying the inequalities $a \leqslant x < b$ or $a < x \leqslant b$ is denoted respectively as $[a, b)$ or $(a, b]$ and called a *half-open* (*semi-open*) or *half-closed interval* (also, simply, a *half-interval*). The half-interval $[a, b)$ contains its left end point and does not contain its right end point

* In this connection see also § 2.7.

while, on the contrary, the interval $(a, b]$ contains the right end point and does not contain the left end point.

We also often consider sets referred to as *infinite intervals* (also *half-intervals*) of the following types: (i) $(-\infty, \infty)$, (ii) $(-\infty, a]$, (iii) $(-\infty, a)$, (iv) (a, ∞) and (v) $[a, \infty)$.

The first of them is the set of all real numbers (of all points of the real line) while the others are respectively the sets containing the numbers x satisfying the inequalities (ii) $x \leq a$, (iii) $x < a$, (iv) $a < x$ and (v) $a \leq x$.

Let A and B be two sets of arbitrary nature. By the *sum* or *union* of A and B is meant the set, denoted $A+B$ or $A \cup B$, which is the collection of all the elements of A and of B.

The *difference* $A \backslash B$, also denoted $A-B$, between two sets A and B (in that order) is the set consisting of all the elements of A not belonging to B.

The *intersection* (or *product* or *meet*) of A and B is the set, denoted by AB or $A \cap B$, containing all the elements which simultaneously belong to A and B.

There holds the set-theoretic equality

$$(A \pm B)C = AC \pm BC \tag{1}$$

where A, B and C are arbitrary sets. For example, in the case of the sign "+" we can prove it as follows. If an element x belongs to the left-hand member of (1) it simultaneously belongs to $A+B$ and C. But then x necessarily belongs to at least one of the sets A and B. For definiteness, let $x \in A$, then $x \in AC$ and consequently the element x also belongs to the right-hand member of (1). Conversely, let x belong to the right-hand member of the equality; then x belongs to at least one of the sets AC and BC. For definiteness, let $x \in AC$; then x belongs both to A and to C and hence x belongs both to $A+B$ and to C, that is it belongs to the left-hand member of (1).

The definition of a sum of sets is readily extended to any finite and even infinite number of summands (sets).

The expressions

$$\bigcup_{k=1}^{N} A_k = A_1 + \ldots + A_N \quad \text{and} \quad \bigcup_{k=1}^{\infty} A_k = A_1 + A_2 + \ldots$$

denote respectively the collection of all the elements of the sets $A_1, \ldots, A_N$ and the collection of all the elements of the sets $A_1, A_2, \ldots$ and are referred to as the *unions* or *sums* of the indicated sets.

There hold the equalities

$$C \bigcup_{k=1}^{N} A_k = \bigcup_{k=1}^{N} CA_k \quad \text{and} \quad C \bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} CA_k$$

(analogous to (1) in the case of the sign "+") where C is an arbitrary set.

Examples

(i) $[0, 2]+[1, 3] = [0, 3]$; (ii) $[0, 2]-[1, 3] = [0, 1)$; (iii) $\bigcup_{k=-\infty}^{\infty} [k, k+1] =$
$= \bigcup_{k=-\infty}^{\infty} [k, k+1) = R$ where R is the set of all real numbers.

§ 1.3. Function

Let E be a set of numbers and let there be given a certain law according to which with every number x belonging to E a (single) number y is associated. Then we say that there is a (*single-valued* or *one-valued*) *function* defined on E. In this case we write

$$y = f(x) \qquad (x \in E) \tag{1}$$

This definition of a function was suggested by N. I. Lobachevsky and P. G. L. Dirichlet*. The set E is called the *domain of definition* (or, simply, the *domain*) of the function $f(x)$. We also say that we are given an *independent variable* x which can assume the particular values x belonging to the set E and that, according to the given law, to each $x \in E$ there corresponds a certain value (number) of the other variable y called a *function* or a *dependent variable*. The independent variable is also called the *argument*.

The concept of a function can be interpreted in geometrical terms. We say that there is a set E of points x of the real line called, as was said, the domain of definition of the function and a law associating with every point $x \in E$ a number $y = f(x)$.

If we speak of a function as a certain law of correspondence associating with each number $x \in E$ a number y it is sufficient to denote the function by a single letter f. The symbol $f(x)$ denotes the number y which, in accordance with that law f, corresponds to the value $x \in E$. For instance, if the number 1 belongs to the domain of definition E of a function f then $f(1)$ is the *particular value* of the function f at the point $x = 1$. If 1 does not belong to E (i.e. $1 \overline{\in} E$) we say that the function f *is not defined* at the point $x = 1$.

The set E_1 of all the values $y = f(x)$ where $x \in E$ is called the *image* of the set E (produced by the function f) or the *range* of the function $y = f(x)$ for $x \in E$. In such a case we sometimes write $E_1 = f(E)$. In using the latter notation care should be taken not to confuse it with the notation $y = f(x)$ where x is an arbitrary number (point) belonging to the set E and y is the corresponding point, of the set E_1, assigned to x by the function (law) f; to this end it is advisable to make the necessary stipulation every time the symbolic relation $E_1 = f(E)$ is used. We also say that the function f *maps* the set E on the set E_1 and sometimes synonymously call f a *mapping*.

If $E_1 = f(E) \subset A$ where A is a set of numbers which, in the general case, may not coincide with E_1 we say that f performs a *mapping of E into A*.

---

*N. I. Lobachevsky (1792-1856), the great Russian mathematician, the creator of the non-Euclidean geometry. P. G. L. Dirichlet (1805-1859), a German mathematician.

For two functions f and φ defined on one and the same set E we define the *sum* $f+\varphi$, the *difference* $f-\varphi$, the *product* $f\varphi$ and the *quotient* $\frac{f}{\varphi}$. These are new functions expressed respectively by the formulas

$$f(x)+\varphi(x), \quad f(x)-\varphi(x), \quad f(x)\varphi(x), \quad \frac{f(x)}{\varphi(x)}, \quad x \in E \tag{2}$$

where in the case of the quotient it is supposed that $\varphi(x) \neq 0$ on E.

Any other letters $F, \Phi, \Psi, \ldots$ can also be used to denote functions, and instead of x and y we can write $z, u, v, w, \ldots$.

If a function f maps a set E into E_1 and a function F maps the set E_1 into a set E_2, the function $z = F(f(x))$ is called a *composite function* or a *function of a function* or the *superposition* of the functions f and F. This new function is defined on the set E and performs a mapping of E into E_2.

A composite function can also be constructed as a superposition of any number n of functions: $z = F_1(F_2(F_3\ldots(F_n(x))\ldots))$.

In practice we encounter many examples of functions. For instance, the area S of a circle of radius r is a function of the radius r expressed by the formula $S = \pi r^2$. This function is obviously defined on the set of all positive numbers r.

The dependence of a variable S on a variable r expressed by the formula $S = \pi r^2$ can be considered irrespective of the question of computing the area of a circle. The function $S = \varphi(r)$ specified by this formula is defined throughout the real axis, that is for all real numbers r, not necessarily for positive ones.

Below are examples of functions determined by formulas:

(i) $y = \sqrt{1-x^2}$ (iv) $y = \dfrac{x^2-1}{x-1}$

(ii) $y = \log(1+x)$ (v) $y = \arcsin x$

(iii) $y = x-1$

Here we mean real functions assuming real values y for the corresponding real values of the argument x. It is readily seen that the domains of definition of these functions are, respectively,

(i) the closed interval $[-1, 1] = \{-1 \leqslant x \leqslant 1\}$;

(ii) the set $x > -1$;

(iii) the whole real axis;

(iv) the whole real axis with the point $x = 1$ deleted;

(v) the closed interval $[-1, +1]$.

The functions in examples (i) and (ii) can be regarded as composite functions: (i) $y = \sqrt{u}$, $u = 1-v$, $v = x^2$; (ii) $y = \log u$, $u = 1+x$.

The construction of the graphs of functions is an important method of representing functions. Let us take a rectangular coordinate system x, y (Fig. 1.1). Marking a closed interval $[a, b]$ on the x-axis we plot an arbitrary curve Γ possessing the following property: for any point $x \in [a, b]$ the straight line parallel to the y-axis and passing through this point meets the

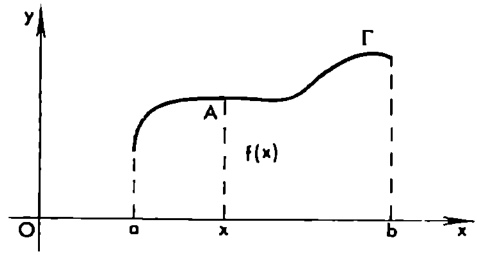

Fig. 1.1

curve Γ at exactly one point A. A curve of this kind specified in a rectangular (Cartesian) coordinate system will be referred to as a *graph*. The given graph determines a function $y = f(x)$ on the interval $[a, b]$: if x is an arbitrary point of the interval $[a, b]$ the corresponding value $y = f(x)$ of the function is determined as the ordinate of the point A (see Fig. 1.1). Thus, by means of the graph we have obtained a definite law of correspondence between x and $y = f(x)$.

We have defined, by means of a graph, a function on the set E which is a closed interval $[a, b]$. In other cases E can be an open interval, a half-interval, the whole real axis, the set of rational points belonging to a given interval and the like.

Let us take a function $f(x)$ defined on an interval (a, b) and an arbitrary (constant) number $\alpha \neq 0$. Using α and f we can construct a number of other functions: (i) $\alpha f(x)$; (ii) $f(x)+\alpha$; (iii) $f(x-\alpha)$; (iv) $f(\alpha x)$. Functions (i) and (ii) are defined on the same interval (a, b). The ordinates of the graph of function (i) are α times the corresponding ordinates of the function $f(x)$. The graph of function (ii) is obtained from that of f by moving the latter graph α units of length upward*; as to the graph of function (iii), it is obtained from the graph of f by shifting the latter by a distance of α to the right. Finally, function (iv) is obviously defined, for $\alpha > 0$, on the interval $\left(\frac{a}{\alpha}, \frac{b}{\alpha}\right)$; its graph is obtained by an α-fold uniform contraction of the graph of f.

A function f is said to be *even* or *odd* if it is defined on a set symmetric with respect to the origin and if it possesses, respectively, the property $f(-x) = f(x)$ or $f(-x) = -f(x)$.

The graph of an even function is apparently symmetric about the y-axis while the graph of an odd function is symmetric about the origin. For example, x^{2k} (where k is a natural number), $\cos x$, $\ln|x|$, $\sqrt{1+x^2}$, $f(|x|)$ are even functions while x^{2k+1} (where $k \geqslant 0$ is an integer), $\sin x$, $x\sqrt{1+x^2}$ and $xf(|kx|)$ are odd ones.

* In the case $\alpha < 0$ the motion of the graph upward by a distance of α (the shift to the right by a distance of α) should be of course understood as the motion downward by $|\alpha|$ (the shift to the left by $|\alpha|$).

It is easily seen that *the product of two even or of two odd functions is an even function and that the product of an even function by an odd function is an odd function.*

Most of the functions are of course neither even nor odd.

A function can be specified by different formulas on different parts of its domain of definition. For instance, suppose that a train departs from station A at time moment $t = 0$ and travels for two hours with a speed of 100 km per hour, arrives at station B where it has an hour stop after which it travels for three hours at 80 km per hour. Then the function $s = f(t)$ expressing the distance (measured in kilometres) from the train to the station A at time moment t is obviously determined with the aid of the following three formulas:

$$f(t) = \begin{cases} 100\,t & (0 \leqslant t \leqslant 2) \\ 200 & (2 \leqslant t \leqslant 3) \\ 200+80(t-3) & (3 \leqslant t \leqslant 6) \end{cases}$$

A function can be specified by a table. For instance, if we measure the air temperature T every hour during 24 hours then to each time moment $t = 0, 1, 2, \ldots, 24$ there corresponds a definite number T, the variation of T being determined by a table of the form

t	0	1	2	3	...
T	T_0	T_1	T_2	T_3	...

We thus obtain a function $T = f(t)$, defined on the set E of integers from 0 to 24 represented by this table.

If a function $y = f(x)$ is defined on a set E by means of a formula we can always speak of a definite graph specifying this function geometrically. The converse is by far not always so clear: the question as to whether a function specified by an arbitrary graph can be expressed by a formula is rather difficult. To answer it we should define more precisely what is meant by the word "formula". Above, when we spoke of a given function $y = f(x)$ determined by a formula, we supposed tacitly that y can be obtained from x by a finite number of such operations as addition, subtraction, multiplication, division, extraction of the kth root with a certain index k, taking logarithms, performing the operations symbolized by sin, cos, arcsin and other algebraic and trigonometric operations.

Mathematical analysis provides certain means for an essential generalization of the notion of a formula. An important operation of that kind is the expansion of a function in an infinite series with respect to elementary functions.

Many, if not all, functions encountered in practice can be represented by a formula of the type of an infinite series whose terms are elementary functions defined later; the notion of a series will be defined later on and now we shall not speak of it since we are not yet ready to discuss this question.

In any event, irrespective of the way in which a function $f(x)$ is specified, by a formula, or a graph or in some other way, it can serve as the object to which the methods of mathematical analysis can be applied if it satisfies some additional general requirements, such as continuity, monotonicity, convexity, differentiability and the like. But these questions will be discussed later.

An important notion which is used in the investigation of functions is that of a limit; the notion of a limit is one of the most important in mathematical analysis. The next chapter is devoted to the study of this notion.

If to each number x belonging to a given set E of numbers there corresponds, in accordance with a certain law, a definite set e_x of numbers y we say that this law of correspondence determines a *many-valued* (*multiple-valued*) function $y = f(x)$. If it turns out that for every $x \in E$ the set e_x only consists of a single number y we obtain, as a special case, a single-valued (one-valued) function.

A one-valued function is often simply spoken of as a "function" without the adjective "one-valued" when this does not lead to a misunderstanding.

In algebra and trigonometry we often deal with multiple-valued functions; such are, for instance, $\sqrt{x}$, Arcsin x, Arctan x, etc.

The function $\sqrt{x}$ is defined for $x \geqslant 0$. It is two-valued* since to every positive number x there correspond two real numbers differing in sign whose squares are equal to x. As to the function Arcsin x, it is infinite-valued. In the functional relationship established by the latter function to each value x belonging to the closed interval $[-1, +1]$ there corresponds an infinite set of numbers y which can be written by means of the formula

$$y = (-1)^k \arcsin x + k\pi \quad (k = 0, 1, 2, \ldots)$$

Our discussion concerned functions of one variable but we can also speak of functions of two, three and, generally, n variables.

The notion of a function of two variables is introduced in the following way. We consider a set E consisting of pairs of numbers (x, y). Here are meant *ordered* pairs: two pairs (x_1, y_1) and (x_2, y_2) are regarded as equal (coinciding) if and only if $x_1 = x_2$ and $y_1 = y_2$. If there is a certain law which assigns to every pair $(x, y) \in E$ a number z we say that this correspondence specifies a *function* $z = f(x, y)$ *of the two variables* x *and* y *on the set* E.

Since to every pair of numbers (x, y) there corresponds a definite point with coordinates x and y, in the plane where the Cartesian coordinates x and y are introduced and, conversely, to each point there corresponds a

---

* By the way, in what follows the symbol $\sqrt[k]{x}$ $(k = 2, 3, \ldots)$ will always be understood, unless otherwise stated, as the arithmetic kth root of $x \geqslant 0$, that is as a nonnegative number whose kth power equals x.

pair (x, y), we can say that the function $z = f(x, y)$ is defined on a set E of points lying in the plane.

Taking a three-dimensional space with rectangular coordinates x, y and z we can represent a function $z = f(x, y)$ of two variables by the locus of points $(x, y, f(x, y))$ whose projections (x, y) on the coordinate plane xy belong to the domain of definition E of the function f.

For instance, if we take the function

$$z = \sqrt{1-x^2-y^2}, \quad x^2+y^2 \leqslant 1$$

the corresponding locus is the upper hemisphere of radius 1 with centre at the origin.

A function of three independent variables can be defined in a similar way. In this case the domain of definition is a set E consisting of ordered triples of numbers (x, y, z) or, which is the same, the set of the corresponding points of a three-dimensional space where a Cartesian coordinate system is introduced.

If to every triple of numbers (i.e. to every point of the three-dimensional space) (x, y, z) belonging to E there corresponds, in accordance with a certain law, a number u we say that this correspondence determines a function $u = F(x, y, z)$ defined on the set E.

Similarly, we can consider a set E of ordered n-tuples $(x_1, \ldots, x_n)$ consisting of n numbers where n is a given natural number. Again, if to each such n-tuple belonging to E there corresponds, according to a certain law, a number z we say that z is a *function of the variables* $x_1, \ldots, x_n$ defined on the set E and write $z = F(x_1, \ldots, x_n)$.

In the case $n > 3$ we no longer have at our disposal a real (geometrical) n-dimensional space which can be used for the visual representation of the n-tuples $(x_1, \ldots, x_n)$ as points belonging to that space. But mathematicians invented the concept of an n-dimensional space which successfully serves this purpose and not worse than the real geometrical three-dimensional space. Namely, by an n-dimensional space is meant the collection of all n-tuples $(x_1, \ldots, x_n)$ (see § 6.1).

If two functions f and φ of n variables are defined on one and the same set E of n-tuples $(x_1, \ldots, x_n)$ (which are regarded as points of the n-dimensional space) the sum $f+\varphi$, the difference $f-\varphi$, the product $f\varphi$ and the quotient f/φ are specified, as functions defined on E, by means of equalities analogous to (2), the only distinction being that the numbers x should be replaced by the n-tuples $(x_1, \ldots, x_n)$. Composite functions such as $f(\varphi(x, y), \psi(x, y, z)) = F(x, y, z)$ where (x, y, z) are triples of numbers belonging to a certain set of triples are also defined in a natural manner in the case of three and, generally, n variables.

Example 1. The expression $u = Ax+By+Cz+D$ where A, B, C and D are given constant real numbers is a linear function of three variables x, y and z. It is defined throughout the three-dimensional space. A more general linear function of n variables $x_1, \ldots, x_n$ is specified by the formula $u =$

$= \sum_{i=1}^{n} a_i x_i + b$ where $a_1, \ldots, a_n$ and b are given constant numbers. This function is defined at any point $(x_1, \ldots, x_n)$ of the n-dimensional space or, as we say, throughout the n-dimensional space.

Example 2. Consider the formula $z = \log \sqrt{1-x^2-y^2}$. It determines a function defined in the domain of the shape of a circle (disc) of radius 1 with centre at (0, 0) with all the boundary points, which are the points of the circumference of the circle, deleted. For the latter points the function is not defined since log 0 does not make sense.

Example 3. The function

$$z = f(x, y) = \begin{cases} 0 & \text{for} \quad y \geqslant 0 \\ 1 & \text{for} \quad y < 0 \end{cases}$$

is represented geometrically by two disjoint parallel half-planes, their disposition with respect to the coordinate system being quite obvious.

A function of one variable can be determined implicitly by an equality

$$F(x, y) = 0 \tag{3}$$

where F is a function of the two variables x and y.

Let a function F be defined on a set G of points (x, y). Equality (3) specifies a subset Ω of the set G on which the function F is equal to zero. Of course the subset Ω can be empty. Suppose that Ω is a nonempty set, and let E be a set (obviously nonempty) of those values (numbers) x to each of which there corresponds at least one y such that the pair (x, y) belongs to Ω. The set E thus consists of all numbers x to each of which there corresponds a nonempty set e_x of numbers y such that $(x, y) \in \Omega$ or, which is the same, such that for the pair (x, y) equality (3) is fulfilled. This determines on the set E a function $y = \varphi(x)$ of x which, in the general case, is many-valued. In such a case we say that the function φ *is defined implicitly* by means of equality (3). It obviously satisfies the identity

$$F(x, \varphi(x)) \equiv 0 \quad \text{for all } x \in E$$

By analogy with the above, we can also define a function $x = \psi(y)$ of the variable y specified implicitly by equality (3). For this function there holds the identity

$$F(\psi(y), y) \equiv 0 \quad \text{for all } y \in E_1$$

where E_1 is a set of numbers y. We also say that the function $y = \varphi(x)$ (or $x = \psi(y)$) satisfies equation (3). The function $x = \psi(y)$ is called the *inverse function* of the function $y = \varphi(x)$.

Example 4. The equation

$$x^2+y^2 = r^2 \tag{4}$$

where $r > 0$ determines implicitly the two-valued function of one variable

$$y = \pm\sqrt{r^2-x^2} \quad (-r \leqslant x \leqslant r)$$

which, by the way, is one-valued for $x = \pm r$.

It is natural to regard this two-valued function as splitting into two continuous one-valued functions $y = +\sqrt{r^2-x^2}$ and $y = -\sqrt{r^2-x^2}$ $(-r \leqslant x \leqslant r)$. The locus formed of their graphs (which are the upper and the lower semi-circles) is the circle of radius r with centre at the origin. This circle is the collection of the points whose coordinates x and y satisfy equation (4).

Let us proceed to the general case of n dimensions. Let a function $F(x_1, \ldots, x_n)$ be defined on a set G of points $(x_1, \ldots, x_n)$ of the n-dimensional space of n-tuples $(x_1, \ldots, x_n)$. The equality

$$F(x_1, x_2, \ldots, x_n) = 0 \tag{5}$$

determines a subset Ω of the set G on which the function F is equal to zero. Let the set Ω be nonempty and let E be the set (also nonempty!) of those n-tuples $(x_1, \ldots, x_{n-1})$ to each of which there corresponds at least one value x_n so that the point $(x_1, \ldots, x_n)$ belongs to Ω. Thus, E is the set of all $(n-1)$-tuples $(x_1, \ldots, x_{n-1})$ to each of which there corresponds a nonempty set $e_{x_1, \ldots, x_{n-1}}$ of numbers x_n such that $(x_1, \ldots, x_n) \in \Omega$ or, which is the same, such that for $(x_1, \ldots, x_n)$ there holds equality (5). This specifies on the set E a function $x_n = \varphi(x_1, \ldots, x_{n-1})$ dependent on $(x_1, \ldots, x_{n-1})$ which is mutiple-valued in the general case. In such a case we say that the function φ is defined implicitly by equality (5). For this function the identity

$$F(x_1, \ldots, x_{n-1}, \varphi(x_1, \ldots, x_{n-1})) \equiv 0, \quad (x_1, \ldots, x_{n-1}) \in E$$

obviously holds.

Elementary Functions

1. *The constant function C.* Here to each real number x there corresponds the value y equal to one and the same number C. The graph of this function (in a rectangular coordinate system) is a straight line parallel to the x-axis passing at a distance of $|C|$ from the x-axis and lying above the x-axis if $C > 0$ and below it if $C < 0$.

2. *The power function* x^n $(n = 0, \pm 1, \pm 2, \ldots)$. For positive integral n the function x^n is defined throughout the real axis. For negative integral n it is defined over the whole real axis with the point $x = 0$ deleted. It should be noted that it is inconvenient to regard 0^0 as a definite number (see § 5.14). Of course, if we deal with the function $y = x^0$ it may turn out to be convenient to put $0^0 = 1$ because, under this convention, this function will possess a continuous graph (which is a straight line parallel to the x-axis) for all values of x.

In Fig. 1.2 we see the graphs of the functions $y = x$, x^2, x^3 and x^4.

3. *A polynomial of the nth degree* is a function of the form

$$P(x) = a_0 + a_1x + \ldots + a_nx^n$$

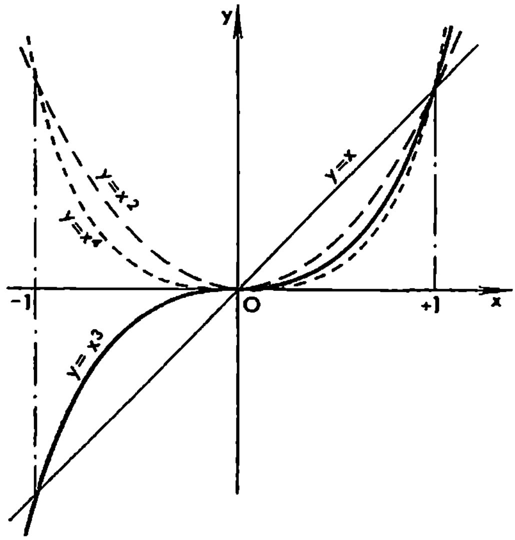

Fig. 1.2

where $a_0, a_1, \ldots, a_n$ are *constant coefficients* and n is a given natural number. A polynomial of degree n is obtained from the constants $a_k (k = 0, \ldots, n)$ and from the functions $x, x^2, \ldots, x^n$ by means of a finite number of arithmetical operations, namely by means of addition, subtraction, and multiplication.

A polynomial is also termed an *entire rational function (of degree n)*. Its domain of definition is the entire real axis.

4. By a *rational function* is meant a function of the type

$$R(x) = \frac{P(x)}{Q(x)}$$

where $P(x) = a_0 + \ldots + a_n x^n$ and $Q(x) = b_0 + b_1 x + b_m x^m$ are some polynomials $(b_m \neq 0)$.

A rational function is defined for all x lying on the real axis for which $Q(x) \neq 0$. Thus, it is not defined only at the zero points (roots) of the polynomial Q, that is at the points x such that $Q(x) = 0$. The number of such points does not exceed the number m.

A rational function is obtained from some constants and from the functions of the form x^k (k is a natural number) by applying to them (a finite number of) arithmetical operations, that is by means of addition, subtraction, multiplication and division.

5. *The power function* x^a (where a is a constant number, not necessarily integral) is known to the reader from elementary mathematics. But in elementary courses of algebra not all properties of this function are given a rigorous justification. For example, the definition of x^a in the case when a is

an irrational number involves some subtle notions of the theory of limits. After the theory of limits has been studied we shall return to the function x^a and give its rigorous definition and the proofs of its properties.

6. The *exponential function* a^x $(a > 0)$. This function is also studied in elementary courses of mathematics. It should be noted that, as in the case of the power function, not all definitions and properties connected with the exponential function are rigorously treated in elementary courses. Therefore later on we shall come back to this function. The inverse function of a^x is the logarithmic function $\log_a x$.

7. *The function* $\sin x$ is known to the reader from trigonometry where it is defined on the basis of geometrical considerations. We shall remind the reader of the definition of $\sin x$. Let us set a number x and lay off the arc length $|x|$ on the circle of radius 1, from the initial point A (Fig. 1.3), in the counterclockwise direction if $x > 0$ and in the clockwise direction if $x < 0$. The arc length is meant to be measured in radians. Let B be the end point of this arc. Then the length of the perpendicular BC to the line OA taken with the sign "+" if B lies above OA and with the sign "−" if B lies below OA is equal to $y = \sin x$.

As the reader knows, the functions $\cos x$, $\tan x$, $\operatorname{cosec} x$, $\sec x$ and $\cot x$ are defined and studied in a similar way.

Proceeding from the trigonometric function we define the inverse *trigonometric functions* Arcsin x, Arccos x, etc.

The functions enumerated in 1-7 are called the *basic elementary functions*. Any function formed of *basic elementary functions* with the aid of the operations of addition, subtraction, multiplication, division and superposition of functions (i.e. forming composite functions) is called an *elementary function* if the number of the indicated operations is finite. It is a function of this type which is said to be specified by a formula.

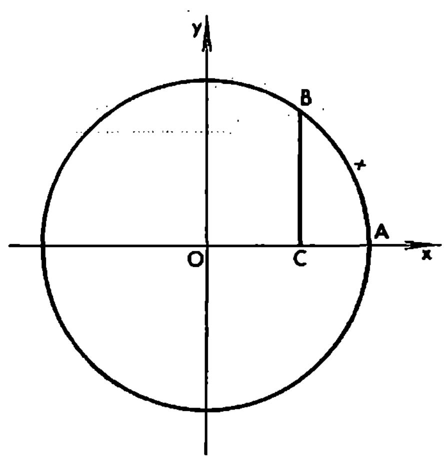

Fig. 1.3

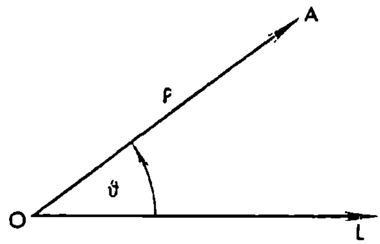

Fig. 1.4

Examples of elementary functions are $\sin x^2$, $(\sin x)^2$, $\tan \log \sqrt{1-x^2}$, $\cos n \arccos x$ and $x^x = a^{x \log_a x}$ $(a > 0)$.

Polar Coordinates. Let us choose in a plane a ray OL (the *polar axis*) issued from a point O (the *pole*; see Fig. 1.4).

Now an arbitrary point A of the plane can be specified by the pair of numbers (ϱ, θ) called its *polar coordinates* where ϱ is the distance from A to O and θ is the angle between OL and OA measured in radians. The pole O is a singular point: it is specified by the pair $(0, \theta)$ where θ can be an arbitrary angle. The angle θ (the *polar angle*) is counted counterclockwise; ϱ is called the *polar radius*. A function $\varrho = f(\theta)$ defined on an open interval (or on a closed interval or on an arbitrary set E) of values of θ can be interpreted as the set of points (ϱ, θ) in the plane for which $\theta \in E$ and $\varrho = f(\theta)$. There are many curves that can be conveniently described in polar coordinates by means of the corresponding functions $\varrho = f(\theta)$ (many-valued or one-valued). Let us consider some examples: (i) the function $\varrho = a^\theta$ $(a > 0, -\infty < \theta < \infty)$ describes in polar coordinates the so-called Archimedes spiral (Fig. 1.5); (ii) the function

$$\varrho = \frac{\varrho_0}{\cos(\theta - \theta_0)}, \quad \theta \in \left(\theta_0 - \frac{\pi}{2}, \theta_0 + \frac{\pi}{2}\right), \quad \varrho_0 > 0$$

describes a straight line such that the perpendicular dropped on it from the pole O is of length ϱ_0 and forms an angle of θ_0 radians with the polar axis

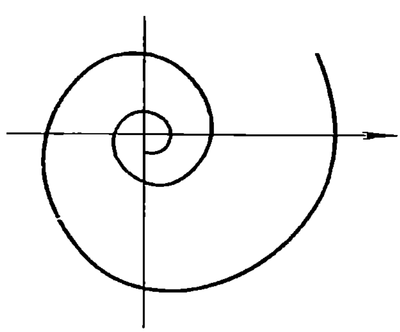

Fig. 1.5

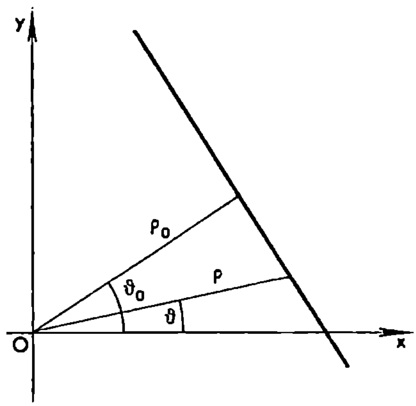

Fig. 1.6

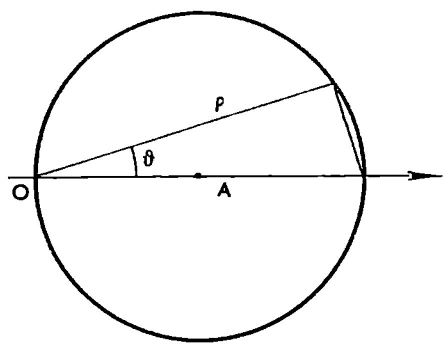

Fig. 1.7

(Fig. 1.6); (iii) the function $\varrho = 2\cos\theta \left(-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}\right)$ describes a circle of radius 1 with centre at the point A (1, 0) (Fig. 1.7).

§ 1.4. The Concept of Continuity of a Function

In Fig. 1.8 we see the graph of a function $y = f(x)$ $(a \leqslant x \leqslant b)$. It is natural to call this graph continuous because it can obviously be drawn by a continuous motion of the pencil without leaving the paper. Let us choose an arbitrary point (number) $x \in [a, b]$. Another point $x' \in [a, b]$ lying close to the former can be written as $x' = x+\Delta x$ where Δx is a positive or negative number called the *increment of the independent variable* x. The difference

$$\Delta f = \Delta y = f(x+\Delta x) - f(x)$$

is called the *increment of the function* f, at the point x, corresponding to the

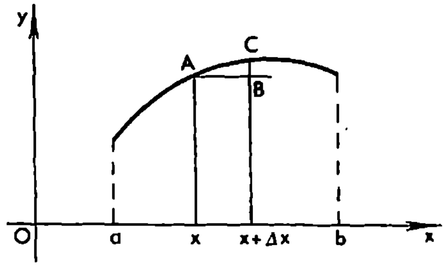

Fig. 1.8

increment Δx of the independent variable x. In Fig. 1.8 the quantity Δy is equal to the length of the line segment BC. Let us make Δx continuously tend to zero; then, for the function in question, Δy will also tend to zero:

$$\Delta y \to 0 \quad (\Delta x \to 0) \tag{1}$$

Now let us consider the graph shown in Fig. 1.9. It consists of two disjoint continuous parts PA and QR. But these parts themselves are not connected continuously and therefore it appears natural to call this graph discontinuous. For this graph to represent a one-valued function $y = F(x)$ at the point x_0 let

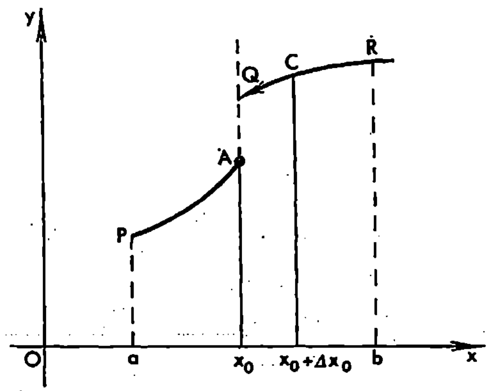

Fig. 1.9

us agree that $F(x_0)$ is equal to the length of the line segment joining A and x_0; to indicate this convention we have shown A in the figure as bold-face point while the point Q is shown as placed at the tip of an arrow, which symbolizes that the latter point does not belong to the graph. If the point Q belonged to the graph the function F would be two-valued at the point x_0.

Now let us give x_0 an increment Δx_0 and compute the corresponding increment of the function:

$$\Delta F = F(x_0+\Delta x_0)-F(x_0)$$

If Δx_0 is made to tend to zero continuously we can no longer assert that ΔF also tends to zero. This is the case for the negative Δx_0 tending to zero while for the positive Δx_0 this is not so; as is seen in the figure, if Δx_0 tends to zero retaining positive sign the corresponding increment ΔF tends to a positive number equal to the length of the line segment AQ.

These considerations lead us in a natural way to the following definition (suggested by A. Cauchy). *A function f defined on an interval $[a, b]$ is said to be continuous at a point x belonging to that interval if the increment of the function at the point x, corresponding to the increment Δx* of the argument, tends to zero as Δx is made to tend to zero in an arbitrary way.* This property of continuity of f at x is expressed by relation (1) and can also be written as

$$\lim_{\Delta x \to 0} \Delta y = 0 \tag{2}$$

Relation (2) reads: the limit of Δy is equal to zero as Δx tends to zero according to any law. By the way, "according to any law" is usually omitted but is of course tacitly implied.

If a function f defined on $[a, b]$ is not continuous at a point $x \in [a, b]$, that is if property (2) does not hold at x for at least one way of making Δx tend to zero, the function f is said to be *discontinuous* at the point x.

The function whose graph is shown in Fig. 1.8 is continuous at any point $x \in [a, b]$ while the function with the graph depicted in Fig. 1.9 is obviously continuous at any point $x \in [a, b]$ except at the point x_0 since for the latter point relation (2) does not hold when Δx tends to zero retaining positive sign.

The above definition of continuity is itself stated quite properly but the explanation we have given is based only on an intuitive idea of the concept of a limit. After the theory of limits has been presented this definition, which can also be extended to the case of many independent variables, will receive full justification.

A function continuous at every point of a closed (or open) interval is said to be *continuous on that interval.*

The continuity of a function expresses mathematically a property which is often encountered in practice: to a small increment of an independent variable there also corresponds a small increment of another variable (function) dependent on the former. Perfect examples of continuous functions are various laws of motion of physical bodies $s = f(t)$ expressing the dependence of the paths s travelled by the bodies in time t. Time and space are continuous, and a given law of motion $s = f(t)$ establishes between them a certain continuous dependence whose characteristic feature is that to a small increment of time there corresponds a small increment of the path.

We come to the abstract notion of continuity observing the so-called continuous media, i.e. rigid bodies, liquids and gases, for instance, such as metals, water or air. Every real physical medium is in fact a collection of a large

* Here is meant an increment Δx such that $x + \Delta x \in [a, b]$.

number of discrete moving particles. But these particles and the distances between them are so small, compared with the volumes occupied by the media we deal with in considering macroscopic physical phenomena, that many of such phenomena can be approximated sufficiently well if we consider the masses of the media in question as being distributed in space continuously without any "gaps" between the particles. Such an approach forms the basis of many physical disciplines, for instance, hydrodynamics, aerodynamics and the theory of elasticity. The mathematical concept of continuity naturally plays an important role in these and many other sciences.

Continuous functions constitute the basic class of functions dealt with in mathematical analysis.

Examples of continuous functions are the elementary functions defined in § 1.3. They are all continuous on the intervals of variation of x where they are defined.

Discontinuous functions which are also studied in mathematics describe jump-like processes observed in reality. For instance, the velocities of the colliding bodies change in a jump-like manner. Many processes of qualitative transition are accompanied with jump phenomena. For example, the dependence $Q = f(t)$ between the temperature t of 1 g of water (ice) and the number Q of calories of heat contained in it corresponding to the variation of t from $-10°$ to $+10°$ is described, if we agree, conditionally, that $Q = 0$ for $t = -10°$, by the formulas

$$Q(t) = \begin{cases} 0.5t+5 & -10 \leqslant t < 0 \\ t+85 & 0 < t < 30 \end{cases}$$

Here we have assumed that the specific heat of ice is equal to 0.5. The function $Q = f(t)$ is not defined in a unique manner (that is it is multiple-valued) for

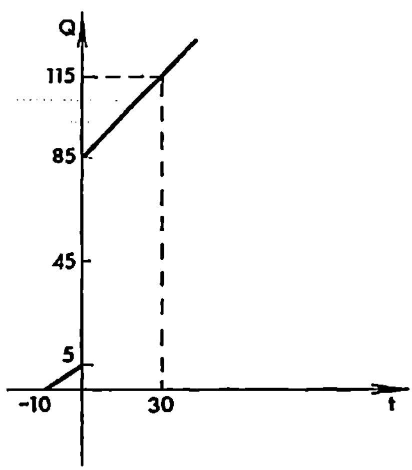

Fig. 1.10

$t = 0$. For the sake of convenience we can agree that at the point $t = 0$ it takes on a definite value, for instance, $f(0) = 45$. The function $Q = f(t)$ in question is obviously discontinuous at the point $t = 0$; its graph is shown in Fig. 1.10.

§ 1.5. Derivative

The notion of the derivative appears in the solution of the two old problems of drawing the tangent line to a curve and computing the velocity of a nonuniform motion. Mathematicians have been interested in these problems and also in the problem of computing the area of a curvilinear figure since ancient times. In the 17th century I. Newton and G. W. Leibniz completed in a certain sense the solution of these problems. They created the general methods of differentiation and integration of functions and proved an important theorem (known as the Newton-Leibniz theorem) establishing a close relationship between the operations of differentiation and integration. But it should be noted that the modern understanding of these questions differs essentially from that in the times of Newton and Leibniz. In the considerations and notions used in those times there were many obscure and ambiguous ideas; the mathematicians of those times themselves realized this situation, which is testified by their heated argument about these questions.

Modern mathematical analysis is based on the concept of a limit which assumed its rigorous and clear form comparatively not long ago—in the first half of the 19th century. The prominent French mathematician A. Cauchy should be particularly mentioned among those who contributed much to the development of this theory.

The concept of a limit is essentially applied in the definitions of the notions of continuity, derivative and integral.

Instantaneous velocity. Let a point move in a straight line and let a function $s = f(t)$ express the dependence of its distance s, from an initial point O, taken with the corresponding sign*, on time $t \in (a, b)$. At time moment $t \in (a, b)$ the point is at a distance of $s = f(t)$ from O. But at another time moment $t+\Delta t \in (a, b)$ $(\Delta t \neq 0)$ it is at a distance $s+\Delta s = f(t+\Delta t)$ from O. Its average velocity during the time interval $(t, t+\Delta t)$ is expressed as

$$v_{av} = \frac{\Delta s}{\Delta t} = \frac{f(t+\Delta t)-f(t)}{\Delta t}$$

The *instantaneous* ("true") *velocity* v of the point at time moment t can naturally be defined as the limit to which v_{av} tends as $\Delta t \to 0$, that is

$$v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}$$

* More precisely, s is the coordinate of the moving point of the line on which an initial point O (the origin), a unit length and a positive direction have been indicated.

The average current intensity during this time interval is expressed as

$$I_{av} = \frac{\Delta Q}{\Delta t} = \frac{f(t+\Delta t)-f(t)}{\Delta t}$$

The limit of this ratio as $\Delta t \to 0$ gives us the current intensity at the moment t:

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t}$$

Derivative. All the three problems we have considered lead us to one and the same mathematical operation on a certain function although they belong to three different branches of knowledge—mechanics, geometry and the theory of electricity. This operation reduces to finding the limit of the ratio of the increment of a function to the corresponding increment of the independent variable as the latter tends to zero. The list of problems reducing to the same operation can be easily extended. In particular, the problem of computing the reaction rate or the problem of finding the density of a nonuniform mass distribution also leads to it.

It appears natural that this operation is subject to a careful study in mathematics and is called by a special name: the operation of *differentiation of a function*. The result of this operation is called the *derivative*.

Thus, *the derivative of a function f, defined on an interval (a, b), at a point x of that interval, is the limit to which tends the ratio of the increment of the function f, at this point, to the corresponding increment of the argument as the latter increment tends to zero.* To denote the derivative we use the following symbol*:

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$$

Some other types of notation are also used, for instance, y', $\frac{dy}{dx}$ and $\frac{df(x)}{dx}$. The expedience of these symbols and their usage in various cases will become clear to the reader later on.

The results of the examples considered above can now be stated as follows:

The velocity of a point which covers a path s whose dependence on time t is expressed by a function $s = f(t)$ is equal to the derivative of this function with respect to time: $s' = f'(t)$.

The tangent of the angle β between the tangent line to a curve, represented by a function $y = f(x)$, at a point with abscissa x, and the positive direction of the x-axis is equal to the derivative $f'(x)$.

* If we consider the limit $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ only for $\Delta x > 0$ or only for $\Delta x < 0$ the corresponding value of the limit is respectively called the *right-hand* or the *left-hand derivative* of f at the point x. When a function f defined on a closed interval $[a, b]$ is said to have a derivative at every point of that interval it is meant that the function possesses a derivative at every point of the open interval (a, b) and, besides, has a right-hand derivative at the point a and a left-hand derivative at the point b.

If a function $Q = f(t)$ expresses the charge of electricity passing through a section of a conductor in time t, the intensity of current I in the conductor at a time moment t is equal to the derivative of Q with respect to t: $I = Q' = f'(t)$.

Some formulas. For any natural $n = 1, 2, \ldots$ we have

$$(x^n)' = nx^{n-1} \tag{1}$$

Indeed, denoting $\Delta x = h$ we can write, according to Newton's binomial formula (see also the remark at the end of this section),

$$\begin{aligned}(x^n)' &= \lim_{h\to 0} \frac{\Delta y}{h} = \lim_{h\to 0} \frac{(x+h)^n - x^n}{h} = \\ &= \lim_{h\to 0} \frac{1}{h}\left(x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \ldots + h^n - x^n\right) = \\ &= nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}\lim_{h\to 0} h + \ldots + \left(\lim_{h\to 0} h\right)^{n-1} = nx^{n-1}\end{aligned}$$

where we have again used some elementary properties of limits which will be justified later on.

There also hold the formulas

$$(\sin ax)' = a\cos ax \tag{2}$$

$$(\cos ax)' = -a\sin ax \tag{3}$$

where a is a constant. We shall prove the first of these equalities leaving the other to the reader:

$$\lim_{h\to 0} \frac{\sin a(x+h) - \sin ax}{h} = \lim_{h\to 0}\left[a\frac{\sin\frac{ah}{2}}{\frac{ah}{2}}\cos\left(ax + \frac{ah}{2}\right)\right] =$$

$$= a\lim_{h\to 0}\frac{\sin\frac{ah}{2}}{\frac{ah}{2}}\lim_{h\to 0}\cos\left(ax + \frac{ah}{2}\right) = a\cdot 1\cdot\cos ax = a\cos ax$$

In the last equality we have used the property $\lim_{\alpha\to 0}\frac{\sin\alpha}{\alpha} = 1$ and the fact that the function $\cos ax$ is continuous. Both assertions will be proved later on (see §§ 4.2 and 4.9).

The derivative of a function $f(x)$ is in its turn a function $f'(x)$ of the variable x and therefore we can try to differentiate $f'(x)$ as well. If the derivative of $f'(x)$ exists it is called the *second derivative* (the *derivative of the second order*) of $f(x)$ and is denoted as $f''(x)$.

The *derivatives of higher orders* $f^{(n)}(x)$ of $f(x)$ where n is any natural number are defined in a similar manner ($f^{(n)}(x)$ is called the nth derivative or the *derivative of the nth order* of $f(x)$).

The *second derivative* of a function $s = f(t)$ expressing the law of motion of a point in a straight line is obviously equal to the acceleration of the point at time moment t.

What has been said indicates that the notion of the derivative plays an extremely important role in applied sciences and is one of the fundamental notions of mathematics. This will be confirmed later on.

It should be noted that a *constant number* C regarded as a function of x (see § 1.3) *possesses the zero derivative at all the points* x (i.e. *its derivative is identically equal to zero for all* x). Indeed,

$$f(x) = C, \quad f(x+\Delta x) = C, \quad C' = \lim_{\Delta x \to 0} \frac{C-C}{\Delta x} = \lim_{\Delta x \to 0} 0 = 0 \tag{4}$$

The converse is also true: *if it is known that a function $f(x)$ has a derivative identically equal to zero then*

$$f'(x) \equiv 0$$

that is the function is equal to a constant. The rigorous mathematical proof of this simple assertion involves some comparatively complicated techniques, and we shall discuss them later on (see § 5.8). On the other hand, this is quite obvious from the point of view of mechanics: if, for instance, a function $s = f(t)$ expresses a law of motion of a point in a line and if it is known that the velocity of the point is identically equal to zero, that is $v = f'(t) \equiv 0$, it is evident that the point is in a state of rest and its distance s from the origin O is constant for any t. In this argument we have replaced x by t but this is inessential since the time can be denoted by x as well.

We also note that if two functions $u(x)$ and $v(x)$ possess derivatives at a point x and A and B are constant numbers, then the function

$$f(x) = Au(x)+Bv(x) \tag{5}$$

also possesses a derivative equal to

$$f'(x) = Au'(x)+Bv'(x) \tag{6}$$

Indeed,

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{Au(x+h)+Bv(x+h)-[Au(x)+Bv(x)]}{h} =$$

$$= \lim_{h\to 0} \left(A\frac{u(x+h)-u(x)}{h}\right) + \lim_{h\to 0} \left(B\frac{v(x+h)-v(x)}{h}\right) =$$

$$= A\lim_{h\to 0} \frac{u(x+h)-u(x)}{h} + B\lim_{h\to 0} \frac{v(x+h)-v(x)}{h} = Au'(x)+Bv'(x) \tag{7}$$

In the second equality in this chain of equalities we have used the fact that the limit of a sum is equal to the sum of the corresponding limits and in the third equality we have relied on the rule that a constant can be taken outside the sign of limit.

By induction we can readily prove the general assertion*

$$\left(\sum_{j=1}^{n} a_j u_j(x)\right)' = \sum_{j=1}^{n} a_j u_j'(x)$$

where a_j are constant numbers and the functions $u_j(x)$ are supposed to have derivatives.

In particular, for the derivative of a polynomial we obtain

$$\left(\sum_{k=0}^{n} a_k x^k\right)' = \sum_{1}^{n} k a_k x^{k-1}$$

(a_k are constants).

Remark. The formula $(x^n)' = nx^{n-1}$ $(n = 1, 2, \ldots)$ can be proved without resorting to Newton's binomial formula. For $n = 1$ we have

$$x' = \lim_{h\to 0} \frac{x+h-x}{h} = 1 = x^0$$

If we now suppose that the formula $(x^{n-1})' = (n-1)x^{n-2}$ $(n = 2, 3, \ldots)$ is valid we readily obtain (see § 5.1 (5))

$$(x^n)' = (xx^{n-1})' = x'x^{n-1} + x(x^{n-1})' = nx^{n-1}$$

We also advise the reader to look through formulas (4)-(9) in § 5.1 and the table in § 5.5 since they can be of use to the reader before he reaches these sections in the gradual study of the subject-matter.

§ 1.6. Antiderivative. Indefinite Integral

Let f be a continuous function defined on an open interval (a, b). By definition, a function F is called an *antiderivative* of f on the interval (a, b)** if the derivative of F is equal to f on that interval:

$$F'(x) = f(x) \qquad (x \in (a, b))$$

It is quite obvious that if F is an antiderivative of f on (a, b) then the function $F(x)+C$ where C is a constant is also an antiderivative of f because

$$(F(x)+C)' = F'(x)+C' = F'(x) = f(x)$$

Conversely, if F and F_1 are two antiderivatives of $f(x)$ on (a, b) their difference is necessarily equal to a constant C throughout the interval (a, b):

$$F_1(x) = F(x)+C \tag{1}$$

* Here we use the notation

$$\alpha_1+\alpha_2+\ldots+\alpha_n = \sum_{j=1}^{n} \alpha_j = \sum_{1}^{n} \alpha_j$$

** An antiderivative of f on a closed interval $[a, b]$ is defined similarly (but in this connection one should take into account the footnote on p. 34).

Indeed, $(F_1(x)-F(x))' = F_1'(x)-F'(x) = f(x)-f(x) \equiv 0$. Now remember (see the foregoing section) that $(F_1(x)-F(x))' \equiv 0$ implies that there is a (constant) number C such that $F_1(x)-F(x) = C$ on (a, b) whence follows (1).

Thus, using purely mechanical considerations we have established the important fact that *if F is an antiderivative of f on an interval (a, b) then all the possible antiderivatives of f on that interval are expressed by the formula $F(x)+C$ where any number can be substituted for C.*

Now we state the following definition.

The *indefinite integral* of a continuous function f defined on an interval (a, b) is an expression which gives the general form of all the antiderivatives of f, that represents any arbitrary antiderivative. The indefinite integral is denoted as

$$\int f(x)\,dx$$

It follows from what has been said that *if F is a certain antiderivative of f on an interval (a, b) then the indefinite integral of f on that interval is written as*

$$\int f\,dx = F(x)+C \tag{2}$$

where C is an appropriately chosen constant.

If f_1 and f_2 are continuous functions on an interval (a, b) and A_1 and A_2 are constants then there holds the following equality expressing the basic property of the indefinite integral:

$$\int (A_1 f_1(x)+A_2 f_2(x))\,dx = A_1 \int f_1(x)\,dx + A_2 \int f_2(x)\,dx + C \tag{3}$$

where C is a constant.

Indeed, by the definition of the indefinite integral, the left member of (3) is one of the antiderivatives of $A_1 f_1(x)+A_2 f_2(x)$. On the other hand we have the equality

$$\begin{aligned}\left(A_1 \int f_1(x)\,dx + A_2 \int f_2(x)\,dx\right)' &= A_1\left(\int f_1(x)\,dx\right)' + A_2\left(\int f_2(x)\,dx\right)' = \\ &= A_1 f_1(x)+A_2 f_2(x)\end{aligned} \tag{4}$$

because the integrals $\int f_1\,dx$ and $\int f_2\,dx$ denote, respectively, some antiderivatives of f_1 and f_2. Therefore the right member of (3), taken without the last term C, is also an antiderivative of $A_1 f_1(x)+A_2 f_2(x)$ and hence it differs from the left member of (3) by a constant.

By induction, we readily extend property (3) to any number of continuous functions $f_1, \ldots, f_n$ defined on the interval (a, b) and any constants $A_1, \ldots, A_n$:

$$\int \left(\sum_{j=1}^{n} A_j f_j(x)\right) dx = \sum_{j=1}^{n} A_j \int f_j(x)\,dx + C \tag{5}$$

As consequences of (5), we obtain for $A_1 = 1$, $A_2 = \pm 1$ and $n = 2$ the equality

$$\int (f_1 \pm f_2)\, dx = \int f_1\, dx \pm \int f_2\, dx + C$$

and for $A_1 = A$, $A_2 = 0$ and $f_1 = f$ the equality

$$\int Af\, dx = A \int f\, dx + C$$

Examples

$$\int x^{n-1}\, dx = \frac{x^n}{n} + C, \qquad n = 1, 2, \ldots \tag{6}$$

$$\int \cos ax\, dx = \frac{\sin ax}{a} + C, \qquad a \neq 0 \tag{7}$$

$$\int \sin ax\, dx = -\frac{\cos ax}{a} + C, \qquad a \neq 0 \tag{8}$$

Indeed (see §1.5 (1), (2), (3)) we have

$$\left(\frac{x^n}{n}\right)' = \frac{1}{n}(x^n)' = x^{n-1}$$

$$\left(\frac{\sin ax}{a}\right)' = \frac{1}{a}(\sin ax)' = \frac{1}{a} a \cos ax = \cos ax$$

$$\left(-\frac{\cos ax}{a}\right)' = -\frac{1}{a}(\cos ax)' = \sin ax$$

From (5) and (6) it follows that the indefinite integral of a polynomial $P_n(x) = \sum_0^n a_k x^k$ of degree n (a_k are constants) is expressed by the formula

$$\int P_n(x)\, dx = \sum_0^n a_k \frac{x^{k+1}}{k+1} + C$$

§ 1.7. The Concept of Definite Integral. The Area of a Curvilinear Figure

Let us take a nonnegative continuous function $f(x)$ defined on an interval $[a, b]$ (a and b are finite numbers). Its graph is shown in Fig. 1.14. Now we pose the following problem: it is required to define the notion of the area of the figure bounded by the curve $y = f(x)$, by the x-axis and by the straight lines $x = a$ and $x = b$ and to compute this area. It is natural to solve this problem as follows.

Let us partition the closed interval $[a, b]$ into n parts by means of points of division

$$a = x_0 < x_1 < \ldots < x_n = b \tag{1}$$

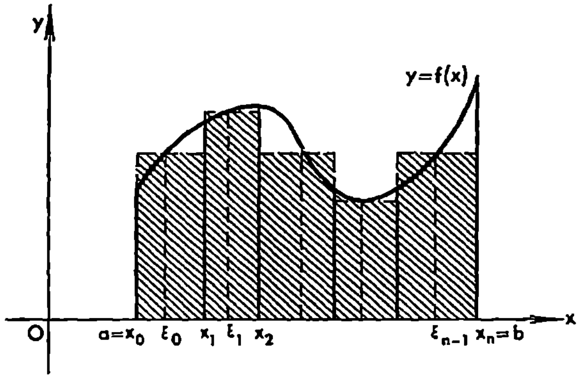

Fig. 1.14

and choose on each of the subintervals

$$[x_j, x_{j+1}] \qquad (j = 0, 1, \ldots, n-1) \tag{2}$$

thus obtained an arbitrary point $\xi_j \in [x_j, x_{j+1}]$. Next we find the values $f(\xi_j)$ of the function f at these points and form the sum

$$S_n = \sum_0^{n-1} f(\xi_j)\, \Delta x_j, \qquad (\Delta x_j = x_{j+1} - x_j) \tag{3}$$

which is termed an *integral sum* and which is obviously equal to the sum of the areas of the shaded rectangles in Fig. 1.14.

Now let us make all Δx_j's tend to zero in such a way that the maximum subinterval (i.e. that having the greatest length) tends to zero. If in this limiting process the sum S_n tends to a definite limit S independent of the ways in which partitions (1) are made and of the choice of the points ξ_j on the subintervals, it appears natural to regard the quantity S as the *area of the curvilinear figure in question*. Hence,

$$S = \lim_{\max \Delta x_j \to 0} \sum_0^{n-1} f(\xi_j)\, \Delta x_j \tag{4}$$

We have thus stated the definition of the area of the curvilinear figure. Now there arises the question as to whether every such figure possesses area, that is whether the corresponding integral sum S_n does in fact tend to a finite limit when $\max |\Delta x_j| \to 0$. Later on we shall show that the answer to the question is affirmative: every curvilinear figure of the type defined above which corresponds to a continuous function $f(x)$ does in fact have area, in the sense of the definition we have stated, which is expressed by a number S thus depending on that figure.

Another important question is to what extent this definition is natural from the point of view of human experience. As usual, such a question should be answered on the basis of practice. Here we limit ourselves to saying that this definition is really justified by practice. In what follows we shall many times have a chance to see that the stated definition is expedient.

Now let us examine expression (4) itself abstracting from the concrete conditions of the problem of computing area. We can regard (4) as an operation by means of which a certain number S is found from the given function f defined on $[a, b]$. This operation is called *definite integration*, and its result, provided it exists, is called the *definite integral* of f on $[a, b]$ and is written as

$$S = \lim_{\max \Delta x_j \to 0} \sum_{j=0}^{n-1} f(\xi_j)\,\Delta x_j = \int_a^b f(x)\,dx \tag{5}$$

Thus, by definition, *the definite integral of the function f on the interval $[a, b]$ is the limit of integral sum* (4) *as the maximum subinterval of partition* (1) *tends to zero.*

In this definition which is no longer connected with the problem of computing area we do not suppose that the function f is necessarily continuous and nonnegative on $[a, b]$. It should be noted that the definition does not assert the existence of the definite integral for any function f defined on $[a, b]$, that is it does not guarantee the existence of limit (5). It only says that if this limit exists for the given function f defined on the interval $[a, b]$ it is called the definite integral of f on $[a, b]$.

We must also take into account that when this limit S is said to exist it is meant to be independent of the ways in which the interval $[a, b]$ is partitioned and of the choice of the points ξ_j belonging to the subintervals obtained. For instance, if it is known that the definite integral $S = \int_0^1 f(x)\,dx$ of a function f on the interval $[0, 1]$ exists, it can be, for example, found by evaluating the limit

$$S = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f\left(\frac{j}{n}\right)$$

of the integral sums corresponding to the partitions of $[0, 1]$ into n equal parts with the aid of the points $x_j = j/n$ $(j = 0, 1, \ldots, n)$ and to the choice of the left end points of the subintervals of the partitions as the points ξ_j.

At the same time this number S can also be obtained as the limit

$$S = \lim_{n\to\infty} \sum_{j=0}^{n-1} f\left(\left(\frac{j+1}{n}\right)^2\right)\left[\left(\frac{j+1}{n}\right)^2 - \left(\frac{j}{n}\right)^2\right]$$

of the integral sums corresponding to the partitions of $[0, 1]$ by means of the points $x_j = (j/n)^2$ $(j = 0, 1, \ldots, n-1)$ and to the choice of the right end points of the subintervals of the partitions as ξ_j. In this case the length of the jth subinterval satisfies the relations

$$\left(\frac{j+1}{n}\right)^2 - \left(\frac{j}{n}\right)^2 = \frac{2j+1}{n^2} \leqslant \frac{2(n-1)+1}{n^2} = \frac{2n-1}{n^2} = \frac{2}{n} - \frac{1}{n^2} \to 0 \quad (n \to \infty)$$

showing that the maximum of the subintervals (the rightmost) is of length which tends to zero together with an indefinite increase of n.

In the theory of the definite integral it is proved that every function continuous on a closed interval $[a, b]$ is *integrable* on that interval, that is limit (5) exists for it. This implies that, as was mentioned, every figure of the type under consideration possesses area.

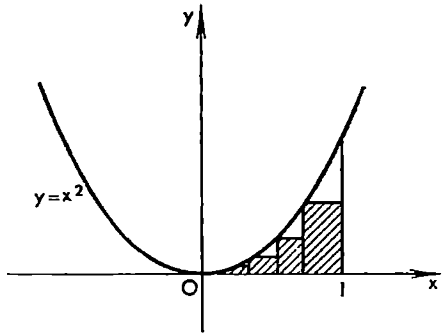

Fig. 1.15

Example. The area S bounded by the parabola $y = x^2$, the x-axis and the straight line $x = 1$ (see Fig. 1.15) is equal to

$$S = \lim_{n\to\infty} \frac{1}{n} \sum_0^{n-1} \left(\frac{k}{n}\right)^2 = \lim_{n\to\infty} \frac{1}{n^3} \sum_0^{n-1} k^2 = \lim_{n\to\infty} \frac{1}{n^3}\, \frac{2n^3 - 3n^2 - n}{6} =$$

$$= \lim_{n\to\infty} \left(\frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}\right) = \frac{1}{3}$$

We have shown that the integral sum of the function $y = x^2$ corresponding to the partition of $[0, 1]$ into equal parts tends to the number $1/3$.

It is by far not so simple to prove by using elementary methods that the sum $\sum\limits_0^{n-1} (x_j)^2 \Delta x_j$ corresponding to an arbitrary partition of $[0, 1]$ tends to $1/3$ when $\max \Delta x_j \to 0$. But this assertion is a consequence of the general proposition stated above which guarantees the existence of the definite integral of every continuous function on any (finite) closed interval.

Let us consider some other examples of practical problems whose solution reduces to the evaluation of definite integrals.

Work. Let a point move in a straight line and let $F = f(x)$, where $f(x)$ is a continuous function of the abscissa x of the moving point, be a variable force directed along that line and applied to that point. The work performed by the force F as the point moves from a to b is equal to

$$W = \lim_{\max \Delta x_j \to 0} \sum_{j=0}^{n-1} f(x_j)\, \Delta x_j = \int_a^b f(x)\, dx$$

where $a = x_0 < x_1 < \ldots < x_n = b$, $\Delta x_j = x_{j+1} - x_j$. Indeed, by the continuity of f, the product $f(x_j)\,\Delta x_j$ is close to the true work performed on $[x_j, x_{j+1}]$ and the sum of these products is close to the true work, on the whole path $[a, b]$, and tends to the latter as $\max_j \Delta x_j$ decreases.

The mass of a rod with variable density. Let us suppose that an interval $[a, b]$ of the x-axis carries a mass distributed with a variable linear density $\varrho(x) \geqslant 0$ where $\varrho(x)$ is a function continuous on $[a, b]$.

The total mass of this interval (of the rod) is expressed by the integral:

$$M = \lim_{\max \Delta x_j \to 0} \sum_{0}^{n-1} \varrho(x_j)\,\Delta x_j = \int_a^b \varrho(x)\,dx \tag{6}$$

$$a = x_0 < x_1 < \ldots < x_n = b, \qquad \Delta x_j = x_{j+1} - x_j$$

The direct computation of a definite integral by means of formula (5) is connected with essential difficulties because the integral sums of even slightly complex functions usually have extremely lengthy and complicated expressions and it is often difficult to reduce them to a form admitting of the computation of the limits because there are no general methods for doing this. It is interesting to note that Archimedes was the first to solve a problem of this kind. Using some considerations which to a certain extent resemble the modern method of limits he managed to evaluate the area of a parabolic segment. Since then many mathematicians have been solving various problems on the computation of areas of figures and volumes of solids. Nevertheless, even in the 17th century the formulation of such problems and the methods of their solution were of particularly special character. An essential achievement was made by Newton and Leibniz who elaborated the general method for solving such problems. They showed that the computation of the definite integral of a function can be reduced to finding its antiderivative.

Suppose that $f(x)$ is a continuous function defined on $[a, b]$, and let $F(x)$ be its antiderivative. The *Newton-Leibniz theorem* asserts that there holds the equality

$$\int_a^b f(x)\,dx = F(b) - F(a) \tag{7}$$

showing that if an antiderivative F of the function f is known the evaluation of the definite integral of f on $[a, b]$ reduces to the simple substitution of a and b into F.

The proof of this theorem will be given in § 9.9; here we shall present its simple mechanical interpretation. Let us regard x as time and suppose that a function $y = F(x)$ expresses the law of motion of a point in a line, that is let y be the distance, taken with the corresponding sign, from the moving point to a fixed point (the origin) at time moment x.

The path travelled by the point* during the time period $a \leqslant x \leqslant b$ is obviously equal to

$$A = F(b) - F(a) \tag{8}$$

On the other hand, the path can be evaluated by integrating the velocity $f(x) = F'(x)$ of the point:

$$A = \lim_{\max \Delta x_j \to 0} \sum_{j=0}^{n-1} f(x_j)\, \Delta x_j = \int_a^b f(x)\, dx \tag{9}$$

Indeed, the product $f(x_j)\, \Delta x_j$ is an approximation to the path travelled by the point during the time interval $[x_j, x_{j+1}]$ where x_j are defined as in formula (1). Therefore (8) and (9) imply (7).

Examples

$$\int_a^b x^{n-1}\, dx = \frac{x^n}{n}\Big|_a^b = \frac{1}{n}(b^n - a^n) \qquad (n \neq 0)$$

$$\int_a^b \cos \alpha x\, dx = \frac{\sin \alpha x}{\alpha}\Big|_a^b = \frac{1}{\alpha}(\sin \alpha b - \sin \alpha a) \qquad (\alpha \neq 0)$$

Here the notation $F(x)\Big|_a^b$ symbolizes the expression $F(b) - F(a)$, that is $F(x)\Big|_a^b = F(b) - F(a)$.

The number of such examples can be essentially increased after the reader has studied §§ 5.1-5.5 which treat of the basic differentiation techniques for elementary functions.

* It should be noted that the expression the "path travelled by the point" does not describe sufficiently precisely the phenomenon in question. For instance, if the law of motion is such that the point first moves to the right travelling a path A_1 and then moves to the left travelling a path A_2 then $A = A_1 - A_2$.

CHAPTER 2

Real Numbers

§ 2.1. Rational and Irrational Numbers

In this chapter we give a review of the basic properties (expressed by axioms) of real numbers. In elementary courses of mathematics, where only arithmetical operations on constant numbers are considered, not all of these properties are adequately treated and therefore it is expedient to consider them here in more detail. These properties are particularly important for the study of *variable numbers* traditionally called *variables*. The investigation of functions involves all the properties of real numbers, not only those treated in elementary mathematics.

The integers

$$\ldots, -2, -1, 0, 1, 2, \ldots \tag{1}$$

can be added together, subtracted from one another and multiplied by each other, the results of the operations being again integers.

The rational numbers will be written as $\pm p/q$ $(+p/q = p/q)$ where p and q are integers, $p \geqslant 0$ (i.e. p is greater than or equal to zero) and $q > 0$. Thus, unless otherwise stated, we regard p and q in an expression p/q as nonnegative integers. By definition, two rational numbers $\pm p_1/q_1$ and $\pm p_2/q_2$ are equal if and only if they have common sign and $p_1q_2 = q_1p_2$. The expression $\pm 0/q$ specifies one and the same number 0 irrespective of the sign "+" or "−" and of the number q. Two nonnegative numbers p_1/q_1 and p_2/q_2 are connected by the relation

$$\frac{p_1}{q_1} < \frac{p_2}{q_2}$$

if $p_1q_2 < q_1p_2$. It is well known how rational numbers with arbitrary signs are compared with each other and how the four arithmetical operations on them are performed and we shall not remind the reader of them*.

For practical calculations it is quite sufficient to use only rational numbers. But numbers are also needed for measuring geometrical and physical magni-

* The reader is supposed to be familiar with basic properties of rational numbers presented in elementary courses and therefore here we confine ourselves to a brief review without presenting the proofs.

tudes such as lengths of line segments, areas, volumes, temperatures and the like. Here we mean not practical approximate measurements but the precise (theoretical) representation of these quantities by numbers, and it is impossible to give a theoretical justification of such measurements using only rational numbers. For example, let us consider a line segment representing the hypotenuse of a right isosceles triangle with unit legs. If we suppose that the length of this line segment is expressible by a positive rational fraction p/q (we suppose that it is in its lowest terms) the area of the square constructed on it will be equal to p^2/q^2 while the areas of the two squares constructed on the legs of the triangle will be equal to 1. By Pythagoras' theorem we then obtain the equality $p^2 = 2q^2$.Its right member is an integer divisible by 2 and hence the left member must also be divisible by 2, and therefore p is an even number. It follows that the left member is divisible by 4; hence q^2 is divisible by 2, and so is q. Hence p and q have a common factor 2 which contradicts the assumption that the fraction p/q is in its lowest terms. Thus, there exist line segments whose lengths are inexpressible by rational numbers. Such line segments are said to be *incommensurable with unit length.* To represent their lengths* we need new numbers called *irrational.* This leads, for instance, to the appearance of the number $\sqrt{2}$ expressing the length of the hypotenuse of the right triangle considered above.

There are different ways of formal (logical) introduction of irrational numbers. Here we shall show how they can be defined by means of infinite decimals.

Let us take an arbitrary positive rational number p/q. Applying the well-known rules of arithmetic we can bring it to the form of a decimal fraction. This results in

$$\begin{array}{l} q\,\overline{)\,p\qquad}\;\alpha_0.\,\alpha_1\alpha_2\ldots \\ \quad\underline{} \\ \quad\;\beta_0 \\ \quad\underline{} \\ \qquad\beta_1 \\ \qquad\ldots \end{array} \tag{2}$$

where α_0 is a nonnegative integer and α_k $(k = 1, 2, \ldots)$ are decimal digits.

We shall write

$$\frac{p}{q} = \alpha_0.\,\alpha_1\alpha_2\ldots = +\alpha_0.\,\alpha_1\alpha_2\ldots \tag{3}$$

and call the decimal fraction on the right-hand side of (3) the *decimal representation of the number* p/q.

It can readily be shown that the decimal representation of a positive rational number is independent of the way in which the latter is written, that is if p and q in (2) are replaced, respectively, by p_1 and q_1 where $pq_1 = p_1q$ the resultant decimal representation $\alpha_0.\,\alpha_1\alpha_2\ldots$ is exactly the same.

---

* *A priori* the length and the positive number are different notions but they are closely interrelated by the so-called isomorphism (see § 2.7).

As is well known from arithmetic, procedure (2) can lead to a decimal representation of only one of the following two types; it must be either a *finite* (*terminating*) decimal

$$\frac{p}{q} = \alpha_0.\alpha_1\ldots\alpha_m = \alpha_0.\alpha_1\ldots\alpha_m 00\ldots \quad (\alpha_m > 0) \tag{4}$$

or an *infinite* (also called *nonterminating*) *repeating* (or *periodic*) decimal

$$\frac{p}{q} = \alpha_0.\alpha_1\ldots\alpha_m\beta_1\ldots\beta_k\beta_1\ldots\beta_k\ldots = \alpha_0.\alpha_1\ldots\alpha_m \quad (\beta_1\ldots\beta_k) \tag{5}$$

In the latter case there appears, beginning with some decimal place, in expression (5) beginning with the $(m+1)$th place, a finite block of digits $\beta_1\ldots\beta_k$ (written in the parentheses on the right-hand side of (5)) repeating indefinitely, not all the digits β_j being equal to zero. The periodicity of the decimal follows from the fact that the partial remainders β_k in expression (2) satisfy the inequality $\beta_k < q$ and therefore among the first q of them, i.e. among $\beta_0, \beta_1, \ldots, \beta_{k-1}$, there must be two equal numbers.

The first case can always be reduced to the second by putting

$$\frac{p}{q} = \alpha_0.\alpha_1\ldots\alpha_{m-1}(\alpha_m - 1)99\ldots \tag{6}$$

It follows that procedure (2), with the application, if necessary, of procedure (6), sets up a one-to-one correspondence between the positive numbers p/q and infinite decimal fractions.

We call a fraction $\alpha_0.\alpha_1\alpha_2\ldots$ *infinite if for every natural k there is a natural number* $l > k$ *such that* $\alpha_l > 0$.

On the other hand, it is proved in arithmetic that *to an arbitrary periodic decimal* (5) *there corresponds a uniquely determined positive rational number* p/q *such that procedure* (2) *results in exactly that decimal representation of this number*. It should only be stipulated that if there appears a finite decimal fraction it must be transformed according to (6).

For instance, if the fraction 103/330 is subjected to procedure (2) the result is the (mixed) infinite repeating decimal

$$0.31212\ldots = 0.3(12) = \frac{3}{10} + \frac{12}{990} = \frac{103}{330}$$

Conversely, by the first of the equalities considered above, the latter decimal is transformed into the original fraction 103/330.

With a negative rational number $-p/q$ we associate the representation of the number p/q as infinite decimal taken with the sign "$-$".

Thus, there is a one-to-one correspondence* $\pm p/q = \pm\alpha_0.\alpha_1\alpha_2\alpha_3\ldots$ between the nonzero rational numbers and repeating infinite decimals.

---

* If to each element x of a set A there corresponds a definite element y of a set B so that every element $y \in B$ corresponds to exactly one $x \in A$ we say that there is a *one-to-one correspondence* between the elements of A and B and write $A \ni x \rightleftharpoons y \in B$.

These procedures assign to every nonzero rational number a uniquely determined representation of the number as periodic decimal. Conversely, every such decimal corresponds, in accordance with the procedures, to a nonzero rational number.

It appears natural to attribute to the number zero (which is also rational) *the decimal representation* $0 = \pm 0.00\ldots = 0.00\ldots$.

Besides periodic decimals there exist nonperiodic ones, for instance, $0.1010010001\ldots$ and $0.121122111222\ldots$.

Here is another example. If we extract the square root of 2 following the well-known arithmetical rule the result is the infinite nonrepeating decimal $\sqrt{2} = 1.41\ldots$. This decimal is completely defined in the sense that to any natural number k there corresponds a definite kth decimal digit α_k of the number $\sqrt{2}$ which is uniquely determined by the rule for extracting square roots.

Mathematical analysis provides many methods for computing the number π with any preassigned accuracy. This results in a quite definite decimal representation of π which turns out to be a mixed nonperiodic decimal fraction.

Now we state (at the present stage—quite formally) the definition of an irrational number. *By an irrational number is meant an arbitrary nonperiodic infinite decimal fraction*

$$a = \pm\alpha_0.\alpha_1\alpha_2\alpha_3\ldots \tag{7}$$

where α_0 is a nonnegative integer and α_k ($k = 1, 2, \ldots$) are decimal digits, the meaning of the equality sign "$=$" being that the right member of (7) *is denoted by a.* On the other hand, it is convenient to say, simply, that the right member of (7) is the decimal representation of the number a.

The rational and irrational numbers are called real numbers.

It follows that *every nonzero real number is representable as an infinite decimal of type* (7). *If it is rational its decimal representation is a repeating infinite decimal. If otherwise, then, according to the definition stated, expression* (7) *itself specifies an irrational number.*

A nonzero decimal fraction can be finite but, according to the convention expressed by equality (6), it does not represent a new rational number since it can be replaced by a periodic infinite decimal equal to it.

A number a for which not all α_k are equal to zero is called positive or negative depending on whether (7) *involves the sign* "$+$" *or* "$-$". As usual, the sign "$+$" can be omitted.

At present we only have a formal definition of a real number. It now remains to define the arithmetical operations on them, to introduce the relation expressed by the symbol "$>$" and to check that the relation "$>$" is coherent with the corresponding operations and the relation "$>$" already known for rational numbers and also to verify that the ordinary conditions imposed on numbers are satisfied.

The definition of the relation "$>$" is given in § 2.2 and the arithmetical operations are defined in § 2.3. In § 2.4 we state and prove the basic properties of real numbers. These properties are divided into five groups I-V. The first three groups include the well-known properties which are usually used when we perform arithmetical calculations and solve inequalities. Group IV consists of one property (the so-called *Archimedean property*). Finally, Group V also contains only one property expressed by the so-called *principle of continuity* for real numbers. For our further aims it suffices to know that the real numbers (the decimals) are objects for which the relation "$>$" and also the arithmetical operations are defined and that conditions I-V are satisfied for them. Therefore there is an alternative way of studying the book in which the material given in normal type is read carefully and systematically while the material given in small print (the proofs of Properties I-V) may be omitted or looked through.

From Properties I-V all the other properties of real numbers can be derived logically.

There is also an axiomatic approach to the definition of real numbers in which by numbers are meant some (abstract) objects a, b, c, ... satisfying conditions I-V. *In this approach Properties* I-V *are understood as axioms of real numbers.*

At first sight an axiomatic construction of real numbers may seem simpler. But in this connection there arises the question as to whether Axioms I-V are consistent. To prove their consistency it is necessary to construct some formal symbols for which it is possible to define the arithmetical operations and the relation "$>$" and to check that they satisfy Axioms I-V. It turns out that the infinite decimals can be chosen as such symbols.

§ 2.2. Definition of Inequality

Let us take two numbers $a = \pm\alpha_0.\,\alpha_1\alpha_2\ldots$ and $b = \pm\beta_0.\,\beta_1\beta_2\ldots$ represented by *infinite* decimals. They will be considered equal if and only if they are of one sign and

$$\alpha_k = \beta_k \qquad (k = 0, 1, 2, \ldots)$$

By definition, $a < b$ or, which is the same, $b > a$, if $\alpha_0 < \beta_0$ or if there is an index (a nonnegative integer) l such that $\alpha_k = \beta_k$ $(k = 0, 1, 2, \ldots, l)$ and $\alpha_{l+1} < \beta_{l+1}$.

Further, we write, by definition, $a > 0$ or $a < 0$ depending on whether a is positive or negative; we write $a < b$ if $a < 0$ and $b > 0$ or if $a, b < 0$ and $|a| > |b|$.

If $a = \pm\alpha_0.\,\alpha_1\alpha_2\ldots$, then, by definition, $-a = \mp\alpha_0.\,\alpha_1\alpha_2\ldots$, and the absolute value $|a|$ of a is $+\alpha_0.\,\alpha_1\alpha_2\ldots = \alpha_0.\,\alpha_1\alpha_2$. Thus,

$$|-a| = |a| = \begin{cases} a & \text{for } a \geqslant 0 \\ -a & \text{for } a \leqslant 0 \end{cases}$$

§ 2.3. Definition of Arithmetical Operations

Suppose that there is a law according to which with every index (nonnegative integer) n a number x_n is associated. The collection of these numbers

$$x_0, x_1, x_2, \ldots \tag{1}$$

is called a (*number*) *sequence. A sequence is said to be nondecreasing* (*nonincreasing*) *if* $x_k \leqslant x_{k+1}$ ($x_k \geqslant x_{k+1}$) *for all* $k = 0, 1, 2, \ldots$.

We say that *sequence* (1) *is bounded above* (*by a number M*) *if there exists an integer M such that* $x_k \leqslant M$ *for all* $k = 0, 1, 2, \ldots$.

If the numbers x_k in sequence (1) are integers and if there is an integer ξ such that $x_k = \xi$ for all $k > k_0$ we say that *the given sequence is stabilized.*

It is obvious that *if a sequence of integers is nondecreasing and bounded above by a number M it is stabilized, the corresponding integer* ξ *satisfying the inequality* $\xi \leqslant M$.

Let us consider a sequence of decimals (which are all either finite or infinite)

$$\left.\begin{array}{l} a_1 = \alpha_{10}.\ \ \alpha_{11}\ \ \alpha_{12}\ \ \alpha_{13} \ldots \\ a_2 = \alpha_{20}.\ \ \alpha_{21}\ \ \alpha_{22}\ \ \alpha_{23} \ldots \\ a_3 = \alpha_{30}.\ \ \alpha_{31}\ \ \alpha_{32}\ \ \alpha_{33} \ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \end{array}\right\} \tag{2}$$

The right members of (2) form a table (an *infinite matrix*).

We shall say that *sequence* (2) *is stabilized if there is a real number* $a = \gamma_0 . \gamma_1\gamma_2\gamma_3 \ldots$ *such that, for any* $k = 0, 1, 2, \ldots$, *the kth column of table* (2) *regarded as a sequence of integers is stabilized, the corresponding integer being* γ_k (*i.e. if* $\alpha_{nk} = \gamma_k$ *for sufficiently large n*). *In this case we shall write*

$$a_n \doteq a \tag{3}$$

It turns out automatically that γ_0 is a nonnegative integer while γ_k ($k = 1, 2, \ldots$) are decimal digits.

Remark. The sequence of numbers $a_1, a_2, a_3, \ldots$ where

$$a_{2k} = 1.\underbrace{0\ldots0}_{k \text{ times}}11 \ldots \qquad (k = 1, 2, \ldots)$$

and

$$a_{2k+1} = 0.\underbrace{9\ldots9}_{k \text{ times}}11 \ldots \qquad (k = 0, 1, 2, \ldots)$$

is not stabilized. From § 3.1 where the notion of the limit will be introduced it will become clear that this sequence possesses the limit 1 (written $a_n \to 1$). We thus see that a sequence of decimals possessing a limit may not be stabilized. At the same time $a_n \doteq a$ implies $a_n \to a$ (see the example in § 3.1).

For an arbitrary number $a = \alpha_0 . \alpha_1\alpha_2 \ldots$ and an arbitrary natural n we now define the truncated number $a^{(n)} = \alpha_0 . \alpha_1 \ldots \alpha_n$ which is a finite decimal fraction. The reader is supposed to be familiar with the operations on finite decimals known from arithmetic.

Let us set two positive numbers $a = \alpha_0.\alpha_1\alpha_2\ldots$ and $b = \beta_0.\beta_1\beta_2\ldots$ represented as infinite decimals.

Now we introduce the following sequence of numbers:

$$a^{(n)}+b^{(n)} = \alpha_0.\alpha_1\ldots\alpha_n+\beta_0.\beta_1\ldots\beta_n = \lambda_0^{(n)}\lambda_1^{(n)}\ldots\lambda_n^{(n)} \tag{4}$$
$$(n = 1, 2, \ldots)$$

Below we shall prove a lemma implying that this sequence is stabilized, that is $a^{(n)}+b^{(n)} \rightleftarrows \gamma$ where $\gamma = \gamma_0.\gamma_1\gamma_2\ldots$ is a definite real number. It is natural to call this number the *sum of the numbers a and b* and write $a+b = \gamma_0.\gamma_1\gamma_2\ldots$.

Thus, by the sum $a+b$ is meant a number such that

$$a^{(n)}+b^{(n)} \rightleftarrows a+b \tag{5}$$

The product ab, the difference $a-b$ and the quotient a/b of two numbers a and b are defined as follows:

$$(a^{(n)}b^{(n)})^{(n)} \rightleftarrows ab \tag{6}$$

$$a^{(n)}-(b^{(n)}+10^{-n}) \rightleftarrows a-b \qquad (a > b > 0)^{*} \tag{7}$$

and

$$\left(\frac{a^{(n)}}{b^{(n)}+10^{-n}}\right)^{(n)} \rightleftarrows \frac{a}{b} \tag{8}$$

The expressions on the left-hand sides of (6), (7) and (8) do not decrease as n increases; by virtue of this fact and by the boundedness above of these expressions, it follows from the lemma proved below that they are stabilized, the corresponding numbers being denoted, respectively, as ab, $a-b$ and a/b. It should be taken into account that $a^{(n)}$ does not decrease as n increases indefinitely while $b^{(n)}+10^{-n}$ does not increase; besides, there hold the inequalities

$$(a^{(n)}b^{(n)})^{(n)} \leqslant (\alpha_0+1)(\beta_0+1), \qquad a^{(n)}-(b^{(n)}+10^{-n}) \leqslant (\alpha_0+1)$$

and

$$\left(\frac{a^{(n)}}{b^{(n)}+10^{-n}}\right)^{(n)} \leqslant \frac{\alpha_0+1}{\beta_0.\beta_1\ldots\beta_s}$$

(where s is such that $\beta_s > 0$) showing that their left members are bounded above.

Let us also put

$$0+a = a\pm 0 = a \quad \text{and} \quad a\cdot 0 = 0\cdot a = a-a = \frac{0}{b} = 0 \tag{9}$$
$$(a \geqslant 0,\ b > 0)$$

* Here $n > n_0$ where n_0 is so large that the difference on the left-hand side of (7) is positive. We also note that the equality $(a-b)+b = a$ $(a > b > 0)$ will be proved in § 2.8 (after formula (4)) and that equality $a\cdot 1/a = 1$ $(a > 0)$ follows from § 2.8 (12).

We have defined for any two nonnegative numbers a and b their sum, difference, product and quotient under the assumption that in the case of the difference $a \geqslant b$ and in the case of the quotient $b > 0$. Next, these definitions are extended in an ordinary way to any numbers a and b of arbitrary signs. For instance, if $a, b \leqslant 0$ we put $a+b = b+a = -(|a|+|b|)$. If a and b are of different signs we put $a+b = b+a = \pm||a|-|b||$ where the sign "+" or "−" should be chosen so that it coincides with the sign of the greatest, in its absolute value, number among a and b. In particular, we have

$$a+(-a) = 0 \tag{10}$$

for any a.

Similar rules can be derived for the remaining arithmetical operations but we shall not dwell on this question since these rules are known to the reader from elementary mathematics.

Now, to justify what was said we must prove the following lemma:

Lemma 1. *If a nondecreasing sequence of type* (2) *is bounded above by an integer M it is necessarily stabilized, that is $a_n \to a$, where a is a number satisfying the inequalities*

$$a_n \leqslant a \leqslant M \qquad (n = 0, 1, 2, \ldots) \tag{11}$$

Indeed, under the conditions of the lemma, the integers forming the zeroth column of matrix (2) are nondecreasing and bounded above by the number M. Therefore they are stabilized, i.e. $a_{n0} = \gamma_0$ for sufficiently large n, the corresponding integer γ_0 satisfying the inequality $\gamma_0 \leqslant M$. Let us suppose that the columns of matrix (2) with indices not exceeding k are stabilized, the corresponding integers $\gamma_0, \gamma_1, \ldots, \gamma_k$ being such that

$$\gamma_0 . \gamma_1 \ldots \gamma_k \leqslant M \qquad \text{(where } \gamma_1, \ldots, \gamma_k \text{ are decimal digits)} \tag{12}$$

Let us prove that, under this assumption, the $(k+1)$th column of (2) is also stabilized (i.e. $a_{n,k+1} = \gamma_{k+1}$ for sufficiently large n), the corresponding integer γ_{k+1} being a decimal digit such that the inequality

$$\gamma_0 . \gamma_1 \ldots \gamma_k \gamma_{k+1} \leqslant M \tag{13}$$

holds.

Indeed, the decimal representation of the number a_n taken for $n > n_1$ where n_1 is sufficiently large is of the form

$$a_n = \gamma_0 . \gamma_1 \ldots \gamma_k \alpha_{n,k+1} \alpha_{n,k+2} \ldots \leqslant M$$

and, besides, a_n does not decrease as n increases. Therefore, for the indicated values of n, the digits $\alpha_{n,k+1}$ (which do not exceed 9) do not decrease and hence they are stabilized, i.e. $a_{n,k+1} = \gamma_{k+1}$ for $n \geqslant n_2$ where n_2 is sufficiently large and γ_{k+1} is a decimal digit. We also obviously have

$$\gamma_0 . \gamma_1 \ldots \gamma_{k+1} \leqslant a_n \leqslant M \qquad (n \geqslant n_2)$$

Now we conclude, by induction, that inequality (13) and the relation $a_n \to a = \gamma_0 . \gamma_1 \gamma_2 \ldots$ hold, whence obviously follows the second inequality (11). Now it only remains to prove the first inequality (11). If we suppose that it does not hold there must be a natural n such that $a < a_n$, which means that

$$a_n = \gamma_0 . \gamma_1 \ldots \gamma_k \alpha_{n,k+1} \alpha_{n,k+2} \ldots$$

for some k and $\gamma_{k+1} < \alpha_{n,k+1}$. But this is impossible since the digit $\alpha_{n,k+1}$ does not decrease as n is given further increase and therefore it must be stabilized, which leads to the contradictory inequality $\gamma_{k+1} < \gamma_{k+1}$ for the corresponding decimal digit γ_{k+1}.

§ 2.4. Basic Properties of Real Numbers

I. Order Properties.

I_1. For every pair of real numbers a and b there holds one and only one of the relations

$$a = b,\ a > b \quad \text{and} \quad a < b$$

I_2. $a < b$ and $b < c$ imply $a < c$ (the *transitivity* of the relation expressed by the symbol "$<$").

I_3. If $a < b$ there exists a number c such that $a < c < b$.

Properties I_1 and I_2 follow immediately from the definitions of the symbols "$=$" and "$<$". If two positive numbers a and b are represented as infinite decimals $a = \alpha_0.\alpha_1\alpha_2\ldots$ and $b = \beta_0.\beta_1\beta_2\ldots$ and if $a < b$, then, for some s_0, we have

$$\alpha_k = \beta_k \qquad k \leq s_0 - 1 \text{ (if } s_0 = 0 \text{ these equalities should be deleted)}$$

and

$$\alpha_{s_0} < \beta_{s_0}$$

There also exists $s_1 > s_0$ such that $\beta_{s_1} > 0$. If we put $c = \alpha_0.\alpha_1\ldots\alpha_{s_0-1}\,\beta_{s_0}\ldots$ $\ldots\beta_{s_1-1}(\beta_{s_1}-1)\beta_{s_1+1}\ldots$ then, obviously, c is a number satisfying the inequalities $a < c < b$ and $c_1 = \alpha_0.\alpha_1\ldots\alpha_{s_0-1}\,\beta_{s_0}\ldots\beta_{s_1}$ is a rational number satisfying similar inequalities $a < c_1 < b$.

We have thus proved I_3 for positive a and b. The extension of this proof to the case of arbitrary a and b is performed quite easily.

II. Properties of the Operations of Addition and Subtraction*.

II_1. $a+b = b+a$ (commutativity).

II_2. $(a+b)+c = a+(b+c)$ (associativity).

II_3. $a+0 = a$.

II_4. $a+(-a) = 0$.

II_5. $a < b$ implies that $a+c < b+c$ for any c.

It suffices to carry out the proofs of these properties for positive numbers a, b and c since after that, as is known from arithmetic, they can be extended automatically to numbers of any sign. Thus, let $a, b, c > 0$.

II_1 follows from § 2.3 (5) because it holds for the finite decimals $a^{(n)}+b^{(n)} = b^{(n)}+a^{(n)}$ $(n = 1, 2, \ldots)$.

Properties II_3 and II_4 immediately follow from the definitions stated above (see § 2.3 (9) and (10)).

The proof of II_2 see in § 2.8. Property II_5 is quite obvious in the case when a and b are finite decimals. Now suppose that a and b are arbitrary, and let d' and d'' be finite decimals

* When approaching the theory of real numbers axiomatically we must add the requirement that to each pair of numbers a, b there should correspond, in accordance with a certain law, a number $a+b$ called their sum, and conditions II_1-II_5 should hold. Conditions II_3 and II_4 should then be modified to read: there exists a number 0 (zero) such that $a+0 = a$ for all a, and for every a there exists a number $-a$ such that $a+(-a) = 0$. The uniqueness of the zero can be drawn logically from the axioms stated: if we suppose that there is another zero $0'$ then $0' = 0'+0 = 0+0' = 0$. The axioms also imply the existence of the difference $a-b$, that is of a number which must be added to b to obtain a. This number is $a+(-b)$ since $a+(-b)+b = a+0 = a$. It is determined uniquely because if $b+c = b+c'$ then $c = (-b)+b+c = (-b)+b+c' = c'$.

such that $a < d' < d'' < b$; then, for sufficiently large n, we have

$$a^{(n)} < d' < d'' < b^{(n)} \quad \text{and} \quad a^{(n)}+c^{(n)} < d'+c^{(n)} < d''+c^{(n)} < b^{(n)}+c^{(n)}$$

It obviously follows that $a+c \leqslant d'+c < d''+c \leqslant b+c$, i.e. $a+c < b+c$.

Equality II_1 becomes trivial when a or b is equal to zero (see § 2.3 (9)).

III. Properties of the Operations of Multiplication and Division*.

III_1. $ab = ba$ (commutativity).
III_2. $ab(c) = a(bc)$ (associativity).
III_3. $a \cdot 1 = a$.
III_4. $a \cdot \dfrac{1}{a} = 1 \quad (a \neq 0)$.
III_5. $(a+b)c = ac+bc$ (distributivity).
III_6. $a < b$, $c > 0$ implies $ac < bc$.

The same argument as in the case of properties II shows that it is sufficient to verify III only for positive a, b and c (see § 2.8).

IV. Archimedean Property.

Given any number $c > 0$, there exists a natural number $n > c$.

Indeed, if $c = \alpha_0.\alpha_1\alpha_2 \ldots$ we can put $n = \alpha_0+2$.

The Archimedean property and some of the preceding properties imply that, given any positive number ε, there always is a natural number n such that the inequality $1/n < \varepsilon$ is fulfilled.

Indeed, according to IV, for the number $1/\varepsilon$ there is a natural number n such that $1/\varepsilon < n$, which, by III_6, implies the required inequality.

Note that for every given number $c \geqslant 0$ there is a single number m in the sequence 0, 1, 2, ... of nonnegative integers for which the inequalities $m \leqslant c < m+1$ hold.

V. Continuity Property of Real Numbers.

Let there be given a sequence of closed intervals (i.e. of sets of numbers x for which $a_n \leqslant x \leqslant b_n$)

$$\sigma_n = [a_n, b_n] \qquad (n = 1, 2, \ldots)$$

A system of such intervals is said to be *nested* if $\sigma_{n+1} \subset \sigma_n$ $(n = 1, 2, \ldots)$

---

* In the axiomatic approach we should add the following: to every pair a, b there corresponds, in accordance with a certain law, a number ab called the product of a and b, and properties III_1-III_6 hold. It is also necessary to modify III_3 and III_4 to read: there exists a number 1 (unity) different from 0 and such that $a \cdot 1 = a$ for all a; for any $a \neq 0$ there is a number $1/a$ such that $a \cdot 1/a = 1$. The uniqueness of 1 can be drawn logically from the axioms stated, together with the existence and uniqueness of the quotient a/b $(b \neq 0)$ which is a number by which b must be multiplied to obtain a. The derivation is completely analogous to that in the footnote concerning II: the only difference is that 0 should be replaced by 1 and the operation of addition by the multiplication. Then it turns out, automatically, that $1 > 0$. Indeed, if we suppose that $1 < 0$ then $0 = 1+(-1) < < -1$ (see II_4 and II_5) and $1(-1) < 0(-1)$ (see III_6), that is $-1 < 0$ (see III_3), whence follows a contradiction: $-1 < 0 < -1$. It should be taken into account that

$$0\cdot(-1) = 0\cdot(-1)+0\cdot1+0\cdot1 = 0\cdot(-1+1+1) = 0\cdot(0+1) = 0\cdot1 = 0$$

and the lengths of the intervals tend to zero*:

$$d_n = b_n - a_n \to 0 \qquad (n \to \infty)$$

Then there is exactly one point (number) which simultaneously belongs to all the closed intervals σ_n.

Let us prove this property. First suppose that $a_1 \geqslant 0$. Obviously, $a_1 \leqslant a_2 \leqslant \ldots \leqslant b_m$ for any m, which shows that the numbers a_n do not decrease and are bounded above by any number b_m with arbitrary index m. Therefore, by Lemma 1, § 2.3, there is a number c such that $a_n \Rightarrow c$, that is the sequence of numbers a_n is stabilized. Furthermore, $a_n \leqslant c \leqslant$ $\leqslant b_m$. Since n and m in these inequalities are arbitrary we have, in particular,

$$a_n \leqslant c \leqslant b_n \qquad (n = 1, 2, \ldots)$$

and consequently $c \in \sigma_n$ for any n.

Let us show that c is the only point possessing the indicated property. Suppose that there is another point c_1 of that kind. Then we have the inequalities $a_n \leqslant c$ and $c_1 \leqslant b_n$ whence $b_n - a_n \geqslant |c_1 - c| = \varepsilon > 0$ for any n, which contradicts the condition $b_n - a_n \to 0$.

Now we take the case $a_1 < 0$. Putting $\alpha_n = a_n + |a_1|$ and $\beta_n = b_n + |a_1|$ $(n = 1, 2, \ldots)$ we conclude, on the basis of what has been proved above, that there exists a single point γ belonging to all intervals $[\alpha_n, \beta_n]$, which means that $c = \gamma - |a_1|$ is a single point belonging simultaneously to all intervals $[a_n, b_n]$.

Property V has thus been proved.

Property V is spoken of as the *nested interval theorem*. In § 2.6 we shall show that this property can be replaced by some other properties equivalent to it (on condition that all the remaining properties hold). In the axiomatic approach to the theory of real numbers IV and V are naturally stated as axioms.

It should be noted that if σ_n are open intervals of the type (a_n, b_n) property V may not hold in the general case. For instance, the intervals $(0, 1/n)$, $n = 1, 2, \ldots$, form a nested family and their lengths tend to zero as n increases indefinitely but at the same time a point (number) which belongs to all of them does not exist.

Indeed, zero and any negative number do not belong to any of the intervals. If $a > 0$ is an arbitrary positive number we can find, using the Archimedean property, a natural number n_0 such that $1/n_0 < a$, and consequently the interval $(0, 1/n)$ does not contain a for $n > n_0$.

§ 2.5. Supremum and Infimum of a Set

A set E of real numbers x is said to be *bounded* if there is a positive number M such that the inequality $|x| < M$ is fulfilled for all $x \in E$ or, which is the same, if $-M < x < M$ for all $x \in E$.

If E does not satisfy this requirement, that is if for any positive number M, however large, there is $x_0 \in E$ such that $|x_0| > M$, the set E is said to be *unbounded*.

---

* A sequence of numbers x_n is said to tend to zero (synonymously, the values entering into the sequence are said to tend to zero) if, given an arbitrary positive number ε, there is $N > 0$ such that $|x_n| < \varepsilon$ for all $n > N$.

A set E is said to be *bounded above* (*below*) if there is a number K (k) such that $x \leqslant K$ ($k \leqslant x$) for all $x \in E$. The number K (k) is called an *upper bound* (a *lower bound*) of the set E.

It is evident that a *bounded set is both bounded above and bounded below*. The set R of all real numbers is obviously unbounded and has neither upper nor lower bound; the set R_+ of all positive numbers is bounded below but is not bounded above; a closed interval $[a, b]$ and an open interval (a, b) are examples of bounded sets (for finite a and b).

A number M (m) is called the *least upper bound* or *supremum* (the *greatest lower bound* or *infimum*) of a set A of numbers if the following conditions hold:

(i) $x \leqslant M$ ($x \geqslant m$) for all $x \in A$.

(ii) Given any $\varepsilon > 0$, however small, there is a number $x_0 \in A$ such that $M - \varepsilon < x_0$ ($x_0 < m + \varepsilon$).

For the supremum of A we use the notation

$$M = \sup A = \sup_{x \in A} x$$

and for the infimum we write

$$m = \inf A = \inf_{x \in A} x$$

("sup" and "inf" are, respectively, the abbreviations of the Latin *supremum* "the greatest" and *infimum* "the lowest"). In the next section we shall prove that every set bounded above possesses the supremum and every set bounded below possesses the infimum. It is quite obvious that the supremum and the infimum, provided they exist, are determined uniquely.

For an unbounded set A having no upper bound we shall write

$$\sup A = \sup_{x \in A} x = +\infty$$

and for a set A having no lower bound

$$\inf A = \inf_{x \in A} x = -\infty$$

In these cases we shall say that $+\infty$ and $-\infty$ are, respectively, the supremum and the infimum of A.

The supremum of a closed interval $[a, b]$ or of an open interval (a, b) is obviously the point (number) b. In the case of a closed interval the supremum (the number b) belongs to it while in the case of an open interval it does not. The supremum of the set $(-\infty, 0)$ is obviously the number 0 while its infimum is understood as the symbol $-\infty$.

We also note the obvious equality

$$\inf_{x \in A} x = -\sup_{x \in A} (-x)$$

§ 2.6. Other Statements of Property V

Property V can be expressed equivalently in one of the following forms:

V′. *Any nonempty set* $\mathfrak{M}$ *of real numbers bounded above (below) by a number* M (m) *possesses the supremum (the infimum)* $c \leqslant M$ $(c \geqslant m)$.

V″. *If the set* R *of all real numbers is divided into two nonempty disjoint sets, that is*

$$R = A+B, \quad A \cap B = O \ (O \text{ is an empty set})$$

so that every $a \in A$ *is less than every* $b \in B$, *then either there exists a number* c *which is the greatest among the numbers belonging to* A, *and then* B *has no least number, or there exists a number* c *which is the least among the numbers belonging to* B, *and then* A *has no greatest number.*

Let us prove that properties V, V′ and V″ are equivalent (provided that all the other properties hold).

Let V hold and let $\mathfrak{M}$ be a set of numbers bounded above by a number M. If $\mathfrak{M}$ is a finite set the existence of its supremum is obvious: it is simply equal to the greatest of the numbers belonging to $\mathfrak{M}$. Therefore we suppose that $\mathfrak{M}$ is an infinite set. Let us denote by a_0 $(a_0 < M)$ one of the points of $\mathfrak{M}$ and take the closed interval $\sigma_0 = [a_0, M]$. Next, dividing σ_0 into two equal closed subintervals we choose one of them containing points of the set $\mathfrak{M}$ on condition that if both subintervals contain points belonging to $\mathfrak{M}$ we take the rightmost of them. Let this subinterval be denoted as $\sigma_1 = [a_1, b_1]$. Now we divide σ_1 into two equal subintervals and denote by $\sigma_2 = [a_2, b_2]$ the rightmost of these subintervals containing points of $\mathfrak{M}$. Proceeding in this way we obtain, by induction, the nested family of closed intervals

$$\sigma_n = [a_n, b_n] \qquad (n = 1, 2, \dots)$$

whose lengths tend to zero. Every such closed interval contains at least one point belonging to $\mathfrak{M}$ and there are no points of $\mathfrak{M}$ to the right of it. By V, there is a point c belonging to all σ_n. Let x be an arbitrary point belonging to $\mathfrak{M}$. Then $x \leqslant c$ because, if we suppose that $c < x$, then there exists n such that x lies to the right of σ_n since $b_n - a_n \to 0$ $(n \to \infty)$, which is impossible. On the other hand, for any point $c_1 < c$ there is n such that c_1 lies to the left of σ_n, and in σ_n there is a point $x_0 \in \mathfrak{M}$. Consequently $c_1 < x_0$. We have thus proved that $\mathfrak{M}$ has the supremum which is the point c.

The fact that a set bounded below possesses the infimum is proved in a similar way. Hence, V implies V′.

Now suppose that V′ holds and that the set R of all real numbers is divided into two classes (sets) A and B as described in V″.

Let b be a point belonging to B. Then, by V″, we have $a < b$ for all $a \in A$, and therefore, by virtue of V', there exists the supremum of A:

$$\sup_{a \in A} a = c \tag{1}$$

and $a \leqslant c \leqslant b$ for any $a \in A$ and $b \in B$. By condition V″ the number c itself belongs to one of the classes A and B. If $c \in A$ then, obviously, c is the greatest number in the set A.

Let us suppose that B also contains the least number which we denote by b_0. Then for the arithmetic mean of the numbers c and b_0 we have

$$(c+b_0)/2 = d < b_0$$

and therefore $d \in A$ (since b_0 is the least number in the set B). On the other hand, $c < d$ and, by (1), the number d cannot belong to A, and we thus arrive at a contradiction.

Now, if we suppose that $c \in B$ an analogous argument shows that c is the least number in the class B and then A has no greatest number. Hence we have proved that V′ implies V″.

Finally, suppose that V″ holds and let $\sigma_n = [a_n, b_n]$ be closed intervals with $b_n - a_n \to 0$ forming a nested family. Let us divide R into two classes by means of the following construction: the class A consists of the numbers x each of which satisfies the inequality $x < a_n$ for at least one n while B consists of all the other real numbers which we denote by y. The set A is nonempty since, for instance, it contains the number $a_1 - 1$. The set B is also nonempty because, for example, any number b_n belongs to it. Now we take two arbitrary points $x \in A$ and $y \in B$. By definition, $y \geqslant a_n$, for any n, and for the given x there is $n = n_1$ such that $x < a_{n_1}$. Therefore $x < a_{n_1} \leqslant y$ and $x < y$.

We have shown that the sets A and B thus constructed satisfy the conditions of V″. Hence, according to V″, there is a number c which is either the greatest number in A or the least number in B. But the set A cannot have the greatest number because if $x \in A$ then, for some n_1, the inequality $x < a_{n_1}$ holds, and consequently the number $x' = (x + a_{n_1})/2$ exceeds x and at the same time belongs to A. Thus, c is the least of the numbers belonging to the set B. But then $a_n \leqslant c \leqslant b_n$ for any $n = 1, 2, \ldots$

Our argument shows that there exists a number c belonging to all σ_n. The uniqueness of c is quite apparent.

We have proved the implications V $\to$ V′ $\to$ V″ $\to$ V, that is V, V′ and V″ are equivalent. The construction of the division of R into two sets described in V″ is known as a *Dedekind* cut.*

Remark. The rational numbers possess properties I-IV but Property V′ does not hold for them. For instance, consider the set e of all rational numbers x less than a given irrational number $a = \alpha_0.\alpha_1\alpha_2, \ldots$. The reader can easily prove that $\sup_{x \in e} x = a$, and thus the supremum of a set of rational numbers may not be a rational number.

§ 2.7. Isomorphism of Various Representations of Real Numbers. Length of an Interval. Physical Quantities

In the foregoing sections the real numbers $a, b, c, \ldots$ were defined as infinite decimals and it was shown that they possess the properties enumerated in Groups I-V (which will be briefly referred to as Properties I-V).

Arguing in the same manner we can define the real numbers as infinite binary or ternary fractions or, generally, use an arbitrary number system with base n to construct the corresponding symbols $a', b', c', \ldots$ representing the real numbers. The relation "$>$" and the operations of addition "$+$" and multiplication "$\cdot$" can be introduced by analogy with the above, and the verification shows that these new objects satisfy Properties I-V as well.

It should also be noted that the Dedekind cuts of the set of all rational numbers can also be used as a basis for the construction of the real numbers (e.g. see P. S. Alexandrov, A. N. Kolmogorov, *An Introduction to the Theory of Functions of a Real Variable*, GTTI, 1938 and G. M. Fikhtengolts, *A Course of Differential and Integral Calculus*, Vol. I, "Nauka", 1970 (in Russian)).

All the possible Dedekind cuts $a', b', c', \ldots$ are nothing but the divisions of the set of rational numbers into two nonempty sets (classes) $\mathfrak{A}$ and $\mathfrak{B}$

* J. W. R. Dedekind (1831-1916), a German mathematician. In an axiomatic approach property V is termed the *Dedekind cut postulate.*

such that any number belonging to the set $\mathfrak{U}$ is less than any number belonging to the set $\mathfrak{B}$. It turns out that it is possible to define the relation "$>$" and the operations "$+$" and "$\cdot$" for the Dedekind cuts and to show that they possess Properties I-V.

Finally, the theory of real numbers can also be constructed by defining a real number as a *Cauchy* (*fundamental*) *sequence* of rational numbers. In the construction, sequences of this kind essentially different (in a certain sense) from each other are denoted by symbols a', b', c', ... For these symbols the notions "$>$", "$+$" and "$\cdot$" are then introduced after which it is shown that they possess Properties I-V (e.g. see V. V. Nemytsky, M. I. Sludskaya, A. N. Cherkasov, *A Course of Mathematical Analysis*, Vol. I, Fizmatgiz, 1957, in Russian).

All the indicated methods of defining real numbers do not differ from each other from the formal, logical point of view. This follows from the theorem below which can be called a *theorem on the isomorphism of the sets satisfying Conditions* I-V.

The notion of an *isomorphism* or, more precisely, of an *isomorphism with respect to the notions* "$>$", "$+$", "$\cdot$" and "sup" (supremum) will be clarified below in the course of the statement of the theorem.

Theorem. *Let E be a set of decimal fractions a, b, c, ... and E' be a set of some elements a', b', c', ... for which the relation "greater than" ("$>$") and the operations of addition ("$+$") and multiplication ("$\cdot$") are defined so that Conditions* I-V *are fulfilled.*

Then it is possible to set up a one-to-one correspondence

$$a \sim a'$$

between the elements $a \in E$ and $a' \in E'$ which is an isomorphism with respect to the relation "$>$", the arithmetical operations and the operation of taking supremum.

This means that if

$$a \sim a' \quad \text{and} \quad b \sim b'$$

then

$$\left.\begin{aligned} a \pm b &\sim a' \pm b' \\ ab &\sim a'b' \\ \frac{a}{b} &\sim \frac{a'}{b'} \qquad (b \neq 0) \end{aligned}\right\} \tag{1}$$

and if $a < b$ then $a' < b'$.

Finally, if E is a set of decimals a bounded above and E' is the set of the corresponding elements a' then

$$\sup_{a \in E} a \sim \sup_{a' \in E'} a' \tag{2}$$

This theorem says that, irrespective of whether we operate on decimal fractions a, b, c, ... or on the corresponding elements a', b', c', ..., if these operations reduce to a finite number of arithmetical operations or to the determination of a supremum, the results of the operations are always a certain number d and a certain element d', respectively, which are in the relation $d \sim d'$ indicated above.

As will be seen later, we can also include into the formulation of the theorem the operation of passing to a limit.

Hence all the concrete sets defined above (i.e. the sets of the Dedekind cuts, fundamental sequences, infinite binary fractions, etc.) are isomorphic to each other in the indicated sense.

What has been said indicates that, as was explained at the end of § 2.1, the theory of real numbers admits of a correct logical construction on the basis of an abstract axiomatic approach.

It also indicates that from the formal point of view the definitions of real numbers as decimal fractions and as certain symbols satisfying the axioms are equivalent. The latter approach is of course preferable from the philosophical point of view since numbers are abstract notions expressing quantitative relations observed in reality while decimal fractions are their formal representing symbols.

We shall describe the basic stages of the proof of the isomorphism theorem.

Let E denote the set of all real numbers and E' a set of some other objects, different from numbers in the general case, satisfying Axioms I-V.

Then E' contains a zero element $0'$ and a unity $1'$ ($0' < 1'$), and the elements $2' = 1'+1'$, $3' = 2'+1'$, ... and $-1'$, $-2'$, $-3'$, ... make sense. As a result we obtain a "two-way" sequence* of (pairwise different) elements of E':

$$\ldots -2', -1', 0, 1', 2', \ldots \tag{3}$$

These elements correspond to the integers

$$\ldots -2, -1, 0, 1, 2, \ldots$$

Elements (3) can be divided by each other with the exception of the division by $0'$. Therefore E' contains (rational) elements of the form $\pm p'/q' = \pm n'p'/n'q' = a'$ ($q' > 0$, $p' \geqslant 0$) which are in a one-to-one correspondence with the rational numbers $\pm p/q = a$, which we write briefly as $a \sim a'$. Here n' corresponds to the natural number n. This correspondence is an *isomorphism with respect to the symbol "$>$" and the arithmetical operations*, that is relation (1) is fulfilled (at present only for the rational elements).

But this isomorphism $a \sim a'$ can be in fact extended to all the elements of the sets E and E'. Let us prove this. By the isomorphism between the rational elements established above, we have the correspondence

$$\pm\alpha_0.\alpha_1 \ldots \alpha_n \sim \pm\alpha_0'.\alpha_1' \ldots \alpha_n' = \pm(\alpha_0\alpha_1 \ldots \alpha_n)'/10^{n'}$$

between the finite decimals formed in E and E' where in the parentheses on the right-hand side stands the integer $\alpha_0\alpha_1 \ldots \alpha_n = \alpha_0 10^n + \alpha_1 10^{n-1} + \ldots + \alpha_n$. This correspondence is isomorphic with respect to the symbol "$>$" and with respect to the arithmetical operations. Let $a = \alpha_0.\alpha_1\alpha_2 \ldots$ be an arbitrary positive number represented by the corresponding decimal. It is readily seen that it is the supremum of the set of all the truncated

* Numbers a_k depending on the integral index k arranged as

$$\ldots a_{-2}, a_{-1}, a_0, a_1, a_2, \ldots$$

are said to form a *two-way sequence*.

numbers $a^{(n)}$: $a = \sup\limits_n a^{(n)}$. Since $a^{(n)} \leqslant \alpha_0 + 1$ for any $n = 1, 2, \ldots$ we also have $a^{(n)'} \leqslant \leqslant (\alpha_0+1)'$, which shows that the set of the elements $a^{(n)'}$ $(n = 1, 2, \ldots)$ is bounded above, and consequently among the elements of E' there is an element a' serving as the supremum $a' = \sup\limits_n a^{(n)'}$.

It is natural to write a' as the expression $a' = \alpha_0'.\, \alpha_1'\alpha_2' \ldots$ and to call it an infinite decimal in E' while the elements $a^{(n)'}$ can be called the corresponding truncated elements. This construction assigns to each real number a an element $a' \in E$ $(a \to a')$. This correspondence is coherent with the correspondence $\alpha_0.\, \alpha_1 \ldots \alpha_k \sim \alpha_0'.\, \alpha_1' \ldots \alpha_k'$ because, together with the equality

$$\alpha_0.\, \alpha_1 \ldots \alpha_k = \alpha_0.\, \alpha_1 \ldots \alpha_{k-1}\, (\alpha_k - 1)\, 99 \ldots$$

we have the equality

$$\alpha_0'.\, \alpha_1' \ldots \alpha_k' = \alpha_0'.\, \alpha_1' \ldots \alpha_{k-1}'(\alpha_k - 1)'\, 9'9' \ldots \tag{4}$$

Indeed, by definition, the infinite decimal on the right-hand side of (4) is the supremum set of the corresponding truncated elements, and it is readily seen that the element $\alpha_0'.\, \alpha_1' \ldots \alpha_k'$ also serves as this supremum, and therefore they must coincide because a set can only have one supremum.

Now let us choose a positive element $\lambda \in E'$. According to the Archimedean property, there is a natural element $n' > \lambda$. Thus, $0 < \lambda < n'$, and, since $0' < 1'$, we have $0' < 1' < \ldots \ldots < n'$. In this finite sequence there is obviously a single (integral nonnegative) element α_0' such that $\alpha_0' < \lambda \leqslant (\alpha_0 + 1)'$. Testing, in succession, the elements of the finite sequence

$$\alpha_0' < (\alpha_0.\, 1)' < (\alpha_0.\, 2)' < \ldots < (\alpha_0.\, 9)' < (\alpha_0 + 1)'$$

we find a unique element $(\alpha_0.\, \alpha_1)'$ such that

$$(\alpha_0.\, \alpha_1)' < \lambda \leqslant (\alpha_0.\, \alpha_1 + 1)'$$

(if $\alpha_1 = 9$ then $\alpha_0.\, (\alpha_1 + 1) = \alpha_0 + 1$). Proceeding in this way we obtain, by induction, a sequence of digits $\alpha_1, \alpha_2, \alpha_3, \ldots$, such that

$$(\alpha_0.\, \alpha_1 \ldots \alpha_k)' < \lambda \leqslant \big(\alpha_0.\, \alpha_1 \ldots \alpha_{k-1}\, (\alpha_k + 1)\big)' \tag{5}$$

for any $k = 1, 2, \ldots$ It cannot happen that for some natural l we have $\alpha_k = 0$ for all $k > l$. Indeed, if we suppose that this is so, the element λ satisfies the inequalities

$$(\alpha_0.\, \alpha_1 \ldots \alpha_l)' < \lambda < (\alpha_0.\, \alpha_1 \ldots \alpha_l)' + (10^{-k})' \qquad (k = l+1, l+2, \ldots)$$

which indicate that it is a single element belonging to all the closed intervals forming a nested family whose lengths tend to zero as $k \to \infty$:

$$\lambda \in [(\alpha_0.\, \alpha_1 \ldots \alpha_l)', \quad (\alpha_0.\, \alpha_1 \ldots \alpha_l)' + (10^{-k})']$$

But the element $(\alpha_0.\, \alpha_1 \ldots \alpha_l)$ is sure to belong to all these closed intervals and therefore $\lambda = (\alpha_0.\, \alpha_1 \ldots \alpha_l)'$; we thus arrive at a contradiction to the first inequality (5).

Inequalities (5) imply that if $a = \alpha_0.\, \alpha_1\, \alpha_2 \ldots$ (here we mean by a a number represented by an infinite decimal) then

$$\lambda = a' = \sup_k\, (a^{(k)})' = \sup_k\, (\alpha_0.\, \alpha_1 \ldots \alpha_k)'$$

In what follows we consider infinite decimal representations of numbers without stipulating that they are infinite.

If $0 < a < b$ then there are natural k and l such that $a < b^{(k)} < b^{(l)}$ and therefore $0 < a^{(n)} < b^{(k)} < b^{(l)}$ and $0' < a^{(n)'} < b^{(k)'} < b^{(l)'}$ for any natural n, and consequently,

$$0' < a' = \sup\, (a^{(n)})' \leqslant b^{(k)'} < \sup b^{(s)} = b'$$

that is $0 < a' < b'$.

We have thus proved (by now for positive a and a' only) that the correspondence $a \to a'$ is an isomorphism with respect to the symbol "$>$". In particular, we have shown that to different positive a there correspond different positive a'. Thus, the correspondence $a \to a'$ is in fact a one-to-one correspondence $a \rightleftharpoons a'$ (also written $a \sim a'$) between the positive elements of E and E'

Now we shall prove that for positive a and b there holds the relation $(a+b)' = a'+b'$. In the case when a and b are finite decimals this equality is sure to hold. Let

$$a = \alpha_0.\,\alpha_1\alpha_2\ldots,\quad b = \beta_0.\,\beta_1\beta_2\ldots \quad \text{and} \quad a+b = \gamma_0.\,\gamma_1\gamma_2\ldots$$

where the representation of $a+b$ is, as we know, the result of the stabilization of $a^{(n)}+b^{(n)}$. Therefore for any natural n there is a natural $l > n$ such that $(a+b)^{(n)} \leqslant a^{(l)}+b^{(l)}$ and $(a+b)^{(n)'} \leqslant (a^{(l)}+b^{(l)})' = a^{(l)'}+b^{(l)'} < a'+b'$. But then

$$(a+b)' = \sup_n (a+b)^{(n)'} \leqslant a'+b' \tag{6}$$

On the other hand, $a < a^{(n)} + (10)^{-n}$ and $b < b^{(n)}+(10)^{-n}$, and consequently

$$a' < a^{(n)'}+(10^{-n})', \quad b' < b^{(n)'}+(10^{-n})'$$

and

$$a'+b' < a^{(n)'}+b^{(n)'}+2(10^{-n})' = (a^{(n)}+b^{(n)})'+2(10^{-n})'$$

But there exists a natural $s > n$ such that $a^{(n)'}+b^{(n)'} < (a+b)^{(s)'} < (a+b)'$, and thus

$$a'+b' < (a+b)'+2(10^{-n})'$$

for any natural n. Hence, $a'+b' \leqslant (a+b)'$, and taking into account (6) we obtain

$$(a+b)' = a'+b' \tag{7}$$

In the same manner we prove for positive a and b the equalities

$$(a-b)' = a'-b' \quad (a > b) \tag{8}$$

and

$$(ab)' = a'b', \quad \left(\frac{a}{b}\right)' = \frac{a'}{b'} \tag{9}$$

For negative numbers a we put $a' = -(-a)'$. As a result, we obtain, together with the correspondence $0 \sim 0'$, a one-to-one correspondence $a \sim a'$ between the real numbers and all the elements of E'. It is obviously isomorphic with respect to the symbol "$>$" and also with respect to the arithmetical operations. We shall limit ourselves to the proof of equality (7) for the case when $a > 0$, $b < 0$ and $a > |b|$ without presenting all the details of the argument justifying the last assertion.

In the case under consideration we have

$$(a+b)' = (a-|b|)' = a'-|b|' = a'+b'$$

The first of these equalities holds by virtue of the well-known properties of numbers, the second equality follows from (8) and the third one is implied by the fact that $a'-|b'|$ is an element of E' such that its addition to $|b|'$ results in a'. This (uniquely determined) element is equal to $a'+(-|b|') = a'+(-(-b)') = a'+b'$ (see the footnote concerning II in § 2.4).

Let us discuss in more detail the representation of real numbers by means of the points of a straight line which is an extremely convenient and commonly used method of representing numbers.

The considerations below are not quite rigorous from the point of view of formal logic; they rather describe a scheme which can be followed to construct a rigorous theory.

Let us consider all the possible line segments belonging to various straight lines lying in a given plane. Among them we choose an arbitrary line segment which will be said to have *unit length*. Congruent line segments possess a common property which we shall denote by the letter a (or b, c, ...) and call the *length* of any of the congruent segments. Let σ and δ be respectively line segments of lengths a and b. If, when these segments are applied to each other, it turns out that one of them, say a, is a *proper* part of δ, that is if it turns out that a is a part of δ and a does not coincide with δ, we say that their lengths satisfy the relation $a < b$.

A line segment obtained from the unit segment by dividing the latter into q equal parts and taking the geometric sum p of such parts is said to be *rational* or *commensurable with unit measure*.

It is clear that the lengths of the line segments satisfy Axioms I.

By definition, the sum $a+b$ and the difference $a-b$ $(a > b)$ are respectively the lengths of the geometrical sum and the geometrical difference of the given line segments. It is readily seen that Axioms II (at present we exclude II_3 and II_4) are satisfied for the lengths.

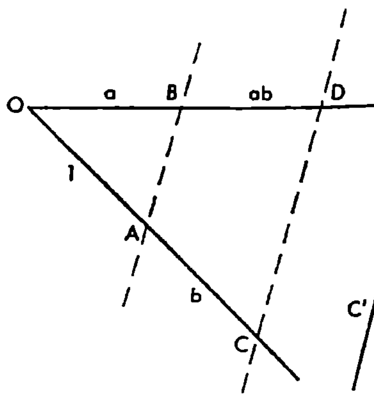

Fig. 2.1

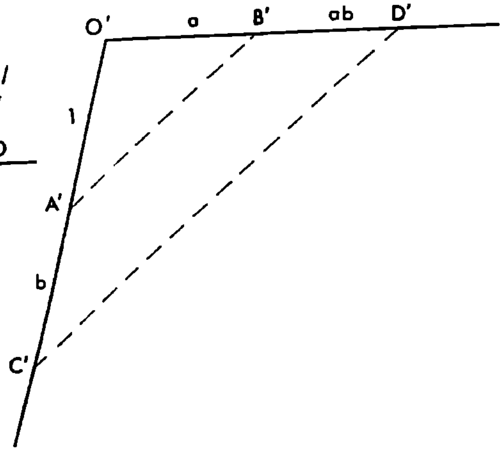

Fig. 2.2

In Fig. 2.1 we see an arbitrary angle on one of whose sides line segments OA and AC of lengths 1 and b respectively are set off from the vertex while on the other side a line segment OB of length a is set off. We also draw through the point C the line CD parallel to AB. It cuts the other side of the angle at D; the length of the segment BD will be called, by definition, the product ab. This definition is properly stated (i.e. it is not contradictory) because if we made the same construction for another angle O' (Fig. 2.2) the resultant line segment $B'D'$ would be congruent to BD.

Now it becomes clear how a/b (the result of the division of a by b) should be defined. We lay off, in succession, from the vertex of an angle (see Fig. 2.3) line segments of lengths b and a on one side and the unit segment on the other side and also draw the line CD parallel to AB. Then, by definition, the length of AC is a/b. This definition is also properly stated, and the operation of division thus defined turns out to be inverse with respect to the multiplication: $b(a/b) = a$.

It can readily be verified geometrically that Axioms III are satisfied for the lengths. For instance, the distributive law III_5 can be verified with the aid of Fig. 2.4. The relation $(a-b)c = ac-bc$ $(a > b)$ is verified similarly.

Now let us take a straight line with a chosen initial point O. We have a one-to-one correspondence between the points of this line lying to the right of O and the lengths of the line segments joining these points to the point O. These lengths will be called positive. Now we introduce formally a negative length $-a$ corresponding to the point of the line symmetric to a with respect to the origin O (i.e. symmetric to the point corresponding to the length a). To the point O we purely formally assign the number zero (0). This results in a one-to-one correspondence between all the points of the line and the new symbols

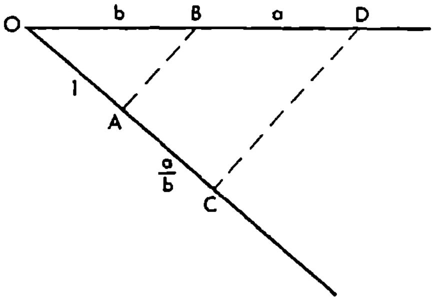

Fig. 2.3

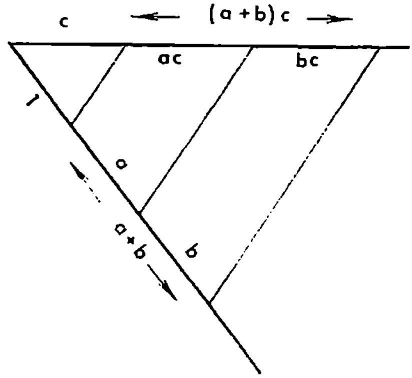

Fig. 2.4

$a, b, \ldots$ which can now be positive or negative or zero. Following the ordinary rules which we shall not enumerate here we can define for the new symbols the arithmetical operations and the symbol "$>$". As is proved in elementary algebra, they satisfy Axioms I-III because the positive symbols $a, b, \ldots$ satisfy them. On the *number line* thus constructed we can (mentally) mark the rational points $\pm p/q$ corresponding to the positive and negative lengths commensurable with the chosen unit measure. They themselves satisfy Axioms I-III.

Axioms IV-V are also satisfied for the new objects. Axiom V expresses mathematically the fact that we think of a straight line as a continuous geometrical entity.

Thus, the lengths of the line segments supplied with the signs "$+$" and "$-$" (as was indicated above) satisfy Axioms I-V and hence their set is isomorphic to the set of real numbers. This allows us to identify, in practice, the notion of the length of a line segment and the notion of the number corresponding to this segment in accordance with the indicated isomorphism.

In conclusion we note that physics gives us many examples of notions isomorphic to real numbers; they are called physical quantities. Temperature, mass and also velocity of motion, when it is directed along a definite line, are all examples of physical quantities. It can of course turn out that they are isomorphic not to all real numbers but only to those belonging to an interval. For instance, in the case of the mass such an interval is $(0, \infty)$.

§ 2.8. Supplement

In this section we present the proofs (sometimes the schemes of the proofs) of some of the assertions stated in § 2.4. Perhaps it would be easier for the reader to understand them after he has studied the next Chapter 3. This does not mean that § 2.8 is based on the material of Chapter 3; simply, the questions considered in these sections of the book involve analogous ideas but the representation of the subject-matter in § 2.8 is rather concise.

A nondecreasing sequence of numbers

$$a_n = \alpha_0^{(n)}.\alpha_1^{(n)}\alpha_2^{(n)}\ldots \tag{1}$$

will be called *essentially nondecreasing* if for any n_0 there is $n > n_0$ such that $a_{n_0} < a_n$.

There hold the following properties:

(i) *If a_n is an essentially nondecreasing sequence and $a_n \Rightarrow a$* ($\Rightarrow$ denotes the stabilization) *where a is a decimal fraction*

$$a = \alpha_0.\alpha_1\alpha_2\ldots \tag{2}$$

the latter is infinite.

For if we had $a = \alpha_0.\alpha_1\ldots\alpha_s00\ldots$ then there would exist n_0 such that $a_n = \alpha_0.\alpha_1\ldots\alpha_s\alpha_{s+1}^{(n)}\alpha_{s+2}^{(n)}\ldots \le \alpha_0.\alpha_1\ldots\alpha_s00\ldots$ $(n > n_0)$. But this would only be possible if we had $0 = \alpha_{s+1}^{(n)} = \alpha_{s+2}^{(n)} = \ldots$ for any n, and then sequence (1) could not be essentially decreasing.

(ii) *Let two infinite fractions $a = \alpha_0.\alpha_1\alpha_2\ldots$ and $b = \beta_0.\beta_1\beta_2\ldots$ be such that the equality*

$$b^{(n)} = a^{(n)} + \lambda_n \tag{3}$$

holds for any n where $a^{(n)}$ and $b^{(n)}$ are the truncated fractions corresponding to a and b and λ_n tends to zero as $n \to \infty$. Then the decimal fractions a and b coincide and, consequently, $\alpha_k = \beta_k$, $k = 0, 1, 2, \ldots$

Indeed, if, for instance, we had $a < b$ then for some k there would be

$$a = \alpha_0.\alpha_1\ldots\alpha_{k-1}\alpha_k\alpha_{k+1}\ldots \quad \text{and} \quad b = \alpha_0.\alpha_1\ldots\alpha_{k-1}\beta_k\beta_{k+1}\ldots, \qquad (\beta_k > \alpha_k)$$

and, besides, $\beta_s > 0$ for some $s > k$. Then for $n > s$ we would have

$$\begin{aligned}b^{(n)} - a^{(n)} &= \alpha_0.\alpha_1\ldots\alpha_{k-1}\beta_k\beta_{k+1}\ldots\beta_s\beta_{s+1}\ldots\beta_n - \alpha_0.\alpha_1\ldots\alpha_{k-1}\alpha_k\alpha_{k+1}\ldots\alpha_n \ge \\ &\ge \alpha_0.\alpha_1\ldots\alpha_{k-1}\beta_k\beta_{k+1}\ldots\beta_s - \alpha_0.\alpha_1\ldots\alpha_{k-1}(\alpha_k+1) \ge \beta_s\cdot 10^{-s} \qquad (\beta_s > 0)\end{aligned}$$

for any $n > s$. But this is impossible since, by the hypothesis, we have $b^{(n)} - a^{(n)} = \lambda_n \to 0$.

Now let us prove that if $a > b > 0$ then

$$(a-b)+b = a \tag{4}$$

Suppose that a and b are represented as infinite decimals: $a = \alpha_0.\alpha_1\ldots$ and $b = \beta_0.\beta_1\ldots$. By definition (see § 2.3 (7)), $a-b$ is a fraction obtained as the result of the stabilization:

$$a^{(n)} - (b^{(n)} + 10^{-n}) \Rightarrow a-b \qquad (a > b) \tag{5}$$

Therefore for each n there is $s > n$ such that

$$(a-b)^{(n)} = [a^{(s)} - (b^{(s)} + 10^{-s})]^{(n)} = a^{(n)} - b^{(n)} + \gamma_n$$

where, obviously, $\gamma_n \to 0$ $(n \to \infty)$. Note that the sequence of numbers on the left-hand side of (5) is essentially nondecreasing and hence, by property (i) proved above, the fraction $a-b$ in (5) obtained by stabilization is infinite. The number $(a-b)+b$ is, by definition, such that $(a-b)^{(n)} + b^{(n)} = a^{(n)} + \gamma_n \Rightarrow (a-b)+b$. But we have $a^{(n)} \Rightarrow a$ and $\gamma_n \to 0$; consequently, by property (ii), equality (4) does in fact hold.

Let us show that

$$(a+b)+c = a+(b+c) \tag{6}$$

Indeed, we have the obvious relations

$$a^{(n)} + b^{(n)} + c^{(n)} \le (a+b)^{(n)} + c^{(n)} \le a^{(n)} + b^{(n)} + 10^{-n} + c^{(n)}$$

and

$$a^{(n)} + b^{(n)} + c^{(n)} \le a^{(n)} + (b+c)^{(n)} \le a^{(n)} + b^{(n)} + c^{(n)} + 10^{-n}$$

They show that the numbers $(a+b)^{(n)} + c^{(n)}$ and $a^{(n)} + (b+c)^{(n)}$ differ from each other by a quantity λ_n not exceeding, in its absolute value, the number $(10)^{-n}$ and thus tending to zero as $n \to \infty$. Therefore, by virtue of (ii), relation (6) holds.

A quantity of the type of λ_n ($\lambda_n \to 0$ for $n \to \infty$) is also denoted as

$$o(1) \qquad (n \to \infty) \tag{7}$$

It should be noted that if, together with λ_n, we deal with some other quantities of this kind (called *infinitesimals*) they are also denoted by the same symbol (7).

Properties III_1-III_5 are immediate consequences of the following equalities:

$$(a^{(n)}b^{(n)})^{(n)} = (b^{(n)}a^{(n)})^{(n)} \tag{8}$$

$$((ab)^{(n)}c^{(n)})^{(n)} = (ab)^{(n)}c^{(n)}+o(1) = a^{(n)}b^{(n)}c^{(n)}+o(1) \qquad (n \to \infty) \tag{9}$$

$$(a^{(n)}(bc)^{(n)})^{(n)} = a^{(n)}b^{(n)}c^{(n)}+o(1) \qquad (n \to \infty) \tag{10}$$

$$(a^{(n)}\cdot(1)^{(n)})^{(n)} = a^{(n)}+o(1) \qquad (n \to \infty) \tag{11}$$

$$\left(\left(\frac{1}{a}\right)^{(n)}a^{(n)}\right)^{(n)} = \left(\frac{1}{a}\right)^{(n)}a^{(n)}+o(1) = \frac{1}{a^{(n)}+10^{-n}}a^{(n)}+o(1) = 1+o(1) \tag{12}$$

$$((a+b)^{(n)}c^{(n)})^{(n)} = (a+b)^{(n)}c^{(n)}+o(1) = a^{(n)}c^{(n)}+b^{(n)}c^{(n)}+o(1) \tag{13}$$

and

$$(ac)^{(n)}+(bc)^{(n)} = a^{(n)}c^{(n)}+b^{(n)}c^{(n)}+o(1) \qquad (n \to \infty) \tag{14}$$

Property (8) is obvious. As an example, we shall elucidate the proof of property (9). The first equality in (9) holds because the removal of the outer subscript (n) leads to an increase of the first term in (9) by a quantity (by a finite decimal fraction) not exceeding $10^{-n} = o(1)$.

Furthermore, ab is a number such that

$$(a^{(n)}b^{(n)})^{(n)} \rightrightarrows ab$$

Therefore for any n there is $s > n$ dependent on n such that

$$(ab)^{(n)} = (a^{(s)}b^{(s)})^{(n)} = \{(a^{(n)}+\mu_n)(b^{(n)}+\nu_n)\}^{(n)} = \{a^{(n)}b^{(n)}+(a^{(n)}\nu_n+b^{(n)}\mu_n+\mu_n\nu_n)\}^{(n)} =$$
$$= \{a^{(n)}b^{(n)}+o(1)\}^{(n)} = (a^{(n)}b^{(n)})^{(n)}+o(1) = a^{(n)}b^{(n)}+o(1) \qquad (n \to \infty)$$

(where $\mu_n = o(1)$ and $\nu_n = o(1)$, $n \to \infty$). In the third equality it should be taken into account that $a^{(n)} \leqslant a$ and $b^{(n)} \leqslant b$, and therefore $a^{(n)}\nu_n = o(1)$ and $b^{(n)}\mu_n = o(1)$ and also that $o(1)+o(1) = o(1)$. By virtue of the above-proved relation (2), from (9)-(14) follow, respectively, III_1-III_5. The verification of the fact that the operations on infinite fractions are coherent with the corresponding operations on rational fractions is left to the reader.

For instance, in the case of addition it is necessary to check that the decimal representation

$$\frac{p_1}{q_1}+\frac{p_2}{q_2} = \frac{p_1q_2+p_2q_1}{q_1q_2}$$

of the sum exactly coincides with the sum of the infinite decimal representations of the summands.

§ 2.9. Inequalities for Absolute Values

The inequality

$$|a| < \varepsilon \tag{1}$$

is equivalent to the two inequalities

$$-\varepsilon < a < \varepsilon \tag{1'}$$

Hence the inequality

$$|a-b| < \varepsilon \tag{2}$$

is equivalent to the inequalities

$$b-\varepsilon < a < b+\varepsilon \tag{2'}$$

Analogously, the inequality

$$|a-b| \leqslant \varepsilon \tag{3}$$

is equivalent to the inequalities $b-\varepsilon \leqslant a \leqslant b+\varepsilon$.

There also hold the inequalities

$$|a+b| \leqslant |a|+|b| \tag{4}$$

and

$$|a-b| \geqslant ||a|-|b|| \tag{5}$$

Inequality (4) can be derived by considering separately the four possible cases: (i) $a, b \geqslant 0$, (ii) $a, b \leqslant 0$, (iii) $a \leqslant 0 \leqslant b$ and (iv) $b \leqslant 0 \leqslant a$.

For instance, in Case (ii) we have

$$a+b \leqslant b \leqslant 0 \quad \text{and} \quad |a+b| = -(a+b) = -a-b = |a|+|b|$$

while in Case (iii), assuming that $|b| \geqslant |a|$, we obtain

$$|a+b| = b+a \leqslant |a|+|b|$$

Let the reader consider Case (iii) by analogy with Case (i) under the assumption $|b| \leqslant |a|$. Case (iv) is reducible to (iii).

Further, by (4) we have

$$|a| \leqslant |b|+|a-b| \quad \text{and} \quad |b| \leqslant |a|+|a-b|$$

that is

$$|a|-|a-b| \leqslant |b| \leqslant |a|+|a-b|$$

whence follows (5).

CHAPTER 3

Limit of Sequence

§ 3.1. Concept of Limit of Sequence

The method of limits is the basic method lying in the foundation of mathematical analysis.

Suppose that there is a certain law of correspondence which assigns to every natural number $n = 1, 2, \ldots$ a number x_n.* Then we say that there is defined a *sequence of numbers* $x_1, x_2, x_3, \ldots$ or, briefly, a *sequence* $\{x_n\}$.

The numbers x_n entering into a sequence $\{x_n\}$ are called its *elements*. They can be real or complex. Here we shall consider the case of real sequences.

Below are examples of sequences:

(i) $\{n\} = \{1, 2, 3, \ldots\}$,

(ii) $\left\{\frac{1}{n}\right\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \ldots\right\}$,

(iii) $\left\{\frac{(-1)^n}{n}\right\} = \left\{-1, \frac{1}{2}, -\frac{1}{3}, \ldots\right\}$,

(iv) $\left\{\frac{n-1}{n}\right\} = \left\{0, \frac{1}{2}, \frac{2}{3}, \ldots\right\}$,

(v) $\left\{1+\frac{1}{10^n}\right\} = \{1.1; 1.01; 1.001; \ldots\}$,

(vi) $\{(-1)^n\} = \{-1, +1, -1, \ldots\}$,

(vii) $\{1; 2; \ldots; 10; 0.1; 0.01; 0.001; \ldots\}$.

In Case (vii) we cannot write a single general formula for an arbitrary element x_n but the law according to which the numbers x_n are formed is clear:

$$x_n = \begin{cases} n & \text{for } n = 1, \ldots, 10 \\ \dfrac{1}{10^{n-10}} & \text{for } n = 11, 12, \ldots \end{cases}$$

* Thus, x_n is a function defined on the set of natural numbers.

When dealing with a sequence $\{x_n\}$ we shall call x_n a *variable* and say that *the variable x_n runs through the sequence $\{x_n\}$ or that it runs through the sequence of values x_n.*

A variable x_n whose all values are equal to one and the same number a is called a *constant* and is usually simply denoted as a.

By definition, a number a is called the *limit of a sequence* $\{x_n\}$ if, given any arbitrary positive number ε, there is a natural number N (dependent on ε) such that the inequality

$$|x_n - a| < \varepsilon \qquad (n > N)$$

holds for all natural $n > N$.

If a is the limit of $\{x_n\}$ we write

$$\lim x_n = \lim_{n\to\infty} x_n = a$$

or

$$x_n \to a$$

and say that *the variable x_n tends to a* or that *the sequence $\{x_n\}$ converges to the number a.*

Let us show that the variable in Example (ii) has a limit equal to zero. Indeed, setting an arbitrary ε we write the inequality

$$|x_n| = \frac{1}{n} < \varepsilon$$

which obviously holds for all $n > \frac{1}{\varepsilon}$ and hence for all $n > N$ where N is an arbitrary natural number exceeding $\frac{1}{\varepsilon}$: $N > \frac{1}{\varepsilon}$. Thus, for any $\varepsilon > 0$ there is a natural number N such that $|x_n| < \varepsilon$ for all $n > N$.

In exactly the same manner we can prove that the sequence in Example (iii) has zero limit. The variable in Example (iv) tends to 1 because in this case

$$|1 - x_n| = \left|1 - \frac{n-1}{n}\right| = \frac{1}{n} < \varepsilon$$

for all $n > N$ where N is any natural number greater than $1/\varepsilon$.

It can readily be shown that the variable in Example (v) tends to 1. The variable in Example (vii) obviously tends to zero. The fact that the values of the elements of the sequence in Example (vii) first increase does not affect its convergence to zero: for the convergence only its values corresponding to sufficiently large n are essential.

If x_n satisfies the inequality

$$|a - x_n| < \varepsilon$$

this is equivalent to the fact that it satisfies the inequalities

$$a - \varepsilon < x_n < a + \varepsilon$$

or, geometrically, to the fact that the point (number) x_n belongs to the interval $(a-\varepsilon, a+\varepsilon)$. Therefore we can restate the definition of a limit in geometrical terms: *a variable x_n has as its limit a number (point) a if, given any $\varepsilon > 0$, there is a natural N such that $x_n \in (a-\varepsilon, a+\varepsilon)$ for all $n > N$.*

An arbitrary interval (c, d) containing a point a, or, equivalently, such that $c < a < d$, is called a *neighbourhood of the point a*. It is obvious that for an arbitrary neighbourhood (c, d) of a there is $\varepsilon > 0$ such that the interval $(a-\varepsilon, a+\varepsilon)$ is contained in (c, d), that is $(a-\varepsilon, a+\varepsilon) \subset (c, d)$.

Therefore the fact that $x_n \to a$ can also be expressed as follows: given an arbitrary neighbourhood (c, d) of the point a, all the points x_n, beginning with some value of the index n, are in that neighbourhood, that is there must exist a natural N such that $x_n \in (c, d)$ for $n > N$. As to the points $x_1, \ldots, x_N$ with the indices $n \leq N$, they may or may not belong to (c, d). Hence, if there are some points x_n lying outside (c, d) their number is finite.

On the other hand, if it is known that there are only a finite number of points $x_{n_1}, x_{n_2}, \ldots, x_{n_s}$ lying outside (c, d) we can put

$$N = \max\{n_1, n_2, \ldots, n_s\}$$

and then $x_n \in (c, d)$ for all $n > N$. Thus, the definition of limit can also be stated as follows: *a variable x_n has as its limit a number a if the set of points x_n lying outside any neighbourhood of the point a is either finite or empty.*

The variable in Example (vi) has no limit at all because if we suppose that it has a limit equal to a then any neighbourhood of the point a, however small in its length, contains all the elements x_n with the exception of a finite number of x_n. But it is clear that outside any interval of length $1/2$, irrespective of its position on the real line, there are an infinite number of elements x_n of the sequence in question, and hence the sequence has no limit.

It is readily seen that the sequence in Example (i) does not tend to any limit either. In what follows we shall say in such a case that the sequence has an infinite limit (tends to infinity); the meaning of this statement which, to some extent, is different from the above definition will be clarified later on.

It is quite obvious that if a variable x_n has a limit this limit is determined uniquely. For if x_n possessed two different limits a and b (for definiteness, let $a < b$) then each of the intervals $(a-\varepsilon, a+\varepsilon)$ and $(b-\varepsilon, b+\varepsilon)$ where $\varepsilon = (b-a)/3$ would contain all the points of the sequence $\{x_n\}$ except possibly a finite number of them. But this is obviously impossible because these intervals have no points in common (do not intersect).

Example (vi) shows that there can occur equal numbers among the elements of a sequence $\{x_n\}$ having different indices n_1 and n_2: $x_{n_1} = x_{n_2}$. But even in this case x_{n_1} and x_{n_2} should be regarded as different elements of the sequence.

It is clear that *if two sequences $\{x_n\}$ and $\{x'_n\}$ have only a finite number of different elements with the same indices n they simultaneously either have no limits or converge to equal limits.*

We shall prove a number of theorems describing some properties of the variables possessing limits.

Theorem 1. *If a variable x_n has a limit then x_n is bounded.*

Proof. Let $x_n \to a$. Then for $\varepsilon = 1$ there is a natural number N such that

$$1 > |x_n - a| \qquad \text{for } n > N$$

It follows that $1 > |x_n - a| \geq |x_n| - |a|$, that is

$$|x_n| < |a| + 1 \qquad \text{for } n > N$$

Let us put $M = \max\{|a|+1, |x_1|, \ldots, |x_N|\}$. Then, obviously,

$$|x_n| < M \qquad (n = 1, 2, \ldots)$$

that is the variable x_n is bounded.

Theorem 2. *If a variable x_n has a nonzero limit a there is N such that*

$$|x_n| > \frac{|a|}{2} \qquad \text{for } n > N$$

Moreover, for these n, if $a > 0$ then $x_n > a/2$ and if $a < 0$ then $x_n < a/2$. Thus, beginning with some value of the index n, the variable x_n is of the same sign as a.

Proof. Let $x_n \to a$. Then for $\varepsilon = |a|/2$ there exists a natural N such that

$$\frac{|a|}{2} > |a - x_n| \geq |a| - |x_n| \qquad (n > N)$$

whence $|x_n| > |a| - \frac{|a|}{2} = \frac{|a|}{2}$, and the first assertion of the theorem has thus been proved. On the other hand, the inequality $|a|/2 > |a - x_n|$ is equivalent to the two inequalities

$$a - \frac{|a|}{2} < x_n < a + \frac{|a|}{2} \qquad (n > N)$$

and therefore if $a > 0$ then

$$\frac{a}{2} = a - \frac{|a|}{2} < x_n \qquad (n > N)$$

and if $a < 0$ then

$$x_n < a + \frac{|a|}{2} = a - \frac{a}{2} = \frac{a}{2} \qquad (n > N)$$

which completes the proof of the second assertion of the theorem.

Theorem 3. *If $x_n \to a$, $y_n \to b$ and $x_n \leq y_n$ for all $n = 1, 2, \ldots$ then $a \leq b$.*

Proof. Suppose that $b < a$. Let us set a number $\varepsilon < \frac{a-b}{2}$ and find natural numbers N_1 and N_2 such that

$$a - \varepsilon < x_n \quad (n > N_1) \quad \text{and} \quad y_n < b + \varepsilon \quad (n > N_2)$$

This is possible since $x_n \to a$ and $y_n \to b$.

Putting $N = \max\{N_1, N_2\}$ we obviously obtain $y_n < b+\varepsilon < a-\varepsilon < x_n$ for $n > N$, and thus we arrive at a contradiction because, by the hypothesis, $x_n \leqslant y_n$ for all n.

Theorem 4. *If variables x_n and y_n tend to one and the same limit a and if $x_n \leqslant z_n \leqslant y_n$ $(n = 1, 2, \ldots)$ the variable z_n also tends to a.*

Proof. Given $\varepsilon > 0$, there are N_1 and N_2 such that

$$a-\varepsilon < x_n \quad (n > N_1) \quad \text{and} \quad y_n < a+\varepsilon \quad (n > N_2)$$

and therefore, for $n > N = \max\{N_1, N_2\}$ we have

$$a-\varepsilon < x_n \leqslant z_n \leqslant y_n < a+\varepsilon$$

whence

$$|z_n - a| < \varepsilon \qquad (n > N)$$

which is what we set out to prove.

Theorem 5. *If $x_n \to a$ then $|x_n| \to |a|$.*

Proof. The assertion of the theorem follows from the inequality

$$||x_n|-|a|| \leqslant |x_n - a|$$

§ 3.2. Arithmetical Operations on Limits

Let x_n and y_n be variables which respectively run through two sequences $\{x_n\}$ and $\{y_n\}$. The *sum* x_n+y_n, the *difference* x_n-y_n, the *product* x_ny_n and the *quotient* x_n/y_n of x_n and y_n are defined as variables running respectively through the sequences $\{x_n+y_n\}$, $\{x_n-y_n\}$, $\{x_ny_n\}$ and $\{x_n/y_n\}$. In the case of quotient it is supposed that $y_n \neq 0$ for all $n = 1, 2, \ldots$

If $x_n = c$ for $n = 1, 2, \ldots$ we write $c\pm y_n$, cy_n and c/y_n instead of $x_n \pm y_n$, x_ny_n and x_n/y_n.

There hold the following assertions:

$$\lim (x_n \pm y_n) = \lim x_n \pm \lim y_n \tag{1}$$

$$\lim (x_ny_n) = \lim x_n \lim y_n \tag{2}$$

$$\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}, \qquad \text{if} \quad \lim y_n \neq 0 \tag{3}$$

These assertions should be understood in the sense that *if x_n and y_n possess limits then the limits of their sum, difference, product and quotient also exist (in the latter case under the assumption that* $\lim y_n \neq 0$*) and equalities* (1)-(3) *hold.*

Proof. Let $x_n \to a$ and $y_n \to b$. Setting an arbitrary $\varepsilon > 0$ we choose N such that

$$|x_n - a| < \frac{\varepsilon}{2} \quad \text{and} \quad |y_n - b| < \frac{\varepsilon}{2} \qquad (n > N)$$

Then

$$|(x_n \pm y_n)-(a\pm b)| \leqslant |x_n - a|+|y_n - b| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon \qquad (n > N)$$

and we have thus proved (1).

To prove (2) note that

$$|x_ny_n-ab| = |x_ny_n-ay_n+ay_n-ab| \leqslant |x_ny_n-ay_n|+|ay_n-ab| \leqslant$$
$$\leqslant |y_n|\,|x_n-a|+|a|\,|y_n-b| \tag{4}$$

Since y_n has a limit (by Theorem 1 proved in the foregoing section) there exists a positive number M such that

$$|y_n| < M \qquad (n = 1, 2, \ldots) \tag{5}$$

We can choose M so that the inequality

$$|a| < M \tag{6}$$

also holds. Let us take a natural N such that

$$|x_n-a| < \frac{\varepsilon}{2M} \quad \text{and} \quad |y_n-b| < \frac{\varepsilon}{2M} \qquad (n > N) \tag{7}$$

Then it follows from (4)-(7) that

$$|x_ny_n-ab| < \frac{M\varepsilon}{2M}+\frac{M\varepsilon}{2M} = \varepsilon \qquad (n > N)$$

This proves equality (2).

Let us add (to the conditions $x_n \to a$ and $y_n \to b$) the supplementary condition $b \neq 0$. Then

$$\left|\frac{x_n}{y_n}-\frac{a}{b}\right| = \left|\frac{x_nb-y_na}{y_nb}\right| = \frac{|(x_n-a)b+(b-y_n)a|}{|y_n|\,|b|} \leqslant \frac{|x_n-a|}{|y_n|}+\frac{|b-y_n|\,|a|}{|y_n|\,|b|} \tag{8}$$

Now we can use Theorem 2 of the foregoing section according to which

$$|y_n| > \frac{|b|}{2} \qquad (n > N_1) \tag{9}$$

for sufficiently large N_1. Setting an arbitrary $\varepsilon > 0$ we choose N_2 and N_3 such that

$$|x_n-a| < \frac{\varepsilon|b|}{4} \qquad (n > N_2) \tag{10}$$

and

$$|a|\,|y_n-b| < \frac{\varepsilon b^2}{4} \qquad (n > N_3) \tag{11}$$

Finally, putting $N = \max\{N_1, N_2, N_3\}$ we obtain, by virtue of (8)-(11), the inequalities

$$\left|\frac{x_n}{y_n}-\frac{a}{b}\right| < \frac{\varepsilon|b|}{4}\,\frac{2}{|b|}+\frac{\varepsilon}{2} < \varepsilon \qquad (n > N)$$

whence follows equality (3).

It should be noted that the limits of the variables on the left-hand sides of equalities (1)-(3) may exist even when the limit of x_n or the limit of y_n (or both) does not exist. For instance, if $x_n = y_n = n$ then x_n and y_n have no (finite) limits while $\lim (x_n-y_n) = 0$ and $\lim x_n/y_n = 1$.

The theorems on the limits of sums, differences, products and quotients make it possible to find out in many cases whether a given variable has a limit and what this limit is provided that this variable is the result of a finite number of arithmetical operations on some other variables the existence of whose limits is established and the values of the limits are known.

But there are also many cases when the above theorems are inapplicable. To find such limits a mathematician should apply some other methods.

§ 3.3. Infinitesimals and Infinities

A variable α_n possessing zero limit is called an *infinitely small quantity (magnitude)* or, simply, an *infinitesimal*.

Thus, a variable α_n is an infinitesimal if, given any $\varepsilon > 0$, there is N such that $|\alpha_n| < \varepsilon$ for $n > N$. It is obvious that *for a variable x_n to have a number a as its limit it is necessary and sufficient that $x_n = a+\alpha_n$ where α_n is an infinitesimal.*

A variable β_n is called an *infinitely large quantity (magnitude)* or, simply, an *infinity*, if for any $M > 0$ there is N such that $|\beta_n| > M$ for $n > N$. For an infinitely large variable β_n we write

$$\lim \beta_n = \infty \quad \text{or} \quad \beta_n \to \infty \tag{1}$$

and say that β_n *tends to infinity*. This terminology proves convenient although the symbol ∞ does not denote any number, and an *infinitely large variable does not tend to any finite limit (number)*.

If, beginning with some $n = N$, an infinity β_n assumes only positive values or only negative values we write, respectively,

$$\lim \beta_n = +\infty \qquad (\text{equivalently, } \beta_n \to +\infty) \tag{2}$$

or

$$\lim \beta_n = -\infty \qquad (\text{equivalently, } \beta_n \to -\infty) \tag{3}$$

We see that from (2) (and also from (3)) follows (1). The example of the variable $\{(-1)^n n\}$ shows that it may happen that relation (1) holds while neither (2) nor (3) holds.

We state the following obvious properties:

1. *If a variable x_n is bounded while y_n is infinitely large then $x_n/y_n \to 0$.*

2. *If the absolute value of x_n is bounded below by a positive number and y_n is an infinitesimal such that $y_n \neq 0$ then $x_n/y_n \to \infty$.*

We shall confine ourselves to the proof of the second property.

By the hypothesis, there is a number $a > 0$ such that the inequality $|x_n| > a$ holds for $n = 1, 2, \ldots$, and, given any $\varepsilon > 0$, there is N such that

$$|y_n| < \varepsilon \qquad (n > N) \tag{4}$$

Then

$$\left|\frac{x_n}{y_n}\right| > \frac{a}{\varepsilon} \qquad (n > N)$$

Taking an arbitrary positive number M we can choose ε such that $M = a/\varepsilon$ and then find, for the given ε, a natural N so that Property (4) holds. Then

$$\left|\frac{x_n}{y_n}\right| > M \qquad (n > N)$$

which is what we intended to prove.

The above two assertions imply corollaries

$$\lim_{y_n \to \infty} \frac{c}{y_n} = 0 \quad \text{and} \quad \lim_{y_n \to 0} \frac{c}{y_n} = \infty \qquad (c \neq 0)$$

Let $a \geq 0$ and let k be a natural number. By $\sqrt[k]{a}$ we shall mean, unless the contrary is stated*, the arithmetic kth root of a, that is a nonnegative number whose kth power is equal to a. Such a number exists and is unique. It will be more convenient to prove this assertion later (at the end of § 4.5) but we shall use it now. This is a way in which roots are treated in elementary mathematics: their properties are proved without justifying their existence**.

Examples

1. $\lim\limits_{n \to \infty} \sqrt[k]{n} = \infty$ $(k = 1, 2, 3 \ldots)$ because the inequalities $\sqrt[k]{n} > N$ and $n > N^k$ (where $N > 0$) imply each other, and therefore, given any N, there is n_0 (namely, any $n_0 > N^k$) such that $\sqrt[k]{n} > N$ for all $n > n_0$.

2. At the same time, we have $\lim\limits_{n \to \infty} \sqrt[n]{n} = 1$. Indeed, $\sqrt[n]{n} = 1 + \varepsilon_n$ where $\varepsilon_n > 0$. Hence***, $n = (1 + \varepsilon_n)^n > \dfrac{n(n-1)}{1 \cdot 2} \varepsilon_n^2$; therefore $\varepsilon_n^2 < \dfrac{2}{n-1}$ and (see Example 1) $\varepsilon_n < \sqrt{2}/\sqrt{n-1} \to 0$ $(n \to \infty)$.

3. For $a > 1$ and natural k we have $\lim\limits_{n \to \infty} (n^k/a^n) = 0$ since if we put $a = 1 + \varepsilon$ then $\varepsilon > 0$ and, for $n > k$, we can write

$$\frac{n^k}{a^n} = \frac{n^k}{(1+\varepsilon)^n} < \frac{n^k}{C_n^{k+1}\varepsilon^{k+1}} = \frac{(k+1)!}{\varepsilon^{k+1}} \frac{n^k}{n(n-1)\ldots(n-k)} =$$

$$= \frac{(k+1)!}{\varepsilon^{k+1}} \frac{1}{\left(1-\frac{1}{n}\right)\ldots\left(1-\frac{k}{n}\right)} \frac{1}{n} \to 0 \qquad (n \to \infty)$$

---

* If $k > 2$ then among the values of the kth root of a there are complex values.

** Here are meant the properties enumerated at the beginning of § 4.6 (including relation (1)).

*** Here we have used Newton's binomial formula. It is usually derived in elementary mathematical course. The derivation of the formula based on the notion of the derivative of x^n is presented in § 5.9.

§ 3.4. Existence of Limit of Monotone Bounded Sequence

Not every variable has a limit. It often happens that it is necessary to find out whether a given variable possesses a limit. The theorem below provides a simple existence criterion for the limit of a variable.

Theorem 1. *Let a variable x_n, $n = 1, 2, \ldots$, be nondecreasing (nonincreasing), that is let x_n satisfy the condition $x_n \leqslant x_{n+1}$ ($x_n \geqslant x_{n+1}$) for any $n = 1, 2, \ldots$ If it is bounded above (below) by a number B (A) there exists a limit* $\lim x_n$ *equal to a number M (m) satisfying the inequality $M \leqslant B$ ($A \leqslant m$). If the variable is not bounded above (below) then* $\lim x_n = +\infty$ ($\lim x_n = -\infty$).

Proof. We shall present the proof for the case of a nondecreasing variable. In the case of a nonincreasing variable the proof is quite analogous.

Let a variable x_n be nondecreasing and bounded above by a number B. Then, as is known, there exists the supremum $\sup\limits_n x_n = M \leqslant B$.

Therefore $x_n \leqslant M$ ($n = 1, 2, \ldots$) and, given any $\varepsilon > 0$, there is $n = n_0$ such that

$$M - \varepsilon < x_{n_0}$$

Since the variable x_n is nondecreasing we have

$$x_{n_0} \leqslant x_n$$

for all $n > n_0$. Consequently

$$M - \varepsilon < x_n \leqslant M < M + \varepsilon \qquad \text{for } n > n_0$$

that is

$$|x_n - M| < \varepsilon \qquad \text{for } n > n_0$$

which means that x_n has a limit equal to M:

$$\lim x_n = M$$

Now suppose that a nondecreasing variable x_n is not bounded above. Then, given an arbitrary positive number M, however large, there is x_{n_0} such that $M < x_{n_0}$. Since x_n is nondecreasing we have

$$x_{n_0} \leqslant x_n \qquad \text{for } n > n_0$$

Hence, for any positive number M there exists $N = n_0$ such that

$$M < x_n \qquad \text{for } n > N$$

which means that $\lim x_n = +\infty$.

Nondecreasing and nonincreasing sequences are both called *monotone sequences.*

Example 1. The variable q^n ($n = 1, 2, \ldots$) where $0 < q < 1$ satisfies the condition $q^{n+1} < q^n$, which means that it is monotone decreasing. It is bounded below because $0 \leqslant q^n$ for any n. Consequently, by Theorem 1 it has a limit $\lim\limits_{n\to\infty} q^n = A$.

Obviously, q^{n+1} must have the same limit A; hence

$$\lim q^{n+1} = q \lim q^n = qA \quad \text{and} \quad A = qA$$

Since $q \neq 1$ this can only be the case when $A = 0$. Thus

$$\lim_{n\to\infty} q^n = 0 \qquad (0 < q < 1)$$

It follows that

$$\lim_{n\to\infty} a^n = \lim_{n\to\infty} \frac{1}{(1/a)^n} = +\infty$$

for $a > 1$.

Example 2. Let us show that $\lim\limits_{n\to\infty} \frac{a^n}{n!} = 0$.

By virtue of the equality $|a^n/n!| = |a|^n/n!$, it suffices to consider the case $a > 0$. Let m be a natural number such that $m+1 > a$. Then (see Example 1)

$$\frac{a^n}{n!} < \frac{a^m}{m!}\left(\frac{a}{m+1}\right)^{n-m} \to 0 \qquad (n \to \infty)$$

Example 3. If a sequence $a_n = \gamma_{n1}.\gamma_{n2}\gamma_{n3}\dots$ $(n = 1, 2, 3, \dots)$ of real numbers represented as infinite decimals is stabilized, i.e. $a_n \rightrightarrows a$ where $a = \gamma_0.\gamma_1\gamma_2\dots$ is a definite number, then $a = \lim a_n$. Indeed, given any $\varepsilon > 0$, there is m such that $2\cdot 10^{-m} < \varepsilon$, and for this m there is n_0 such that

$$a_n = \gamma_0.\gamma_1\dots\gamma_m\gamma_{n,\,m+1}\gamma_{n,\,m+2}\dots$$

for $n > n_0$. Since the first $m+1$ decimal digits of a and a_n coincide we have

$$\begin{aligned} |a-a_n| &= |0.0\dots 0\gamma_{m+1}\gamma_{m+2}\dots - 0.0\dots 0\gamma_{n,\,m+1}\gamma_{n,\,m+2}\dots| \leqslant \\ &\leqslant |0.0\dots 0\gamma_{m+1}\dots| + |0.0\dots 0\gamma_{n,\,m+1}\dots| \leqslant \\ &\leqslant 10^{-m} + 10^{-m} < \varepsilon \end{aligned}$$

§ 3.5. The Number e

Let us consider the variable

$$\alpha(n) = \left(1+\frac{1}{n}\right)^n \qquad (n = 1, 2, \dots)$$

We have

$$\begin{aligned} \alpha(n) &= 1 + n\frac{1}{n} + \frac{n(n-1)}{1\cdot 2}\,\frac{1}{n^2} + \dots = \\ &= 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) + \dots \end{aligned}$$

and, similarly,

$$\alpha(n+1) = 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n+1}\right) + \frac{1}{3!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right) + \dots$$

The terms in $\alpha(n)$ are less than the corresponding terms in $\alpha(n+1)$ and, besides, $\alpha(n+1)$ includes an extra positive term (the last one). Therefore

$\alpha(n) < \alpha(n+1)$ $(n = 1, 2, \ldots)$ and thus the variable $\alpha(n)$ is monotone increasing.

Furthermore,

$$\alpha(n) = 2+\frac{1}{2!}\left(1-\frac{1}{n}\right)+\frac{1}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)+\ldots <$$

$$< 2+\frac{1}{2!}+\frac{1}{3!}+\ldots+\frac{1}{n!} < 2+\frac{1}{2}+\frac{1}{2^2}+\ldots+\frac{1}{2^{n-1}} < 2+1 = 3$$

This means that the variable $\alpha(n)$ is bounded above.

Consequently the variable $\alpha(n)$ is monotone increasing and bounded above. Hence, by Theorem 1, it possesses a limit which does not exceed 3.

This limit is a definite number called the *number e*. Thus,

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e \tag{1}$$

The number e plays an extremely important role in mathematics, which very soon will become clear to the reader. In a certain sense it is natural to choose it as the base of logarithms. This number is also called the *Napier number* after J. Napier (1550-1617), a Scotch mathematician. It is irrational. Here we write the number e to six correct decimal places:

$$e = 2.718281$$

In § 5.10 we shall show how the number e can be evaluated with an arbitrary preassigned accuracy.

After the notion of a limit of a function has been introduced we shall see that the above limit exists and is equal to e even when n tends to plus infinity and to minus infinity assuming continuously the corresponding real values.

§ 3.6. Cauchy's* Criterion for Existence of Limit

Let a variable x_n $(n = 1, 2, \ldots)$ tend to a finite limit a. Then given an arbitrary positive ε, there is N such that

$$|x_n-a| < \frac{\varepsilon}{2} \qquad (n > N)$$

Let n and m be any two natural numbers exceeding N. Then

$$|x_n-a| < \frac{\varepsilon}{2} \quad \text{and} \quad |x_m-a| < \frac{\varepsilon}{2} \qquad (n, m > N)$$

whence

$$|x_n-x_m| = |x_n-a+a-x_m| \leqslant |x_n-a|+|a-x_m| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon$$

and we thus arrive at the following assertion:

* A. L. Cauchy (1789-1857), a famous French mathematician. He was the first to state the definitions of the basic notions of mathematical analysis (limit, continuity, integral, etc.) in the way characteristic of modern mathematics.

If a variable x_n $(n = 1, 2, \ldots)$ has a finite limit it satisfies the condition known as Cauchy's condition: for any $\varepsilon > 0$ there is N such that the inequality

$$|x_n - x_m| < \varepsilon$$

*is fulfilled for all n, $m > N$.**

It turns out that the converse is also true:

If a variable x_n $(n = 1, 2, \ldots)$ satisfies Cauchy's condition it tends to a finite limit, that is there exists a number a such that

$$\lim x_n = a$$

We shall prove this proposition. Let there be given a variable x_n $(n = 1, 2, \ldots)$ satisfying Cauchy's condition. Let us put $\varepsilon = 1$ and find N such that

$$|x_n - x_m| < 1 \qquad (n, m > N)$$

Fixing some $m > N$ we obtain from the last inequality

$$1 > |x_n - x_m| \geqslant |x_n| - |x_m|$$

that is

$$|x_n| < 1 + |x_m| \qquad (n > N)$$

and hence the variable x_n is bounded.

Now, for an arbitrary natural number n we define the following two numbers

$$\alpha_n = \inf_{k>n} x_k \quad \text{and} \quad \beta_n = \sup_{k>n} x_k$$

They are respectively the supremum and the infimum of the numbers x_k for $k > n$. We can also say that the numbers α_n and β_n are defined so that $[\alpha_n, \beta_n]$ is the smallest (in its length) closed interval containing the set of the numbers x_k with $k > n$. The intervals $[\alpha_n, \beta_n]$ form a nested family:

$$[\alpha_n, \beta_n] \supset [\alpha_{n+1}, \beta_{n+1}]$$

because

$$\alpha_n = \inf_{k>n} x_k \leqslant \inf_{k>n+1} x_k = \alpha_{n+1} \leqslant \beta_{n+1} = \sup_{k>n+1} x_k \leqslant \sup_{k>n} x_k = \beta_n$$

It may turn out that $\alpha_n = \beta_n$ for some n. Then the interval $[\alpha_n, \beta_n]$ degenerates into a point but this does not affect the validity of our further argument.

The length of the interval $[\alpha_n, \beta_n]$ tends to zero as $n \to \infty$ and therefore Cauchy's condition implies that for any $\varepsilon > 0$ there is n_0 such that

$$x_{n_0} - \varepsilon < x_k < x_{n_0} + \varepsilon$$

for all $k > n_0$ whence

$$x_{n_0} - \varepsilon \leqslant \alpha_{n_0} \leqslant \beta_{n_0} \leqslant x_{n_0} + \varepsilon$$

---

* A sequence satisfying Cauchy's condition is called a *Cauchy sequence* or a *fundamental sequence*. — *Tr.*

i.e.

$$2\varepsilon \geqslant \beta_{n_0}-\alpha_{n_0} \geqslant \beta_n-\alpha_n \qquad (n > n_0)$$

According to the nested interval theorem, there is a point a belonging to all the intervals $[\alpha_n, \beta_n]$. Let us prove that a is the limit of x_n. To this end, let us take an arbitrary interval (c, d) containing the point a.

Since the lengths of the intervals $[\alpha_n, \beta_n]$ tend to zero there is an interval $[\alpha_n, \beta_n]$ which is entirely contained in (c, d).

It is obvious that for $k > n$ we have $x_k \in [\alpha_n, \beta_n] \subset (c, d)$ which shows that

$$a = \lim_{n\to\infty} x_n$$

Combining the direct and the converse propositions we arrive at the following theorem expressing *Cauchy's criterion for the existence of a (finite) limit*:

Theorem. *For a variable x_n $(n = 1, 2, 3, \ldots)$ to tend to a (finite) limit it is necessary and sufficient that it should satisfy Cauchy's condition.*

We also note that Cauchy's condition can be stated in the following alternative form: *for any $\varepsilon > 0$ there is N such that*

$$|x_{n+p}-x_n| < \varepsilon$$

for all $n > N$ and any natural p.

§ 3.7. Subsequences. Limit Superior and Limit Inferior

Consider an arbitrary sequence $\{x_n\} = \{x_1, x_2, x_3, \ldots\}$ consisting of real numbers. We can choose in infinitely many ways a new sequence consisting of some elements of the given sequence:

$$\{x_{n_k}\} = \{x_{n_1}, x_{n_2}, x_{n_3}, \ldots\}$$

Here the index n_k runs through an increasing (infinite!) sequence of natural numbers $n_1 < n_2 < n_3 \ldots$. The sequence $\{x_{n_k}\}$ is termed a *subsequence of the original sequence* $\{x_n\}$.

We shall only be interested in subsequences which converge to finite numbers or to $+\infty$ or to $-\infty$ (that is have finite limits or infinite limits of the type of $+\infty$ or $-\infty$). All the sequences (and subsequences) of this kind will be referred to as *convergent sequences* and their limits will be conditionally called *numbers* (*finite* or *infinite*). This convention extends the term "number" to the symbols $-\infty$ and $+\infty$. If α is an arbitrary (finite) number we usually write $-\infty < \alpha < +\infty$. Under the above convention $+\infty$ can be regarded as the *greatest number* and $-\infty$ as the *least number.* It is obvious that for the set of numbers thus extended all the axioms of real numbers forming Group I (see § 2.4) hold.

It should be stressed (this is extremely important for our further argument) that the elements x_n (but not the numbers x_n!) of a sequence $\{x_n\}$ are regarded as different if they are supplied by different indices n. Thus, one should distinguish between the numbers (points) corresponding to the values through which the general element of a sequence runs and the elements of the sequence themselves.

For instance, the sequence

$$1, 2, 3, 1, 2, 3, 1, \ldots \tag{1}$$

(like any other sequence) consists of an infinite number of elements x_1, x_2, x_3, ... but its general element x_n runs through a set consisting of only three numbers (points) 1, 2 and 3.

It is evident that if a sequence is convergent its every subsequence is also convergent to the same number (finite or equal to $+\infty$ or to $-\infty$). But if a given sequence $\{x_n\}$ has a convergent subsequence this does not imply that the given sequence is itself convergent.

At the same time there holds an important theorem often called the *Weierstrass* theorem* which we state and prove below.

Theorem 1. *Every bounded sequence $\{x_n\}$ contains a convergent subsequence $\{x_{n_k}\}$ whose limit is a finite number.*

Proof Let the values x_n of the sequence in question belong to a closed interval $\Delta_0 = [c, d]$. Dividing this interval into two equal parts we can take the rightmost of these parts containing an infinite number of elements x_n (let it be denoted as Δ_1).

This means that if both parts contain an infinitude of elements then Δ_1 is the rightmost of them and if only one of these parts contains an infinite number of elements x_n then Δ_1 denotes that very part.

Let x_{n_1} be one of the elements belonging to the interval Δ_1. Dividing the interval Δ_1 into two equal parts we denote by Δ_2 the rightmost of the parts containing an infinite number of elements x_n. Among these elements there is obviously an element x_{n_2} with $n_2 > n_1$. Generally, if the closed intervals $\Delta_1 \supset \Delta_2 \supset \ldots \supset \Delta_{k-1}$ and the corresponding elements $x_{n_1}, \ldots, x_{n_{k-1}}$ belonging to them have been already constructed we denote by Δ_k the rightmost of the halves of the interval Δ_{k-1} containing an infinitude of elements x_n. Among these elements there is obviously an element x_{n_k} with $n_k > n_{k-1}$. Let a be the point belonging to all Δ_k $(k = 1, 2, \ldots)$. The subsequence $\{x_{n_k}\}$ which has thus been constructed is obviously convergent to a.

The theorem has been proved.

Now we state the following definition: *a number α* (finite or equal to $+\infty$ or to $-\infty$) *is called the limit superior (the limit inferior) of a sequence of real numbers $\{x_n\}$, or, equivalently, of the variable x_n, if there exists a subsequence $\{x_{n_k}\}$ convergent to α and if any other convergent subsequence of the sequence $\{x_n\}$ tends to a number which is not greater (not less) than α.*

For example, sequence (1) obviously has 3 as its limit superior and 1 as its limit inferior while the limit superior of the sequence $1, -2, 3, -4, \ldots$ is equal to $+\infty$, its limit inferior being equal to $-\infty$.

---

* K. I. W. Weierstrass (1815-1897), a noted German mathematician.

The limit superior and the limit inferior are respectively denoted as $\overline{\lim}\, x_n$, and $\underline{\lim}\, x_n$ and also as l m sup x_n and lim inf x_n (in connection with the second notation also see the exercise at the end of this section).

A sequence $\{x_n\}$ can have only one limit superior (limit inferior) because if we suppose that a_1 and a_2 are two such limits and $a_1 < a_2$ then there must exist a subsequence $\{x_{n_k}\}$ convergent to a_2 which contradicts the assumption that a_1 is the limit superior.

It should be noted that the construction used in the proof of Theorem 1 which is based on the nested interval theorem leads to a subsequence $\{x_{n_k}\}$ convergent to a number a coinciding with the limit superior of the sequence $\{x_n\}$:

$$a = \overline{\lim}\, x_n$$

Indeed, suppose that $a' > a$. We can choose a natural n so large that a' lies to the right of Δ_n. The number of elements x_n lying to the right of Δ_n can only be finite and consequently there cannot exist a subsequence of the sequence $\{x_n\}$ convergent to a'.

This construction thus proves the existence of the limit superior of a bounded sequence.

If we changed this construction and, for every n, denoted by Δ_n not the rightmost but the leftmost of the halves of Δ_{n-1} containing an infinite number of elements x_n we should obtain a number (point) a equal to the limit inferior of the sequence $\{x_n\}$.

Next we shall show that the limit superior (the limit inferior) a of a bounded sequence $\{x_n\}$ possesses the following property: *given an arbitrary $\varepsilon > 0$, the interval $(a-\varepsilon,\ a+\varepsilon)$ contains an infinite number of elements x_n and there are not more than a finite number of elements x_n lying to the right (to the left) of this interval.*

Indeed, there exists n such that $\Delta_n \subset (a-\varepsilon,\ a+\varepsilon)$; since Δ_n contains an infinite number of elements x_n so does $(a-\varepsilon,\ a+\varepsilon)$. To the right (to the left) of Δ_n there are not more than a finite number of elements x_n and therefore the same applies to $(a-\varepsilon,\ a+\varepsilon)$.

For a bounded sequence $\{x_n\}$ of real numbers this property is equivalent to the definition of the limit superior (of the limit inferior) stated above.

Indeed, for definiteness, let us consider the case of the limit superior. If a is a number possessing the indicated property and a' is another number exceeding a $(a' > a)$ then choosing ε so that $a < a+\varepsilon < a'$ we see that there are not more than a finite number of elements x_n to the right of $a+\varepsilon$. Consequently there cannot exist a subsequence of the sequence $\{x_n\}$ convergent to a'. At the same time there exists a subsequence of the sequence $\{x_n\}$ convergent to a. To obtain it we fix $\varepsilon > 0$ and choose n_1 so that $x_{n_1} \in (a-\varepsilon,\ a+\varepsilon)$ after which we choose $n_2 > n_1$ so that $x_{n_2} \in \left(a-\frac{\varepsilon}{2},\ a+\frac{\varepsilon}{2}\right)$, which is possible since the interval $\left(a-\frac{\varepsilon}{2},\ a+\frac{\varepsilon}{2}\right)$ contains an infinite number of elements x_n. Next we choose $n_3 > n_2$ so that $x_{n_3} \in \left(a-\frac{\varepsilon}{3}, a+\frac{\varepsilon}{3}\right)$, etc.

If a sequence is not bounded above it is clear that it contains a subsequence convergent to $+\infty$, and since $+\infty$ is greater than any number we have

$$\overline{\lim}\, x_n = +\infty$$

If a sequence is bounded above by a finite number (we denote it b) but is not bounded below the following two cases are possible:

Case 1. There is a finite number $a < b$ such that the inequalities $a \leqslant x_n \leqslant b$ are satisfied for an infinite number of values of the index n. If these values are arranged in an increasing order they form an infinite subsequence of natural numbers $\{n_k\} = \{n_1 < n_2 < \ldots\}$. To this subsequence corresponds the subsequence $\{x_{n_k}\}$ of the original sequence which is obviously bounded. The existence of the limit superior of such a subsequence has already been proved. It is readily seen that

$$\overline{\lim_{n\to\infty}}\, x_n = \overline{\lim_{k\to\infty}}\, x_{n_k}$$

and we have thus proved that in Case 1 $\overline{\lim}\, x_n$ also exists.

Case 2. For any finite a, inequality $a \leqslant x_n \leqslant b$ or, which is the same in this case, the inequality $a \leqslant x_n$ holds for a finite number of values of the index n. This obviously means that

$$\lim x_n = -\infty$$

But then for the limit superior (and also for the limit inferior!) we have $\overline{\lim}\, x_n = -\infty$, that is this limit exists in the latter case as well.

Combining these results with the results concerning a bounded sequence established above we arrive at

Theorem 2. *Any sequence of real numbers $\{x_n\}$ has the limit superior (the limit inferior) equal to a finite number or to $+\infty$ or to $-\infty$.*

If a sequence is bounded its limit superior and limit inferior are finite numbers.

There also holds

Theorem 3. *Every infinite sequence $\{x_n\}$ of real numbers contains a subsequence convergent to a finite number or to $+\infty$ or to $-\infty$.*

(If a sequence is bounded the limit of its every subsequence is a finite number.)

Indeed, by Theorem 2, any sequence has the limit superior and hence there exists a subsequence $\{x_{n_k}\}$ convergent to it.

Theorem 4. *The inequality $\underline{\lim}\, x_n \leqslant \overline{\lim}\, x_n$ holds in all the cases; the equality sign appears in this relation if and only if there exists the limit (equal* *to a finite number or to $+\infty$ or to $-\infty$) of the variable x_n. Then*

$$\underline{\lim}\, x_n = \overline{\lim}\, x_n = \lim x_n \tag{2}$$

Indeed, if the limit of x_n exists all the subsequences $\{x_n\}$ tend to it, which implies (2). Conversely, let

$$\underline{\lim}\, x_n = \overline{\lim}\, x_n = A \tag{3}$$

If A is a finite number it follows from (3) that, given any $\varepsilon > 0$, the inequality

$$A-\varepsilon < x_n < A+\varepsilon$$

is fulfilled for all the values of the index n except a finite number of them, which means that $x_n \to A$. If $A = +\infty$ the inequality $x_n \leqslant M$ can only be satisfied by a finite number of elements x_n for any finite M; then $\lim x_n = +\infty$. The case $A = -\infty$ is treated similarly.

Theorem 5. *If a sequence $\{x_n\}$ is such that its every subsequence contains a subsequence convergent to one and the same number A there exists the limit $\lim x_n = A$.*

Indeed, since any convergent subsequence $\{x_n\}$ tends to A we have $\underline{\lim}\, x_n = \overline{\lim}\, x_n = A$, and hence, by Theorem 4, the limit $\lim x_n = A$ exists.

We also note the following useful equality:

$$\underline{\lim}\, x_n = -\overline{\lim}\, (-x_n) \tag{4}$$

For bounded variables x_n and y_n we can also write the inequalities

$$\overline{\lim}\, (x_n+y_n) \leqslant \overline{\lim}\, x_n + \overline{\lim}\, y_n \tag{5}$$

and

$$\underline{\lim}\, (x_n+y_n) \geqslant \underline{\lim}\, x_n + \underline{\lim}\, y_n \tag{6}$$

where x_n and y_n are bounded variables.

Inequality (5) is proved as follows: there is a convergent subsequence $\{x_{n_k}+y_{n_k}\}$ such that

$$\overline{\lim}\, (x_n+y_n) = \lim\, (x_{n_k}+y_{n_k}) \tag{7}$$

and it is possible to choose a subsequence $\{n_k'\}$ of $\{n_k\}$ such that the limit $\lim x_{n_k'}$ exists. Further, the subsequence $\{n_k'\}$, in its turn, contains a subsequence $\{n_k''\}$ such that the limit $\lim y_{n_k''}$ exists, and therefore the limit $\lim x_{n_k''}$ also exists. Consequently,

$$\lim\, (x_{n_k}+y_{n_k}) = \lim\, (x_{n_k''}+y_{n_k''}) = \lim x_{n_k''} + \lim y_{n_k''} \leqslant \overline{\lim}\, x_n + \overline{\lim}\, y_n \tag{8}$$

From (7) and (8) follows (5).

By (4) and (5), we have

$$\begin{aligned}\underline{\lim}\, (x_n+y_n) &= -\overline{\lim}\, \big(-x_n+(-y_n)\big) \geqslant -\big(\overline{\lim}\, (-x_n)+\overline{\lim}\, (-y_n)\big) = \\ &= \underline{\lim}\, x_n + \underline{\lim}\, y_n\end{aligned}$$

that is (6) holds.

Example. Let us consider the sequence $E = \{\sin n\alpha\}$ $(n = 1, 2, \ldots)$. If $\alpha = \lambda\pi$ where $\lambda = p/q$ is a rational number $(p, q > 0)$ the values of the sequence are of a periodic character:

$$\sin\alpha, \sin 2\alpha, \ldots, \sin 2q\alpha, \sin\alpha, \sin 2\alpha, \ldots \tag{9}$$

The limits of its various convergent subsequences can only be equal to one of the first $2q$ numbers (9). The greatest of them is obviously $\overline{\lim}\sin n\alpha$ while the least is $\underline{\lim}\sin n\alpha$.

Now suppose that $\lambda > 0$ is an irrational number. Let us mark the numbers $n\alpha$ as arc lengths on the unit circle γ, as is done in trigonometry. Then, for any different natural numbers n_1 and n_2, the points $n_1\alpha$ and $n_2\alpha$ are, geometrically, different because, if otherwise, there would hold the equality

$$n_2\alpha = n_1\alpha + 2k\pi \qquad (\alpha = \lambda\pi)$$

where k is an integer, that is $(n_2 - n_1)\lambda = 2k$, and λ would be a rational number. Consequently the points $n\alpha$ $(n = 0, 1, 2, \ldots)$ form an infinite set which we denote by $\mathfrak{M}$. Now it can readily be shown that, given any $\varepsilon > 0$, there is a pair of points $n_1\alpha$ and $n_2\alpha$ such that the distance between them (measured along γ) is less than ε. This means that

$$(n_2 - n_1)\alpha = 2k\pi + \omega = \beta \qquad (n_1 < n_2)$$

where $|\omega| < \varepsilon$ and k is an integer.

The points $0, \beta, 2\beta, 3\beta, \ldots$ obviously belong to $\mathfrak{M}$. Furthermore, the distance between any two neighbouring points of this sequence is equal to one and the same constant number not exceeding ε. It follows that for any point $t \in \gamma$ there is a point on γ belonging to the set $\mathfrak{M}$ and lying at a distance (measured along γ) less than ε from the former point. This shows that any point $t \in \gamma$ is a limit point* of the set $\mathfrak{M}$.

What has been said implies that for any t there is a subsequence of the sequence of natural numbers $n_1 < n_2 < n_3 \ldots$ such that

$$\lim_{k\to\infty} \sin n_k\alpha = \sin t$$

But $\sin t$ assumes all the values lying within the closed interval $[-1, +1]$ whence follows that

$$\underline{\lim}_{n\to\infty} \sin n\alpha = -1 \quad \text{and} \quad \overline{\lim}_{n\to\infty} \sin n\alpha = 1$$

Exercise. Let the reader prove that for any variable x_n there hold the relations

$$\overline{\lim}\, x_n = \lim_{n\to\infty} \sup_{k>n} x_k \quad \text{and} \quad \underline{\lim}\, x_n = \lim_{n\to\infty} \inf_{k>n} x_k$$

* The definition of a limit of a set is given in § 3.8. As follows from this definition, a set possessing a limit point contains a sequence convergent to that point (which may not belong to the set). — *Tr.*

Hint. For a variable not bounded above (below) we have $\sup_{k>n} x_k = +\infty$ $\left(\inf_{k>n} x_k = -\infty\right)$ and therefore it is allowable to write $\lim_{n\to\infty} (+\infty) = +\infty$ $\left(\lim_{n\to\infty} (-\infty) = -\infty\right)$.

§ 3.8. Weierstrass' Theorem

Let E be a set of numbers (of points of the real line). By definition, a point a is called a *limit point* (also an *accumulation point*) of E if its any neighbourhood, that is any interval (c, d) such that $c < a < d$, contains at least one point x belonging to E and distinct from a.

It is readily shown that *every neighbourhood* (c, d) *of a limit point a of a set E does in fact contain an infinite number of points of the set E, and it is possible to choose from them an infinite sequence of different points* $x_1, x_2, x_3, \ldots$ $(x_n \neq x_k$ *for* $n \neq k)$ *convergent to a.* Indeed, according to the definition, (c, d) contains a point $x_1 \in E$ such that $x_1 \neq a$. Now we can take an interval (c_1, d_1) of length $d_1 - c_1 < 1$ containing the point a and not containing x_1. Again, according to the definition of the point a, there is a point belonging to the latter interval such that $x_2 \in E$ and $x_2 \neq a$. Next we can find an interval (c_2, d_2) of length $d_2 - c_2 < 1/2$, containing the point a and not containing x_1 and x_2, in which there is a point $x_3 \in E$ such that $x_3 \neq a$, etc. As a result, we obtain the required sequence.

Thus, the definition of a limit point can be restated in the equivalent alternative form: *a point a is a limit point of a set E if its any neighbourhood contains an infinitude of points belonging to E.*

The set E' of all limit points of the given set E is called its *derived set.*

Example 1. The set E of the points of the sequence $\{1/n\}$ has the point 0 as its limit point.

Indeed, any neighbourhood of the point 0 contains an infinite number of points belonging to the sequence in question. On the other hand, if $a \neq 0$ the interval $(a-\varepsilon, a+\varepsilon)$ where $0 < \varepsilon < |a|$ contains either a finite number of points of the set E or no points of E at all. Hence, the only limit point of the set E is the point 0. This point itself does not belong to E.

We see that in this example the derived set E' consists of only one point 0 not belonging to E. The set E_1 consisting of the point 0 and of the points of the form $1/n$ $(n = 1, 2, \ldots)$ obviously also has 0 as its limit point but in this case the latter belongs to E_1. Thus, a limit point of a set may or may not belong to that set.

Example 2. The set R of all rational numbers has as its limit points all the points of the real axis, both rational and irrational, since any interval contains a point of R. The collection of all the limit points of the set of irrational points x satisfying the inequality $0 \leqslant x \leqslant 1$ is obviously the closed interval $[0, 1]$.

A set A consisting of a finite number of points obviously cannot possess a limit point, that is in this case A' is an empty set. An unbounded infinite set may have no limit point either; this is confirmed by the example of the set of natural numbers 1, 2, 3, ... On the contrary, for a bounded infinite set there holds the following

Weierstrass' Theorem. *Every bounded infinite set of points E has at least one limit point, that is in this case E' is a nonempty set.*

This theorem is proved in § 7.9 (Theorem 2) for the general case of n dimensions. But the reader may study this proof even now interpreting it for the one-dimensional case.

§ 3.9. Countable Sets. Countability of the Set of Rational Numbers. Uncountability of the Set of Real Numbers

A set E of arbitrary elements x is said to be *infinite* if, given any natural number n, however large, there is a subset of E containing more than n elements.

A set E is called *countable* if it is *infinite* and if its elements can be *numbered.* This means that it is possible to set up a one-to-one correspondence between all the elements $x \in E$ and the natural numbers

$$1, 2, 3, \ldots \tag{1}$$

If, under this correspondence, an element $x \in E$ is associated with a natural number n it appears natural to denote it x_n. As a result, the elements of the set E can be arranged as a *sequence*

$$E = \{x_1, x_2, x_3, \ldots\} \tag{2}$$

In particular, set (1) of natural numbers is a trivial example of a countable set. It is obvious that the set of all even natural numbers is also countable since it is infinite and its elements x can be numbered by putting $x_n = 2n$ $(n = 1, 2, \ldots)$.

Let E be a countable set and let its elements be numbered and arranged as a sequence of type (2). *If A is a nonempty subset of E then it contains an element with the smallest number (index).* Indeed, in (2) there is an element x_n, with number $n = n_1$, belonging to A. There are only a finite number of elements $x_n \in A$ with indices $n \leqslant n_1$; among them there is an element x_{n_0} with the smallest number which is obviously an element of A having the smallest number in comparison with all the other elements of A.

If E is a countable set and A is its infinite subset then A is a countable set whose elements can be numbered as follows: we denote by z_1 the element of A having the smallest number in the set E, delete this element from A and take the element of the remaining infinite set A_1 having the smallest number in E, this element being denoted by z_2; then we delete z_2 from A_1, etc.

A countable (set-theoretic) sum (union) of sets of the type

$$E = \bigcup_{k=1}^{\infty} E^k = E^1 + E^2 + \cdots$$

where each set E^k is countable is also a countable set. Indeed, let us arrange the elements $x_j^k \in E^k$ $(j = 1, 2, \ldots)$ as the table

$$\begin{array}{l} E^1 = \{x_1^1, \quad x_2^1, \quad x_3^1, \ \ldots\} \\ E^2 = \{x_1^2, \quad x_2^2, \quad x_3^2, \ \ldots\} \\ E^3 = \{x_1^3, \quad x_2^3, \quad x_3^3, \ \ldots\} \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots \end{array}$$

Now we can number them in the following order:

$$x_1^1, x_2^1, x_1^2, x_3^1, x_2^2, x_1^3, x_4^1, \ldots$$

Here at each stage of the numbering process we should delete those elements which have already been numbered at the foregoing stages; the matter is that E^k and E^l may have some common elements. This procedure results in an infinite sequence of elements $\{y_1, y_2, y_3, \ldots\}$ which obviously exhausts the set E. This shows that E is a countable set.

We can similarly prove that *a finite sum $E = E^1 + \ldots + E^N$ of countable or finite sets among which there is at least one countable set is also countable.*

Let us prove that *the set of all positive rational numbers and also the set of all negative rational numbers are countable sets;* this means that *the set of all rational numbers is also countable.*

There are no rational numbers p/q with $q \geqslant 0$ and $q > 0$ (p and q are integers) such that $p+q = 1$. There is exactly one number p/q with $p+q = 2$ which is $1 = 1/1$; let us denote it y_1. Now among the positive rational numbers yet remaining not numbered there are two numbers p/q with $p+q = 3$ which are $1/2$ and $2 = 2/1$; let us denote them respectively as y_2 and y_3. This process can be extended indefinitely by induction. As a result, all the positive rational numbers will be supplied with natural indices, which is what we intended to prove.

On the other hand, *the set of all real numbers is uncountable.*

To show this we shall prove that even the set of real numbers contained in the interval $(0, 1)$ is uncountable. This will obviously mean that the whole set of real numbers is uncountable as well since, as we know, a subset of a countable set can be either finite or countable (or empty). Let us write the points (numbers) x belonging to the interval $(0, 1)$ as infinite decimals. Suppose that the interval $(0, 1)$ (as a point set) is countable. Then all its points can be numbered:

$$\begin{array}{l} x^1 = 0.\alpha_1^1\alpha_2^1\alpha_3^1 \ldots \\ x^2 = 0.\alpha_1^2\alpha_2^2\alpha_3^2 \ldots \\ x^3 = 0.\alpha_1^3\alpha_2^3\alpha_3^3 \ldots \\ \ldots\ldots\ldots\ldots\ldots \end{array} \tag{3}$$

We shall show that this conclusion leads to a contradiction. Let us take, for every natural n, a decimal digit α_n such that the inequalities $\alpha_n > 0$ and $\alpha_n \neq \alpha_n^n$ hold, which is obviously possible. Now we construct the number

$a = 0.\alpha_1\alpha_2\alpha_3\ldots$. It belongs to the interval (0, 1) and therefore must have a certain number n_0 in Table (3): $a = x^{n_0}$. But then there must be $\alpha_{n_0} = \alpha_{n_0}^{n_0}$ which is impossible.

Exercises

1. Prove that the set of points in the plane with rational coordinates (x, y) is countable.
2. Prove the same assertion for the set of points $(x_1, \ldots, x_n)$ of the n-dimensional space of n-tuples having rational coordinates x_j.

CHAPTER 4

Limit of Function

§ 4.1. Concept of Limit of a Function

A number A is called the *limit of a function f at a point a* if the function is defined in a neighbourhood of the point a, that is on an interval (c, d), $c < a < d$, except possibly at the point a itself, and if for any $\varepsilon > 0$ there is $\delta > 0$ (dependent on ε) such that the inequality

$$|f(x)-A| < \varepsilon$$

holds for all x satisfying the condition $0 < |x-a| < \delta$.

The fact that A is the limit of f at a point a is symbolized as

$$\lim_{x\to a} f(x) = A \quad \text{or} \quad f(x) \to A \quad (x \to a)$$

An alternative definition of the limit of a function at a point can be stated in terms of limits of sequences.

A number A is called the *limit of a function f at a point a* if the function is defined in a neighbourhood of the point a, except possibly at the point a itself, and if the limit of the sequence $\{f(x_n)\}$ exists and is equal to A for any sequence $\{x_n\}$ convergent to a such that $x_n \neq a$ for all n. Thus,

$$\lim_{\substack{x_n\to a\\ x_n\neq a}} f(x_n) = A$$

Here, as in all similar cases, it is tacitly implied that the variable x_n tending to a runs through a set of values for which $f(x)$ is defined.

These two definitions are equivalent. Indeed, let a function f have a limit in the sense of the first definition and let there be a variable x_n, whose values do not coincide with the number a for any n tending to a. Setting an arbitrary ε we can choose, according to the first definition, a number δ such that $|f(x)-A| < \varepsilon$ for $0 < |x_n-a| < \delta$. Since $x_n \to a$ there is a natural number N such that $|x_n-a| < \delta$ for $n > N$; therefore we have

$$|f(x_n)-A| < \varepsilon \qquad \text{for} \quad n > N$$

that is the sequence $\{f(x_n)\}$ converges to A. Since this property holds for every sequence $\{x_n\}$ tending to a and such that $x_n \neq a$ and all x_n belong to the domain of definition of the function, we have thus proved that the first definition implies the second one.

Conversely, let a function $f(x)$ have a limit in the sense of the second definition. Suppose that it has no limit in the sense of the first definition. Then there exists at least one value of ε, say $\varepsilon = \varepsilon_0$, for which there is no δ mentioned in the first definition; this means that for any δ there is a value $x = x^{(\delta)}$ belonging to the set of numbers x satisfying the condition $0 < |x-a| < \delta$ such that $|f(x^{(\delta)})-A| \geqslant \varepsilon_0$.

Let us consecutively take as δ all the numbers $\delta_k = 1/k$ $(k = 1, 2, \ldots)$. For each of them there is a point $x_k = x_k^{(\delta)}$ such that

$$0 < |x_k-a| < \frac{1}{k} \qquad (x_k \neq a)$$

and

$$|f(x_k)-A| \geqslant \varepsilon_0 \qquad (k = 1, 2, \ldots)$$

These relations show that x_k $(x_k \neq a)$ tends to a while $f(x_k)$ does not tend to the number A. Hence, the assumption that the second definition does not imply the first one leads to a contradiction.

We have thus proved the equivalence of the two definitions.

The expression "the limit of a function at a point a" is often replaced by "the limit of a function for x tending to a" or, briefly, "the limit of a function as $x \to a$". The last two expressions are in a certain sense closer to the idea of the limit because the number $A = \lim\limits_{x\to a} f(x)$ has nothing in common with the value of f at the point $x = a$ itself. Moreover, the function may not be defined at $x = a$. The number A provides some information on the behaviour of the function in a small neighbourhood of the point a from which this point is deleted. It says that when x assumes values different from a and approaches a according to an arbitrary law, the corresponding value $f(x)$ approaches A, that is becomes arbitrarily close to A.

Example 1. Let us consider the function $f(x) = \frac{x^2-4}{x-2}$. It is defined for all $x \neq 2$. We shall find its limit as $x \to 2$. For any $x \neq 2$ we have $\frac{x^2-4}{x-2} = x+2$, and since the definition of the limit as $x \to 2$ does not involve the value of the function f at the point $x = 2$, we obtain

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} (x+2)$$

The meaning of this equality is that if one of the limits entering into it exists the other also exists and is equal to the former. Hence, instead of evaluating the limit of a more complex function $\frac{x^2-4}{x-2}$, it is sufficient to find the limit of the simpler function $x+2$ as $x \to 2$. The latter limit is obviously equal to 4. Indeed, if we substitute any variable x_n tending to 2 for x

into $x+2$ then, irrespective of the way in which x_n tends to 2, we obtain

$$\lim_{x_n\to 2} (x_n+2) = 2+2 = 4$$

The calculations connected with the evaluation of the given limit can be written in the following traditional form:

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} (x+2) = \lim_{x\to 2} x+2 = 4$$

It should be stressed that $f(x) = \dfrac{x^2-4}{x-2}$ and $\varphi(x) = x+2$ are different functions. The former is defined for $x \neq 2$ while the latter is defined for all x. But when evaluating the limit of the given function $f(x)$ as $x \to 2$ we are not interested in whether or not the function is defined at the point $x = 2$ itself, and, since $f(x) = \varphi(x)$ for $x \neq 2$, we have

$$\lim_{x\to 2} f(x) = \lim_{x\to 2} \varphi(x) = \varphi \tag{2}$$

Example 2. It is evident that $\lim\limits_{x\to 1} x^2 = 1$ because if $x_n \to 1$ and $x_n \neq 1$ then

$$\lim x_n^2 = \lim x_n \lim x_n = 1\cdot 1 = 1$$

On the other hand, this fact can also be proved by using the "ε, δ language".

Let us consider an interval containing the point 1, for instance $(-1/2, 3/2)$. For any x belonging to it we can write the obvious inequalities

$$|x^2-1| = |x+1|\,|x-1| \leqslant \frac{5}{2}|x-1|$$

Now, taking an arbitrary $\varepsilon > 0$ and putting $\delta = \min\left\{\frac{1}{2}, \frac{2}{5}\varepsilon\right\}$, we obtain the relation

$$|x^2-1| \leqslant \frac{5}{2}\cdot\frac{2}{5}\varepsilon = \varepsilon$$

for all x satisfying the inequality $|x-1| < \delta$, which completes the proof.

Example 3. The function $\sin (1/x)$ (its graph is shown in Fig. 4.1) is defined for all $x \neq 0$ and hence it is defined in any neighbourhood of the point $x = 0$ except at the point $x = 0$ itself. This function has no limit as $x \to 0$ because the sequence of values $x_k = \dfrac{2}{\pi(2k+1)}$, $k = 0, 1, 2, \ldots$, which are different from zero, tends to zero while the values

$$f(x_k) = (-1)^k$$

do not tend to any limit as $k \to \infty$.

Let us state the following definition. We shall write

$$A = \lim_{x\to\infty} f(x)$$

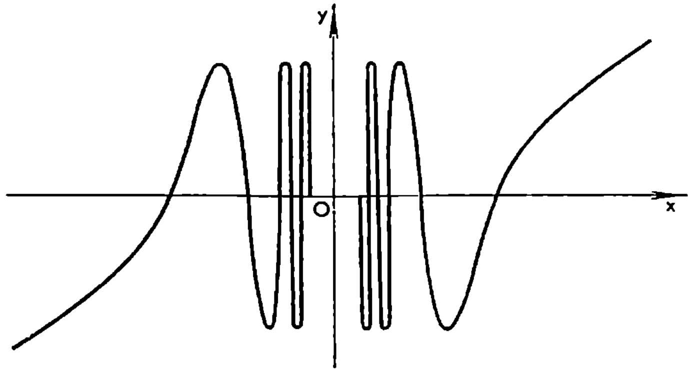

Fig. 4.1

and say that *A is the limit of the given function $f(x)$ for x tending to infinity* if f is defined for all x satisfying the inequality $|x| > K$ with some $K > 0$ and if for any $\varepsilon > 0$ there is a number $M > K$ such that $|f(x)-A| < \varepsilon$ for all x satisfying the inequality $|x| > M$.

It can be proved that this definition is equivalent to the following one:

A number A is the limit of a function $f(x)$ for $x \to \infty$ if the function $f(x)$ is defined for all x satisfying the condition $|x| > M$ for some M and if

$$\lim_{x_n \to \infty} f(x_n) = A$$

for any sequence $\{x_n\}$ tending to ∞.

The proof of the equivalence of these two definitions is carried out according to the scheme used above in the case of the limit of f at a finite point a.

Generally, many properties of limits of functions for $x \to a$ where a is a finite number and for $x \to \infty$ are analogous. These properties admit of a unified presentation which simultaneously covers the cases when $x \to a$ where a is a finite number and when $x \to \infty$. To this end we shall agree to interpret the symbol a as any (finite*) number or ∞. If a is a number then by a neighbourhood of the point a is meant any interval (c, d) containing the point a. Hence, a *neighbourhood of a (finite) point a* is the set of all the points x satisfying the inequalities $c < x < d$. If $a = \infty$ (or $a = +\infty$ or $a = -\infty$) we shall agree to understand a neighbourhood of a as the set of all x satisfying the inequalities

$$|x| > M \text{ (or } x > M \text{ or } x < -M) \quad \text{where} \quad M > 0$$

We shall write

$$\lim_{x \to a} f(x) = A$$

* The symbols ∞, $+\infty$ and $-\infty$ are often called *infinite numbers* and then the ordinary numbers are termed *finite numbers*.

where a can be a finite number or ∞ (or $+\infty$ or $-\infty$) if the function $f(x)$ is defined in a neighbourhood of the point a except possibly at the point a itself* and if for any $\varepsilon > 0$ there is a neighbourhood of the point a such that the inequality

$$|f(x)-A| < \varepsilon$$

is fulfilled for all x belonging to this neighbourhood and different from a.

The last definition of the limit of f unifies all the cases treated above, i.e. the cases when x tends to a finite number a and when x tends to ∞ or $+\infty$ or $-\infty$.

Now we proceed to the study of the properties of a function $f(x)$ having the limit as $x \to a$ where a is a (finite) number or ∞ or $+\infty$ or $-\infty$. Let us agree that an arbitrary neighbourhood of a will be denoted by the symbol $U(a)$. It is readily checked that *the intersection of two neighbourhoods $U_1(a)$ and $U_2(a)$ is again a neighbourhood $U(a)$.*

Theorem 1. *If $\lim\limits_{x\to a} f(x) = A$ where A is a finite number then the function $f(x)$ is bounded in some neighbourhood $U(a)$, that is there exists a positive number M such that*

$$|f(x)| \leqslant M \quad \textit{for all} \quad x \in U(a), \qquad x \neq a$$

Proof. The conditions of the theorem guarantee the existence of a neighbourhood $U(a)$ such that

$$1 > |f(x)-A| \geqslant |f(x)|-|A| \qquad (x \in U(a),\ x \neq a)$$

Whence for the indicated values of x it follows that

$$|f(x)| \leqslant 1+|A|$$

where $1+|A|$ plays the role of M. The theorem has been proved.

Theorem 2. *If $\lim\limits_{x\to a} f(x) = A$ where $A \neq 0$ is a finite number then there is a neighbourhood $U(a)$ such that*

$$|f(x)| > \frac{|A|}{2} \qquad (x \in U(a),\ x \neq a)$$

Moreover, for the indicated values of x we have

$$f(x) > \frac{A}{2} \quad \textit{if} \quad A > 0$$

and

$$f(x) < \frac{A}{2} \quad \textit{if} \quad A < 0$$

* This stipulation is only necessary when a is a finite point (number).

Proof. It follows from the conditions of the theorem that there is a neighbourhood $U(a)$ such that

$$\frac{|A|}{2} > |A-f(x)| \geqslant |A|-|f(x)| \qquad (x \in U(a),\ x \neq a)$$

which becomes quite clear if we denote $\varepsilon = \frac{A}{2}$. Therefore for the indicated values of x we have $|f(x)| > |A|/2$. The first of the above inequalities can be equivalently replaced by the inequalities

$$A-\frac{|A|}{2} < f(x) < A+\frac{|A|}{2}$$

It follows that for $A > 0$ we have

$$\frac{A}{2} = A-\frac{|A|}{2} < f(x)$$

and for $A < 0$ we have

$$f(x) < A+\frac{|A|}{2} = \frac{A}{2}$$

which is what we had to prove.

Theorem 3. *If*

$$\lim_{x\to a} f_1(x) = A_1 \quad \textit{and} \quad \lim_{x\to a} f_2(x) = A_2$$

and if there is a neighbourhood $U(a)$ such that

$$f_1(x) \leqslant f_2(x)$$

for $x \in U(a)$ and $x \neq a$ then $A_1 \leqslant A_2$.

Proof. Let $x_n \to a$, $x_n \neq a$; then for sufficiently large n_0 we can write the inequality

$$f_1(x_n) \leqslant f_2(x_n) \qquad (n > n_0)$$

which, after the passage to the limit, results in $A_1 \leqslant A_2$.

Theorem 4. *If*

$$\lim_{x\to a} f_1(x) = A \quad \textit{and} \quad \lim_{x\to a} f_2(x) = A \tag{1}$$

and if

$$f_1(x) \leqslant \varphi(x) \leqslant f_2(x) \tag{2}$$

for the values of x belonging to a neighbourhood $U(a)$, $x \neq a$, then

$$\lim_{x\to a} \varphi(x) = A \tag{3}$$

Proof. Let $x_n \to a$ and $x_n \neq a$, then for $n > n_0$ we have

$$f_1(x_n) \leqslant \varphi(x_n) \leqslant f_2(x_n)$$

if n_0 is sufficiently large. By virtue of (1), it follows that the limit of $\varphi(x_n)$ exists and is equal to A. Since $\{x_n\}$ is an arbitrary sequence convergent to a, condition (3) does in fact hold.

Theorem 5 (Cauchy's Criterion for Existence of Limit). *For the (finite) limit* $\lim\limits_{x\to a} f(x)$ *to exist it is necessary and sufficient that the function* $f(x)$ *be defined in a neighbourhood of* a *except possibly at the point* a *itself and that for any* $\varepsilon > 0$ *there exist a neighbourhood* $U(a)$ *such that the inequality*

$$|f(x')-f(x'')| < \varepsilon$$

holds for any two points x' *and* x'' *belonging to* $U(a)$ *and different from* a, *that is* $x', x'' \in U(a)$ *and* $x', x'' \neq a$.

Proof. Let $\lim\limits_{x\to a} f(x) = A$ where A is a finite number. Then there is a neighbourhood of a such that $f(x)$ is defined throughout this neighbourhood with the possible exception of the point a itself. Furthermore, given any $\varepsilon > 0$, there is a neighbourhood $U(a)$ such that if $x \in U(a)$, $x \neq a$ then $|f(x)-A| < \varepsilon/2$. Let $x', x'' \in U(a)$ and let $x', x'' \neq a$; then

$$|f(x')-f(x'')| \leqslant |f(x')-A|+|A-f(x'')| < \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon$$

This shows the necessity of the condition of the theorem.

Let us prove the sufficiency of this condition. Let a function $f(x)$ be defined in a neighbourhood of a except possibly at the point a itself and let for any $\varepsilon > 0$ there exist a neighbourhood $U(a)$ such that $|f(x')-f(x'')| < \varepsilon$ for all $x', x'' \in U(a)$ $(x', x'' \neq a)$. Taking an arbitrary sequence $\{x_n\}$, $x_n \neq a$ $(n = 1, 2, \ldots)$ tending to a we can assert that, according to Cauchy's criterion for convergence of a sequence, there is a natural number N such that $x_n, x_m \in U(a)$ for $n, m > N$. Then

$$|f(x_n)-f(x_m)| < \varepsilon \qquad (n, m > N)$$

Consequently the sequence $\{f(x_n)\}$ satisfies Cauchy's criterion and thus has a limit.

We have proved the following property of the function f under consideration: for any sequence of numbers $x_n \neq a$ convergent to a there exists the limit $\lim f(x_n)$. This property implies automatically that all these limits $\lim f(x_n)$ corresponding to all the possible different sequences tending to a are equal to each other. This means that the limit $\lim\limits_{x\to a} f(x)$ exists. Indeed, let $x_n \to a$ and $x_n' \to a$ $(x_n, x_n' \neq a;\ n = 1, 2, \ldots)$. Then, as we have shown, there are numbers A and A' such that $f(x_n) \to A$ and $f(x_n') \to A'$. Let us construct the new sequence $\{x_1, x_1', x_2, x_2', x_3, \ldots\}$. It converges to a. By what has already been proved, the corresponding sequence $\{f(x_1), f(x_1'), f(x_2), f(x_2'), f(x_3), \ldots\}$ should also converge, which is only possible if $A = A'$. Hence, $A = A'$.

The theorem has been proved.

Theorem 6. *Let*

$$\lim_{x\to a} f(x) = A \quad \textit{and} \quad \lim_{x\to a} \varphi(x) = B$$

where A and B are finite numbers. Then

$$\lim_{x\to a} [f(x)\pm\varphi(x)] = A\pm B \quad \textit{and} \quad \lim [f(x)\cdot\varphi(x)] = AB$$

and if $B \neq 0$ then also

$$\lim_{x\to a} \frac{f(x)}{\varphi(x)} = \frac{A}{B}$$

As an instance, let us prove the second equality. Let $x_n \to a$, $x_n \neq a$ $(n = 1, 2, \ldots)$; then

$$\lim f(x_n) = A \quad \text{and} \quad \lim \varphi(x_n) = B$$

Since the limit of the product of two variables running through some sequences is equal to the product of the corresponding limits we have

$$\lim [f(x_n)\varphi(x_n)] = \lim f(x_n) \lim \varphi(x_n) = AB$$

This equality has been proved for any variable $x_n \to a$, $x_n \neq a$ and therefore $\lim\limits_{x\to a} [f(x)\varphi(x)] = AB$.

According to the definition, $\lim\limits_{x\to a} f(x) = \infty$ if the function $f(x)$ is defined in a neighbourhood of a except possibly at the point a itself and if for any positive number M there is a neighbourhood $U(a)$ of the point a such that

$$|f(x)| > M \qquad (x \in U(a),\ x \neq a)$$

If $\lim\limits_{x\to a} f(x) = \infty$ and if the function $f(x)$ satisfies the inequality $f(x) > 0$ $(f(x) < 0)$ in a neighbourhood of a we also write

$$\lim_{x\to a} f(x) = +\infty \qquad \left(\lim_{x\to a} f(x) = -\infty\right)$$

It is easy to prove the following theorems:

Theorem 7. *If a function $f(x)$ satisfies the inequality*

$$|f(x)| > M > 0$$

in a neighbourhood of a and if a function $\varphi(x)$ is such that

$$\lim_{x\to a} \varphi(x) = 0 \qquad (\varphi(x) \neq 0 \ \text{for} \ x \neq a)$$

then

$$\lim_{x\to a} \frac{f(x)}{\varphi(x)} = \infty$$

Theorem 8. *If* $\lim_{x\to a} f(x) = A$ *and* $\lim_{x\to a} \varphi(x) = \infty$ *(where A is a finite number) then*

$$\lim_{x\to a} \frac{f(x)}{\varphi(x)} = 0$$

Corollary. If $\varphi(x) \to 0$ $(x \to a, \varphi(x) \neq 0)$ *then*

$$\lim_{x\to a} \frac{1}{\varphi(x)} = \infty$$

and if $\varphi(x) \to \infty$ $(x \to a, \varphi(x) \neq 0)$ *then*

$$\lim_{x\to a} \frac{1}{\varphi(x)} = 0$$

Theorem 9. *Let a function f defined in a neighbourhood of a point a (which can be finite or infinite) satisfy the following condition: every sequence* $\{x_n\}$ *convergent to a contains a subsequence* $\{x_{n_k}\}$ *such that* $\lim_{k\to\infty} (x_{n_k}) = A$. *Then* $\lim_{x\to a} f(x) = A$.

Proof. Let $x_n \to a$. According to the hypothesis, any subsequence of the sequence $\{f(x_n)\}$ contains a subsequence convergent to A. Therefore, by Theorem 5 in § 3.8, we have $f(x_n) \to A$.

§ 4.2. Continuity of a Function at a Point

By definition, a function f is said to be *continuous at a (finite) point a* if it is defined in a neighbourhood of that point (including the point a itself) and if

$$\lim_{x\to a} f(x) = f(a) \tag{1}$$

By what was said in § 4.1 about the limit of a function at a point, the definition of the continuity of a function at a point stated above can be formulated in full as follows:

A function f is said to be continuous at a point a if it is defined on an interval (c, d) containing the point a and if for any $\varepsilon > 0$ *there is* $\delta > 0$ *such that the inequality*

$$|f(x)-f(a)| < \varepsilon$$

holds for all x satisfying the inequality $|x-a| < \delta$.

By virtue of § 4.1, this statement of the definition is completely equivalent to the following.

A function f is said to be continuous at a point a if it is defined on an interval (c, d) containing a and if for any sequence $\{x_n\}$ *convergent to a there holds the relation*

$$\lim_{x_n\to a} f(x_n) = f(a)$$

If a function $f(x)$ *defined in a neighbourhood of a point a* is not continuous at the point a, that is if the property indicated above does not hold for it, we say that f is *discontinuous at the point a* or that *it has a discontinuity at a* (a point a of this kind is called a *point of discontinuity* of f).

The discontinuity of a function f at a point a can also be defined in a direct manner:

Let a function f be defined in a neighbourhood of a point a and let there be a number $\varepsilon_0 > 0$ *such that for any* $\delta > 0$ *there exists a point* x_δ *such that*

$$|a-x_\delta| < \delta, \quad |f(a)-f(x_\delta)| \geqslant \varepsilon_0$$

Then $f(x)$ *is said to be discontinuous at the point a.*

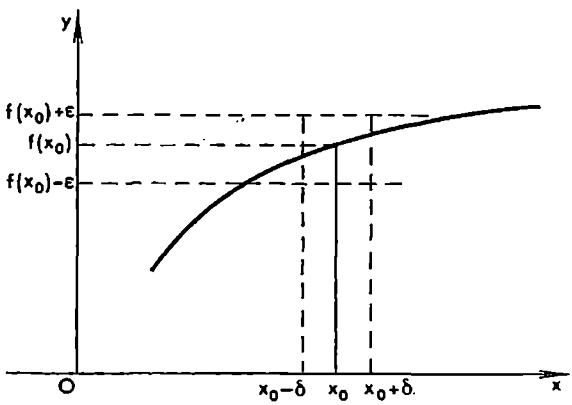

Fig. 4.2

Let us consider a continuous curve C representing graphically a continuous function $y = f(x)$, $x \in [a, b]$ (see Fig. 4.2). The term "continuous curve" is used here in an intuitive sense meaning that it can be entirely drawn with a "continuous" motion of the pencil without leaving the paper.

Let us take an arbitrary value $x_0 \in (a, b)$. The value of the function in question corresponding to x_0 is $f(x_0)$. Setting $\varepsilon > 0$ and drawing the three straight lines parallel to the x-axis and passing respectively at distances $f(x_0)-\varepsilon$, $f(x_0)$ and $f(x_0)+\varepsilon$ from the x-axis (see Fig. 4.2) we readily see that for the (continuous) curve in question it is possible to find $\delta > 0$ (dependent on ε) such that all the ordinates $f(x)$ of the curve corresponding to the values of x belonging to the interval $(x_0-\delta, x_0+\delta)$ satisfy the inequalities

$$f(x_0)-\varepsilon < f(x) < f(x_0)+\varepsilon$$

In other words, for any $\varepsilon > 0$ there is $\delta > 0$ such that the inequality $|f(x)-f(x_0)| < \varepsilon$ is fulfilled for all x satisfying the inequality $|x-x_0| < \delta$.

Hence, the mathematical definition of the continuity of a function is coherent with the intuitive idea of a continuous curve.

Let us consider the graph shown in Fig. 4.3. This graph represents a discontinuous curve L consisting of the two disjoint continuous parts L_1 and L_2. The part L_1 is projected in a one-to-one manner (in the direction of the y-axis) on the interval $[a, c]$. As to the part L_2, we suppose that its left end point is deleted; the latter part is projected in a one-to-one manner on the half-interval $(c, b]$. To each value $x \in [a, b]$ there corresponds a single value $y = f(x)$ equal to the ordinate of the point of the curve L having the abscissa x. The curve L is discontinuous since it consists of two disjoint parts L_1 and L_2

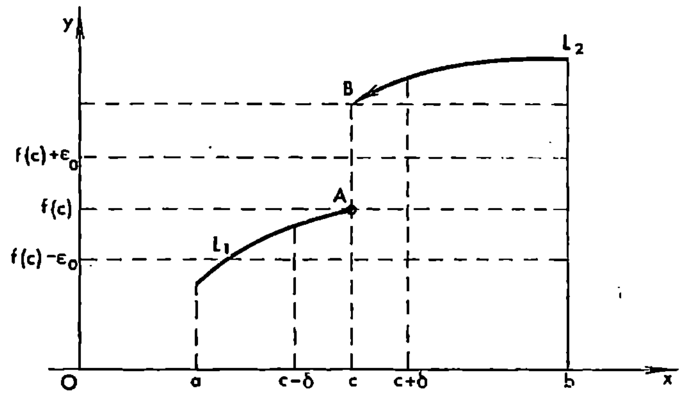

Fig. 4.3

"not pasted together". The discontinuity is encountered as the moving point with abscissa x passes through the point c (or, as we say, as the value of the argument x passes through the value c). As is shown in Fig. 4.3, $f(c) = Ac$. It is readily seen that the function $f(x)$ is also discontinuous at the point c. Indeed, let us take a positive number $\varepsilon_0 < AB$. Observing the figure we conclude that for any $\delta > 0$, however small, there are values of x among those satisfying the inequality $|c-x| < \delta$ (namely, the values exceeding c and differing from it by less than δ) such that

$$|f(x)-f(c)| > \varepsilon_0$$

Thus, to a discontinuous graph there corresponds a discontinuous function. In the case under consideration the function $f(x)$ is discontinuous at the point c (cf. § 1.4).

The difference $\Delta y = \Delta f(x) = f(x+h)-f(x)$ is called *the increment of the function f, at the point x, corresponding to the increment h of the independent variable.*

The concept of continuity of a function f at a point a can also be defined as follows using the notion of increment: *a function $f(x)$ is continuous at a point a if the expression $f(a+h)$ regarded as a function of h is defined in a neighbourhood of the point $h = 0$ and if, given an arbitrary $\varepsilon > 0$, there is*

$\delta > 0$ *such that the inequality*

$$|f(a+h)-f(a)| < \varepsilon$$

holds when $|h| < \delta$.

In other words, a function f is continuous at a point a if its increment at this point corresponding to the increment h of the argument tends to zero together with h.

The properties of limits of functions (see § 4.1) and the definition of the continuity of a function at a point imply

Theorem 1. *If functions $f(x)$ and $\varphi(x)$ are continuous at a point a their sum $f(x)+\varphi(x)$, difference $f(x)-\varphi(x)$, product $f(x)\varphi(x)$ and quotient $f(x)/\varphi(x)$ (the latter is considered under the additional assumption that $\varphi(a) \neq 0$) are also continuous at the point a.*

We shall also prove the theorem on the continuity of a function of a function.

Theorem 2. *If a function $\varphi(x)$ is continuous at a point a and a function $f(y)$ is continuous at $b = \varphi(a)$ then the function of a function (the composite function) $F(x) = f(\varphi(x))$ is continuous at the point a.*

Proof. Let us take an arbitrary $\varepsilon > 0$. By the continuity of the function f at the point b, there is $\sigma > 0$ such that the function $f(y)$ is defined on the interval $(b-\sigma, b+\sigma)$ and the condition

$$|f(y)-f(b)| < \varepsilon \quad \text{for} \quad |y-b| < \sigma \tag{2}$$

is fulfilled. Furthermore, by the continuity of the function φ at the point a there is $\delta > 0$ such that the function $\varphi(x)$ is defined on the interval $(a-\delta, a+\delta)$ and the inequality $|\varphi(x)-\varphi(a)| < \sigma$ holds for all x satisfying the inequality

$$|x-a| < \delta \tag{3}$$

It follows from the relations obtained that the function $f(\varphi(x))$ is defined for all x satisfying inequality (3) and that for these values of x there holds the relation

$$|f(\varphi(x))-f(\varphi(a))| < \varepsilon, \qquad \text{that is} \quad |F(x)-F(a)| < \varepsilon \tag{4}$$

which is what we set out to prove.

To prove the continuity of F at the point $x = a$ we can also use the following argument. Since the function φ is continuous at the point a and the function f is continuous at the point $b = \varphi(a)$ and, besides, $F(a) = f(\varphi(a))$, we must have the relation

$$\lim_{x_n \to a} F(x_n) = \lim_{\varphi(x_n) \to \varphi(a)} f(\varphi(x_n)) = f(\varphi(a)) = F(a)$$

for any sequence $\{x_n\}$ tending to a.

If a function $\Phi(x)$ is obtained from several functions by applying to them only the arithmetical operations and the operations of forming a function of a function, the verification of the continuity of Φ at a given point reduces to a consecutive application of the last two theorems provided that these theorems are used a finite number of times.

Here we also state the following theorems which are a direct consequence of the definition of continuity of a function at a point and of the theorems on the limit of a function in § 4.1.

Theorem 3. *If a function $f(x)$ is continuous at a point a there exists a neighbourhood $U(a)$ of the point a in which the function $f(x)$ is bounded.*

Theorem 4. *If a function $f(x)$ is continuous at a point a and $f(a) \neq 0$ then there is a neighbourhood $U(a)$ of the point a at whose all points the inequality*

$$|f(x)| > \frac{|f(a)|}{2}$$

is fulfilled. Moreover, if $f(a) > 0$ then

$$\frac{f(a)}{2} < f(x) \qquad (x \in U(a))$$

and if $f(a) < 0$ then

$$f(x) < \frac{f(a)}{2} \qquad (x \in U(a))$$

Example 1. The constant function $f(x) = C$ is defined and continuous for any value of x because its increment corresponding to any increment h of the argument is

$$\Delta C = C - C = 0$$

and therefore the condition $\Delta C \to 0$ $(h \to 0)$ is fulfilled automatically.

Example 2. The function $f(x) = x^n$ $(n = 1, 2, \ldots)$ is defined throughout the real axis and is continuous on it.

Indeed, the function $y = x$ is obviously continuous for any x, and hence so is the function $x^2 = xx$. The same argument shows that the function $x^3 = x^2x$ is also continuous: proceeding in this way we conclude, by induction, that the function x^n is continuous for any n.

Example 3. A polynomial

$$P(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_n$$

(where $a_0, \ldots, a_n$ are given numbers and n is a natural number) is obviously a continuous function for any x since, as has been shown above, x^{n-k} $(k = 0, 1, \ldots, n)$ is a continuous function throughout the real axis and so is the function a_kx^{n-k} as the product of two functions a_k and x^{n-k} which are continuous for all x; finally, $P(x)$ is continuous on the real axis as a finite sum of continuous functions.

Example 4. A rational function

$$f(x) = \frac{P(x)}{Q(x)} = \frac{a_0x^n + \ldots + a_n}{b_0x^m + \ldots + b_m}$$

(where n and m are natural numbers and a_k, b_k are given real numbers) is continuous for all the values of x for which $Q(x) \neq 0$. This follows from the fact that $f(x)$ is obtained from the continuous functions x^k and from a finite number of real (constant) coefficients by means of a finite number of arithmetical operations (addition, subtraction, multiplication and division).

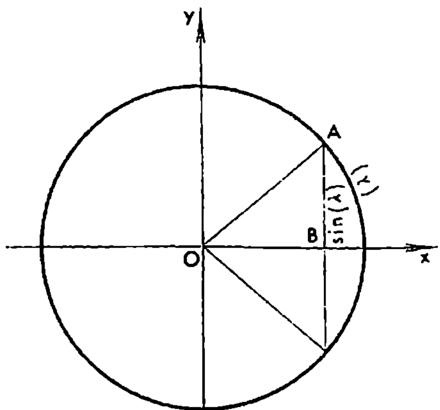

Fig. 4.4

Example 5. The function $f(x) = \sin x$ is continuous at all the points x. This is shown by the following argument.

We can easily prove the inequality $|\sin \lambda| \leqslant |\lambda|$. To this end we first prove it for $|\lambda| \leqslant \pi/2$: on multiplying the desired inequality (its both sides) by 2 we obtain on the left-hand side the length of the chord (see Fig. 4.4) subtending an arc with length $2|\lambda|$ of a circle of unit radius; the inequality thus obtained is obviously valid and is equivalent to the former inequality. Now, taking $|\lambda| \geqslant \pi/2$ we can write $|\lambda| \geqslant \pi/2 > 1 \geqslant \sin \lambda$ whence

$$\left|\sin(x+h)-\sin x\right| = \left|2\sin\frac{h}{2}\cos\left(x+\frac{h}{2}\right)\right| \leqslant 2\left|\frac{h}{2}\right|\cdot 1 = |h|$$

which means that $|\sin(x+h)-\sin x| \to 0$ as $h \to 0$, that is the function $\sin x$ is continuous at any point x.

Example 6. The function $\cos x$ is continuous for all the values of the argument x since

$$|\cos(x+h)-\cos x| = \left|2\sin\frac{h}{2}\sin\left(x+\frac{h}{2}\right)\right| \leqslant 2\left|\frac{h}{2}\right|\cdot 1 = |h| \to 0, \quad h \to 0$$

The inequality obtained shows that for any ε there is δ, namely, in this case $\delta = \varepsilon$, such that $|\cos(x+h)-\cos x| < \varepsilon$ when $|h| < \delta$.

Remark. In our course we rely upon the ordinary geometrical definition of trigonometric functions (see § 1.3, 7). But it is also possible to state some other definitions of the trigonometric functions which are of a purely analytical character (see § 10.11).

Example 7. The function $|x|$ is continuous for all the values of x since

$$||x+h|-|x|| \leqslant |x+h-x| = |h| \to 0, \qquad h \to 0$$

If a function f is not continuous at a point $x = a$ and, at the same time, there exists a finite limit $\lim\limits_{x\to a} f(x)$ we say that the function has a *removable* *discontinuity* at that point. This term is accounted for by the fact that in this case the function f can be redefined at the point a (provided that it is defined at a) or extended to the point a (if it was not originally defined at a) by putting $f(a) = \lim_{x\to a} f(x)$ so that the (nearly defined!) function f is continuous at that point.

Example 8. The function

$$f(x) = \begin{cases} \dfrac{x^2-1}{x-1} & \text{for } x \neq 1, \\ 3 & \text{for } x = 1 \end{cases}$$

is obviously discontinuous at the point $x = 1$. But this discontinuity is removable since if we put $f(1) = 2$ the (new) function f becomes continuous at $x = 1$.

If a function f is continuous for all x in a sufficiently small neighbourhood of a point a except possibly at the point $x = a$ itself and is unbounded in this neighbourhood we say that f has an *infinite discontinuity* at a.

Example 9. The function $\sin(1/x)$ is an example of a bounded function having an unremovable discontinuity at $x = 0$ while $\tan x$ is an example of a function having infinite discontinuities $\left(\text{at the points } x_k = \dfrac{\pi}{2}+k\pi,\right.$ $\left.k = 0, \pm 1, \pm 2, \ldots\right)$.

Example 10. The functions $\cos(\sin x^2)$ and $(\sin x)^2$ are continuous throughout the real axis. This follows from the continuity of the functions x^2, $\sin x$ and $\cos x$ and from the theorem on the continuity of a function of a function.

§ 4.3. Right-hand and Left-hand Limits of a Function. Monotone Functions

Let us state the following definitions. A *left-hand neighbourhood of a point* (number) a is an arbitrary half-interval $(c, a]$ and a *right-hand neighbourhood of* a is an arbitrary half-interval $[a, d)$ $(c < a < d)$. By a *neighbourhood of* ("the point") $+\infty$ is naturally meant an arbitrary (*semi-infinite*) *interval* $(N, +\infty)$ and by a *neighbourhood of* $-\infty$ an *interval* $(-\infty, N)$ where in both cases N is an arbitrary (finite) number. In accordance with this terminology we can say that the neighbourhoods of $+\infty$ and $-\infty$ are respectively left-hand and right-hand neighbourhoods of ("the point") ∞. Right-hand and left-hand neighbourhoods are also termed *one-sided neighbourhoods.*

Proceeding from these definitions we introduce the notions of the *right-hand* and *left-hand limits* of a function f at a point a (which can be finite or infinite). For instance, we call A the *right-hand limit of f at a* (finite or infinite) *point a* if f is defined in a right-hand neighbourhood of a except possibly at the point a itself and if for any $\varepsilon > 0$ there is a right-hand neighbourhood of a such that the inequality $|f(x)-A| < \varepsilon$ is fulfilled for all the points x belonging to the latter neighbourhood.

The right-hand (left-hand) limit of f at ∞ is usually called the *limit of f for $x \to -\infty$ (for $x \to +\infty$)*.

We can also state the following alternative definition of a right-hand (left-hand) limit at a point. *A function f is said to have a right-hand (left-hand) limit at a* (finite or infinite) *point a* equal to a number A if the function f is defined in a right-hand (left-hand) neighbourhood of a except possibly at the point a itself and if $\lim\limits_{x_n \to a} f(x_n) = A$ for any sequence $\{x_n\}$, $x_n \neq a$, convergent to a on condition that the values of the sequence belong to that right-hand (left-hand) neighbourhood.

The equivalence of the definitions stated here is proved by a complete analogy with the case of an ordinary limit (see § 4.1). The theorems on limits in § 4.1 are extended by analogy with right-hand and left-hand limits.

If a is a finite point the right-hand and left-hand limits of f at this point are symbolized respectively as

$$f(a+0) = \lim_{\substack{x \to a \\ x > a}} f(x) \quad \text{and} \quad f(a-0) = \lim_{\substack{x \to a \\ x < a}} f(x)$$

Using the definitions of limits expressed in the "ε, δ language" we can readily prove that for a function f to have a limit at a finite point a it is necessary and sufficient that there exist the right-hand and left-hand limits of f at this point and that these limits coincide; then $f(a+0) = f(a-0) = \lim\limits_{x \to a} f(x)$.

The limits of f for $x \to -\infty$, $x \to +\infty$ and $x \to \infty$ are respectively denoted as

$$\lim_{x \to -\infty} f(x) = f(-\infty), \quad \lim_{x \to +\infty} f(x) = f(+\infty) \quad \text{and} \quad \lim_{x \to \infty} f(x) = f(\infty)$$

Here, like in the case of a finite point, there holds the following obvious assertion: for the limit of f for $x \to \infty$ to exist it is necessary and sufficient that the limits of f for $x \to +\infty$ and for $x \to -\infty$ should exist and coincide; then $f(-\infty) = f(+\infty) = f(\infty)$.

Up till now we have dealt with finite limits of functions (i.e. the number A has been supposed to be finite) but by analogy with the above we can also introduce limits of the type

$$\lim_{x \to a} = A$$

where a and A are finite numbers or $+\infty$ or $-\infty$ or ∞ (all the combinations are allowable).

For example, in the case $a = A = -\infty$ we deal with the symbolic relation $\lim\limits_{x \to -\infty} f(x) = -\infty$ which means that the function $f(x)$ is defined for all the values of x less than a certain finite number (i.e. in a neighbourhood of

$-\infty$) and that for any positive number N there is a number L such that the inequality $f(x) < -N$ holds for all $x < L$.

Right-hand and left-hand limits are of great importance for the study of monotone functions.

Let E be a set of real numbers (of points of the real line). A function f defined on E is called *nondecreasing (nonincreasing)* on E if the relations $x', x'' \in E$ and $x' < x''$ imply $f(x') \leqslant f(x'')$ ($f(x') \geqslant f(x'')$).

Nondecreasing and nonincreasing functions on E are termed *monotone functions on E*.

Theorem 1. *Let a function f be nondecreasing on an interval (a, b) where, in particular, there can be $a = -\infty$ or $b = +\infty$ (or both). If the function is bounded above by a number M then there exists a (finite) limit $\lim\limits_{\substack{x\to b\\ x<b}} f(x) \leqslant M$. If it is not bounded above then $\lim\limits_{\substack{x\to b\\ x<b}} f(x) = +\infty$.*

Proof. The boundedness of f implies the existence of the supremum $\sup\limits_{x\in(a,b)} f(x) = A \leqslant M$. Hence, $f(x) \leqslant A$ for all $x \in (a, b)$, and for any $\varepsilon > 0$ there is $x_1 \in (a, b)$ such that $A-\varepsilon < f(x_1) \leqslant A$. Since f is nondecreasing we have $f(x_1) \leqslant f(x)$ for $x_1 \leqslant x < b$. Thus, for any $\varepsilon > 0$ there is $x_1 < b$ such that $A-\varepsilon < f(x) < A+\varepsilon$ for all x satisfying the inequalities $x_1 < x < b$. This exactly means that

$$A = \lim_{\substack{x\to b\\ x<b}} f(x)$$

Now let a nondecreasing function f be not bounded above. Then for any M there is $x_1 \in (a, b)$ such that $M < f(x_1)$, and, since f is nondecreasing on (a, b), we have

$$M < f(x_1) \leqslant f(x) \quad \text{for} \quad x_1 < x < b$$

which means that

$$\lim_{\substack{x\to b\\ x<b}} f(x) = +\infty$$

The following theorem is proved by a complete analogy with the above.

Theorem 2. *If a function f is nondecreasing on (a, b) (where there can be $a = -\infty$ or $b = +\infty$, or both) and if $f(x)$ is bounded below by a number m, there exists a (finite) right-hand limit of the function f at the point a:*

$$\lim_{\substack{x\to a\\ x>a}} f(x) = A \geqslant m$$

If the function f is not bounded below then

$$\lim_{\substack{x\to a\\ x>a}} f(x) = -\infty$$

Let the reader modify the statements of these theorems for the case of a nonincreasing function defined on (a, b).

Example. Consider the function

$$f(x) = \begin{cases} x & \text{for} \quad 0 \leqslant x < 1, \\ x+1 & \text{for} \quad 1 \leqslant x \leqslant 2 \end{cases}$$

defined on the closed interval $[0, 2]$. It is one-valued and monotone on $[0, 2]$. As is readily seen, $f(1-0) = 1$ and $f(1+0) = f(1) = 2$.

Theorem 3. *If a function f is nondecreasing on a closed interval $[a, b]$ then at each point $x \in (a, b)$ there exist the limits $f(x-0)$ and $f(x+0)$ and the inequalities*

$$f(x-0) \leqslant f(x) \leqslant f(x+0)$$

hold. The limits $f(a+0)$ and $f(b-0)$ also exist and satisfy the inequalities

$$f(a) \leqslant f(a+0), \quad f(b-0) \leqslant f(b)$$

This theorem follows immediately from the foregoing theorems if we take into account that its conditions imply that the function f is nondecreasing on each of the intervals $[a, x]$ and $[x, b]$.

Now we can introduce the notions of *continuity of a function from the right and from the left at a given point.*

A function f is said to be *continuous from the right (from the left) at a finite point a* if $f(a+0)$ exists and $f(a+0) = f(a)$ (if $f(a-0)$ exists and $f(a-0) = f(a)$).

If for a function f there exist the (finite) numbers $f(a-0)$ and $f(a+0)$ at a (finite) point a and if the function is nevertheless discontinuous at a we say that the function has a *discontinuity of the first kind at the point a.*

It should be noted that *if a function f is continuous both from the right and from the left at a point a it is obviously continuous at the point a.*

We can also say that *for a function $f(x)$ to be continuous at a point a it is necessary and sufficient that the three numbers $f(a-0)$, $f(a)$ and $f(a+0)$ should make sense and coincide with each other.*

In Figs. 4.5-4.10 we see the examples of six graphs representing functions having discontinuities of the first kind at a point a. Here the letter A designates the point $A = (a, f(a))$ in the xy-plane. An arrow placed at an end point of a portion of a curve symbolizes that this end point is deleted from the graph. In Figs. 4.5-4.8 we see the cases when all the three numbers $f(a)$, $f(a-0)$ and $f(a+0)$ make sense for the depicted graphs of certain functions f. In Fig. 4.5 the numbers $f(a)$, $f(a-0)$ and $f(a+0)$ are pairwise different; the corresponding function is not only discontinuous at a but is also discontinuous from the right and from the left at a. In Fig. 4.6 the function f is continuous from the left at a. In Fig. 4.7 the function f is continuous from the right at a. In Fig. 4.8 we see the case of a removable discontinuity of f at a. The function f whose graph is shown in Fig. 4.9 is not defined at a and the discontinuity is unremovable. Finally, in Fig. 4.10 the function f is

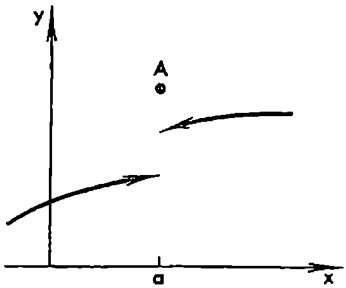

Fig. 4.5

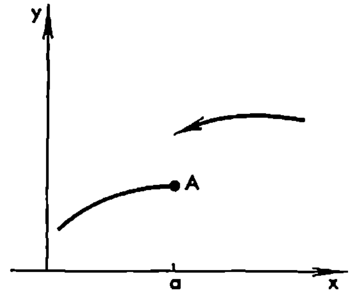

Fig. 4.6

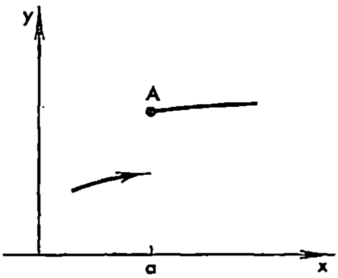

Fig. 4.7

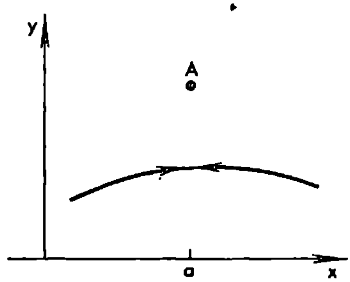

Fig. 4.8

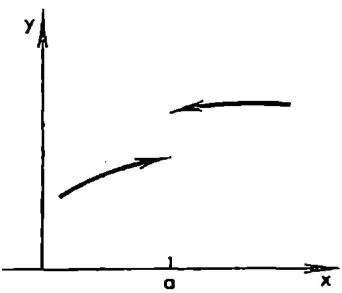

Fig. 4.9

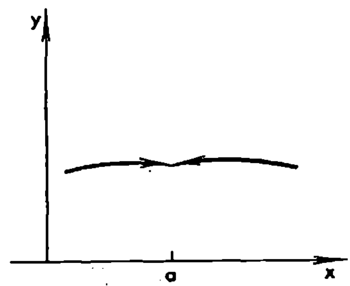

Fig. 4.10

not defined at a but f can be extended to a so that it becomes continuous at a.

Attention should be paid to the following important fact. *If a function f defined on a closed interval $[a, b]$ is monotone on it (i.e. is nondecreasing or nonincreasing) then the function f is either continuous or has a discontinuity of the first kind at each point $x \in [a, b]$.* This proposition is an immediate consequence of Theorems 1 and 2 and of the definition of a point of discontinuity of the first kind*.

* On the number of points of discontinuity of a monotone function see the end of §9.5.

If a function f is defined in a neighbourhood of a point a except possibly at the point a itself and has a discontinuity at a not belonging to the class of discontinuities of the first kind the function is said to have a *discontinuity of the second kind at a.* For instance, the function sin $(1/x)$ has a discontinuity of the second kind at the point $x = 0$. The function

$$\psi(x) = \begin{cases} 0 & \text{for} \quad x \leqslant 0 \\ \sin \frac{1}{x} & \text{for} \quad x > 0 \end{cases}$$

also has a discontinuity of the second kind at the point $x = 0$ since, although the number $\psi(0-0) = 0$ makes sense, the number $\psi(0+0)$ does not.

§ 4.4. Functions Continuous on a Closed Interval

A function f is said to be *continuous on a closed interval* $[a, b]$ (i.e. on the set of points x satisfying the inequalities $a \leqslant x \leqslant b$) if it is continuous at all the points of the open interval (a, b) (i.e. at all the points x for which $a < x < b$), is continuous from the right at the point a and continuous from the left at the point b*.

A function continuous on a closed interval possesses a number of remarkable properties which will be presented in this section. But the extremely important property of uniform continuity of such a function will not be treated here; it will be considered later (§ 7.10, Theorem 4) for the general case of n variables. From the results obtained in § 7.10 the corresponding results concerning functions of one variable continuous on a closed interval $[a, b]$ can be readily derived as special cases.

We begin with the following lemma.

Lemma 1. *If all the values x_n of a sequence $\{x_n\}$ convergent to a number α belong to $[a, b]$ then also $\alpha \in [a, b]$.*

Proof. The assertion of the lemma follows from Theorem 3, § 3.1.

Theorem 1. *If a function f is continuous on a closed interval $[a, b]$ it is bounded on it.*

Proof. Suppose that f is unbounded on $[a, b]$. Then for each natural number n there is a point $x_n \in [a, b]$ such that

$$|f(x_n)| > n \qquad (n = 1, 2, \ldots) \tag{1}$$

The sequence $\{x_n\}$ is bounded (since a and b are finite numbers) and therefore it contains a subsequence $\{x_{n_k}\}$ convergent to a point $\alpha \in [a, b]$ (see the foregoing lemma and Theorem 3, § 3.8). The function f is continuous at the point α and therefore** we must have

$$\lim_{k=\infty} f(x_{n_k}) = f(\alpha) \tag{2}$$

* It should be stressed that when speaking of a closed interval $[a, b]$ we always mean that its end points are finite numbers (points).

** If $\alpha = b$ $(\alpha = a)$ the function f is continuous from the left (from the right) at the point α.

Property (2) obviously contradicts property (1). Therefore f must be bounded on $[a, b]$.

It should be noted that if a function is continuous on an open interval (a, b) or on a half-interval of the type $[a, b)$ or $(a, b]$ it is not necessarily bounded on it. For example, the function $1/x$ is continuous on $(0, 1]$ but not bounded on that interval.

If we extend the function $1/x$ to the point $x = 0$ by putting, for instance, $f(0) = 0$ the new function will be *finite* at any point of the closed interval $[0, 1]$ but *is not bounded on it.*

Theorem 2. *A function f continuous on $[a, b]$ attains at some points of the closed interval $[a, b]$ its maximum and minimum, that is there are points α and β belonging to $[a, b]$ such that*

$$\min_{x \in [a, b]} f(x) = f(\alpha) \quad \text{and} \quad \max_{x \in [a, b]} f(x) = f(\beta)$$

Thus, $f(\alpha) \leqslant f(x) \leqslant f(\beta)$ for all $x \in [a, b]$.

Proof. By the foregoing theorem, a continuous function on a closed interval is bounded on that interval, and hence the function f is bounded above by a number K on $[a, b)$:

$$f(x) \leqslant K \qquad (x \in [a, b])$$

Therefore the supremum of f on $[a, b]$ must exist:

$$\sup_{x \in [a, b]} f(x) = M \tag{3}$$

The number M possesses the property that for any natural number n there is a point x_n on $[a, b]$ such that

$$M - \frac{1}{n} < f(x_n) \leqslant M \qquad (n = 1, 2, \ldots)$$

The sequence $\{x_n\}$ belongs to $[a, b]$ and is therefore bounded. Hence it contains a subsequence $\{x_{n_k}\}$ convergent to a number β which is sure to belong to $[a, b]$ (see Lemma 1). Since the function f is continuous at the point β we must have $\lim_{k \to \infty} f(x_{n_k}) = f(\beta)$. On the other hand, $M - \frac{1}{n_k} < f(x_{n_k}) \leqslant M$ $(k = 1, 2 \ldots)$ and

$$\lim_{k \to \infty} f(x_{n_k}) = M$$

But $f(x_{n_k})$ can have only one limit, and therefore $M = f(\beta)$.

The value equal to supremum (3) is thus attained by the function at the point β; in such a case we say that *the function f attains at the point β its maximum* (i.e. its maximum value) *on the interval* $[a, b]$. We have shown that there exists a point $\beta \in [a, b]$ such that

$$\max_{x \in [a, b]} f(x) = f(\beta)$$

The proof of the remaining part of the theorem concerning the minimum of the function is proved analogously; it can also be reduced to the first part of the theorem because

$$\min_{x \in [a, b]} f(x) = - \max_{x \in [a, b]} \{-f(x)\}$$

Remark. The function $y = x$ is continuous on the open interval (0, 1) and is bounded on it; but its supremum $\sup\limits_{x \in (0, 1)} x = 1$ is not attained on this interval, that is there is no $x_0 \in (0, 1)$ such that the function is equal to 1 at x_0. We see that the requirement of the last theorem that f be continuous on a *closed interval* $[a, b]$ (which includes its both end points a and b) is essential.

We obviously have $\sup\limits_{x \geqslant 0} \arctan x = \pi/2$. But there is no point x on the ray $x \geqslant 0$ at which the function $\arctan x$ assumes the value $\pi/2$; hence it does not attain its maximum for $x \geqslant 0$. The conditions of the above theorem are violated in this case since the domain of definition of the continuous function $\arctan x$ is not bounded.

Theorem 3. *If a function f is continuous on a closed interval $[a, b]$ and the numbers $f(a)$ and $f(b)$ are different from zero and have opposite signs there is at least one point c on the open interval (a, b) such that $f(c) = 0$.*

Proof. Let us denote the closed interval $[a, b]$ as Δ_0. On dividing it into two equal parts we test the value of the function f at the midpoint. If the function is equal to zero at the midpoint of Δ_0 the proof of the theorem is completed. Let us suppose that it is different from zero at that point; then one of the two halves of the interval Δ_0 is such that the function assumes nonzero values of different sighs at its end points. Now we denote that very half by Δ_1 and again divide it into two equal parts. It may happen that the function is equal to zero at the midpoint of Δ_1 and then the theorem is proved. If otherwise, we denote as Δ_2 the half of Δ_1 at whose end points the values of f are different from zero and have different signs, etc. Proceeding in this way we conclude, by induction, that we either encounter a point $c \in (a, b)$ for which $f(c) = 0$, and then the proof of the theorem is completed, or arrive at the family of nested intervals $\Delta_1 \supset \Delta_2 \supset \dots$ on each of which the function f assumes values of both signs. Then there exists a (single) point c belonging to all Δ_n, $n = 1, 2, \dots$, and, consequently, belonging to $[a, b]$. It is evident that $f(c) = 0$ because, if otherwise, for definiteness, if $f(c) > 0$, then there is a neighbourhood U_c of the point c such that the function $f(x)$ is positive for all x simultaneously belonging to $[a, b]$ and to U_c. But this is impossible because we have $\Delta_n \subset U_c$ for sufficiently large n and the function f cannot retain its sign of Δ_n. The theorem has been proved.

Corollary. *If a function f is continuous on $[a, b]$, $f(a) = A$, $f(b) = B$ and C is an arbitrary number lying between the numbers A and B then there is at least one point c belonging to the open interval (a, b) such that $f(c) = C$.*

This corollary can also be stated as follows: *a function continuous on a closed interval* $[a, b]$ *assumes all the intermediate values lying between its end-point values.*

Proof. Let us take the auxiliary function $F(x) = f(x) - C$ where C is a constant number lying between $A = f(a)$ and $B = f(b)$. Since f is a continuous function on $[a, b]$, so is the function F. The function F obviously attains values of opposite signs at the end points of the interval $[a, b]$. Therefore, by the above theorem, there is a point c lying inside $[a, b]$, i.e. $c \in (a, b)$, such that $F(c) = 0$, that is $f(c) - C = 0$, which means that $f(c) = C$. This is what we wished to prove.

Example. The equation

$$\cos x - x = 0$$

possesses a root lying within the interval $(0, \pi)$.

Indeed, the function $f(x) = \cos x - x$ is continuous on the closed interval $[0, \pi]$ and assumes the values $f(0) = 1$ and $f(\pi) = -(1+\pi)$ having opdosite signs at its end points.

Remark. For a discontinuous function defined on $[a, b]$ the theorem stated above does not hold; this is readily confirmed by the example of the function

$$f(x) = \begin{cases} -1 & \text{for } -1 \leqslant x < 0 \\ 1 & \text{for } \quad 0 \leqslant x \leqslant 1 \end{cases}$$

§ 4.5. Inverse Function

Let us consider an arbitrary function $y = f(x)$ defined on an arbitrary set E of numbers (points of the real line) and denote by $E_1 = f(E)$ the image of E (see § 1.3).

Let us assign to each $y \in E_1$ the set of all $x \in E$ for which $y = f(x)$. This is a nonempty set; we shall denote it e_y.

Thus, we have a function $x = \varphi(y)$ defined on E_1 which is many-valued in general case. The function $\varphi(y)$ is called the *inverse function of* $f(x)$.

The cases when the inverse function is one-valued are of particular importance. This is always so when the function f is strictly monotone, that is when it is strictly increasing or strictly decreasing in its domain of definition E.

A function f is said to be *strictly increasing (decreasing)* on E if $x', x'' \in E$ and $x' < x''$ imply $f(x') < f(x'')$ $(f(x') > f(x''))$.

If a (one-valued) function $f(x)$ *is strictly increasing (decreasing) on* E *its inverse function* $x = \varphi(y)$ *is obviously also a one-valued and srictly increasing (decreasing) function on the image* $E_1 = f(E)$.

In this case we have the obvious identities

$$\varphi[f(x)] = x, \qquad x \in E \quad \text{and} \quad f[\varphi(y)] = y, \; y \in E_1$$

It is sometimes convenient to denote the inverse function of f by the symbol f^{-1}:

$$f^{-1}f(x) = x, \quad x \in E \quad \text{and} \quad ff^{-1}(y) = y, y \in E_1$$

Theorem 1. *Let $y = f(x)$ be a continuous strictly increasing function on a closed interval $[a, b]$ and let $A = f(a)$ and $B = f(b)$.*

Then the image of the interval $[a, b]$ is the interval $[A, B]$, and the function $x = \varphi(y)$, inverse to f, is one-valued, strictly increasing and continuous on $[A, B]$.

Remark. In this theorem "increasing" can be replaced by "decreasing" with the simultaneous replacement of $[A, B]$ by $[B, A]$.

Proof. Let $E_1 = f([a, b])$. By the hypothesis, $A, B \in E_1$, and, since the function f is continuous on $E = [a, b]$, every point of $[A, B]$ also belongs to E_1 (see the corollary of Theorem 3 on the intermediate values of a continuous function in § 4.4).

If a point y does not belong to $[A, B]$ it cannot be the image of a point $x \in [a, b]$ since the given function is strictly increasing. This shows that the image of the interval $[a, b]$, produced by the function f, is the closed interval $[A, B]$. The (strict) monotonicity of the function $y = f(x)$ on the interval $[a, b]$ implies immediately that the inverse function $x = \varphi(y)$ defined on the interval $[A, B]$ is one-valued and strictly monotone. Now it only remains to prove the continuity of the function $x = \varphi(y)$ at every point $y_0 \in [A, B]$.

Let y_0 be an interior point of $[A, B]$, i.e. $y_0 \in (A, B)$. As is already known, there is a single point $x_0 \in (a, b)$ corresponding to y_0 such that $y_0 = f(x_0)$ or, equivalently, $x_0 = \varphi(y_0)$.

Let us take an arbitrary positive number ε on condition that it is so small that $[x_0 - \varepsilon, x_0 + \varepsilon] \subset [a, b]$, and let $y_1 = f(x_0 - \varepsilon)$ and $y_2 = f(x_0 + \varepsilon)$. The strict monotonicity of f implies that the values $x = \varphi(y)$ corresponding to all $y \in (y_1, y_2)$ belong to the interval $(x_0 - \varepsilon, x_0 + \varepsilon)$: $x = \varphi(y) \in (x_0 - \varepsilon, x_0 + \varepsilon)$.

Thus we see that for any sufficiently small ε (i.e. satisfying the condition $[x_0 - \varepsilon, x_0 + \varepsilon] \subset [a, b]$) there is a neighbourhood (y_1, y_2) of the point y_0 such that $|x - x_0| = |\varphi(y) - \varphi(y_0)| < \varepsilon$ for all $y \in (y_1, y_2)$.

The property of the function $\varphi(y)$ stated here has been proved for sufficiently small ε. But it is clear that it also holds for any ε. This property exactly means that the function $\varphi(y)$ is continuous at the point y_0.

For the end point $y_0 = B$ the corresponding point is $x_0 = b = \varphi(y_0)$. Putting $x_1 = b - \varepsilon > a$ and $y_1 = f(x_1)$ we obviously obtain $|\varphi(y_0) - \varphi(y)| < \varepsilon$ for all $y \in (y_1, y_0]$.

The case $y_0 = A$ is treated similarly.

The proof of the continuity of the function $x = \varphi(y)$ (the inverse of the function $y = f(x)$, the latter being supposed to be continuous and strictly monotone on $[a, b]$) can also be carried out in another way. Let us set a value $y_0 \in [A, B]$ and choose an arbitrary sequence of points $y_n \in [A, B]$ such that $y_n \to y_0$. Putting $x_0 = \varphi(y_0)$ and $x_n = \varphi(y_n)$ we can write $y_0 = f(x_0)$ and $y_n = f(x_n)$. The continuity of φ at the point y_0 can now be proved if we show that $x_n \to x_0$.

Suppose the contrary. Then there must exist a subsequence $\{x_{n_k}\}$ convergent to a point $x' \in [a, b]$ distinct from x_0. By the strict monotonicity of f we then have $f(x_0) \neq f(x')$.

But, according to the hypothesis, $f(x_{n_k}) = y_{n_k} \to y_0 = f(x_0)$, and, by the continuity of f,

$$f(x_{n_k}) \to f(x'), \qquad x_{n_k} \to x'$$

We have thus arrived at a contradiction because the given sequence $\{f(x_{n_k})\}$ cannot converge to different limits.

Example. The function $y = \sin x$ is continuous and strictly increasing on the interval $[-\pi/2, \pi/2]$. The image of this interval under the mapping specified by the function $\sin x$ is the interval $[-1, +1]$. By the above theorem, there is a one-valued continuous and strictly increasing function defined on the interval $[-\pi/2, \pi/2]$ which is the inverse of $\sin x$. As is known, this function is designated as $x = \arcsin y$ ($[-1 \leqslant y \leqslant 1]$).

The inverse function corresponding to the function $y = \sin x$ regarded on the whole real axis is no longer one-valued; it is infinitely valued and to each $y \in [-1, +1)$ there corresponds the set e_y of values of x determined by the formula

$$x = \operatorname{Arcsin} y = (-1)^k \arcsin y + k\pi \qquad (k = 0, \pm 1, \pm 2, \ldots) \qquad (1)$$

Theorem 2. *Let $y = f(x)$ be a continuous strictly increasing function defined on an open interval (a, b) and let*

$$A = \inf f(x) \quad \text{and} \quad B = \sup f(x), \qquad x \in (a, b) \qquad (2)$$

where, in particular, a or A (or both) can be equal to $-\infty$ while b or B (or both) can be equal to $+\infty$.

Then the image of (a, b) is the open interval (A, B), and the function $x = \varphi(y)$ inverse to f, is one-valued, strictly increasing and continuous on (A, B).

Remark. In this theorem "increasing" can be replaced by "decreasing"; then the image of (a, b) will be (B, A).

From the definition of the number B it follows directly that if it is finite then the point $y > B$ cannot belong to the image $f((a, b))$. The number B itself cannot belong to $f((a, b))$ either, because, if otherwise, there would exist a point $x_1 \in (a, b)$ such that $B = f(x_1)$ and, since we can take a point $x_2 > x_1$ belonging to the interval (a, b), the strict monotonicity of f would imply the inequality $f(x_2) > B = f(x_1)$ which contradicts the definition of B.

In just the same way we can prove that the number A does not belong to $f((a, b))$ either if it is finite. Hence, the image $f((a, b))$ is contained in (A, B). But these two sets do in fact coincide. To prove this let us take an arbitrary point $y \in (A, B)$. By definitions (2), there must exist $x_1, x_2 \in (a, b)$ such that

$$y_1 = f(x_1) < y < f(x_2) = y_2$$

and, since f if strictly increasing, $x_1 < x_2$. The function $f(x)$ is continuous on (a, b) and, moreover, it is continuous on $[x_1, x_2]$. Therefore when x runs through the closed interval $[x_1, x_2]$ the function f should assume all the values between y_1 and y_2 and, consequently, the value y as well.

This means that there is a value $x = \varphi(y)$ (which, by virtue of the strict monotonicity of f is unique) such that $y = f(x)$. Hence we have proved that the image of the interval (a, b) is the interval (A, B) and that the function $x = \varphi(y)$ defined above is the inverse function of f.

The function φ is continuous at the point y since φ can be regarded as the inverse function of f considered as a function defined on the closed interval $[x_1, x_2]$, and to the latter function we can apply the foregoing theorem. The fact that φ is a strictly increasing function is quite obvious. The theorem has been proved.

Remark. The intervals (a, b) and (A, B) in Theorem 2 can be respectively replaced by half-intervals, for instance, by $[a, b)$ and $[A, B)$; in this case a and A are finite numbers.

Example. Let us consider the expression $\sqrt[n]{a}$ where $a > 0$ and n is a natural number. The *nth arithmetic root of a* is a positive number whose nth power is equal to a. This number is also denoted as

$$\sqrt[n]{a} = a^{1/n} \tag{3}$$

The existence and the uniqueness of this number are established as follows. The function

$$y = x^n \tag{4}$$

is continuous and strictly increasing on the half-interval $[0, \infty)$ and, besides, it is equal to zero for $x = 0$ and tends to $+\infty$ together with x. On the basis of Theorem 2 and the remark to this theorem we conclude that the function $y = x^n$ possesses an inverse function $x = \varphi(y)$ $(0 \leq y < \infty)$ which is one-valued, continuous and strictly increasing; $x = \varphi(y)$ is equal to zero for $y = 0$ and tends to $+\infty$ together with y.

Consequently, for any $y \in [0, \infty)$ there is a single positive number $x = \varphi(y)$ such that $[\varphi(y)]^n = y$. This exactly means that $\varphi(y) = y^{1/n}$. In particular, putting $y = a$ we conclude that the nth arithmetic root of a exists and its value is determined uniquely.

§ 4.6. Exponential and Logarithmic Functions

The Function a^x. Let $a > 0$ be an arbitrary positive number. If n is a natural number, the number a^n is defined as the product $a^n = a \dots a$ of n factors each of which is equal to a while the number $a^{1/n}$ is understood as the nth arithmetic root of a: $a^{1/n} = \sqrt[n]{a}$.

Now let p/q $(q > 0)$ be a nonnegative rational fraction. By definition, we put

$$a^{p/q} = (a^p)^{1/q} = (a^{1/q})^p \quad \text{and} \quad a^0 = 1$$

The proof of the equality $(a^p)^{1/q} = (a^{1/q})^p$ and of the fact that this definition leads to the same number when the fraction p/q is written as $np/nq = p/q$ where n is an arbitrary natural number is known to the reader from elementary mathematics. Further, we also put, by definition,

$$a^{-p/q} = \frac{1}{a^{p/q}}$$

Thus, we have defined the function a^x $(a > 0)$ for any rational values of x.

Let $\mathfrak{M}$ denote the set of all rational numbers. The function a^x is defined on this set. In elementary mathematics it is proved, on the basis of the axioms of real numbers entering into Groups I-IV only, that

$$a^{x+y} = a^x a^y \tag{1}$$

for any $x, y \in \mathfrak{M}$. In elementary mathematics it is also proved that $a^x < a^y$ $(x < y;\ x, y \in \mathfrak{M},\ a > 1)$.

We shall prove that the function a^x can be extended to all irrational points so that the extended function, which we again denote a^x, is defined and continuous throughout the real axis R. Moreover, the extended function possesses property (1) *for all $x, y \in R$.*

To begin with we shall prove an auxiliary inequality (derived by Jacob Bernoulli*).

If $a > 1$ and N is a natural number then $a^{1/N} = 1+\lambda$ where $\lambda > 0$. Whence, taking into account Newton's binomial formula, we obtain $a = (1+\lambda)^N > 1+N\lambda$, i.e. $a^{1/N}-1 < (a-1)/N$.

Now, if h is an arbitrary positive rational number satisfying the inequality $0 < h \leq 1$, there is a natural number N such that $1/(N+1) < h \leq 1/N$. Therefore we have

$$a^h - 1 \leq a^{1/N} - 1 < \frac{a-1}{N} = \frac{N+1}{N}(a-1)\frac{1}{N+1} < 2(a-1)h$$

for $a > 1$. Further, $1-a^{-h} = (a^h-1)/a^h < a^h - 1 < 2(a-1)h$. The last two inequalities and the fact that $a^0 = 1$ imply *Bernoulli's inequality*

$$|a^h - 1| \leq 2(a-1)|h| \qquad (|h| \leq 1,\ a \geq 1)$$

which holds for all the indicated rational numbers h.

From Bernoulli's inequality we obtain

$$a^y - a^x = a^x(a^{y-x}-1) \leq 2a^x(a-1)(y-x) \quad (x, y \in \mathfrak{M},\ 0 < y-x \leq 1) \tag{2}$$

Let us take an arbitrary number c and denote by $\mathfrak{M}_c$ the set of all $x \in \mathfrak{M}$ satisfying the inequality $x \leq c$.

It follows from (2) that

$$a^y - a^x \leq M(y-x) \qquad \left(x,\ y \in \mathfrak{M}_c,\ 0 < y-x \leq 1,\ M = 2(a-1)a^c\right) \tag{3}$$

where M is thus a constant independent of the values of x and y in question.

---

* Jacob Bernoulli (1654-1705), a Swiss mathematician.

Inequality (3) says that if the difference between x and y is small the difference between the corresponding values of the function is also small. More precisely, given an arbitrarily small $\varepsilon > 0$, the inequality

$$|a^y - a^x| < \varepsilon \tag{4}$$

holds for all $x, y \in \mathfrak{M}_c$ satisfying the inequality $|x-y| < \delta$ where $\delta > 0$ is sufficiently small (namely, for $\delta < \min(\varepsilon/M, 1)$).

The property of the function a^x which we have established describes its *uniform continuity* on the set $\mathfrak{M}_c$. The general notion of the uniform continuity of a function will be studied in §§ 7.10, 7.11; here it is only mentioned in connection with the concrete function under investigation.

Let x_0 be a rational point such that $c > x_0$, that is $x_0 \in \mathfrak{M}_c$, and let the variable x_n run through some values belonging to $\mathfrak{M}_c$ and tend to x_0. Then it follows from (3) that $|a^{x_n} - a^{x_0}| \leq M|x_n - x_0|$ for sufficiently large n, i.e. $|a^{x_n} - a^{x_0}| \to 0$, which means $\lim a^{x_n} = a^{x_0}$.

Now let x_0 be an arbitrary irrational point such that $c > x_0$ and let a variable x_n run through some values $x_n \in \mathfrak{M}_c$ and tend to x_0. We shall show that there exists the limit of a^{x_n} as $x_n \to x_0$ which is independent of the way in which x_n approaches x_0. This property justifies the notation a^{x_0} for this limit:

$$a^{x_0} = \lim_{x_n \to x_0} a^{x_n} \tag{5}$$

Indeed, since $x_n \to x_0$, Cauchy's criterion for the existence of a limit implies that, given any ε, there is N such that $|x_n - x_m| < \varepsilon/M$ for all n, $m > N$. Therefore, by virtue of (3), we have

$$|a^{x_n} - a^{x_m}| < \varepsilon \qquad \text{for all } n, m > N$$

The last condition shows that, according to Cauchy's criterion, limit (5) exists. It is independent of the way in which x_n tends to x_0 and therefore if x_n' is another variable assuming some values belonging to $\mathfrak{M}_c$ and tending to x_0, inequality (3) indicates that

$$|a^{x_n} - a^{x_n'}| \leq M|x_n - x_n'| \to 0$$

Thus, the function a^x $(a > 1)$ has been extended to the whole set R of all the points of the real line. Let us dwell on the properties of this function.

If $x, y < c$, $|x-y| \leq 1$ are two arbitrary points (rational or irrational) and x_n and y_n are two variables, assuming some values belonging to $\mathfrak{M}_c$ and such that $x_n \to x$ and $y_n \to y$, we have, by inequality (3), the relation

$$|a^{y_n} - a^{x_n}| \leq M\,|y_n - x_n| \tag{6}$$

which holds for all $n = 1, 2, \ldots$ On passing to the limit in (6) for $n \to \infty$ we arrive at the inequality

$$|a^y - a^x| \leq M\,|y-x| \tag{7}$$

which is fulfilled for all $x, y < c$, $|x-y| \leq 1$. The passage to the limit is legitimate since the function $|x|$ is continuous. It follows directly from inequality (7) that the function a^x is continuous for any $x < c$ and, consequently, for any x since c can be chosen quite arbitrarily.

The function a^x also possesses the properties

$$a^x < a^y, \quad \text{if} \quad x < y \tag{8}$$

$$\lim_{x\to-\infty} a^x = 0, \qquad \lim_{x\to+\infty} a^x = +\infty \tag{9}$$

and

$$a^{x+y} = a^x a^y \tag{10}$$

We shall prove these properties under the assumption that for rational x and y they are known from elementary mathematics.

Let λ and μ be constant rational numbers such that $x < \lambda < \mu < y$ and let $x_n, y_n \in \mathfrak{M}$ be an increasing variable and a decreasing variable tending respectively to x and y. Then $a^{x_n} < a^\lambda < a^\mu < a^{y_n}$ whence, on passing to the limit, we obtain $a^x \leq a^\lambda < a^\mu \leq a^y$, which implies (8). Property (9) is an immediate consequence of the facts that it holds when x tends to $+\infty$ or to $-\infty$ running through rational numbers and that the function a^x is monotone (see (8)). Finally, (10) follows from the equality $a^{x_n+y_n} = a^{x_n}a^{y_n}$ after the passage to the limit in it.

Up till now we have supposed that $a > 1$. If $0 < a < 1$ we put

$$a^x = \frac{1}{\left(\frac{1}{a}\right)^x}, \qquad -\infty < x < \infty \tag{11}$$

In this case property (10) of the function a^x and its continuity also hold but the function a^x is strictly decreasing. Finally, we put

$$1^x = 1 \tag{12}$$

for all x.

We also note that for a natural exponent m we have

$$a^{xm} = a^x a^{(m-1)x} = (a^x)^2 a^{(m-2)x} = \ldots = (a^x)^m,$$

$$(a^{x/m})^m = a^x \quad \text{and} \quad a^{x/m} = (a^x)^{1/m}$$

and therefore

$$(a^x)^{p/q} = (a^x)^{(1/q)p} = (a^{x/q})^p = a^{x(p/q)}$$

for a rational number $p/q > 0$. Furthermore if y is an arbitrary positive number and $y_n \to y$ where y_n are rational, then

$$a^{xy} = \lim_{n\to\infty} a^{xy_n} = \lim_{n\to\infty} (a^x)^{y_n} = (a^x)^y$$

and thus, for positive y, we have shown that $a^{xy} = (a^x)^y$. By virtue of (10), this equality obviously extends to the case of arbitrary y (since $a^{-y}a^y = a^{y-y} = a^0 = 1$).

The Function $\log_a x$ $(a > 0,\ a \neq 1)$. For definiteness, let $a > 1$. Then $y = a^x$ is a continuous and strictly increasing function defined throughout the real axis, and we have

$$\inf_{x \in (-\infty, +\infty)} a^x = 0 \quad \text{and} \quad \sup_{x \in (-\infty, +\infty)} a^x = +\infty$$

Hence, the function a^x determines a mapping of the real axis $(-\infty, +\infty)$ onto the infinite interval $(0, \infty)$, and, by Theorem 2 in § 4.5, its inverse function is one-valued, strictly increasing and continuous on $(0, \infty)$. This function is called the *logarithm of* y *to the base* a and is denoted

$$\log_a y$$

It follows that

$$\lim_{x \to +\infty} \log_a x = +\infty \quad \text{and} \quad \lim_{\substack{x \to 0 \\ x > 0}} \log_a x = -\infty$$

(here we replace y by x). In the case $a < 1$ the argument is quite similar. If $a < 1$ the function a^x also maps the real axis $(-\infty, +\infty)$ onto the half-axis $(0, +\infty)$ but is strictly decreasing. The inverse function $\log_a x$ defined on $(0, +\infty)$ is also strictly decreasing and we have

$$\lim_{x \to \infty} \log_a x = -\infty \quad \text{and} \quad \lim_{\substack{x \to 0 \\ x > 0}} \log_a x = +\infty$$

Thus, there hold the identities

$$a^{\log_a x} = x \ (0 < x < +\infty) \quad \text{and} \quad \log_a a^x = x \ (-\infty < x < +\infty)$$

where $a \neq 1$, $a > 0$. From the properties of the function a^x for $x, y > 0$ it now follows that

$$a^{\log_a (xy)} = xy = a^{\log_a x} a^{\log_a y} = a^{\log_a x + \log_a y}$$

and

$$\log_a (xy) = \log_a x + \log_a y$$

Replacing x by x/y in the latter equality we derive

$$\log_a x - \log_a y = \log_a \frac{x}{y}$$

We also have

$$a^{\log_a x^y} = x^y = (a^{\log_a x})^y = a^{y \log_a x} \qquad (x > 0)$$

and therefore

$$\log_a x^y = y \log_a x \qquad (a \neq 1,\ a > 0,\ x > 0)$$

Finally, for positive numbers a and b distinct from 1 the equality

$$a^{\log_a b \cdot \log_b a} = (a^{\log_a b})^{\log_b a} = b^{\log_b a} = a$$

onsequently

$$\log_a b \cdot \log_b a = 1$$

the base e is called the *natural* (also *Napierian*)

where $\varepsilon_2(x,$ nd is denoted as $\log_e a = \ln a$.

§ 4.7. The Power Function x^b

Let us consider the expression x^b where b is a constant and x is a variable. For any given b this function is defined and positive for at least all points of the positive half-axis $x > 0$ (since in § 4.6 we justified the definition of the number a^x for $a > 0$ and arbitrary x).

Using the formula

$$x^b = e^{b \ln x} \qquad (x > 0) \tag{1}$$

(see § 4.6) we can derive all the properties of the power function from the properties of the exponential and logarithmic functions which are already known. The function x^b is obviously continuous. In the case $b > 0$ it is strictly increasing and possesses the properties

$$\lim_{\substack{x \to 0 \\ x > 0}} x^b = 0 \quad \text{and} \quad \lim_{x \to +\infty} x^b = +\infty$$

In the case $b > 0$ it appears natural to put $0^b = 0$; then the function x^b becomes continuous from the right at the point $x = 0$.

In the case $b < 0$ the function x^b is continuous and strictly decreasing on the positive half-axis and possesses the properties

$$\lim_{\substack{x \to 0 \\ x > 0}} x^b = +\infty \quad \text{and} \quad \lim_{x \to +\infty} x^b = 0$$

Formula (1) implies a characteristic property of the power function:

$$(xy)^b = e^{b \ln (xy)} = e^{b \ln x} e^{b \ln y} = x^b y^b \qquad (x, y > 0)$$

In Figs. 1.2, 4.11 and 4.12 we see the graphs of the function x^b, $x > 0$, for several positive and negative values of b.

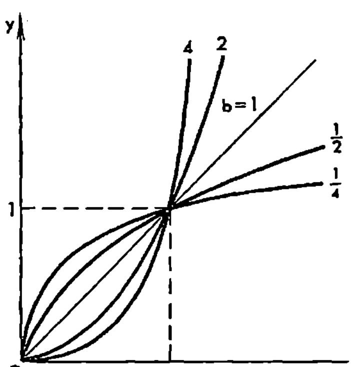

Fig. 4.11

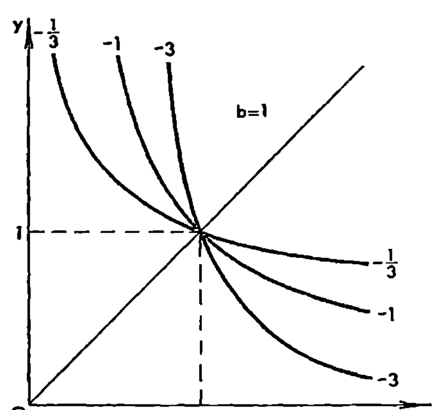

Fig. 4.12

For the negative values of x the power function x^b makes sense as a real function when b is an integer or a rational number p/q with an odd denominator q.

§ 4.8. More on the Number e

In § 3.5 we considered the function

$$\alpha(n) = \left(1+\frac{1}{n}\right)^n$$

of the integral argument n and showed that if $n \to \infty$ running through the natural numbers the variable $\alpha(n)$ tends to a limit which is called the number e. But the expression $\alpha(n)$ does in fact make sense for any nonzero real values of n, and we shall show (this is important for applications) that

$$\lim_{n\to\infty} \alpha(n) = e \tag{1}$$

where the limit is understood as the limit of the function $\alpha(n)$ defined for all (real) $n \neq 0$.

To prove (1) it suffices to justify (1) for the following two cases: (1) $n \to +\infty$ and (2) $n \to -\infty$ (in both cases n runs through any real values, not necessarily integral ones).

Let us take an arbitrary $\varepsilon > 0$ and choose $N > 0$ such that

$$e-\varepsilon < \left(1+\frac{1}{n}\right)^n < e, \quad \left(1+\frac{1}{n}\right)^2 < 1+\frac{\varepsilon}{e}$$

for all natural $n > N$. Now, if $n > N+1$ is an arbitrary real number and $[n]$ is its integral part then $[n] > N$, and we can obviously write

$$e-\varepsilon < \left(1+\frac{1}{[n]+1}\right)^{[n]+1} < \left(1+\frac{1}{[n]}\right)^{n+1} < \left(1+\frac{1}{[n]}\right)^{[n]+2} < e\left(1+\frac{1}{[n]}\right)^2 <$$
$$< e\left(1+\frac{\varepsilon}{e}\right) = e+\varepsilon$$

Therefore $\left(1+\frac{1}{n}\right)^{n+1} \to e$ as $n \to +\infty$ and, since $1+\frac{1}{n} \to 1$, we see that (1) holds when n tends to plus infinity in an arbitrary way.

Now let us take the case when $n \to -\infty$. Putting $m = -n \to +\infty$ we see that

$$\lim_{n\to-\infty} \left(1+\frac{1}{n}\right)^n = \lim_{m\to+\infty} \left(1-\frac{1}{m}\right)^{-m} = \lim_{n\to+\infty} \left(\frac{m}{m-1}\right)^m =$$
$$= \lim_{m\to+\infty} \left(1+\frac{1}{m-1}\right)^{m-1}\left(1+\frac{1}{m-1}\right) = e$$

whence follows that (1) holds for $n \to -\infty$. Thus, we have proved (1) for the general case.

Putting $h = 1/n$ we also obtain

$$\lim_{h\to 0} (1+h)^{1/h} = \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e$$

§ 4.9. $\lim\limits_{x\to 0} \dfrac{\sin x}{x}$

In this section we shall prove that

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \tag{1}$$

The function $\varphi(x) = \dfrac{(\sin x)}{x}$ is defined for all the values $x \neq 0$. Let $0 < x < \pi/2$; then (see Fig. 4.13) we have $\sin x < x < \tan x$ because half the chord subtending a circular arc is less than half the arc which, in its turn, is less than half the length of the polygonal line circumscribed about the arc. Hence, $1 < x/\sin x < < 1/\cos x$, i.e.

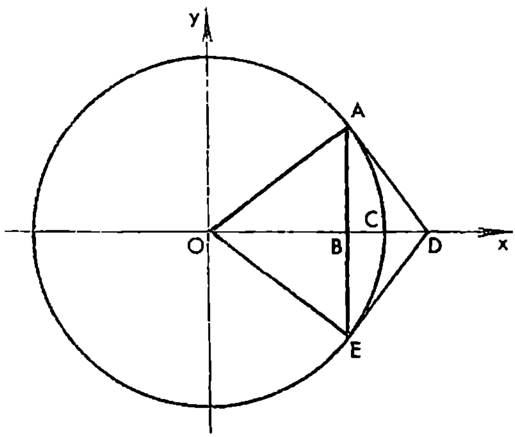

Fig. 4.13

$$\cos x < \frac{\sin x}{x} < 1 \tag{2}$$

These inequalities obviously hold not only for the positive values of x but also for the negative ones satisfying the inequalities $0 < |x| < \pi/2$ since the functions entering into (2) are even. The function $\cos x$ is continuous (see § 4.2, Example 6) and consequently

$$\lim_{x\to 0} \cos x = \cos 0 = 1$$

On passing to the limit in relations (2) as $x \to 0$ we see that the limits of the leftmost and rightmost members of (2) coincide with 1 and hence the limit of the middle term in (2) also exists and is equal to 1.

§ 4.10. Order of a Variable. Equivalence (Asymptotic Equality)

We say that *f is of the order of φ* on E and write

$$f(x) = O(\varphi(x)) \quad \text{on } E \tag{1}$$

if

$$|f(x)| \leq C\,|\varphi(x)| \quad \text{on } E \tag{2}$$

where C is a positive constant independent of x.

In particular, the symbolic relation

$$f(x) = O(1) \quad \text{on } E$$

means that f is bounded on E.

It is obvious that if $f(x) = O(\varphi_1(x))$ on E and $\varphi_1(x) = O(\varphi_2(x))$ on E then $f(x) = O(\varphi_2(x))$ on E.

Examples

1. $\sin x = O(x)$ on $(-\infty, +\infty)$.

2. $x = O(x^2)$ on $[1, \infty]$ (but not on $[0, 1]$); it is evident that x^2 and x cannot be interchanged here. On the other hand, $x^2 = O(x) = O(1)$ on $[0, 1]$.

We shall write

$$f(x) = o(\varphi(x)) \qquad (x \to a) \tag{3}$$

and say that *the function f is of higher order (of smallness) than φ as $x \to a$* if

$$f(x) = \varepsilon(x)\varphi(x) \tag{3'}$$

where the function $\varepsilon(x)$ tends to zero as $x \to a$.

We shall write

$$f(x) = O(\varphi(x)) \qquad (x \to a) \tag{4}$$

if there is a neighbourhood $U(a)$ of the given (finite or infinite) point a such that

$$f(x) = O(\varphi(x)) \qquad (x \in U(a), x \neq a) \tag{4'}$$

It is of course tacitly implied here that both functions f and φ in definitions (3) and (4) are defined in some common neighbourhood of the point a except possibly at the point a itself. If there is such a neighbourhood on which $\varphi(x) \neq 0$ everywhere except possibly at the point a itself, definitions (3) and (4) are obviously equivalent to the following ones: *$f(x) = o(\varphi(x))$ as $x \to a$ if*

$$\lim_{x \to a} \frac{f(x)}{\varphi(x)} = 0 \tag{5}$$

and $f(x) = O(\varphi(x))$ as $x \to a$ if there is a neighbourhood $U(a)$ at whose all points, except possibly at the point a, the ratio $f(x)/\varphi(x)$ is bounded. We can also consider the cases when x approaches a only from the left ($x < a$) or only from the right ($x > a$); for the point at infinity we must then write $x \to +\infty$ in the former case and $x \to -\infty$ in the latter. Obviously, in these cases by a neighbourhood of a we mean its corresponding one-sided (i.e. a left-hand or a right-hand) neighbourhood. Finally, we can also include the cases when x in (3) and (4) tends to a finite or an infinite limit running through a definite sequence of values $x_1, x_2, \ldots$

It is evident that if $f(x) = o(\varphi(x))$ $(x \to a)$ and $\varphi(x) = o(\psi(x))$ $(x \to a)$ then $f(x) = o(\psi(x))$ $(x \to a)$ since

$$f(x) = \varepsilon(x)\varphi(x) = \varepsilon(x)\,\varepsilon_1(x)\,\psi(x) = \varepsilon_2(x)\,\psi(x)$$

where $\varepsilon_2(x) = \varepsilon(x)\,\varepsilon_1(x) \to 0$ $(x \to a)$ because $\varepsilon(x) \to 0$ and $\varepsilon_1(x) \to 0$.

Examples

1. $x^n = o(e^x)$ $\quad (x \to +\infty), (n = 1, 2, 3, \ldots)$.
2. $x^2 = o(x)$ $\quad (x \to 0)$.
3. $x = o(x^2)$ $\quad (x \to \infty)$.
4. $\ln x = o(x)$ $\quad (x \to +\infty)$.
5. $x = O(\sin x)$ $\quad (x \to 0)$.

Two functions $f_1(x)$ and $f_2(x)$ are said to be *equivalent* (or *asymptotically equal*) for $x \to a$ if both functions are defined and different from zero in a neighbourhood of the point a, except possibly at the point a itself, and if

$$\lim_{x\to a} \frac{f_1(x)}{f_2(x)} = 1 \tag{6}$$

For equivalent functions we shall write

$$f_1(x) \approx f_2(x) \qquad (x \to a)$$

Here the condition that x tends to a can also be understood as was mentioned above (i.e. in the sense of one-sided limits).

Theorem 1. *For two functions $f_1(x)$ and $f_2(x)$ defined on E to be equivalent (asymptotically equal) for $x \to a$ $(x \in E)$ it is necessary and sufficient that the conditions*

$$f_1(x) = f_2(x)+o(f_2(x)) \quad (x \to a) \text{ and } f_2(x) \neq 0 \quad (x \neq a) \tag{7}$$

hold.

Proof. It follows from (6) that

$$\frac{f_1(x)}{f_2(x)} = 1+\varepsilon(x) \qquad (\varepsilon(x) \to 0,\ x \to a)$$

whence

$$f_1(x) = f_2(x)+\varepsilon(x)f_2(x) = f_2(x)+o(f_2(x)) \qquad (x \to a)$$

that is (7) holds.

Conversely, let (7) take place. Then $f_1(x) = f_2(x)+\varepsilon(x)f_2(x)$ where $\varepsilon(x) \to 0$ $(x \to a)$ whence

$$\frac{f_1(x)}{f_2(x)} = 1+\varepsilon(x)$$

Now, passing to the limit as $x \to a$ we arrive at (6).

Note that if $f_1(x) \approx f_2(x)$ $(x \to a)$ then we also obviously have $f_2(x) \approx f_1(x)$ $(x \to a)$.

Theorem 2. *Let there be a neighbourhood of a at whose all points, except possibly at the point a itself, three functions $f_1(x)$, $f_2(x)$, and $\Lambda(x)$ are defined. If $f_1(x) \approx f_2(x)$ for $x \to a$ then*

$$\lim_{x\to a} \{f_1(x)\Lambda(x)\} = \lim_{x\to a} \{f_2(x)\Lambda(x)\} \tag{8}$$

This equality should be understood in the sense that *if the limit on its right-hand side exists then the limit on its left-hand side also exists and coincides with the former, and vice versa.*

It follows that if one of the limits does not exist the other does not exist either.

Proof. Let the limit on the left-hand side of (8) exist and be equal to A. Then obviously

$$\lim_{x\to a}\{f_1(x)A(x)\} = \lim_{x\to a}\frac{f_1(x)}{f_2(x)}\lim_{x\to a}\{f_2(x)A(x)\} = 1\cdot A = A$$

The existence of the limit on the right-hand side of (8) and its equality to the limit on the left-hand side, provided that the latter exists, are proved analogously.

The theorem we have proved is quite simple but is very important. Its practical application requires the knowledge of as many pairs of equivalent functions as possible.

Below are some examples of this kind.

(i) $\sin x \approx x$ $(x\to 0)$ because $\lim\limits_{x\to 0}\dfrac{\sin x}{x} = 1$.

(ii) $1-\cos x \approx \dfrac{1}{2}x^2$ $(x\to 0)$ since

$$\lim_{x\to 0}\frac{1-\cos x}{x^2} = \lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{x^2} = \lim_{x\to 0}\frac{2\left(\frac{x}{2}\right)^2}{x^2} = \frac{1}{2}$$

The second equality in this chain holds on the basis of Theorem 2 and due to the fact that $\sin x/2 \approx x/2$ $(x\to 0)$.

(iii) $e^h-1\approx h$ $(h\to 0)$ since if we put $e^h-1 = z$ then $e^h = 1+z$, $h = \ln(1+z)$ and

$$\lim_{h\to 0}\frac{e^h-1}{h} = \lim_{z\to 0}\frac{z}{\ln(1+z)} = \lim_{z\to 0}\frac{1}{\ln(1+z)^{1/z}} = \frac{1}{\ln e} = 1$$

The equality $\lim\limits_{z\to 0}\dfrac{1}{\ln(1+z)^{1/z}} = \dfrac{1}{\ln e} = 1$ holds because of the continuity of $\ln u$ for $u > 0$ and, in particular, for $u = e$.

(iv) $\ln(1+u)\approx u$ $(u\to 0)$. Indeed,

$$\lim_{u\to 0}\frac{\ln(1+u)}{u} = \lim_{u\to 0}\ln(1+u)^{1/u} = \ln e = 1$$

(v) $\sqrt[n]{1+u}-1 \approx \dfrac{u}{n}$ $(u\to 0)$, $n = 1, 2, \ldots$ Indeed, taking into account that the function $(1+u)^\alpha$ is continuous at the point $u = 0$ we can write

$$\frac{\sqrt[n]{1+u}-1}{\frac{u}{n}} = \frac{u}{\frac{u}{n}\left[(1+u)^{(n-1)/n}+(1+u)^{(n-2)/n}+\ldots+1\right]} \to 1 \qquad (u\to 0)$$

(vi) $\tan x \approx x$ $(x \to 0)$ since

$$\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \left(\frac{\sin x}{x}\cdot\frac{1}{\cos x}\right) = 1$$

because $\cos x$ is a continuous function.

As an instance, consider the following example involving (ii), (v) and Theorem 2:

$$\lim_{x\to 0} \frac{\sqrt[3]{1+x^2}-1}{1-\cos x} = \lim_{x\to 0} \frac{\dfrac{x^2}{3}}{\dfrac{x^2}{2}} = \frac{2}{3} \tag{9}$$

The following definition is also of use. If for a function $\varphi(x)$ there are numbers $\alpha \neq 0$ and m such that $\varphi(x) \approx \alpha x^m$, $x \to 0$, we say that *the function* αx^m *is the principal (power) asymptotic term of the function* $\varphi(x)$ (*for* $x \to 0$).

The right-hand members of asymptotic equalities (i)-(vi) are obviously the principal (asymptotic) terms of their left-hand members. The general methods of determining principal terms in more complicated cases are based on the application of Taylor's formula (see § 5.11, Examples 4.5, and § 5.14).

If αx^m and βx^n $(\alpha, \beta \neq 0)$ are respectively the principal terms of some functions φ and ψ, Theorem 2 indicates that

$$\lim_{x\to 0} \frac{\varphi(x)}{\psi(x)} = \lim_{x\to 0} \frac{\alpha x^m}{\beta x^n} = \begin{cases} \dfrac{\alpha}{\beta} & \text{for} \quad m = n \\ 0 & \text{for} \quad m > n \\ \infty & \text{for} \quad m < n \end{cases} \tag{10}$$

Above we used this general property in the special case of the computation of limit (9).

CHAPTER 5

Differential Calculus. Functions of One Variable

§ 5.1. Derivative

Before proceeding to study the present chapter it is advisable to reread § 1.5 where the notion of the derivative was discussed. Here we begin with the formal definition of the derivative.

The *derivative of a function f at a point x* is the limit to which tends the ratio of the increment Δy of the function at that point to the corresponding increment Δx of the argument as Δx tends to zero:

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x} \tag{1}$$

It should be noted that for any fixed x the ratio $\frac{\Delta y}{\Delta x}$ is a definite function of Δx:

$$\psi(\Delta x) = \frac{\Delta y}{\Delta x}$$

If the given function f is defined in a neighbourhood of the given point x, the function $\psi(\Delta x)$ is defined for all sufficiently small Δx distinct from zero, that is for Δx satisfying an inequality $0 < |\Delta x| < \delta$ where δ is a sufficiently small positive number. For $\Delta x = 0$ it is not of course defined. The question of the existence of the derivative of the function f at the point x is equivalent to the question of the existence of the limit of the function $\psi(\Delta x)$ at the point $\Delta x = 0$.

Theorem 1. *If a function f possesses the derivative f' at a point x it is continuous at that point.*

Proof. The existence of the finite limit (1) implies that

$$\frac{\Delta y}{\Delta x} = f'(x)+\varepsilon(\Delta x)$$

where $\varepsilon(\Delta x) \to 0$ as $\Delta x \to 0$. It follows that

$$\Delta y = f'(x)\,\Delta x+\varepsilon(\Delta x)\,\Delta x = f'(x)\,\Delta x+o(\Delta x) \qquad (\Delta x \to 0)$$

whence

$$\lim_{\Delta x \to 0} \Delta y = 0$$

The latter relation means that the function f is continuous at the point x.

The converse of Theorem 1 is not true: a function f continuous at a point x may have no derivative at that point (see the discussion below).

We say that *f has an infinite derivative at a point x equal to $+\infty$ or to $-\infty$* (the case ∞ is excluded) if, respectively, we have $f'(x) = \lim\limits_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = +\infty$ or $f'(x) = \lim\limits_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = -\infty$ at the point x.

Finally, we introduce the notions of the *right-hand* and *left-hand derivatives of f at a point x* which are, respectively, determined by the relations

$$f'_+(x) = \lim_{\substack{\Delta x \to 0 \\ \Delta x > 0}} \frac{\Delta y}{\Delta x} = \psi(0+0) \quad \text{and} \quad f'_-(x) = \lim_{\substack{\Delta x \to 0 \\ \Delta x < 0}} \frac{\Delta y}{\Delta x} = \psi(0-0)$$

The right-hand and left-hand derivatives are also termed *one-sided derivatives*.

For the derivative $f'(x)$ to exist it is obviously necessary and sufficient that the right-hand and left-hand derivatives of f at the point x should exist and be equal to each other; then the value of each of them automatically coincides with $f'(x)$.

This assertion continues to hold if "derivative" is replaced by "infinite derivative".

The function whose graph is shown in Fig. 5.1*a* possesses the derivative at the point x_0 since at this point the graph has a (uniquely determined) tangent

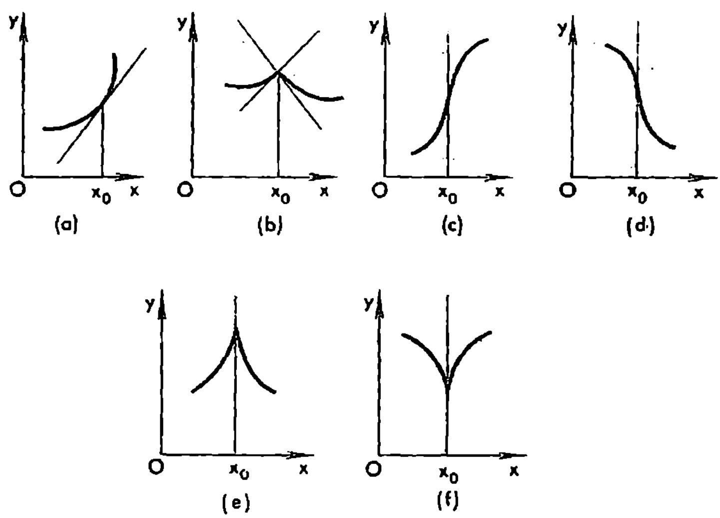

Fig. 5.1

line. The function shown in Fig. 5.1b has no derivative at x_0 but the derivatives $f'_+(x_0)$ and $f'_-(x_0)$ exist and are different from each other. The functions in Figs. 5.1c and d have, respectively, the infinite derivatives $f'(x_0) = +\infty$ and $f'(x_0) = -\infty$ while the functions in Figs 5.1e and f have no derivatives at the point x_0. In the case shown in Fig. 5.1e we have $f'_-(x_0) = +\infty$ and $f'_+(x_0) = -\infty$ and in the case of Fig. 5.1f we have $f'_-(x_0) = -\infty$ and $f'_+(x_0) = +\infty$.

It should be stressed that the derivative of a function f at x is a (new) function of x. Therefore the symbol $f'(x)$ is convenient for the notation of the derivative as function of x while $f'(a)$ denotes a concrete number equal to the value of the derivative of the function f at the point a.

In § 1.5 we derived formulas (1)-(4) for the derivatives of x^n ($n = 0, 1, \ldots$), $\sin x$ and $\cos x$. Below we demonstrate the computation of the derivative of the exponential function a^x ($a > 0$):

$$(a^x)' = \lim_{h\to 0} \frac{a^{x+h}-a^x}{h} = \lim_{h\to 0} a^x \frac{a^h-1}{h} = a^x \lim_{z\to 0} \frac{z}{\log_a(1+z)} =$$

$$= a^x \frac{1}{\lim\limits_{z\to 0} \log_a (1+z)^{1/z}} = \frac{a^x}{\log_a e} = a^x \ln a \tag{2}$$

Here we have used the substitution $a^h - 1 = z \to 0\,(h \to 0)$ and the fact that the function $\log_a u$ is continuous for $u > 0$ and, in particular, for $u = e$.

Putting $a = e$ in (2) we obtain

$$(e^x)' = e^x \tag{3}$$

Let us consider the function

$$y = |x| = \begin{cases} x & \text{for } x \geqslant 0 \\ -x & \text{for } x < 0 \end{cases}$$

(see Fig. 5.2). For $x = 0$ we have

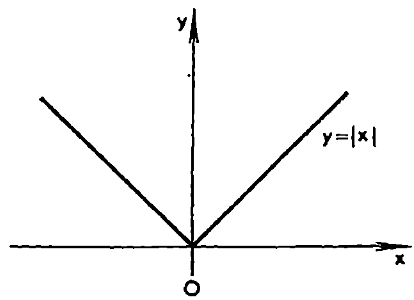

Fig. 5.2

$$\frac{\Delta y}{\Delta x} = \frac{|0+h|-|0|}{h} = \frac{|h|}{h} \to \begin{cases} 1 & \text{for } h > 0,\ h \to 0 \\ -1 & \text{for } h < 0,\ h \to 0 \end{cases}$$

The derivative of $|x|$ at the point $x = 0$ does not exist since the right-hand and left-hand derivatives at this point are different; at all the remaining points the derivative of $|x|$ exists and is expressed by the formula

$$|x|' = \operatorname{sgn} x = \begin{cases} +1 & \text{for } x > 0 \\ -1 & \text{for } x < 0 \end{cases}$$

The graph of the function $|x|$ clearly shows that this function is continuous for any x and, in particular, at the point $x = 0$ as well. This is also seen from the following calculations:

$$||x+h|-|x|| \leqslant |x+h-x| = |h| \to 0 \qquad (h \to 0)$$

The function $|x|$ gives us an example of a function continuous at a point and having no derivative at that point, namely, it has no derivative at the point $x = 0$ but is continuous at it. The graph of this function has no (definite) tangent line at the point $x = 0$. This example shows that the converse of Theorem 1 does not hold.

In mathematics there are examples of functions f continuous throughout a closed interval $[a, b]$ and having no derivative at every point of the interval (a well-known example of such a function was constructed by K.T.W. Weierstrass). The graphs of such functions cannot be plotted in an ordinary way but the functions themselves can be represented by certain formulas which we do not give here.

Example 1. The function specified by the formula

$$f(x) = \begin{cases} 0 & \text{for rational } x \\ x^2 & \text{for irrational } x \end{cases}$$

is discontinuous at all points $x \neq 0$ while at the point $x = 0$ it possesses the derivative $f'(0) = 0$ because we have $\frac{f(h)-f(0)}{h} = \frac{0-0}{h} = 0$ for rational h and $\frac{f(h)-f(0)}{h} = \frac{h^2-0}{h} = h \to 0$ for irrational h tending to zero.

Example 2. The function

$$f(x) = \begin{cases} x \sin \frac{1}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

is continuous on $(-\infty, \infty)$ and has the derivative for all $x \neq 0$; at the point $x = 0$ it even has neither the right-hand nor the left-hand derivative since the expression $\frac{f(h)-f(0)}{h} = \sin \frac{1}{h}$ has no limit as h tends to zero retaining positive or negative values.

If two functions $u(x)$ and $v(x)$ possess derivatives at a point x, their sum, difference, product and quotient (in the last case it is additionally supposed that $v(x) \neq 0$) also possess derivatives; these derivatives are expressed by the formulas

$$(u \pm v)' = u' \pm v' \tag{4}$$

$$(uv)' = uv' + u'v \tag{5}$$

and

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2} \qquad (v \neq 0) \tag{6}$$

The proof of (4) was presented in Chapter 1, § 5. Let us derive (5) and (6). If the independent variable x is given an increment Δx the functions u and v

gain the corresponding increments Δu and Δv. Now we write

$$(uv)' = \lim_{\Delta x\to 0} \frac{\Delta(uv)}{\Delta x} = \lim_{\Delta x\to 0} \frac{(u+\Delta u)(v+\Delta v)-uv}{\Delta x} =$$

$$= \lim_{\Delta x\to 0} u\frac{\Delta v}{\Delta x} + \lim_{\Delta x\to 0} v\frac{\Delta u}{\Delta x} + \lim_{\Delta x\to 0} \frac{\Delta u}{\Delta x}\Delta v = uv' + vu'$$

since the existence of the derivative of v implies that the function v is continuous, that is $\Delta v \to 0$ for $\Delta x \to 0$.

In particular, if $u = C$ where C is a constant we obtain $(Cu)' = Cu' + C'u = Cu'$ because $C' = 0$.

Formula (6) is obtained as follows:

$$\left(\frac{u}{v}\right)' = \lim_{\Delta x\to 0} \frac{1}{\Delta x}\left(\frac{u+\Delta u}{v+\Delta v} - \frac{u}{v}\right) = \lim_{\Delta x\to 0} \frac{v\frac{\Delta u}{\Delta x} - u\frac{\Delta v}{\Delta x}}{v(v+\Delta v)} = \frac{vu'-uv'}{v^2}$$

Some Basic Differentiation Formulas

1. $\left(\frac{1}{x}\right)' = \frac{x\cdot 1' - 1\cdot x'}{x^2} = \frac{-1}{x^2} \qquad (x \neq 0)$

A more general formula is

$$\left(\frac{1}{x^n}\right)' = \frac{x^n\cdot 1' - 1(x^n)'}{x^{2n}} = \frac{-nx^{n-1}}{x^{2n}} = -\frac{n}{x^{n+1}} \qquad (n = 1, 2, \ldots; x \neq 0)$$

Thus, there holds the formula

$$(x^n)' = nx^{n-1} \qquad (n = 0, \pm 1, \pm 2, \ldots) \tag{7}$$

generalizing formula (1) in § 1.5 to the case of any integral n.

As will be shown, formula (7) continues to hold for nonintegral values of n as well.

2. $$(\tan x)' = \left(\frac{\sin x}{\cos x}\right)' = \frac{\cos x(\sin x)' - \sin x(\cos x)'}{\cos^2 x} =$$

$$= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x \tag{8}$$

3. $$(\cot x)' = \left(\frac{\cos x}{\sin x}\right)' = \frac{\sin x(\cos x)' - \cos x(\sin x)'}{\sin^2 x} =$$

$$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = -\csc^2 x \tag{9}$$

§ 5.2. Differential of a Function

If a function $f(x)$ has the derivative $f'(x)$ at a point x there exists the limit

$$\lim_{\Delta x\to 0} \frac{\Delta y}{\Delta x} = f'(x), \qquad \Delta y = f(x+\Delta x) - f(x)$$

It follows that

$$\frac{\Delta y}{\Delta x} = f'(x)+\varepsilon(\Delta x) \qquad (\Delta x \to 0)$$

where $\varepsilon(\Delta x) \to 0$ as $\Delta x \to 0$. Hence

$$\Delta y = f'(x)\,\Delta x+\varepsilon(\Delta x)\,\Delta x, \qquad \varepsilon(\Delta x) \to 0 \qquad \text{for } \Delta x \to 0 \tag{1}$$

that is

$$\Delta y = f'(x)\,\Delta x+o(\Delta x) \qquad (\Delta x \to 0) \tag{1'}$$

Denoting $A = f'(x)$ we rewrite equality (1) as

$$\Delta y = A\Delta x+o(\Delta x) \qquad (\Delta x \to 0) \tag{2}$$

A function f is said to be differentiable at a point x if its increment Δy at that point can be represented in form (2) *where A is a constant independent of Δx (but, generally, dependent on x).*

From the above it follows that *if a function $f(x)$ has the derivative $f'(x)$ at a point x it is differentiable at that point and $A = f'(x)$.*

The converse also holds: *if a function f is differentiable at a point x, that is if its increment at the point x is representable in form* (2), *it possesses the derivative at the point x equal to the number A.*

Indeed, let the increment Δy at the point x be representable as (2). Dividing both members of (2) by Δx and passing to the limit we obtain

$$f'(x) = A+\lim_{\Delta x\to 0}\frac{o(\Delta x)}{\Delta x} = A+\lim_{\Delta x\to 0} o(1) = A$$

Thus, *for a function $f(x)$ to have the derivative $f'(x)$ at a point x it is necessary and sufficient for it to be differentiable at that point.*

Equality (2) shows that if $A = f'(x) \neq 0$, the increment of the function is equivalent to the first term on the right-hand side of (2) for $\Delta x \to 0$:

$$\Delta y \approx A\,\Delta x \qquad (\Delta x \to 0)$$

In this case (when $A \neq 0$) the term $A\,\Delta x$ is called the *principal linear part* of the increment. The principal part depends on Δx linearly (more precisely, it is directly proportional to Δx). *Approximately, discarding the infinitesimal of higher order $o(\Delta x)$, we can put Δy equal to the principal part for small values of Δx.*

The principal linear part of the increment is called the *differential* of the function f, at the point x, corresponding to the given increment Δx of the independent variable x and is denoted as

$$dy = df = f'(x)\,\Delta x$$

For the sake of the symmetry of the notation the increment Δx of the independent variable is also denoted as dx and called the *differential of x*; we thus put, by definition, $\Delta x = dx$. This convention does not contradict the expression $dx = x' \Delta x = \Delta x$ of the differential of the function $x = x$ dependent on the argument x.

Thus, the differential of the function f at the point x is written as

$$dy = f'(x)\, dx \tag{3}$$

It follows from this equality that *the derivative* $f'(x)$ *of* $f(x)$ *at the point* x *is equal to* $\frac{dy}{dx}$, *that is it is equal to the ratio of the differential of the function* f *at the point* x *to the corresponding differential of the independent variable* x.

It should be noted that *the differential* dx *of the independent variable is independent of* x *and is equal to* Δx, *i.e. to an arbitrary increment of the argument* x. *As to the differential* dy *of the function* y *(different from* $y = x$*), it depends both on* x *and on* dx (see (3)).

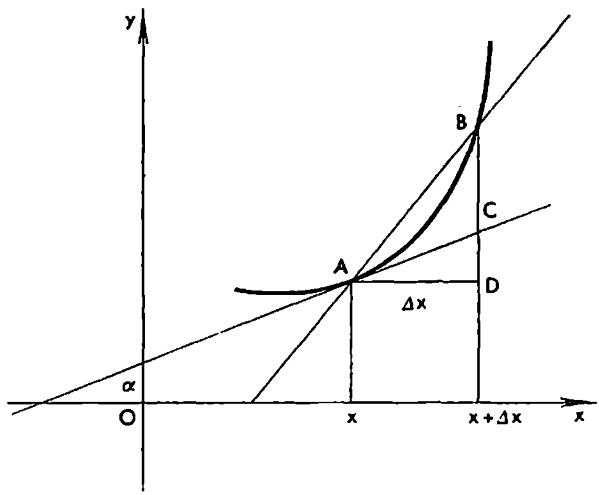

Fig. 5.3

The notions we have discussed can be given a simple geometrical interpretation. Consider the graph of the function $y = f(x)$ shown in Fig. 5.3; the points A and B of the graph correspond to the values x and $x+\Delta x$ of the independent variable. The ordinates of the points A and B are respectively equal to $f(x)$ and $f(x+\Delta x)$. The increment of the function at the point x is equal to $\Delta y = f(x+\Delta x) - f(x)$ and is represented by the length of the line segment BD; the latter can be written as the sum $\Delta y = BD = DC + CB$ where $DC = \tan\alpha\, \Delta x = f'(x)\, \Delta x$ and α is the angle between the tangent to the graph at the point A and the positive direction of the x-axis.

As is seen, the line segment DC is the differential of the function f at the point x:

$$DC = dy = f'(x)\, \Delta x$$

Thus, the second term CB entering into the expression of the increment Δy is equal to the quantity $o(\Delta x)$. The latter can even be greater than the principal part of the increment for large Δx but if $\Delta x \to 0$ it becomes an infinitesimal of higher order of smallness relative to Δx. If $f'(x) \neq 0$ then for any $\varepsilon > 0$ there is $\delta > 0$ such that the inequality $CB/DC < \varepsilon$ holds for all Δx satisfying the inequality $|\Delta x| < \delta$.

Below are some evident formulas:

$$d(u \pm v) = (u \pm v)'\,dx = u'\,dx \pm v'\,dx = du \pm dv \tag{4}$$

$$d(uv) = (uv)'\,dx = (uv' + u'v)\,dx = u\,dv + v\,du \tag{5}$$

and

$$d\left(\frac{u}{v}\right) = \left(\frac{u}{v}\right)'dx = \frac{v\,du - u\,dv}{v^2} \tag{6}$$

Example. Let it be required to estimate the amount of the material needed for manufacturing a cubic box if it is known that the internal edge of the box is of 10 cm and that the width of its walls is of 0.1 cm.

The volume of a cube with edge a is expressed by the function $V(a) = a^3$. The volume of the material needed for the walls of the box is expressed approximately by the increment of this function:

$$\Delta V = V(10+0.1) - V(10) \approx V'(10)\cdot 0.1 = [3a^2]_{a=10}0.1 =$$
$$= 300\cdot 0.1 = 30 \ (\text{cm}^3)$$

§ 5.3. Derivative of Composite Function

Theorem. *Let $z = F(x) = f(\varphi(x))$ be a function of a function where $y = \varphi(x)$ and $z = f(y)$, and let the function φ have the derivative $\varphi'(x)$ at a point x and the function f have the derivative $f'(y)$ with respect to y at the point $y = \varphi(x)$. Then the derivative of F at the point x exists and is expressed as**

$$F'(x) = f'(y)\,\varphi'(x) \tag{1}$$

Proof. Since the function φ has the derivative φ' at the point x it is differentiable at this point (see the foregoing section), that is, for sufficiently small Δx we have

$$\Delta y = \varphi'(x)\,\Delta x + \varepsilon(\Delta x)\,\Delta x \qquad (\varepsilon(\Delta x) \to 0 \text{ for } \Delta x \to 0) \tag{2}$$

Similarly, the existence of the derivative of f at the point y implies, for sufficiently small Δy, the relation

$$\Delta z = f'(y)\,\Delta y + \varepsilon_1(\Delta y)\,\Delta y \qquad (\varepsilon_1(\Delta y) \to 0,\ \Delta y \to 0) \tag{3}$$

If we put $\varepsilon_1(0) = 0$ equality (3) continues to hold for $\Delta y = 0$ as well.

---

* Formula (1) expresses the so-called *chain rule for differentiation*. — *Tr.*

Therefore for sufficiently small Δx and, consequently, for sufficiently small Δy we have

$$\Delta z = f'(y)\varphi'(x)\,\Delta x + f'(y)\,\varepsilon(\Delta x)\,\Delta x + \Delta y\,\varepsilon_1(\Delta y)$$

$$\left(\frac{\Delta y}{\Delta x} \to y'_x,\ \Delta x \to 0\right)$$

On dividing both members of this equality by Δx and passing to the limit as $\Delta x \to 0$ we receive $F'(x) = f'(y)\varphi'(x)$ because $\Delta x \to 0$ implies $\Delta y \to 0$ (see (2)); then (see (3)) we also have $\varepsilon_1(\Delta y) \to 0$. The reason why the convention $\varepsilon_1(0) = 0$ has been stipulated is that there can be cases when $\Delta y = 0$ for some $\Delta x \neq 0$.

Formula (1) can be extended to the more general case of a composite function formed of more than two constituent functions. For instance, if $z = f(y)$, $y = \varphi(x)$ and $x = \psi(\xi)$ and if all the three functions have derivatives at the corresponding points we have $z'_\xi = z'_y y'_x x'_\xi$.

Example. To compute the derivative of the function $z = \cos(\sin^3 x^2)$ with respect to the variable x we introduce the chain of auxiliary functions

$$z = \cos u,\quad u = v^3,\quad v = \sin w \quad \text{and} \quad w = x^2$$

Now we can write

$$\frac{dz}{dx} = (\cos u)'\,(v^3)'\,(\sin w)'\,(x^2)' = -\sin u\,(3v^2)'\cos w\cdot 2x =$$
$$= -6x\cos x^2 \sin^2 x^2 \sin(\sin^3 x^2)$$

The functions

$$\sinh x = \frac{e^x - e^{-x}}{2},\quad \cosh x = \frac{e^x + e^{-x}}{2},$$

$$\tanh x = \frac{\sinh x}{\cosh x},\quad \coth x = \frac{\cosh x}{\sinh x}$$

are respectively called the ***hyperbolic sine***, ***cosine***, ***tangent*** and ***cotangent***. We obviously have

$$(\sinh x)' = \frac{1}{2}(e^x + e^{-x}) = \cosh x \tag{4}$$

$$(\cosh x)' = \frac{1}{2}(e^x - e^{-x}) = \sinh x \tag{5}$$

$$(\tanh x)' = \frac{\cosh x(\sinh x)' - \sinh x(\cosh x)'}{(\cosh x)^2} = \frac{(\cosh x)^2 - (\sinh x)^2}{(\cosh x)^2} = \frac{1}{(\cosh x)^2} \tag{6}$$

$$(\coth x)' = \left(\frac{1}{\tanh x}\right)' = -\frac{(\tanh x)'}{(\tanh x)^2} = -\frac{1}{(\cosh x)^2}\,\frac{(\cosh x)^2}{(\sinh x)^2} = -\frac{1}{(\sinh x)^2} \tag{7}$$

Alternative Proof of the Theorem. Let us consider a sequence of values of Δx tending to zero ($\Delta x \to 0$). If the values Δy corresponding to these Δx's are different from zero then

$$\lim_{\Delta x \to 0} \frac{\Delta z}{\Delta x} = \lim_{\Delta x \to 0} \left(\frac{\Delta z}{\Delta y} \cdot \frac{\Delta y}{\Delta x}\right) = z'_y y'_x$$

If to the values of Δx entering into the sequence in question there correspond $\Delta y = 0$ we have $\Delta z = 0$ and

$$\lim_{\Delta x \to 0} \frac{\Delta z}{\Delta x} = \lim_{\Delta x \to 0} \frac{0}{\Delta x} = 0 = z'_y y'_x$$

The last equality holds since in this case we necessarily have $y'_x = 0$.

Now, if the sequence of values $\Delta x \to 0$ is arbitrary we can always split it into two sequences of the above two types. The limit $\frac{\Delta z}{\Delta x}$ ($\Delta x \to 0$) exists and is equal to $z'_y y'_x$ for both of the subsequences. Therefore the limit $\frac{\Delta z}{\Delta x}$ also exists for the entire sequence and is obviously equal to $z'_y y'_x$ (see § 4.1, Theorem 9).

§ 5.4. Derivative of Inverse Function

Let a strictly monotone (that is a strictly increasing or strictly decreasing) function $y = f(x)$ be defined on an interval (a, b). If an interval (A, B) is the image of (a, b) (corresponding to this function), the inverse function $x = \varphi(y)$ of the function f exists and is defined on (A, B); it is one-valued, continuous and strictly monotone (see § 4.5).

If $x \in (a, b)$ is fixed and then is given an increment Δx $(x+\Delta x \in (a, b))$ the function f receives the corresponding increment Δy $(y, y+\Delta y \in (A, B))$ such that $y+\Delta y = f(x+\Delta x)$.

Conversely, $\varphi(y+\Delta y) = x+\Delta x$.

By the continuity of the given function and its inverse function the following assertion holds for the indicated Δx and Δy: $\Delta x \to 0$ implies $\Delta y \to 0$ and vice versa.

Now, let the function φ have a nonzero derivative $\varphi'(y)$ at the point y. We shall show that in this case the function f also has the derivative f' at the corresponding point x. Indeed,

$$\frac{\Delta y}{\Delta x} = \frac{1}{\dfrac{\Delta x}{\Delta y}}$$

and, since $\Delta x \to 0$ implies $\Delta y \to 0$, we have

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{1}{\lim\limits_{\Delta y \to 0} \dfrac{\Delta x}{\Delta y}} = \frac{1}{\varphi'(y)}$$

We thus receive

$$f'(x) = \frac{1}{\varphi'(y)} \tag{1}$$

or, equivalently,

$$\frac{dy}{dx} = \frac{1}{\dfrac{dx}{dy}} \tag{1'}$$

We have proved that *if $y = f(x)$ is a strictly monotone continuous function and $x = \varphi(y)$ is its inverse function possessing the derivative $\varphi'(y) \neq 0$ at a point y then the function f possesses at the corresponding point x the derivative specified by formula* (1).

It may turn out that $\lim\limits_{\Delta y \to 0} \dfrac{\Delta x}{\Delta y} = \infty$ at the point y. In this case the function f obviously has the derivative $f'(x) = 0$ at the point x.

If $\lim\limits_{\Delta y \to 0} \dfrac{\Delta x}{\Delta y} = 0$ we have $\dfrac{\Delta x}{\Delta y} > 0$ for the strictly increasing function and $\dfrac{\Delta x}{\Delta y} < 0$ for the strictly decreasing function. In the former case $f'(x) = +\infty$ and in the latter $f'(x) = -\infty$.

Derivative of $\log_a x$. On the basis of what has been proved, putting $y = \log_a x$ we get

$$(\log_a x)' = \frac{1}{(a^y)'} = \frac{1}{a^y \ln a} = \frac{1}{x \ln a} = \frac{\log_a e}{x} \qquad (a > 0)$$

For the natural logarithm the derivative is expressed particularly simply:

$$(\ln x)' = \frac{1}{x}$$

This formula accounts for the fact that in mathematical analysis, at least in theoretical studies, it is preferable to deal with the logarithmic function to the base e.

The function $\ln x = \log_e x$ is defined as a real function only for positive values of x.*

The function $\ln |x|$ can of course be considered both for positive and negative values of x. Its graph is symmetric about the y-axis and coincides with the graph of $\ln x$ for positive x (Fig. 5.4).

The function $\ln |x|$ plays an important role in the integral calculus. For $x \neq 0$ its derivative is given by the formula

$$(\ln |x|)' = \frac{1}{|x|} \operatorname{sgn} x = \frac{1}{x} \qquad (x \neq 0)$$

---

* For negative values of x the function $\ln x$ can also be derived in a natural way as a complex function. Here we shall not discuss these questions.

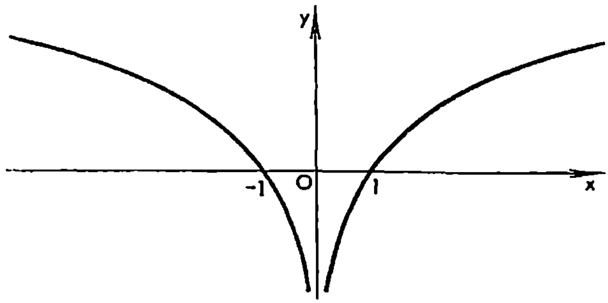

Fig. 5.4

where

$$\operatorname{sgn} x = \begin{cases} 1 & \text{for } x > 0 \\ -1 & \text{for } x < 0 \end{cases}$$

(In connection with the function $y = \ln |x|$ see the second example in the table of indefinite integrals given in § 8.1.)

For the derivative of the power function x^n $(x > 0)$ where n is an arbitrary real number we now obtain the formula

$$(x^n)' = (e^{n \ln x})' = \frac{n}{x} e^{n \ln x} = nx^{n-1} \tag{8}$$

generalizing formula (1) in § 5 of Chapter 1.

Derivatives of Inverse Trigonometric Functions. The function $y = \arcsin x$ is strictly increasing on the closed interval $[-1, +1]$ and maps this interval on the closed interval $[-\pi/2, \pi/2]$. Its inverse function $x = \sin y$ has the derivative $(\sin y)' = \cos y$ different from zero within the open interval $(-\pi/2, \pi/2)$. Therefore

$$(\arcsin x)' = \frac{1}{(\sin y)'} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}} \qquad (-1 < x < 1)$$

Consequently,

$$(\arccos x)' = \left(\frac{\pi}{2} - \arcsin x\right)' = \frac{-1}{\sqrt{1-x^2}} \qquad (-1 < x < 1)$$

Here we take the arithmetic root (with the sign "+").

The function $y = \arctan x$ is strictly increasing throughout the real axis $(-\infty, +\infty)$ and maps it onto the open interval $(-\pi/2, \pi/2)$. Its inverse function $x = \tan y$ has the derivative $(\tan y)' = \sec^2 y$ different from zero on that interval. Therefore

$$(\arctan x)' = \frac{1}{(\tan y)'} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2} \qquad (-\infty < x < \infty)$$

Exercise. Prove the equality $(\log_x a)' = -\dfrac{(\log_x a)^2}{x \ln a}$.

§ 5.5. Table of the Derivatives of the Basic Elementary Functions

$(C)' = 0 \quad (C \text{ is a constant});$

$(x^n)' = nx^{n-1} \quad (n = 0, \pm 1, \pm 2, \ldots);$

$(a^x)' = \ln a \cdot a^x \quad (a > 0);$

$(e^x)' = e^x;$

$(x^a)' = ax^{a-1} \quad (x > 0);$

$(\log_a x)' = \dfrac{\ln_a e}{x} \quad (x > 0,\ a > 0);$

$(\ln x)' = \dfrac{1}{x} \quad (x > 0);$

$(\ln |x|)' = \dfrac{1}{x} \quad (x \neq 0);$

$$(|x|)' = \operatorname{sgn} x = \begin{cases} 1 & \text{for } x > 0; \\ -1 & \text{for } x < 0; \end{cases}$$

$(\sin x)' = \cos x;$

$(\cos x)' = -\sin x;$

$(\tan x)' = \sec^2 x;$

$(\cot x)' = -\csc^2 x;$

$(\arcsin x)' = \dfrac{1}{\sqrt{1-x^2}};$

$(\arccos x)' = \dfrac{1}{\sqrt{1-x^2}};$

$(\arctan x)' = \dfrac{1}{1+x^2};$

$(\sinh x)' = \cosh x;$

$(\cosh x)' = \sinh x;$

$(\tanh x)' = \dfrac{1}{(\cosh x)^2};$

$(\coth x)' = -\dfrac{1}{(\sinh x)^2}.$

Exercises

Prove that*

1. $\dfrac{d}{dx}\sqrt{ax^2+bx+c} = \dfrac{2ax+b}{2\sqrt{ax^2+bx+c}}.$

2. $\dfrac{d}{dx}\ln\left(x+\sqrt{a^2+x^2}\right) = \dfrac{1}{\sqrt{a^2+x^2}}.$

3. $\left(\arcsin\dfrac{x}{a}\right)' = \dfrac{1}{\sqrt{a^2-x^2}}.$

4. $\left(\arcsin\dfrac{1}{x}\right)' = \mp\dfrac{1}{x\sqrt{x^2-1}}$ (the upper sign corresponds to $x > 0$ and the lower to $x < 0$).

5. $(x^x)' = (e^{x \ln x})' = x^x\,(1+\ln x).$

6. $(\ln|\tan x|)' = \dfrac{1}{\sin x \cos x}$ whence $\left(\ln\left|\tan\dfrac{x}{2}\right|\right)' = \dfrac{1}{\sin x}.$

7. $\dfrac{a^2}{2}\left(\arcsin\dfrac{x}{a}+\dfrac{x}{a^2}\sqrt{a^2-x^2}\right)' = \sqrt{a^2-x^2}.$

8. $$(|x|^p)' = \begin{cases} p|x|^{p-2}x & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \text{ where } p > 1. \end{cases}$$

---

* Formulas 1-7 are of use for the computation of indefinite integrals.

§ 5.6. Derivatives and Differentials of Higher Orders

The derivative of a function f is a new function of the same argument. Therefore we can try to find its derivative. The resultant function (provided it exists) is called the *second derivative of* $f(x)$ and is denoted as $f''(x)$. Thus,

$$f''(x) = (f'(x))'$$

By induction, the *derivative* $f^{n}(x)$ *of the nth order* (or, simply, the *nth derivative*) is defined as the (first) derivative of the derivative $f^{(n-1)}(x)$ of the $(n-1)$th order:

$$f^{(n)}(x) = (f^{(n-1)}(x))'$$

Of course, the nth derivative of a given function f at a given point x may or may not exist.

If a function f is said to possess the nth derivative at a point x it is meant that it possesses the derivative $f^{(n-1)}(x)$ of the $(n-1)$th order in at least a sufficiently small neighbourhood of the point x and that this derivative itself has the derivative at the point x. It is the latter derivative that is denoted $f^{(n)}(x)$ and is called the derivative of the nth order, or the nth derivative, of f at the point x.

The function x^m where m is a positive integer possesses the derivatives of any order throughout the real axis:

$$(x^m)^{(n)} = m(m-1) \ldots (m-n+1)\, x^{m-n}$$

For $n > m$ we have $(x^m)^{(n)} \equiv 0$.

The power function x^a where a is an arbitrary real number possesses for $x > 0$ the derivative of any order n which is determined by the analogous formula

$$(x^a)^{(n)} = a(a-1) \ldots (a-n+1)\, x^{a-n} \tag{1}$$

Obviously,

$$(a^x)^{(n)} = (\ln a)^n a^x \qquad (-\infty < x < \infty) \tag{2}$$

and, in particular,

$$(e^x)^{(n)} = e^x \qquad (-\infty < x < \infty) \tag{3}$$

It is also easy to verify the formulas

$$(\sin x)^{(n)} = \sin\left(x+n\frac{\pi}{2}\right) \qquad (n = 1, 2, \ldots) \tag{4}$$

and

$$(\cos x)^{(n)} = \cos\left(x+n\frac{\pi}{2}\right) \tag{5}$$

If $s = f(t)$ is a function expressing the dependence of the path travelled by a point in a rectilinear motion on time t, the second derivative $s'' = f''(t)$ is the acceleration of the moving point at time moment t. As will be seen, the second derivative of a function is extremely important for the study of the geometrical properties of the graph of the function.

Leibniz Formula. If we take a function of the form $f = uv$ where u and v are certain functions possessing the nth derivatives at a given point, the nth derivative of f exists at that point and is expressed by the *Leibniz formula*

$$f^{(n)} = uv^{(n)} + \binom{n}{1} u'v^{(n-1)} + \binom{n}{2} u''v^{(n-2)} + \dots + u^{(n)} v =$$

$$= \sum_{l=0}^{n} \binom{n}{l} u^{(l)} v^{(n-l)} \tag{6}$$

where $\binom{n}{l}$ are the *binomial coefficients* and $u^{(0)} = u$ (see § 5.9, formulas (6) and (7)).

The formula is proved by induction. It evidently holds for $n = 1$. If we suppose that it holds for n, its validity for $n+1$ is proved as follows:

$$f_{(n+1)} = \frac{d}{dx} \sum_{l=0}^{n} \binom{n}{l} u^{(l)} v^{(n-l)} = \sum_{l=0}^{n} \binom{n}{l} (u^{(l+1)} v^{(n-l)} + u^{(l)} v^{(n-l+1)}) =$$

$$= \sum_{l=1}^{n+1} \binom{n}{l-1} u^{(l)} v^{(n+1-l)} + \sum_{l=0}^{n} \binom{n}{l} u^{(l)} v^{(n+1-l)} =$$

$$= \sum_{0}^{n+1} \binom{n+1}{l} u^{(l)} v^{(n+1-l)}$$

Here we have used the relations $\binom{n+1}{0} = \binom{n}{0} = \binom{n}{n} = \binom{n+1}{n+1} = 1$ and $\binom{n+1}{l} = \binom{n}{l} + \binom{n}{l-1}$ $(l = 1, \dots, n)$*.

Example.

$(x \sin x)^{(100)} = x \sin\left(x + 100\frac{\pi}{2}\right) + 100 \cdot 1 \cdot \sin\left(x + 99\frac{\pi}{2}\right) = x \sin x - 100 \cos x$

Let us consider a function $y = f(x)$ defined on an interval (a, b). There is an infinite number of ways in which it can be represented as

$$y = \varphi(\psi(x)) = f(x) \qquad (x \in (a, b)) \tag{7}$$

In the considerations below we use the following terminology for such a representation: the variable y is a function $(y = f(x))$ of the *independent variable* x; this same variable y is a function of the *dependent variable* u $(y = \varphi(u), u \not\equiv x)$. The latter variable depends on the independent variable x $(u = \psi(x))$. Thus, *the variable x plays an exceptional role here: in all the considerations it only enters as an independent variable.*

The differential

$$dy = f'(x)\, dx \tag{8}$$

* A simple derivation of Newton's binomial formula based on the differentiation formula $(x^n)' = nx^{n-1}$ for natural n is presented in § 5.9, Example 1.

of the function f will also be called *the first differential of f, at the point x, corresponding to the differential (increment) $dx = \Delta x$ of the independent variable x.*

The nth differential (the differential of the nth order) of the function f, at the point x, corresponding to the differential $dx = \Delta x$ of the independent variable x is defined by induction:

$$d^n y = d(d^{n-1}y) = d\big(f^{(n-1)}(x)\,dx^{n-1}\big) = f^{(n)}(x)\,dx^n \tag{9}$$

Hence, the differentials d and d^{n-1} in these equalities are taken for one and the same differential dx, of the independent variable x, which is regarded as a constant quantity (independent of x).

It follows from equality (9) that the nth derivative of f at the point x is equal to the ratio of $d^n y$ to $(dx)^n$:

$$f^{(n)}(x) = \frac{d^n y}{dx^n} \tag{10}$$

where the expression $d^n y$ in the numerator is the nth differential of y (at the point x) corresponding to the value of dx in the denominator.

Now let us regard the variable y as a function of the *dependent* variable u ($u \not\equiv x$), that is $y = \varphi(u)$ and $u = \psi(x)$, where φ and ψ possess a sufficient number of derivatives. Then

$$dy = f'(x)\,dx = \varphi'(u)\psi'(x)\,dx = \varphi'(u)\,du \tag{11}$$

We have thus expressed the first differential dy in terms of u.

Equality (11) describes a remarkable property of the (first) differential. Namely, the differential dy was defined originally as the product of the derivative of y with respect to the *independent* variable x by the differential dx. It now turns out that, according to (11), the differential dy can also be defined as the product of the derivative of y with respect to the *dependent* variable u by the differential du, and there hold the equalities

$$dy = y'_x\,dx = y'_u\,du \tag{12}$$

provided that the differential du in the third term of (12) corresponds to that very dx which enters into the second term of (12).

To express the meaning of (11) ((12)) we say that the form $dy = \varphi'(u)\,du$ in which the first differential is written is *invariant with respect to any variable u*. It turns out that for differentials of the second and higher orders the invariance property no longer holds in the general case. More precisely, if $n > 1$ and $u \not\equiv x$ then, generally, $d^n y$ is not equal to $\varphi^{(n)}(u)\,du^n$, in contrast to the case when $u = x$ is an independent variable.

Indeed, we have

$$\begin{aligned} d^2 y &= d(dy) = d\big(\varphi'(u)\,du\big) = d\varphi'(u)\,du + \varphi'(u)\,d^2 u = \\ &= \varphi''(u)\,(du)^2 + \varphi'(u)\,d^2 u \end{aligned} \tag{13}$$

Here $d^2 u \neq 0$, and therefore the term $\varphi'(u)\,d^2 u$ cannot be discarded since

$d^2u = \varphi''(x)(dx)^2$. Thus, the second differential, in contrast to the first differential, is not invariant in the above sense.

The same situation (the absence of the invariance) occurs in the case of higher-order differentials. For instance,

$$\begin{aligned} d^3y &= d(d^2y) = \varphi'''(u)(du)^3+\varphi''(u)\,d(du)^2+\varphi''(u)\,du\,d^2u+\varphi'(u)\,d^3u = \\ &= \varphi'''(u)(du)^3+\varphi''(u)\,2du\,d^2u+\varphi''(u)\,du\,d^2u+\varphi'(u)\,d^3u = \\ &= \varphi'''(u)(du)^3+3\varphi''(u)\,du\,d^2u+\varphi'(u)\,d^3u \end{aligned} \tag{14}$$

The differentials of the fourth, fifth, etc. orders are computed in the same manner. As n grows the expression of d^ny becomes more and still more complicated.

Let y be a function of u where u is a dependent variable, that is u, in its turn, is a function of a third variable x. We may not express the latter dependence explicitly. Moreover, the precise dependence may be unknown and we may only suppose that such a dependence does in fact exist. Under these assumptions we shall compute the derivatives of y with respect to u: y'_u, y''_u, y'''_u, ...

As is already known,

$$y'_u = \frac{dy}{du} \tag{15}$$

that is the derivative of y with respect to the variable u is equal to the ratio of the corresponding differentials: $y'_u = dy : du$.

In the computation of higer-order derivatives we use the above rule for the first derivative and the rules for the computation of the differentials of sums, differences, products and quotients (see § 5.2, formulas (4), (5) and (6)). We should also take into account that $d^ny = d(d^{n-1}y)$.

Following this course we find the second derivative:

$$y''_u = \frac{d\left(\frac{dy}{du}\right)}{du} = \frac{1}{du}\,\frac{du\,d^2y-dy\,d^2u}{du^2} = \frac{du\,d^2y-dy\,d^2u}{du^3} \tag{16}$$

The left member of (16) is the second derivative of y with respect to u while the right member is a definite rational expression containing the differentials du, dy, d^2u and d^2y. If $u = x$, that is if u is an independent variable, we have $du = dx \neq 0$ and $d^2u = 0$; in this case (16) implies the equality $y''_u = \frac{d^2y}{du^2}$ already known. But if u is a function of x distinct from x then y''_u should be computed by formula (16).

We similarly find

$$\begin{aligned} y'''_{u^3} &= \frac{d\left(\frac{du\,d^2y-dy\,d^2u}{du^3}\right)}{du} = \\ &= \frac{du^3(d^2u\,d^2y+du\,d^3y-d^2y\,d^2u-dy\,d^3u)-(du\,d^2y-dy\,d^2u)\,3du^2\,d^2u}{du^7} \end{aligned} \tag{17}$$

Supplementary Remark. *The derivative of an even function is an odd function:*

$$f'(-x) = \lim_{h\to 0} \frac{f(-x+h)-f(-x)}{h} = \lim_{h\to 0} \frac{f(x-h)-f(x)}{h} = -f'(x)$$

Analogously, *the derivative of an odd function is an even function.* It follows that the derivative of order k of an even function is an even or an odd function depending on whether k is even or odd.

§ 5.7. Increase and Decrease of a Function on an Interval and at a Point. Local Extremum

A function f is said to be *strictly increasing* on an open interval (a, b) (or on a closed interval $[a, b]$) if, given any two points x_1 and x_2 belonging to (a, b) (or to $[a, b]$) and satisfying the inequality $x_1 < x_2$, the inequality $f(x_1) < f(x_2)$ holds.

A function f is said to be *nondecreasing* on (a, b) (or on $[a, b]$) if $x_1, x_2 \in (a, b)$ (or $x_1, x_2 \in [a, b]$) and $x_1 < x_2$ imply $f(x_1) \leqslant f(x_2)$.

Similarly, a function f is said to be *strictly decreasing (nonincreasing)* on (a, b) or on $[a, b]$ if $x_1 < x_2$ and $x_1, x_2 \in (a, b)$ or $x_1, x_2 \in [a, b]$ imply $f(x_1) > f(x_2)$ $\big(f(x_1) \geqslant f(x_2)\big)$.

Let a function f be defined in a neighbourhood of a point x. Then the expression

$$\Delta y = f(x+\Delta x) - f(x)$$

of its increment at the point x makes sense for all sufficiently small Δx.

By definition, we say that the function

(i) *increases at the point* x if there is $\delta > 0$ such that

$$\frac{\Delta y}{\Delta x} > 0 \qquad (0 < |\Delta x| < \delta) \tag{1}$$

(ii) *decreases at the point* x if there is $\delta > 0$ such that

$$\frac{\Delta y}{\Delta x} < 0 \qquad (0 < |\Delta x| < \delta) \tag{2}$$

(iii) *attains a local maximum at the point* x if there is $\delta > 0$ such that

$$\Delta y \leqslant 0 \qquad (|\Delta x| < \delta) \tag{3}$$

(iv) *attains a local minimum at the point* x if there is $\delta > 0$ such that

$$\Delta y \geqslant 0 \qquad (|\Delta x| < \delta) \tag{4}$$

It should be stressed that all the inequalities (1)-(4) must hold for all positive and negative values of Δx which are sufficiently small in their absolute values.

The above four properties can also be expressed as follows: for all the points $x' \in (x-\delta, x)$ and for all the points $x'' \in (x, x+\delta)$ the relations

$$f(x') < f(x) < f(x'')$$

and

$$f(x') > f(x) > f(x'')$$

hold, respectively, in the first and second cases, and for all the points $x' \in \in (x-\delta, x+\delta)$ the relations

$$f(x') \leqslant f(x)$$

and

$$f(x') \geqslant f(x)$$

hold, respectively, in the third and fourth cases. Thus, in Case (iii) the value of f at the point x *is its maximum value in a sufficiently small neighbourhood of* x and in Case (iv) the value of f at the point x *is its minimum value in a sufficiently small neighbourhood of* x.

Local maxima and minima are called *local extrema.*

We shall be interested in the problem of finding out which of the above four cases takes place provided that the derivatives of f of the first and higher orders are known at the point x or in the vicinity of x.

Suppose that the function f has the derivative f' at the point x which is positive: $\lim\limits_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(x) > 0$. This means that the ratio $\Delta y/\Delta x$, which is a function of Δx for the fixed value of x, tends to a positive number. Therefore (see Theorem 2 in § 4.1) this ratio itself must be positive for all Δx satisfying the inequality $|\Delta x| < \delta$ for a sufficiently small δ; hence, according to Definition (i), the function f increases at the point x.

It is similarly proved that if $f'(x) < 0$ then the function f decreases at the point x. We have thus proved the following theorem:

Theorem. *If a function $f(x)$ has the positive derivative $f'(x) > 0$ (the negative derivative $f'(x) < 0$) at a point x it increases (decreases) at that point.*

This theorem immediately implies

Fermat's Theorem. *If a function f attains a local extremum (a local maximum or a local minimum) at a point x and if the derivative $f'(x)$ exists at this point it is equal to zero: $f'(x) = 0$.*

Indeed, if $f'(x) \neq 0$ then, by the foregoing theorem, the function increases or decreases at the point x and therefore the extremum of the function cannot be attained at that point.

This theorem can be restated as follows.

For a function f possessing the derivative at a point x to attain a local extremum at this point it is necessary that the derivative of f at the point x be equal to zero.

The condition $f'(x) = 0$ is not of course sufficient for the function f to attain a local extremum at the point x. If $f'(x) = 0$ the function f may have no local extremum at x. It can increase at that point (for instance, this is the case for the function x^3 at the point $x = 0$) or decrease (which is the case for the function $f(x) = -x^3$ at the point $x = 0$); moreover, the point x may neither be a point of increase or decrease nor a point of extremum. The last possibility is clearly demonstrated by the example of the function

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases} \tag{5}$$

It has the derivative $f'(0) = 0$ because

$$\lim_{x \to 0} \frac{f(x)-f(0)}{x-0} = \lim_{x \to 0} \frac{f(x)}{x} = \lim_{x \to 0} \frac{x^2 \sin (1/x)}{x} = \lim_{x \to 0} x \sin \frac{1}{x} = 0$$

$\left(\text{since} \left|\sin \frac{1}{x}\right| \leq 1\right)$. On the other hand, in any neighbourhood $|x| < \delta$ of the point 0, however small, the function f assumes positive and negative values both to the right and to the left of the point $x = 0$. Therefore the point 0 is neither a point of increase or decrease nor the point of extremum of the function f.

In the next section we shall present some extremely important theorems known as the *mean-value theorems* (or *laws of means*) *for derivatives*. Later on hey will be conveniently applied to the study of local extrema.

§ 5.8. Mean-value Theorems for Derivatives. Tests for Increase and Decrease of a Function on an Interval. Sufficient Tests for Local Extrema

Rolle's* Theorem. *Let a function f be continuous on a closed interval $[a, b]$, have the derivative in the open interval (a, b) and assume equal values at the end points $(f(a) = f(b))$.*

Then there is at least one point c belonging to the open interval (a, b) at which the derivative of f turns into zero $(f'(c) = 0)$.

Proof. Let M and m be, respectively, the maximum and the minimum values of f on the interval $[a, b]$. They are sure to exist since the function f is continuous on $[a, b]$. If the equalities $M = m = f(a)$ hold then $f(x) = M$ for all $x \in [a, b]$ and $f'(c) = 0$ at any point $c \in (a, b)$. If these equalities do not hold simultaneously then at least one of the numbers M and m is distinct from the number $f(a) = f(b)$; for definiteness, let $M \neq f(a) = f(b)$. This means that the maximum value of the function f is attained on the closed interval $[a, b]$ at a point c belonging to the open interval (a, b), and consequently f has a local maximum at that point. Since the derivative $f'(c)$ of f at the point c exists, Fermat's theorem implies that it is equal to zero. The case $m \neq f(a)$ is treated analogously.

The theorem has been proved.

---

* M. Rolle (1652-1719), a French mathematician who was the first to prove this theorem but only for polynomials.

Cauchy's Mean-value Theorem*. *Let two functions $\varphi(x)$ and $\psi(x)$ be continuous on a closed interval $[a, b]$ and possess the derivatives within the open interval (a, b) which do not turn to zero simultaneously. Also let $\varphi(b) - \varphi(a) \neq$ $\neq 0$***.

Then there is a point c belonging to the open interval (a, b) at which the equality

$$\frac{\psi(b)-\psi(a)}{\varphi(b)-\varphi(a)} = \frac{\psi'(c)}{\varphi'(c)} \qquad (a < c < b) \tag{1}$$

holds.

Proof. Consider the auxiliary function

$$F(x) = [\psi(b)-\psi(a)]\,\varphi(x)-[\varphi(b)-\varphi(a)]\,\psi(x)$$

Obviously, it is continuous on $[a, b]$ and has the derivative on the interval (a, b). Besides, $F(a) = F(b)$. Therefore, by Rolle's theorem, there is a point $c \in (a, b)$ such that $F'(c) = 0$, that is

$$[\varphi(b)-\varphi(a)]\,\psi'(c) = [\psi(b)-\psi(a)]\,\varphi'(c) \tag{2}$$

There must be $\varphi'(c) \neq 0$ because, if otherwise, there would be $\psi'(c) = 0$ since $\varphi(b)-\varphi(a) \neq 0$ while, by the hypothesis, $\varphi'(c)$ and $\psi'(c)$ do not vanish simultaneously. Therefore the product $[\varphi(b)-\varphi(a)]\,\varphi'(c)$ is different from zero. On dividing the left-hand and right-hand members of (2) by this product we arrive at (1).

Putting $\varphi(x) = x$ we obtain, as a consequence of Cauchy's theorem, Lagrange's theorem.

Lagrange's*** Mean-value Theorem.** *Let a function $\varphi(x)$ be continuous on a closed interval $[a, b]$ and possess the derivative on the open interval (a, b). Then there is a point c belonging to the interval (a, b) such that the equality*

$$\varphi(b)-\varphi(a) = (b-a)\,\varphi'(c) \qquad (a < c < b) \tag{3}$$

holds.

Lagrange's theorem admits of a simple geometrical interpretation. Let us write it as

$$\frac{\varphi(b)-\varphi(a)}{b-a} = \varphi'(c) \qquad (a < c < b)$$

The left member of this equality is the tangent of the angle of inclination of the chord, subtending the arc of the graph of the function $y = \varphi(x)$ with end points $(a, \varphi(a))$ and $(b, \varphi(b))$, to the x-axis; the right member is equal to the tangent of the angle of inclination of the tangent line to the graph drawn

* This theorem is also called the *second mean-value theorem* or the *double law of the mean*; sometimes it is referred to as the *generalized* (or *extended*) *mean-value theorem*, the latter name not being commonly used. – *Tr.*

** It should be noted that if, for instance, we introduced the condition $\varphi'(x) \neq 0$ on (a, b) this would imply $\varphi(b)-\varphi(a) \neq 0$.

*** J. L. Lagrange (1736-1813), a famous French mathematician.

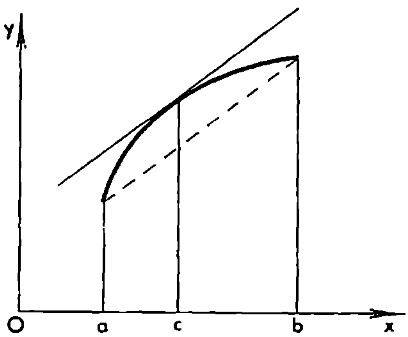

Fig. 5.5

through the point with an intermediate abscissa $c \in (a, b)$. Lagrange's theorem thus means that if a curve (see Fig. 5.5) is the graph of a continuous function defined on $[a, b]$ and having the derivative on (a, b) then there is a point on the curve corresponding to an abscissa $c\,(a < c < b)$ such that the tangent line to the curve at this point is parallel to the chord joining the end points $(a, \varphi(a))$ and $(b, \varphi(b))$ of the curve.

Equality (3) is called *Lagrange's formula of finite increments*. The intermediate value c can be conveniently written as

$$c = a+\theta(b-a)$$

where θ is a number satisfying the inequalities $0 < \theta < 1$. Then Lagrange's formula turns into

$$\varphi(b)-\varphi(a) = (b-a)\,\varphi'(a+\theta(b-a)) \qquad (0 < \theta < 1). \tag{4}$$

It obviously holds not only for $a < b$ but also for $a \geqslant b$.

Theorem 1. *If a function is continuous on a closed interval* $[a, b]$ *and has a nonnegative (positive) derivative on the open interval* (a, b) *it is nondecreasing (strictly increasing) on* $[a, b]$.

Indeed, let $a \leqslant x_1 < x_2 \leqslant b$; then the conditions of Lagrange's theorem are fulfilled for the closed interval $[x_1, x_2]$ and therefore there is a point c belonging to the open interval (x_1, x_2) such that

$$f(x_2)-f(x_1) = (x_2-x_1)f'(c) \qquad (x_1 < c < x_2)$$

If $f' \geqslant 0$ on (a, b) then $f'(c) \geqslant 0$ and

$$f(x_2)-f(x_1) \geqslant 0 \tag{5}$$

If $f' > 0$ on (a, b) then $f'(c) > 0$ and

$$f(x_2)-f(x_1) > 0 \tag{6}$$

Inequalities (5) and (6) holding for any x_1 and x_2 such that $a \leqslant x_1 < x_2 \leqslant b$, the function f is nondecreasing on the interval $[a, b]$ in the first case and strictly increasing on the interval $[a, b]$ in the second case.

Theorem 2. *If a function has a derivative identically equal to zero in an interval* (a, b) *the function is constant on* (a, b).

Indeed, by Lagrange's theorem, we have

$$f(x)-f(x_1) = (x-x_1)f'(c)$$

where x_1 is a fixed point of the interval (a, b), x is the variable point (which can be to the right or to the left of x_1) and c is a point lying between x_1 and x whose location is dependent on x_1 and x. By the hypothesis, $f'(x) \equiv 0$ on (a, b), and therefore we have $f'(c) = 0$; hence $f(x) = f(x_1) = C = \text{const}$ for all $x \in (a, b)$.

It should be noted that the conditions imposed on the functions in these theorems cannot be weakened because this may lead to the failure of the assertions of the theorems.

For instance, the function $f(x)$ specified by the equalities

$$f(x) = \begin{cases} x & \text{for } 0 \leqslant x < \frac{1}{2} \\ 1-x & \text{for } \frac{1}{2} \leqslant x < 1 \end{cases}$$

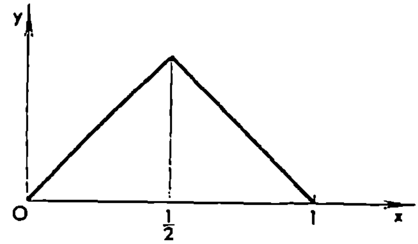

Fig. 5.6

(see Fig. 5.6) is obviously continuous on the interval $[0, 1]$, takes on zero values at its end points and has the derivative at all the points of $(0, 1)$ except at the single point $x = \frac{1}{2}$; it is readily seen that neither Rolle's theorem nor Lagrange's theorem any longer hold for this function.

Now we proceed to a theorem providing some sufficient criterion for the existence of a local extremum of a function.

Theorem 3. *Let a function f be continuous in a neighbourhood of a point x_0 and have the derivative $f'(x)$ in this neighbourhood. If*

$$f'(x) \geqslant 0 \ (\leqslant 0) \qquad \text{to the right of } x_0$$

and

$$f'(x) \leqslant 0 \ (\geqslant 0) \qquad \text{to the left of } x_0$$

then x_0 is a point of local minimum (maximum) of f.

The expression "to the right (to the left) of x_0" should be understood here in the sense "on a sufficiently small open interval with left (right) end point x_0". The proof of the theorem is a direct consequence of the formula of finite increments

$$\Delta f = f(x) - f(x_0) = (x - x_0) f'(x_0 + \theta(x - x_0))$$

because the conditions of the theorem imply that the right-hand side of the formula is nonnegative (nonpositive) in a sufficiently small neighbourhood of the point x_0 irrespective of the sign of the difference $x - x_0$.

Note that the existence of the derivative at the point x_0 itself is not supposed in this theorem. Of course, if the derivative $f'(x_0)$ exists then, by Fermat's theorem, it is equal to zero.

The following theorem provides a sufficient test for the existence of a local extremum using the sign of the second derivative.

Theorem 4. *If a function f satisfies the conditions $f'(x_0) = 0$ and $f''(x_0) > 0$ ($f''(x_0) < 0$) then x_0 is a point of local minimum (maximum) of the function f.*

Proof. The existence of the second derivative at the point x_0 implies the existence of the first derivative $f'(x)$ in a neighbourhood of the point x_0 and, of course, the continuity of f in this neighbourhood. The condition $f''(x_0) > 0$ (< 0) means that $f'(x)$ increases (decreases) at the point x_0 and, since $f'(x_0) = 0$, we have $f' < 0$ (> 0) to the left of x_0 and $f' > 0$ (< 0) to the right of x_0. The assertion of the theorem now follows from the foregoing theorem.

As is known, a function continuous on a closed interval $[a, b]$ and having a positive derivative throughout the open interval (a, b) is strictly increasing on $[a, b]$. On the other hand, in the example presented below we have a function f which is continuous in a neighbourhood of the point $x = 0$ and has the positive derivative $f'(0) > 0$ at that point but at the same time is not increasing in any neighbourhood of $x = 0$, however small.

Thus, generally, the increase of a function at a point does not imply its increase in a neighbourhood of that point.

Example 1. The function $F(x) = \frac{x}{2}+f(x)$ where f is defined by equality (5) in the foregoing section has the derivative $F'(0) = \frac{1}{2}+f'(0) = \frac{1}{2} > 0$ at the point $x = 0$ and consequently it increases at that point. At the same time the function is not increasing in any interval containing this point. Indeed, for $x \neq 0$ we have

$$F'(x) = \frac{1}{2}-\cos\frac{1}{x}+2x\sin\frac{1}{x}$$

If $x_k = 1/k\pi$ ($k = 1, 2, \ldots$) then

$$F'(x_k) = \frac{1}{2}-(-1)^k$$

whence we see that any interval enveloping the origin contains points at which the derivative F' assumes values of different signs and consequently F is not monotone on such an interval.

Example 2. Consider the function

$$\psi(x) = \begin{cases} x\ln|x| & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

Let us regard it as being defined on the closed interval $[-1, e]$. The function is continuous and has a finite derivative throughout $[-1, e]$ except at $x = 0$ where

$$\psi'(0) = \lim_{h\to 0}\frac{\psi(h)-\psi(0)}{h} = \lim_{h\to 0}\ln|h| = -\infty \tag{7}$$

It follows from (7) that ψ decreases at the point $x = 0$. The equation $\psi'(x) = 1+\ln|x| = 0$ has two roots: $x_1 = -1/e$ and $x_2 = 1/e$. We also have $\psi''(x) = 1/x$ $(x \neq 0)$, $\psi''(-1/e) < 0$ and $\psi''(1/e) > 0$; consequently $-1/e$ is a point of local maximum and $1/e$ a point of local minimum.

Example 3. The graph of the function

$$x = \frac{t^2-a}{b+2t\sqrt{c}} \qquad (b^2-4ac < 0,\ c > 0)$$

(see § 8.9) splits into two continuous branches corresponding respectively to the variation of t within the intervals $(-\infty, -b/2\sqrt{c})$ and $(-b/2\sqrt{c}, \infty)$. On each of the intervals the function is monotone and increases from $-\infty$ to $+\infty$. This can readily be seen if we take into account that the condition $b^2-4ac < 0$ implies that $x' > 0$ and that $(t^2-a) = \frac{b^2}{4c}-a < 0$ for $t = -b/2\sqrt{c}$.

§ 5.9. Taylor's* Formula

Taylor's formula which we study in this section makes it possible to determine approximately, sometimes with a high accuracy, the values of a given function f in a neighbourhood of a point a from the given values $f(a), f'(a), \ldots, f^{(n-1)}(a)$ of the function f and its derivatives up to order $n-1$ inclusive at that point a and from some information on the values of the nth derivative $f^{(n)}$ in this neighbourhood.

To this end some special approximating polynomials known as *Taylor's polynomials of the given function* are constructed from the given values of the function and its derivatives.

We shall begin with the derivation of Taylor's formula for an arbitrary polynomial

$$P(x) = b_0+b_1x+ \ldots +b_nx^n \tag{1}$$

Let us choose an arbitrary number a and replace x by $(x-a)+a$ in the right-hand side of equality (1):

$$P(x) = b_0+b_1[(x-a)+a]+ \ldots +b_n[(x-a)+a]^n$$

On opening the square brackets and combining similar terms containing the same powers of $x-a$ we obtain the equality

$$P(x) = \beta_0+\beta_1(x-a)+ \ldots +\beta_n(x-a)^n = \sum_{k=0}^{n} \beta_k(x-a)^k \tag{2}$$

where β_k are constants dependent on the original coefficients b_k.

Equality (2) is called the *expansion of the polynomial $P(x)$ in powers of $x-a$* and the numbers β_k are *the coefficients of this expansion.*

The original equality (1) can be interpreted, from this point of view, as the expansion of $P(x)$ in powers of x, that is in powers of the difference $x-a$ with $a = 0$.

* B. Taylor (1685-1731), an English mathematician.

Let us differentiate in succession equality (2):

$$P'(x) = \beta_1+2\beta_2(x-a)+3\beta_3(x-a)^2+\dots$$
$$P''(x) = 2\beta_2+3\cdot 2\beta_3(x-a)+4\cdot 3\beta_4(x-a)^2+\dots$$
$$\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots$$
$$P^{(k)}(x) = k!\beta_k+(k+1)k\dots 2\beta_{k+1}(x-a)+\dots$$

Putting $x = a$ in the last equality specifying the kth derivative we see that all the terms on the right-hand side, beginning with the second, turn into zero, and we thus obtain $P^{(k)}(a) = k!\beta_k$ $(k = 0, 1, 2, \dots)$. Here, as usual, we put $P^{(0)}(a) = P(a)$ and $0! = 1$. Thus, the coefficients β_k of expansion (2) of the polynomial $P(x)$ in powers of $x-a$ are necessarily expressed by the formulas

$$\beta_k = \frac{P^{(k)}(a)}{k!} \qquad (k = 0, 1, 2, \dots) \tag{3}$$

In particular, it follows that one and the same polynomial $P(x)$ of degree n can be represented as an expansion in powers of $x-a$ in a unique way, that is if

$$P(x) = \sum_{k=0}^{n} \beta_k(x-a)^k = \sum_{k=0}^{n} \beta'_k(x-a)^k$$

for all the values of x where β_k and β'_k are constants then

$$\beta_k = \beta'_k \qquad (k = 0, 1, \dots, n)$$

since both numbers β_k and β'_k are expressed by the same formulas (3).

Thus,

$$P(x) = P(a)+\frac{P'(a)}{1}(x-a)+\frac{P''(a)}{2}(x-a)^2+\dots+\frac{P^{(n)}(a)}{n!}(x-a)^n =$$
$$= \sum_{k=0}^{n} \frac{P^{(k)}(a)}{k!}(x-a)^k \tag{4}$$

Formula (4) is called *Taylor's formula, in powers of* $x-a$, *for the given polynomial* $P(x)$ *of the nth degree* or, simply, *Taylor's formula for* $P(x)$ *at the point* a.

Taylor's formula in powers of x, that is the representation

$$P(x) = \sum_{k=0}^{n} \frac{P^{(k)}(0)}{k!}x^k \tag{5}$$

is also called *Maclaurin's* formula*.

Example 1. *Newton's binomial formula.* Let us consider the polynomial

$$P(x) = (a+x)^n$$

of the nth degree where a is an arbitrary number and n is a natural number.

* C. Maclaurin (1698-1746), a Scotch mathematician. The name "Maclaurin's formula" is commonly used but is historically incorrect. — *Tr.*

The kth derivative of the polynomial is

$$P^{(k)}(x) = n(n-1) \dots (n-k+1)(a+x)^{n-k}$$

whence $P^{(k)}(0) = n(n-1) \dots (n-k+1)a^{n-k}$; consequently, by Maclaurin's formula, we have

$$(a+x)^n = a^n+na^{n-1}x+\frac{n(n-1)}{2!}a^{n-2}x^2+\frac{n(n-1)(n-2)}{3!}a^{n-3}x^3+\dots+$$
$$+\frac{n(n-1)\dots 2}{(n-1)!}ax^{n-1}+x^n \tag{6}$$

Equality (6) is known as *Newton's binomial formula.*

Denoting

$$\binom{n}{k} = \frac{n(n-1)\dots(n-k+1)}{k!} \quad \text{and} \quad \binom{n}{n} = \binom{n}{0} = 1 \tag{7}$$

we can rewrite Newton's binomial formula in a more compact form:

$$(a+x)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}x^k \tag{6'}$$

The numbers $\binom{n}{k}$ are called *binomial coefficients.*

If we multiply both the numerator and the denominator of the fraction in (7) by $(n-k)!$ we obtain

$$\binom{n}{k} = \frac{n(n-1)\dots(n-k+1)(n-k)\dots 1}{k!\,(n-k)!}$$

i.e.

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} \qquad (k = 0, 1, 2, \dots) \tag{7'}$$

The case $k = 0$ is also embraced by this formula since $0! = 1$.

Another important property of binomial coefficients is expressed by the formula

$$\binom{n+1}{k+1} = \binom{n}{k}+\binom{n}{k+1}$$

whose proof is left to the reader. Using the last equality and taking into account that $\binom{n}{0} = \binom{n}{n} = 1$ we can compute in succession the numbers $\binom{n}{k}$ for any n and k using only the operation of addition.

Above we have derived Taylor's formula for polynomials. Now we shall generalize it to the case of an arbitrary function f defined in a neighbourhood of a point a; the function f may not be a polynomial but is supposed to possess the derivatives up to order n inclusive in this neighbourhood*.

---

* The considerations in this section require in fact less restrictive conditions on f (see the statement of Theorem 1 below).

Let us compute the numbers $f(a)$, $f'(a)$, $\ldots$, $f^{(n-1)}(a)$ and use them to construct the function

$$Q(x) = f(a)+\frac{f'(a)}{1}(x-a)+\ \ldots\ +\frac{f^{(n-1)}(a)}{(n-1)!}(x-a)^{n-1} =$$
$$= \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k \tag{8}$$

Obviously, Q is a polynomial of degree not higher than $n-1$. It is called *Taylor's polynomial* or, more precisely, *Taylor's polynomial of degree* $(n-1)$ *for the function* f *at the point* a.

If the original function f itself is a polynomial of the $(n-1)$th degree then, as was shown, we have the identity $f(x) = Q(x)$ for all the values of x belonging to the neighbourhood in question. But in the case under consideration the identity may fail to hold since, by the hypothesis, the function f may not be a polynomial of degree $n-1$. But nevertheless the polynomial Q turns out to be closely related to f in its properties. Indeed, representing the polynomial Q by Taylor's formula we obtain

$$Q(x) = \sum_{k=0}^{n-1} \frac{Q^{(k)}(a)}{k!}(x-a)^k \tag{9}$$

Since (8) and (9) are expansions of one and the same polynomial in powers of $x-a$ we must have

$$f^{(k)}(a) = Q^{(k)}(a) \qquad (k = 0, 1, \ldots, n-1) \tag{10}$$

Hence, Taylor's polynomial of the $(n-1)$th degree for function f can also be defined as a polynomial of degree $(n-1)$ such that the n equalities (10) hold for it.

Example 2. In Fig. 5.7 we see a curve representing the graph of a function $y = f(x)$. As a zeroth approximation to this curve in the vicinity of the point $a = 0$ we can naturally take the graph of Taylor's polynomial of zeroth degree for the function $f(x)$; this polynomial $y = Q_0(x) = f(0) = \text{const}$ represents the straight line $y = f(0)$ parallel to the x-axis. As a first approximation to the given curve it appears natural to take its tangent line at the point $x = 0$. The equation of the tangent is $y = Q_1(x)$ where $Q_1(x) = f(0)+f'(0)x$. The next (second) approximation computed in this manner is $y = Q_2(x)$ where $Q_2(x) = f(0)+f'(0)x+\frac{f''(0)}{2}x^2$. This is Taylor's polynomial of degree 2 for f at the point

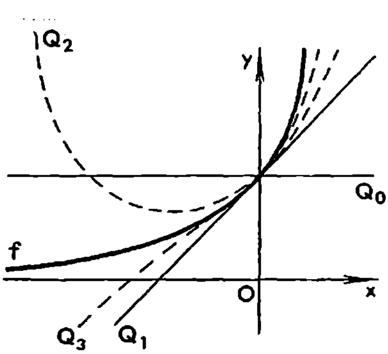

Fig. 5.7

$x = a = 0$, that is a polynomial of the second degree such that its value at the point 0 and the values of its derivatives of the first and second orders at that point coincide respectively with $f(0)$, $f'(0)$ and $f''(0)$.

We see that, at least in a sufficiently small neighbourhood of $a = 0$, the graphs of the subsequent Taylor polynomials of the function f at the point 0 become closer and still closer to the graph of f as their degree increases provided that the function is differentiable at the point $a = 0$ a sufficiently large number of times.

Let us put

$$f(x) = Q_{n-1}(x)+R_n(x) \tag{11}$$

where Q_{n-1} is Taylor's polynomial of degree $n-1$ for the function f at the point $x = a$.

Equality (11) is called *Taylor's formula for the function f in the neighbourhood of the point a* (or simply, *Taylor's formula for f at the point a*), and $R_n(x)$ is called *the remainder term* or *the nth remainder of Taylor's formula.*

It is remarkable that the remainder can be expressed in terms of the nth derivative of f. In this section we shall derive two such expressions, namely *Lagrange's form* and *Cauchy's form of the remainder.*

The remainder of Taylor's formula in Lagrange's form is written as

$$R_n(x) = \frac{(x-a)^n}{n!}f^{(n)}(\xi), \qquad \xi \in (a, x)$$

where ξ is a point belonging to the interval (a, x) (and dependent on x and n). The value of x in this formula can be not only greater than a but also smaller than a*. The exact value of ξ is usually unknown and the only information is that ξ lies somewhere in the interval (a, x).

It is also convenient to represent the number ξ as $\xi = a+\theta(x-a)$ where θ is a number satisfying the inequality $0 < \theta < 1$. Then Lagrange's form of the remainder becomes

$$R_n(x) = \frac{(x-a)^n}{n!} f^{(n)}(a+\theta(x-a)), \qquad 0 < \theta < 1$$

The expression of the remainder of Taylor's formula in Cauchy's form is written

$$R_n(x) = \frac{(x-a)^n (1-\theta)^{n-1}}{(n-1)!} f^{(n)}(a+\theta(x-a)), \qquad 0 < \theta < 1$$

where θ is a number dependent on x and n.

Note that in the case $n = 1$ Taylor's formula for a function $f(x)$ with the remainder in Lagrange's (or Cauchy's) form turns into Lagrange's formula of finite increments established in § 5.8:

$$f(x)-f(a) = (x-a)f'(a+\theta(x-a)), \qquad 0 < \theta < 1$$

The theorem on the remainder reads:

* If $x < a$ then (a, x) and $[a, x]$ denote, respectively, the sets of points t satisfying the inequalities $x < t < a$ and $x \leqslant t \leqslant a$.

Theorem 1. *Let a function f be continuous on a closed interval $[a, x]$ together with its derivatives up to order $n-1$ inclusive and let the nth derivative of f exist in the open interval (a, x).*

Then the nth remainder for Taylor's formula can be written in Lagrange's form and in Cauchy's form.

Proof. Let us set an arbitrary number x belonging to the interval indicated in the statement of the theorem and also an arbitrary natural number p. The number x will remain unchanged in the course of the proof. For our aims it will be convenient to introduce a new auxiliary variable u relative to which x will be regarded as a constant.

Our purpose is to find a convenient expression for the remainder $R_n(x)$ in Taylor's formula

$$f(x) = \sum_{k=0}^{n-1} \frac{(x-a)^k}{k!} f^{(k)}(a) + R_n(x)$$

We shall find $R_n(x)$, that is the value of the remainder at the point x. To this end we represent $R_n(x)$ as the product $R_n(x) = (x-a)^p H$; the problem is thus reduced to determining the quantity H. The quantity H depends on x and, according to the hypothesis, it will be regarded as constant.

Hence, we proceed from the equality

$$f(x) = \sum_{k=0}^{n-1} \frac{(x-a)^k}{k!} f^{(k)}(a) + (x-a)^p H$$

Now we formally replace the constant a entering into the right member of this equality by the variable u. This leads to the function

$$\Phi(u) = \sum_{0}^{n-1} \frac{(x-u)^k}{k!} f^{(k)}(u) + (x-u)^p H =$$

$$= f(u) + \frac{x-u}{1} f'(u) + \ldots + \frac{(x-u)^{n-1}}{(n-1)!} f^{(n-1)}(u) + (x-u)^p H \quad (12)$$

which is defined for at least all the values of u belonging to the interval $[a, x]$ and is continuous for these values of u since the original function $f(u)$ is continuous on that interval together with its derivatives up to the order $n-1$ inclusive. The definition of the function $\Phi(u)$ implies that for $u = a$ it assumes the value $f(x)$ $(\Phi(a) = f(x))$ and for $u = x$ it turns into $f(x)$ $(\Phi(x) = f(x))$, which immediately follows from the expression on the right-hand side of (12). Indeed, if we put $u = x$ in this expression, all the terms beginning with the second vanish while the first term turns into $f(x)$. Finally, the function $\Phi(u)$ possesses the derivative in the interval (a, x) because the original function f has the nth derivative in that interval.

We see that the auxiliary function $\Phi(u)$ satisfies the conditions of Rolle's theorem: it is continuous on the closed interval $[a, x]$, is differentiable on the open interval (a, x) and assumes equal values at its end points. Therefore, by Rolle's theorem, there is an intermediate point $u = a + \theta(x-a)$ lying between a and x and such that the derivative Φ' is equal to zero at it.

Let us compute this derivative:

$$\Phi'(u) = f'(u) - f'(u) + (x-u) f''(u) - (x-u) f''(u) + \dots$$
$$\dots - \frac{(x-u)^{n-2}}{(n-2)!} f^{(n-1)}(u) + \frac{(x-u)^{n-1}}{(n-1)!} f^{(n)}(u) - p(x-u)^{p-1} H$$

All the terms in this expression, except the last two summands, cancel out. If the indicated value $u = a+\theta(x-a)$ is substituted into the remaining terms their sum, as was said, turns into zero.

This results in an equation in the unknown H. On finding H from this equation and multiplying it by $(x-a)^p$ we arrive at the expression for the remainder*:

$$R_n(x) = \frac{(x-a)^n}{(n-1)!\,p} (1-\theta)^{n-p} f^{(n)}(a+\theta(x-a))$$

Here $R_n(x)$ depends on p where p can be any natural number. Putting $p = n$ we obtain Lagrange's form of the remainder and putting $p = 1$ we obtain Cauchy's form.

In the case $a = 0$ Taylor's formula is also called *Maclaurin's formula* which has the form

$$f(x) = \sum_0^{n-1} \frac{f^{(k)}(0)}{k!} x^k + R_n(x) \tag{13}$$

where the remainder is written as

$$R_n(x) = \frac{x^n}{n!} f^{(n)}(\theta x)$$

in Lagrange's form and as

$$R_n(x) = \frac{x^n}{(n-1)!} (1-\theta)^{n-1} f^n(\theta x)$$

in Cauchy's form.

Now let us suppose that the function f has the continuous nth derivative $f^{(n)}(x)$ at the point a. This means that there is a neighbourhood of the point a in which the function f has the derivative $f^{(n)}$ and therefore has the continuous derivative $f^{(n-1)}$. Hence the conditions for the expansion of the function f by formula (11) with the remainder in Lagrange's form are fulfilled in this case, and taking into account the continuity of $f^{(n)}$ at the point $x = a$ we can write

$$R_n(x) = \frac{(x-a)^n}{n!} f^{(n)}(a+\theta(x-a)) = \frac{(x-a)^n}{n!} f^{(n)}(a) + r_n(x) \tag{14}$$

where

$$r_n(x) = \frac{(x-a)^n}{n!} \left[f^{(n)}(a+\theta(x-a)) - f^{(n)}(a)\right] =$$
$$= (x-a)^n o(1) = o((x-a)^n) \qquad (x \to a)$$

* This expression is known as *Schlömilch's form of the remainder*. O.X. Schlömilch (1823-1901), a German mathematician. – *Tr.*

Consequently,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o((x-a)^n) \qquad (x \to a) \tag{15}$$

Expansion (15) is termed *Taylor's formula for the function f at the point a with remainder in Peano's* form.* The result we have obtained can be stated as

Theorem 2. *If a function f possesses the continuous derivative of order n at a point a it is representable by Taylor's formula* (15) *at the point a with remainder in Peano's form.*

Now we shall prove the following lemma.

Lemma. *The equality*

$$\alpha_0+\alpha_1(x-a)+ \dots +\alpha_n(x-a)^n+o((x-a)^n) = \alpha_0'+$$
$$+\alpha_1'(x-a)+ \dots +\alpha_n'(x-a)^n+o((x-a)^n), \qquad x \to a \tag{16}$$

where α_k and α_k' are constant numbers independent of x implies that

$$\alpha_k = \alpha_k' \qquad (k = 0, 1, \dots, n) \tag{17}$$

Indeed, passing to the limit as $x \to a$ in the right-hand and left-hand sides of (16) we obtain the equality $\alpha_0 = \alpha_0'$. On deleting the terms with α_0 and α_0' and cancelling (16) by $x-a$ we arrive at the equality

$$\alpha_1+\alpha_2(x-a)+ \dots +\alpha_n(x-a)^{n-1}+o((x-a)^{n-1}) = \alpha_1'+$$
$$+\alpha_2'(x-a)+ \dots +\alpha_n'(x-a)^{n-1}+o((x-a)^{n-1})$$

whence, on passing to the limit as $x \to a$, we obtain $\alpha_1 = \alpha_1'$. Proceeding in this way we arrive at (17), and the lemma has thus been proved.

This lemma and what was said above imply *the uniqueness of the expansion of the function f by Taylor's formula with remainder in Peano's form.* This must be understood in the following sense. If a function f possessing the continuous nth derivative at a point $x = a$ is representable in the form

$$f(x) = \alpha_0+\alpha_1(x-a)+ \dots +\alpha_n(x-a)^n+o((x-a)^n), \qquad x \to a \tag{18}$$

where α_k's are constant numbers then these numbers are uniquely determined by the formulas

$$\alpha_k = \frac{f^{(k)}(a)}{k!}, \qquad k = 1, \dots, n$$

that is (18) is nothing but the expansion of $f(x)$ by Taylor's formula with remainder in Peano's form.

Taylor's formula at the point $x = 0$ for an even (odd) function f only involves terms containing even (odd) powers of x:

$$f(x) = a_0+a_2x^2+a_4x^4+ \dots,$$
$$(f(x) = a_1x+a_3x^3+ \dots)$$

* G. Peano (1858-1932), an Italian mathematician.

This follows from the fact that the derivatives of odd orders of an even function and the derivatives of even orders of an odd function are themselves odd functions (see the end of § 5.6) and, since these derivatives are supposed to be continuous at the point $x = 0$, they necessarily vanish at that point.

In particular, it follows from the last assertion that *for a polynomial*

$$P(x) = \sum_{0}^{n} \alpha_k x^k$$

to be even (odd), that is to be an even (odd) function, it is necessary and sufficient that all its terms involve x only in even (odd) powers.

Example 3. From the equality $1+x^2+\ldots+x^{2m} = (1-x^{2m+2})/(1-x^2)$ and from the fact that $x^{2m+2}/(1-x^2) = o(x^{2m})$ $(x \to 0)$ it follows that

$$\frac{1}{1-x^2} = 1+x^2+\ldots+x^{2m}+o(x^{2m}) \qquad (x \to 0) \tag{19}$$

Therefore (19) is nothing but Maclaurin's formula for the function $(1-x^2)^{-1}$ with remainder in Peano's form.

§ 5.10. Taylor's Formula for Most Important Elementary Functions

The Function $f(x) = e^x$. For this function we have

$$f^{(n)}(x) = e^x, \quad f^{(n)}(0) = 1, \quad f^{(n)}(\theta x) = e^{\theta x} \qquad (x = 0, 1, 2, \ldots)$$

Therefore Taylor's formula in powers of x (Maclaurin's formula) for the function e^x with remainder in Lagrange's form is written as

$$e^x = 1+\frac{x}{1}+\frac{x^2}{2!}+\ldots+\frac{x^{n-1}}{(n-1)!}+R_n(x),$$

$$R_n(x) = \frac{x^n}{n!}e^{\theta x} \qquad (0 < \theta < 1)$$

Putting $x = 1$ in it we obtain the following approximation to the number e:

$$e \approx 1+1+\frac{1}{2!}+\ldots+\frac{1}{(n-1)!}$$

The error of this approximation is estimated by the inequality $|R_n(1)| \leqslant \leqslant \frac{1}{n!}e < \frac{3}{n!}$.

For any $x \geqslant 0$ we have

$$|R_n(x)| \leqslant \frac{x^n}{n!}e^x \to 0 \qquad (n \to \infty)$$

and for $x < 0$ we have

$$|R_n(x)| \leqslant \frac{|x|^n}{n!} \to 0 \qquad (n \to \infty)$$

The Function $f(x) = \sin x$. In this case we have

$$f^{(n)}(x) = \sin\left(x+n\frac{\pi}{2}\right), \quad f^{(n)}(0) = \sin\frac{n\pi}{2}$$

and

$$f^{(n)}(\theta x) = \sin\left(\theta x+\frac{n\pi}{2}\right) \qquad (n = 1, 2, \ldots)$$

Taylor's formula in powers of x with remainder in Lagrange's form for the function $\sin x$ is thus written

$$\sin x = x-\frac{x^3}{3!}+\ldots+(-1)^{\nu+1}\frac{x^{2\nu-1}}{(2\nu-1)!}+R_{2\nu+1}(x)$$

$$R_{2\nu+1}(x) = \frac{x^{2\nu+1}}{(2\nu+1)!}\sin\left(\theta x+(2\nu+1)\frac{\pi}{2}\right)$$

The remainder tends to zero as $\nu\to\infty$ for any x:

$$|R_{2\nu+1}(x)| \leqslant \frac{|x|^{2\nu+1}}{(2\nu+1)!}\to 0 \qquad (\nu\to\infty)$$

The Function $f(x) = \cos x$. For this function

$$f^{(n)}(x) = \cos\left(x+\frac{n\pi}{2}\right), \quad f^{(n)}(0) = \cos\frac{n\pi}{2}$$

and

$$f^{(n)}(\theta x) = \cos\left(\theta x+\frac{n\pi}{2}\right) \qquad (n = 1, 2, \ldots)$$

In this case Taylor's formula in powers of x with remainder in Lagrange's form is

$$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots+(-1)^{\nu-1}\frac{x^{2(\nu-1)}}{(2(\nu-1))!}+R_{2\nu}(x)$$

$$R_{2\nu}(x) = \frac{x^{2\nu}}{(2\nu)!}\cos\left(\theta x+2\nu\frac{\pi}{2}\right)$$

The remainder behaves like in the case of $\sin x$:

$$|R_{2\nu}(x)| \leqslant \frac{|x|^{2\nu}}{(2\nu)!}\to 0 \qquad (\nu\to\infty)$$

The remainders in Taylor's formulas for the functions $\sin x$ and $\cos x$ tend to zero particularly fast when $|x| \leqslant 1$. It should be added that for practical computations it is sufficient to know the values of these functions for the arcs x lying within the limits from 0 to $\frac{\pi}{4} < 1$.

The Function $f(x) = \ln(1+x)$. This function is defined and infinitely differentiable for $x > -1$. Hence, Taylor's formula can be written for this function for any $n = 1, 2, \ldots$ and $x > -1$. Since

$$f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}, \quad f^{(n)}(0) = (-1)^{n-1}(n-1)!$$

Taylor's formula is

$$\ln(1+x) = x-\frac{x^2}{2}+\ \dots\ +(-1)^n\frac{x^{n-1}}{n-1}+R_n(x)$$

The remainder can be taken both in Lagrange's form

$$R_n(x) = \frac{(-1)^{n+1}x^n}{n(1+\theta x)^n} \qquad (0<\theta<1)$$

and in Cauchy's form

$$R_n = (-1)^{n+1}\frac{x^n}{1+\theta x}\left(\frac{1-\theta}{1+\theta x}\right)^{n-1} \qquad (0<\theta<1)$$

Let $0 \leqslant x \leqslant 1$; then using Lagrange's form we obtain $|R_n(x)| \leqslant \frac{1}{n}x^n \to 0$ $(n\to\infty)$. We see that for $0<x<1$ the remainder tends to zero fast while for $x = 1$ it tends to zero very slowly.

In the case $-1<x<0$ Lagrange's form does not provide any information on the rate at which R_n tends to zero because we only know that θ satisfies the inequalities $0<\theta<1$, and one should bear in mind that θ depends on both x and n. But if we apply Cauchy's form we obtain the estimation

$$|R_n| < \frac{|x|^n}{1-|x|} \to 0 \qquad (n\to\infty)$$

for the remainder valid for $0<|x|<1$ because $\frac{1-\theta}{1+\theta x} < \frac{1-\theta}{1-\theta} = 1$.

For $x=-1$ the expression $\ln(1+x)$ loses sense. When $x>1$ Taylor's formula makes sense for any n but its remainder $R_n(x)$ no longer tends to zero as $n\to\infty$. This can be shown with the aid of the following artificial technique*. Let us put

$$S_n(x) = x-\frac{x^2}{2}+\ \dots\ +(-1)^n\frac{x^{n-1}}{n-1}$$

Then

$$S_n(x)+R_n(x) = S_{n+1}(x)+R_{n+1}(x)$$

and

$$R_n(x)-R_{n+1}(x) = (-1)^{n+1}\frac{x^n}{n}$$

For $x>1$ and $n\to\infty$ the right-hand member of this equality does not tend to zero. Therefore $R_n(x)$ cannot tend to zero for $n\to\infty$ since the conditions of Cauchy's necessary and sufficient criterion for the existence of limit are violated here.

Thus, the remainder in Taylor's formula for the function $\ln(1+x)$ in powers of x tends to zero as $n\to\infty$ only when x satisfies the inequalities $-1<x\leqslant 1$.

---

* This also follows from the divergence for $|x|>1$ of the series with general term $(-1)^n x^{n-1}/(n-1)$ (see § 11.1).

The Function $f(x) = (1+x)^m$. In this case

$$f^{(n)}(x) = m(m-1)\dots(m-n+1)(1+x)^{m-n}$$

and

$$f^{(n)}(0) = m(m-1)\dots(m-n+1)$$

Taylor's expansion at the point $x = 0$ takes the form

$$(1+x)^m = 1+mx+\frac{m(m-1)}{1\cdot 2}x^2+\frac{m(m-1)(m-2)}{3!}x^3+\dots$$

$$\dots+\frac{m(m-1)\dots(m-n+2)}{(n-1)!}x^{n-1}+R_n(x)$$

Lagrange's form of the remainder in this formula is

$$R_n = \frac{m(m-1)\dots(m-n+1)}{n!}x^n(1+\theta x)^{m-n}$$

and Cauchy's form is

$$R_n = \frac{m(m-1)\dots(m-n+1)}{(n-1)!}x^n(1+\theta x)^{m-1}\left(\frac{1-\theta}{1+\theta x}\right)^{n-1}$$

For any natural m and any x all the terms of this formula beginning with the $(m+1)$th vanish and thus in this case Taylor's formula turns into Newton's binomial formula (see § 5.9, formula (6)).

For all the other values of m the formula involves infinitely many nonzero terms; it makes sense at least for $x > -1$.

Let $0 \leqslant x < 1$. Then, using Lagrange's form of the remainder, we obtain for $n > m$ the inequality

$$|R_n(x)| \leqslant \frac{|m(m-1)\dots(m-n+1)|}{n!}|x|^n \to 0 \qquad (n\to\infty)$$

(see the remark below).

In the case $-1 < x < 0$ we make use of Cauchy's form of the remainder to obtain (see the remark below)

$$|R_n(x)| \leqslant C\frac{|m(m-1)\dots(m-n+1)|}{(n-1)!}|x|^n \to 0 \qquad (n\to\infty)$$

where C is a number which in the general case depends on x but is independent of n since $((1-\theta)/(1+\theta x))^{n-1} \leqslant ((1-\theta)/(1-\theta))^{n-1} = 1$ and

$$(1+\theta x)^{m-1} \leqslant 2^{m-1}$$

for $m-1 > 0$ and

$$(1+\theta x)^{m-1} < \frac{1}{(1-|x|)^{m-1}}$$

for $m-1 < 0$.

Thus, the remainder of Taylor's formula for the function $(1+x)^m$ tends to zero as $n\to\infty$ for $-1 < x < 1$. For $x > 1$ the remainder no longer tends to zero as $n \to \infty$* because if we denote by $S_n(x)$ the sum of the first n terms of the expansion of $(1+x)^m$ we obtain (see the remark below)

$$R_n(x)-R_{n+1}(x) = S_{n+1}(x)-S_n(x) = \frac{m(m-1)\dots(m-n+1)}{n!}x^n \to \infty$$
$$(n \to \infty)$$

which shows that $R_n(x)$ does not satisfy the conditions of Cauchy's criterion for the existence of limit.

Remark. The expression

$$u_n = \frac{m(m-1)\dots(m-n+1)}{n!}x^n$$

where m is an arbitrary real number satisfies the condition

$$\left|\frac{u_{n+1}}{u_n}\right| = \frac{m-n}{n+1}|x| \to |x|, \qquad n \to \infty$$

It follows (let the reader prove this) that

$$u_n \to 0, \quad \text{if} \quad |x| < 1 \quad \text{and} \quad u_n \to \infty \qquad \text{if } |x| > 1$$

(in this connection also see § 11.3, Theorem 2).

§ 5.11. Taylor's Series

An expression

$$u_0+u_1+u_2+u_3+\dots \tag{1}$$

where u_k are numbers dependent, in accordance with a certain law, on the natural index k ($k = 0, 1, 2, \dots$), is called a *(number) series*.

Let $S_n = \sum\limits_0^{n-1} u_k$ denote the sum of its first n terms (it is called the nth *partial sum*). The numbers S_n form a sequence $\{S_n\} = \{S_1, S_2, S_3 \dots\}$. If this sequence is convergent, that is if there exists a finite limit $\lim\limits_{n\to\infty} S_n = S$, *series* (1) *is said to be convergent and to have a sum equal to S*. We then write $S = u_0+u_1+u_2+\dots$.

If a function f is infinitely differentiable in a neighbourhood of a point a (i.e. it has derivatives of all orders) we can formally write for it the series

$$f(a)+\frac{f'(a)}{1}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\dots \tag{2}$$

called *Taylor's series for the function f in powers of* $(x-a)$ (or *Taylor's series for f at the point* $x = a$). For given values of a and x it can be convergent or divergent. The case when Taylor's series of the function f is convergent

* This also follows from the divergence for $x > 1$ of the series with the general term $\dfrac{m(m-1)\dots(m-n+1)}{n!}x^n$ (see § 11.1).

to that very function is of particular importance; in this case the sum of the series is equal to $f(x)$.

Taylor's series is convergent to f if and only if the remainder in Taylor's formula

$$f(x) = \sum_{0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x) = S_n(x) + R_n(x) \tag{3}$$

tends to zero as $n \to \infty$. Indeed, if $\lim_{n\to\infty} R_n(x) = 0$ then (3) implies

$$\lim_{n\to\infty} S_n(x) = f(x)$$

and, since $S_n(x)$ is the sum of the first n terms of series (2), this series is convergent and has $f(x)$ as its sum:

$$f(x) = f(a) + \frac{f'(a)}{1}(x-a) + \frac{f''(a)(x-a)^2}{2!} + \dots \tag{4}$$

Conversely, if it is known that equality (4) is fulfilled for some value of x, that is if it is known that series (2) is convergent and has the number $f(x)$ as its sum, this means that

$$\lim_{n\to\infty} S_n(x) = f(x)$$

for the indicated value of x. Then (3) implies $R_n(x) \to 0$, $n \to \infty$.

Now the results established in the foregoing section allow us to assert that the following Taylor series expansions take place:

$$\left.\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots && (-\infty < x < \infty)\\
\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots && (-\infty < x < \infty)\\
\cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots && (-\infty < x < \infty)\\
(1+x)^m &= 1 + mx + \frac{m(m-1)}{2!}x^2 + \\
&\quad + \frac{m(m-1)(m-2)}{3!}x^3 + \dots && (-1 < x < 1)\\
\ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots && (-1 < x \leqslant 1)
\end{aligned}\right\} \tag{5}$$

In these examples the sets E of points x on which Taylor's series in powers of x are convergent are intervals or half-intervals with centre at 0. As will be shown in § 11.11, this fact reflects a general property of series of the form

$$a_0 + a_1 x + a_2 x^2 + \dots \tag{6}$$

where a_k are some given constant numbers. It turns out that if such a series is convergent at a point x_1 it is sure to converge for all x satisfying the inequality $|x| < |x_1|$. Series of type (6) are termed *power series.*

It may turn out that with a given function f we can formally associate its Taylor series, at a point a, of the form

$$f(a)+\frac{f'(a)}{1}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\ldots \tag{7}$$

(in other words, this means the derivatives $f^{(k)}(a)$ make sense for this function for any $k = 0, 1, 2, \ldots$) and that series (7) is convergent for some values of x but its sum is not equal to $f(x)$.

Example 1. An example of a function of the type mentioned above is

$$\psi(x) = \begin{cases} e^{1/x^2-1} & \text{for } |x| < 1 \\ 0 & \text{for } |x| \geqslant 1 \end{cases} \tag{8}$$

If $|x| < 1$ and if we put $u = 1-x^2$ then

$$\psi'(x) = -2x(1-x^2)^{-2}e^{1/x^2-1} = -2xu^{-2}e^{-1/u} \to 0, \qquad x \to 0$$

It is readily proved by induction that

$$\psi^{(k)}(x) = P(x)(1-x^2)^{-l}e^{1/x^2-1} = P(x)u^{-l}e^{-1/u} \to P(1)\cdot 0 = 0, \qquad x \to 1$$

for $k = 0, 1, 2, \ldots$ where $P(x)$ is a polynomial and the number $l > 0$ depends on k. Since $\psi(x) = 0$ for $|x| \geqslant 1$ we have thus shown that $\lim\limits_{x\to 1} \psi^{(k)}(x) = 0$. Further, we have $\psi(1) = 0$. If it is already known that $\psi^{(k)}(1) = 0$ for some k then

$$\psi^{(k+1)}(1) = \lim_{x\to 1}\frac{\psi^{(k)}(x)-\psi^{(k)}(1)}{x-1} = \lim_{x\to 1}\psi^{(k+1)}\big(1+\theta(x-1)\big) = 0 \quad (0 < \theta < 1)$$

Thus, the derivatives $\psi(1)$, $\psi'(1)$, $\psi''(1)$, ... make sense for the function ψ and, as has been proved by induction, they are numbers equal to zero. Therefore we can formally write Taylor's series for this function at the point $x = 1$. All its terms are equal to zero for any x and thus it is convergent; its sum is equal to zero for any x and is distinct from $\psi(x)$ for $|x| < 1$. A similar situation takes place in the case $x = -1$.

The function ψ is an example of an infinitely differentiable function defined throughout the real axis which is equal to zero outside an interval (see § 14.4).

A function $f(x)$ whose Taylor series in powers of $(x-a)$ is convergent to $f(x)$ in a neighbourhood of the point a is called an *analytic function at all the points of this (open) neighbourhood*. In particular, *it is analytic at the point a.*

It follows that the functions e^x, $\sin x$ and $\cos x$ are analytic throughout the real axis while the functions $\ln(1+x)$ and $(1+x)^m$ are analytic on the interval $(-1, +1)$.

It can be shown (see Example 2 below) that the functions $\ln(1+x)$ and $(1+x)^m$ can in fact be expanded into Taylor's series at any point $a > 0$; these series are convergent to the corresponding functions for sufficiently small $x-a$ whence it follows that the functions $\ln(1+x)$ and $(1+x)^m$ are analytic for any $x>0$. Analytic functions are studied in a special division of higher mathematics termed the *theory of functions of a complex argument* and also the *theory of analytic functions.*

The functions defined on an interval can be classified as

(i) arbitrary and, generally, discontinuous functions;

(ii) continuous functions;

(iii) functions having the derivative $f^{(n)}$ for some $n = 1, 2, 3, \ldots$;

(iv) functions having the continuous derivative $f^{(n)}$ for some $n = 1, 2, \ldots$;

(v) infinitely differentiable functions, that is having the derivatives $f^{(n)}$ of any order and thus having the continuous derivatives $f^{(n)}$ of any order;

(vi) analytic functions.

Each of these classes is contained in the preceding class and consists of functions "with a better behaviour".

The function specified by equalities (8) is infinitely differentiable on $(-\infty, \infty)$ but is not analytic on this interval. By the way, at the same time it is analytic on $(1, \infty)$, $(-\infty, -1)$ and $(-1, 1)$.

Example 2. Let $f(x) = \ln x$. Then $f^{(k)}(x) = (-1)^{k-1}(k-1)!\, x^{-k}$ $(x>0;$ $k = 1, 2, \ldots)$ and

$$\ln x = \ln a + \sum_{1}^{n-1} \frac{(-1)^{k-1}(x-a)^k}{ka^k} + R_n(x)$$

where

$$R_n(x) = \frac{(-1)^{n-1}}{n}\,\frac{(x-a)^n}{(a+\theta(x-a))^n}, \qquad |x-a| < a$$

If $|x-a| < a/2$ then $|a+\theta(x-a)| > a-|x-a| > a-(a/2) = a/2$, $|(x-a)/$ $/(a+\theta(x-a))| < 1$ and $|R_n(x)| < 1/n \to 0$.

Thus, we have the expansion into the convergent series

$$\ln x = \ln a + \frac{x-a}{1\cdot a} - \frac{(x-a)^2}{2a^2} + \frac{(x-a)^3}{3a^3} + \ldots$$

for any $a>0$ and $|x-a| < a/2$. This shows that $\ln x$ is an analytic function for any $x>0$.

Example 3. Let us write down the principal term of the expansion of the function $\ln(1+x+x^2)$ in powers of x:

$$\ln(1+x+x^2) = (x+x^2)+o(x+x^2) = x+o(x)+o(x) = x+o(x), \qquad x\to 0$$

Indeed, we have $\ln(1+u) = u+o(u)$ for $u\to 0$, $u = x+x^2 \to 0$ for $x\to 0$, $x^2 = o(x)$ for $x\to 0$ and $o(x+x^2) = o(x)$ for $x\to 0$, because $\frac{o(x+x^2)}{x} =$ $= \frac{o(x+x^2)}{x+x^2}\cdot\frac{x+x^2}{x}$, $\frac{o(x+x^2)}{x+x^2} \to 0$ and $\frac{x+x^2}{x} \to 0$ for $x\to 0$ and thus

$$\frac{o(x+x^2)}{x} \to 0, \qquad x\to 0$$

Example 4. Let us consider the same problem for the function

$$\ln(1+x+x^2)-x$$

The application of the foregoing result does not enable us to determine the principal term because this yields

$$\ln(1+x+x^2)-x = x+o(x)-x = o(x), \qquad x \to 0$$

But nevertheless we have obtained some useful information: if the principal term exists its degree is $n > 1$. Let us use Taylor's formula with the remainder $o(u^2)$, $u \to 0$, in Peano's form. We have $\ln(1+u) = u-\frac{u^2}{2}+o(u^2)$, $\quad u \to 0$ and therefore

$$\ln(1+x+x^2)-x = x+x^2-\frac{(x+x^2)^2}{2}+o((x+x^2)^2)-x =$$

$$= x^2-\frac{x^2}{2}+o(x^2)+o(x^2) = \frac{x^2}{2}+o(x^2), \qquad x \to 0$$

§ 5.12. Convexity of a Curve at a Point. Points of Inflection

*A curve $y = f(x)$ is said to be convex upward (downward) at a point x_0** if there is a neighbourhood of x_0 such that the tangent to the curve at the point x_0 (that is the tangent drawn through the point of the curve with abscissa x_0) lies above (below) the points of the curve for all abscissas x belonging to that neighbourhood (see Fig. 5.8; in the figure the curve is convex downward at the point x_1 and convex upward at the point x_2).

A point x_0 is called a *point of inflection of the curve* $y = f(x)$ if the moving point of the curve having abscissa x passes from one side of the tangent (drawn through the point with abscissa x_0) to the other as x increases and

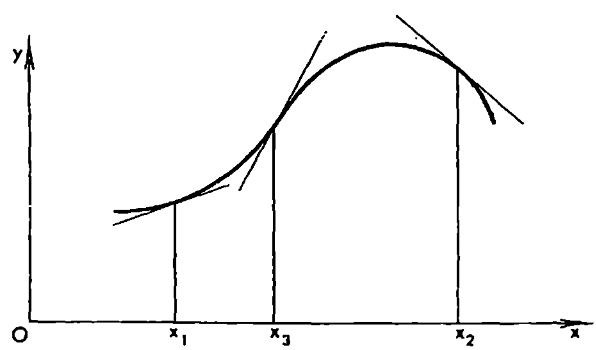

Fig. 5.8

* "Convex upward (downward)" is sometimes equivalently replaced by the expression "concave downward (upward)". — *Tr.*

passes through the value x_0 (in Fig. 5.8 the point x_3 is a point of inflection). In other words, there exists a sufficiently small $\delta > 0$ such that the curve lies on one side of the tangent line at the point x_0 for all $x \in (x_0-\delta, x_0)$ and on the other side for all $x \in (x_0, x_0+\delta)$.

These definitions describe some characteristic elements of the behaviour of a curve in the sense of its disposition relative to the tangent line in a sufficiently small neighbourhood of the point of tangency. But it should be stressed that these definitions do not exhaust all the possible configurations of a curve and its tangent line. For instance, let us return to the function

$$f(x) = \begin{cases} 0 & \text{for } x = 0 \\ x^2 \sin \dfrac{1}{x} & \text{for } x \neq 0 \end{cases}$$

The x-axis intersects the graph of this function and is tangent to it at the point $x = 0$ but $x = 0$ is not a point of inflection.

Theorem 1. *If a function f has the continuous second derivative f'' at a point x_0 and $f''(x_0) > 0$ (< 0) then the curve $y = f(x)$ is convex downward (upward) at x_0.*

Proof. Let us represent f by Taylor's formula at the point $x = x_0$:

$$f(x) = f(x_0)+f'(x_0)(x-x_0)-R(x)$$

$$R(x) = \frac{(x-x_0)^2}{2} f''(x_0+\theta(x-x_0)) \qquad (0 < \theta < 1)$$

The value of the remainder $R(x)$ at a point x is equal to the elevation of the curve f over its tangent drawn through the point $(x_0, f(x_0))$. By the continuity of f'', we conclude that if $f''(x_0) > 0$ then also $f''(x_0+\theta(x-x_0)) > 0$ for all x belonging to a sufficiently small neighbourhood of the point x_0 and therefore we obviously have $R(x) > 0$ for any value of x different from x_0 and belonging to this neighbourhood.

The case $f''(x_0) < 0$ is considered similarly.

Theorem 2. *If a function f is such that the derivative f''' is continuous at x_0 and $f''(x_0) = 0$ while $f'''(x_0) \not\equiv 0$ then the curve $y = f(x)$ has a point of inflection for $x = x_0$.*

Proof. In this case we have

$$f(x) = f(x_0)+f'(x_0)(x-x_0)+R(x)$$

$$R(x) = \frac{(x-x_0)^3}{3!} f'''(x_0+\theta(x-x_0))$$

By the continuity of f''' at x_0 and by the fact that $f'''(x_0) \neq 0$, we conclude that $f'''(x_0+\theta(x-x_0))$ retains sign in a neighbourhood of the point x_0 (i.e. the sign is the same to the right and to the left of the point x_0 within this neighbourhood). On the other hand, the factor $(x-x_0)^3$ changes sign as x passes through x_0 and, together with it, quantity $R(x)$ (equal to the elevation of the point of the curve over the tangent at x_0) also changes sign. This completes the proof of the theorem.

Now we state a more general theorem.

Theorem 3. *Let a function f possess the following properties:*

$$f''(x_0) = \ldots = f^{(k)}(x_0) = 0$$

$f^{(k+1)}(x)$ is continuous at x_0 and $f^{(k+1)}(x_0) \neq 0$.

Then if k is an odd number the curve $y = f(x)$ is convex upward or downward at x_0 depending on whether $f^{(k+1)}(x_0) < 0$ or $f^{(k+1)}(x_0) > 0$; if k is an even number then x_0 is a point of inflection of the curve.

If, in addition to the enumerated requirements, the function satisfies the condition

$$f'(x_0) = 0 \tag{1}$$

then if k is an odd number the function f attains a maximum or a minimum at the point x_0 depending on whether $f^{(k+1)}(x_0) < 0$ or $f^{(k+1)}(x_0) > 0$.

Proof. Here we use the expansion by Taylor's formula

$$f(x) = f(x_0) + (x - x_0) f'(x_0) + \frac{(x-x_0)^{k+1}}{(k+1)!} \cdot f^{(k+1)}(x_0 + \theta(x - x_0))$$

which is valid under the conditions stated. Under the additional condition (1) this expansion turns into

$$f(x) = f(x_0) + \frac{(x - x_0)^{k+1}}{(k+1)!} f^{(k+1)}(x_0 + \theta(x - x_0))$$

The course of the proof is obvious.

In conclusion we note that when a function $y = f(x)$ has an infinite derivative equal to $+\infty$ or to $-\infty$ at a point x_0 (see Fig. 5.1*c* and *d* on page 128 and the corresponding remarks) we also say that the graph of the function has a point of inflection at x_0.

§ 5.13. Convexity of a Curve on an Interval

By definition, a curve $y = f(x)$ is said to be *convex upward (downward)* on an interval $[a, b]$ if any arc of this curve with end points having abscissas x_1 and x_2 ($a \leq x_1 < x_2 \leq b$) lies not lower (not higher) than the chord subtending it (see Figs. 5.9 and 5.10).

Theorem 1. *Let a function f be continuous on $[a, b]$ and have the second derivative f'' for $x \in (a, b)$.*

Then for the curve $y = f(x)$ to be convex upward (downward) on $[a, b]$ it is necessary and sufficient that the inequality $f''(x) \leq 0$ $(f''(x) \geq 0)$ hold for all $x \in (a, b)$.

Proof. Suppose that the curve is convex upward on $[a, b]$. Then, for any x and $h > 0$ such that $x, x+2h \in [a, b]$, we have $f(x+h) \geq (f(x)+f(x+2h))/2$ whence $f(x+h)-f(x) \geq f(x+2h)-f(x+h)$.

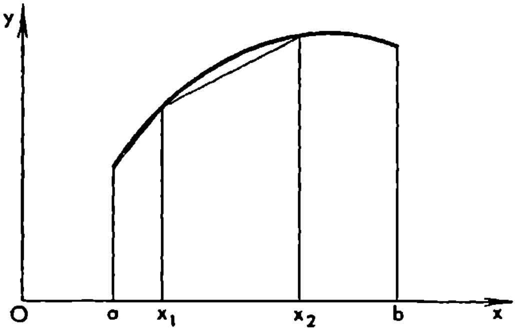

Fig. 5.9

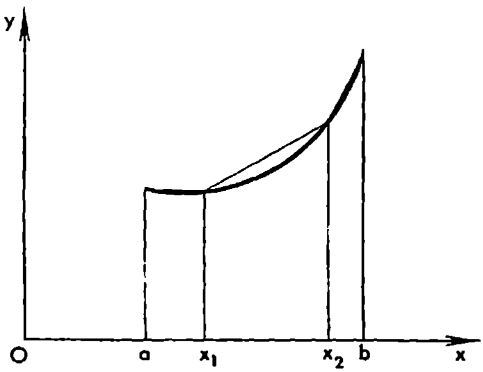

Fig. 5.10

Now, if x_1 and x_2 are two arbitrary points of the interval (a, b) we can write

$$f(x_1+h)-f(x_1) \geqslant f(x_1+2h)-f(x_1+h) \geqslant \ldots \geqslant f(x_2)-f(x_2-h)$$

where $h = (x_2-x_1)/n$.

Hence, $\big(f(x_1+h)-f(x_1)\big)/h \geqslant \big(f(x_2-h)-f(x_2)\big)/(-h)$, and passing to the limit for $h \to 0$ we obtain the inequality

$$f'(x_1) \geqslant f'(x_2)$$

showing that the derivative f' does not increase on the interval (a, b). This means that $f''(x) \leqslant 0$ on (a, b).

Conversely, let $f''(x) \leqslant 0$ and $a < x_1 < x_2 < b$. We have to show that the function $F(x) = f(x)-f(x_1)-m(x-x_1)$ where $m = \big(f(x_2)-f(x_1)\big)/ /(x_2-x_1)$ satisfies the inequality $F(x) \geqslant 0$ on $[x_1, x_2]$.

Suppose the contrary. Then $\min\limits_{x_1 \leqslant x \leqslant x_2} F(x) = F(x_0) < 0$ and $x_1 < x_0 < x_2$. Therefore $F'(x_0) = 0$.

Applying Taylor's formula we receive

$$0 = F(x_2) = F(x_0)+\frac{(x_2-x_0)^2}{2!}F''(x_0+\theta(x_2-x_0)) =$$
$$= F(x_0)+\frac{(x_2-x_0)^2}{2!}f''(x_0+\theta(x_2-x_0))$$

By the hypothesis, the first term in the rightmost member of this chain of equalities is negative while the second term is nonpositive; consequently the rightmost member is less than zero, and we have thus arrived at a contradiction.

In the case $f''(x) \geqslant 0$ the course of the proof is quite analogous.

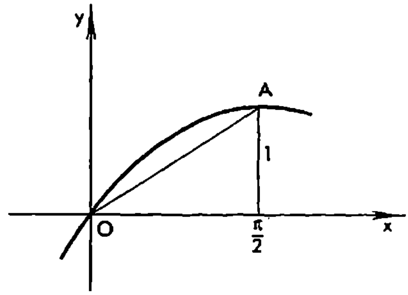

Fig. 5.11

Example. The function $y = \sin x$ has the continuous first derivative $(\sin x)' = \cos x$ and the second derivative $(\sin x)'' = -\sin x \leqslant 0$ on $[0, \pi/2]$. Therefore the chord OA (see Fig. 5.11) subtending the arc of the curve $y = \sin x$ on $[0, \pi/2]$ is below the sinusoid. The equation of the chord being $y = (2/\pi)x$, we get the inequality

$$\frac{2}{\pi}x \leqslant \sin x, \qquad 0 \leqslant x \leqslant \frac{\pi}{2}$$

which is often used in mathematical analysis.

§ 5.14. Evaluation of Indeterminate Forms

The laws of the mean for derivatives and Taylor's formula we have established provide some powerful techniques of differential calculus which make it possible to compute automatically many limits, namely those which lead, when ordinary rules are applied, to the *indeterminate forms* of the type $\frac{0}{0}$, $\frac{\infty}{\infty}$, $\infty-\infty$, $0\cdot\infty$, 0^0, ∞^0 and 1^∞.

The Case 0/0. Let it be necessary to find the limit $\lim\limits_{x\to a}\frac{f(x)}{\varphi(x)}$ on condition that $\lim\limits_{x\to a} f(x) = 0$, $\lim\limits_{x\to a}\varphi(x) = 0$ and $\varphi(x) \neq 0$ in a neighbourhood of a.

Let a be a finite number and let the principal terms of the expansions of f and φ in powers of $x-a$ be known:

$$f(x) = \alpha_p(x-a)^p+o((x-a)^p) \qquad (x \to a), \alpha_p \neq 0$$

and

$$\varphi(x) = \beta_q(x-a)^q+o((x-a)^q) \qquad (x \to a), \beta_q \neq 0$$

Then (see § 4.10, formula (10)) we have

$$\lim_{x\to a}\frac{f(x)}{\varphi(x)} = \lim_{x\to a}\frac{\alpha_p(x-a)^p}{\beta_q(x-a)^q} = \begin{cases} \dfrac{\alpha_p}{\beta_p} & \text{for } p = q \\ 0 & \text{for } p > q \\ \infty & \text{for } p < q \end{cases}$$

Examples

1. $$\lim_{x\to 0}\frac{x-\tan x}{x-\sin x} = \lim_{x\to 0}\frac{x-\left(x+\frac{1}{3}x^3+o(x^3)\right)}{x-\left(x-\frac{x^3}{3!}+o(x^3)\right)} = \lim_{x\to 0}\frac{-\frac{1}{3}+o(1)}{\frac{1}{6}+o(1)} = -2.$$

2. $$\lim_{x\to 0}\frac{1-\cos x}{\sin x} = \lim_{x\to 0}\frac{1-\left(1-\frac{x^2}{2}+o(x^2)\right)}{x+o(x)} = \lim_{x\to 0}\frac{\frac{x^2}{2}+o(x^2)}{x+o(x)} = \lim_{x\to 0}\frac{\frac{x^2}{2}}{x} = 0.$$

3. $$\lim_{x\to 0}\frac{e^x-1-x}{\sqrt{1-x}-\cos\sqrt{x}} = \lim_{x\to 0}\frac{1+x+\frac{x^2}{2}+o(x^2)-1-x}{\left(1-\frac{x}{2}-\frac{x^2}{8}+o(x^2)\right)-\left(1-\frac{x}{2}+\frac{x^2}{4!}+o(x^2)\right)} =$$

$$= \lim_{x\to 0}\frac{\frac{x^2}{2}+o(x^2)}{-\left(\frac{1}{8}+\frac{1}{24}\right)x^2+o(x^2)} = -3.$$

It may also turn out that the functions f and φ have no derivatives at the point a or that for some reason it is difficult or inexpedient to compute the derivatives at that point. In such cases the following general theorem whose proof is based on the application of Cauchy's mean-value theorem may be of use:

Theorem 1. *Let functions f and φ be continuous and have the derivatives in a neighbourhood of a point a (where a is a finite number or ∞) except possibly at the point a itself; also let φ and φ' be different from zero in this neighbourhood and*

$$\lim_{x\to 0} f(x) = \lim_{x\to a}\varphi(x) = 0$$

Then if the limit

$$\lim_{x\to a}\frac{f'(x)}{\varphi'(x)} = A \tag{1}$$

(finite or infinite) exists the limit of the ratio $f(x)/\varphi(x)$ as $x \to a$ also exists and is equal to the former:

$$\lim_{x\to a}\frac{f(x)}{\varphi(x)} = \lim_{x\to a}\frac{f'(x)}{\varphi'(x)} = A \tag{2}$$

In particular, this assertion also applies to right-hand and left-hand limits; in these cases a neighbourhood of a should be understood as a one-sided (respectively, right-hand or left-hand) neighbourhood.

Proof. Let a be a (finite) number. Putting $f(a) = 0 = \lim_{x\to a} f(x)$ and $\varphi(a) = 0 = \lim_{x\to a} \varphi(x)$ we obtain the (new) functions f and φ defined and continuous throughout the neighbourhood in question including the point a itself. This property together with the properties enumerated in the statement of the theorem allow us to apply Cauchy's theorem to the functions f and φ. Thus, for any point x belonging to the neighbourhood there is a point $\xi = a+\theta(x-a)$ $(0 < \theta < 1)$ lying between a and x such that

$$\frac{f(x)}{\varphi(x)} = \frac{f(x)-f(a)}{\varphi(x)-\varphi(a)} = \frac{f'(\xi)}{\varphi'(\xi)} \tag{3}$$

If limit (1) exists then obviously the limit

$$\lim_{x\to a} \frac{f'(\xi)}{\varphi'(\xi)} = A$$

also exists and hence so does limit (2).

We see that the existence of the second limit in (2) implies the existence of the first limit in (2) equal to the former. But the converse is not true.

Example 4. Since $\sin x \sim x$ $(x \to 0)$ we have

$$\lim_{x\to 0} \frac{x^2 \sin(1/x)}{\sin x} = \lim_{x\to 0} x \sin\frac{1}{x} = 0$$

On the other hand, the corresponding ratio of the derivatives is

$$\frac{2x\sin(1/x)-\cos(1/x)}{\cos x} = 2x\frac{1}{\cos x}\sin\frac{1}{x} - \frac{\cos(1/x)}{\cos x} \tag{4}$$

It obviously has no limit as $x \to 0$. This follows from the fact that the first term on the right-hand side tends to zero while the second term has no limit. But this does not contradict the fact that when the function $\xi = \xi(x)$ appearing in Cauchy's theorem is substituted for x into (4) the resultant expression has a limit for $x \to 0$.

Now we proceed to the case $a = \infty$ (or $a = +\infty$ or $a = -\infty$). Let us make the substitution $x = 1/u$. This results in the functions $F(u) = f(1/u)$ and $\Phi(u) = \varphi(1/u)$ of the argument u. They are continuous in a neighbourhood of the point $u = 0$ (in the cases $a = +\infty$ and $a = -\infty$ they are respectively continuous in a right-hand and a left-hand neighbourhood of the point $u = 0$), have the derivatives with respect to u in that neighbourhood and both Φ and Φ' are different from zero in it.

We also have $\lim_{u\to 0} F(u) = \lim_{x\to\infty} f(x) = 0$ and $\lim_{u\to 0} \Phi(u) = \lim_{x\to\infty} \varphi(x) = 0$. If the limit $\lim_{x\to\infty} \frac{f'(x)}{\varphi'(x)}$ exists then obviously the limit of the ratio $F'(u)/\Phi'(u)$

as $u \to 0$ also exists and is equal to the former:

$$\lim_{u\to 0} \frac{F'(u)}{\Phi'(u)} = \lim_{u\to 0} \frac{f'(1/u)(-1/u^2)}{\varphi'(1/u)(-1/u^2)} = \lim_{x\to\infty} \frac{f'(x)}{\varphi'(x)}$$

Hence, by what was proved above (for a finite a), we conclude that

$$\lim_{x\to\infty} \frac{f(x)}{\varphi(x)} = \lim_{u\to 0} \frac{F(u)}{\Phi(u)} = \lim_{u\to 0} \frac{F'(u)}{\Phi'(u)} = \lim_{x\to\infty} \frac{f'(x)}{\varphi'(x)}$$

The theorem has been proved.

The Case ∞/∞. There holds the following theorem.

Theorem 2. *Let functions $f(x)$ and $\varphi(x)$ be continuous and possess the derivatives f' and φ' in a neighbourhood (in particular, in a right-hand or a left-hand neighbourhood) of a point a (finite or infinite) except at the point a itself. Also let φ and φ' be different from zero in that neighbourhood and*

$$\lim_{x\to a} f(x) = \lim_{x\to a} \varphi(x) = \infty \qquad (+\infty \text{ or } -\infty) \tag{5}$$

Then if the limit $\lim\limits_{x\to a} \dfrac{f'(x)}{\varphi'(x)} = A$ exists the limit of the ratio $f(x)/\varphi(x)$ as $x \to a$ also exists and is equal to the former:

$$\lim_{x\to a} \frac{f(x)}{\varphi(x)} = \lim_{x\to a} \frac{f'(x)}{\varphi'(x)} = A \tag{6}$$

The proof of the theorem is based on the following lemma.

Lemma. *Let $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a} \varphi(x) = \infty$. Then there is a neighbourhood of a in which there exists a function $x' = \lambda(x)$ such that $\lim\limits_{x\to a} x' = \lim\limits_{x\to a} \lambda(x) = a$ and, simultaneously,*

$$\lim_{x\to a} \frac{f(x')}{f(x)} = 0 \quad \text{and} \quad \lim_{x\to a} \frac{\varphi(x')}{\varphi(x)} = 0 \tag{7}$$

that is

$$f(x) \approx f(x) - f(x') \quad \text{and} \quad \varphi(x) \approx \varphi(x) - \varphi(x') \qquad (x \to a)$$

Proof. In the course of the proof we shall suppose, for the sake of definiteness, that a is a finite number and that x tends to a so that $x < a$ all the time. The other possible cases are treated similarly.

Let us choose $x_1 < a$ such that $|f(x)| > 0$ and $|\varphi(x)| > 0$ for $x \geqslant x_1$ and also take x_2 $(x_1 < x_2 < a)$ such that

$$\left|\frac{f(x)}{f(x_1)}\right| > 1, \quad \left|\frac{\varphi(x)}{\varphi(x_1)}\right| > 1 \qquad (x_2 < x < a) \text{ and } a - x_2 < \frac{1}{2}$$

Next we choose x_3 $(x_2 < x_3 < a)$ so that

$$\left|\frac{f(x)}{f(x_2)}\right| > 2, \quad \left|\frac{\varphi(x)}{\varphi(x_2)}\right| > 2 \qquad (x_3 < x < a) \text{ and } a - x_3 < \frac{1}{3}$$

Proceeding in this way we obtain an increasing sequence $\{x_n\}$ convergent to a such that

$$\left|\frac{f(x)}{f(x_n)}\right| > n, \quad \left|\frac{\varphi(x)}{\varphi(x_n)}\right| > n \qquad (x_{n+1} < x < a),$$
$$a - x_{n+1} < \frac{1}{n+1} \qquad (n = 1, 2, \ldots)$$

Now we set an arbitrary value $x \in (x_2, a)$, choose a (uniquely determined) natural number $n = n(x)$ corresponding to it such that $x_{n+1} < x \leqslant x_{n+2}$ and then put

$$x' = \lambda(x) = x_n$$

Obviously, $x' \to a$, $n \to \infty$ for $x \to a$, $|f(x)/f(x')| > n$ and $|\varphi(x)/\varphi(x')| > n$. It now follows that (7) takes place.

The lemma has been proved.

The Proof of the Theorem. We have to compute the limit $\lim\limits_{x \to a} \frac{f(x)}{\varphi(x)}$. As is already known,

$$f(x) \sim f(x) - f(x') \quad \text{and} \quad \varphi(x) \sim \varphi(x) - \varphi(x') \text{ as } x \to a$$

Hence (see § 4.10, (8)) we have

$$\lim_{x \to a} \frac{f(x)}{\varphi(x)} = \lim_{x \to a} \frac{f(x) - f(x')}{\varphi(x) - \varphi(x')}$$

By Cauchy's mean-value theorem,

$$\frac{f(x) - f(x')}{\varphi(x) - \varphi(x')} = \frac{f'(\xi)}{\varphi'(\xi)}, \qquad \xi = x' + \theta(x - x') \qquad (0 < \theta < 1)$$

Now we see that if the limit $\lim\limits_{x \to a} \frac{f'(x)}{\varphi'(x)}$ (finite or infinite) exists the limit $\lim\limits_{x \to a} \frac{f'(\xi)}{\varphi'(\xi)}$ also exists and is equal to the former limit: $\lim\limits_{x \to a} \frac{f'(\xi)}{\varphi'(\xi)} = \lim\limits_{x \to 0} \frac{f'(x)}{\varphi'(x)}$.

Consequently,

$$\lim_{x \to a} \frac{f(x)}{\varphi(x)} = \lim_{x \to a} \frac{f'(x)}{\varphi'(x)}$$

which completes the proof of the theorem.

Again, the existence of the second limit in equality (6) implies the existence of the first limit equal to the former but, as the following example shows, the converse is not true: the limit

$$\lim_{x \to \infty} \frac{x - \sin x}{x} = \lim_{x \to \infty} \left(1 - \frac{\sin x}{x}\right) = 1$$

exists while the limit of the ratio $(1 - \cos x)/1$ of the derivatives does not exist as $x \to \infty$.

The rule stated in Theorems 1 and 2 according to which the computation of the limit of the ratio of functions can be reduced, under certain conditions, to the computation of the limit of the ratio of their derivatives is called *L'Hospital's rule* after G.F.A. L'Hospital* who stated this rule for some simpler cases; this rule was in fact known to Johann Bernoulli** before L'Hospital.

Other Types of Indeterminate Forms. It now remains to consider the other types of indeterminate forms which turn out to be reducible to the two types treated above.

If $f \to \infty$ and $\varphi \to \infty$ we write $f-\varphi = \left(\frac{1}{\varphi} - \frac{1}{f}\right) : \frac{1}{f\varphi}$ and thus obtain an indeterminate form of the type $0/0$.

If $f \to 0$ and $\varphi \to \infty$ we write $f\varphi = \frac{f}{1/\varphi}$, which leads to a form $0/0$.

The expressions of the form u^v which lead to the indeterminate forms 0^0, ∞^0 and 1^∞ are found by taking logarithms; this results in indeterminate forms of the type $0 \cdot \infty$.

For example, we have $\lim\limits_{x\to a} u^v = e^{\lim\limits_{x\to a} v \ln u}$ if the limit of the exponent in the right member is finite. If the latter limit is equal to $+\infty$ or $-\infty$ the limit of the left member is respectively equal to $+\infty$ or 0.

Examples

5. $\lim\limits_{x\to\infty} \frac{\ln x}{x} = \lim\limits_{x\to\infty} \frac{\frac{1}{x}}{1} = 0.$

6. $\lim\limits_{\substack{x\to 0\\ x>0}} x \ln x = \lim\limits_{\substack{x\to 0\\ x>0}} \frac{\ln x}{\frac{1}{x}} = \lim\limits_{\substack{x\to 0\\ x>0}} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = 0.$

7. $\lim\limits_{x\to+\infty} x^k e^{-x} = \lim\limits_{x\to+\infty} \frac{x^k}{e^x} = \lim\limits_{x\to+\infty} \frac{kx^{k-1}}{e^x} = \ldots = \lim\limits_{x\to+\infty} \frac{k!}{e^x} = 0$

$(k = 1, 2, \ldots).$

§ 5.15. Piecewise Continuous and Piecewise Smooth Functions

We say that a function f is *smooth* on a closed interval $[a, b]$ if it possesses a continuous derivative in this interval.

As usual, it is meant in this definition that at the points a and b the function f has, respectively, the right-hand and the left-hand derivatives. A smooth function on $[a, b]$ is automatically continuous on $[a, b]$ since its derivative exists throughout $[a, b]$.

---

* G.F. A. L'Hospital (1661-1704), a French mathematician.

** Johann Bernoulli (1667-1748), a Swiss mathematician.

An equivalent form of this definition reads: *a function f is smooth on $[a, b]$ if it is continuous on $[a, b]$ and has the continuous derivative $f'(x)$ in the open interval (a, b) such that the limits*

$$f'(a+0) = A \quad \text{and} \quad f'(b-0) = B \tag{1}$$

exist.

It is clear that the first definition implies the second. Now suppose that f is smooth in the sense of the second definition. Then

$$\frac{f(a+h)-f(a)}{h} = f'(a+\theta h) \to A \ (h > 0, h \to 0, 0 < \theta < 1) \tag{2}$$

and consequently f has the right-hand derivative at the point a equal to $f'(a) = A$. By virtue of the first equality (1), the derivative f' is continuous from the right at that point. The existence and the continuity of the derivative of f at the point b and the equality $f'(b) = B$ are proved similarly. Hence, the function f turns out to be smooth in the sense of the first definition as well.

A function f is said to be *piecewise continuous* on a closed interval $[a, b]$ if it is defined and continuous throughout $[a, b]$ except possibly at a finite number of points x_j $(a \leqslant x_1 < \ldots < x_N \leqslant b)$ at which the right-hand and the left-hand limits of f exist, that is the finite numbers $f(x_j-0)$ and $f(x_j+0)$ exist. Here of course for $x = a$ and $x = b$ only $f(a+0)$ and $f(b-0)$ are respectively meant to have sense.

Thus, a piecewise continuous function f on $[a, b]$ is continuous on each of the intervals (x_j, x_{j+1}). Furthermore, irrespective of whether or not it is defined at the points x_j and x_{j+1} the function can be redefined or defined at these points so that the resultant function is continuous on the chosen closed interval $[x_j, x_{j+1}]$.

Example 1. The function $[x]$ defined as the greatest integer not exceeding x is an example of a piecewise continuous function on any closed interval $[a, b]$. Its graph is shown in Fig. 5.12. The points of discontinuity of the function $[x]$ are the integral values $x = 0, \pm 1, \pm 2, \ldots$ The discontinuities at these points are of the first kind, that is the right-hand and the left-hand limits of the function exist at these points. Let us consider one of the inter-

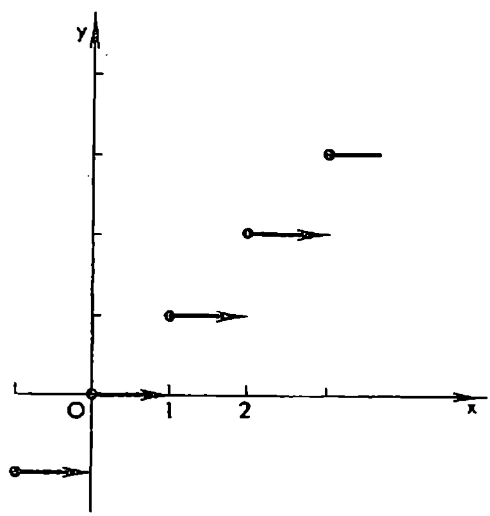

Fig. 5.12

vals of continuity of the function with maximum length 1, for definiteness, the interval (1, 2). The function $[x]$ is continuous on this interval. On the corresponding closed interval [1, 2] this function $f(x) = [x]$ is no longer continuous since $f(2-0) = [2-0] \neq [2] = f(2)$. But if we redefine the function by putting it equal to 1 for $x = 2$ the resultant function is continuous on [1, 2].

We shall say that a function f is *piecewise smooth* on a closed interval $[a, b]$ if it is itself piecewise continuous and if its derivative f' exists and is also piecewise continuous on $[a, b]$. This means that the interval $[a, b]$ can be partitioned with the aid of points of division

$$a = x_0 < x_1 < \ldots < x_N = b \tag{3}$$

so that f is continuous together with its derivative f' on each of the intervals (x_j, x_{j+1}) and, besides, the one-sided limits of both f and f' exist at the end points of these intervals.

Choosing a definite subinterval $[x_j, x_{j+1}]$ we can define (if f is not defined at some of the points of division) or redefine f at the end points $x = x_j$ and $x = x_{j+1}$ so that it assumes, respectively, the values $f(x_j+0)$ and $f(x_{j+1}-0)$ at these points; then, as was shown at the beginning of this section, the (new) function f is smooth on the interval $[x_j, x_{j+1}]$.

For instance, the function $\psi(x) = [x]$ is obviously not only piecewise continuous but also piecewise smooth on any closed interval $[a, b]$ because on the intervals $(m, m+1)$ where m is an integer it is continuous together with its derivative and at the end points of these intervals there exist the corresponding one-sided limits of ψ and ψ'. If we replace the value $\psi(m+1) = m+1$ by the new value $\psi(m+1) = m$, the (new) function ψ turns out to be constant and, consequently, smooth on the closed interval $[m, m+1]$ (see Fig. 5.12).

An important special case of a piecewise smooth function is a *continuous piecewise smooth function f on a closed interval* $[a, b]$. Characteristic features of such a function are:

(i) f is continuous on $[a, b]$; (ii) there is a (finite) partition (3) of the interval $[a, b]$ such that f is a smooth function on each of the subintervals $[x_j, x_{j+1}]$.

Example 2. The function $|x|$ is not smooth on $[-1, 1]$ because it has no derivative at the point $x = 0$. On the other hand, $|x|$ is a continuous and piecewise smooth function on $[-1, +1]$ since it is continuous throughout $[-1, +1]$ and has the continuous derivatives -1 and $+1$ in, respectively, the open intervals $(-1, 0)$ and $(0, 1)$, and the derivatives themselves have the corresponding one-sided limits at the end points of these intervals.

It is obvious that

$$|x|' = \operatorname{sgn} x = \begin{cases} +1 & \text{for } x > 0 \\ -1 & \text{for } x < 0 \end{cases}$$

The function $\operatorname{sgn} x$ is considered as not being defined at the point $x = 0$.

Exercises

1. Show that the function whose graph is depicted in Fig. 5.6 is piecewise smooth. Also show that the functions with the graphs presented in Fig. 5.1, *c-f*, are not piecewise smooth.

Hint. Take into account that these functions have infinite (at least right-hand and left-hand) derivatives at the point x_0.

2. Show that the function

$$f(x) = \begin{cases} 0 & \text{for } x = 0 \\ x^2 \sin \dfrac{1}{x} & \text{for } 0 < |x| \leq 1 \end{cases}$$

is not smooth on the interval $[-1, +1]$ in spite of the fact that its derivative exists at all the points of the interval.

3. Prove that if a function f is continuous but not smooth on a closed interval $[a, b]$ and if at the same time it is smooth on each of the subintervals $[a, c]$ and $[c, b]$ then f has no derivative at the point c although its right-hand and left-hand derivatives at this point exist.

4. Show that if f is continuous and possesses the derivative f' at all the points of $[a, b]$ the derivative cannot have discontinuities of the first kind (in Exercise 2 the derivative of f, although it exists everywhere on $[-1, +1]$, has a discontinuity of the second kind at $x = 0$).

CHAPTER 6

n-dimensional Space. Geometrical Properties of Curves

§ 6.1. n-dimensional Space. Linear Space

An arbitrary ordered system $x = (x_1, \ldots, x_n)$ of n real (complex) numbers x_j (a *number n-tuple*) is called a *vector* or a *point* or an *element of the real (complex) n-dimensional space* R_n. Hence, R_n (also called the *n-space* or the *n-dimensional Cartesian space*) is the set of all such x's.

For the vectors (points) $x, y \in R_n$ we shall define the operations of addition and subtraction and also the operation of multiplication of the vectors by real (complex) numbers according to the following rule: if $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ are arbitrary vectors and α, β are some real (complex*) numbers then

$$\alpha x \pm \beta y = (\alpha x_1 \pm \beta y_1, \ldots, \alpha_n x_n \pm \beta_n y_n)$$

The vector (point) $\mathbf{0} = (0, \ldots, 0)$ is called the *zero vector* (the *origin*) of R_n. Obviously, $x + \mathbf{0} = x$ for any $x \in R_n$. We also put $(-1)x = -x$; then, apparently, $x - y = x + (-y)$.

In mechanics and geometry an element $x = (x_1, \ldots, x_n)$ of the space R_n is spoken of as a vector with point of application at the origin and terminus at the point $x = (x_1, \ldots, x_n)$. In the cases of dimensions 2 and 3 this terminology has a visual geometrical sense.

The properties enumerated below are checked directly on the basis of the above rule:

(i) $x + y = y + x$;

(ii) $(x + y) + z = x + (y + z)$;

(iii) $x + y = x + z$ implies $y = z$;

(iv) $\alpha x + \alpha y = \alpha(x + y)$;

(v) $\alpha x + \beta x = (\alpha + \beta)x$;

(vi) $\alpha(\beta x) = (\alpha\beta)x$;

(vii) $1 \cdot x = x$

(here $x, y, z \in R_n$ are arbitrary vectors and α, β are any real or complex numbers).

---

* On complex numbers see § 8.2.

An abstract set E consisting of elements $x, y, z \ldots$ of any nature is called a *real (complex) linear space* if there is a law according to which, for any two elements $x, y \in E$ there is defined an element $x+y \in E$ called the *sum of* x *and* y and if, for any real or complex number α and any element $x \in E$, there is defined an element $\alpha x \in E$ (called the *product of* x *by* α) so that the enumerated properties (axioms) (i)-(vii) hold.

What has been said shows that R_n *can be regarded as an example of a linear space*, but there are also many other examples of this kind. For instance, the set of all (infinite) sequences $x = \{x_1, x_2, \ldots\}$ of real or complex numbers becomes a linear space if we put

$$\alpha x+\beta y = \{\alpha x_1+\beta y_1, \alpha x_2+\beta y_2 \ldots\} \quad \text{and} \quad y = \{y_1, y_2 \ldots\}$$

The subset of this set consisting of all sequences convergent to finite limits is also a linear space if the operations of addition of sequences and their multiplication by numbers are defined in the same manner as above. Further, the set C of all (real or complex) functions f defined and continuous on a closed interval $[a, b]$ is another example of a linear space if, as usual, the operations of addition of functions and their multiplication by numbers are performed according to the ordinary rule

$$\alpha f+\beta\varphi = \alpha f(x)+\beta\varphi(x) \qquad (f, \varphi \in C)$$

Finally, the set of all polynomials $P_n(x) = \sum_0^n a_k x^k$ of degree not higher than n is also a linear space if the addition of polynomials and their multiplication by numbers are understood in the ordinary algebraic sense.

The subtraction of elements of a linear space and the zero element are not mentioned explicitly in axioms (i)-(vii) but these notions do in fact arise on the basis of these axioms. If we put $\theta_x = 0\cdot x$ then $x+\theta_x = x+0\cdot x = 1\cdot x = x$. Introducing similarly the element $\theta_y = 0\cdot y$ we obtain the relation $y+\theta_y = y$. For θ_x and θ_y we then have

$$x+y+\theta_y = x+(y+\theta_y) = x+y,$$
$$x+y+\theta_x = x+(\theta_x+y) = (x+\theta_x)+y = x+y$$

whence, by Axiom (iii) it follows that $\theta_x = \theta_y = \theta$ for any $x, y \in E$. Consequently, θ is the zero element of E since the relation $x+\theta = x$ is fulfilled for any $x \in E$.

Next we put $-x = (-1)x$; then $x+(-x) = (1-1)x = 0x = \theta$, and the subtraction of any two elements $x, y \in E$, that is their difference $x-y$, is defined with the aid of the equality $x-y = x+(-y)$. The latter operation is inverse to the operation of addition:

$$(x-y)+y = x+(-y)+y = x+[(-y)+y] = x+\theta = x$$

§ 6.2. n-dimensional Euclidean Space. Spaces with Scalar Product

Let R_n be a real or complex n-dimensional linear space of n-tuples. Taking its two arbitrary points (vectors)

$$x = (x_1, \ldots, x_n) \quad \text{and} \quad y = (y_1, \ldots, y_n)$$

we associate with them the number

$$(x, y) = \sum_{j=1}^{n} x_j \bar{y}_j \tag{1}$$

which is called the *scalar product* of the vectors x and y.

Here the bar over y_j designates the operation of passing to the complex conjugate of y_j. In the case of real space the numbers y_j are real and $\bar{y}_j = y_j$.

The scalar product obviously possesses the following properties:

(i) $(x, y) = \overline{(y, x)}$;

(ii) (x, y) is a *linear form in* x, that is, given any vectors x, y, z and numbers α and β, there holds

$$(\alpha x + \beta y, z) = \alpha(x, z) + \beta(y, z)$$

By Property (i), it follows that

$$(x, \alpha y + \beta z) = \bar{\alpha}(x, y) + \bar{\beta}(x, z)$$

(iii) $(x, x) \geqslant 0$ for any vector x; equality $(x, x) = 0$ implies $x = \theta$ because it means that $x_j = 0, j = 1, \ldots, n$.

Now we state the following definition: a (real or complex) linear space E to whose any two elements (also called *generalized vectors*) x and y there corresponds, in accordance with a definite law, a number (x, y) so that conditions (i)-(iii) hold is called a *linear space with scalar product.*

If E is a real linear space then, evidently, the bars designating the complex conjugates in conditions (i)-(iii) can be omitted.

Now we can say that the *n-dimensional space R_n for which relation* (1) *is introduced is a space with scalar product.*

Mathematics also deals with many other linear spaces with scalar product. Some of them will be studied in our course (see Chapter 14).

Let x and y be two elements of a linear space E with scalar product and let λ be an arbitrary number which, depending on whether E is a real or complex space, can be real or complex.

By properties (i)-(iii), we then have

$$0 \leqslant (x + \lambda y, x + \lambda y) = (x, x) + \bar{\lambda}(x, y) + \lambda\overline{(x, y)} + |\lambda|^2 (y, y) \tag{2}$$

If $(y, y) \neq 0$ we can form the number

$$\lambda = -\frac{(x, y)}{(y, y)} \tag{3}$$

and substitute it into (2). Since we have $a\bar{a} = |a|^2$ for any complex number a the substitution results in

$$\bar{\lambda}(x, y) = \lambda\overline{(x, y)} = -\frac{|(x, y)|^2}{(y, y)} = -|\lambda|^2 (y, y)$$

That is

$$0 \leqslant (x, x) - \frac{|(x, y)|^2}{(y, y)} \tag{4}$$

Relation (4) leads to *Bunyakovsky's* inequality*

$$|(x, y)| \leqslant (x, x)^{1/2}(y, y)^{1/2} \tag{5}$$

It continues to hold for the case $(y, y) = 0$, that is when $y = \theta$ is the zero element, since $(x, \theta) = (x, 0\cdot\theta) = 0(x, \theta) = 0$.

Taking two arbitrary elements $x, y \in E$ we can write

$$(x+y, x+y) = (x, x)+(x, y)+(y, x)+(y, y) \leqslant (x, x)+2|(x, y)|+(y, y) \leqslant$$
$$\leqslant (x, x)+2\sqrt{(x, x)}\sqrt{(y, y)}+(y, y) = \left(\sqrt{(x, x)}+\sqrt{(y, y)}\right)^2 \tag{6}$$

and thus arrive at another important inequality:

$$(x+y, x+y)^{1/2} \leqslant (x, x)^{1/2}+(y, y)^{1/2} \tag{7}$$

For any pair of elements of the form $x = \alpha y, y$ where α is a number and y is arbitrary or $y = \theta$ and x is arbitrary, inequality (5) turns into the exact equality

$$|(x, y)| = (x, x)^{1/2}(y, y)^{1/2} \tag{8}$$

Conversely, if equality (8) is fulfilled then either $y = \theta$ or relation (2) holds with the equality sign and with λ defined by formula (3); in the latter case Property (iii) of the scalar product implies $x+\lambda y = \theta$.

It is evident that inequality (7) becomes an exact equality if either $y = \theta$ or $x = \alpha y$ where α is a nonnegative number.

Conversely, if (7) turns into equality all the relations in (6) turn into equalities whence it follows, in particular, that (8) holds. Consequently, either $y = \theta$ or $x = \alpha y$ where α is a certain number. Next, putting $x = \alpha y$ in equality (7) and taking y such that $(y, y) > 0$ we obtain, after cancelling by (y, y), the equality $|1+\alpha| = 1+|\alpha|$ whence $\alpha \geqslant 0$.

The arithmetic square root of (x, x) is termed the *norm* of x and is denoted $||x||$: $||x|| = (x, x)^{1/2}$ (see the next section).

The n-dimensional space R_n in which scalar product (1) and, together with it, the norm $||x|| = \left(\sum_1^n |x_j|^2\right)^{1/2}$ are introduced for the element $x=(x_1, \ldots, x_n)$ is called the *n-dimensional Euclidean space***. The norm $||x||$ of an element x of the n-dimensional Euclidean space will also be denoted as $|x|$. For $n = 3$ the norm $|x| = (x_1^2+x_2^2+x_3^2)^{1/2}$ of a vector $x = (x_1, x_2, x_3)$ coincides with its length (the same is of course true for $n = 2$ and $n = 1$).

---

* V. Ya. Bunyakovsky (1804-1889), a distinguished Russian mathematician who deduced this inequality in 1859. For number sequences this important inequality was established in 1821 by A. Cauchy who used a terminology different from that of the theory of linear spaces (see below). For some other cases (5) was derived in 1884 by H. Schwarz (1843-1921), a German mathematician. That is why (5) is sometimes referred to as the *Cauchy-Bunyakovsky-Schwarz inequality*.—*Tr.*

** An arbitrary linear space E with scalar product is also called a Euclidean space (in the general case it can be infinite-dimensional).—*Tr.*

For the elements of the n-dimensional Euclidean space inequalities (5) and (7) turn, respectively, into the corresponding inequalities

$$\left|\sum_1^n x_j\bar{y}_j\right| \leqslant \left(\sum_1^n |x_j|^2\right)^{1/2} \left(\sum_1^n |y_j|^2\right)^{1/2} \tag{9}$$

and

$$\left(\sum_1^n |x_j+y_j|^2\right)^{1/2} \leqslant \left(\sum_1^n |x_j|^2\right)^{1/2} + \left(\sum_1^n |y|^2\right)^{1/2} \tag{10}$$

It follows from (9) that for number n-tuples $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$

$$\sum_1^n |x_jy_j| \leqslant \left(\sum_1^n |x_j|^2\right)^{1/2} \left(\sum_1^n |y_j|\right)^{1/2} \tag{11}$$

because we can apply inequality (9) to the nonnegative numbers $|x_j|$ and $|y_j| = |\bar{y}_j|$.

We also mention the inequalities

$$\frac{1}{n^{1/2}}\sum_1^n |x_j| \leqslant \left(\sum_{j=1}^n |x_j|^2\right)^{1/2} \leqslant \sum_1^n |x_j| \tag{12}$$

The first of them follows from (11) if we put $y_j = 1$ $(j = 1, \ldots, n)$ and the second can be checked directly by squaring its both members.

Relation (9) is called *Cauchy's inequality* and (10) is a special case of a relation known as *Minkowski's inequality* (sec § 14.2, formula (12)).

§ 6.3. Normed Linear Space

A linear space E of elements $\boldsymbol{x}, \boldsymbol{y}, \ldots$ with each element $\boldsymbol{x}$ of which is associated a number $||\boldsymbol{x}||$ satisfying conditions (i)-(iii) stated below is called a *normed linear space* and the number $||\boldsymbol{x}||$ is called the *norm of the element* $\boldsymbol{x}$. The conditions mentioned include the following properties:

(i) $||\boldsymbol{x}|| \geqslant 0$ for any $\boldsymbol{x} \in E$; equality $||\boldsymbol{x}|| = 0$ implies $\boldsymbol{x} = \boldsymbol{\theta}$, that is in this case $\boldsymbol{x}$ coincides with the zero element $\boldsymbol{\theta}$ of the linear space E.

(ii) $||\alpha\boldsymbol{x}|| = |\alpha|\,||\boldsymbol{x}||$ for any $\boldsymbol{x} \in E$ and any number α (which can be real or complex depending on whether E is a real or complex space).

(iii) $||\boldsymbol{x}+\boldsymbol{y}|| \leqslant ||\boldsymbol{x}||+||\boldsymbol{y}||$ for any $\boldsymbol{x}, \boldsymbol{y} \in E$.

Now we see that the Euclidean space R_n is a normed space with the norm

$$||\boldsymbol{x}|| = |\boldsymbol{x}| = \left(\sum_1^n |x_j|^2\right)^{1/2} \tag{1}$$

It is also possible to introduce some other (non-Euclidean) norms in the space R_n. For instance, we can define the norm by putting

$$||\boldsymbol{x}|| = \max\{|x_1|, |x_2|, \ldots, |x_n|\} \tag{2}$$

for the points (vectors) $x = (x_1, \ldots, x_n) \in R_n$ or we can take

$$\|x\| = \left(\sum_1^n |x_j|^p\right)^{1/p} \qquad (1 \leqslant p < \infty) \tag{3}$$

The reader can easily check that relation (2) defines a norm and that for $p = 1$ relation (3) also defines a norm (the general case of an arbitrary p is treated in § 14.2).

Relation (iii) is termed the *triangle inequality*. In the cases of two-dimensional and three-dimensional Euclidean spaces it does in fact express the well-known geometrical property: the length of a side of a triangle does not exceed the sum of the lengths of its other two sides; thus, in particular, (iii) can be regarded as an analytical proof of this geometrical theorem.

Replacing x by $x-y$ or y by $y-x$ in (iii) we obtain

$$\|x-y\| \geqslant \|x\|-\|y\|, \quad \|x-y\| \geqslant \|y\|-\|x\|$$

and hence

$$\|x-y\| \geqslant |\|x\|-\|y\|| \tag{4}$$

For a normed space E we can define the *notion of limit*. A sequence of elements $x_n \in E$ is said to be *convergent* (to *tend*) to an element $x \in E$ if $\|x_n-x\| \to 0$ $(n \to \infty)$. In this case we shall write $x_n \to x$ or $\lim x_n = x$ $(n \to \infty)$.

If a sequence of elements $x_n \in E$ has a limit $x \in E$ the latter is determined uniquely because $x_n \to x$ and $x_n \to y$ imply

$$\|x-y\| = \|(x-x_n)+(x_n-y)\| \leqslant \|x-x_n\|+\|x_n-y\| \to 0$$

whence $\|x-y\| = 0$, that is $x = y$.

Since $|\|x_n\|-\|x\|| \leqslant \|x_n-x\| \to 0$, the convergence of x_n to x implies that $\|x_n\|$ tends to $\|x\|$:

$$\|x_n\| \to \|x\| \qquad (n \to \infty)$$

If x_n, y_n, x, y are elements of E and α_n, α are numbers and if $x_n \to x$, $y_n \to y$ and $\alpha_n \to \alpha$ then

$$\lim_{n\to\infty} (x_n \pm y_n) = x \pm y \quad \text{and} \quad \lim_{n\to\infty} (\alpha_n x_n) = \alpha x$$

Indeed, we have

$$\|(x \pm y)-(x_n \pm y_n)\| \leqslant \|x-x_n\|+\|y-y_n\| \to 0 \qquad (n \to \infty)$$

and

$$\|\alpha x-\alpha_n x_n\| = \|(\alpha-\alpha_n)x+\alpha_n(x-x_n)\| \leqslant \|(\alpha-\alpha_n x)\|+$$
$$+\|\alpha_n(x-x_n)\| \leqslant |\alpha-\alpha_n|\,\|x\|+|\alpha_n|\,\|x-x_n\| \to 0 \qquad (n \to \infty)$$

§ 6.4. Vector Function in n-dimensional Euclidean Space

Let E be a set of real numbers t. If there is a definite law according to which to every $t \in E$ there corresponds a vector*

$$x = x(t) = \big(x_1(t),\ x_2(t),\ \ldots,\ x_n(t)\big) \tag{1}$$

we shall say that the correspondence (1) defines a *vector function* $x(t)$ *on* E.

An ordinary function $\alpha(t)$ (which assigns to each $t \in E$ a number $\alpha(t)$) is also called a *scalar function.*

We shall say that a vector function $x(t)$ has a limit at a point t_0 equal to a vector $y = (y_1, \ldots, y_n)$ and shall write

$$\lim_{t \to t_0} x(t) = y \quad \text{or} \quad x(t) \to y \qquad (t \to t_0) \tag{2}$$

if

$$\lim_{t \to t_0} |y - x(t)| = 0 \tag{3}$$

or, which is the same (see the explanations below), if

$$\lim_{t \to t_0} x_j(t) = y_j \qquad (j = 1, \ldots, n) \tag{4}$$

Equality (3) says that the scalar function $|y - x(t)|$ of the argument t tends to zero limit as $t \to t_0$, and, as we know, this means that the function is defined in a neighbourhood of the point t_0 except possibly at the point t_0 itself; therefore all its components $x_j(t)$ are also defined in this neighbourhood.

As follows from formula (11) in § 6.2, there hold the inequalities

$$\frac{1}{\sqrt{n}}|y_j - x_j(t)| \leqslant \frac{1}{\sqrt{n}} \sum_{j=1}^{n} |y_j - x_j(t)| \leqslant \left[\sum_{j=1}^{n} (y_j - x_j(t))^2\right]^{1/2} = |y - x(t)|$$

which imply that if (3) holds then (4) holds as well for all $j = 1, \ldots, n$ and vice versa.

By definition, we say that a vector function $x(t)$ has a right-hand (left-hand) limit at a point t_0 equal to $y = (y_1, \ldots, y_n)$ if

$$|x(t) - y| \to 0, \qquad t \to t_0, \qquad t > t_0$$

$\big(|x(t) - y| \to 0,\ t \to t_0,\ t < t_0$ respectively$\big)$. These limits are denoted as

$$x(t_0+0) = \lim_{\substack{t \to t_0 \\ t > t_0}} x(t) \quad \text{and} \quad x(t_0-0) = \lim_{\substack{t \to t_0 \\ t < t_0}} x(t)$$

Arguing as above we readily see that

$$x(t_0+0) = \big(x_1(t_0+0), \ldots, x_n(t_0+0)\big)$$

---

* Here we mean vectors x belonging to the real space R_n but all the considerations remain valid for the complex space R_n as well.

and

$$x(t_0-0) = \big(x_1(t_0-0), \ldots, x_n(t_0-0)\big)$$

and that the existence of the vector limits on the left-hand sides of these equalities implies the existence of the corresponding limits of the components of the vectors and vice versa.

We say that a vector function $x(t)$ is *continuous* or *continuous from the right* or *continuous from the left* at the point t_0 if, respectively, the limit $\lim\limits_{t\to t_0} x(t)$, or $x(t_0+0)$ or $x(t_0-0)$ exists and is equal to $x(t_0)$.

These definitions are obviously equivalent to the requirements that the components $x_j(t)$ ($j = 1, \ldots, n$) should be, respectively, continuous or continuous from the right or continuous from the left at the point t_0.

It appears evident that $x(t)$ is continuous at $t = t_0$ if and only if the limits $x(t_0)$, $x(t_0+0)$ and $x(t_0-0)$ exist and the equalities $x(t_0) = x(t_0+0) = x(t_0-0)$ hold.

The *derivative of a vector function $x(t)$ at a point t* is defined as the limit

$$\dot{x}(t) = \frac{dx}{dt} = \lim_{h\to 0} \frac{x(t+h)-x(t)}{h} = \lim_{h\to 0} \frac{\Delta x}{h}$$

provided that it exists. The mth derivative of $x(t)$ is defined by induction:

$$\frac{d^m x}{dt^m} = \frac{d}{dt}\,\frac{d^{m-1}x}{dt^{m-1}} \qquad (m = 2, 3, \ldots)$$

Its existence obviously implies the existence of the derivatives of the mth order of the components of x and vice versa. We also have the equality

$$\frac{d^m x}{dt^m} = \big(x_1^{(m)}(t), \ldots, x_n^{(m)}(t)\big) \qquad (m = 1, 2, \ldots)$$

The derivatives of the first and of the second order (and sometimes higher-order derivatives as well) are also designated by dots:

$$\dot{x} = \dot{x}(t), \qquad \ddot{x} = \ddot{x}(t)$$

If $x(t)$ and $y(t)$ are vector functions and $\alpha(t)$ is a scalar function we have the relations

$$\lim_{t\to t_0} [x(t)\pm y(t)] = \lim_{t\to t_0} x(t) \pm \lim_{t\to t_0} y(t)$$

$$\lim_{t\to t_0} [\alpha(t)\, x(t)] = \lim_{t\to t_0} \alpha(t) \lim_{t\to t_0} x(t)$$

and

$$\frac{d}{dt}(x\pm y) = \frac{dx}{dt} \pm \frac{dy}{dt}, \quad \frac{d}{dt}(\alpha x) = \alpha \frac{dx}{dt} + \frac{d\alpha}{dt} x$$

where it is of course supposed that the limits and the derivatives entering into the right-hand sides exist. The proof of these equalities is quite simple and is based on the passage to the corresponding components (*coordinates*) of the vectors. For instance,

$$\lim_{t\to t_0}[x(t)\pm y(t)] = \left(\lim_{t\to t_0}[x_1(t)\pm y_1(t)], \ldots, \lim_{t\to t_0}[x_n(t)\pm y_n(t)]\right) =$$

$$= \left(\lim_{t\to t_0} x_1(t)\pm\lim_{t\to t_0} y_1(t), \ldots, \lim_{t\to t_0} x_n(t)+\lim_{t\to t_0} y_n(t)\right) =$$

$$= \left(\lim_{t\to t_0} x_1(t), \ldots, \lim_{t\to t_0} x_n(t)\right)\pm\left(\lim_{t\to t_0} y_1(t), \ldots, \lim_{t\to t_0} y_n(t)\right) =$$

$$= \lim_{t\to t_0} x(t)\pm\lim_{t\to t_0} y(t)$$

It is also possible, of course, to retain the vector notation in this proof. For example, putting $\lim_{t\to t_0}\alpha(t) = \beta$ and $\lim_{t\to t_0} x(t) = y$ we obtain

$$|\alpha(t)x(t)-\beta y| \leqslant |[\alpha(t)-\beta]x(t)|+|\beta(x(t)-y)| \leqslant |\alpha(t)-\beta|\,|x(t)|+$$
$$+|\beta|\,|x(t)-y| \to 0\cdot|y|+|\beta|\cdot 0 = 0, \qquad t\to t_0$$

§ 6.5. Curve in *n*-dimensional Space

If t increases continuously and runs through $[a, b]$ (or (a, b)) the point specified by the continuous vector function

$$x(t) = (\varphi_1(t), \ldots, \varphi_n(t)) \in R_n \tag{1}$$

describes a *set of points belonging to R_n which is ordered by means of the variable t; this set is called the hodograph (trajectory) of the vector function* $x(t)$. It is not excluded that the moving point $x(t)$ may return to a point of the space R_n through which it has already passed for some other value of the parameter t. In the cases of dimensions $n = 2$ and $n = 3$ such trajectories have a real geometrical meaning.

Proceeding from a given vector function (1) we can construct a new vector function defined by the vector equalities

$$x_*(\tau) = x(\lambda(\tau)) = (\varphi_1(\lambda(\tau)), \ldots, \varphi_n(\lambda(\tau)))$$
$$(\tau\in[c, d] \quad\text{or}\quad \tau\in(c, d)) \tag{2}$$

or by the equivalent system of scalar equalities

$$x_1 = \varphi_{*1}(\tau) = \varphi_1(\lambda(\tau)), \ldots, x_n = \varphi_{*n}(\tau) = \varphi_n(\lambda(\tau)) \tag{2'}$$

where $t = \lambda(\tau)$ is an arbitrary strictly monotone (real!) function specifying a (one-to-one!) mapping of the closed interval $[c, d]$ (open interval (c, d)) of variation of the new variable τ onto the closed interval $[a, b]$ (open interval (a, b)) of variation of the old variable t.

It is clear that equation (2) specifies the same trajectory Γ as equation (1) and that the orderings of its points induced by the variables t and τ are the same if $\lambda(\tau)$ is strictly increasing or have opposite senses if $\lambda(\tau)$ is strictly decreasing.

Equation (1) is said to determine a *continuous curve* Γ with the aid of the parameter t while equation (2) is said to determine the same curve Γ but with the aid of the parameter τ. Thus, to different strictly monotone continuous functions $\lambda(\tau)$ there correspond different *parametric representations* of one and the same *continuous curve* Γ.

The function $\lambda(\tau)$ can always be chosen so that $c = 0$ and $d = 1$.

The functions $\lambda(\tau)$ can be divided into two classes: the class of strictly increasing functions and the class of strictly decreasing functions. The first class specifies the ordering of the points of Γ in one sense and the second class the ordering in the other, opposite sense. In this connection there appears the notion of an *oriented curve* Γ. For instance, we can denote by Γ_+ the curve Γ oriented with the aid of the parameter t. Γ_+ also determines all the possible equations (2) in which $\lambda(\tau)$ are continuous strictly increasing functions. The same curve Γ supplied with the opposite orientation is then naturally denoted by Γ_-. The oriented curve Γ_- is specified by the class of all possible equations (2) with continuous strictly decreasing functions λ (τ).

A curve Γ is said to be *smooth on* $[a, b]$ $\big($*on* $(a, b)\big)$ if it admits* of a parametric representation (*parametrization*) with the aid of a *smooth vector function* $x(t)$, the latter being a continuous function possessing a *nonzero derivative* on $[a, b]$ $\big($on $(a, b)\big)$ or, which is the same, a vector function $x(t)$ whose all components $x_j(t)$ are smooth scalar functions on $[a, b]$ $\big($on $(a, b)\big)$ whose derivatives do not vanish simultaneously. The last property is equivalent to the fact that

$$|x'(t)|^2 = \sum_{j=1}^{n} [x_j'(t)]^2 > 0, \qquad t \in [a, b] \quad \big((a, b)\big) \tag{3}$$

In this formulation, the continuity of $x(t)$ at the end points a and b is of course understood as one-sided continuity: from the right at a and from the left at b. As to the derivative $x'(t)$ at a and b, it is understood as the right-hand derivative at a and as the left-hand derivative at b.

We shall say that τ is an *admissible parameter* for the smooth curve Γ represented by equation (1) if it is connected with t by means of an equality $t = \lambda(\tau)$, $\tau \in [c, d]$ $\big((c, d)\big)$ where $\lambda(\tau)$ is a function which is not only strictly monotone and continuous but also possesses a continuous derivative on $[c, d]$ $\big($on $(c, d)\big)$ different from zero. Thus, the derivative $\lambda'(\tau)$ must in fact have constant sign on $[c, d]$ $\big($on $(c, d)\big)$, i.e. "+" or "−". If τ is an admissible parameter then the defining property of a smooth curve stated above in terms of the parameter t continues to hold if it is stated in terms of τ because the vector function $x(\lambda(\tau)) = x_*(\tau)$ has a continuous derivative on $[c, d]$

* This definition does not exclude the possibility that a smooth curve can be represented parametrically by a function not satisfying the stated requirements. For instance, we can introduce a new parameter τ by means of a substitution $t = \lambda(\tau)$ where $\lambda(\tau)$ is a strictly monotone continuous function whose derivative vanishes or does not exist for some τ.

(on (c, d)) which, in addition, does not vanish:

$$\sum_{j=1}^{m} x_{*j}'(\tau)^2 = \lambda'(\tau)^2 \sum_{j=1}^{n} x_j'(t)^2 > 0$$

Let us choose a point $(x_1^0, \ldots, x_n^0) = \mathbf{x}^0 \in \Gamma$ corresponding to a value $t_0 \in (a, b)$ of the parameter t. At least one of the terms in the sum on the right-hand side of (3) must be positive; for definiteness, let it be the nth summand: $x_n'(t_0)^2 > 0$. Then there exists a neighbourhood* of the point t_0 in which $x_n'(t)$ retains sign, and the equation $x_n = x_n(t)$ can be resolved with respect to t in this neighbourhood:

$$t = \mu(x_n), \qquad (x_n^0 - \eta_1 < x_n < x_n^0 + \eta_2)$$

where $\eta_1, \eta_2 > 0$ are some numbers and μ is the inverse function of $x_n(t)$ which is continuous and has a continuous derivative. This means that the portion of the curve (1) in question which corresponds to the indicated neighbourhood and is determined by equations (1) can also be determined by the equations

$$x_j = \mu_j(x_n) = x_j[\mu(x_n)] \qquad (j = 1, \ldots, n-1)$$
$$x_n = x_n \qquad (x_n^0 - \eta_1 < x_n < x_n^0 + \eta_2)$$

These equations can be differentiated once (in the indicated neighbourhood):

$$\frac{dx_j}{dx_n} = \frac{\dfrac{dx_j}{dt}}{\dfrac{dx_n}{dt}} \qquad (j = 1, \ldots, n-1)$$

Summing up we state the following theorem.

Theorem 1. *For any point $\mathbf{x}^0$ of a given smooth curve Γ corresponding to a value $t = t_0 \in (a, b)$ of the parameter t there is a sufficiently small number $\delta > 0$ such that the portion of Γ corresponding to the variation of t in the interval $(t_0 - \delta, t_0 + \delta)$ can be represented parametrically with the aid of at least one of the coordinates x_j:*

$$x_j = \varphi_j(x_i) \qquad (j = 1, \ldots, i-1, i+1, \ldots, n; \quad x_i^0 - \eta_1 < x_i < x_i^0 + \eta_2)$$

where the functions $x_j = \varphi_j(x_i)$ possess the continuous derivatives

$$\frac{dx_j}{dx_i} = \frac{\dfrac{dx_j}{dt}}{\dfrac{dx_i}{dt}} \qquad (j = 1, \ldots, i-1, i+1, \ldots, n)$$

*If $t_0 = a$ or $t_0 = b$ then by a neighbourhood is meant a one-sided (i.e. a right-hand or left-hand) neighbourhood of the point t_0.

In particular, in the two-dimensional case a smooth curve is determined by two equations

$$x = \varphi(t), \quad y = \psi(t) \qquad \bigl(t \in (a, b)\bigr) \tag{4}$$

where φ and ψ possess continuous derivatives not vanishing simultaneously. If, for instance, $\varphi'(t_0) \neq 0$, then there exists an interval $(t_0-\delta, t_0+\delta)$ on which φ has the inverse function $t = \varphi^{-1}(x)$ and $y = f(x) = \psi(\varphi^{-1}(x))$. In such a case we usually say that the function $y = f(x)$ is specified parametrically by equalities (4), and formula

$$\frac{dy}{dx} = \frac{\dfrac{dy}{dt}}{\dfrac{dx}{dt}} \tag{5}$$

is interpreted as providing a *parametric representation of the derivative of the function* $f(x)$. It is also evident that

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{y'_t}{x'_t}\right) = \frac{d}{dt}\left(\frac{y'_t}{x'_t}\right)\frac{dt}{dx} = \frac{x'_t y''_t - y'_t x''_t}{x'^3_t} \tag{6}$$

if we additionally assume that the second derivatives x''_t and y''_t exist.

We say that Γ is a *piecewise smooth continuous curve* on $[a, b]$ (on (a, b)) if it can be represented by a continuous vector function $\boldsymbol{x}(t)$ on $[a, b]$ (on (a, b)) such that the closed interval $[a, b]$ (the open interval (a, b)) can be split into a finite number of parts by means of points of division $a = t_0 < t_1 < \ldots < t_N = b$ so that $\boldsymbol{x}(t)$ is a smooth curve on each of the subintervals* $[a, t_1]$, $[t_1, t_2]$, $\ldots$, $[t_{N-1}, b]$.

In should be taken into account that, generally, at a point of division t_k $(k = 1, \ldots, N-1)$ the left-hand derivative $\dot{\boldsymbol{x}}(t_k-0)$ may not be equal to the right-hand derivative $\dot{\boldsymbol{x}}(t_k+0)$ but at the same time both derivatives are different from zero (that is among the components of each of them there is at least one different from zero).

The various parametric representations of a piecewise smooth continuous curve Γ are given by equation (2) with functions $t = \lambda(\tau)$ having on $[c, d]$ (on (c, d)) continuous derivatives $\lambda'(\tau)$ not turning into zero.

A smooth curve of type (1) is called a *Jordan curve* after C. Jordan (1838-1922), a noted French mathematician. If we have $\boldsymbol{x}(a) = \boldsymbol{x}(b)$ for such a curve it is called a *closed* (*Jordan*) *curve*. If a curve of this kind is such that the fact that $\boldsymbol{x}(t_1) = \boldsymbol{x}(t_2)$ implies that either $t_1 = t_2$ or one of the numbers t_1 and t_2 is equal to a while the other is equal to b, the curve Γ is said to be a *closed Jordan curve not intersecting itself* or a *continuous closed curve without self-intersection* or a *simple closed curve*.

If equalities $\boldsymbol{x}(t_1) = \boldsymbol{x}(t_2)$ $(t_1, t_2 \in [a, b]$ or $t_1, t_2 \in (a, b))$ imply $t_1 = t_2$ we say that Γ is a *nonclosed curve not intersecting itself*.

---

* In the case of an open interval (a, b) the subintervals $[a, t_1]$ and $[t_{N-1}, b]$ are respectively replaced by $(a, t_1]$ and $[t_{N-1}, b)$.

In the case $n = 2$ we have a plane continuous curve

$$x_1 = x_1(t),\ x_2 = x_2(t),\ t \in [a, b] \quad \text{or} \quad t \in (a, b) \tag{7}$$

For instance, the equations

$$x = \cos\theta, \quad y = \sin\theta \qquad (-\infty < \theta < \infty) \tag{8}$$

specify a smooth plane curve. When θ varies continuously from $-\infty$ to $+\infty$ the corresponding point (x, y) traverses infinitely many times the circle

$$x^2+y^2 = 1 \tag{9}$$

Therefore equations (8) are termed parametric equations of circle (9). In this example the parameter θ has an obvious geometrical meaning: it is equal to the angle between the radius of the point (x, y) and the positive direction of the x-axis.

The equations of circle (8) can be written more "economically":

$$x = \cos\theta, y = \sin\theta \qquad (0 \leqslant \theta \leqslant 2\pi) \tag{10}$$

Here θ runs through the closed interval $[0, 2\pi]$. Curve (10) is an example of a smooth closed curve not intersecting itself. The circle Γ understood as the locus of points (x, y) satisfying equation (9) is also usually said to be a closed curve. This should in fact be understood in the following sense: there exists a continuous closed curve without self-intersection represented parametrically (namely, curve (10)) whose moving point runs through the points of Γ and only through these points.

C. Jordan proved the following proposition which is geometrically obvious while its rigorous proof involves some intricate considerations: *a simple closed curve Γ lying in a plane R divides the set $R-\Gamma$ into two nonempty and nonintersecting regions one of which is the interior of Γ while the other is the exterior of Γ*: $R-\Gamma = A_i+A_e$. Any two points belonging to A_i can be joined by a continuous curve lying entirely within A_i and any two points of A_e can be joined by a continuous curve belonging to A_e. Every curve connecting an arbitrary point of A_i with an arbitrary point of A_e has at least one common point with Γ (intersects Γ). The region A_i is bounded; the region A_e is unbounded.

It should be noted that the definition of a continuous curve is so general that there are examples of mathematical objects satisfying the conditions of this definition which are essentially different from a curve understood in the sense of our ordinary intuitive idea, especially when self-intersections are allowed.

For instance, it was proved that it is possible to define two functions

$$x = \varphi(t), \ y = \psi(t) \qquad (0 \leqslant t \leqslant 1)$$

continuous on the closed interval $[0, 1]$ such that when the parameter t continuously increases from $t = 0$ to $t = 1$ the variable point $(\varphi, (t), \psi(t))$ moves in such a way that it starts from the position $(0, 0)$ corresponding to $t = 0$, runs through exactly all the points of the square $0 \leqslant x, \ y \leqslant 1$

and arrives at its right upper corner (1, 1) for $t = 1$. Hence, this curve (known as *Peano's curve*) literally passes through each point of the square $0 \leqslant x, y \leqslant 1$ (through some of the points the curve passes not once).

In Fig. 6.1 we see a plane curve Γ. Let us suppose that it is representable by continuously differentiable functions

$$x = \varphi(t), \quad y = \psi(t), \qquad \varphi'^2+\psi'^2 > 0, \quad 0 < t < 1$$

When t increases continuously on the interval (0, 1) the point (x, y) moves along the curve Γ starting from the point A, passes through the points B and C and again tends to B as $t \to 1$. We see that in spite of the continuous differentiability of the functions φ and ψ the curve Γ has a singularity at the point B in the sense that the part of Γ lying within any rectangle Δ with centre at B, however small, cannot be projected in a one-to-one manner on any of the coordinate axes.

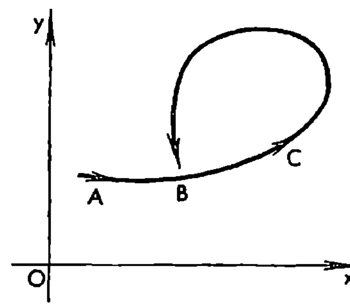

Fig. 6.1

If a smooth curve Γ is not "deficient" in this sense at its every point, that is if its every point can be covered with a rectangle Δ with edges parallel to the coordinate axes such that $\Gamma\Delta$ (the intersection of Γ and Δ) is projected in a one-to-one manner on at least one of the coordinate axes, Γ is called a *one-dimensional differentiable manifold.*

In § 17.1 we shall prove Lemma 1 from which, as a special case, the following assertion follows:

If a smooth curve Γ defined on a closed interval $[a, b]$ *by equations*

$$x_i = \varphi_i(t), \quad \sum_{i=1}^{n} (\varphi_i')^2 > 0, \qquad a \leqslant t \leqslant b$$

does not intersect itself, that is if the points of Γ and of $[a, b]$ *are in a one-to-one correspondence specified by equations of type* (1) *the curve Γ obtained from Γ by deleting both end points of the latter (Γ is defined on the open interval* (a, b)) *is a one-dimensional differentiable manifold.*

Example 1. The ellipse Γ determined by the equation

$$\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1 \qquad (a, b > 0) \tag{11}$$

is a bounded closed smooth curve not intersecting itself since Γ can also be described by the parametric equations

$$x = a\cos\theta, y = b\sin\theta \qquad (0 \leqslant \theta \leqslant 2\pi) \tag{12}$$

which define a bounded closed smooth curve in the sense of the definitions of smoothness and closedness stated in this section.

Example 2. The astroid Γ determined by the equation

$$|ax|^{2/3}+|by|^{2/3} = (a^2-b^2)^{2/3} \qquad (0 < b < a) \tag{13}$$

is a bounded piecewise smooth continuous closed curve because equation (13) is equivalent to the two equations

$$x = \frac{a^2-b^2}{a}\cos^3\theta, \quad y = \frac{a^2-b^2}{a}\sin^3\theta \qquad (0 \leqslant \theta \leqslant 2\pi) \tag{14}$$

and because there is only one pair of values of θ ($\theta = 0$ and $\theta = 2\pi$) to which there corresponds one and the same point of Γ. Equation (13) shows that the curve Γ is symmetric with respect to the coordinate axes, and (14) indicates that it is continuous. The derivatives of x and y with respect to θ are also continuous; they do not vanish simultaneously except at the points $0, \pi/2, \pi$ and $3\pi/2$. Therefore the parts of Γ corresponding to the intervals $(0, \pi/2)$, $(\pi/2, \pi)$, $(\pi, 3\pi/2)$ and $(3\pi/2, 2\pi)$ are smooth (see Fig. 6.11).

§ 6.6. Geometrical Interpretation of the Derivative of a Vector Function

Let us consider (ordinary) three-dimensional geometrical space with rectangular Cartesian coordinates (x, y, z). Suppose that in this space a smooth vector function (see p. 189)

$$\boldsymbol{r}(t) = (\varphi(t), \psi(t), \chi(t)), \qquad t \in (a, b) \tag{1}$$

is defined. The hodograph of the vector $\boldsymbol{r} = \boldsymbol{r}(t)$ is shown in Fig. 6.2; in the figure we also see the two points A and B which are respectively the termini of the vectors $\boldsymbol{r}(t)$ and $\boldsymbol{r}(t+\Delta t)$.

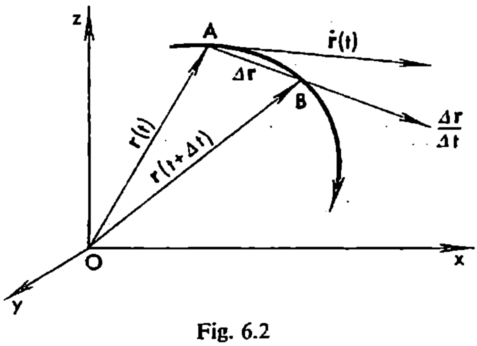

Fig. 6.2

It is obvious that the vector $\overline{AB}$ can be represented as $\overline{AB} = \Delta\boldsymbol{r} = \boldsymbol{r}(t+\Delta t)-\boldsymbol{r}(t)$. When $\Delta t \to 0$ the point B moves along the hodograph and tends to the point A while the secant passing through the points A and B tends to occupy the position of a definite (limiting) straight line which is called the *tangent to the hodograph at the point A*. It follows that the limiting vector

$$\dot{\boldsymbol{r}} = \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t}$$

(it is different from zero!) lies in the tangent line to the hodograph at the point A. The length $|\dot{\boldsymbol{r}}|$ of the vector $\dot{\boldsymbol{r}}$ is equal to the limit of the length of the vector $\frac{\Delta r}{\Delta t}$ as $\Delta t \to 0$ since

$$\left| |\dot{r}| - \left|\frac{\Delta r}{\Delta t}\right| \right| \leq \left|\dot{r} - \frac{\Delta r}{\Delta t}\right| \to 0 \qquad (\Delta t \to 0)$$

If t is time and the terminus of the vector $r(t)$ describes the motion of a point, then $\dot{r}(t)$ is the vector representing the velocity of that point at time moment t. Its length $|\dot{r}|$ is the absolute value of the velocity. The vector $\dot{r}$ determines the direction of motion of the point at the moment t. Further, the vector $\ddot{r}$ is nothing but the acceleration of the point at the moment t.

In § 6.4 we discussed some properties of the derivative of a vector function. Here we state some further obvious properties.

If $(a, b) = a_x b_x + a_y b_y + a_z b_z$ is the scalar product and $[a \times b] = (a_y b_z - - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x)$ the vector product of the vectors a and b then

$$\frac{d}{dt}(a, b) = \left(a, \frac{db}{dt}\right) + \left(\frac{da}{dt}, b\right)$$

and

$$\frac{d}{dt}[a \times b] = \left[a \times \frac{db}{dt}\right] + \left[\frac{da}{dt} \times b\right]$$

We also note the following fact. Let a vector function $b = b(t)$ have a constant norm (length): $|b(t)| = c = \text{const} > 0$. Then

$$\frac{d}{dt}(b, b) = 2\left(b, \frac{db}{dt}\right) = 0$$

It follows that the vectors b and $\frac{db}{dt}$ are mutually perpendicular for any t or $\frac{db}{dt} = 0$ or $b = 0$.

An arbitrary vector function $a = a(t)$ whose length is positive for any value of t in question ($|a| > 0$) can be written in the form $a = \alpha\omega$ where

$$\omega(t) = \frac{a}{|a|} = \left(\frac{a_1(t)}{|a|}, \frac{a_2(t)}{|a|}, \frac{a_3(t)}{|a|}\right)$$

and

$$\alpha(t) = |a| = \sqrt{a_1(t)^2 + a_2(t)^2 + a_3(t)^2}$$

It is obvious that if the vector a possesses the derivative with respect to t for the values of t under consideration then the functions ω and α also have derivatives for these t. The derivative of the vector a can be resolved along the two vectors ω and $\frac{d\omega}{dt}$:

$$\frac{da}{dt} = \frac{d\alpha}{dt}\omega + \alpha\frac{d\omega}{dt} \tag{1}$$

The first component $\frac{d\alpha}{dt}\omega$ in this resolution is directed along a or, which is the same, along ω and its length is equal to the rate of change of the length of a; the second component $\alpha \frac{d\omega}{dt}$ is orthogonal to ω. This formula is used in mechanics for the resolution of the velocity vector of a moving point into two components one of which goes in the direction of motion while the other is perpendicular to the former.

§ 6.7. Arc Length of a Curve

Let Γ be a continuous curve determined by equations

$$r(t) = (\varphi(t),\ \psi(t),\ \chi(t)) \qquad (t \in [a, b]) \tag{1}$$

Let us break up the interval $[a, b]$ into parts with the aid of points of division

$$a = t_0 < t_1 < \ldots < t_n = b \tag{2}$$

the corresponding points of the curve Γ being respectively $A_0 = A, A_1, \ldots, A_{n-1}, A_n = B$. On connecting them in succession with line segments (see Fig. 6.3) we obtain a polygonal line inscribed in Γ.

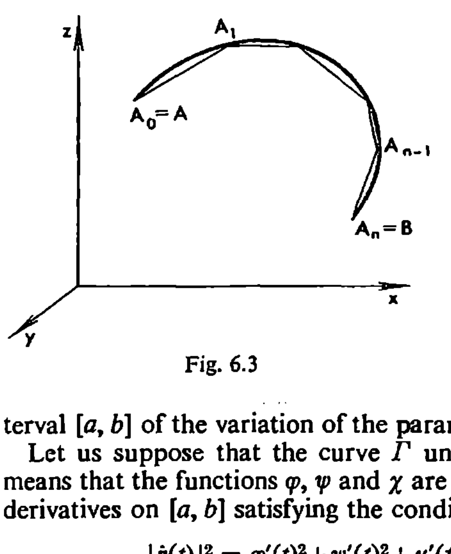

Fig. 6.3

By the *length* of the curve Γ is meant the limit to which tends the sum of the lengths of the segments of this polygonal line when the maximum of the subintervals in the partition (2) tends to zero:

$$|\breve{AB}| = \lim \sum_{1}^{n} |A_{k-1}A_k|,$$
$$\max (t_k - t_{k-1}) \to 0 \tag{3}$$

If limit (3) exists the curve Γ is said to be *rectifiable* (on the interval $[a, b]$ of the variation of the parameter t).

Let us suppose that the curve Γ under consideration is smooth. This means that the functions φ, ψ and χ are continuous and possess continuous derivatives on $[a, b]$ satisfying the condition

$$|\dot{r}(t)|^2 = \varphi'(t)^2 + \psi'(t)^2 + \chi'(t)^2 > 0 \qquad (t \in [a, b]) \tag{4}$$

Later on, when studying the integral calculus, we shall prove that a smooth curve Γ is rectifiable on any interval of variation of the parameter t and that the arc length of the curve Γ possesses the additivity property in the sense that if P_1, P_2 and P_3 are three arbitrary points of Γ corresponding to some values t_1, t_2 and t_3 of the parameter and $t_1 < t_2 < t_3$ then

$$|\breve{P_1P_3}| = |\breve{P_1P_2}| + |\breve{P_2P_3}|$$

Let us introduce the new function $s = F(t)$ equal to the length of the arc $\overset{\smile}{AC}$ corresponding to the variation of the parameter on the interval $[a, t]$. It will be proved in the integral calculus that the function $F(t)$ possesses the following properties: it is continuous and has the continuous derivative on $[a, b]$ determined by the formula

$$F'(t) = \frac{ds}{dt} = \sqrt{\varphi'(t)^2+\psi'(t)^2+\chi'(t)^2} > 0 \tag{5}$$

Besides, $F(a) = 0$. It follows that s is a strictly increasing function which specifies a mapping of the interval $[a, b]$ of variation of t onto the corresponding interval $[0, l]$ of variation of s and that its inverse function

$$t = \Lambda(s) \qquad (0 \leqslant s \leqslant l)$$

exists, is continuous and has the continuous derivative $\Lambda'(s) > 0$.

We see that s can be chosen as one of the admissible parameters for the smooth curve Γ in question:

$$x = \varphi(\Lambda(s)),\;\; y = \psi(\Lambda(s)),\;\; z = \chi(\Lambda(s)) \qquad (0 \leqslant s \leqslant l)$$

Now let τ be an arbitrary admissible parameter for the curve Γ connected with t by means of a function $t = \lambda(\tau)$ possessing a continuous nonzero derivative. Then the sign of $\dfrac{ds}{d\tau} = \dfrac{ds}{dt}\,\lambda'(\tau)$ depends on the sign of $\lambda'(\tau)$ and hence, taking into account the formula for the derivative of a function of a function, we can write

$$\frac{ds}{d\tau} = \lambda'(\tau)\sqrt{\varphi'(t)^2+\psi'(t)^2+\chi'(t)^2} = \pm\sqrt{\varphi_1'(\tau)^2+\psi_1'(\tau)^2+\chi_1'(\tau)^2} \tag{6}$$

where

$$\begin{aligned} x &= \varphi_1(\tau) = \varphi(\lambda(\tau)),\;\; y = \psi_1(\tau) = \psi(\lambda(\tau)) \\ z &= \chi_1(\tau) = \chi(\lambda(\tau)) \qquad (\tau \in (c, d)) \end{aligned} \tag{7}$$

are the equations of Γ written in terms of the parameter τ, and the square root in (6) is taken with the sign "+" or "−" depending on whether s increases or decreases as τ increases.

It follows that

$$ds = \pm\sqrt{dx^2+dy^2+dz^2} \tag{8}$$

If $d\tau > 0$ we should take the sign "+" in (8) in the first case (see above) and "−" in the second case. But if $d\tau < 0$ then, conversely, in the first case the sign "−" is taken and in the second case the sign "+".

If we put $\tau = s$ in equality (6) then in front of the root on the right-hand side of (8) we should place the sign "+" since s regarded as a function of s is increasing; we thus obtain the equality $1 = \sqrt{\left(\frac{dx}{ds}\right)^2+\left(\frac{dy}{ds}\right)^2+\left(\frac{dz}{ds}\right)^2}$. It should be noted that in these considerations we assume that $s = 0$ for $t = a$ and that s increases together with t.

The definition of the arc length of a curve stated in this section involves explicitly a definite parametric representation of the curve. But it can be shown (see § 10.3, formula (3)) that *the arc length is in fact an invariant independent of the choice of the variable t by means of which the curve is parametrized.*

§ 6.8. Tangent Line. Normal to a Plane Curve

Let us consider a smooth curve determined by a vector function $r(t) = (x, (t), y(t), z(t))$, $t \in (a, b)$ defined in the (ordinary, geometrical) three-dimensional space with rectangular Cartesian coordinates (x, y, z) (see Fig. 6.4). Suppose that the direction in which the arc length is reckoned is chosen so that it increases together with the parameter t (like in § 6.7).

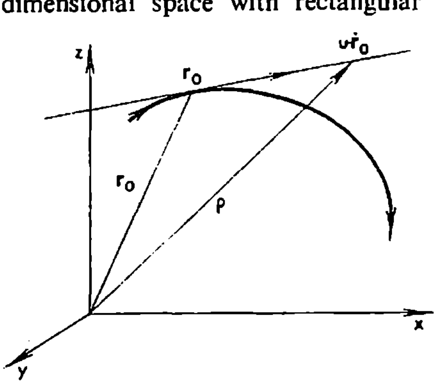

Fig. 6.4

Let us put $r_0 = r(t_0) = (x_0, y_0, z_0)$ and $\dot{r}_0 = \dot{r}(t_0) = (x_0', y_0', z_0')$. The vector $\dot{r}_0$ goes in the direction of the tangent line to the curve at the point t_0 and therefore the variable point $\varrho = (x, y, z)$ of the tangent (the tangent line) is determined by the equation

$$\varrho = r_0 + u\dot{r}_0 \tag{1}$$

where u is an arbitrary number (a new parameter providing a representation of the tangent line).

Equality (1) is the equation of the *tangent line to the curve at the point t_0* written in vector form.

It follows from (1) that in Cartesian coordinates the equations of the tangent are written as

$$x - x_0 = ux_0', \quad y - y_0 = uy_0', \quad z - z_0 = uz_0'$$

or, equivalently,

$$\frac{x-x_0}{x_0'} = \frac{y-y_0}{y_0'} = \frac{z-z_0}{z_0'} \tag{2}$$

Let us denote by α, β and γ the angles formed by the positive direction of the tangent (i.e. the direction of $\dot{r}_0$) with the positive directions of the coordinate axes x, y and z respectively. Then, obviously,

$$\cos\alpha = \frac{x_0'}{\sqrt{x_0'^2+y_0'^2+z_0'^2}} = \left(\frac{dx}{ds}\right)_0$$

$$\cos\beta = \frac{y_0'}{\sqrt{x_0'^2+y_0'^2+z_0'^2}} = \left(\frac{dy}{ds}\right)_0$$

$$\cos\gamma = \frac{z_0'}{\sqrt{x_0'^2+y_0'^3+z_0'^2}} = \left(\frac{dz}{ds}\right)_0$$

where $\left(\frac{dx}{ds}\right)_0$ symbolizes that the value $s = s_0$ corresponding to $t = t_0$ is substituted into $\frac{dx}{ds}$. In front of the square root stands the sign "+" because we agree that the arc length increases together with t.

A curve lying in the xy-plane can be treated as a special case of as pace curve with $z(t) \equiv 0$. Therefore in the case of a plane curve relations (2) reduce to one equation

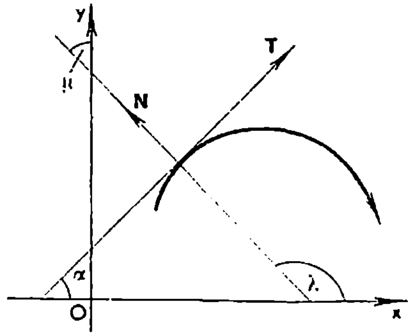

Fig. 6.5

$$\frac{x-x_0}{x_0'} = \frac{y-y_0}{y_0'}$$

In this case the positive direction of the tangent forms an angle α with the x-axis for which

$$\cos\alpha = \frac{x_0'}{\sqrt{x_0'^2+y_0'^2}} = \left(\frac{dx}{ds}\right)_0 \quad \text{and} \quad \sin\alpha = \frac{y_0'}{\sqrt{x_0'^2+y_0'^2}} = \left(\frac{dy}{ds}\right)_0$$

For a plane curve we can also define the notion of the *normal to the curve at the point* t_0 which is a straight line lying in the plane under consideration and passing through the point of the curve corresponding to $t = t_0$ perpendicularly to the tangent. For some problems it is important to define the *positive direction of the normal* specified by a (normal) vector $\boldsymbol{N}$. It is chosen so that the direction along the tangent $\boldsymbol{T}$ corresponding to the increase of t and the direction of $\boldsymbol{N}$ form a rectangular system oriented like the coordinate axes x and y, that is it *must be possible* to make the angle formed by $\boldsymbol{T}$ and $\boldsymbol{N}$ coincident with the first coordinate angle by means of a continuous motion in the plane so that $\boldsymbol{T}$ goes along the positive x-axis and $\boldsymbol{N}$ along the positive y-axis (Figs. 6.5 and 6.6).

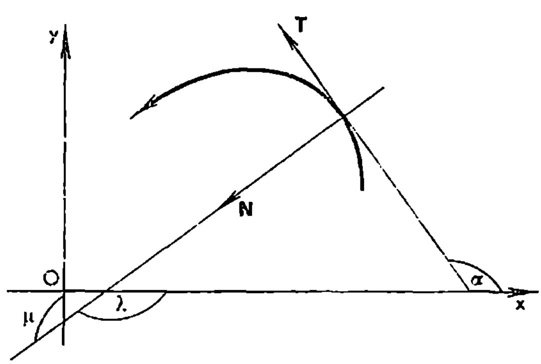

Fig. 6.6

Let λ and μ be respectively the angles formed by the positive direction of the normal with the axes x and y. It is seen from the figures that the convention we have made leads to the formulas*

$$\cos\lambda = -\sin\alpha = -\left(\frac{dy}{ds}\right)_0 \quad \text{and} \quad \cos\mu = \cos\alpha = \left(\frac{dx}{ds}\right)_0$$

§ 6.9. Curvature and Radius of Curvature of a Curve. Plane Curve. Evolute and Evolvent

By the *curvature of a circle* of radius R is meant the number $1/R$. This number can be expressed as the ratio of the angle (measured in radians) between the tangent lines drawn through the end points of an arc of the circle to the length of that arc. This property provides an idea of a curvature which can be generalized to arbitrary smooth curves.

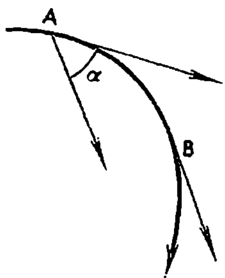

Fig. 6.7

Let us consider a smooth curve Γ (Fig. 6.7). It is rectifiable, and the length of its arbitrary arc $\breve{AB}$ makes sense. The angle α $(0 \leqslant \alpha \leqslant \pi)$ between the positive directions of the tangent lines to the curve at its points A and B is called the *angle of contingence of the arc* $\breve{AB}$. The ratio of the angle of contingence (measured in radians) of the arc $\breve{AB}$ to its length is called the *average curvature of the arc* $\breve{AB}$ (see Fig. 6.7). The *curvature K of the curve Γ* at its point A is the limit (finite or infinite) to which tends the ratio of the angle of contingence α of the arc $\breve{AB}$ to its length $|\Delta s|$, i.e. the average curvature of the arc AB, as the latter tends to zero:

$$K = \lim_{\Delta s \to 0} \frac{\alpha}{|\Delta s|} \tag{1}$$

Thus, $0 \leqslant K \leqslant \infty$. By definition, the quantity $R = 1/K$ (here we agree that $0 = 1/\infty$ and $1/0 = \infty$) is called the *radius of curvature of Γ* at the point A.

Note that the angle of contingence α $(\alpha \geqslant 0)$ of the arc $\breve{AB}$ is equal to the angle between the vectors $\dot{r}(t)$ and $\dot{r}(t+\Delta t) = \dot{r}+\Delta\dot{r}$ where $r(t)$ is the radius vector of the moving point of Γ or, which is the same, to the angle between the unit vectors $\tau(t) = \dot{r}(t)/|\dot{r}(t)|$ and $\tau(t+\Delta t)$. Therefore the cosine of the angle α is obviously equal to the scalar product $(\tau(t), \tau(t+\Delta t))$ and the angle α itself can be written as

$$\alpha = \arccos\left(\tau(t), \ \tau(t+\Delta t)\right) \qquad (0 \leqslant \alpha \leqslant \pi)$$

whence we see that, for a smooth curve, $\Delta t \to 0$ implies $\alpha \to 0$.

---

* To memorize these formulas note that the vector product of the vectors $(0, 0, 1)$ and $(\cos\alpha, \sin\alpha, 0)$ is expressed as $(0, 0, 1)\times(\cos\alpha, \sin\alpha, 0) = (-\sin\alpha, \cos\alpha, 0)$.

According to what is known from vector algebra,

$$\sin\alpha = \frac{|\dot{r}\times(\dot{r}+\Delta\dot{r})|}{|\dot{r}|\,|\dot{r}+\Delta\dot{r}|} = \frac{|\dot{r}\times\Delta\dot{r}|}{|\dot{r}|\,|\dot{r}+\Delta\dot{r}|}, \qquad \Delta\dot{r} = \dot{r}(t+\Delta t)-\dot{r}(t) \tag{2}$$

because $\dot{r}\times\dot{r} = 0$. Here the denominator is different from zero since we have $\dot{r} \neq 0$ for a smooth curve. When $\Delta t \to 0$ the denominator tends to $|\dot{r}|^2 > 0$ while the numerator tends to zero. Let us consider the arc length $s = s(t)$ along the curve in question. The length of its part $\overset{\frown}{AB}$ is equal to

$$|\Delta s| = |s(t+\Delta t)-s(t)|$$

From $\Delta s \to 0$ it follows that $\Delta t \to 0$ because t and s are both admissible parameters for the smooth curve Γ (see § 6.8).

Let us suppose now that the radius vector $r(t)$ of the smooth curve Γ under consideration possesses the second derivative $\ddot{r}(t)$; under this assumption we shall prove the existence of a finite curvature of Γ at the point A corresponding to the value t of the parameter.

By virtue of (1) and (2), the curvature of Γ at the point t (i.e. at the moving point of the curve corresponding to the value t of the parameter) is expressed by the formula

$$K = \lim_{\Delta s\to 0}\frac{\alpha}{\Delta s} = \lim_{\Delta t\to 0}\frac{\sin\alpha}{\Delta s} = \lim_{\Delta t\to 0}\frac{\left|\dot{r}\times\dfrac{\Delta\dot{r}}{\Delta t}\right|}{|\dot{r}|\,|\dot{r}(t+\Delta t)|\,\dfrac{\Delta s}{\Delta t}} \tag{3}$$

(see the explanations given below), that is

$$K = \frac{1}{R} = \frac{|\dot{r}(t)\times\ddot{r}(t)|}{|\dot{r}(t)|^3} = \frac{\sqrt{(y'z''-z'y'')^2+(z'x''-x'z'')^2+(x'y''-y'x'')^2}}{(x'^2+y'^2+z'^2)^{3/2}} \tag{4}$$

In the third term in (3) we have replaced α by $\sin\alpha$ under the sign of limit. This is legitimate because if the values of α corresponding to a sequence of values of Δs tending to zero are positive ($\alpha > 0$) then $\sin\alpha \approx \alpha$ ($\alpha \to 0$), and therefore Theorem 2 in § 4.10 applies and if, beginning with a certain term of the sequence, we have $\alpha = 0$ then $\sin\alpha = \alpha = 0$ for these values and the second equality (3) is again valid.

If the parameter $t = s$ is the arc length of Γ then, as we know, $|\dot{r}(s)| = 1$ and the vector $\ddot{r}(s)$ is perpendicular to $\dot{r}(s)$, and hence

$$K = |\ddot{r}(s)| \quad \text{and} \quad R = \frac{1}{|\ddot{r}(s)|} \tag{5}$$

In the case of a plane curve ($z \equiv 0$) the expression of the curvature in terms of the coordinates is

$$K = \frac{|x'y''-y'x''|}{(x'^2+y'^2)^{3/2}} \tag{6}$$

If a plane curve is determined by an equation $y = f(x)$ where the function f possesses a continuous derivative with respect to x in a neighbourhood of a point x and the second derivative of $f(x)$ exists at the point x itself then, putting $t = x$ in the last formula, we obtain

$$K = \frac{\left|\frac{d^2y}{dx^2}\right|}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}} \tag{7}$$

(for the expression of the curvature in polar coordinates see Exercise 1 in § 7.26).

Let $A = (x, f(x))$ be a point of a curve Γ. Consider the point O lying at a distance of $R = 1/K$ from the point A of Γ on the normal to Γ at the point A drawn in the direction of concavity of Γ. This point is called the *centre of curvature of Γ at the point A*.

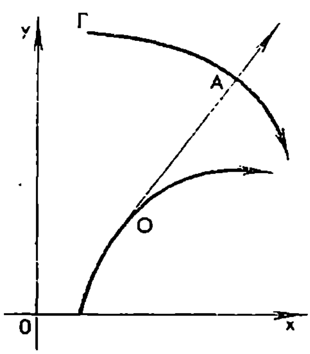

Fig. 6.8

The curve γ serving as the locus of the centres O of curvature of a plane curve Γ is termed the *evolute* of Γ. The curve Γ itself is called an *evolvent* (or *involute*) of γ.

In Fig. 6.8 we see a plane curve Γ on which the direction of increase of its arc length s is marked by an arrow. Here the second derivative $f''(x)$ is negative and therefore the radius of curvature at the point A is equal to

$$R = -\frac{(1+y'^2)^{3/2}}{y''} \tag{8}$$

The direction cosines of the tangent line (drawn in the direction of increase of s) are equal to $\frac{dx}{ds}$ and $\frac{dy}{ds}$ while the direction cosines of unit normal vector ν going from A in the direction of convexity of Γ are $-\frac{dy}{ds}$ and $\frac{dx}{ds}$ so that

$$\nu = \left(-\frac{dy}{ds}, \frac{dx}{ds}\right) \tag{9}$$

The coordinates ξ and η of the centre of curvature $O = (\xi, \eta)$ of the curve Γ at the point A are obviously given by the formulas

$$\xi = x+\frac{dy}{ds}R \quad \text{and} \quad \eta = y-\frac{dx}{ds}R \tag{10}$$

Relations (10) can thus be regarded as *equations of the evolute.*

In the case when the curve Γ is located and oriented as shown in Fig. 6.8 we have $y''_x < 0$ and $x'_t > 0$. Therefore from the formula

$$y''_x = \frac{x'_t y''_t - y'_t x''_t}{x'^3_t}$$

it follows that the numerator of its right member is negative, and hence (see (6))

$$R = \frac{(x_t'^2+y_t'^2)^{3/2}}{y_t'x_t''-x_t'y_t''}, \qquad s_t' = (x_t'^2+y_t'^2)^{1/2} \tag{11}$$

Consequently (10) shows that the equations of the evolute of the curve Γ can be written in parametric form as

$$\xi = x-y_t' \frac{x_t'^2+y_t'^2}{x_t'y_t''-y'x_t''}, \qquad \eta = y+x_t' \frac{x_t'^2+y_t'^2}{x_t'y_t''-y_t'x_t''} \tag{12}$$

It can be proved that these equations continue to hold for all the other possible locations and orientations of the curve relative to the coordinate axes.

The two basic facts below are characteristic of the relationship between a given curve (evolvent) and its evolute:

(i) The normal to the evolvent at its every point s coincides with a tangent line to the evolute.

Indeed (see (11)), we have $1/R = y_s'x_s''-x_s'y_s''$ and hence property (i) follows from the fact that the tangent vectors to Γ and γ at the corresponding points are mutually perpendicular (see (10)):

$$x_s'\xi_s'+y_s'\eta_s' = (x_s'^2+y_s'^2)+(y_s''x_s'-x_s''y_s')\,R+(x_s'y_s'-x_s'y_s')\,R_s' = 1-\frac{1}{R}\,R+0 = 0$$

(ii) There holds the equality

$$\sigma' = \pm R' \tag{13}$$

where R is the radius of curvature of Γ at A and σ is the length of the arc of γ joining O to an arbitrary fixed point of γ, the points A and O being two points, corresponding to each other, of the evolvent Γ and the evolute γ respectively. The signs "+" and "−" depend on the direction of the reckoning of σ.

Indeed, we can rewrite equality (10) in the form

$$\boldsymbol{r}-\boldsymbol{\varrho} = R\boldsymbol{\nu}$$

where $\boldsymbol{r}$ and $\boldsymbol{\varrho}$ are, respectively, the radius vectors of A and O whence $(\boldsymbol{r}-\boldsymbol{\varrho}, \boldsymbol{r}-\boldsymbol{\varrho}) = R^2$. On differentiating with respect to t (see the explanations below) we obtain

$$RR' = (\boldsymbol{r}-\boldsymbol{\varrho}, \dot{\boldsymbol{r}}-\dot{\boldsymbol{\varrho}}) = -(\boldsymbol{r}-\boldsymbol{\varrho}, \dot{\boldsymbol{\varrho}}) = -R|\dot{\boldsymbol{\varrho}}| = \mp R\sigma'$$

which implies (13).

The second equality in the above chain of relations follows from the fact that, by virtue of property (i) already proved, we have $(\boldsymbol{r}-\boldsymbol{\varrho}, \dot{\boldsymbol{r}}) = 0$; the third equality is implied by the relation $|\boldsymbol{r}-\boldsymbol{\varrho}| = R$ and by the fact that the vectors $\boldsymbol{r}-\boldsymbol{\varrho}$ and $\dot{\boldsymbol{\varrho}}$ go in one direction; the fourth equality follows from $|\dot{\boldsymbol{\varrho}}| = \pm\dot{\sigma}$ where the sign "+" or "−" depends on the choice of the direction of reckoning σ along the evolute.

If, for instance, σ increases together with t then $R' = -\sigma'$ whence $\int_{t_1}^{t_2} R'\,dx = -\int_{t_1}^{t_2} \sigma'\,dt$ and, by the Newton-Leibniz formula, $R_2-R_1 = \sigma_1-\sigma_2$ where σ_1

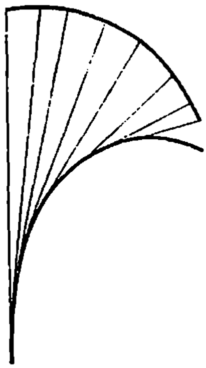

Fig. 6.9

and R_1 correspond to the value t_1 of the parameter and σ_2 and R_2 correspond to the value t_2. Thus, in the case under consideration the increase of the length of the arc of the evolute results in the decrease, equal to the former, of the radius of curvature of the evolvent.

Imagine that a flexible, inextensible string is stretched about the evolute. Now, let the string be unwound, under tension, from the evolute. At every moment in this process the string is tangent to the evolute and its free end point describes a curve which is nothing but an evolvent (Fig. 6.9). Since the string can be of arbitrary length the given evolute generates an infinite number of evolvents.

Example 1. Consider the cycloid

$$x = t-\sin t, \qquad y = 1-\cos t \tag{14}$$

The curve $\xi = t+\sin t$, $\eta = -1+\cos t$ serves as its evolute. Putting $t = \tau+\pi$ we arrive at the equations

$$\xi-\pi = \tau-\sin\tau, \qquad \eta+2 = 1-\cos\tau$$

We thus obtain the same curve (the cycloid) but shifted relative to the original

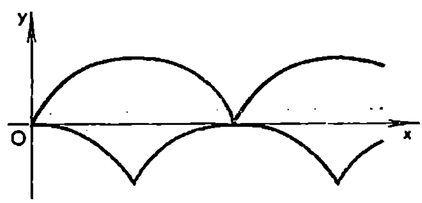

Fig. 6.10

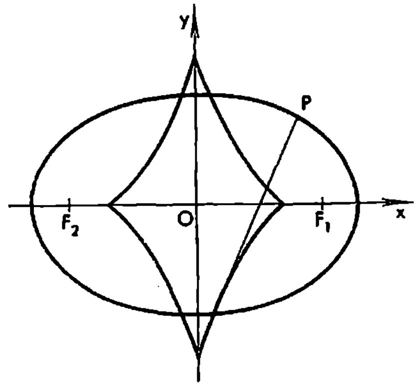

Fig. 6.11

curve; thus, an evolute of the cycloid is a cycloid congruent to the former (see Fig. 6.10).

Example 2. As an evolute of the ellipse $x = a\cos t$, $y = b\sin t$ ($a \geqslant b > 0$) serves the astroid (Fig. 6.11)

$$\xi = \frac{a^2-b^2}{a}\cos^3 t, \qquad \eta = -\frac{a^2-b^2}{b}\sin^3 t$$

(see § 6.5, Example 2).

§ 6.10. Osculating Plane and Moving Trihedron of a Curve

By the *osculating plane* of a curve Γ at its point A is meant the limiting position of the plane passing through the tangent line to Γ at the point A parallel to the tangent drawn through another point B of the curve as the latter moves along the curve tending to A.

Let us show that if a curve $r(t)$ has the continuous derivative $\dot{r}$ in a neighbourhood of a point t_0 and, besides, if the second derivative $\ddot{r}(t_0)$ at the point t_0 itself is such that

$$\dot{r}_0 \times \ddot{r}_0 = \dot{r}(t_0) \times \ddot{r}(t_0) \neq 0$$

then the osculating plane of the given curve at the point $t = t_0$ exists and is described by the equation

$$(\varrho - r_0)[\dot{r}_0 \times \ddot{r}_0] = 0 \tag{1}$$

where ϱ is the radius vector of the variable point of the osculating plane.

To this end we consider the increment of $\dot{r}(t)$ at $t = t_0$:

$$\Delta\dot{r}_0 = \dot{r}(t_0 + \Delta t) - \dot{r}(t_0)$$

The vector $\dot{r}_0 \times \frac{\Delta\dot{r}_0}{\Delta t} = \dot{r}(t_0) \times \frac{\dot{r}(t_0+\Delta t)}{\Delta t}$ is orthogonal (perpendicular) to the vectors $\dot{r}(t_0)$ and $\dot{r}(t_0+\Delta t)$ applied at the point A and, consequently, to the plane passing through them. Since this vector tends to the vector $\dot{r} \times \ddot{r}_0 \neq 0$ as $\Delta t \to 0$ the indicated plane tends to occupy the position of the plane passing through A perpendicularly to $\dot{r}_0 \times \ddot{r}_0$; therefore this limiting plane is nothing but the osculating plane of Γ at A, and its equation is obviously (1).

There is also an alternative definition of the osculating plane: *the osculating plane of a curve Γ at its point A is the limiting position of the moving plane passing through the point A and two other points A_1 and A_2 ($A_1 \neq A_2$) of the curve Γ (distinct from the point A) when the latter points tend to A moving along Γ.*

It can be shown that under the requirements imposed above on $r(t)$ in the neighbourhood of t_0 there exists the osculating plane of Γ at the point $t = t_0$ in the sense of the second definition as well and it is determined by equation (1). Thus, it coincides with the osculating plane understood in the sense of the first definition.

In Cartesian coordinates equation (1) is written in the form

$$\begin{vmatrix} x-x_0 & y-y_0 & z-z_0 \\ x_0' & y_0' & z_0' \\ x_0'' & y_0'' & z_0'' \end{vmatrix} = 0 \tag{2}$$

where x, y and z are the coordinates of the variable point of the osculating plane, $r_0 = (x_0, y_0, z_0)$, $\dot{r}_0 = (x_0', y_0', z_0')$ and $\ddot{r}_0 = (x_0'', y_0'', z_0'')$.

The vectors $\dot{r}$ and $\ddot{r}$ issued from the point $A = (x, y, z)$ are obviously in the osculating plane S at A. If $t = s$ is the arc length of Γ then $\dot{r}(s)$ is unit vector and the vector $\ddot{r}(s)$ is perpendicular to $\dot{r}(s)$.

From the moving point A of the curve under consideration (satisfying the conditions imposed) we can issue the following three unit vectors α, β and γ determining the so-called *natural* rectangular coordinate system in the vicinity of A:

$$\left.\begin{array}{ll}\alpha = \dfrac{\dot{r}}{|\dot{r}|} = \dot{r}(s) & \text{which is the unit vector of the tangent} \\[2ex] \beta = \dfrac{\ddot{r}(s)}{|\ddot{r}(s)|} = R\ddot{r}(s) = R\dfrac{d\alpha}{ds} & \text{which is the unit vector of the so-called } \textit{principal normal} \\[2ex] \gamma = \alpha \times \beta & \text{which is the unit vector of the so-called } \textit{binormal.}\end{array}\right\} \quad (3)$$

It should be noted that the direction of α depends on the parameter t in the sense that the replacement of t by $-t$ changes the direction of α to the opposite.

As to the vector β, it is defined by means of the parameter s which is the arc length of Γ. The replacement of s by $-s$ or by $s+s_0$ where s_0 is a constant does not lead to any change in $\ddot{r}(s)$ (since the differentiation is performed twice); therefore β is an *invariant*: its direction is not connected with the parametrization of a curve.

By a *normal* to a curve Γ at a point A we naturally mean a straight line passing through the point A perpendicularly to the tangent to Γ at that point. Among these normals there is exactly one belonging to the osculating plane S (of the curve Γ at the point A). It is called the *principal normal.* The vector $\ddot{r}(s)$ is obviously in the plane S and is perpendicular to the tangent and therefore lies in the principal normal. It is convenient to agree that the vector $\ddot{r}(s)$ or, which is the same, the vector β determines the positive direction along the principal normal. We can also say that β is a unit vector issued from the point A, lying in S and directed toward the concavity of the curve Γ (more precisely, of its projection on S).

Let us lay off from the point A a vector of length R (R is the radius of curvature of Γ at A) in the direction of β. Its terminus O is then called the *centre of curvature of* Γ at A. In the case of a plane curve Γ this definition coincides with the one stated in § 6.10 for the centre of curvature of a plane curve. The centre of curvature O is obviously determined by the vector

$$\varrho = r + R\beta = r + \frac{\ddot{r}(s)}{|\ddot{r}(s)|^2}$$

(see § 6.10, formula (5)).

Finally, the vector γ is defined as a unit vector perpendicular to α and β and directed so that the triad (α, β, γ) is oriented like the Cartesian coordinate system (x, y, z) in which the given curve is considered.

The straight line in which the vector γ (applied at the point A) lies is called the *binormal* of Γ at A. The vector γ determines the positive direction along the binormal.

The three vectors α, β and γ applied at the variable point A of the curve Γ form the so-called *moving* (or *natural*) *trihedron* of Γ.

In the case of a plane curve the normal to the curve defined in § 6.8 obviously coincides with the principal normal (on condition that $\ddot{r}(s) \neq 0$). At the same time the positive directions along these lines (along the normal and along the principal normal which coincide in the case of a plane curve) were, respectively, defined in § 6.8 and in the present section on the basis of different principles and therefore they may not coincide.

To investigate the behaviour of the vector $r(s)$ in the vicinity of a point s_0 it is convenient to consider the difference $r(s)-r(s_0)$ in the rectangular coordinates determined by the unit vectors α, β and γ.

Let us assume that $r = r(s)$ where s is the arc length of Γ and that $r_0 = r(s_0)$ is the radius vector of the point $A_0 \in \Gamma$ in whose vicinity the behaviour of Γ is studied.

Then α, β and γ are functions of s. From the equalities $\gamma\alpha = 0$ and $\gamma\gamma = 1$ it follows that the projections of $\frac{d\gamma}{ds}$ on the directions of α and γ are equal to zero:

$$\frac{d\gamma}{ds}\alpha = -\gamma\frac{d\alpha}{ds} = -|\ddot{r}|\,\gamma\beta = 0, \qquad \frac{d\gamma}{ds}\gamma = 0$$

Therefore

$$\frac{d\gamma}{ds} = \frac{1}{T}\beta \tag{4}$$

where

$$\frac{1}{T} = \frac{d\gamma}{ds}\beta \tag{5}$$

The number $1/T$ is called the *torsion* (or the *second curvature*) of Γ at the point $A \in \Gamma$ under consideration. It is obvious that it can also be defined as a number whose absolute value $\left|\frac{1}{T}\right|$ is equal to $\left|\frac{d\gamma}{ds}\right|$, that is to the rate of change of unit vector of the binormal relative to s. As to the sign of $1/T$, *it is positive or negative depending on whether the projection of* $\frac{d\gamma}{ds}$ *on the direction of* β *is positive or negative.*

Now we write down the so-called *Frenet-Serret** *formulas:*

$$\frac{d\alpha}{ds} = \frac{\beta}{R}, \qquad \frac{d\gamma}{ds} = \frac{\beta}{T}, \qquad \frac{d\beta}{ds} = -\frac{\alpha}{R} - \frac{\gamma}{T} \tag{6}$$

The first two of them are already proved (see formulas (3) and (4)); the third formula is derived as follows. On differentiating the identities $\beta\alpha =$

* J. F. Frenet (1816-1900) and J. A. Serret (1819-1885), French mathematicians. — *Tr.*

$= \gamma\beta = 0$ and $\beta\beta = 1$ with respect to s we obtain

$$\frac{d\beta}{ds}\alpha = -\beta\frac{d\alpha}{ds} = -\frac{1}{R}, \qquad \frac{d\beta}{ds}\gamma = -\beta\frac{d\gamma}{ds} = -\frac{1}{T} \quad \text{and} \quad \frac{d\beta}{ds}\beta = 0$$

Consequently

$$\frac{d\beta}{ds} = \left(\frac{d\beta}{ds}\alpha\right)\alpha + \left(\frac{d\beta}{ds}\beta\right)\beta + \left(\frac{d\beta}{ds}\gamma\right)\gamma = -\frac{\alpha}{R} - \frac{\gamma}{T}$$

From (6) it follows that if the curvature of Γ is identically equal to zero ($1/R \equiv 0$) then $\frac{d\alpha}{ds} \equiv 0$ whence we conclude that $r(s) = r(s_0) + (s-s_0)\dot{r}(s_0)$, that is Γ is a straight line. If the torsion of Γ is identically equal to zero ($1/T \equiv 0$) then $\frac{d\gamma}{ds} = 0$, $(\gamma r)' = \gamma\alpha + \dot{\gamma} r = 0$ and $\gamma r = \text{const}$; hence, in this case Γ is a plane curve.

Example 1. The screw line (circular helix) Γ determined by the equations

$$x = a\cos\theta, \qquad y = a\sin\theta, \qquad z = h\theta, \qquad (a, h > 0)$$

has the arc length s whose derivative with respect to θ is $\frac{ds}{d\theta} = \sqrt{x_\theta'^2 + y_\theta'^2 + + t_\theta'^2} = \sqrt{a^2+h^2}$. Therefore the unit vector α of the tangent to Γ has the projections

$$\frac{dx}{ds} = \frac{\frac{dx}{d\theta}}{\frac{ds}{d\theta}} = -\frac{a\sin\theta}{\sqrt{a^2+h^2}}, \qquad \frac{dy}{ds} = \frac{a\cos\theta}{\sqrt{a^2+h^2}} \quad \text{and} \quad \frac{dz}{ds} = \frac{h}{\sqrt{a^2+h^2}}$$

Further, we have

$$\frac{d^2x}{ds^2} = -\frac{a\cos\theta}{a^2+h^2}, \quad \frac{d^2y}{ds^2} = \frac{-a\sin\theta}{a^2+h^2}, \quad \frac{d^2z}{ds^2} = 0$$

$$K = |\ddot{r}| = \frac{a}{a^2+h^2}, \quad \beta = \frac{\ddot{r}}{|\ddot{r}|} = -\cos\theta\, i - \sin\theta\, j + 0\cdot k$$

which shows that the principal normal of Γ is parallel to the xy-plane and goes in the direction toward the axis of the circular cylinder around which the screw line is wound. Finally,

$$\gamma = \frac{h\sin\theta}{\sqrt{a^2+h^2}}\, i - \frac{h\cos\theta}{\sqrt{a^2+h^2}}\, j + \frac{a}{\sqrt{a^2+h^2}}\, k$$

$$\frac{d\gamma}{ds} = \frac{h\cos\theta}{a^2+h^2}\, i + \frac{h\sin\theta}{a^2+h^2}\, j + 0k \quad \text{and} \quad \frac{1}{T} = \frac{d\gamma}{ds}\beta = -\frac{h}{a^2+h^2}$$

Let us suppose that $r(s)$ possesses the continuous derivatives up to the third order inclusive in a neighbourhood of a point $s = s_0$. Then we can write Taylor's formula

$$r - r_0 = (s-s_0)\dot{r}_0 + \frac{(s-s_0)^2}{2!}\ddot{r}_0 + \frac{(s-s_0)^3}{3!}\dddot{r}_0 + o((s-s_0)^3) \qquad (s \to s_0) \quad (7)$$

in which the remainder is a vector whose length tends to zero faster than $|s-s_0|^3$ when $s \to s_0$.

Since the unit vectors

$$\alpha = \dot{r}_0, \quad \beta = \frac{\ddot{r}_0}{|\ddot{r}_0|} \quad \text{and} \quad \gamma = \alpha \times \beta \quad (\ddot{r} \neq 0)$$

are pairwise mutually orthogonal we have the equalities

$$r(s)-r(s_0) = \lambda(s)\alpha+\mu(s)\beta+\nu(s)\gamma \tag{8}$$

$$\lambda(s) = (r-r_0, \alpha), \quad \mu(s) = (r-r_0, \beta), \quad \nu(s) = (r-r_0, \gamma)$$

$$\lambda'(s_0) = \frac{d}{ds}(r-r_0, \alpha)\Big|_{s=s_0} = (\dot{r}(s), \alpha)\Big|_{s=s_0} = 1 \tag{9}$$

$$\mu'(s_0) = (\dot{r}_0, \beta) = 0, \quad \mu''(s_0) = (\ddot{r}_0, \beta) \neq 0 \tag{10}$$

$$\nu'(s_0) = \nu''(s_0) = 0, \quad \nu'''(s_0) = (\dddot{r}(s_0), \gamma) \neq 0 \tag{11}$$

The last condition that $\nu'''(s_0) \neq 0$ has been introduced additionally. Usually it holds (the case $\nu'''(s_0) = 0$ can be regarded as exceptional).

If we look at the curve Γ in the direction of the binormal we see the projection Γ_γ of Γ on the plane of the vectors α and β (the osculating plane), the equation of the projection being

$$(r-r_0)_\gamma = \lambda(s)\alpha+\mu(s)\beta$$

By virtue of properties (9) and (10), the curve Γ_γ has the tangent T at the point A_0, and in a sufficiently small neighbourhood of A_0 it lies entirely either above T or below T (Fig. 6.12).

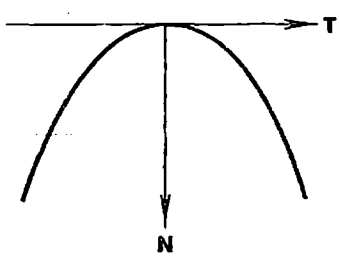

Fig. 6.12

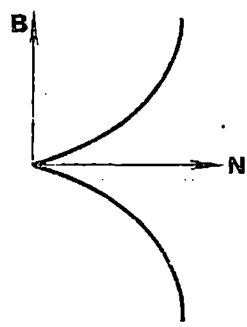

Fig. 6.13

If we look at Γ in the direction of the tangent we see its projection Γ_α on the plane of the vectors β and γ*, the equation of the projection being $(r-r_0)_\alpha = \mu(s)\beta+\nu(s)\gamma$. Hence, Γ_α is determined by the equations

$$\eta = \mu(s), \qquad \zeta = \nu(s) \tag{12}$$

* This is the so-called *normal plane of Γ at the point A*. — *Tr.*

where (η, ζ) are the rectangular coordinates connected with the unit vectors $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$.

By (10), we have

$$\eta = \frac{(s-s_0)^2}{2}\mu''(s_0)+O((s-s_0)^3) \qquad (s \to s_0)$$

and by virtue of (11),

$$\zeta = \frac{(s-s_0)^3}{3!}\nu'''(s_0)+o((s-s_0)^3) \qquad (s \to s_0)$$

Thus, for small $|s-s_0|$ the sign of $\eta = \mu(s)$ is retained and is independent of the sign of $s-s_0$. We also have

$$\left(\frac{d\zeta}{d\eta}\right)_0 = \lim_{s\to s_0}\frac{\nu'(s)}{\mu'(s)} = \lim_{s\to s_0}\frac{\nu''(s)}{\mu''(s)} = \frac{\nu''(s_0)}{\mu''(s_0)} = 0$$

and the sign of $\zeta = \nu(s)$ varies together with the change of the sign of the difference $s-s_0$; hence the curve (12) has a cuspidal point at the origin of the coordinates (η, ζ) (see Fig. 6.13; also see § 7.23).

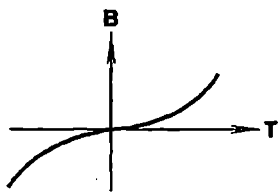

Fig. 6.14

Finally, if we look at Γ along the principal normal we see its projection Γ_β on the plane of the vectors $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$*, the corresponding equation being

$$(r-r_0)_\beta = \lambda(s)\boldsymbol{\alpha}+\nu(s)\boldsymbol{\gamma}$$

By virtue of (9) and (11), the curve $\xi = \lambda(s)$, $\zeta = \nu(s)$ possesses the properties

$$\left(\frac{d\zeta}{d\xi}\right)_0 = \frac{\nu'(s_0)}{\lambda'(s_0)} = 0, \quad \left(\frac{d^2\zeta}{d\xi^2}\right)_0 = \frac{\lambda'(s_0)\nu''(s_0)-\lambda''(s_0)\nu'(s_0)}{\lambda'(s_0)^3} = 0$$

and

$$\left(\frac{d^3\zeta}{d\xi^3}\right)_0 = \nu'''(s_0) \neq 0$$

This shows that the curve Γ_β has a point of inflection at A_0 (Fig. 6.14).

* This plane is called the *rectifying plane of Γ at A*. – *Tr.*

§ 6.11. Asymptote

Let us consider a curve (or a branch of a curve) Γ determined by an equation

$$y = f(x) \qquad (x > N) \tag{1}$$

where $f(x)$ is a function continuous for any $x > N$. The variable point $A = (x, f(x))$ of this curve can then be regarded as being dependent on x.

Besides, let there be a straight line L specified by an equation

$$y = ax+b \tag{2}$$

where a and b are constant numbers. If the distance from the variable point A of the curve to the line L tends to zero as x increases indefinitely the line L is called an *asymptote of the curve* Γ (corresponding to $x \to +\infty$).

Now, let L be the asymptote of Γ for $x \to +\infty$. The equation of L in normal form is written

$$\frac{y-ax-b}{\sqrt{1+a^2}} = 0$$

Therefore the distance from the point $A = (x, f(x))$ of the curve Γ to L is equal to $\varrho(x) = |f(x)-ax-b|/\sqrt{1+a^2}$. By the hypothesis, L is an asymptote of Γ for $x \to +\infty$, and therefore $\lim\limits_{x\to+\infty} \varrho(x) = 0$ whence

$$\lim_{x\to+\infty} [f(x)-ax-b] = 0 \tag{3}$$

Consequently $\lim\limits_{x\to+\infty} \left[\frac{f(x)}{x} - a - \frac{b}{x}\right] = \lim\limits_{x\to+\infty} \left[\frac{f(x)}{x} - a\right] = 0$, i.e.

$$\lim_{x\to+\infty} \frac{f(x)}{x} = a \tag{4}$$

What has been said provides the following rule for finding an asymptote of Γ for $x \to +\infty$. We must begin with computing limit (4). If it does not exist the asymptote of Γ for $x \to \infty$ does not exist either. If limit (4) exists and is equal to a finite number a we must compute the limit

$$\lim_{x\to+\infty} \{f(x)-ax\} = b \tag{5}$$

If limit (5) does not exist the curve Γ has no asymptote for $x \to +\infty$. If it exists the constants a and b thus obtained determine the straight line $y = ax+b$ which is the sought-for asymptote of Γ for $x \to +\infty$. Since limits (4) and (5) are unique (provided they exist) the continuous curve (or a branch of a curve) specified by equality (1) either has no asymptote for $x \to +\infty$ or possesses a unique asymptote of that kind.

An asymptote is defined analogously for the case $x \to -\infty$ when a continuous curve (or a branch of a curve) is defined by an equation

$$y = f(x) \qquad (x < -N) \tag{6}$$

The notion of an asymptote for $x \to \infty$ of a curve

$$y = f(x) \qquad (N \leqslant |x|) \tag{7}$$

(consisting of two branches corresponding to $x > N$ and $x < -N$) is treated similarly (in the arguments above we should take $x \to -\infty$ in Case (6) and $x \to \infty$ in Case (7)).

If a curve (or a branch of a curve) Γ is represented by an equation $y = f(x)$ $(a < x < b)$ where $f(x)$ is a continuous function on an interval (a, b) possessing the property that $\lim\limits_{\substack{x \to a \\ x > a}} f(x) = +\infty$, it is natural to call the straight line $x = a$ a *(vertical) asymptote* of Γ. Usually the line $x = a$ is called an *asymptote* of Γ if the continuous function $f(x)$ tends to infinity as $x \to a$ and is strictly monotone in a left-hand or right-hand (or both) neighbourhood of the point $x = a$. In such a case the curve Γ can be represented by an equation $x = \varphi(y)$ where y is positive or negative and is sufficiently large in its absolute value, and then the straight line $x = a$ can obviously be interpreted as an asymptote of Γ in the sense indicated at the beginning of the present section.

Example 1. Let us analyse the shape of the graph Γ of the function $f(x) = \dfrac{1}{x} + x + e^{-x}$.

We have $\lim\limits_{x \to +\infty} \dfrac{f(x)}{x} = \lim\limits_{x \to +\infty} \left(\dfrac{1}{x^2} + 1 + \dfrac{e^{-x}}{x}\right) = 1$. But in the case $x \to -\infty$ the limit of this ratio is equal to $+\infty$.

Furthermore, we have $\lim\limits_{x \to +\infty} [f(x) - x] = 0$. Hence $y = x$ is the asymptote of Γ for $x \to +\infty$. The straight line $x = 0$ is also an asymptote of Γ for x tending to 0 from the right and from the left:

$$\lim_{\substack{x \to 0 \\ x > 0}} f(x) = +\infty \quad \text{and} \quad \lim_{\substack{x \to 0 \\ x < 0}} f(x) = -\infty$$

The roots of the equation $f'(x) = 0$ cannot be found analytically. At the same time it is evident that

$$f''(x) = \frac{2}{x^3} + e^{-x} > 0 \qquad (x > 0)$$

$$\lim_{x \to 0+0} f'(x) = -\infty \quad \text{and} \quad \lim_{x \to +\infty} f'(x) = 1$$

Thus, $f'(x)$ is strictly increasing on $(0, \infty)$ and there exists exactly one value $x_0 > 0$ for which $f'(x_0) = 0$. The function $f(x)$ obviously decreases on $(0, x_0)$ from $+\infty$ to $f(x_0)$ and then increases, its graph having the asymptote $y = x$ for $x \to +\infty$ and lying entirely above the latter.

On the interval $(-\infty, 0)$ we have

$$f'(x) = -\frac{1}{x^2} + 1 - e^{-x} < 0$$

because $-1/x^2 < 0$ and $1-e^{-x} < 0$. Taking this into account we readily see that $f(x)$ is strictly decreasing on $(-\infty, 0)$ from $+\infty$ to $-\infty$. Further, we have

$$f''(x) = \frac{2}{x^3}+e^{-x}, \qquad f'''(x) = -\frac{6}{x^4}-e^{-x} < 0 \text{ on } (-\infty, 0)$$

$$\lim_{x\to-\infty} f''(x) = +\infty \quad \text{and} \quad \lim_{x\to 0,\, x<0} f''(x) = -\infty$$

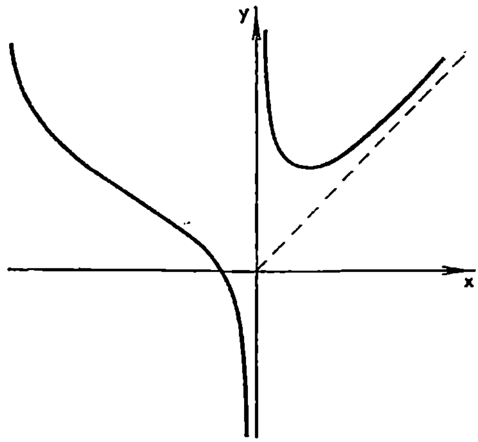

Fig. 6.15

Therefore there is exactly one point x_1 on $(-\infty, 0)$ at which the graph of $f(x)$ has a point of inflection. On $(-\infty, x_1)$ the graph of f is convex downward and on $(x_1, 0)$ convex upward (see the schematic graph in Fig. 6.15).

Example 2. The curve $y = \sqrt{x}$ $(x \geqslant 0)$ has no asymptote because, although the limit

$$\lim_{x\to+\infty} \frac{\sqrt{x}}{x} = 0$$

exists, the limit

$$\lim_{x\to+\infty} (\sqrt{x}-0\cdot x) = \infty$$

is not equal to a finite number.

§ 6.12. Change of Variables

Let y be a function of x and $x = \varphi(t)$ be a given function of t. Then y is a (composite) function of t. The derivatives $\frac{dy}{dx}, \frac{dy^2}{dx^2}, \ldots$ of y with respect to x are expressed in terms of the derivatives $\frac{dy}{dt}, \frac{d^2y}{dt^2}, \ldots$ and in terms of the known derivatives $\frac{dx}{dt} = \varphi'(t), \frac{d^2x}{dt^2} = \varphi''(t), \ldots$ according to the formulas (see § 6.5, formulas (5) and (6))

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \frac{dx}{dt} \neq 0 \tag{1}$$

and

$$\frac{d^2y}{dx^2} = \frac{\frac{dx}{dt}\frac{d^2y}{dt^2} - \frac{dy}{dt}\frac{d^2x}{dt^2}}{\left(\frac{dx}{dt}\right)^3} \tag{2}$$

A more complicated situation arises when it is necessary to express the derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots$ in terms of $\frac{dv}{dt}, \frac{d^2v}{dt^2}, \ldots$ where $v = \lambda(y)$ and $x = \mu(t)$ are given functions. In this case the function $\lambda(y)$ is supposed to have the inverse function $(y = \lambda^{-1}(v))$.

It is obvious that

$$\frac{dy}{dv}\frac{dv}{dt} = \frac{dy}{dx}\frac{dx}{dt} \tag{3}$$

whence

$$\frac{dy}{dx} = \frac{\frac{dy}{dv}\frac{dv}{dt}}{\frac{dx}{dt}} \tag{4}$$

and we thus have expressed $\frac{dy}{dx}$ in terms of $\frac{dv}{dt}$ and the given functions of v and t.

On differentiating (3) with respect to t we obtain

$$\frac{d^2y}{dv^2}\left(\frac{dv}{dt}\right)^2 + \frac{dy}{dv}\frac{d^2v}{dt^2} = \frac{d^2y}{dx^2}\left(\frac{dx}{dt}\right)^2 + \frac{dy}{dx}\frac{d^2x}{dt^2} \tag{5}$$

Now, replacing $\frac{dy}{dx}$ in (5) by (4) and solving the resultant equation with respect to $\frac{d^2y}{dx^2}$ we arrive at the expression of $\frac{d^2y}{dx^2}$ in terms of $\frac{dv}{dt}, \frac{d^2v}{dt^2}$ and the known functions of v and t.

To obtain $\frac{d^3y}{dx^3}$ we should differentiate (5) with respect to t, replace $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ by the expressions already found and solve the resultant equation with respect to $\frac{d^3y}{dx^3}$. The same technique is used for the computation of higher-order derivatives $\frac{d^ky}{dx^k}$.

CHAPTER 7

Differential Calculus. Functions of Several Variables

§ 7.1. Open Set

Let us choose an arbitrary point $x^0 = (x_1^0, \ldots, x_n^0)$ in the n-dimensional space $R_n = R$. By a *closed sphere* (or a *closed ball*) of radius $r > 0$ with centre at the point x^0 is meant the set of points $x = (x_1, \ldots, x_n) \in R$ satisfying the inequality $|x-x^0| = \left[\sum_1^n (x_j-x_j^0)^2\right]^{1/2} \leq r$.

By an *open sphere* (or an *open ball*) *of radius r with centre at* x^0 we shall mean the set of points x for which the strict inequality $|x-x^0| < r$ is fulfilled*.

A closed *rectangle* in R (or, which is the same, a *closed rectangular parallelepiped in* R) is a set of points $x \in R$ whose coördinates satisfy inequalities $a_j \leq x_j \leq b_j$ ($a_j < b_j$; $j = 1, \ldots, n$). In the case $n = 3$ this is a real rectangular parallelepiped with faces parallel to the axes of the rectangular Cartesian coordinates (x_1, x_2, x_3).

We can also define an *open rectangle* in R as a set of points x satisfying strict inequalities $a_j < x_j < b_j$ ($j = 1, \ldots, n$).

A set of points x whose coordinates satisfy inequalities $|x_j-x_j^0| \leq a$ ($j = 1, \ldots, n$) where $a > 0$ is a given number is naturally called a (*closed*) *cube* in R with centre at the point x^0 and side of length $2a$. For $n = 3$ this is of course an ordinary cube with faces parallel to the coordinate axes of the (Cartesian) rectangular coordinate system (x_1, x_2, x_3).

Finally, an *open cube* (in R) is defined by means of inequalities $|x_j-x_j^0| < a$ ($j = 1, \ldots, n$).

The obvious inequalities $|x_j-x_j^0| \leq \left(\sum_1^n (x_j-x_j^0)^2\right)^{1/2} < r \quad (j = 1, \ldots, n)$

mean that if a point x belongs to a ball of radius r with centre at x^0 it also belongs to the cube with side $2r$ and centre at the same point. Thus, *the cube with side $2r$ and centre at x^0 entirely contains the ball of radius r and centre at the same point.* On the other hand, if a point x belongs to a cube $|x_j-x_j^0| < a$

* Here "sphere" is meant in the sense of a solid sphere (or ball). The confusion with "sphere" understood in the sense of a spherical surface can always be avoided by the use of the adjectives "closed" or "open". – *Tr.*

$(j = 1, \ldots, n)$ it satisfies the inequality $\left(\sum_{1}^{n} (x_j - x_j^0)^2\right)^{1/2} < \sqrt{n}a$ showing that *the ball of radius* $a\sqrt{n}$ *and centre at* x^0 *entirely contains the cube with side* $2a$ *and centre at the same point* (see § 6.2 (12)).

We have considered open balls and cubes; the same argument applies of course to closed balls and cubes.

Let us consider an arbitrary set E of points $x \in R$. By definition, x^0 is called an *interior* point of the set E if there is an open ball with centre at that point entirely contained in E. Here "ball" can be replaced by "cube" since every ball contains a cube with the same centre and vice versa.

A set is said to be *open* if all its points are interior points. This definition can be stated alternatively as follows: *a set E is open if the fact that a given point belongs to it implies that it is an interior point of E.*

It follows that *an empty set is an open set.*

An open ball

$$|x - x^0| < r \tag{1}$$

is also an open set. Indeed, let y be a point belonging to the given ball, i.e. $|y - x^0| = \varrho < r$, and let x be an arbitrary point belonging to the ball

$$|x - y| < \varepsilon \qquad (\varepsilon < r - \varrho) \tag{2}$$

For this point we have

$$|x - x^0| = |x - y + y - x^0| \leqslant |x - y| + |y - x^0| < \varepsilon + \varrho < r$$

which shows that ball (2) is contained in ball (1).

Let the reader prove that *an open rectangle and*, in particular, *an open cube are open sets.*

The intersection G_1G_2 *of two open sets* G_1 *and* G_2 *is an open set.* Indeed, let a point x^0 belong to G_1G_2. Since x^0 is an interior point of both G_1 and G_2 there exist two concentric open balls with centre at x^0 one of which belongs to G_1 and the other to G_2. Their intersection is obviously an open ball (coinciding with the smallest of them) belonging to G_1G_2.

It is readily seen that *the sum (union) of a finite or countable number of open sets is an open set.* But it should be stressed that the intersection of a *countable* number of open sets is not necessarily open, for instance, the intersection of the open balls $|x| < 1/k$ $(k = 1, 2, \ldots)$ is a single point (the origin).

By a *neighbourhood* of a point $x^0 \in R_n$ is meant an arbitrary open set containing that point. It is obvious that *the intersection of two neighbourhoods of* x^0 *is again a neighbourhood of* x^0.

What has been said allows us to restate the definition of an interior point of a set E: x^0 *is an interior point of E if there is a neighbourhood of* x^0 *belonging to E.* Indeed, if x^0 is an interior point in the sense of the first definition then there is an open ball with centre at x^0 belonging to E; such a ball is a neighbourhood of x^0 contained in E. Conversely, if x^0 is an interior point in the sense of the second definition there exists a neighbourhood of $\boldsymbol{x}^0$, belonging to E, which contains an open ball with centre at $\boldsymbol{x}^0$ since this neighbourhood is an open set.

In our course we shall deal with many examples of open sets defined in a strict mathematical manner but now, using solely the geometrical intuition, we can roughly say that if the boundary of a geometrical object is deleted from it the resultant object is an open set.

In the sections that follow we shall deal with functions $f(\boldsymbol{x}) = f(x_1, \ldots, x_n)$ of n variables $x_1, \ldots, x_n$ or, which is the same, of the point (n-tuple) $\boldsymbol{x} = (x_1, \ldots, x_n)$, defined on open sets in the n-dimensional space R_n.

A set E is said to be *(arcwise) connected* if its any two points x' and x'' can be joined with a continuous curve entirely lying within E, that is if there exists a continuous vector function $\boldsymbol{x} = \boldsymbol{x}(t)$, $0 \leq t \leq 1$, such that $\boldsymbol{x}(0) = \boldsymbol{x}'$, $\boldsymbol{x}(1) = \boldsymbol{x}''$ and $\boldsymbol{x}(t) \in E$ (see § 6.5).

By a *line segment* $\overline{\boldsymbol{x}'\boldsymbol{x}''}$ is meant the curve with parametric representation $\boldsymbol{x}(t) = t\boldsymbol{x}' + (1-t)\boldsymbol{x}''$, $t \in [0, 1]$, which is obviously a continuous curve connecting $\boldsymbol{x}'$ and $\boldsymbol{x}''$.

A set E is said to be *convex* if together with its any two points $\boldsymbol{x}'$ and $\boldsymbol{x}''$ the entire line segment connecting them belongs to E. See the examples at the end of § 7.3.

Remark 1. A cube Δ with side $2d$ ($d > 0$) in R_n can be defined by means of inequalities of the form

$$\Delta = \{a_j \leq x_j \leq b_j;\ j = 1, \ldots, n\}$$

where $b_j - a_j = 2d$ ($j = 1, \ldots, n$). It is easily seen that Δ can be represented as the union of 2^n cubes of the form $\{\lambda_j \leq x_j \leq \mu_j; j = 1, \ldots, n\}$ where we put $\lambda_j = a_j$, $\mu_j = (a_j + b_j)/2$ or $\lambda_j = (a_j + b_j)/2$, $\mu_j = b_j$ in all the possible combinations. In this case we say that the cube Δ is partitioned into 2^n equal (congruent) cubes with edges of length d.

Remark 2. A cube in R_n defined above reduces in the case $n = 3$ to an ordinary (three-dimensional) cube with edges parallel to the coordinate axes. The general definition of an n-dimensional cube involves the introduction of orthogonal transformations of coordinates of the form

$$x_j - x_j^0 = \sum_{k=1}^{n} \alpha_{jk}\xi_k$$

where $\boldsymbol{x}^0 = (x_1^0, \ldots, x_n^0) \in R_n$ and α_{jk} are real numbers for which $\sum_{k=1}^{n} \alpha_{jk}^2 = 1$ and $\sum_{k=1}^{n} \alpha_{ik}\alpha_{jk} = 0$ ($i \neq j$; $i, j = 1, \ldots, n$). For example, by a closed cube Δ in R_n with centre at $\boldsymbol{x}^0 \in R_n$ and sides of length $2d$ is meant a set of points $\boldsymbol{x} \in R_n$ which turns into a set of the form $\{|\xi_k| \leq d; k = 1, \ldots n\}$ after an appropriate (dependent on Δ) orthogonal transformation of coordinates. Similar remarks should be made for n-dimensional rectangles.

§ 7.2. Limit of a Function

A function $f(x) = f(x_1, \ldots, x_n)$ *is said to have a limit at a point* $x^0 = (x_1^0, \ldots, x_n^0)$ equal to a number A and designated as

$$\lim_{x \to x^0} f(x) = \lim_{\substack{x_j \to x_j^0 \\ (j=1, \ldots, n)}} f(x_1, \ldots, x_n) = A \tag{1}$$

(also written $f(x) \to A$, $x \to x^0$) if the function is defined in a neighbourhood of the point x^0 except possibly at that point itself and if for any sequence of points x^k $(k = 1, 2, \ldots)$ belonging to this neighbourhood, different from x^0 and tending to x^0 (see § 6.3), there exists the limit

$$\lim_{\substack{|x^k - x^0| \to 0 \\ x^k \neq x^0}} f(x^k) = A \tag{2}$$

Another equivalent definition reads: *a function f has a limit at a point x^0 equal to A* if it is defined in a neighbourhood of x^0 except possibly at the point x^0 itself and if for any $\varepsilon > 0$ there is $\delta > 0$ such that

$$|f(x) - A| < \varepsilon \tag{3}$$

for all x satisfying the inequality

$$0 < |x - x^0| < \delta \tag{4}$$

Inequality (4) in this definition can be replaced by

$$0 < \sum_{j=1}^{n} |x_j - x_j^0| < \delta \qquad (j = 1, \ldots, n)$$

In other words, given any $\varepsilon > 0$, there is a neighbourhood $U(x^0)$ such that inequality (3) holds for all $x \neq x^0$ belonging to this neighbourhood.

The proof of the equivalence of the first and second definitions for the general case of dimension n is analogous to the proof given in § 4.1 for the one-dimensional case.

Now we state *Cauchy's criterion* for the existence of limit (which is proved like in the one-dimensional case; see § 4.1, Theorem 5).

For a function f to have a (finite) limit at a point x^0 it is necessary and sufficient that for any $\varepsilon > 0$ there exists a neighbourhood $U(x^0)$ of the point x^0 (in particular, a cube or a ball with centre at x^0) *such that the inequality*

$$|f(x) - f(x')| < \varepsilon$$

holds for all the points $x, x' \in U(x^0)$ distinct from x^0.

If a number A is the limit of $f(x)$ at a point x^0 then obviously A serves as the limit of the function $f(x^0 + h)$ of the argument h at the point $h = 0$!, that is $\lim_{h \to 0} f(x^0 + h) = A$, and vice versa.

Let us consider a function f defined at all points of a neighbourhood of a point x^0 except possibly at the point x^0 itself, an arbitrary vector $\omega = (\omega_1, \ldots, \omega_n)$ of unit length $(|\omega| = 1)$ and a scalar $t \geq 0$. The collection of the points of the form $\boldsymbol{x}^0+t\omega$ $(0 \leqslant t)$ is spoken of as a *ray issued from* $\boldsymbol{x}^0$ *in the direction of the vector* ω.

For every ω we can take the function

$$f(\boldsymbol{x}^0)+t\omega) = f(x_1^0+t\omega_1, \ \ldots, \ x_n^0+t\omega_n) \qquad (0 < t < \delta_\omega)$$

dependent on the scalar variable t, the number δ_ω being dependent on ω. The limit of this function (of one variable t) as $t \to 0$, provided it exists, will be called the *limit of* f *at the point* x^0 *in the direction of the vector* ω:

$$\lim_{t\to 0,\ t>0} f(\boldsymbol{x}^0+t\omega) = \lim_{t\to 0,\ t>0} f(x_1^0+t\omega_1, \ \ldots, \ x_n^0+t\omega_n)$$

In particular, if ω is the unit vector $e^j = (0, \ldots, 0, 1, 0, \ldots, 0)$ of the x_j-axis we can speak of the *limit of* f *at the point* x^0 *in the direction of the positive* x_j*-axis*, that is

$$\lim_{\substack{t\to 0\\ t>0}} f(\boldsymbol{x}^0+te^j) = \lim_{\substack{t\to 0\\ t>0}} f(x^0, \ \ldots, \ x_{j-1}^0, x_j^0+t, x_{j+1}^0, \ \ldots, \ x_n^0)$$

or *in the direction of the negative* x_j*-axis*:

$$\lim_{\substack{t\to 0\\ t>0}} f(x^0-te^j) = \lim_{\substack{t\to 0\\ t>0}} f(x_1^0, \ \ldots, \ x_{j-1}^0, x_j^0-t, x_{j+1}^0, \ \ldots, \ x_n^0)$$

If a function f has a limit equal to A at a point $\boldsymbol{x}^0$ then obviously it has a limit at this point in any direction and all these limits coincide with the former limit. The converse is not true: in the general case a function can have a limit at $\boldsymbol{x}^0$ in any direction equal to one and the same number A and at the same time it can have no limit at $\boldsymbol{x}^0$.

Example 1. Let us consider the two functions

$$\text{(i)}\ f(x,y) = \frac{x^3+y^3}{x^2+y^2} \quad \text{and} \quad \text{(ii)}\ \varphi(x,y) = \frac{x^2-y^2}{x^2+y^2}$$

The functions f and φ are defined throughout the xy-plane except at the origin (0, 0). We have

$$|f(x,y)| \leqslant \frac{2(x^2+y^2)^{3/2}}{x^2+y^2} = 2(x^2+y^2)^{1/2}$$

whence

$$\lim_{x,\ y\to 0} f(x,y) = 0$$

(for any $\varepsilon > 0$ we put $\delta = \varepsilon/2$; then $|f(x,y)| < \varepsilon$ provided that $(x^2+y^2)^{1/2} < \delta$).

As for the function $\varphi(x, y)$, taking an arbitrary constant number k we obtain

$$\varphi(x, kx) = \frac{1-k^2}{1+k^2}$$

whence follows that the limits of φ at (0, 0) in different directions are, generally, different from each other, and therefore the function φ has no limit at (0, 0).

Example 2. Let us consider the spiral in the xy-plane described by the equation $\varrho = \theta$ $(0 < \theta \leqslant 2\pi)$ where ϱ is the polar radius of the moving point (x, y) and θ is its polar angle. Let us take the function $\psi(x, y)$ defined by means of the following rule (see Fig. 7.1):

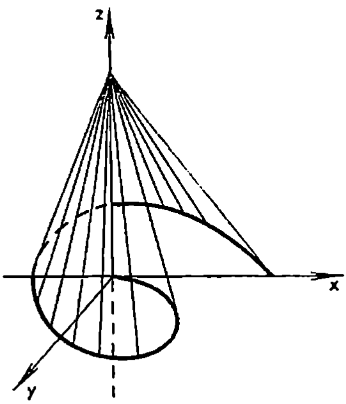

Fig. 7.1

$$\psi(0, 0) = 1, \quad \psi(x, y) = 0 \quad \text{for}$$
$$\varrho = \sqrt{x^2+y^2} \geqslant \theta > 0$$

and ψ is a linear function on any line segment connecting the point (0, 0) with the variable point of the spiral.

It is readily seen that $\lim\limits_{t\to 0} \psi(tx, ty) = 1$ for any point $(x, y) \neq (0, 0)$, that is the limit of ψ at (0, 0) in any direction exists, whereas the limit of ψ at (0, 0) does not exist. For, if the variable point (x, y) approaches the point (0, 0) along a curve lying between the spiral and the x-axis in the first quadrant of the xy-plane, then $\psi(x, y) = 0$ along this curve.

We shall write $\lim\limits_{x\to x^0} f(x) = \infty$ if the function f is defined in a neighbourhood of x^0 except possibly at x^0 and if, given any $N > 0$, there is $\delta > 0$ such that $|f(x)| > N$ when $0 < |x-x^0| < \delta$.

We can also introduce the notion of the limit of f for $x \to \infty$:

$$\lim_{x\to\infty} f(x) = A \tag{5}$$

For instance, if A is a finite number, equality (5) should be understood in the sense that, for any $\varepsilon > 0$, there exists $N > 0$ such that the function f is defined at all the points x for which $|x| > N$ and that the inequality $|f(x)-A| < \varepsilon$ is fulfilled for such x.

There hold the equalities

$$\lim_{x\to x^0} (f(x)\pm\varphi(x)) = \lim_{x\to x^0} f(x) \pm \lim_{x\to x^0} \varphi(x) \tag{6}$$

$$\lim_{x\to x^0} (f(x)\,\varphi(x)) = \lim_{x\to x^0} f(x) \lim_{x\to x^0} \varphi(x) \tag{7}$$

and

$$\lim_{x\to x^0} \frac{f(x)}{\varphi(x)} = \frac{\lim\limits_{x\to x^0} f(x)}{\lim\limits_{x\to x^0} \varphi(x)} \qquad \left(\lim_{x\to x^0} \varphi(x) \neq 0\right) \tag{8}$$

where, in particular, there can be $x^0 = \infty$. As usual, these relations mean that the (finite) limits on their left-hand sides exist if the limits of f and φ exist. As an instance, let us prove (7).

Suppose that $x^k \to x^0$ $(x^k \neq x^0)$; then

$$\lim_{x^k \to x^0} (f(x^k)\varphi(x^k)) = \lim_{x^k \to x^0} f(x^k) \lim_{x^k \to x^0} \varphi(x^k) = \lim_{x \to x^0} f(x) \lim_{x \to x^0} \varphi(x) \quad (9)$$

Hence, the limit on the left-hand side of (9) exists and is equal to the right-hand member of (9), and, since the sequence $\{x^k\}$ has been taken quite arbitrarily, this limit is equal to the limit of the function $f(x)\varphi(x)$ at the point x^0.

Theorem 1. *If a function f has a nonzero limit at a point x^0, that is*

$$\lim_{x \to x^0} f(x) = A \neq 0$$

then there exists $\delta > 0$ such that for all x satisfying the inequalities

$$0 < |x - x^0| < \delta \quad (10)$$

the given function satisfies the inequality

$$|f(x)| > |A|/2 \quad (11)$$

and the sign of $f(x)$ coincides with that of A for these values of x.

Indeed, putting $\varepsilon = |A|/2$ we can find $\delta > 0$ such that the inequality

$$|f(x) - A| < |A|/2 \quad (12)$$

is fulfilled for all x satisfying inequalities (10). Therefore we have $|A|/2 > |A - f(x)| \geqslant |A| - |f(x)|$ for such x's, i.e. inequality (11) holds.

For the indicated x it follows from (12) that

$$A - \frac{|A|}{2} < f(x) < A + \frac{|A|}{2}$$

whence $A/2 < f(x)$ for $A > 0$ and $f(x) < A/2$ for $A < 0$, which means that the signs of $f(x)$ and A coincide.

Remark. In § 7.10 we shall state a more general definition of the limit for a function defined on an arbitrary set.

§ 7.3. Continuous Functions

A function $f(x) = f(x_1, \ldots, x_n)$ is said to be *continuous at a point* $x^0 = (x_1^0, \ldots, x_n^0)$ if it is defined in a neighbourhood of x^0 including the point x^0 itself and if the limit of the function at the point x^0 exists and is equal to the value of $f(x)$ at x^0:

$$\lim_{x \to x^0} f(x) = f(x^0) \quad (1)$$

The condition of the continuity of f at x^0 can be rewritten in the equivalent form

$$\lim_{h\to 0} f(x^0+h) = f(x^0) \tag{1'}$$

This means that the function $f(x)$ is continuous at the point x^0 if the function $f(x^0+h)$ of the argument h is continuous at the point $h = 0$ (and vice versa).

We can also introduce the notion of the *increment of f at the point x^0* corresponding to the increment $h = (h_1, \ldots, h_n)$ of the vector argument $x = (x_1, \ldots, x_n)$ by putting

$$\Delta_h f(x^0) = f(x^0+h)-f(x^0)$$

and restate the definition of the continuity of f at x^0 in terms of the increments: *the function f is continuous at x^0 if*

$$\lim_{h\to 0} \Delta_h f(x^0) = \lim_{h_1, \ldots, h_n \to 0} [f(x_1^0+h_1, \ldots, x_n^0+h_n)-f(x_1^0, \ldots, x_n^0)] = 0 \tag{1''}$$

From formulas (6)-(8) of § 7.2 we directly deduce the following theorem.

Theorem 1. *The sum, the difference, the product and the quotient of two functions $f(x)$ and $\varphi(x)$ continuous at a point x^0 are continuous functions at that point (provided that $\varphi(x^0) \neq 0$ in the case of the quotient).*

A constant c can be regarded as the function $f(x) = c$ of $x = (x_1, \ldots, x_n)$. It is continuous for any x since

$$f(x+h)-f(x) = c-c = 0 \to 0 \qquad (h \to 0)$$

A more complicated function is $f_j(x) = x_j$ ($j = 1, \ldots, n$) where the index j can be equal to one of the numbers $1, \ldots, n$. It is also continuous (as function of $x = (x_1, \ldots, x_n)$!). Indeed, let $h = (h_1, \ldots, h_n)$; then

$$|f_j(x+h)-f_j(x)| = |(x_j+h_j)-x_j| = |h_j| \leq |h| \to 0 \qquad (h \to 0)$$

A finite number of operations of addition, subtraction and multiplication performed on the functions x_j and on constants leads to a function called a *polynomial of x* (or a *polynomial in the variables $x_1, \ldots, x_n$*). On the basis of the above-stated properties we conclude that the polynomials are continuous functions in R_n (for all $x \in R_n$). The ratio P/Q of two polynomials is called a *rational function* which is obviously continuous throughout R_n except at the points x at which $Q(x) = 0$.

The function

$$P(x) = x_1^3-x_2^3+x_1^2x_3+2x_1^2x_2-3x_3^2+4$$

is an example of a polynomial of the third degree in the variables x_1, x_2 and x_3.

There holds the following general theorem.

Theorem 2. *Let $f(x_1, \ldots, x_m)$ be a continuous function at a point $(x_1^0, \ldots, x_m^0)$ in the space R_m and let $m < n$.*

If this function is regarded as the function

$$F(x_1, \ldots, x_n) = f(x_1, \ldots, x_m)$$

of $x = (x_1, \ldots, x_n)$ *then the latter function F is continuous* (*as function of* $x = (x_1, \ldots, x_n)$ *in the space* R_n) *at any point of the form* $(x_1^0, \ldots, x_m^0, x_{m+1}^0, \ldots, x_n^0)$ *where the numbers* $x_{m+1}^0, \ldots, x_n^0$ *are arbitrary.*

Indeed, if $h = (h_1, \ldots, h_n)$ then

$$\Delta_h F(x^0) = F(x_1^0+h_1, \ldots, x_n^0+h_n) - F(x_1^0, \ldots, x_n^0) =$$
$$= f(x_1^0+h_1, \ldots, x_m^0+h_m) - f(x_1^0, \ldots, x_m^0) \to 0 \qquad (h \to 0)$$

Let $k = (k_1, \ldots, k_n)$ be a nonzero vector with nonnegative integral components k_j $(j = 1, \ldots, n)$. Such a vector will be briefly spoken of as a *nonnegative integral vector*. Now we shall use the following notation: if $x = (x_1, \ldots, x_n)$ is a point of R_n then

$$x^k = x_1^{k_1} x_2^{k_2}, \ldots, x_n^{k_n} \tag{2}$$

This expression is a continuous function for all $x \in R_n$ because it is equal to the product of a finite number of factors of the form x_j each of which is a continuous function of x. We also introduce the notation

$$|k| = \sum_{j=1}^{n} k_j \tag{3}$$

which will be used only for nonnegative integral vectors k and which must not be confused with the length $|k| = \left(\sum_{j=1}^{n} k_j^2\right)^{1/2}$ of an arbitrary n-dimensional vector. Let us form the sum

$$P_n(x) = \sum_{|k| \leq n} a_k x^k = \sum_{|k| \leq n} a_{k_1, \ldots, k_n} x_1^{k_1}, \ldots, x_n^{k_n}$$

extended over all the possible integral vectors k with $|k| \leq n$ where $a_k = a_{k_1}, \ldots, {}_{k_n}$ are constant coefficients supplied with nonnegative integral vector indices k. Such a function $P_n(x)$ will be spoken of as a *polynomial in* x *of degree* n.

Next we state the following theorem.

Theorem 3. *Let a function* $f(x) = f(x_1, \ldots, x_m)$ *be continuous at a point* $x^0 = (x_1^0, \ldots, x_m^0)$ *belonging to the space* R_m *(consisting of the points* x*) and let a function* $\varphi_j(u) = \varphi_j(u_1, \ldots, u_n)$ *be continuous at a point* $u^0 = (u_1^0, \ldots, u_n^0)$ *belonging to the space* R_n *(consisting of the points* u*). Also let* $\varphi_j(u^0) = x_j^0$ $(j = 1, \ldots, m)$. *Then the (composite) function*

$$F(u) = f(\varphi_1(u), \varphi_2(u), \ldots, \varphi_m(u))$$

is continuous (with respect to u*) at the point* u^0.

Proof. The function f being continuous at x^0, for any $\varepsilon > 0$ there is $\delta > 0$ such that f is defined for all x satisfying the condition $|x_j - x_j^0| < \delta$

$(j = 1, \ldots, m)$ and such that the inequality $|f(x)-f(x^0)| < \varepsilon$ holds for these x's. Further, since the functions φ_j are continuous at the point u^0 of the space R_n there is $\eta > 0$ such that for the points $u \in R_n$ belonging to the ball $|u-u^0| < \eta$ there hold the inequalities

$$|\varphi_j(u)-\varphi_j(u^0)| < \delta \qquad (j = 1, \ldots, m)$$

Then the inequality

$$|F(u)-F(u^0)| = |f(\varphi_1(u), \ldots, \varphi_n(u))-f(\varphi_1(u^0), \ldots, \varphi_m(u^0))| < \varepsilon$$

also holds, which completes the proof of the theorem.

A function will be spoken of as an *elementary function* of the variables $x_1, \ldots, x_n$ if it can be obtained from these variables and from constant numbers with the aid of a finite number of operations of addition, subtraction, multiplication, division and operations φ where φ symbolizes elementary functions of one variable (see § 1.3). The functions

$$\text{(i) } \sin \ln \sqrt{1+x^2+y^2} = \varphi_1$$

$$\text{(ii) } \sin^2 x+\cos 3(x+y) = \varphi_2$$

and

$$\text{(iii) } \ln \frac{x-y}{x+y} = \varphi_3$$

can serve as examples of elementary functions of several variables.

Using Theorems 1-3 we can readily check that the functions φ_1 and φ_2 are continuous throughout the xy-plane while the function φ_3 is obviously defined and continuous at the points (x, y) for which the fraction $(x-y)/(x+y)$ is finite and positive.

From Theorem 1 in § 7.2 and from the definition of the continuity of a function at a point follows directly

Theorem 4. *If a function $f(x) = f(x_1, \ldots, x_n)$ is continuous at a point x^0 and assumes a nonzero value at x^0 it retains sign in a neighbourhood of that point which coincides with the sign of $f(x^0)$.*

Corollary. *Let a function $f(x)$ be defined and continuous throughout R_n (i.e. at all the points of R_n). Then the set G of the points x at which the function satisfies the inequality $f(x) > c$ (or $f(x) < c$) where c is an arbitrary constant is an open set.*

Indeed, the function $F(x) = f(x)-c$ is continuous on R_n and the set of all points x where $F(x) > 0$ coincides with G. Let $x^0 \in G$; then there is a ball

$$|x-x^0| < \delta$$

on which $F(x) > 0$; this means that this ball belongs to G and therefore $x^0 \subset G$ is an interior point of G.

The case $f(x) < c$ is treated similarly.

Example. Consider the three functions

$$\text{(i) } f_1(x) = \sum_1^n \frac{x_k^2}{a_k} \ (a_k > 0), \quad \text{(ii) } f_2(x) = \sum_1^n |x_k|$$

and

$$\text{(iii) } f_3(x) = \max_k |x_k|$$

They are defined and continuous in the space R_n. The continuity of f_1 and f_2 is obvious and the continuity of f_3 follows from the following calculations:

$$|f_3(x+h)-f_3(x)| = |\max_k |x_k+h_k| - \max_k |x_k|| \leq$$

$$\leq \max_k |x_k+h_k-x_k| = \max_k |h_k| \to 0 \qquad (|h| \to 0)$$

We see now that the sets of points x for which the inequalities $f_i(x) < c$ ($i = 1, 2, 3$) hold are open. The first of them is the interior of an ellipsoid in the n-dimensional space; in the case $n = 2$ the second and the third sets are the interiors of the squares shown, respectively, in Figs. 7.2 and 7.3.

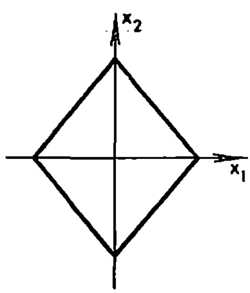

Fig. 7.2

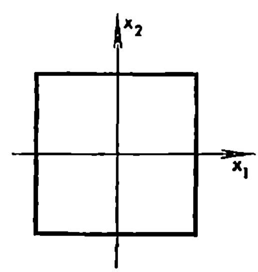

Fig. 7.3

These three sets are convex since the inequalities $f_i(x) < c$ and $f_i(y) < c$ imply $f_i(tx+(1-t)y) < c$, $0 \leq t \leq 1$.

The inequalities $f_i(x) > c > 0$ specify the exteriors of these figures.

§ 7.4. Partial Derivatives and Directional Derivative

In this section we shall consider functions f defined on an arbitrary open set $G \subset R_n$.

By the *increment of f at a point $x = (x_1, \ldots, x_n) \in G$ with respect to the variable x_j with step h* we shall mean the difference

$$\Delta_{x_j h} f(x) = f(x_1, \ldots, x_{j-1}, x_j+h, x_{j+1}, \ldots, x_n) - f(x_1, \ldots, x_n)$$

where h is a real number which is sufficiently small so that this difference makes sense.

The *partial derivative of f with respect to x_j at the point x* is defined as the limit

$$f'_{x_j} = \frac{\partial f}{\partial x_j} = \frac{\partial f(x)}{\partial x_j} = \lim_{h\to 0} \frac{\Delta_{x_j h} f(x)}{h} \qquad (j = 1, \ldots, n)$$

provided that it exists. The partial derivative $\frac{\partial f(x)}{\partial x_j}$ is nothing but the ordinary derivative of the function $f(x_1, \ldots, x_n)$ regarded as a function of the variable x_j alone for fixed $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_n$.

A function $z = f(x, y)$ of two variables is represented geometrically by the locus of points $(x, y, f(x, y))$ where $(x, y) \in G$, that is by a surface in the three-dimensional space in which rectangular Cartesian coordinates (x, y, z) are introduced. It is evident that *the derivative $f'_x(x_0, y_0)$* (provided it exists) *is equal to the tangent of the angle of inclination (to the x-axis) of the tangent line to the section of that surface by the plane $y = y_0$ drawn through the point with abscissa x_0.*

The derivatives $\frac{\partial f}{\partial x_j}$ $(j = 1, \ldots, n)$ are also termed the *partial derivatives of the first order (the first partial derivatives) of f.*

The expressions $\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f = \frac{\partial^2 f}{\partial x_i \partial x_j}$ $(i, j = 1, \ldots, n)$ are called *partial derivatives of the second order (the second partial derivatives).* For $i = j$ they are denoted as

$$\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_i} f = \frac{\partial^2 f}{\partial x_i^2} \qquad (i = 1, \ldots, n)$$

The expressions $\frac{\partial}{\partial x_k}\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j} f = \frac{\partial^3 f}{\partial x_k \partial x_i \partial x_j}$ are *partial derivatives of the third order (the third partial derivatives)* and so on. To denote partial derivatives we write

$$\underbrace{\frac{\partial}{\partial x_k} \cdots \frac{\partial}{\partial x_k}}_{m \text{ times}} = \frac{\partial^m}{\partial x_k^m} \text{ and } \frac{\partial}{\partial z}\frac{\partial}{\partial y}\frac{\partial}{\partial y}\frac{\partial}{\partial z}\frac{\partial}{\partial z}\frac{\partial}{\partial x} = \frac{\partial^6}{\partial z\, \partial y^2\, \partial z^2\, \partial x}$$

and the like (also see § 7.8 (14)).

As will be shown later, in many important cases these operations of successive partial differentiation can be interchanged without affecting the ultimate result.

Another important notion connected with differentiation is the *directional derivative* (it is not used in the case of one independent variable).

Let $\omega = (\omega_1, \ldots, \omega_n)$ be an arbitrary unit vector. The *(directional) derivative of a function f at a point x in the direction of ω* (along ω) is defined as the limit

$$\frac{\partial f}{\partial \omega} = \lim_{\substack{t\to 0\\ t>0}} \frac{f(x+t\omega)-f(x)}{t}$$

(provided it exists). It should be stressed that in the computation of this limit it is assumed that t tends to zero retaining *positive* sign; therefore we can also say that $\frac{\partial f(x)}{\partial \omega}$ is the *right-hand derivative of the function $f(x+\omega t)$ with respect to t at the point $t = 0$.*

By analogy with functions of one variable we can speak of the right-hand and left-hand partial derivatives with respect to x_j. It should also be taken into account that *the directional derivative along the positive x_j-axis coincides with the right-hand partial derivative with respect to x_j while the directional derivative along the negative x_j-axis has the sign opposite to that of the left-hand derivative with respect to x_j.*

§ 7.5. Differentiable Function. Tangent Plane

For the sake of simplicity we shall consider the three-dimensional case; in the general case of dimension n the argument is quite analogous. The case $n = 1$ was considered separately in § 5.2.

Let a function

$$u = f(x, y, z)$$

be defined in an open set $G \subset R_3$ and let it possess the continuous partial derivatives of the first order $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$ at a point $(x, y, z) \in G$. This implies automatically that these partial derivatives also exist in some neighbourhood of (x, y, z) although they may not be continuous at points different from (x, y, z). Let us consider the increment of f at (x, y, z) corresponding to an increment $(\Delta x, \Delta y, \Delta z)$ where $|\Delta x|, |\Delta y|, |\Delta z| < \delta$ and δ is sufficiently small so that the point $(x+\Delta x, y+\Delta y, z+\Delta z)$ does not fall outside the indicated neighbourhood. Then there hold the equalities

$$\Delta u = f(x+\Delta x, y+\Delta y, z+\Delta z)-f(x, y, z) = \tag{1}$$

$$= f(x+\Delta x, y+\Delta y, z+\Delta z)-f(x, y+\Delta y, z+\Delta z)+ \tag{2}$$

$$+f(x, y+\Delta y, z+\Delta z)-f(x, y, z+\Delta z)+ \tag{3}$$

$$+f(x, y, z+\Delta z)-f(x, y, z) = \tag{4}$$

$$= f'_x(x+\theta_1\Delta x, y+\Delta y, z+\Delta z)\,\Delta x+f'_y(x, y+\theta_2\Delta y, z+\Delta z)\,\Delta y+$$
$$+f'_z(x, y, z+\theta_3\Delta z)\,\Delta z = \tag{5}$$

$$= (f'_x(x, y, z)+\varepsilon_1)\,\Delta x+(f'_y(x, y, z)+\varepsilon_2)\,\Delta y+$$
$$+(f'_z(x, y, z)+\varepsilon_3)\,\Delta z = \tag{6}$$

$$= f'_x(x, y, z)\,\Delta x+f'_y(x, y, z)\,\Delta y+f'_z(x, y, z)\,\Delta z+o(\varrho)\ (\varrho \to 0), \tag{7}$$

$$0 < \theta_1, \theta_2, \theta_3 < 1,\ \varrho = \sqrt{\Delta x^2+\Delta y^2+\Delta z^2},\ \varepsilon_1, \varepsilon_2, \varepsilon_3 \to 0\ (\varrho \to 0) \tag{8}$$

(see the explanations below).

The passage from (2) to the first term in (5) is justified as follows. By the hypothesis, the function $f(\xi, y+\Delta y, z+\Delta z)$ of the variable ξ (considered for fixed $y+\Delta y$ and $z+\Delta z$) has the derivative (with respect to ξ) on the interval $[x, x+\Delta x]$, and Lagrange's mean-value theorem is applicable to it. The second and third terms in (5) are explained similarly. The passage from (5) to (6) is purely formal; for instance, we simply put

$$f'_x(x+\theta_1\Delta x, y+\Delta y, z+\Delta z) = f'_x(x, y, z)+\varepsilon_1$$

But the fact that $\varepsilon_1 \to 0$ as $\varrho \to 0$ is not formal; it is a consequence of the assumption that f'_x is continuous at (x, y, z). Finally, the passage from (6) to (7) reduces to the assertion that there holds the relation

$$\varepsilon_1 \Delta x+\varepsilon_2 \Delta y+\varepsilon_3 \Delta z = o(\varrho) \qquad (\varrho \to 0)$$

which is proved (see § 6.2 (8)) as follows:

$$|\varepsilon_1 \Delta x+\varepsilon_2 \Delta y+\varepsilon_3 \Delta z|/\varrho \leqslant \varrho\sqrt{\varepsilon_1^2+\varepsilon_2^2+\varepsilon_3^2}/\varrho = \sqrt{\varepsilon_1^2+\varepsilon_2^2+\varepsilon_3^2} \to 0 \qquad (\varrho \to 0)$$

We have thus proved the following important theorem.

Theorem 1. *If a function $u = f(x, y, z)$ has continuous first-order partial derivatives at a point (x, y, z) its increment at this point corresponding to a sufficiently small increment $(\Delta x, \Delta y, \Delta z)$ is representable in the form*

$$\Delta u = \frac{\partial f}{\partial x}\Delta x+\frac{\partial f}{\partial y}\Delta y+\frac{\partial f}{\partial z}\Delta z+o(\varrho)\ (\varrho \to 0),\ \varrho = \sqrt{\Delta x^2+\Delta y^2+\Delta z^2} \quad (9)$$

where the values of the partial derivatives are taken at the point (x, y, z).

The values of the partial derivatives on the right-hand side of (9) being independent of Δx, Δy, Δz, the conditions of Theorem 1 imply that the increment of f at (x, y, z) corresponding to the increment $(\Delta x, \Delta y, \Delta z)$ can be written in the form

$$\Delta u = A\,\Delta x+B\,\Delta y+C\,\Delta z+o(\varrho) \qquad (\varrho \to 0) \qquad (10)$$

where the numbers A, B and C are independent of Δx, Δy and Δz.

Let us state the following definition: if the increment of a function u at a point (x, y, z) can be represented as sum (10) for sufficiently small $(\Delta x, \Delta y, \Delta z)$ where A, B and C are constant numbers independent of Δx, Δy and Δz the function f is said to be *differentiable* at the point (x, y, z).

Thus, *the differentiability of f at (x, y, z) means that the increment Δf of the function f at this point can be written as the sum of two terms one of which is the linear function $A\,\Delta x+B\,\Delta y+C\,\Delta z$* of $(\Delta x, \Delta y, \Delta z)$ *called the principal linear part of the increment Δf and the other is an expression dependent on the increments Δx, Δy and Δz (which can be a complicated function of Δx, Δy, Δz in the general case) such that when Δx, Δy and Δz are made to tend to zero it tends to zero faster than $\varrho = \sqrt{\Delta x^2+\Delta y^2+\Delta z^2}$.*

It is easily seen that if a function f is differentiable at a point (x, y, z), that is if its increment is representable in form (10), it has the derivatives at that point equal respectively to A, B and C:

$$\frac{\partial f}{\partial x} = A, \quad \frac{\partial f}{\partial y} = B, \quad \frac{\partial f}{\partial z} = C \qquad (11)$$

As an instance, let us prove the first relation (11). Suppose that the increment of f at (x, y, z) is representable in form (10). Putting $\Delta x = h$ and $\Delta y =$

$= \Delta z = 0$ in (10) we obtain the equality $\Delta_{xh} u = Ah + o(h)$ $(h \to 0)$. On dividing the latter by h and passing to the limit we arrive at

$$\lim_{h \to 0} \frac{\Delta_{xh} u}{h} \frac{\partial f}{\partial x} = A$$

From what has been said follows

Theorem 2. *For a function f to be differentiable at a point it is necessary that it possess partial derivatives at that point and it is sufficient that it have continuous partial derivatives at the point.*

It follows from (10) that *a function differentiable at a point is continuous at that point.*

Example 1. The function $f(x, y, z)$ equal to zero on the coordinate planes $x = 0$, $y = 0$ and $z = 0$ and equal to unity at the other points of R_3 obviously has zero partial derivatives at the point $(0, 0, 0)$ but is discontinuous at that point and therefore cannot be differentiable at it. We see that the existence of partial derivatives of a function at a point is not sufficient for the differentiability and even for the continuity of the function at that point.

We should also stress a distinction between the one-dimensional and many-dimensional cases. In the case $n = 1$ the property of differentiability of f at x is expressed in the form of the equality $\Delta f = A\,\Delta x + o(\Delta x)$ and consequently if $A \neq 0$ the second term on the right-hand side tends to zero as $\Delta x \to 0$ faster than the principal part. But this is no longer the case for $n > 1$; for instance, if $n = 3$ then for any three numbers A, B and C not equal to zero simultaneously we can always make Δx, Δy and Δz tend to zero in such a way that the equality $A\,\Delta x + B\,\Delta y + C\,\Delta z = 0$ constantly holds; then the term $o(\varrho)$ in (10) is even greater than the principal linear part. By the way, if we make Δx, Δy and Δz tend to zero so that the proportionality relation $\Delta x : \Delta y : \Delta z = A : B : C$ is fulfilled, the principal part of the increment will be exactly of the order of ϱ and the term $o(\varrho)$ will tend to zero faster than the principal part.

If a function f is differentiable at a point (x, y, z) then the principal linear part of its increment at that point is also called the *differential of f at the point (x, y, z) corresponding to the increments Δx, Δy and Δz of the independent variables*. The differential is denoted $df = \frac{\partial f}{\partial x}\,\Delta x + \frac{\partial f}{\partial y}\,\Delta y + \frac{\partial f}{\partial z}\,\Delta z$.

Let us consider a surface S represented by a function $z = f(x, y)$ defined in a neighbourhood of a point (x_0, y_0).

A plane L_0 is termed the *tangent plane* to the surface S at its point $P_0 = (x_0, y_0, z_0)$ $\big(z_0 = f(x_0, y_0)\big)$ if the distance $r(P, L_0)$ from its moving point $P = (x, y, z)$ to L_0 tends to zero faster than the distance ϱ from P to P_0:

$$r(P, L_0) = o(\varrho) \qquad (\varrho \to 0) \tag{12}$$

Theorem 3. *If a function f is differentiable at a point (x_0, y_0), the surface represented by this function has the uniquely determined tangent plane at the* *point $P_0(x_0, y_0, z_0)$ specified by the equation*

$$z-z^0 = \left(\frac{\partial f}{\partial x}\right)_0 (x-x_0)+\left(\frac{\partial f}{\partial y}\right)_0 (y-y_0) \tag{13}$$

where $(\ )_0$ symbolizes the fact that the expression in the parentheses is taken for $x = x_0, y = y_0$.

Proof. Let f be a differentiable function at a point (x_0, y_0), S be the surface represented by this function and L_0 be the plane described by equation (13). The variable point $P \in S$ has the coordinates $(x, y, f(x, y))$. As is known from analytic geometry, the distance between P and L_0 is expressed by the formula (see the explanations below)

$$r(P, L_0) = \frac{1}{M}\left|f(x, y)-f(x_0, y_0)-\left(\frac{\partial f}{\partial x}\right)_0 (x-x_0)-\left(\frac{\partial f}{\partial y}\right)_0 (y-y_0)\right| =$$
$$= o(r) = o(\varrho), \qquad r \to 0,\ \varrho \to 0 \tag{14}$$

where

$$r = \sqrt{(x-x_0)^2+(y-y_0)^2}$$
$$\varrho = \sqrt{(x-x_0)^2+(y-y_0)^2+[f(x, y)-f(x_0, y_0)]^2}$$

Here $M = \sqrt{\left(\frac{\partial f}{\partial x}\right)_0^2+\left(\frac{\partial f}{\partial y}\right)_0^2+1}$. The second equality (14) is fulfilled because of the assumption that f is differentiable at the point (x_0, y_0). As for the last equality, it says that the quantity $o(r)$ $(r \to 0)$ in (14) possesses the property that its ratio to ϱ tends to zero as $\varrho \to 0$. Indeed, if $\varrho \to 0$ then also $r \to 0$ $(0 \leqslant r \leqslant \varrho!)$ and therefore

$$\left|\frac{o(r)}{\varrho}\right| = \left|\frac{o(r)}{r}\cdot\frac{r}{\varrho}\right| \leqslant \left|\frac{o(r)}{r}\right| \to 0$$

We have proved that the plane specified by equation (13) is the tangent plane to S at the point (x_0, y_0).

Now we must show that there are no other tangent planes to S at the point (x_0, y_0) distinct from L_0. To this end suppose that there is a tangent plane to S at the point P_0 described by an equation

$$A(x-x_0)+B(y-y_0)-C(z-z_0) = 0,\ A^2+B^2+C^2 = 1 \tag{15}$$

Then

$$A(x-x_0)+B(y-y_0)-C(f-f_0) = \underset{\varrho\to 0}{o(\varrho)} = \underset{r\to 0}{o(r)} \tag{16}$$

The last equality in (16) is explained as follows. By the differentiability of f at the point (x_0, y_0), we have

$$|f-f_0| = \left|\left(\frac{\partial f}{\partial x}\right)_0 (x-x_0)+\left(\frac{\partial f}{\partial y}\right)_0 (y-y_0)+\varepsilon r\right| \leqslant$$
$$\leqslant \sqrt{\left(\frac{\partial f}{\partial x}\right)_0^2+\left(\frac{\partial f}{\partial y}\right)_0^2}\, r+\varepsilon r < Cr$$

where C is a constant independent of r. Taking into account that ε is bounded (since $\varepsilon \to 0$ for $r \to 0$) we can write

$$\varrho = \sqrt{r^2+|f-f_0|^2} < C_1 r$$

where C_1 is independent of r. Consequently,

$$\left|\frac{o(\varrho)}{r}\right|_{r\to 0} = \left|\frac{o(\varrho)}{\varrho}\,\frac{\varrho}{r}\right| \leqslant C_1 \left|\frac{o\varrho}{\varrho}\right|_{\varrho\to 0} \to 0$$

On putting $y = y_0$ in (16), dividing by $x-x_0$ and passing to the limit as $x \to x_0$ we obtain

$$A - C\left(\frac{\partial f}{\partial x}\right)_0 = 0$$

Similarly, putting $x = x_0$ in (16), dividing by $y-y_0$ and passing to the limit as $y \to y_0$ we receive

$$B - C\left(\frac{\partial f}{\partial y}\right)_0 = 0$$

We now see that $C \neq 0$ because, if otherwise, there would be $A = B = C = 0$; hence, the plane in question is described by equation (13).

Example 2. Let us consider the function

$$f(x, y) = \begin{cases} (x^2+y^2)^{(1+\alpha)/2} & \text{at rational points} \\ 0 & \text{at irrational points} \end{cases}$$

where $\alpha > 0$. It is obviously discontinuous at every point different from the origin; at the origin it is differentiable:

$$f(x, y)-f(0, 0) = f(x, y) = 0x+0y+\varrho^{1+\alpha}$$

where $\varrho^{1+\alpha} = o(\varrho)$ ($\varrho \to 0$). Thus, f is an example of a function differentiable at a point but having no continuous partial derivatives at that point.

Examples 1 and 2 show that the requirement that a function should be differentiable at a point is weaker than the requirement that this function should have continuous partial derivatives at the point but is stronger than the requirement that it should have partial derivatives at the point.

§ 7.6. Derivative of a Composite Function. Directional Derivative. Gradient

We shall limit ourselves to considering functions of three variables defined in an open set $G \subset R_3$. The extension of the facts established here to the n-dimensional case is carried out analogously.

Theorem 1. *Let a function*

$$u = f(x, y, z) \tag{1}$$

be differentiable at a point $(x, y, z) \in G$ *and let functions*

$$x = \varphi(t),\ y = \psi(t) \quad \text{and} \quad z = \chi(t) \tag{2}$$

dependent on the scalar parameter t have derivatives with respect to t. Then the derivative with respect to t of the composite function $u = F(t) = f(\varphi(t), \psi(t), \chi(t))$ (*that is the derivative of f along the curve specified by equation* (2)) *is expressed by the formula*

$$F'(t) = f'_x(\varphi(t), \psi(t), \chi(t))\, \varphi'(t) + f'_y(\varphi(t), \psi(t), \chi(t))\, \psi'(t) + \\ + f'_z(\varphi(t), \psi(t), \chi(t))\, \chi'(t)$$

or, in the abbreviated form,

$$\frac{du}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \tag{3}$$

Indeed, by the differentiability of f at (x, y, z), we have

$$\Delta u = f(x+\Delta x, y+\Delta y, z+\Delta z) - f(x, y, z) = \\ = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \frac{\partial f}{\partial z}\Delta z + o(\varrho) \\ \left(\varrho = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} \to 0\right) \tag{4}$$

for any sufficiently small increment $(\Delta x, \Delta y, \Delta z)$. Let us give an increment Δt to the value t of the parameter to which, in accordance with equalities (2), there corresponds the point (x, y, z). This results in some increments Δx, Δy and Δz of functions (2). If these very increments are substituted into (4) the left member of (4) becomes equal to the increment $F(t+\Delta t) - F(t) = \Delta u$ of the function F at the point t. On dividing (4) by Δt and passing to the limit we obtain

$$F'(t) = \lim_{\Delta t \to 0} \frac{\Delta u}{\Delta t} = \lim_{\Delta t \to 0} \left(\frac{\partial f}{\partial x}\frac{\Delta x}{\Delta t} + \frac{\partial f}{\partial y}\frac{\Delta y}{\Delta t} + \frac{\partial f}{\partial z}\frac{\Delta z}{\Delta t} + \frac{o(\varrho)}{\Delta t}\right) = \\ = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

which proves (3) because functions (2) possess derivatives and

$$\frac{o(\varrho)}{\Delta t} = \varepsilon(\varrho) \cdot \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2 + \left(\frac{\Delta z^2}{\Delta t}\right)} \to 0 \cdot \sqrt{x_t'^2 + y_t'^2 + z_t'^2} = 0 \qquad (\Delta t \to 0)$$

($\Delta t \to 0$ implies $\varrho \to 0$).

Theorem 2. *If a function f is differentiable at a point* (x, y, z) *its derivative in the direction of any unit vector* $\boldsymbol{n} = (\cos\alpha, \cos\beta, \cos\gamma)$ *exists and is expressed by the formula*

$$\frac{\partial f}{\partial n} = \frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta + \frac{\partial f}{\partial z}\cos\gamma \tag{5}$$

Proof. According to the definition of the directional derivative (see § 7.4) and by virtue of the foregoing theorem we have

$$\frac{\partial f}{\partial n} = \lim_{\substack{t\to 0\\ t>0}} \frac{f(x+t\cos\alpha, y+t\cos\beta, z+t\cos\gamma)-f(x,y,z)}{t} =$$

$$= \left[\frac{d}{dt}f(x+t\cos\alpha,\ y+t\cos\beta,\ z+t\cos\gamma)\right]_{t=0} =$$

$$= \frac{\partial f}{\partial x}\cos\alpha+\frac{\partial f}{\partial y}\cos\beta+\frac{\partial f}{\partial z}\cos\gamma$$

where the partial derivatives are taken at (x, y, z).

If $x = \varphi(s)$, $y = \psi(s)$, $z = \chi(s)$ are equations of a smooth curve Γ where the parameter s is the arc length then the derivatives

$$\frac{dx}{ds} = \varphi'(s), \qquad \frac{dy}{ds} = \psi'(s) \quad \text{and} \quad \frac{dz}{ds} = \chi'(s)$$

are equal to the direction cosines of the tangent vector to Γ. Therefore the expression

$$\frac{\partial f}{\partial s} = \frac{\partial f}{\partial x}\varphi'(s)+\frac{\partial f}{\partial y}\psi'(s)+\frac{\partial f}{\partial z}\chi'(s) = \frac{d}{ds}f(\varphi,(s),\psi(s),\chi(s))$$

where f is a differentiable function is the directional derivative of f along the indicated tangent vector. In such a case we also say that $\frac{\partial f}{\partial s}$ is the derivative of f along Γ.

Let us consider the vector

$$\operatorname{grad} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) \tag{6}$$

It is called the *gradient* of the function f at the point (x, y, z).

The plane passing through a point (x_0, y_0, z_0) perpendicularly to the gradient of f at this point, provided that the gradient is different from zero, has the equation

$$\left(\frac{\partial f}{\partial x}\right)_0(x-x_0)+\left(\frac{\partial f}{\partial y}\right)_0(y-y_0)+\left(\frac{\partial f}{\partial z}\right)_0(z-z_0) = 0 \tag{7}$$

$$((\psi)_0 = \psi(x_0, y_0, z_0))$$

A remarkable property of this plane is that, according to (5), it can be interpreted as the locus of rays issued from (x_0, y_0, z_0) along which the derivative of f is equal to zero. In § 7.19 we shall prove that this is the tangent plane at the point (x_0, y_0, z_0) to the surface specified by the equation

$$f(x, y, z) = A \qquad (A = f(x_0, y_0, z_0)) \tag{8}$$

Formula (5) means that *the derivative of f in the direction of unit vector n at the point (x, y, z) is equal to the projection of the gradient of f at this point on that direction:*

$$\frac{\partial f}{\partial n} = (\operatorname{grad} f, n) = \operatorname{grad}_n f \tag{9}$$

For any vector n there holds the obvious inequality

$$\frac{\partial f}{\partial n} \leqslant |\operatorname{grad} f| \tag{10}$$

If $\operatorname{grad} f = 0$ (usually this only occurs at some singular points) then $\frac{\partial f}{\partial n} = 0$ for any vector n. If $\operatorname{grad} f \neq 0$, that is if at least one of the partial derivatives of f is different from zero, then (10) is a strict inequality for all unit vectors n except for the single vector $n_0 = (\cos\alpha_0, \cos\beta_0, \cos\gamma_0)$ going in the direction of $\operatorname{grad} f$. Thus,

$$\left.\begin{aligned} \cos\alpha_0 &= \frac{\dfrac{\partial f}{\partial x}}{\sqrt{\left(\dfrac{\partial f}{\partial x}\right)^2+\left(\dfrac{\partial f}{\partial y}\right)^2+\left(\dfrac{\partial f}{\partial z}\right)^2}} \\ \cos\beta_0 &= \frac{\dfrac{\partial f}{\partial y}}{\sqrt{\left(\dfrac{\partial f}{\partial x}\right)^2+\left(\dfrac{\partial f}{\partial y}\right)^2+\left(\dfrac{\partial f}{\partial z}\right)^2}} \\ \cos\gamma_0 &= \frac{\dfrac{\partial f}{\partial z}}{\sqrt{\left(\dfrac{\partial f}{\partial x}\right)^2+\left(\dfrac{\partial f}{\partial y}\right)^2+\left(\dfrac{\partial f}{\partial z}\right)^2}} \end{aligned}\right\} \tag{11}$$

It follows that *the gradient of the function f at a point (x, y, z) can be defined as a vector possessing the following two properties:*

(i) *its length is equal to the maximum possible value of the directional derivative $\frac{\partial f}{\partial n}$ at (x, y, z) (this maximum value exists for any function differentiable at (x, y, z) and is a nonnegative number);*

(ii) *if its length is different from zero it goes along the vector n in whose direction the derivative $\frac{\partial f}{\partial n}$ attains the maximum value.*

The new definition of the gradient is completely equivalent to its formal definition by formula (6); it shows that $\operatorname{grad} f$ is an invariant in the sense that it can be defined irrespective of the choice of the coordinate system in which the function u dependent on the moving point in the space is considered (see (1)). To elucidate what has been said let us consider a physical example. Suppose that G is a part of space occupied by a physical body and $u = u(P)$ is the temperature at its variable point P; in the general case the temperature varies from point to point. If we introduce rectangular coordinates (x, y, z) in space this "physical" function $u = u(P)$ can be replaced by the "mathematical" function $u = f(x, y, z)$ where (x, y, z) are the coordinates of the point $P \in G$. If we take another rectangular system (x', y', z') the same physical function is represented by another mathematical function

$$\begin{aligned} u &= f_1(x', y', z') = \\ &= f(\alpha_1 x'+\alpha_2 y'+\alpha_3 z', \beta_1 x'+\beta_2 y'+\beta_3 z', \gamma_1 x' \gamma_2 y'+\gamma_3 z') \end{aligned} \tag{12}$$

where

$$x = \alpha_1 x' + \alpha_2 y' + \alpha_3 z',\ y = \beta_1 x' + \beta_2 y' + \beta_3 z',\ z = \gamma_1 x' + \gamma_2 y' + \gamma_3 z' \quad (13)$$

are the transformation formulas from the old coordinates (x, y, z) to the new ones (x', y', z'), the functions $f(x, y, z)$ and $f_1(x', y', x')$ being different in the general case.

It is natural to define the gradient of the physical function $u = u(P)$ in accordance with the second definition stated above, that is as a vector in whose direction the rate of increase of the temperature at the given point P is the greatest and whose length is equal to the maximum rate of increase of the temperature.

We know that if the mathematical function f describing the physical function in question in the coordinate system (x, y, z) is differentiable at a point $P = (x, y, z)$ the gradient of this function at that point makes sense, and its three components are $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$, i.e. the gradient is

$$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) \quad (14)$$

In the other coordinate system (x', y', z') the same vector is represented by another triple of quantities:

$$\left(\frac{\partial f_1}{\partial x'}, \frac{\partial f_1}{\partial y'}, \frac{\partial f_1}{\partial z'}\right) \quad (15)$$

What has been said can be interpreted as a "physical" proof of the fact that if f is a differentiable function at a point (x, y, z) then the vector represented by the triple of numbers (14) in the original coordinate system (x, y, z) is specified in any new coordinate system (x', y', z') by another triple given by formula (15) in which f_1 is determined by formulas (12), (13). But this fact can also be proved formally as a mathematical proposition.

Indeed, according to the formulas known from analytic geometry, the components of vector (14) in the new system (x', y', z') are

$$\left.\begin{aligned}\alpha_1 \frac{\partial f}{\partial x} + \beta_1 \frac{\partial f}{\partial y} + \gamma_1 \frac{\partial f}{\partial z} &= \frac{\partial f_1}{\partial x'}\\ \alpha_2 \frac{\partial f}{\partial x} + \beta_2 \frac{\partial f}{\partial y} + \gamma_2 \frac{\partial f}{\partial z} &= \frac{\partial f_1}{\partial y'}\\ \alpha_3 \frac{\partial f}{\partial x} + \beta_3 \frac{\partial f}{\partial y} + \gamma_3 \frac{\partial f}{\partial z} &= \frac{\partial f_1}{\partial z'}\end{aligned}\right\} \quad (16)$$

The fact that these components are, respectively, equal to $\frac{\partial f_1}{\partial x'}$, $\frac{\partial f_1}{\partial y'}$ and $\frac{\partial f_1}{\partial z'}$ follows from Theorem 1 on the differentiation of a composite function. When applying this theorem one should take into account that the ordinary derivative with respect to t can be replaced by the partial derivative with respect to t.

The gradient of f is also written in the form

$$\operatorname{grad} f = \nabla f \quad (17)$$

where the symbol ∇ designates the operator which associates the vector grad f with every function f differentiable at (x, y, z). This operator is called the *Hamiltonian operator* or *nabla* or *del**. It is convenient to regard it as a symbolic (formal) vector with components $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$ and $\frac{\partial}{\partial z}$:

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right) \tag{18}$$

The gradient grad f can then be interpreted as the symbolic product of the vector ∇ by the scalar f.

In the physical interpretation of a function as a quantity u dependent on the moving point in space it is natural to rewrite formulas (16) without introducing the functional symbols f and f_1 for different coordinate systems:

$$\left.\begin{aligned} \alpha_1\frac{\partial u}{\partial x}+\beta_1\frac{\partial u}{\partial y}+\gamma_1\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial x'} \\ \alpha_2\frac{\partial u}{\partial x}+\beta_2\frac{\partial u}{\partial y}+\gamma_2\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial y'} \\ \alpha_3\frac{\partial u}{\partial x}+\beta_3\frac{\partial u}{\partial y}+\gamma_3\frac{\partial u}{\partial z} &= \frac{\partial u}{\partial z'} \end{aligned}\right\} \tag{19}$$

In the formal mathematical theory of vectors by a (three-dimensional) *vector* is meant an abstract object denoted by a symbol $\boldsymbol{a}$ which is represented in every orthogonal coordinate system (x, y, z) by a triple of numbers $\boldsymbol{a} = (a_x, a_y, a_z)$ (which are the components of $\boldsymbol{a}$ in the system) on condition that the components a_x, a_y, a_z of the vector $\boldsymbol{a}$ in any other orthogonal coordinate system (x', y', z') $\left(\boldsymbol{a} = (a_{x'}, a_{y'}, a_{z'})\right)$ are obtained from (a_x, a_y, a_z) by means of the transformation formulas

$$a_{x'} = \alpha_1 a_x+\beta_1 a_y+\gamma_1 a_z, \quad a_{y'} = \alpha_2 a_x+\beta_2 a_y+\gamma_2 a_z,$$
$$a_{z'} = \alpha_3 a_x+\beta_3 a_y+\gamma_3 a_z$$

These formulas are analogous to the transformation formulas from the old coordinates (x, y, z) to the new ones (x', y', z'):

$$x' = \alpha_1 x+\beta_1 y+\gamma_1 z, \quad y' = \alpha_2 x+\beta_2 y+\gamma_2 z, \quad z' = \alpha_3 x+\beta_3 y+\gamma_3 z$$

Thus, formulas (19) provide a formal proof of the fact that grad f is a vector.

For further formalization of operations with the symbol ∇ let us consider the operator ∇ as a symbolic vector having the components $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$ in the rectangular coordinate system (x, y, z) and the components $\left(\frac{\partial}{\partial x'}, \frac{\partial}{\partial y'}, \frac{\partial}{\partial z'}\right)$ in any other orthogonal system (x', y', z') and let us agree

* After W. R. Hamilton (1805-1865), an Irish mathematician. The term nabla introduced by Hamilton originates from the name (Greek $\nu\alpha\beta\lambda\alpha$) of an ancient musical instrument of triangular form.

that the new components are expressed in terms of the old ones by means of the (symbolic) equalities

$$\left.\begin{aligned}\frac{\partial}{\partial x'} &= \alpha_1\frac{\partial}{\partial x}+\beta_1\frac{\partial}{\partial y}+\gamma_1\frac{\partial}{\partial z}\\ \frac{\partial}{\partial y'} &= \alpha_2\frac{\partial}{\partial x}+\beta_2\frac{\partial}{\partial y}+\gamma_2\frac{\partial}{\partial z}\\ \frac{\partial}{\partial z'} &= \alpha_3\frac{\partial}{\partial x}+\beta_3\frac{\partial}{\partial y}+\gamma_3\frac{\partial}{\partial z}\end{aligned}\right\} \tag{20}$$

Formulas (20) are written as if the symbolic vector ∇ were a real vector. If we multiply (symbolically) the left-hand and right-hand sides of (20) by a scalar u (i.e. by a differentiable function u) the resultant relation will coincide with equalities (19) expressing the well-known relationship between the partial derivatives of the function u.

Thus, if the components of the gradient of f in a system (x, y, z) are known and if it is required to find its components in another system (x', y', z') we can use the following rule: grad $f = \nabla f$ is interpreted as the product of the vector ∇ by the scalar f, the components of ∇ in the system (x', y', z') are computed by formulas (20) and, finally, these components are multiplied by the scalar f. In other words, the transformation of the components of the vector ∇f to new coordinates is performed as if ∇ were an ordinary vector and f were a number (scalar) by which this vector is multiplied.

Example 1. Let $f(r) = F(Q)$ be a function dependent on the distance $r = r(P, Q)$ from a fixed point $P = (x_0, y_0, z_0)$ to the variable point $Q = (x, y, z)$. Then

$$\operatorname{grad} F = \left(f'(r)\frac{x-x_0}{r},\ f'(r)\frac{y-y_0}{r},\ f'(r)\frac{z-z_0}{r}\right)$$

is a vector going in the direction of the vector PQ and having the length $|\operatorname{grad} F| = |f'(r)|$. Therefore

$$\frac{\partial F}{\partial n} = |f'(r)|\cos(PQ, \boldsymbol{n})$$

In particular, if $F(Q) = f(r) = \ln(1/r)$ then

$$\frac{\partial F}{\partial n} = \frac{\cos(PQ, \boldsymbol{n})}{r}$$

Example 2. Let us consider the expressions

$$(\nabla u, \nabla u) = \left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2 \tag{21}$$

and

$$\Delta u = \nabla\nabla u = \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2} \tag{22}$$

They are invariant with respect to the transformations of rectangular coordinates because $\nabla u = \operatorname{grad} u$ is a vector and thus is an invariant, the left member of (21) is the square of its length (i.e. the scalar product of the vector by itself) and $\nabla\nabla$ is a symbolic invariant (i.e. a symbolic scalar equal to the product of the symbolic vector ∇ by itself) multiplied by a scalar.

§ 7.7. Independence of Mixed Derivatives of the Order of Differentiation

If $f(x_1, \ldots, x_n)$ is a function of n variables $x_1, \ldots, x_n$, its partial derivatives of higher orders involving differentiation with respect to different variables, that is the derivatives of the type $\frac{\partial^2 f}{\partial x_i \partial x_j}$, $\frac{\partial^3 f}{\partial x_i \partial x_j^2}$ $(i \neq j)$ and the like, are termed *mixed partial derivatives.*

Theorem 1. *Let a function $z = f(x, y)$ be defined in an open set G in the xy-plane. If it has the continuous mixed partial derivatives $\frac{\partial^2 f}{\partial x \partial y}$ and $\frac{\partial^2 y}{\partial y \partial x}$ at a point (x, y) belonging to G these derivatives are equal to each other at that point:*

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x} \tag{1}$$

To prove the theorem we shall use the obvious relation

$$\begin{aligned}\Delta_{xh}\,\Delta_{yh} f = \Delta_{xh}[f(x, y+h)-f(x, y)] = f(x+h, y+h)-\\ -f(x+h, y)-f(x, y+h)+f(x, y) = \Delta_{yh}\,\Delta_{xh} f\end{aligned} \tag{2}$$

We can also write (see the explanations below)

$$\begin{aligned}\Delta_{yh}\,\Delta_{xh} f = \Delta_{yh}[f(x+h, y)-f(x, y)] = h[f'_y(x+h, y+\theta h)-f'_y(x, y+\theta h)] =\\ = h^2 f''_{xy}(x+\theta_1 h, y+\theta h) = h^2[f''_{xy}(x, y)+\varepsilon] \quad (\varepsilon \to 0 \text{ for } h \to 0)\end{aligned} \tag{3}$$

Since the derivative $\frac{\partial^2 z}{\partial x\,\partial y}$ is continuous at the point (x, y) it exists in a sufficiently small neighbourhood of this point and the derivative $\frac{\partial z}{\partial y} = f'_y$ exists automatically in this neighbourhood. If h is sufficiently small the points we deal with do not fall outside this neighbourhood and therefore it is legitimate to apply the mean-value theorem (with respect to y) to the function $[f(x+h, y)-f(x, y)]$ as was done in the second equality (3). The third equality (3) results from the application of the same theorem to f'_y (with respect to x) which is also legitimate since in the neighbourhood in question there exists the partial derivative $\frac{\partial f'_y}{\partial x} = f''_{xy}$. The last equality in which we indicate that $\varepsilon \to 0$ as $h \to 0$ expresses the continuity of the derivative f''_{xy} at the point (x, y). It follows from (3) that

$$\lim_{h \to 0} \frac{\Delta_{yh}\,\Delta_{xh} f}{h^2} = f''_{xy}(x, y)$$

Similarly, using the continuity of f''_{yx} we readily prove the equality

$$\lim_{h\to 0} \frac{\Delta_{xh}\Delta_{yh}f}{h^2} = f''_{yx}(x, y)$$

Now, since $\Delta_{xh}\Delta_{yh}f = \Delta_{yh}\Delta_{xh}f$ for any h, we conclude that (1) holds.

It should be noted that the requirement that both partial derivatives entering into (1) should be continuous is only a sufficient condition for the validity of equality (1). Some less restrictive conditions on f which imply (1) are also known but they are rarely used and we do not discuss them here.

Let us take an arbitrary nonnegative integral vector $\boldsymbol{k} = (k_1, \ldots, k_n)$ $(k_j \geqslant 0)$. We shall say that a partial derivative is associated with the vector $\boldsymbol{k}$ if in the computation of the derivative the operation $\frac{\partial}{\partial x_j}$ is performed not more than k_j times for any $j = 1, \ldots, n$. In particular, if $k_j = 0$ the operation $\frac{\partial}{\partial x_j}$ is not performed. Now we state the following theorem.

Theorem 2. *If all partial derivatives of a function $f(x) = f(x_1, \ldots, x_n)$ associated with a given vector $\boldsymbol{k}$ are continuous (in the space R_n) at a point x then it is allowable to change arbitrarily the order of differentiation in any of these derivatives without affecting the ultimate result.*

The proof of this general theorem is rather simple but involves a lengthy induction argument and therefore we shall confine ourselves to an example. The derivative $\frac{\partial^4 f}{\partial z\,\partial x\,\partial z\,\partial y}$ is obviously associated with the vector (1, 1, 2). Assuming that not only this derivative but also all the other derivatives of f associated with the vector (1, 1, 2) are continuous with respect to (x, y, z) we can obtain, by using the definition of the partial derivative and Theorem 1, the equalities

$$\frac{\partial^4 f}{\partial z\,\partial x\,\partial z\,\partial y} = \frac{\partial^2}{\partial z\,\partial x}\,\frac{\partial^2 f}{\partial z\,\partial y} = \frac{\partial^2}{\partial z\,\partial x}\left(\frac{\partial^2 f}{\partial y\,\partial z}\right) = \frac{\partial^2}{\partial x\,\partial z}\left(\frac{\partial^2 f}{\partial y\,\partial z}\right) =$$

$$= \frac{\partial}{\partial x}\,\frac{\partial^2}{\partial z\,\partial y}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial}{\partial x}\,\frac{\partial^2}{\partial y\,\partial z}\left(\frac{\partial f}{\partial z}\right) = \frac{\partial^4 f}{\partial x\,\partial y\,\partial z^2}$$

For instance, to derive the second equality we argue as follows: by the hypothesis, the derivatives $\frac{\partial^2 f}{\partial z\,\partial y}$ and $\frac{\partial^2 f}{\partial y\,\partial z}$ are continuous with respect to (x, y, z); therefore they are sure to be continuous with respect to (y, z) for a fixed x and consequently they are equal to each other. In the fifth equality the same argument is applied to $\frac{\partial f}{\partial z}$ (instead of f).

Exercise. Prove that the function

$$v = \frac{1}{8\pi^{3/2}(t_0 - t)^{3/2}}\, e^{-\frac{r^2}{4(t_0 - t)}}$$

where

$$r = \sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}$$

satisfies the partial differential equations

$$\Delta_0 v - \frac{\partial v}{\partial t_0} = 0 \quad \text{and} \quad \Delta v + \frac{\partial v}{\partial t} = 0$$

where

$$\Delta_0 = \frac{\partial^2}{\partial x_0^2} + \frac{\partial^2}{\partial y_0^2} + \frac{\partial^2}{\partial z_0^2} \quad \text{and} \quad \Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

(The function v termed the *fundamental solution of the heat conductivity equation* is encountered in mathematical physics.)

§ 7.8. Differential of a Function. Differentials of Higher Orders

Let us consider a function

$$W = f(\boldsymbol{x}) = f(x_1, \ldots, x_n) \tag{1}$$

defined in an open set $G \subset R_n$. It can be represented in infinitely many ways in the form

$$W = \varphi(\boldsymbol{u}) = \varphi(u_1, \ldots, u_m) \tag{2}$$

where

$$u_j = \psi_j(\boldsymbol{x}) \qquad (j = 1, \ldots, m, \ \boldsymbol{x} \in G) \tag{3}$$

We shall use the following terminology: the variable W will be spoken of as a function of the *independent vector variable* $\boldsymbol{x}$, and the same variable W will be referred to as a function of the *dependent vector variable* $\boldsymbol{u}$. The latter variable ($\boldsymbol{u}$) depends on the independent variable $\boldsymbol{x}$; to each vector $\boldsymbol{x}$ belonging to G there corresponds the vector $\boldsymbol{u} = (\psi_1(\boldsymbol{x}), \ldots, \psi_m(\boldsymbol{x}))$.

Thus, the vector variable $\boldsymbol{x}$ will play an exceptional role: in all the considerations below it will only enter as an *independent variable.*

Let the function f possess continuous partial derivatives of the first order at a point $\boldsymbol{x} \in G$. Then, as is known from § 7.5, it is differentiable, that is its increment at this point can be represented in the form

$$\Delta W = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \Delta x_j + o(\varrho) \qquad (\varrho \to 0) \tag{4}$$

where

$$\varrho = |\Delta \boldsymbol{x}| = \left(\sum_{j=1}^{n} \Delta x_j^2\right)^{1/2}$$

The sum

$$dW = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} \Delta x_j \tag{5}$$

is called the *principal linear part of the increment of W at the point* $\boldsymbol{x}$ (and also the *differential of W at that point*) *corresponding to the increments* $\Delta x_1, \ldots, \Delta x_n$ *of the independent variables.*

For the independent variables $x_1, \ldots, x_n$ we put

$$\Delta x_j = dx_j \qquad (j = 1, \ldots, n) \tag{6}$$

and call these quantities the *differentials* of $x_1, \ldots, x_n$. The differentials dx_j will also be called *independent differentials* in order to stress that they are independent of $x = (x_1, \ldots, x_n)$. The "independence" of the quantities dx_j is displayed formally in the differentiation process: when differentiating with respect to $x_1, \ldots, x_n$ we regard them as constants, i.e. $d(dx_j) = 0$, $j = 1, \ldots, n$.

By virtue of the convention expressed by (6), the differential of W can be written in the form

$$dW = \sum_{j=1}^{n} \frac{\partial f}{\partial x_j} dx_j \tag{7}$$

It is clear that dW is a quantity which, generally speaking, depends both on $x_1, \ldots, x_n$ and on $dx_1, \ldots, dx_n$.

The general properties expressed by the relations below apply to any two functions u and v having continuous partial derivatives at x:

$$d(u \pm v) = du \pm dv \tag{8}$$

$$d(uv) = u\,dv + v\,du \tag{9}$$

$$d\left(\frac{u}{v}\right) = \frac{v\,du - u\,dv}{v^2} \qquad (v \neq 0) \tag{10}$$

In these formulas the partial derivatives of the functions in the parentheses are continuous at the point x.

As an example, let us prove the third of these equalities:

$$d\left(\frac{u}{v}\right) = \sum_{j=1}^{n} \frac{\partial}{\partial x_j}\left(\frac{u}{v}\right) dx_j = \sum_{1}^{n} \frac{v\dfrac{\partial u}{\partial x_j} - u\dfrac{\partial v}{\partial x_j}}{v^2} dx_j =$$

$$= \frac{1}{v^2}\left(v \sum_{1}^{n} \frac{\partial u}{\partial x_j} dx_j - u \sum_{1}^{n} \frac{\partial v}{\partial x_j} dx_j\right) = \frac{v\,du - u\,dv}{v^2}$$

The continuity of $\dfrac{\partial}{\partial x_j}\left(\dfrac{u}{v}\right)$ is seen from the third term in this chain of equalities.

The differential of W is also called the *differential of the first order* (below we also consider higher-order differentials).

Now let us suppose that the function W has continuous partial derivatives of the second order. The *second differential of* W corresponding to the independent increments (differentials) $dx_1, \ldots, dx_n$ is defined by the equality

$$d^2W = d(dW) \tag{11}$$

where the operation d used twice in the right-hand side of (11) is performed for the given independent increments $dx_1, \ldots, dx_n$ which are regarded as

20006–16

constants independent of $x_1, \ldots, x_n$. Thus,

$$d^2W = d\sum_{i=1}^{n}\frac{\partial f}{\partial x_i}dx_i = \sum_{i=1}^{n} d\left(\frac{\partial f}{\partial x_i}dx_i\right) = \sum_{i=1}^{n}\left(d\frac{\partial f}{\partial x_i}\right)dx_i =$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2 f}{\partial x_j\,\partial x_i}dx_j\,dx_i \tag{12}$$

Since $\frac{\partial^2 f}{\partial x_i\,\partial x_j} = \frac{\partial^2 f}{\partial x_j\,\partial x_i}$, the second differential is a quadratic form in the independent differentials $dx_1, \ldots, dx_n$.

Generally, the differential of the lth order (or, simply, the lth differential) of W taken for the independent differentials $dx_1, \ldots, dx_n$ is defined by induction with the aid of the recurrence formula

$$d^lW = d(d^{l-1}W) \qquad (l = 2, 3, \ldots) \tag{13}$$

where d^l, d and d^{l-1} are taken for the indicated independent differentials dx_i, the latter being regarded as constants (independent of $x_1, \ldots, x_n$) in the computation process.

Arguing as in the deduction of (12) we readily see that

$$d^3W = \sum_{k=1}^{n}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^3 f}{\partial x_k\,\partial x_i\,\partial x_j}dx_k\,dx_i\,dx_j$$

The expression of the lth differential can be conveniently written in the form

$$d^lW = l!\sum_{|k|=l}\frac{1}{k!}w^{(k)}\,dx^k,\ w^{(k)} = \frac{\partial^{|k|}f}{\partial x_1^{k_1}\ldots\partial x_n^{k_n}}$$

$$k = (k_1, \ldots, k_n),\ |k| = \sum_{1}^{n}k_j,\ dx^k = dx_1^{k_1}\ldots dx_n^{k_n}$$

$$k! = k_1!\ldots k_n!\ (0! = 1) \tag{14}$$

where the sum is extended over all the nonnegative integral vectors k with $|k| = l$. Formula (14) is proved by induction.

We have defined the notion of the differential of the function W in terms of the independent variables $x_1, \ldots, x_n$ (or, which is the same, in terms of the vector variable x). Now let us suppose that the variable W is regarded as a function of the *dependent* vector variable $u = (u_1, \ldots, u_m)$ (see the beginning of the section). There arises the question as to what are the expressions of the first and higher differentials in terms of the variable u. We begin the study of this question with the case of the first differential.

We shall assume that the functions $\varphi(u)$ and $\psi_j(x)$ $(j = 1, \ldots, n)$ mentioned at the beginning of the section have continuous partial derivatives of the first order. Then

$$dW = \sum_{j=1}^{n}\frac{\partial f}{\partial x_j}dx_j = \sum_{j=1}^{n}\frac{\partial}{\partial x_j}\varphi\,dx_j = \sum_{j=1}^{n}\sum_{i=1}^{m}\frac{\partial\varphi}{\partial u_i}\frac{\partial\psi_i}{\partial x_j}dx_j =$$

$$= \sum_{i=1}^{m}\frac{\partial\varphi}{\partial u_i}\sum_{j=1}^{n}\frac{\partial\psi_i}{\partial x_j}dx_j = \sum_{i=1}^{m}\frac{\partial\varphi}{\partial u_i}du_i \tag{15}$$

We thus see that, as in the case of one independent variable, the first differential of W is expressed in terms of the dependent variables in the same way as in terms of the independent variables. This demonstrates *the invariance of the form of the first differential.*

To answer the question stated above for the case of the second differential we shall suppose that the functions φ and ψ_j possess continuous partial derivatives of the second order.

On differentiating both members of (15) and taking into account properties (8) and (9) we obtain (see the explanations below)

$$d^2W = d(dW) = \sum_{i=1}^{m} d\left(\frac{\partial\varphi}{\partial u_i} du_i\right) = \sum_{i=1}^{m}\left(\sum_{j=1}^{m} \frac{\partial^2\varphi}{\partial u_i\,\partial u_j} du_i\,du_j + \frac{\partial\varphi}{\partial u_i} d^2u_i\right) =$$

$$= \sum_{i=1}^{m}\sum_{j=1}^{m} \frac{\partial^2\varphi}{\partial u_i\,\partial u_j} du_i\,du_j + \sum_{i=1}^{m} \frac{\partial\varphi}{\partial u_i} d^2u_i \tag{16}$$

In the second equality (16) we have applied property (8) and in the third equality we have used property (9) and also the fact that the form of the first differential remains unchanged when we pass to the dependent variables u_j. We see that, in contrast to the first differential, the second differential of W expressed in terms of the dependent variables u_j is the sum of two summands of essentially different nature. The first summand is a quadratic form analogous to form (12) expressing d^2W in terms of the independent variables. As to the second summand, it is an additional expression which is different from zero when $u_i \not\equiv x_i$ and therefore should be taken into account in the general case.

It should be noted that, as follows from our argument, if expression (16) is taken for the differentials $dx_1, \ldots, dx_n$ which enter into expression (12) then both expressions coincide identically for any point x and any dx_i provided that the indicated continuous partial derivatives of the second order exist at x.

The expressions of the differentials d^3W, d^4W, ... in terms of the dependent variables u_j are found in a similar way but they become more and still more complicated as the order of the differential increases.

§ 7.9. Limit Point. Weierstrass' Theorem. Closed and Open Sets

Let us consider an arbitrary set E of points $x = (x_1, \ldots, x_n)$ belonging to the space $R_n = R$.

A point $x^0 = (x_1^0, \ldots, x_n^0)$ is called a *limit point* (or *accumulation point*) of E if its every neighbourhood contains at least one point x belonging to E and distinct from x^0.

This definition does in fact imply that any neighbourhood of x^0 contains an infinitude of points x belonging to E and that it is possible to choose a sequence of points $x^k \in E$, $k = 1, 2, \ldots$, such that $x^k \neq x^0$ and $|x^k - x^0| \to 0$.

Indeed, by the above definition of a limit point, for any natural $k = 1, 2, \ldots$ there is a point $x^k \in E$ for which $0 < |x^k - x^0| < 1/k$ where $k = 1, 2, \ldots$.

Let us state an alternative definition of a limit point which is obviously equivalent to the former.

A point x^0 is a limit point of E if any open ball (open cube) with centre at x^0 contains at least one point $x \in E$, $x \neq x^0$ or if there exists a sequence of points $\{x^k\}$ such that $x^k \in E$, $x^k \neq x^0$ and $x^k \to x^0$ $(k \to \infty)$.

The set of all limit points of E is denoted by E' and is called the *derived set* of E.

Example 1. Let E be the set of points $x = (x_1, \ldots, x_n)$ with rational coordinates. Then any point $x^0 = (x_1^0, \ldots, x_n^0) \in R$ is a limit point of E because any cube $|x_j - x_j^0| < \delta$ $(j = 1, \ldots, n)$ contains points of E different from x^0. Thus, $E' = R_n$.

Example 2. A finite set $x^1, x^2, \ldots, x^N$ has no limit points (i.e. $E' = 0$); this follows, for instance, from the fact that any neighbourhood of a limit point must contain an infinite number of points belonging to E.

Example 3. The set E consisting of the points $x^k = (k, 0, \ldots, 0)$ $(k = 1, 2, \ldots)$ has no limit points $(E' = 0)$ because the distance between its any two points satisfies the inequality $|x^k - x^l| = |k - l| \geqslant 1$ $(k \neq l)$. If a point $x \in R_n$ were a limit point of E its any sufficiently small neighbourhood would contain two points belonging to E, which means that such points would lie at a distance less than any arbitrarily small $\delta > 0$ from each other.

Example 4. Let $V_1 = \left\{x : \sum_1^n x_k^2 < r^2\right\}$, $\Gamma = \left\{x : \sum_1^n x_k^2 = r^2\right\}$

and $V_2 = \left\{x : \sum_1^n x_k^2 > r^2\right\}$.

Extending our usual terminology used in the case $n = 3$ to an arbitrary dimension n we shall say that V_1 is an open ball (sphere) in R_n with centre at the origin and that Γ is its boundary and V_2 is its exterior. There hold the following relations:

$$V_1' = V_1 + \Gamma,\ (V_1 + \Gamma)' = V_1 + \Gamma,\ V_2' = V_2 + \Gamma,$$
$$(V_2 + \Gamma)' = V_2 + \Gamma,\ \Gamma' = \Gamma \tag{1}$$

Visual geometrical considerations (see Fig. 7.4) which can be easily expressed in terms of inequalities show that any point $x^1 \in V_1$ can be enclosed in a sufficiently small ball with centre at this point contained entirely in V_1 whence it follows that x^1 is a limit point of V_1 but is not a limit point either of Γ or of V_2. Any ball with centre at an arbitrary point $x^2 \in \Gamma$ contains some points belonging to V_1, Γ and V_2 different from $x^2 \in \Gamma$ and therefore x^2 is a limit point of V_1, Γ, V_2, $V_1 + \Gamma$ and $V_2 + \Gamma$. Finally, an arbitrary point $x^3 \in V_2$ can be enclosed into a ball entirely contained in V_2 and hence it is a limit point of V_2 but is not a limit point of any of the sets V_1, Γ and $V_1+\Gamma$. We have thus proved equalities (1).

A set E is said to be *bounded* if it is contained in some ball (cube). If otherwise, the set E is said to de *unbounded*. The ball (cube) mentioneb in these definitions can be taken with centre at the origin because if all the points $x \in E$ satisfy the inequality $|x-x^0| < \varrho_1$ they also satisfy the inequality $|x| \leqslant |x-x^0|+|x^0| < \varrho_2$ where $\varrho_2 = \varrho_1+|x^0|$.

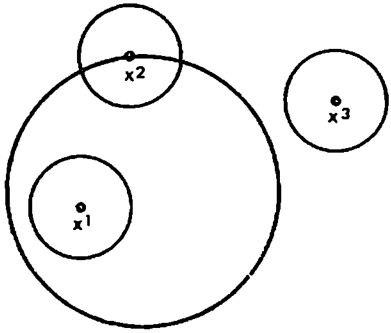

Fig. 7.4

The following theorem generalizes the corresponding one-dimensional theorem and is based on it.

Theorem 1. *Every bounded sequence of points $x^k=(x_1^k, \ldots, x_n^k)$ $(k=1, 2, \ldots)$ contains a subsequence $\{x^{k_l}\}$ $(l = 1, 2, \ldots)$ convergent to a point x^0:*

$$||x^{k_l}-x^0|| \to 0 \qquad (l \to \infty)$$

Proof. The sequence $\{x^k\}$ being bounded, there is a number M such that

$$M > |x^k| \geqslant |x_j^k| \qquad (j = 1, \ldots, n; k = 1, 2, \ldots)$$

This shows that the coordinates of the points x^k are also bounded. The first coordinates of these points form a bounded sequence $\{x_1^k\}$ $(k = 1, 2, \ldots)$, and, by virtue of the analogous theorem for the one-dimensional case, there is a sequence $\{k_{l_1}\}$ of natural numbers and a number x_1^0 such that $x_1^{k_{l_1}} \to x_1^0$ $(l_1 \to \infty)$. Now let us consider the second coordinates x_2^k but only for the natural k_{l_1} indicated above. The subsequence $\{x_2^{k_{l_1}}\}$ is bounded and, by virtue of the one-dimensional theorem, there is a subsequence $\{x_2^{k_{l_2}}\}$ and a number x_2^0 such that $x_2^{k_{l_2}} \to x_2^0$. Since $\{k^{l_2}\}$ is a subsequence of $\{k^{l_1}\}$ we simultaneously have $x_1^{k_{l_2}} \to x_1^0$ and $x_2^{k_{l_2}} \to x_2^0$. Proceeding to argue in this way and taking into account the boundedness of the set of the third coordinates we obtain a subsequence $\{k^{l_3}\}$ of the subsequence $\{k^{l_2}\}$ for which, simultaneously,

$$x_1^{k_{l_3}} \to x_1^0, \quad x_2^{k_{l_3}} \to x_2^0 \quad \text{and} \quad x_3^{k_{l_3}} \to x_3^0$$

where x_3^0 is a certain number. At the nth stage of this process we arrive at a subsequence of natural numbers $k_{l_n}=k_l$ and at a system of numbers $x_1^0, \ldots, x_n^0$ such that the relations $x_j^{k_l} \to x_j^0$ $(l \to \infty;\ j = 1, \ldots, n)$ hold simultaneously. Finally, putting $x^0=(x_1^0, \ldots, x_n^0)$ we complete the proof of the theorem.

Let us come back to limit points. A finite set (i.e. a set consisting of a finite number of points) has no limit points (Example 2). There are also infinite and unbounded sets having no limit points (Example 3). For the bounded infinite sets there holds the following theorem.

Theorem 2 (Weierstrass). *Every bounded infinite set E has at least one limit point.*

Proof. Since the set E is infinite it contains an (infinite) sequence $\{x^k\}$ of pairwise distinct points. According to the foregoing theorem, there exists a subsequence $\{x^{k_l}\}$ convergent to a point $x^0 \in R_n$. Since the points x^{k_l} corresponding to different l are distinct the point x^0 is obviously a limit point of E because its any neighbourhood contains some points $x^{k_l} \in E$ different from x^0 (of course, x^0 may or may not belong to E).

A set $E \subset R_n$ is said to be *closed* if all its limit points belong to it, i.e. if $E' \subset E$.

It should be stressed that this definition does not require that E should have limit points; it only requires that if E has such points they should belong to E. Thus, *every set which has no limit points is closed. An empty set, a finite set and a set of integral points* (that consists of points having integral coordinates) *in the space R_n are all examples of closed sets.*

The set in Example 1 is not closed because the irrational points (i.e. points having irrational coordinates) are its limit points but do not belong to it.

A ball with boundary (Example 4), the exterior of a ball taken together with its boundary and the boundary of the ball itself are examples of closed sets. An open sphere (ball), the exterior of a closed ball are examples of nonclosed sets. This is geometrically evident and can be justified rigorously by means of the corresponding inequalities.

We shall also state an alternative definition of a closed set equvalient to the former: a set E is closed if the fact that the points of a sequence $\{x^k\}$ convergent to x^0 belong to E implies that x^0 also belongs to the set E.

We shall show that these definitions are equivalent. Let E be a closed set in the sense of the former definition and let us suppose that the conditions of the latter definition of closedness are not fulfilled for E. Then there must exist a sequence $\{x^k\}$ and a point x^0 such that $x^k \in E$, $x^k \to x^0$ and $x^0 \notin E$. Obviously, this is only possible when the elements x^k of the sequence run over an infinite set of (pairwise distinct) points; therefore x^0 is a limit point of E and, according to the hypothesis, it must belong to E. We have thus arrived at a contradiction. Hence, the former definition implies the latter.

Conversely, let E be a closed set in the sense of the latter definition and let x^0 be a limit point of E; then, as is known, there is a sequence of points $x^k \in E$ such that $x^k \to x^0$. By the latter definition, we have $x^0 \in E$ and consequently every limit point of E belongs to E, which means that E is a closed set in the sense of the former definition as well.

The addition of all limit points of a set E to this set results in a new set which is denoted $\bar{E}$ and is called the *closure* of E; hence, $\bar{E} = E+E'$.

Theorem 3. *The closure $\bar{E}$ of a set E is a closed set.*

Proof. Let x^0 be a limit point of $\bar{E}$. Let us take an arbitrary $\varepsilon > 0$ and consider the ball $|x-x^0| < \varepsilon$. It must contain a point $x' \in \bar{E}$, $x' \neq x^0$, which can be enclosed in a ball lying within the former ball and having a sufficiently small radius so that it does not contain the point x^0. In the latter ball there must be a point belonging to E which thus belongs to the former ball. Consequently, any ball with centre at x^0 contains a point belonging to E and distinct from x^0. This means that x^0 is a limit point of E, that is $x^0 \in \bar{E}$.

There is a close relationship between closed and open sets (see § 7.1): *if E is closed then $R-E$ is open and vice versa.* Indeed, let E be closed and let $x^0 \in R-E$; then there is a ball with centre at x^0 belonging to $R-E$. For, if we suppose that there is no such ball then there must exist a sequence of points $x^k \in E$ convergent to x^0 ($x^k \to x^0$) and, by the closedness of E, this implies $x^0 \in E$, which leads to a contradiction. Now let us assume that E is an open set. Taking an arbitrary sequence of points x^k belonging to $R-E$ and convergent to a point x^0 we conclude that the latter cannot belong to E because in its every neighbourhood there are points of $R-E$; therefore x^0 belongs to $R-E$, and hence $R-E$ is a closed set.

Now we state the following three definitions:

1. A point x^0 is called a *boundary point* of the given set E if its any neighbourhood contains both points belonging to E and points not belonging to E.

2. A point x^0 is called an *interior point* of E (see § 7.1) if there is a neighbourhood of x^0 entirely contained in the set E.

3. A point x^0 is called an *exterior point* relative to a set E if it does not belong to E and if there is a neighbourhood of x^0 not belonging to E.

It is obvious that in these definitions "neighbourhood" can be replaced by "open ball with centre at x^0" or by "open cube with centre at x^0".

Note that definitions 1-3 are mutually exclusive in the sense that every point $x \in R$ can only satisfy the conditions of one of these definitions.

Hence, if we are given an arbitrary set $E \subset R$ then, relative to this set, the whole space R splits into the following three pairwise disjoint (non-intersecting) sets:

(i) the set E_1 of all interior points of E called the *interior of the set E*. This is an open set because if $x \in E_1$ then there is an open sphere (ball) V with centre at x_0 lying entirely within E and, since all the points of V are its interior points, they are also interior points of E; consequently $V \in E_1$;

(ii) the set E_2 of all exterior points of E called the *exterior of E*. This is also an open set, which is proved analogously;

(iii) the set Γ of all boundary points of E called the *boundary of E*. This is a closed set because $\Gamma = R-(E_1+E_2)$ and the set E_1+E_2 is the sum of the two open sets E_1 and E_2 and thus is an open set.

The sets $E_1+\Gamma$ and $E_2+\Gamma$ are obviously closed since their complements (relative to R), i.e. the sets $R-(E_1+\Gamma)$ and $R-(E_2+\Gamma)$, are open.

There hold the (set-theoretic) equalities

$$E_1+\Gamma = E+\Gamma = \bar{E} \quad \text{and} \quad E_2+\Gamma = (R-E)+\Gamma = \overline{R-E}$$

whose proof is left to the reader.

In the equality $R = E_1+E_2+\Gamma$ one or two summands on the right-hand side may be an empty set. It should also be noted that an empty set and the entire space R are simultaneously open and closed.

Example 5. Let a function $f(x)$ be defined and continuous throughout R and let c be a given number; then, as can easily be proved, the sets $\{f(x) = c\}$, $\{f(x) \leqslant c\}$ and $\{f(x) \geqslant c\}$ (i.e. the sets of points x for which the corresponding relations written in the braces are fulfilled) are closed while the sets $\{f(x) > c\}$ and $\{f(x) < c\}$ are open. Some of these sets may of course be empty; this does not contradict the assertions since an empty set is simultaneously open and closed.

The results of Example 4 follow immediately from these assertions. It should also be taken into account that if the function $f(x)$ were only defined on a part of R the validity of these assertions might be violated.

Theorem 4. *Every open (one-dimensional) set G in the real line is a union of a finite or countable number of pairwise disjoint (nonintersecting) open intervals:*

$$G = \sum \delta_k \tag{1}$$

where δ_k are open intervals.

Indeed, let x^0 be a point belonging to G; then there exist open intervals $\delta_{x^0} = (\lambda, \mu)$ covering x^0 (i.e. containing x^0) and lying entirely within the set G. Let

$$\alpha = \inf \lambda \quad \text{and} \quad \beta = \sup \mu$$

where the infimum and the supremum are taken over all the possible intervals δ_{x^0}. In particular, it may happen that $\alpha = -\infty$ or $\beta = +\infty$ or both (in the last case $G = (-\infty, \infty)$). To each point $x^0 \in G$ thus corresponds the maximum interval $\delta = (\alpha, \beta)$ which contains x^0 and is entirely contained in G.

Now let A be the collection of all pairwise distinct intervals of this kind. It is clear that the (set-theoretic) sum (that is the union) of all open intervals entering into A exactly coincides with the set G. Finally, in each such interval we can choose a rational number whence follows that the system of intervals A is finite or countable since it is equivalent to a subject of the set of rational numbers. This completes the proof of the theorem.

Exercises

1. Prove that the sum of a finite or countable number of open sets and also the intersection of a finite number of open sets is again an open set. Give an example of a countable system of open sets whose intersection is not an open set.

2. Prove that the sum of a finite number of closed sets and also the intersection of a finite or countable number of closed sets is a closed set. Give an example of a countable system of closed sets whose sum is not a closed set.

§ 7.10. Function on a Set. Properties of Continuous Functions on Closed Sets

Let a function $f(\boldsymbol{x}) = f(x_1, \ldots, x_n)$ be defined on an arbitrary set A in the n-dimensional space R_n $(A \subset R = R_n)$ and let $\boldsymbol{x}^0$ be a limit point of A. We say that *a number A is the limit of f at the point $\boldsymbol{x}^0$ on the set A if* $\lim\limits_{\boldsymbol{x}^k \to \boldsymbol{x}^0} f(\boldsymbol{x}^k) = A$ *for any sequence of points $x^k \in A$ convergent to $\boldsymbol{x}^0$ and distinct from $\boldsymbol{x}^0$.*

This definition is equivalent to the following: given any $\varepsilon > 0$, there is a cube or a sphere (ball) with centre at $\boldsymbol{x}^0$ for whose all points $\boldsymbol{x}$ contained in A and distinct from $\boldsymbol{x}^0$ the inequality $|f(\boldsymbol{x}) - A| < \varepsilon$ holds. The equivalence of the two definitions is proved by analogy with the case of the limit of a function f at a point $\boldsymbol{x}^0$ without the stipulation that the function is only considered on a given set (see §§ 4.1 and 7.2).

Cauchy's criterion for the existence of the limit of f at a point $\boldsymbol{x}^0$ on a set A is also proved similarly; in this case it reads: *for a (finite) limit of f at $\boldsymbol{x}^0$ on A to exist it is necessary and sufficient that, given any $\varepsilon > 0$, there should exist $\delta > 0$ such that the inequality $|f(\boldsymbol{x}) - f(\boldsymbol{x}')| < \varepsilon$ is fulfilled for all $\boldsymbol{x}, \boldsymbol{x}' \in A$ satisfying the conditions $0 < |\boldsymbol{x} - \boldsymbol{x}^0| < \delta$, $0 < |\boldsymbol{x}' - \boldsymbol{x}^0| < \delta$.*

We shall say that a function f defined on a set A is *continuous on A at a point $\boldsymbol{x}^0 \in A$* if

$$\lim_{\substack{\boldsymbol{x}^k \to \boldsymbol{x}^0 \\ \boldsymbol{x}^k \in A}} f(\boldsymbol{x}^k) = f(\boldsymbol{x}^0) \tag{1}$$

for any sequence of points $\boldsymbol{x}^k \in A$ convergent to $\boldsymbol{x}^0$.

Note that if $\boldsymbol{x}^0$ is an *isolated point* of A, that is a point which is not a limit point of A, then there exists a sphere with centre at $\boldsymbol{x}^0$ containing only a single point of the set A, namely the point $\boldsymbol{x}^0$ itself. In this case the relations $\boldsymbol{x}^k \in A$ and $\boldsymbol{x}^k \to \boldsymbol{x}^0$ imply that $\boldsymbol{x}^k = \boldsymbol{x}^0$ for all $k > N$ where N is sufficiently large and hence equality (1) is fulfilled automatically.

Thus, *if $\boldsymbol{x}^0$ is an isolated point of A then any function defined on A is continuous on A at that point.*

A function f defined on A and continuous at every point of A is said to be *continuous on the set A.*

We shall prove two theorems expressing some remarkable properties of functions continuous on bounded closed sets; they generalize the corresponding properties of continuous functions of one variable defined on closed intervals.

Theorem 1. *A function f continuous on a bounded closed set A is bounded on A.*

Proof. Suppose the contrary: let f be unbounded on A; then for any natural k there is a point $\boldsymbol{x}^k \in A$ such that

$$|f(\boldsymbol{x}^k)| > k \qquad (k = 1, 2, \ldots) \tag{2}$$

The sequence $\{\boldsymbol{x}^k\}$ thus obtained is bounded and consequently it contains a subsequence $\{x^{k'}\}$ convergent to a point $x^0 \in R_n$. Since the set A is closed the point x^0 belongs to it and, by the continuity of f on A at x^0, we have $\lim\limits_{x^{k'} \to x^0} f(x^{k'}) = f(x^0)$, which contradicts inequalities (2).

Theorem 2. *A function f continuous on a bounded closed set A attains its maximum and minimum values on that set.*

Proof. By the foregoing theorem, f is bounded on A. Consequently the function has a finite infimum and a finite supremum on A:

$$m = \inf_{x \in A} f(x), \qquad M = \sup_{x \in A} f(x)$$

The properties of the supremum imply that, given any natural k, there is a point $x^k \in A$ such that

$$M - \frac{1}{k} < f(x^k) \leqslant M \qquad (k = 1, 2, \ldots) \tag{3}$$

The sequence $\{x^k\}$ thus obtained is bounded and therefore it contains a subsequence $\{x^{k_j}\}$ convergent to some point x^0. Since, by the hypothesis, the set A is closed it contains the point x^0, and the continuity of f on A implies $\lim\limits_{x^{k_j} \to x^0} f(x^{k_j}) = f(x^0)$. On the other hand, (3) implies that this limit must be equal to the number M. This means that

$$f(x^0) = M = \max_{x \in A} f(x)$$

The existence of a point $y^0 \in A$ at which f attains its minimum value on A, that is

$$f(y^0) = m = \min_{x \in A} f(x)$$

is proved similarly.

Now let us again consider an arbitrary set $A \subset R$ (not necessarily bounded or closed), and let a function f be defined and bounded on A (it may not be continuous).

For an arbitrary $\delta > 0$ we now construct the function

$$\omega(\delta) = \sup_{\substack{|x'-x''| < \delta \\ x', x'' \in A}} |f(x')-f(x'')| \tag{4}$$

(a more precise notation is $\omega(\delta, f)$) called the *modulus of continuity of f on the set A*. The supremum on the right-hand side of (4) is taken over the absolute values of the differences of the values of f corresponding to all the possible pairs of points $x', x'' \in A$ lying at a distance less than δ from one another.

The modulus of continuity of $f(x)$ is a function of δ which is obviously nonnegative. It is nondecreasing because if $0 < \delta < \delta_1$ then

$$\omega(\delta) = \sup_{\substack{|x'-x''| < \delta \\ x', x'' \in A}} |f(x')-f(x'')| \leqslant$$

$$\leqslant \sup_{\substack{|x'-x''| < \delta_1 \\ x', x'' \in A}} |f(x')-f(x'')| = \omega(\delta_1)$$

Consequently there exists the limit

$$\omega(0+0) = \lim_{\substack{\delta \to 0 \\ \delta > 0}} \omega(\delta) = \lambda \geqslant 0 \tag{5}$$

Theorem 3. *Given an arbitrary $\varepsilon > 0$, there is $\delta > 0$ such that the inequality*

$$|f(x')-f(x'')| < \lambda+\varepsilon$$

holds for any $x', x'' \in A$, $|x'-x''| < \delta$, where λ is the number defined in (5).

Proof. Suppose that the assertion of the theorem is not true. Then there exists $\varepsilon_0 > 0$ such that for any natural k there are points $x'_k, x''_k \in A$ such that $|x'_k-x''_k| < 1/k$ while $|f(x'_k)-f(x''_k)| \geqslant \lambda+\varepsilon_0$. This implies that

$$\omega(1/k) \geqslant |f(x'_k)-f(x''_k)| \geqslant \lambda+\varepsilon_0$$

for any k and that $\lambda = \omega(0+0) = \lim\limits_{k\to\infty} \omega\left(\frac{1}{k}\right) \geqslant \lambda+\varepsilon_0$, which is impossible.

Let us state the following definition:

(i) A function f is said to be *uniformly continuous* on A if its modulus of continuity $\omega(\delta)$ on A tends to zero as $\delta \to 0$, i.e.

$$\omega(0+0) = \lim_{\delta\to 0} \omega(\delta) = 0 \tag{6}$$

We also state an alternative definition of the uniform continuity equivalent to the former:

(ii) A function f is *uniformly continuous on A* if for any $\varepsilon > 0$ there is $\delta > 0$ such that for any $x', x'' \in A$ satisfying the condition $|x'-x''| < \delta$ the inequality $|f(x')-f(x'')| < \varepsilon$ holds.

By Theorem 3 in which we should put $\lambda = 0$, definition (i) implies (ii). Conversely, if (ii) holds then setting an arbitrary $\varepsilon > 0$ and taking the number $\delta > 0$ mentioned in (ii) we obtain

$$\omega(\delta) = \sup_{\substack{x', x'' \in A \\ |x'-x''| < \delta}} |f(x')-f(x'')| \leqslant \varepsilon$$

whence follows (6), that is (i), since ω is a monotone nondecreasing function of δ.

Next we state and prove the following important theorem.

Theorem 4. *A function f continuous on a bounded closed set A is uniformly continuous on A.*

Proof. Let us suppose the contrary. Then there is $\varepsilon_0 > 0$ such that for any natural k there is a pair of points

$$x'_k, x''_k \in A, \qquad |x'_k-x''_k| < 1/k \tag{7}$$

for which

$$|f(x'_k)-f(x''_k)| \geqslant \varepsilon_0 \tag{8}$$

By the boundedness of the sequence $\{x'_k\}$ and by the closedness of the set A, there is a subsequence $\{x'_{k_j}\}$ convergent to a point $x^0 \in A$: $x'_{k_j} \to x^0$. Then

(7) implies that x''_{k_j} also converges to x^0 and, by the continuity of f at x^0, we have

$$\lim_{k_1\to\infty} |f(x'_{k_j})-f(x''_{k_j})| = |f(x^0)-f(x^0)| = 0$$

which contradicts (8).

Let us consider a function f defined on a set $A \in R_n$. We shall suppose that it is bounded on A. Let $x^0 \in A$ and $\delta > 0$. Denoting by V_δ the sphere $|x-x^0| \leq \delta$ of radius δ with centre at x^0 we put $M_\delta = \sup\limits_{x\in AV_\delta} f(x)$ and $m_\delta = \inf\limits_{x\in AV_\delta} f(x)$. It is evident that M_δ is a nondecreasing function of δ and m_δ is a nonincreasing function of δ, and therefore the difference $M_\delta - m_\delta$ is a nondecreasing function of δ. Consequently there exists the limit

$$\lim_{\substack{\delta\to 0\\ \delta>0}} (M_\delta - m_\delta) = \omega(x^0)$$

which is called the *oscillation of the function f at the point* x^0. It is easy to prove the following theorem.

Theorem 5. *A function f defined on a bounded closed set A is continuous on A at a point* $x^0 \in A$ *if and only if its oscillation at that point is equal to zero, i.e.* $(\omega(x^0) = 0)$.

We shall also prove

Theorem 6. *Let A be a bounded closed set and let* $\lambda > 0$. *Then the set* E_λ *of points* $x \in A$ *for which* $\omega(x) \geq \lambda$ *is closed.*

Indeed, if $x^k \to x^0$ and $\omega(x^k) \geq \lambda$ $(k = 1, 2, \ldots)$ then $x^0 \in A$ and, besides, if $V_\delta(x^0)$ is a ball of radius δ with centre at x^0 there is k such that the point x^k lies strictly inside $V_\delta(x^0)$. Therefore there is a ball $V_\sigma(x^k)$ with centre at x^k whose radius is so small that

$$V_\sigma(x^k) \subset V_\delta(x^0)$$

and consequently

$$M_\delta(x^0) - m_\delta(x^0) \geq M_\sigma(x^k) - m_\sigma(x^k) \geq \omega(x^k) \geq \lambda$$

Hence, $M_\delta(x^0) - m_\delta(x^0) \geq \lambda$ for any ball $V_\delta(x^0)$, which implies $\omega(x^0) \geq \lambda$.

Example 1. A set Ω is said to be *convex* if the line segment connecting any two points of Ω entirely belongs to Ω.

For a convex set Ω we have the inequality

$$\omega(\delta_1+\delta_2, f) \leq \omega(\delta_1, f) + \omega(\delta_2, f) \tag{9}$$

Indeed, if $x', x'' \in \Omega$, $|x'-x''| < \delta_1+\delta_2$ then the line segment joining x' and x'' contains a point x such that $|x'-x| < \delta_1$ and $|x''-x| < \delta_2$. Therefore

$$\omega(\delta_1+\delta_2, f) = \sup_{|x'-x''|\leq\delta_1+\delta_2} |f(x')-f(x'')| \leq \sup_{|x'-x|\leq\delta_1} |f(x')-f(x)| + $$
$$+ \sup_{|x''-x|\leq\delta_2} |f(x'')-f(x)| = \omega(\delta_1, f) + \omega(\delta_2, f)$$

where the supremums in the third expression are taken over arbitrary x, x', $x'' \in \Omega$ satisfying the corresponding inequalities.

From (9) follows the inequality

$$\omega(m\delta) \leqslant m\omega(\delta) \tag{10}$$

valid for any natural m.

Example 2. From (9) and from the monotonicity of ω follow the inequalities

$$0 \leqslant \omega(\delta_2, f) - \omega(\delta_1, f) \leqslant \omega(\delta_2 - \delta_1, f), \qquad 0 \leqslant \delta_1 \leqslant \delta_2 \tag{11}$$

whence we see that $\omega(t, f)$ is a continuous function of $t \geqslant 0$ (if f is continuous on the closure of a bounded convex domain Ω).

Example 3. The *Dirichlet function* which is defined on [0, 1] and is equal to zero at all rational points and equal to unity at the irrational points is discontinuous (relative to [0, 1]) at all the points of the closed interval [0, 1]. But at the same time if it is considered as a function defined on the set A of rational points it is continuous (relative to A) throughout the set A.

Exercises

1. Prove that the moduli of continuity $\omega(t)$ of the functions

(i) $\sqrt{x}$ $(0 \leqslant x \leqslant 1)$ (see the beginning of § 15.5); (ii) x^2 $(0 \leqslant x \leqslant 1)$; (iii) $\sin x$ $\left(0 \leqslant x \leqslant \frac{\pi}{2}\right)$; (iv) $\sin \frac{1}{x}$ $(0 < |x| \leqslant 1)$;

(v) $\sin x$ $(-\infty < x < \infty)$

are respectively

(i) $\omega(t) = \begin{cases} \sqrt{t} & (0 \leqslant t \leqslant 1), \\ 1 & (1 \leqslant t); \end{cases}$ (ii) $\omega(t) = \begin{cases} 1-(1-t)^2 & (0 \leqslant t \leqslant 1), \\ 1 & (1 \leqslant t); \end{cases}$

(iii) $\omega(t) = \begin{cases} \sin t & (0 \leqslant t \leqslant \pi/2), \\ 1 & (\pi/2 \leqslant t); \end{cases}$ (iv) $\omega(t) = 2 \quad (0 \leqslant t)$;

(v) $\omega(t) = \begin{cases} 2\sin \frac{t}{2} & (0 \leqslant t \leqslant \pi), \\ 2 & (\pi \leqslant t). \end{cases}$

2. Show that if $\omega(t)$ $(t \geqslant 0)$ is a continuous function at the point $t = 0$ and satisfies the conditions

$$0 \leqslant \omega(\delta_2) - \omega(\delta_1) \leqslant \omega(\delta_2 - \delta_1) \qquad (0 < \delta_1 < \delta_2)$$

then it is the modulus of continuity of itself.

3. By the *distance from a point* $x^0 \in R$ *to a set* $E \subset R$ is meant the number

$$r(x^0, E) = \inf_{x \in E} |x^0 - x| = \inf_{x \in E} \sqrt{\sum_1^n (x_j - x_j^0)^2}$$

Here and below E, E_1, E_2 and F are nonempty sets.

Prove that if E is a closed set (bounded or unbounded) then there is a point $y \in E$ such that the distance $r(x^0, E)$ from x^0 to E is equal to the distance between x^0 and y: $r(x^0, E) = \min\limits_{x \in E} |x^0 - x| = |x^0 - y|$.

4. Prove that the distance $r(x^0, E)$ is a continuous function of x^0.

5. By the *distance between two sets* E_1 and E_2 is meant the number

$$r(E_1, E_2) = \inf_{\substack{x' \in E_1 \\ x'' \in E_2}} |x' - x''|$$

Prove that if E_1 and E_2 are closed sets one of which (or both) is bounded then there are two points $y \in E_1$ and $z \in E_2$ such that $r(E_1, E_2) = \min\limits_{x' \in E_1, x'' \in E_2} |x' - x''| = |y - z|$, i.e. the above infimum is equal to the distance between these points. It follows from this property that if the sets E_1 and E_2 are disjoint (i.e. nonintersecting) then $r(E_1, E_2) > 0$.

6. Prove that if F is a closed set, Ω is a bounded open set and $F \subset \Omega$ then there is $\varepsilon > 0$ such that the set F^ε of points x whose distances to F do not exceed ε belongs to Ω.

§ 7.11. Extension of a Uniformly Continuous Function. Partial Derivatives on the Boundary of the Domain of Definition

Theorem 1. *If a function f is uniformly continuous on a nonclosed set A it can be extended to $\bar{A} - A$ in a unique way so that the resultant (extended) function defined on $\bar{A}$ is continuous on $\bar{A}$.*

Proof. Let $x^0 \in \bar{A} - A$; then x^0 is a limit point of A. By the uniform continuity of f on A, given any $\varepsilon > 0$, there is $\delta > 0$ such that

$$|f(x') - f(x'')| < \varepsilon \tag{1}$$

for all $x', x'' \in A$ satisfying the condition

$$|x' - x''| < \delta \tag{2}$$

For the points $x', x'' \in A$ satisfying the conditions $|x' - x^0| < \delta/2$ and $|x'' - x^0| < \delta/2$ there holds inequality (2) and therefore inequality (1) as well. Consequently, by Cauchy's criterion, the limit of f on A at the point x^0 exists. It appears natural to denote it $f(x^0)$, that is to extend f to the point x^0 by defining its value at x^0 as being equal to this limit.

Now, let x_0' and x_0'' be two arbitrary points of $\bar{A}$ such that

$$|x_0' - x_0''| < \delta \tag{3}$$

Then there are two sequences of points $x_k', x_k'' \in A$ such that $|x_k' - x_k''| < \delta$ and $x_k' \to x_0'$, $x_k'' \to x_0''$ $(k \to \infty)$. For these points the inequality $|f(x_k') - f(x_k'')| < \varepsilon$ holds with any k; on passing to the limit in this inequality as $k \to \infty$ we obtain the relation

$$|f(x_0') - f(x_0'') \leqslant \varepsilon \tag{4}$$

Hence, we have proved that, given any $\varepsilon > 0$, there is $\delta > 0$ such that the relations $x_0', x_0'' \in \bar{A}$ together with inequality (3) imply automatically inequality (4). In other words, we have shown that the extended function f is uniformly continuous on $\bar{A}$ whence it follows that it is continuous at every point belonging to the set $\bar{A}$.

There can be no other extension of f satisfying the conditions of the theorem because if we put f to be equal at a point $x^0 \in \bar{A} - A$ to a number different from the limit $\lim_{\to x^0} f(x)$ the extended function f will be discontinuous at that point. The theorem has been proved.

Remark 1. In § 4.6 we proved a special case of this theorem when we extended the exponential function a^x defined originally for rational x to irrational x. In this construction we first defined a^x on the set of rational x and showed that this function was continuous on the set of all rational x belonging to any closed interval $[c, d]$. The closure of this set coincides with the entire closed interval $[c, d]$ and therefore a^x is continuously extended in a unique manner to $[c, d]$.

Let $G \subset R$ be an open set, $\bar{G}$ be its closure and f be a function defined and continuous on $\bar{G}$. It may happen that for some points of the boundary Γ of the set G it is impossible to speak of partial derivatives of f with respect to some of its arguments. For instance, if G is the circle σ specified by the inequality $x^2+y^2 < 1$ then the derivative f_x' of a function $f(x, y)$ defined on $\bar{\sigma}$ does not make sense at the point $A = (0, 1)$ of the boundary of the circle because the points of the x-axis adjoining A and lying to the right of A do not belong to $\bar{\sigma}$. In such cases it is sometimes possible to define the generalized notion of a partial derivative of f with the aid of continuous extension. If the given function f is not only continuous on $\bar{G}$ but also has on G the partial derivatives $\dfrac{\partial f}{\partial x_j}$ $(j = 1, \ldots, n)$ uniformly continuous on G we can apply Theorem 1 to extend them continuously to the boundary $\Gamma = \bar{G} - G$ of the set G. Such extensions are usually called the corresponding partial derivatives of f on Γ although in this case the notion of a partial derivative is understood in the generalized sense. Further, if a point $x^0 \in \Gamma$ admits of the computation of an ordinary one-sided (i.e. right-hand or left-hand or both) derivative the latter coincides with the generalized derivative at that point. To show this let us come back to the example of a function defined in the circle σ. For the point $B = (x^0, y^0), x_0 > 0, |y^0| < 1$, of the boundary of σ we have

$$\frac{f(x^0-h, y^0)-f(x^0, y^0)}{-h} = f_x'(x^0-\theta h, y^0) \to f_x'(x^0, y^0), \quad h \to 0, \quad h > 0$$

where the limit on the right is the corresponding generalized derivative. It is evident that in this case there exists the ordinary left-hand derivative of f with respect to x at the point (x^0, y^0) equal to the generalized derivative.

Later (see § 19.8, Theorems 1, 3 and 4) we shall prove that if the $(n-1)$-dimensional boundary Γ of a domain G (or a part γ of the boundary) is $r+1$

times continuously differentiable then a function f uniformly continuous on G together with its partial derivatives up to the order $r+1$ inclusive can be extended across Γ (or across γ) with the retention of its differentiability properties. In particular, the extended function $\bar{f}(x)$ will have ordinary continuous partial derivatives up to order r inclusive at all the points of Γ (or γ).

§ 7.12. Nested Rectangle Theorem and Heine-Borel* Lemma

Lemma. *Let*

$$\Delta_k = \{a_j^{(k)} \leq x_j \leq b_j^{(k)}, j = 1, \ldots, n\} \qquad (k = 1, 2, \ldots)$$

be a nested family of closed rectangles ($\Delta_k \supset \Delta_{k+1}$) *with diameters* $d_k = \sqrt{\sum_1^n (b_j^{(k)} - a_j^{(k)})^2}$ *tending to zero* ($d \to 0$). *Then there exists a single point* $x^0 = (x_1^0, \ldots, x_n^0) \in R$ *belonging to all* Δ_k *simultaneously.*

Proof. The conditions of the lemma imply that the closed intervals $[a_j^{(k)}, b_j^{(k)}]$ ($k = 1, 2, \ldots$) form a nested family for every $j = 1, \ldots, n$ and that their lengths tend to zero as $k \to \infty$. Therefore, by the nested interval axiom, for every j there is a single number x_j^0 belonging to all the intervals $[a_j^{(k)}, b_j^{(k)}]$ ($k = 1, 2, \ldots$) simultaneously. Now we see that the point x^0 with these numbers as its coordinates is nothing but the point whose existence is asserted in the lemma.

Heine-Borel Lemma. *Let an infinite system of open sets V (for instance, open cubes or balls) cover a bounded closed set $E \subset R$* (i.e. *the union of these intervals contains E*). *Then this system contains a finite subsystem of sets V which also covers E.*

Proof. The set E being bounded, there exists a cube $\Delta \subset R$ containing E. Suppose that the assertion of the lemma fails to hold. Let us divide Δ into 2^n equal subcubes. Among these smaller cubes there must be at least one (denote it by Δ_1) such that the assertion of the lemma fails to hold for the set $E\Delta_1$ (the intersection of E and Δ_1); this means that any finite subsystem of sets V does not cover $E\Delta_1$ either. Next we divide Δ_1 into 2^n equal cubes; among them there is at least one (denote it Δ_2) such that the assertion of the lemma does not hold for the set $E\Delta_2$. Continuing this process indefinitely we arrive at a nested family of cubes $\Delta_1 \supset \Delta_2 \supset \ldots$ whose diameters tend to zero such that each of the sets $E\Delta_k (k = 1, 2, \ldots)$ cannot be covered by any finite system of sets V. The foregoing lemma indicates that there is a point $x^0 \subset R$ belonging to all Δ_k simultaneously. By the closedness of E, it must belong to E as well and therefore it is contained in some set V_0 belonging to the system in question. Since V_0 is an open set we have $\Delta_k \subset V_0$ for some sufficiently large k. Consequently $E\Delta_k \subset V_0$.

---

* E. Heine (1821-1881), a German mathematician, F. É. É. Borel (1871-1936), a French mathematician.

We have thus arrived at a contradiction because, on the one hand, $E\Delta_k$ belongs to the set V_0 and, on the other hand, there is no finite system of sets V covering $E\Delta_k$.

§ 7.13. Taylor's Formula

Let us consider a function $f(x) = f(x_1, \ldots, x_n)$ defined on an open set $\Omega \subset \subset R_n$ and having on Ω the continuous partial derivatives needed for the formulas below to make sense.

Let us fix a point $x_0 = (x_1^0, \ldots, x_n^0) \subset \Omega$ and choose a sufficiently small number $\delta > 0$ so that all the points $x = (x_1, \ldots, x_n)$ lying within the neighbourhood

$$|x^0 - x| = \sqrt{\sum_1^n (x_j - x_j^0)^2} < \delta \tag{1}$$

of the point x^0 belong to Ω.

Now we construct the auxiliary function

$$F(t) = f\big(x^0 + t(x - x^0)\big) = f\big(x_1^0 + t(x_1 - x_1^0), \ldots, x_n^0 + t(x_n - x_n^0)\big)$$

of the variable $t \in [0, 1]$. We have temporarily fixed one more point x belonging to neighbourhood (1). But later x will be regarded as a variable point. It is evident that

$$F(0) = f(x^0) \text{ and } F(1) = f(x) \tag{2}$$

According to the differentiation theorem for a composite function, we have

$$F'(t) = \sum_{j=1}^n (x_j - x_j^0) \frac{\partial f}{\partial x_j}\big(x^0 + t(x - x^0)\big) \tag{3}$$

Hence

$$F'(0) = \sum_{j=1}^n (x_j - x_j^0) \frac{\partial f}{\partial x_j}(x^0) \tag{4}$$

Further, we have

$$F''(t) = \sum_{j=1}^n (x_j - x_j^0) \sum_{k=1}^n (x_k - x_k^0) \frac{\partial^2 f}{\partial x_j \partial x_k}\big(x^0 + t(x + x^0)\big) =$$

$$= \sum_{j=1}^n \sum_{k=1}^n (x_j - x_j^0)(x_k - x_k^0) \frac{\partial^2 f}{\partial x_j \partial x_k}\big(x^0 + t(x - x^0)\big) \tag{5}$$

and

$$F''(0) = \sum_{j=1}^n \sum_{k=1}^n (x_j - x_j^0)(x_k - x_k^0) \frac{\partial^2 f}{\partial x_j \partial x_k}(x^0) \tag{6}$$

Arguing by induction we arrive at the derivative

$$F^{(l)}(t) = \sum_{j_1=1}^n \cdots \sum_{j_l=1}^n (x_{j_1} - x_{j_1}^0) \ldots (x_{j_l} - x_{j_l}^0) \frac{\partial^l f\big(x^0 + t(x - x^0)\big)}{\partial x_{j_1} \ldots \partial x_{j_l}} \tag{7}$$

and to its value at the point $t = 0$:

$$F^{(l)}(0) = \sum_{j_1=1}^{n} \dots \sum_{j_l=1}^{n} (x_{j_1}-x_{j_1}^0) \dots (x_{j_l}-x_{j_l}^0) \frac{\partial^l f(x^0)}{\partial x_{j_1} \dots \partial x_{j_l}} \tag{8}$$

Let f possess partial derivatives up to the order l inclusive on the set Ω. Then f has continuous ordinary derivatives with respect to t up to the same order l inclusive. Therefore we can write the expasion of the function $F(t)$ of one variable by Taylor's formula:

$$F(t) = \sum_{0}^{l-1} \frac{t^k}{k!} F^{(k)}(0)+r_l(t)$$

where $r_l(t) = (t^l/l!)\, F^{(l)}\ (\theta t)$ $(0 < \theta < 1)$ and θ is dependent on x and l. It follows that

$$F(1) = \sum_{0}^{l-1} \frac{1}{k!} F^{(k)}(0)+R_l, \quad R_l(x) = r_l(1) = \frac{1}{l!} F^{(l)}(\theta) \tag{9}$$

Consequently

$$f(x) = \sum_{k=0}^{l-1} \frac{1}{k!} \sum_{j_1=1}^{n} \dots \sum_{j_k=1}^{n} (x_{j_1}-x_{j_1}^0) \dots (x_{j_k}-x_{j_k}^0) \frac{\partial^k f(x^0)}{\partial x_{j_1} \dots \partial x_{j_k}} + R_l(x) \tag{10}$$

where

$$R_l(x) = \frac{1}{l!} \sum_{j_1=1}^{n} \dots \sum_{j_l=1}^{n} (x_{j_1}-x_{j_1}^0) \dots (x_{j_l}-x_{j_l}^0) \frac{\partial^l f(x^0+\theta(x-x^0))}{\partial x_{j_1} \dots \partial x_{j_l}} \tag{11}$$

Relations (10), (11) present *Taylor's formula for the function* $f(x)$ *of* n *variables at the point* $\mathbf{x}^0$ *with the remainder* R_l *in Lagrange's form.*

It should be noted that among the terms of the multiple sum in (10) there are some coincident with each other. Let us write down several terms of Taylor's formula for the case of two independent variables:

$$\begin{aligned} f(x_1, x_2) &= f(x_1^0, x_2^0)+\left[\left(\frac{\partial f}{\partial x_1}\right)_0 (x_1-x_1^0)+\left(\frac{\partial f}{\partial x_2}\right)_0 (x_2-x_2^0)\right]+ \\ &+\frac{1}{2}\left[\left(\frac{\partial^2 f}{\partial x_1^2}\right)_0 (x_1-x_1^0)^2+2\left(\frac{\partial^2 f}{\partial x_1\, \partial x_2}\right)_0 (x_1-x_1^0)(x_2-x_2^0)+ \right.\\ &\left.+\left(\frac{\partial^2 f}{\partial x_2^2}\right)_0 (x_2-x_2^0)^2\right]+ \dots +\frac{1}{(l-1)!}\left((x_1-x_1^0)\frac{\partial}{\partial x_1}+ \right.\\ &\left.+(x_2-x_2^0)\frac{\partial}{\partial x_2}\right)_0^{l-1} f+R_l, \end{aligned}$$

$$R_l = \frac{1}{l!}\left((x_1-x_1^0)\frac{\partial}{\partial x_1}+(x_2-x_2^0)\frac{\partial}{\partial x_2}\right)_\theta^l f \qquad (0 < \theta < 1)$$

Here we have used the symbolic relation

$$\left(\xi_1 \frac{\partial}{\partial x_1}+\xi_2 \frac{\partial}{\partial x_2}\right)_0^k f = \sum_{j=0}^{k} C_k^j \xi_1^j \xi_2^{k-j} \left(\frac{\partial^j f}{\partial x_1^j\, \partial x_2^{k-j}}\right)_0$$

The symbols $(\quad)_0$ and $(\quad)_\theta$ show that in the expressions in the parentheses x is replaced in f by x^0 and $x^0+\theta(x-x^0)$ respectively.

Similarly, in the case of three independent variables we have

$$f(x_1, x_2, x_3) = \sum_{k=0}^{l-1} \frac{1}{k!}\left[(x_1-x_1^0)\frac{\partial}{\partial x_1}+(x_2-x_2^0)\frac{\partial}{\partial x_2}+\right.$$

$$\left.+(x_3-x_3^0)\frac{\partial}{\partial x_3}\right]_0^k f+R_l,$$

$$R_l = \frac{1}{l!}\left[(x_1-x_1^0)\frac{\partial}{\partial x_1}+\ \dots\ +(x_3-x_3^0)\frac{\partial}{\partial x_3}\right]_\theta^l f, \qquad 0<\theta<1$$

The notation used here should be clear to the reader.

Taylor's formula can also be written in the following compact form:

$$f(\boldsymbol{x}) = \sum_{|k|\leq l-1} \frac{(\boldsymbol{x}-\boldsymbol{x}^0)^k}{k!} f^{(k)}(\boldsymbol{x}^0)+R_l,$$

$$R_l = \sum_{|k|=l} \frac{(\boldsymbol{x}-\boldsymbol{x}^0)^k}{k!} f^{(k)}(\boldsymbol{x}^0+\theta(\boldsymbol{x}-\boldsymbol{x}^0)), \qquad (0<\theta<1)$$

where $|k| = \sum_1^n k_j,\ k! = k_1!\ \dots\ k_n! \qquad (0! = 1),\ \ f^{(k)}(\boldsymbol{x}) = \dfrac{\partial^{|k|}f}{\partial x_1^{k_1}\dots\partial x_n^{k_n}}$,

$(\boldsymbol{x}-\boldsymbol{x}^0)^k = (x_1-x_1^0)^{k_1}\dots(x_n-x_n^0)^{k_n}$.

Taylor's formula is used particularly often in the cases $l = 1, 2$. If $l = 1$ then, on transposing $f(\boldsymbol{x}^0)$ to the left-hand side, we obtain the formula

$$f(\boldsymbol{x})-f(\boldsymbol{x}^0) = \sum_{j=1}^{n}\left(\frac{\partial f}{\partial x_j}\right)_{\boldsymbol{x}^0+\theta(\boldsymbol{x}-\boldsymbol{x}^0)}(x_j-x_j^0) \tag{12}$$

which generalizes Lagrange's one-dimensional formula of finite increments to the case of dimension n.

For $l = 2$ this formula is written as

$$f(\boldsymbol{x}) = f(\boldsymbol{x}^0)+\sum_{j=1}^{n}\left(\frac{\partial f}{\partial x_j}\right)_0(x_j-x_j^0)+R_2,$$

$$R_2 = \frac{1}{2}\sum_{k=1}^{n}\sum_{l=1}^{n}\left(\frac{\partial^2 f}{\partial x_k\,\partial x_l}\right)_{\boldsymbol{x}^0+\theta(\boldsymbol{x}-\boldsymbol{x}^0)}(x_k-x_k^0)(x_l-x_l^0) =$$

$$= \frac{1}{2}\sum_{k=1}^{n}\sum_{l=1}^{n}\left(\frac{\partial^2 f}{\partial x_k\,\partial x_l}\right)_0(x_k-x_k^0)(x_l-x_l^0)+\varepsilon\varrho^2 \tag{13}$$

where $\varepsilon\to 0$ for $\varrho^2 = \sum_{j=1}^{n}(x_j-x_j^0)^2\to 0$.

Indeed, by the continuity of the derivatives we deal with,

$$\left(\frac{\partial^2 f}{\partial x_k\, \partial x_l}\right)_{x^0+\theta(x-x^0)} = \left(\frac{\partial^2 f}{\partial x_k\, \partial x_l}\right)_0 + \varepsilon_{kl}$$

where $\varepsilon_{kl} \to 0$ as $\varrho \to 0$, and therefore, putting $\eta = \max\limits_{k,l} |\varepsilon_{kl}|$, we conclude that the second summand on the right-hand side of (13) does not exceed

$$\eta \sum_k \sum_l |x_k - x_k^0|\, |x_l - x_l^0| = \eta \Big(\sum_k |x_k - x_k^0|\Big)^2 \leqslant \eta n \varrho^2$$

in its absolute value. Therefore this summand can also be written in the form $\varepsilon\varrho^2$ where $|\varepsilon| < n\eta \to 0$ $(\varrho \to 0)$.

The last sum in (13) can be rewritten as $A(\xi) = \sum\limits_{k=1}^{n} \sum\limits_{l=1}^{n} a_{kl}\xi_k\xi_l$ where $a_{kl} = a_{lk} = \left(\frac{\partial^2 f}{\partial x_k\, \partial x_l}\right)_0$ and $\xi_k = (x_k - x_k^0)$ $(k = 1, \ldots, n)$.

Hence, $A(\xi)$ is a quadratic form in n variables. If we denote $\xi_k = x_k - x_k^0 = dx_k$ then

$$A(\xi) = d^2 f_0 = \sum_k \sum_l \left(\frac{\partial^2 f}{\partial x_k\, \partial x_l}\right)_0 dx_k\, dx_l$$

which is nothing but the second differential of f at the point x^0 corresponding to the independent differentials $dx_1, \ldots, dx_n$.

We also note that remainder (9) in Taylor's formula for the function $F(t)$ of one variable can be written in the integral form (see § 9.17):

$$r_l(t) = \frac{1}{(l-1)!} \int_0^t (t-u)^{l-1} F^{(l)}(u)\, du$$

By virtue of (7), to this expression corresponds the following formula for the remainder in Taylor's formula for the function $f(x)$ of several variables:

$$R_l(x) = r_l(1) = \frac{1}{(l-1)!} \int_0^1 (1-u)^{l-1} \sum_{j_1=1}^{n} \ldots \sum_{j_l=1}^{n} (x_{j_1} - x_{j_1}^0) \ldots$$

$$\ldots (x_{j_l} - x_{j_l}^0) \frac{\partial^l f(x^0 + u(x-x^0))}{\partial x_{j_1} \ldots \partial x_{j_l}}\, du =$$

$$= l \sum_{|k|=l} \frac{(x-x^0)^k}{k!} \int_0^1 (1-u)^{l-1} f^{(k)}(x^0 + u(x-x^0))\, du \tag{14}$$

By the continuity (with respect to (u, x)) of the integrands in (14) the integrals themselves are continuous functions of the parameter x (see § 13.14, Theorem 1). Moreover, if the function f has continuous partial derivatives of the $(l+s)$th order in a neighbourhood of x^0 the integrals can be differentiated s times under the integral sign (see § 13.14, Theorem 3).

§ 7.14. Taylor's Formula with Peano's Form of the Remainder. Uniqueness of Taylor's Coefficients

Let a function $f(x) = f(x_1, \ldots, x_n)$ be defined in a neighbourhood $\Omega \subset R_n$ of a point $x^0 = (x_1^0, \ldots, x_n^0)$ and let, for all x belonging to the sphere $|x - x^0| < \delta$ lying within Ω, this function be representable in the form

$$f(x) = P_N(x) + o(\varrho^N) \quad \left(\varrho = \left[\sum_{j=1}^{n} (x_j - x_j^0)^2\right]^{1/2} \to 0\right) \tag{1}$$

where

$$P_N(x) = \sum_{|k| \leq N} a_k (x - x^0)^k$$
$$\left(a_k = a_{k_1, \ldots, k_n}, |k| = \sum_{1}^{n} k_j, \ (x - x^0)^k = \prod_{j=1}^{n} (x_j - x_j^0)^{k_j}\right) \tag{2}$$

In this case we say that the function f is expanded by *Taylor's formula with the remainder in Peano's form at the point* x^0. If, for instance, it is known that the function f has continuous partial derivatives up to order N inclusive in a neighbourhood of the point x^0 it is representable by formula (1).

Let us prove that the representation of the function $f(x)$ by formula (1) is unique; in other words, we shall show that if f admits of representation (1) and, simultaneously, of a representation

$$f(x) = \sum_{|k| \leq N} a'_k (x - x^0)^k + o(\varrho^N) \quad (\varrho \to 0), \ |x - x^0| < \delta) \tag{3}$$

where $a'_k = a'_{k_1, \ldots, k_n}$ are constant coefficients then $a_k = a'_k$ $(|k| \leq N)$. Indeed, on subtracting (3) from (1) we obtain the identity

$$0 \equiv \sum_{|k| \leq N} \alpha_k (x - x^0)^k + o(\varrho^N) \quad (\varrho \to 0, \alpha_k = (a_k - a'_k)) \tag{4}$$

which holds for $|x - x^0| < \delta$. Let us fix the point x and consider the variable point $z_t = x^0 + t(x - x^0)$ dependent on $t \in [0, 1]$. It is obvious that $z_t - x^0 = t(x - x^0)$ and $|z_t - x^0| = t|x - x^0|$. Therefore the substitution of z_t for x into (4) results in

$$0 \equiv \sum_{|k| \leq N} \alpha_k (x - x^0)^k t^{|k|} + o(t^N) \qquad (t \to 0)$$

The relation obtained can be written in the form

$$0 \equiv \sum_{l=0}^{N} t^l \sum_{|k| = l} \alpha_k (x - x^0)^k + o(t^N) \qquad (t \to 0)$$

It follows (see the lemma in § 5.9) that

$$\sum_{|k| = l} \alpha_k (x - x^0)^k = 0 \qquad (l = 0, 1, \ldots, N) \tag{5}$$

for any point x belonging to the ball $|x-x^0| < \delta$. Hence, equality (5) is fulfilled identically for all x belonging to that ball. On finding the derivative of order $k = (k_1, \ldots, k_n)$ of the left-hand side of (5) and substituting $x = x^0$ into it we obtain $k! \cdot \alpha_k = 0$ ($|k| \leqslant N$) whence $\alpha_k = 0$, which is what we set out to prove.

Example. For $x, y \to 0$ we have

$$\begin{aligned}\psi(x, y) &= \frac{1}{(1-x)(1-y)} = [1+x+x^2+o(x^2)]\,[1+y+y^2+o(y^2)] = \\ &= 1+(x+y)+(x^2+xy+y^2)+o(x^2)+x^2y+xy^2+x^2y^2+o(y^2) = \\ &= 1+(x+y)+(x^2+xy+y^2)+o(\varrho^2) \qquad (\varrho \to 0)\end{aligned}$$

and thus the function ψ has been expanded at the point (0, 0) by Taylor's formula with the remainder $o(\varrho^2)$ ($\varrho \to 0$) in Peano's form.

§ 7.15. Local Extremum of a Function

Let a function $f(x) = f(x_1, \ldots, x_n)$ be defined in an open set $\Omega \subset R_n$. We say that *f attains its local (unconditional) maximum at a point $x^0 \in \Omega$* if there is a positive number $\delta > 0$ such that the function $f(x)$ is defined for all the points x satisfying the condition

$$|x-x^0| = \sqrt{\sum_{j=1}^{n} (x_j - x_j^0)^2} < \delta \tag{1}$$

and the inequality $f(x) \leqslant f(x^0)$ is fulfilled for these x. Similarly, we say that *f attains at x^0 its local (unconditional) minimum* if there is a neighbourhood of type (1) in which the function f is defined and satisfies the inequality $f(x) \geqslant f(x^0)$.

Local maxima and minima are referred to as *local extrema*.

If a function f attains at a point x^0 a local extremum and has the partial derivatives of the first order $\frac{\partial f}{\partial x_j}$ ($j = 1, \ldots, n$) at that point these derivatives must vanish at the point x^0:

$$\left(\frac{\partial f}{\partial x_j}\right)_0 = \frac{\partial f}{\partial x_j}(x_1^0, \ldots, x_n^0) = 0 \qquad (j = 1, \ldots, n)$$

Indeed, in this case the function $f(x_1^0, \ldots, x_{j-1}^0, x_j, x_{j+1}^0, \ldots, x_n^0)$ of one variable x_j has a local extremum at x_j^0 for each $j = 1, \ldots, n$ and its derivative with respect to x_j at the point $x_j = x_j^0$, which is equal to $\left(\frac{\partial f}{\partial x_j}\right)_0$, must turn into zero.

It follows that to determine the points $x^0 \in \Omega$ where f attains its local extrema we must seek them among the points $x \in \Omega$ such that f either has no partial derivative at x with respect to some of the variables or has the derivatives at x equal to zero. We shall be interested in the latter case. Let us show

how it can be found whether f has an extremum and what this extremum is (i.e. a maximum or a minimum) at a point x^0 of this kind.

Let f possess continuous partial derivatives of the second order in a neighbourhood $|x-x^0| < \delta$ of the point x^0 in question and let its all partial derivatives of the first order vanish at x^0. Then the expansion of f by Taylor's formula (with $l = 2$) can be written as

$$f(x)-f(x^0) = \sum_{k=1}^{n} \sum_{l=1}^{n} a_{kl}\xi_k\xi_l+\varepsilon\varrho^2 \tag{2}$$

$$a_{kl} = a_{lk} = \left(\frac{\partial^2 f}{\partial x_k\,\partial x_l}\right)_0;\ \xi_k = x_k-x_k^0,\ \varepsilon \to 0 \quad \text{for } \varrho = \sqrt{\sum_1^n \xi_k^2} \to 0$$

The quadratic form $A(\xi) = \sum_{k=1}^{n} \sum_{l=1}^{n} a_{kl}\xi_k\xi_l$ necessarily possesses one of the following properties:

(i) the form $A(\xi)$ is *positive definite*, that is $A(\xi) > 0$ for any $\xi = (\xi_1, \xi_2, \ldots, \xi_n)$ with $\varrho > 0$;

(ii) the form $A(\xi)$ is *negative definite*, that is $A(\xi) < 0$ for any ξ with $\varrho > 0$;

(iii) the form $A(\xi)$ is *positive semidefinite* or *negative semidefinite*, that is $A(\xi) \geq 0$ for all ξ or $A(\xi) \leq 0$ for all ξ, respectively, and there exists a point $\xi' = (\xi_1', \ldots, \xi_n')$ with $\varrho' \sqrt{\sum_1^n \xi_k'^2} > 0$ such that $A(\xi') = 0$;

(iv) the form $A(\xi)$ is *indefinite*, that is there exist ξ' and ξ'' such that $A(\xi') > 0$ and $A(\xi'') < 0$.

We shall prove that *in the first case the function f attains at x^0 its local minimum, in the second case its local maximum, in the fourth case the function f is sure to have no extremum and in the third case the question as to whether f has an extremum at x^0 remains open since there are examples in which, under the conditions in question, a function has an extremum at x^0 or has no extremum.*

Taking an arbitrary ξ with $\varrho > 0$ we put $\eta = \xi/\varrho = (\eta_1, \ldots, \eta_n)$, i.e. $\eta_j = \xi_j/\varrho$ $(j = 1, \ldots, n)$. Then equality (2) can be written as

$$f(x)-f(x^0) = \varrho^2(\Phi(\eta)+\varepsilon) \tag{3}$$

where $\quad \Phi(\eta) = \sum_{k=1}^{n} \sum_{l=1}^{n} a_{kl}\eta_k\eta_l \quad$ and

$$\sum_{j=1}^{n} \eta_j^2 = \frac{1}{\varrho^2}\sum_{j=1}^{n} \xi_j^2 = 1 \tag{4}$$

This means that the function $\Phi(\eta)$ is considered on the spherical surface (described by equation (4)) which is a bounded closed point set (denote it by σ). The function $\Phi(\eta)$ is obviously continuous on σ.

In Case (i) we have $\Phi(\eta) > 0$ on σ. Since the set σ is bounded and closed and the function $\Phi(\eta)$ is continuous on it there exists the minimum value $\Phi(\eta) = m > 0$, $\eta \in \sigma$.

Further, since $\varepsilon \to 0$ as $\varrho \to 0$ there is a sufficiently small $\delta > 0$ such that $|\varepsilon| < m/2$ for all $\varrho < \delta$. On the basis of (3), we have

$$f(x)-f(x^0) > \varrho^2\left(m-\frac{m}{2}\right) = \frac{m}{2}\varrho^2 > 0$$

for the indicated ϱ, that is the function f attains a local minimum at the point x^0.

Assertion (ii) is proved similarly. If the quadratic form in question is negative definite we have $\Phi(\eta) < 0$ for σ, and consequently the function $\Phi(\eta)$ attains its (negative) maximum on σ which we denote $-M$ $(M > 0)$. Now, if $\varrho < \delta$ for a sufficiently small $\delta > 0$ then $|\varepsilon| < M/2$ and therefore we have

$$f(x)-f(x^0) < \varrho^2\left(-M+\frac{M}{2}\right) = -\frac{M}{2}\varrho^2 < 0$$

for $|x-x^0| < \delta$, which means that f has a local maximum at x^0.

In Case (iii) the form turns into zero at some point $\xi' \neq 0$ and therefore, by the homogeneity of the form, it must also vanish for any point of the form $x' = \alpha\xi'$ where α is an arbitrary number. This shows that the form is equal to zero for all the indicated points x' and consequently $f(x')-f(x^0) = \varepsilon\varrho^2$.

But the sign of ε is unknown in this case, and therefore it is impossible to judge upon the existence of the extremum of f at x^0 by the second derivatives of f. The only conclusion that can be drawn in these circumstances is that if the form is not identically equal to zero and is positive semidefinite then there can be no maximum at x^0 and if it is not identically equal to zero and is negative semidefinite then there can be no minimum at x^0.

In Case (iv) it is convenient to return to equality (3). By the hypothesis, in this case there is a point ξ' at which the form is positive and a point ξ'' at which it is negative; therefore at the points η' and η'' corresponding to them we have the inequalities $\Phi(\eta') > 0$ and $\Phi(\eta'') < 0$, and for small ϱ we have $\Phi(\eta')+\varepsilon > 0$ and $\Phi(\eta'')+\varepsilon < 0$. This means that in any neighbourhood of x^0, however small, there are points x' and x'' such that $f(x') > f(x^0)$ and $f(x'') < f(x^0)$, that is there is no extremum of f at x^0.

Let us consider the sequence of the so-called *principal (diagonal) minors* of the quadratic form $A(\xi)$:

$$\Delta_1 = a_1;\ \Delta_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix};\ \ldots;\quad \Delta_n = \begin{vmatrix} a_{11} & \ldots & a_{1n} \\ a_{n1} & \ldots & a_{nn} \end{vmatrix}$$

By the *Sylvester theorem* proved in the theory of quadratic forms, we can assert that

(i) if $\Delta_1 > 0, \Delta_2 > 0, \ldots, \Delta_n > 0$ then the form is positive definite (Case 1);

(ii) if $\Delta_1 < 0$, $\Delta_2 > 0$, $\Delta_3 < 0$, $\ldots$, $(-1)^n\Delta_n > 0$, the form is negative definite (Case 2);

(iii) if $\Delta_1 \geqslant 0, \Delta_2 \geqslant 0, \ldots, \Delta_n \geqslant 0$ or $\Delta_1 \leqslant 0, \Delta_2 \geqslant 0, \ldots, (-1)^n \Delta_n \geqslant 0$ and if there is j for which $\Delta_j = 0$ then the form is positive or negative semi-definite respectively (Case 3);

(iv) in all the other cases the form is indefinite (Case 4).

In the two-dimensional case equality (2) takes the form

$$f(x_1, x_2)-f(x_1^0, x_2^0) = \frac{1}{2}(A\xi_1^2+2B\xi_1\xi_2+C\xi_2^2)+\varepsilon\varrho^2,$$

$$A = \left(\frac{\partial^2 f}{\partial x_1^2}\right)_0, \quad B = \left(\frac{\partial^2 f}{\partial x_1\, \partial x_2}\right)_0, \quad C = \left(\frac{\partial^2 f}{\partial x_2^2}\right)_0$$

and the corresponding sequence of principal minors consists of only two terms:

$$\Delta_1 = A, \quad \Delta_2 = \begin{vmatrix} A & B \\ B & C \end{vmatrix} = AC-B^2$$

Consequently,

(a) if $A > 0$ and $AC-B^2 > 0$ then f has a minimum at x^0;

(b) if $A < 0$ and $AC-B^2 > 0$ then there is a maximum of f at x^0;

(c) if $AC-B^2 < 0$ there is no extremum at $\boldsymbol{x}^0$;

(d) if $AC-B^2 = 0$ then it is impossible to judge upon the existence of an extremum at $\boldsymbol{x}^0$ by the second derivatives of f at $\boldsymbol{x}^0$.

These assertions can also be easily deduced from the representation

$$A(\boldsymbol{\zeta}) = A\xi^2+2B\xi\eta+C\eta^2 = \frac{(A\xi+B\eta)^2+(AC-B^2)\eta^2}{A}$$

where $A \neq 0$ ($\boldsymbol{\zeta} = (\xi, \eta) \neq 0$).

In Case (a) we have $A(\boldsymbol{\zeta}) > 0$ if $|\eta| > 0$, and if $\eta = 0$ then there must be $\xi \neq 0$ which again implies $A(\boldsymbol{\zeta}) > 0$.

In Case (b) we have $A(\boldsymbol{\zeta}) < 0$ if $|\eta| > 0$, and if $\eta = 0$ then $\xi \neq 0$ and again $A(\boldsymbol{\zeta}) < 0$.

In Case (c) with $A \neq 0$ it is possible to find (ξ, η) such that $\eta \neq 0$ and $(A\xi + B\eta) = 0$ and, on the other hand, to put $\eta = 0$ and $\xi > 0$. In both cases we obtain $A(\boldsymbol{\zeta}) \neq 0$ but the signs of the quadratic form are different. If $A = 0$ but $C \neq 0$ the same conclusions can be drawn if the roles of A and C are interchanged.

In Case (d) we have $A(\boldsymbol{\zeta}) = (A\xi+B\eta)^2/A$ for $A \neq 0$, and there is a point $\boldsymbol{\zeta} = (\xi, \eta)$ not coinciding with the origin such that $A(\boldsymbol{\zeta}) = 0$. The same is obtained when $C \neq 0$ if we replace A by C in the argument. Finally, if $A = C = 0$ the form $A(\boldsymbol{\zeta}) = B\xi\eta$ is obviously indefinite.

Example 1. For the function $f(x, y) = x^3+y^3-9xy+1$ the equations $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$ have two solutions: $x = y = 3$ (the case of a minimum) and $x = y = 0$ (the case when there are no extrema).

Example 2. For the function $f(x, y) = x^4+y^4-2x^2+4xy-2y^2+1$ the corresponding equations have three solutions: $x = +\sqrt{2}$, $y = -\sqrt{2}$; $x =$

$= -\sqrt{2}$, $y = +\sqrt{2}$ and $x = y = 0$. Accordingly, the first solution corresponds to a minimum, the second solution also corresponds to a minimum and in the case of the third solution there is no extremum.

Example 3. Equating to zero the derivatives of the function $f(x, y) = x^2 - 2xy^2 + y^4 - y^5$ we find $x = y = 0$ (an ambiguous case). On the other hand, we obviously have $f(x, y) = (x-y^2)^2 - y^5$, and, since $f(\varepsilon^2, \varepsilon) = -\varepsilon^5 < 0$ and $f(\varepsilon, 0) = \varepsilon^2 > 0$ for any $\varepsilon > 0$, there is no extremum at the point (0, 0). At the same time the function $y(t) = f(ht, kt)$ has a minimum for $t = 0$ for any h and k such that $h^2 + k^2 > 0$.

§ 7.16. Existence Theorem for an Implicit Function

Let us take an arbitrary function $f(x, y)$ of two variables x and y and equate it to zero:

$$f(x, y) = 0 \tag{1}$$

Let $\mathfrak{M}$ denote the set of all the points (x, y) for which equality (1) is fulfilled and let (x_0, y_0) be a point belonging to this set, that is $f(x_0, y_0) = 0$.

If no additional conditions are imposed on the function f the set $\mathfrak{M}$ can be of an arbitrary structure. For example, in the case of the function $f(x, y) = (x-x_0)^2 + (y-y_0)^2$ the set $\mathfrak{M}$ consists of a single point (x_0, y_0); if

$$f(x, y) = (x-x_0)^2 - (y-y_0)^2 = (x+y-x_0-y_0)(x-y-x_0+y_0)$$

then $\mathfrak{M}$ is a pair of straight lines passing through (x_0, y_0). We also often encounter the case when, at least in a sufficiently small neighbourhood of (x_0, y_0), the set $\mathfrak{M}$ is a curve described by a continuous (one-valued) function

$$y = \psi(x), \qquad x \in (x_0 - \delta, x_0 + \delta)$$

In this connection there arises the question as to how we can find which of the cases takes place judging by the properties of the function f. Below are two theorems giving answers to this question.

Theorem 1. *Let an equation*

$$f(x, y) = 0 \tag{1}$$

satisfy the following conditions.

The function f is defined in a two-dimensional neighbourhood Ω of the point (x^0, y^0) in the xy-plane and is continuous in that neighbourhood together with its partial derivatives of the first order; the derivative of f with respect to y is different from zero:

$$\frac{\partial f}{\partial y} \neq 0 \tag{2}$$

Besides, let $f(x^0, y^0) = 0$.

Let $\mathfrak{M}$ be the set of all points (x, y) satisfying equation (1) (*in particular, $(x^0, y^0) \in \mathfrak{M}$*).

Then for any $b_0 > 0$ there is a rectangle

$$\Delta = \{|x-x^0| < a, |y-y^0| < b\}, \qquad b < b_0 \tag{3}$$

belonging to Ω such that the set $\mathfrak{M}\Delta$ (the intersection of $\mathfrak{M}$ and Δ) is described by a (uniquely determined) continuously differentiable (one-valued) function

$$y = \varphi(x), \qquad x \in \Delta^0 \tag{4}$$

where

$$\Delta^0 = \{|x-x_0| < a\} \tag{5}$$

In other words, the rectangle Δ possesses the property that on its projection Δ^0 on the x-axis it is possible to define a continuously differentiable function (4) satisfying equation (1):

$$f(x, \varphi(x)) \equiv 0, \qquad x \in \Delta^0 \tag{6}$$

Its graph is entirely contained in Δ. This function is determined uniquely in the sense that the coordinates of any point $(x, y) \in \mathfrak{M}\Delta$ are connected by equation (4). In particular, $y^0 = \varphi(x^0)$ because $(x^0, y^0) \in \mathfrak{M}\Delta$.

Theorem 1'. *Let an equation*

$$f(\boldsymbol{x}, y) \doteq f(x_1, \ldots, x_n, y) = 0 \tag{1'}$$

satisfy the following conditions.

The function f is defined in a neighbourhood Ω of a point $(\boldsymbol{x}^0, y^0) = (x_1^0, \ldots, x_n^0, y^0)$ belonging to the space R_{n+1} of points $(\boldsymbol{x}, y) = (x_1, \ldots, x_n, y)$ and is continuous there together with its first-order partial derivatives; the derivative with respect to y is different from zero and the function itself is equal to zero at $(\boldsymbol{x}^0, y^0)$:

$$\frac{\partial f}{\partial y} \neq 0 \text{ and } f(\boldsymbol{x}^0, y^0) = 0 \tag{2'}$$

Let $\mathfrak{M}$ be the set of all points (x, y) satisfying equation (1') (in particular, $(x^0, y^0) \in \mathfrak{M}$).

Then for any $b_0 > 0$ there is a rectangle

$$\Delta = \{|x_j - x_j^0| < a,\ j = 1, \ldots, n,\ |y-y^0| < b\}, \qquad b < b_0 \tag{3'}$$

lying within Ω such that the set $\mathfrak{M}\Delta$ is described by a (uniquely determined) continuously differentiable (i.e. having continuous partial derivatives) function

$$y = \psi(\boldsymbol{x}) = \psi(x_1, \ldots, x_n), \qquad \boldsymbol{x} \in \Delta^0 \tag{4'}$$

where

$$\Delta^0 = \{|x_j - x_j^0| < a, \quad j = 1, \ldots, n\} \tag{5'}$$

Proof of Theorem 1. Without loss of generality we can suppose that Ω is a *domain*, that is a connected open set. Since $\frac{\partial f}{\partial y}$ is continuous on Ω condition (2') implies that $\frac{\partial f}{\partial y}$ retains sign throughout Ω. For definiteness, let $\frac{\partial f}{\partial y} > 0$ on Ω. Then, by the conditions of the theorem, there is $b > 0$ ($b < b_0$) such that the function $f(x^0, y)$ of one variable y is continuous on the closed interval $[y^0-b, y^0+b]$ ($\boldsymbol{x} = \boldsymbol{x}^0$), is strictly increasing on that interval and turns into zero only at the point $y = y^0$ ($f(x^0, y^0) = 0$). It follows that $f(x^0, y^0-b) <$

< 0 and $f(x^0, y^0+b) > 0$ and, by the continuity of f, there is a sufficiently small number $a > 0$ such that $f(x, y^0-b) < 0$ and $f(x, y^0+b) > 0$ for all $x \in \Delta^0$ and such that $\Delta \subset \Omega$ (see (3) and (5)).

Now we choose an arbitrary point $x \in \Delta^0$, fix it temporarily and consider the function $f(x, y)$ of one variable y on the interval (y^0-b, y^0+b). By the properties of f, this function is continuous, strictly increasing $\left(\text{since } \frac{\partial f}{\partial y} > 0\right)$ and assumes values of opposite signs at the end points of the interval. Therefore there exists a single value $y \in (y^0-b, y^0+b)$ which we denote $y = \psi(x)$ for which $f(x, \psi(x)) = 0$.

We have thus proved the existence of a function $\psi(x)$ defined on Δ^0 and satisfying all the conditions of the theorem except the continuity and the continuous differentiability since they have not yet been proved.

Let us prove that the function ψ is continuous. To this end it is sufficient to show that it is continuous at the point $x = x^0$. Indeed, the same proof can then be extended to any other point x' of the interval Δ^0 because the point $(x', \psi(x'))$ can be enclosed in a neighbourhood $\Omega' \subset \Omega$ such that all the conditions of the theorem are fulfilled for it if Ω, x^0 and y^0 are replaced by Ω', x' and $y' = \psi(x')$ respectively.

Let us take a number $\beta > 0$. Without violating generality we can assume that $\beta < b$. By what has been already proved, there is a rectangle $\Delta_* = \{|x-x^0| < \alpha, |y-y^0| < \beta'\}$, $\alpha < a$, $\beta' < \beta$ and a function $y = \psi_*(x)$, $x \in \Delta^0_* = \{|x-x^0| < \alpha\}$, which describes the set $\mathfrak{M}\Delta_*$.

We have $\Delta_* \subset \Delta$ and therefore, obviously, $\psi(x) = \psi_*(x)$, $x \in \Delta^0_*$. This shows that for any $\beta > 0$ there is $\alpha > 0$ such that $|\psi(x)-\psi(x_0)| < \beta$ provided that $|x-x^0| < \alpha$, which means that ψ is continuous at the point $x = x^0$.

To complete the proof we must show that the function $\psi(x)$ possesses the continuous derivative $\frac{d\psi}{dx}$ with respect to x.

Let $x, x+\Delta x \in \Delta^0$, $y = \psi(x)$ and

$$\Delta y = \psi(x+\Delta x)-\psi(x) \tag{7}$$

Then

$$0 = f(x+\Delta x, y+\Delta y)-f(x, y) = \left(\frac{\partial f}{\partial x}+\varepsilon_1\right)\Delta x+\left(\frac{\partial f}{\partial y}+\varepsilon_2\right)\Delta y \tag{8}$$

If we pass to the limit as $\Delta x \to 0$ in this equality then, as is already known, $\Delta y \to 0$; consequently, by the differentiability of f at the point (x, y), we have $\varepsilon_1, \varepsilon_2 \to 0$. Since $\frac{\partial f}{\partial y} > 0$ we also have $\frac{\partial f}{\partial y}+\varepsilon_2 > 0$ for sufficiently small Δx, and therefore

$$\frac{\Delta y}{\Delta x} = -\frac{\frac{\partial f}{\partial x}+\varepsilon_1}{\frac{\partial f}{\partial y}+\varepsilon_2} \to -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}, \qquad \Delta x \to 0 \tag{9}$$

We have proved the existence of the derivative ψ' at the point x and the equality

$$\psi'(x) = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{f'_x(x, \psi(x))}{f'_y(x, \psi(x))} \tag{10}$$

The continuity of $\psi'(x)$ is directly implied by this formula because f'_x and f'_y are continuous in the rectangle Δ and the curve $y = \psi(x)$ whose continuity has been already established does not fall outside it.

Proof of Theorem 1'. This proof is analogous to the above but we must of course consider the vector variable $\boldsymbol{x}$ instead of x and take $\mathfrak{M}$, Δ and Δ^0 mentioned in Theorem 1'. Further, in place of (7) we should take the increment

$$\Delta y = \psi(x_1, \ldots, x_{j-1}, x_j+h, x_{j+1}, \ldots, x_n)-\psi(x_1, \ldots, x_n)$$

of the function ψ at the point $\boldsymbol{x}$ corresponding to an increment h of the variable x_j ($j = 1, \ldots, n$); then instead of (8)

$$0 = f(x_1, \ldots, x_{j-1}, x_j+h, x_{j+1}, \ldots, x_n, y+\Delta y)-f(\boldsymbol{x}, y) =$$
$$= \left(\frac{\partial f}{\partial x_j}+\varepsilon_1\right) h+\left(\frac{\partial f}{\partial y}+\varepsilon_2\right) \Delta y \tag{8'}$$

we obtain the relation where $\varepsilon_1, \varepsilon_2 \to 0$ for $h \to 0$. Consequently,

$$\frac{\partial \psi}{\partial x_j} = -\lim_{h\to 0} \frac{\frac{\partial f}{\partial x_j}+\varepsilon_1}{\frac{\partial f}{\partial y}+\varepsilon_2} = -\frac{\frac{\partial f}{\partial x_j}}{\frac{\partial f}{\partial y}} \qquad (j = 1, \ldots, n) \tag{10'}$$

Theorem 2. *If a function $f(x,y)$ $(f(\boldsymbol{x}, y))$ satisfies the conditions of Theorem 1 (Theorem 1') and, besides, possesses continuous partial derivatives of order l on Ω, the function $\psi(x)$ $(\psi(\boldsymbol{x}))$ mentioned in Theorem 1 (Theorem 1') possesses continuous partial derivatives of order l on Δ^0.*

Proof. On differentiating (10) with respect to x we get

$$\psi''(x) = -\frac{f'_y[f''_{xx}+\psi' f''_{xy}]-f'_x[f''_{yx}+\psi' f''_{yy}]}{f'^2_y} \tag{11}$$

where (x, y) should of course be replaced by $(x, \psi(x))$ in all partial derivatives. The right member of this equality is a continuous function of x and hence $\psi''(x)$ is continuous.

On differentiating repeatedly the required number of times we obtain as ultimate result that the derivative $\psi^{(l)}(x)$ is a rational expression in the partial derivatives of f up to order l inclusive and in the derivatives $\psi', \ldots, \psi^{(l-1)}$ whose continuity has been established in the foregoing differentiations. The denominator of this rational function is different from zero (by virtue of the condition $f'_y \neq 0$) and therefore the derivative $\psi^{(l)}(x)$ exists and is continuous.

In the case of Theorem 1′ equality (11) is replaced by

$$\frac{\partial^2\psi}{\partial x_i\,\partial x_j} = -\frac{f'_y[f''_{x_ix_j}+\psi'_{x_i}f''_{yx_j}]-f'_{x_j}[f''_{x_iy}+\psi'_{x_i}f''_{yy}]}{f'^2_y} \qquad (i, j = 1, \ldots, n)$$

(see (10′)).

Example. The left-hand side of the equation $(y-x)^2 = 0$ has continuous partial derivatives but the derivative with respect to y turns into zero for $x = y = 0$. Despite this fact the given equation possesses the unique solution $y = x$ assuming the zero value at the point $x = 0$. Hence, Theorem 1 only provides sufficient conditions for the existence of a uniquely determined implicit function whose graph passes through the given point (x^0, y^0), and these conditions are not in fact necessary.

§ 7.17. Existence Theorem for the Solution of a System of Equations

Theorem 1. *Let a system of equations*

$$f_j(x_1, \ldots, x_n, y_1, \ldots, y_m) = f_j(\boldsymbol{x}, \boldsymbol{y}) = 0, \qquad j = 1, \ldots, m \tag{1}$$

satisfy the following properties.

The functions f_j are defined in an $((n+m)$-dimensional$)$ neighbourhood Ω of a point $(\boldsymbol{x}^0, \boldsymbol{y}^0) = (x_1^0, \ldots, x_n^0, y_1^0, \ldots, y_m^0)$ belonging to the space R_{n+m} of points $(\boldsymbol{x}, \boldsymbol{y}) = (x_1, \ldots, x_n, y_1, \ldots, y_m)$ and are continuous there together with their partial derivatives of the first order, and the functional determinant (the Jacobian)*

$$\frac{D(f_1, \ldots, f_m)}{D(y_1, \ldots, y_m)} \equiv \begin{vmatrix} \frac{\partial f_1}{\partial y_1} & \cdots & \frac{\partial f_1}{\partial y_m} \\ \cdots & \cdots & \cdots \\ \frac{\partial f_m}{\partial y_1} & \cdots & \frac{\partial f_m}{\partial y_m} \end{vmatrix}$$

is different from zero:

$$\frac{D(f_1, \ldots, f_m)}{D(y_1, \ldots, y_m)} \neq 0 \tag{2}$$

Besides, let the point $(\boldsymbol{x}^0, \boldsymbol{y}^0)$ satisfy system (1).

Let $\mathfrak{M}$ be the set of all points $(\boldsymbol{x}, \boldsymbol{y})$ satisfying system (1) (in particular, $(\boldsymbol{x}^0, \boldsymbol{y}^0) \in \mathfrak{M}$).

Then for any $b_0 > 0$ there is a rectangle

$$\Delta = \{|x_i-x_i^0| < a,\ i = 1, \ldots, n,\ |y_j-y_j^0| < b,\ j = 1, \ldots, m\},\ b < b_0 \tag{3}$$

belonging to Ω such that the set $\mathfrak{M}\Delta$ is described by (uniquely determined) continuously differentiable functions

$$y_j = \psi_j(\boldsymbol{x}), \qquad j = 1, \ldots, m,\ \boldsymbol{x} \in \Delta^0 \tag{4}$$

* After K. G. J. Jacobi (1804-1851), a German mathematician.

where

$$\Delta^0 = \{|x_i - x_i^0| < a; \qquad i = 1, \ldots, n\} \tag{5}$$

In other words, the rectangle Δ possesses the property that it is possible to define, in its projection Δ^0 on the coordinate subspace $(x_1, \ldots, x_n)$, (one-valued) continuously differentiable functions (4) satisfying equations (1), i.e.

$$f_j(x, \psi_1(x), \ldots, \psi_m(x)) \equiv 0, \; x \in \Delta^0, \qquad j = 1, \ldots, m \tag{6}$$

and the inequalities $|\psi_j(x) - y_j^0| < b$. These functions are determined uniquely in the sense that the coordinates of any point $(x, y) \in \mathfrak{M}\,\Delta$ are connected by equations (4).

In particular, $y_j^0 = \psi_j(x^0)$, $j = 1, \ldots, m$ because $(x^0, y^0) \in \mathfrak{M}\,\Delta$.

Remark 1. We can also use the following formulation which will be convenient for our further aims: the points of the form

$$(x, y) = (x, \psi_1(x), \ldots, \psi_m(x)), \qquad x \in \Delta^0 \tag{7}$$

belong to Δ and satisfy equations (1) (i.e. $(x, y) \in \mathfrak{M}\,\Delta$); there are no other points belonging to Δ and satisfying equations (1), i.e. if $(x, y) \in \mathfrak{M}\,\Delta$ then (x, y) has form (7) for some $x \in \Delta^0$.

Remark 2. It is allowable to assume that the rectangle Δ and its projection Δ^0 mentioned in the theorem are specified by inequalities

$$\Delta = \{|x_i - x_i^0| < a_i, i = 1, \ldots, n; \; |y_j - y_j^0| < b_j, j = 1, \ldots, m\} \tag{3'}$$

and

$$\Delta^0 = \{|x_i - x_i^0| < a_i, i = 1, \ldots, n\} \tag{5'}$$

where, generally speaking, a_i and b_j are different numbers. For if the theorem holds for rectangle (3′) with some a_i and b_j then, putting $b = \min b_j$ and taking into account the continuity of the functions ψ_i, we can find a number $a < a_i$, $i = 1, \ldots, n$, such that the points $(x, \psi_1(x), \ldots, \psi_m(x))$ where $x \in \{|x_i - x_i^0| < a, i = 1, \ldots, n\}$ are in rectangle (3).

However, it should be noted that in the general case it may be impossible to choose a and b in (3) so that they are equal; this is readily confirmed by the example of the equation $F(x, y) = y - x^2 = 0$ considered in a neighbourhood of the point $x_0 = y_0 = 0$.

Proof. For the case $m = 1$ the theorem was already proved (see § 7.16, Theorem 1′). Let us suppose that it holds for $m-1$ ($m > 1$) and prove that it then holds for m.

The Jacobian (2) being different from zero at the point (x^0, y^0), one of its minors of order $m-1$ is also different from zero at that point; since this minor is a continuous function there is a sufficiently small neighbourhood of the point (x^0, y^0) in which it is different from zero (without loss of generality we can assume that this neighbourhood coincides with Ω because, if necessary, the original neighbourhood Ω can be diminished). For definiteness, let us suppose that this minor is

$$\frac{D(f_1, \ldots, f_{m-1})}{D(y_1, \ldots, y_{m-1})} \neq 0 \tag{8}$$

By the hypothesis, the theorem holds for $m-1$ and therefore, taking into account (8), we can apply it to the first $m-1$ equations (1) and thus conclude that for any $b_0 > 0$ there is a rectangle

$$\Delta = \{|x_j - x_j^0| < \alpha,\ j = 1, \ldots, n,\ |y_m - y_m^0| < \beta,\ |y_i - y_i^0| < \gamma,$$
$$i = 1, \ldots, m-1\}, \qquad \gamma < b_0 \tag{9}$$

lying in the space R_{n+m} and belonging to Ω such that the set $\mathfrak{M}'$ of points $(x, y) = (x_1, \ldots, x_n, y_1, \ldots, y_m)$ belonging to Δ and satisfying the first $m-1$ equations (1) is described by (uniquely determined) continuously differentiable functions

$$\begin{aligned} &y_j = \varphi_j(x, y_m), \qquad j = 1, \ldots, m-1, (x, y_m) \in \Delta', \\ &\Delta' = \{|x_i - x_i^0| < \alpha,\ i = 1, \ldots, n,\ |y_m - y_m^0| < \beta\} \end{aligned} \tag{10}$$

Thus, in particular,

$$y_j^0 = \varphi_j(x^0, y_m^0), \qquad j = 1, \ldots, m-1 \tag{11}$$

Remark 3. At the above, first, stage of the argument we can take $\alpha = \beta$ but at the next stage this may be, generally, impossible, and it may happen that the numbers α and β should be decreased different numbers of times. It is easily seen that such a decrease does not violate the validity of what has already been proved.

Thus, functions (10) satisfy on Δ' the first $m-1$ equations (1), that is there hold the identities

$$f_j(x, \varphi_1(x, y_m), \ldots, \varphi_{m-1}(x, y_m), y_m) \equiv 0, (x, y_m) \in \Delta',$$
$$j = 1, \ldots, m-1 \tag{12}$$

Let us substitute these functions into the mth equation of system (1). This leads to the equation

$$F(x, y_m) = f_m(x, \varphi_1(x, y_m), \ldots, \varphi_{m-1}(x, y_m), y_m) = 0 \tag{13}$$

which specifies implicitly a function $y_m = \psi_m(x)$ whose existence, however, is to be proved. In the proof we shall use the following properties:

(i) The function $F(x, y_m)$ is defined in the rectangle Δ' and has continuous partial derivatives there because functions (10) possess these properties; the points (x, y) specified by these functions lie within the rectangle Δ where the function $f_m(x, y)$ is continuously differentiable.

(ii) $F(x^0, y_m^0) = f_m(x^0, \varphi_1(x^0, y_m^0), \ldots, \varphi_{m-1}(x^0, y_m^0), y_m^0) = f_m(x^0, y^0) = 0.$

(iii) The partial derivative of F with respect to y_m is different from zero in some neighbourhood of the point (x^0, y_m^0) of the space (x, y_m):

$$\frac{\partial F}{\partial y_m} \neq 0 \tag{14}$$

Property (iii) follows from the argument below.

Differentiating (on Δ') functions (12) and (13) with respect to y_m we obtain

$$\frac{\partial f_j}{\partial y_1}\frac{\partial \varphi_1}{\partial y_m} + \ldots + \frac{\partial f_j}{\partial y_{m-1}}\frac{\partial \varphi_{m-1}}{\partial y_m} + \frac{\partial f_j}{\partial y_m} \equiv 0, \qquad j = 1, \ldots, m-1$$

$$\frac{\partial F}{\partial y_m} = \frac{\partial f_m}{\partial y_1}\frac{\partial \varphi_1}{\partial y_m} + \ldots + \frac{\partial f_m}{\partial y_{m-1}}\frac{\partial \varphi_{m-1}}{\partial y_m} + \frac{\partial f_m}{\partial y_m}$$

Therefore on adding to the mth column of determinant (2) its jth columns ($j = 1, \ldots, m-1$) multiplied by $\frac{\partial \varphi_j}{\partial y_m}$ we receive

$$\begin{vmatrix} \frac{\partial f_1}{\partial y_1} & \ldots & \frac{\partial f_1}{\partial y_{m-1}} & \frac{\partial f_1}{\partial y_m} \\ \ldots & \ldots & \ldots & \ldots \\ \frac{\partial f_m}{\partial y_1} & \ldots & \frac{\partial f_m}{\partial y_{m-1}} & \frac{\partial f_m}{\partial y_m} \end{vmatrix} = \begin{vmatrix} \frac{\partial f_1}{\partial y_1} & \ldots & \frac{\partial f_1}{\partial y_{m-1}} & 0 \\ \ldots & \ldots & \ldots & \ldots \\ \frac{\partial f_{m-1}}{\partial y_1} & \ldots & \frac{\partial f_{m-1}}{\partial y_{m-1}} & 0 \\ \frac{\partial f_m}{\partial y_1} & \ldots & \frac{\partial f_m}{\partial y_{m-1}} & \frac{\partial F}{\partial y_m} \end{vmatrix} \neq 0$$

whence, by (8), it follows that $\frac{\partial F}{\partial y_m} \neq 0$.

The theorem follows from properties (i)-(iii).

Indeed, let $\mathfrak{M}''$ be the set of points (x, y_m) satisfying equation (13). Since the theorem holds for $m = 1$ (see Theorem 1', § 7.16) we see that, by properties (i)-(iii), there exists a rectangle $\Delta' = \{|x_j - x_j^0| < \alpha, j = 1, \ldots, m-1;\ |y_m - y^0| < \beta\}$ which (see Remark 3) can be regarded as being found at the first stage of the argument (see (10) where α and β should be accordingly reduced), and there exists a (unique) continuously differentiable function $y_m = \psi_m(x)$ defined on

$$\Delta^0 = \{|x_j - x_j^0| < \alpha,\ j = 1, \ldots, n\} \tag{15}$$

which describes the set $\mathfrak{M}''\Delta'$.

Thus,

$$f_m\big(x, \varphi_1(x, \psi_m(x)), \ldots, \varphi_{m-1}(x, \psi_m(x)), \psi_m(x)\big) \equiv 0, \qquad x \in \Delta^0 \tag{16}$$

Let us put

$$\psi_j(x) = \varphi_j(x, \psi_m(x)),\ j = 1, \ldots, m-1,\ \psi_m(x) = \psi_m(x),\ x \in \Delta^0 \tag{17}$$

The functions $\psi_j(x)$ satisfy the conditions of the theorem.

Indeed, the function $y_m = \psi_m(x)$ possesses continuous partial derivatives in Δ^0, and the points $(x, \psi_m(x))$ of the coordinate space (x, y_m) do not fall outside the rectangle Δ' on which $\varphi_j(x, y_m)$ are continuously differentiable, which implies that the functions $\psi_j(x)$ ($j = 1, \ldots, m-1$) are continuously differentiable on Δ^0 as well. Besides, by virtue of (17), it follows from (12) and (16) that $f_j(x, \psi_1(x), \ldots, \psi_m(x)) \equiv 0, j = 1, \ldots, m, x \in \Delta^0$.

Thus, we have proved the theorem under the assumption that Δ and Δ^0 are chosen in accordance with (9) and (15). But, by Remark 2, it is clear that the theorem also holds for some $a, b > 0$ in its original formulation.

Let us show that if a point (x, y) belongs to Δ where Δ is the rectangle defined in (9) and if $\mathfrak{M}$ is the set of points (x, y) satisfying system (1) then the coordinates of the point are connected by the equalities $y_j = \psi_j(x), x \in \Delta^0$, $j = 1, \ldots, m$ (this property follows from the above consideration but we want to present here its independent proof).

If $(x, y) \in \mathfrak{M}\Delta$ then $(x, y) \in \mathfrak{M}'\Delta$ where $\mathfrak{M}'$ is the set of points (x, y) satisfying the first $m-1$ equations of system (1). Therefore there hold the relations

$$y_j = \varphi_j(x, y_m), \quad (x, y_m) \in \Delta', \qquad j = 1, \ldots, m-1 \tag{18}$$

and, since the point $(x, y) \in \mathfrak{M}\Delta$ satisfies the mth equation (1) as well, we see that, by virtue of (18), $(x, y_m) \in \mathfrak{M}''$ where $\mathfrak{M}''$ is the set of points satisfying equation (13), that is $(x, y_m) \in \mathfrak{M}''\Delta'$. Therefore $y_m = \psi_m(x), x \in \Delta^0$ and $y_j = \varphi_j(x, \psi_m(x)) = \psi_j(x), x \in \Delta^0, j = 1, \ldots, m-1$.

Theorem 2. *If, in addition to the conditions of Theorem* 1, *the functions f_j are l times continuously differentiable on Ω the solutions to the given system, that is the functions $\psi_j(x)$, $x \in \Delta^0$, $j = 1, \ldots, m$, are also l times continuously differentiable on Δ.*

This theorem is proved by analogy with Theorem 2 in § 7.16.

§ 7.18. Mappings

Let there be given a system of continuously differentiable functions

$$y_j = \varphi_j(x) = \varphi_j(x_1, \ldots, x_m), \; x \in \Omega, \qquad j = 1, \ldots, m \tag{1}$$

where Ω is an open set of points $x = (x_1, \ldots, x_m)$.

We shall say that system (1) specifies a *continuously differentiable mapping*

$$y = Ax, \qquad x \in \Omega \tag{1'}$$

of the set Ω on the corresponding set Ω' of points $y = (y_1, \ldots, y_m)$. We shall also write $\Omega' = A(\Omega)$ and call Ω' the *image* of Ω and Ω the *preimage* (or the *inverse image*) of Ω' (under the mapping A).

Together with A let us take another continuously differentiable mapping B of an open set Λ of points y on a set of points $z = (z_1, \ldots, z_m)$ determined by some functions

$$z_j = \psi_j(y) = \psi_j(y_1, \ldots, y_m), \qquad y \in \Lambda, \; j = 1, 2, \ldots, m$$

Thus, $z = By, y \in \Lambda$.

If $\Omega' \subset \Lambda$* we can consider the *composite* continuously differentiable mapping $z = BAx$, $x \in \Omega$, specified by the equalities $z_j = \psi_j\,(\varphi_1(x), \ldots$ $\ldots, \varphi_m(x))$, $x \in \Omega\ (j = 1, \ldots, m)$.

The Jacobians of the mappings A, B and BA are connected by the important relations

$$\frac{D(z_1, \ldots, z_m)}{D(x_1, \ldots, x_m)} = \left|\frac{\partial z_i}{\partial x_j}\right| = \left|\sum_{s=1}^{m} \frac{\partial z_i}{\partial y_s}\frac{\partial y_s}{\partial x_j}\right| =$$

$$= \left|\frac{\partial z_i}{\partial y_s}\right| \left|\frac{\partial y_s}{\partial x_j}\right| = \frac{D(z_1, \ldots, z_m)}{D(y_1, \ldots, y_m)}\frac{D(y_1, \ldots, y_m)}{D(x_1, \ldots, x_m)} \quad (2)$$

(where $|\ldots|$ is the symbol of the determinant) whose proof, as we see, is based on the application of the differentiation formula for a composite function and on the multiplication rule for determinants.

In particular, if B is the *inverse* (the *inverse mapping*) of A which maps the set $A(\Omega)$ on the set of points $x \in \Omega$, that is if $x = BAx$ $(x \in \Omega)$ is the *identity mapping*, then, since its Jacobian is equal to 1, we receive the formula

$$1 = \frac{D(x_1, \ldots, x_m)}{D(y_1, \ldots, y_m)}\frac{D(y_1, \ldots, y_m)}{D(x_1, \ldots, x_m)}, \qquad x \in \Omega \quad (3)$$

Now we shall suppose that the Jacobian of the continuously differentiable mapping $y = Ax$ specified by equalities (1) is different from zero** throughout the open set Ω: $\dfrac{D(y_1, \ldots, y_m)}{D(x_1, \ldots, x_m)} \neq 0$, $x \in \Omega$. Our immediate aim is to prove the following properties:

(i) $\Omega' = A(\Omega)$ is an open set (together with Ω!);

(ii) if Ω is a domain (i.e. a connected open set) then Ω' is also a domain;

(iii) the mapping A is *locally one-to-one*, that is for any point $x^0 \in \Omega$ there is a ball $V \in \Omega$ with centre at that point such that the mapping A, considered only on V, is one-to-one.

Let $x^0 \in \Omega$ and $y^0 = Ax^0 \in \Omega'$. Let us denote by R_{2m} the space of points $(x, y) = (x_1, \ldots, x_m, y_1, \ldots, y_m)$ and consider in it the equations

$$f_i(x, y) = \varphi_i(x_1, \ldots, x_m) - y_i = 0, \qquad i = 1, \ldots, m$$

connecting the coordinates $x_1, \ldots, x_m, y_1, \ldots, y_m$. The point (x^0, y^0) is sure to satisfy these equations and the functions f_j are continuously differentiable in a neighbourhood (in the space R_{2m}) of this point, the Jacobian of the functions f_j satisfying the condition

$$\frac{D(f_1, \ldots, f_m)}{D(x_1, \ldots, x_m)} = \frac{D(\varphi_1, \ldots, \varphi_m)}{D(x_1, \ldots, x_m)} \neq 0$$

Therefore, by virtue of Theorem 1 proved in the foregoing section, for any $b_0 > 0$ there are positive numbers a and $b < b_0$ such that the set of points

---

* Note that if $x^0 \in \Omega$ and $y^0 = Ax^0 \in \Lambda$ then, by the continuity of A, there is a neighbourhood V_{x^0} of the point x^0 whose image under the mapping A belongs to Λ. Reducing Ω by putting $\Omega = V_{x^0}$ we then obtain $\Omega' \subset \Lambda$.

** The case when the Jacobian of mapping (1) turns into zero will be studied in § 7.27.

(x, y) belonging to the rectangle

$$\Delta = \Delta_1 \times \Delta_2$$

where

$$\Delta_1 = \{|y_j - y_j^0| < a, j = 1, \ldots, m\} \quad \text{and}$$
$$\Delta_2 = \{|x_j - x_j^0| < b, j = 1, \ldots, m\} \subset \Omega$$

and satisfying equations (1) is described by continuously differentiable functions

$$x_i = \varphi_i(y), \qquad y \in \Delta_1 \ (i = 1, \ldots, m)$$

Consequently $x \in \Delta_2$ when $y \in \Delta_1$. Let us denote as $x = By$ ($y \in \Delta_1$) the continuously differentiable mapping determined by these functions.

What has been said can be expressed as follows:

(a) if $y \in \Delta_1$ then $x = By \in \Delta_2$;

(b) if $y = ABy$ then $y \in \Delta_1$;

(c) relations $x \in \Delta_2$, $y \in \Delta_1$ and $y = Ax$ imply $x = By$.

Let $B(\Delta_1) = \omega$. By (a), we have $\omega \subset \Delta_2 \subset \Omega$ and, by (b), we have $A\omega = \Delta_1 \subset \Omega'$.

Thus, every point $y^0 \in \Omega'$ is contained in an open cube $\Delta_1 \subset \Omega'$ and therefore is an interior point of Ω'. We have proved property (i): Ω' is an open set.

By (b), the Jacobian of the transformation from y to x under B is different from zero (see (3)) on the open set Δ_1. Hence, applying property (i), which has already been proved, to B and Δ_1 instead of A and Ω we conclude that the set ω of points x is open.

Consequently the operator* A maps the open set ω onto $\Delta_1 = \omega'$.

Let x' and x'' be some points belonging to ω for which $Ax' = Ax'' = y$. Then $x' \in \omega \subset \Delta_2$, $y \in \Delta_1$ and, by (c), $x' = By$. Arguing similarly we also obtain $x'' = By$, i.e. $x' = x''$. In particular, this shows that

(d) $BAx = x$, $x \in \omega$.

This means that the mapping of the open set ω onto Δ_1 determined by the operator A is one-to-one. In particular, A specifies a one-to-one mapping of any ball $V \subset \omega$ with centre at x^0, which proves property (iii).

Remark. Properties (c) and (d) express the fact that the operators A and B are mutually inverse.

Now let us suppose that Ω is a domain; then, by property (i) already proved, Ω' is open together with Ω. If we take two arbitrary points y', $y'' \in \Omega'$ then the points $x', x'' \in \Omega$ corresponding to them are such that $Ax' = y'$ and $Ax'' = y''$. The set Ω is connected and therefore there exists a continuous curve $x(t) \in \Omega$, $0 \leqslant t \leqslant 1$ such that $x(0) = x'$ and $x(1) = x''$ which belongs to Ω and joins the points x' and x''. Consequently the curve $y(t) = A[x(t)]$, $0 \leqslant t \leqslant 1$, which is obviously also continuous, belongs to

* In modern mathematics the terms "operator", "transformation" and "mapping" (and also "function") are used synonymously. — *Tr.*

Ω' and connects y' with y''. Hence, Ω' is a connected open set, that is a domain, which proves property (ii).

Property (iii) expresses the fact that the mapping A is only locally one-to-one (as we say, one-to-one "in the small") because, generally speaking, it may not be one-to-one throughout Ω (one-to-one "in the large"). For example, the transformation $x = \varrho \cos \theta$, $y = \varrho \sin \theta$ from polar coordinates ϱ, θ of the points of the xy-plane into the Cartesian coordinates x, y is continuously differentiable and has the nonzero Jacobian equal to ϱ for $\varrho > 0$ and any θ. It maps the points (ϱ, θ) $(\varrho > 0, -\infty < \theta < \infty)$ of the (auxiliary) $\varrho\theta$-plane distinct from the origin onto the points of the xy-plane in a locally one-to-one manner. However, although to each such point (x, y) there corresponds a single value of ϱ the number of the corresponding values of θ which differ from each other by $2k\pi$ $(k = \pm 1, \pm 2, \ldots)$ is infinite.

§ 7.19. Smooth Surfaces

Let R be the three-dimensional space with rectangular Cartesian coordinates (x, y, z).

If G is an open set in the xy-plane and

$$z = f(x, y) \qquad ((x, y) \in G) \tag{1}$$

is a function having continuous partial derivatives of the first order on the set G, then the set $S \subset R$ described by this function is referred to as a *smooth surface*.

We shall say that this surface is projected on the plane $z = 0$. Equality (1) establishes a one-to-one correspondence $S \rightleftarrows G$ between the points $(x, y, z) \in S$ and the points $(x, y) \in G$.

If G is a bounded domain (i.e. a connected open set) with a boundary γ and if the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are not only continuous but also uniformly continuous on G, the function f and its partial derivatives can be continuously extended to γ. The set described by the extended function $z = f(x, y)$, $(x, y) \in \bar{G}$ (we denote it by the same letter f) will be denoted $\bar{S}$. The set $\Gamma = \bar{S} - S$ is called the *edge* (or, sometimes, the *boundary*) of S (and of $\bar{S}$). The set $\bar{S}$ will be referred to as a *smooth surface with edge*. It is obvious that the projection of Γ on the plane $z = 0$ is the set γ.

If γ is a piecewise smooth curve (contour) described by equations

$$x = \varphi(s), \quad y = \psi(s) \qquad (0 \leqslant s \leqslant s_0)$$

then Γ is also a piecewise smooth curve represented by the equations

$$x = \varphi(s), \quad y = \psi(s), \quad z = f(\varphi(s), \psi(s)) \qquad (0 \leqslant s \leqslant s_0)$$

It will be called the *boundary curve* of S (and of $\bar{S}$). In this case we shall call $\bar{S}$ a *smooth (surface) element with (piecewise smooth) boundary curve* or, briefly, a *smooth (surface) element*.

If $A_0 = (x_0, y_0, z_0)$ is an arbitrary point of a smooth surface represented by an equation of type (1) then, since G is an open set and the function f is continuous, for any $\delta_2 > 0$ there is $\delta_1 > 0$ such that the rectangle (rectangular parallelepiped)

$$\Delta = \{|x-x_0| \leqslant \delta_1,\ |y-y_0| \leqslant \delta_1,\ |z-z_0| \leqslant \delta_2\} \tag{2}$$

cuts from S a smooth element σ (with boundary curve) specified by the function

$$z = f(x, y) \ \ ((x, y) \in \Delta'), \quad \Delta' = \{|x-x_0| \leqslant \delta_1, |y-y_0| \leqslant \delta_1\} \subset G$$

where Δ' is the projection of Δ on the plane $z = 0$.

If A_0 is a point of the boundary curve Γ of a smooth surface element S we can only assert that there is a three-dimensional rectangle Δ of type (2) with centre at A_0 cutting from S a portion ω described by an equation $z = f(x, y)$, $(x, y) \in \omega'$ where ω' is a *part* of Δ'.

The definitions of a smooth surface and of a smooth surface element are extended, by analogy with the above, to the cases when the surfaces are described by equations of the form $x = \Psi(y, z)$ or $y = \Phi(z, x)$, that is when they are projected (in a one-to-one manner) on the plane $x = 0$ or $y = 0$.

Now we shall generalize these definitions to the case of surfaces which, generally, cannot be projected as a whole on any of the coordinate planes.

We shall say that a set $S \subset R$ is a *smooth surface* if for its any point $A^0 = (x^0, y^0, z^0)$ there is a (three-dimensional) rectangle

$$\Delta = \{|x-x^0| \leqslant \delta_1,\ |y-y^0| \leqslant \delta_2,\ |z-z^0| \leqslant \delta_3\}$$

cutting from S a smooth element σ which can be described by an equation belonging to at least one of the three types

$$\left.\begin{array}{ll} z = f_1(x, y) & \{|x-x^0| \leqslant \delta_1,\ |y-y^0| \leqslant \delta_2\} \\ x = f_2(y, z) & \{|y-y^0| \leqslant \delta_2,\ |z-z^0| \leqslant \delta_3\} \\ y = f_3(z, x) & \{|z-z^0| \leqslant \delta_3,\ |x-x^0| \leqslant \delta_1\} \end{array}\right\} \tag{3}$$

Since we refer to σ as a smooth surface element we thus mean that the functions f_1, f_2 and f_3 have continuous partial derivatives in the corresponding closed rectangles.

For example, let there be an arbitrary function $F(x, y, z)$ defined on a three-dimensional set $\Omega \subset R$ and continuous together with its partial derivatives of the first order such that

$$\left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2 + \left(\frac{\partial F}{\partial z}\right)^2 > 0 \qquad ((x, y, z) \in \Omega) \tag{4}$$

Let us denote by S the set of points $A = (x, y, z)$ for which

$$F(A) = F(x, y, z) = 0 \tag{5}$$

If the set S is nonempty it is a smooth surface.

Indeed, let $A_0 = (x_0, y_0, z_0) \in S$. By (4), at least one of the partial derivatives of F is different from zero at the point A_0; for definiteness, let $\left(\frac{\partial F}{\partial z}\right)_0 \neq 0$. Then, by the continuity of the partial derivatives of F, we conclude, on the basis of the existence theorem for implicit functions (see § 7.16, Theorem 1′), that there exists a three-dimensional rectangle

$$\Delta = \{|x-x_0|,\ |y-y_0| \leq \delta;\ |z-z_0| \leq \lambda\}, \qquad (\delta, \lambda > 0)$$

cutting from S a portion σ represented explicitly by a continuously differentiable function

$$z = \varphi(x, y), \qquad (x, y) \in \Delta', \quad \Delta' = \{|x-x_0|,\ |y-y_0| \leq \delta\} \tag{6}$$

This means that σ is a smooth surface element with boundary curve.

The smooth element and the surface S have a tangent plane at the point A_0 whose equation is (see § 7.5 (13))

$$z-z_0 = \left(\frac{\partial \varphi}{\partial x}\right)_0 (x-x_0)+\left(\frac{\partial \varphi}{\partial y}\right)_0 (y-y_0) \tag{7}$$

By virtue of the equalities

$$\left(\frac{\partial \varphi}{\partial x}\right)_0 = -\frac{\left(\frac{\partial F}{\partial x}\right)_0}{\left(\frac{\partial F}{\partial z}\right)_0}, \quad \left(\frac{\partial \varphi}{\partial y}\right)_0 = -\frac{\left(\frac{\partial F}{\partial y}\right)_0}{\left(\frac{\partial F}{\partial z}\right)_0}$$

(see § 7.16 (10′)), equation (7) can be rewritten equivalently as

$$\left(\frac{\partial F}{\partial x}\right)_0 (x-x_0)+\left(\frac{\partial F}{\partial y}\right)_0 (y-y_0)+\left(\frac{\partial F}{\partial z}\right)_0 (z-z_0) = 0 \tag{8}$$

The spherical surface

$$x^2+y^2+z^2 = R^2 \qquad (R > 0) \tag{9}$$

is a smooth surface because if a point $A_0 = (x_0, y_0, z_0)$ belongs to S then one of its coordinates, say z_0, is different from zero and therefore, as can be easily shown, there is a three-dimensional rectangle

$$\Delta = \{|x-x_0|,\ |y-y_0| \leq \delta,\ |z-z_0| \leq \lambda\} \qquad (\delta, \lambda > 0)$$

cutting from S a smooth element σ described by the equation

$$z = f(x, y) = \operatorname{sgn} z_0 \sqrt{R^2-x^2-y^2}\ \{|x-x_0|,\ |y-y_0| \leq \delta\} \tag{10}$$

where the function $f(x, y)$ in (10) possesses continuous partial derivatives.

In this example we can also use another argument. Let $\Omega \subset R_n$ be the set determined as

$$\Omega = \{(R')^2 < x^2+y^2+z^2 < (R'')^2\}$$

where $0 < R' < R < R''$. This is an open set entirely containing the spherical surface S. The function on the left-hand side of (9) has continuous partial derivatives on Ω which do not vanish simultaneously. Therefore, according to what was established above, S is a smooth surface.

By the *edge* of a smooth surface S is meant the set $\Gamma = \bar{S} - S$ provided that it is nonempty. Thus, *a smooth surface S has no edge if and only if it is a closed set.*

The functions

$$z = x^2 + y^2 \qquad (-\infty < x, y < \infty),$$

$$z = \tan x \tan y \qquad (-\pi/2 \leqslant x, y < \pi/2),$$

$$z = \sin x \qquad (-\infty < x < \infty)$$

describe unbounded smooth surfaces without edge. The third of them is specified by a bounded function but the corresponding surface (set) is unbounded.

The spherical surface S is a smooth surface and at the same time a closed set; it has no edge. If we delete from S a point A_0 belonging to it the remaining set is obviously a smooth surface $\tilde{S}$ with the edge consisting of that single point.

If a part of a surface S (which may coincide with S) is the closure of a smooth connected surface with piecewise smooth edge we shall say that it is a *smooth surface with edge cut from S.*

The part S_1 of the spherical surface S (represented by (9)) consisting of the points (x, y, z) with $z > 0$ is a smooth surface. Its edge Γ is the circle $x^2+y^2 = R^2$, $z = 0$. The closure $\bar{S}_1 = S_1 + \Gamma$ of S_1 is the upper hemisphere (with edge) described by the function $z = f(x, y) = \sqrt{R^2-x^2-y^2}$, $x^2+y^2 \leqslant R^2$. It is a smooth surface with edge cut from the surface S. The function $z = f(x, y)$ is continuous in the disc (closed circle) $x^2+y^2 \leqslant R^2$ but its partial derivatives are only continuous in the open circle $x^2+y^2 < R^2$ and are unbounded in the vicinity of its boundary. Hence, although S_1 is projected on the xy-plane, the corresponding function representing S_1 (when extended to the closed circle) is not continuously differentiable on the boundary of that circle. Further, S_1 cannot be projected (in a one-to-one manner) on any other coordinate plane. Thus, $\bar{S}_1$ is not a smooth element (with piecewise smooth boundary curve). On the other hand, it is readily seen that S (and S_1) can be cut into a finite number of smooth elements.

A surface which can be cut into a finite number of smooth elements is called *piecewise smooth.*

The surface of a rectangle (rectangular parallelepiped) is an example of a piecewise smooth surface which is not smooth.

Finally, we note that, in accordance with the terminology of Chapter 17, a smooth surface understood in the sense of the definition stated in this section is a two-dimensional differentiable manifold in the space R_3.

§ 7.20. Parametric Representation of a Smooth Surface. Orientable Surface

As we know, a smooth surface can be represented in an explicit form (e.g. see § 7.19 (1)) or in an implicit form (see § 7.19 (5)). The definition of an arbitrary smooth surface says that it can always be represented locally explicitly. There is another important method for the representation of smooth surfaces, namely the *parametric representation*.

Let us consider the three-dimensional space R with rectangular coordinates (x, y, z) and an auxiliary plane W with rectangular coordinates (u, v) where u and v are parameters with the aid of which a smooth surface will be represented. Let $\Omega \subset W$ be an open set and let three functions

$$x = \varphi(u, v), \quad y = \psi(u, v), \quad z = \chi(u, v) \qquad (u, v) \in \Omega \tag{1}$$

of the parameters u and v be defined in it. Relations (1) can be written in the equivalent vector form

$$\boldsymbol{r} = \varphi\boldsymbol{i}+\psi\boldsymbol{j}+\chi\boldsymbol{k} \tag{2}$$

We shall suppose that the functions φ, ψ and χ possess continuous partial derivatives in Ω and that the inequality

$$|\dot{\boldsymbol{r}}_u\times\dot{\boldsymbol{r}}_v|^2 = \left(\frac{D(x, y)}{D(u, v)}\right)^2+\left(\frac{D(y, z)}{D(u, v)}\right)^2+\left(\frac{D(z, x)}{D(u, v)}\right)^2 > 0 \qquad (u, v) \in \Omega \tag{3}$$

is fulfilled throughout Ω.

The locus S of points (x, y, z) specified by functions (1) is called a *(smooth) surface*. To express the fact that S is represented by functions (1) with the indicated properties we shall say that *the surface S admits of a smooth parametric representation with the aid of the parameters* $(u, v) \in \Omega$.

A surface S admitting of a smooth parametric representation in the sense of the above definition is by far not always a smooth surface (differentiable manifold) in the sense of the definition stated in the foregoing section but there is a simple sufficient criterion for S to satisfy both definitions.

Namely, if equations (1) establish a one-to-one correspondence $\Omega \ni (u, v) \rightleftharpoons (x, y, z) \in S$ then the system of functions (1) considered in an arbitrary domain $\Omega' \subset \bar{\Omega}' \subset \Omega$ represents a smooth surface (two-dimensional differentiable manifold).

Here we shall not prove this proposition. It follows as a special case from a more general proposition proved in § 17.1, Vol. 2, Lemma 1).

Inequality (3) implies an important fact. Let $(u^0, v^0) \in \Omega$ be an arbitrary fixed point of the domain Ω in the plane of the parameters u and v. One of the determinants in (3) is different from zero at this point; for definiteness, let it be the first one. Then there is a neighbourhood $\omega \subset \Omega$ of this point such that the first two equations (1) are uniquely resolvable with respect to (u, v) in ω. On substituting the corresponding functions expressing u and v in terms of x and y into the third equation we see that there is a portion $\sigma \subset S$ described by a continuously differentiable function $z = f(x, y)$ defined in a domain λ of the xy-plane $\lambda \ni (x, y) \rightleftharpoons (u, v) \in \omega$.

This shows that for any point (u^0, v^0) there is its neighbourhood $\omega \subset \Omega$ such that the portion $\sigma \subset S$ of the surface S corresponding to it is projected (in a one-to-one fashion) on at least one of the coordinate planes. This local property of a surface having a smooth parametric representation is extremely important but it should be stressed that the condition expressing it is weaker than the condition imposed on a smooth surface (two-dimensional differentiable manifold in the space R_3) in the definition given in the foregoing section (see Example 1 below).

Example 1. Let us consider the surface S (Fig. 7.5). It can be thought of as being obtained from a rectangular sheet of paper Δ (shown as occupying the domain $(0 < u < a, 0 < v < b)$ in the uv-plane in Fig 7.6) by twisting it smoothly so that it intersects itself along the line segments $u = u_1$, $u = u_2$ which are made coincident. A real model of such a surface can be made if

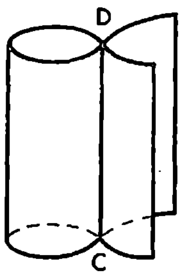

Fig. 7.5

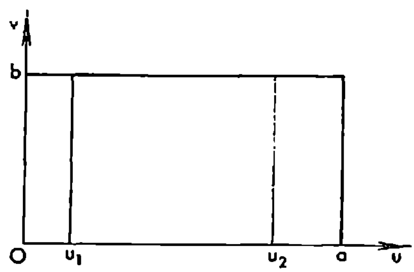

Fig. 7.6

we cut the sheet Δ into two parts along the line $u = u_2$ and then paste them together as shown in Fig. 7.5. To every point of this surface S we assign as its parameters (u, v) the coordinates of the corresponding point of the rectangle Δ in the original uv-plane. Then each point of the line segment $CD \subset S$ will correspond to two pairs of values (u_1, v) and (u_2, v) of the parameters. Therefore, we can say that S is a surface (represented parametrically) intersecting itself. From the point of view of the terminology of the foregoing section the surface S is not smooth since any rectangle (rectangular parallelepiped) with centre at a point $P \in [C, D]$ cuts from S a part which cannot be projected in a one-to-one manner on any of the coordinate planes. On the other hand, every point $P \in [C, D]$ can be regarded as corresponding to two points (u_1, v_0) and (u_2, v_0) of the uv-plane and both points can be enclosed in circles (discs), belonging to the uv-plane, with centres at these points and sufficiently small radii so that each of the portions $\sigma_1, \sigma_2 \subset S$ of the surface S corresponding to them can be projected on at least one of the coordinate planes.

It is also interesting to consider the surface $S' \subset S$ corresponding to the values (u, v) of the parameters running through the rectangle $\Delta' = \{0 < u < u_2, 0 < v < b\}$. As is seen from Fig. 7.5, S' is a surface without self-intersection admitting of a parametric representation (parametrization) because in this case we have the one-to-one correspondence $S' \rightleftharpoons \Delta'$. But nevertheless each $P \in [C, D]$ is a singular point of S' since if we take a (three-dimensional) neighbourhood Ω of P, however small, the part $S'\Omega$ of S' belonging to Ω cannot be projected in a one-to-one manner on any of the coordinate planes. Thus, the set S', like S, is not a smooth surface.

Remark. There is another terminological system in which by a smooth surface S' is meant a set of points (x, y, z) parametrized with the aid of some variables (u, v) by means of equalities (1) where φ, ψ and χ are continuously differentiable functions satisfying condition (3). According to this terminology, the points (x, y, z) corresponding to different pairs $(u, v)'$ are not identified and are considered as different elements of S even if (which is possible) they specify one and the same geometrical point (x, y, z).

Now let us replace the parameters (u, v) of the surface S by some other parameters (u', v'):

$$u = \lambda(u', v'), \qquad v = \mu(u', v') \qquad ((u', v') \in \Omega' = \Omega) \tag{4}$$

where λ and μ are continuously differentiable functions with a nonzero Jacobian

$$\frac{D(u', v')}{D(u, v)} \neq 0 \qquad ((u, v) \in \Omega') \tag{5}$$

We also suppose that mapping (4) sets up a one-to-one correspondence between the points (u', v') of the open set Ω' and the points (u, v) of the set Ω which is also open (see § 7.18). This results in parametric equations of S involving the parameters (u', v'):

$$\left.\begin{aligned} x &= \varphi_1(u', v') = \varphi(\lambda(u', v'), \mu(u', v')), \\ y &= \psi_1(u', v') = \psi(\lambda(u', v'), \mu(u', v')), \\ z &= \chi_1(u', v') = \chi(\lambda(u', v'), \mu(u', v'))\ ((u', v') \subset \Omega'), \end{aligned}\right\} \tag{6}$$

where φ_1, ψ_1 and χ_1 are functions continuously differentiable on Ω' such that the inequalities

$$|\dot{r}_{u'} \times \dot{r}_{v'}|^2 = \left(\frac{D(x, y)}{D(u', v')}\right)^2 + \left(\frac{D(y, z)}{D(u', v')}\right)^2 + \left(\frac{D(z, x)}{D(u', v')}\right)^2 =$$
$$= |\dot{r}_u \times \dot{r}_v| \left(\frac{D(u, v)}{D(u', v')}\right)^2 > 0 \quad ((u', v') \in \Omega') \tag{7}$$

hold.

The new parameters (u', v') with the indicated properties will be referred to as *admissible parameters* for the surface S.

Let us consider an arbitrary smooth surface S represented parametrically by equations (1) satisfying the conditions indicated at the beginning of the present section. This surface has a tangent plane and a normal at its every point. To derive the equation of the normal in terms of the vector function $r = \varphi i + \psi j + \chi k$ we can use the following argument.

For a fixed value of the parameter v the vector $r = r(u, v)$ describes a curve corresponding to the variation of the parameter u. The vector $\dot{r}_u$

goes along the tangent to this curve. Similarly, the vector $\dot{r}_v$ is directed along the tangent to another curve which is described by the vector function r when u is fixed and v varies.

If the vectors $\dot{r}_u$ and $\dot{r}_v$ are issued from the point (u, v) of the surface S they determine the plane containing them which is tangent to S at the point u, v. From condition (3) it follows that the vectors $\dot{r}_u$ and $\dot{r}_v$ are not collinear. The normal to S at its point corresponding to the given values of u and v is therefore specified by the vector

$$\dot{r}_u \times \dot{r}_v = \left(\frac{\partial\psi}{\partial u}\frac{\partial\chi}{\partial v}-\frac{\partial\psi}{\partial v}\frac{\partial\chi}{\partial u}\right)i+\left(\frac{\partial\chi}{\partial u}\frac{\partial\varphi}{\partial v}-\frac{\partial\chi}{\partial v}\frac{\partial\varphi}{\partial u}\right)j+\left(\frac{\partial\varphi}{\partial u}\frac{\partial\psi}{\partial v}-\frac{\partial\varphi}{\partial v}\frac{\partial\psi}{\partial u}\right)k \quad (8)$$

In the normal to S we can take two unit vectors with opposite directions which depend continuously on $(u, v) \in \Omega$:

$$n = \pm\frac{\dot{r}_u \times \dot{r}_v}{|\dot{r}_u \times \dot{r}_v|} \qquad ((u, v) \in \Omega) \quad (9)$$

To the sign "+" there corresponds one side of the surface S with a continuous family (continuously dependent on (u, v)) of unit normal vectors issued from its every point and to the sign "–" there corresponds the other side of S.

Now we state the following definition. If from each point A of a smooth surface S it is possible to draw a unit normal vector $n(A)$ so that the vector function $n(A)$ of the moving point A is continuous throughout the surface S, then S is said to be an *orientable surface*.

A surface for which a definite vector function $n(A)$ of this kind has been constructed is said to be *oriented* (with the aid of $n(A)$). When saying that S is an oriented smooth surface we not only mean that the symbol S designates the surface in question as a point set but also the fact that a definite function $n(A)$ continuous on S has been chosen. We also say that the function $n(A)$ determines a definite *side* of the orientable smooth surface S, i.e. the side from which is issued the family of the unit normal vectors $n(A)$ continuously dependent on A.

The same surface with the opposite orientation should be designated differently. It is convenient to denote two such oppositely oriented surfaces (which coincide as sets of points) by the symbols S_+ and S_-. Of course the choice of the side denoted as S_+ is quite arbitrary but after this choice has been made the other side is automatically supplied with the symbol S_-.

A spherical surface is an example of an orientable surface since if, for instance, we issue a unit normal vector from an arbitrary point of the spherical surface in the direction to the exterior of the ball bounded by the surface this vector can be continuously extended to the whole surface. This continuation of the normal vector determines a definite orientation of the spherical surface. Then the other, opposite, orientation of the spherical surface is specified by the unit normal vector directed to the interior of the ball bounded by the surface.

As was shown above, a smooth surface S represented parametrically by equations (1) satisfying the conditions indicated at the beginning of this section is orientable. To the sign "+" in formula (9) there corresponds one orientation of S and to the sign "−" the opposite orientation.

It should be noted (see the next section) that there also exist nonorientable surfaces.

What was said allows us to assert that *a nonorientable surface cannot be represented parametrically with the aid of equations* (1) *satisfying the conditions imposed on the functions* φ, ψ *and* χ.

On the other hand, the definition of a smooth surface S requires that for its each point A_0 there should exist a rectangle Δ with centre at this point cutting from S a portion σ representable explicitly, say, by a function $z = f(x, y)$, $(x, y) \in \Delta'$, which is continuously differentiable in the corresponding projection of Δ. Such a portion σ clearly has two sides determined by the normals $\boldsymbol{n} = (n_x, n_y, n_z)$ with the components

$$n_x = \mp\frac{p}{\sqrt{1+p^2+q^2}},\ n_y = \mp\frac{q}{\sqrt{1+p^2+q^2}},\ n_z = \pm\frac{1}{\sqrt{1+p^2+q^2}}$$

$$\left(p = \frac{\partial f}{\partial x},\ q = \frac{\partial f}{\partial y}\right)$$

To the upper signs in these equalities there corresponds the upper side of σ and to the lower signs there corresponds its lower side.

Hence, we can say that *every smooth surface is locally orientable.*

Example 2. *Spherical Surface.* The equations

$$x = R\cos\theta\cos\varphi,\ y = R\cos\theta\sin\varphi,\ z = R\sin\theta, -\infty < \theta, \varphi < \infty \quad (10)$$

where $R > 0$ is a given number specify with the aid of the angular parameters θ and φ the spherical surface S of radius R with centre at the origin, which can be readily checked by eliminating θ and φ from these equations.

In this case the domain of variation G of the parameters (θ, φ) is the whole plane (the auxiliary $\theta\varphi$-plane). The right-hand members of equations (10) are continuously differentiable functions of θ and φ. The computations show that

$$|\boldsymbol{r}_\theta \times \boldsymbol{r}_\varphi| = R^2|\cos\theta| \quad (11)$$

It is seen from (11) that we cannot say that the spherical surface S is represented smoothly with the aid of the parameters θ and φ throughout the plane of these parameters because such a representation fails at the points (θ, φ) for which $\cos\theta = 0$. But this is due not to the surface itself but to the chosen parametrization: as we know (see the foregoing section), S is a smooth surface and its poles corresponding to the values $\theta = \pm\pi/2$ of the parameter θ are not singular points.

Parametric representation (10) results in self-intersection because we did not impose any limits on the variation of the parameters. It appears more "economical" to take as the range of the parameters the set $-\pi/2 < \theta < \pi/2$, $a \leq \varphi < a+2\pi$ where a is a real number. This set is mapped in a one-to-one way onto the spherical surface S with its two poles deleted. On this set we have $|\dot{\boldsymbol{r}}_\theta \times \dot{\boldsymbol{r}}_\varphi| > 0$.

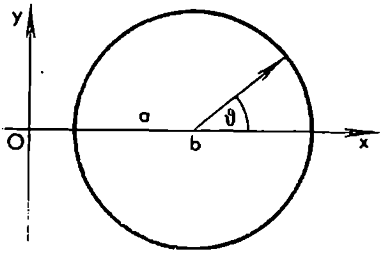

Fig. 7.7

Example 3. *Torus.* Let us take in the xy-plane a circle of radius a with centre at a point $(b, 0)$ $(0 < a < b)$. The rotation of the circle as rigid body in the space (x, y, z) about the y-axis generates a surface T called a *torus* (also *anchor ring*). Let θ be the magnitude of the angle shown in Fig. 7.7 and φ be the angle through which the circle in question is turned about the y-axis. The surface T is represented parametrically with the aid of the parameters θ and φ in the form

$$x = (b+a\cos\theta)\cos\varphi, \quad y = a\sin\theta,$$
$$z = (b+a\cos\theta)\sin\varphi \qquad (0 \leq \theta, \varphi < 2\pi) \tag{12}$$

In the Cartesian coordinates (x, y, z) the equation of T is written as

$$\Phi = (\sqrt{x^2+z^2}-b)^2+y^2-a^2 = 0$$

where

$$\left(\frac{\partial\Phi}{\partial x}\right)^2+\left(\frac{\partial\Phi}{\partial y}\right)^2+\left(\frac{\partial\Phi}{\partial z}\right)^2 = 4(\sqrt{x^2+z^2}-b)^2+4y^2 > 0$$

on T. This shows (see § 7.19) that the torus T is a smooth surface (which is quite clear intuitively).

§ 7.21. Example of a Nonorientable Surface. Möbius* Strip

Suppose that the rectangular sheet of paper shown in Fig. 7.8 (thought of as if the line segments aa' and bb' which are a part of its boundary were removed from it) is twisted once after which its sides ab and $a'b'$ are pasted together so that the point a coincides with b' and b with a' (see Fig. 7.9). The resultant surface is known as the *Möbius strip*. It is intuitively clear that this surface is smooth if the sheet is twisted "continuously" and "smoothly". Such a construction can easily be described analytically with the aid of formulas but we shall not dwell on this question relying upon the intuition of the reader. The line segment cc' marked in Fig. 7.8 is the midline of the rectangular sheet of paper. On the Möbius strip to this line there corresponds the closed contour cc' (it is not shown in Fig. 7.9) whose points c and c' merge. Let us issue a unit normal vector $\boldsymbol{n}(c)$ from the point c in an arbitrary direction. After the direction of $\boldsymbol{n}(c)$ has been definitely chosen (among the two possible directions) the direction of $\boldsymbol{n}(A)$ is predetermined for all the points $A \in cc'$ if it is required that the vector $\boldsymbol{n}(A)$ should depend on A continuously. But at the point c' the vector $\boldsymbol{n}(c')$ has already been chosen

* A. F. Möbius (1790-1868), a German geometer.

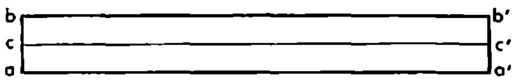

Fig. 7.8

Fig. 7.9

since c and c' coincide. It is easy to see that when the variable point of the midline of the twisted rectangle is moved continuously from c to c' the unit normal $\boldsymbol{n}(A)$ (where A is the variable point on the Möbius strip moving in the midline) tends to $-\boldsymbol{n}(c)$ but not to $\boldsymbol{n}(c)$, and hence the vector function $\boldsymbol{n}(A)$ is discontinuous at the point $c = c' \in S$. We see that the Möbius strip is in fact a nonorientable surface.

§ 7.22. Local Conditional Extremum

Let Ω be an open set in the n-dimensional space R_n and let $f, \varphi_1, \ldots, \varphi_m$ $(1 \leqslant m < n)$ be functions defined on Ω.

Let us denote by E the set of points x for which certain equalities (called *constraints* or *constraint equations* or *coupling equations*)

$$\varphi_j(x) = 0 \qquad (j = 1, \ldots, m;\ m < n) \tag{1}$$

hold simultaneously.

A point $x^0 \in \Omega$ is a *point of local conditional maximum (minimum) of the function f under constraints* (1) if $x^0 \in E$ and if there is $\delta > 0$ such that the inequality $f(x) \leqslant f(x^0)$ $(f(x) \geqslant f(x^0))$ holds for all $x \in E$ satisfying the condition $|x - x^0| = \sqrt{\sum_1^n (x_k - x_k^0)^2} < \delta$.

Points of local conditional maxima and minima are referred to as *points of conditional extremum.*

We shall begin with establishing necessary conditions for a given point x^0 to be a point of conditional extremum.

Let us suppose that in a neighbourhood of the point x^0 the functions f, $\varphi_1, \ldots, \varphi_n$ have continuous partial derivatives. Moreover, we shall assume that at the point x^0 the rank of the (functional) matrix $\left\| \left(\frac{\partial \varphi_j}{\partial x_k}\right)_0 \right\|$ $(j = 1, \ldots, m;\ k = 1, \ldots, n)$ is equal to m: rank $\left\| \left(\frac{\partial \varphi_j}{\partial x_k}\right)_0 \right\| = m$. This means that among the determinants (minors) of the mth order of this matrix (at the point x^0) there is at least one different from zero. For definiteness, let it be the minor

$$\frac{D(\varphi_1, \ldots, \varphi_m)}{D(x_1, \ldots, x_m)_0} = \begin{vmatrix} \left(\frac{\partial \varphi_1}{\partial x_1}\right)_0 & \cdots & \left(\frac{\partial \varphi_1}{\partial x_m}\right)_0 \\ \cdots & \cdots & \cdots \\ \left(\frac{\partial \varphi_m}{\partial x_1}\right)_0 & \cdots & \left(\frac{\partial \varphi_m}{\partial x_m}\right)_0 \end{vmatrix} \neq 0 \tag{2}$$

The symbol $(\ldots)_0$ indicates that x^0 is substituted for x in the expression in the parentheses. By the existence theorem for implicit functions, there is a rectangle

$$\Delta = \Delta' \times \Delta'', \tag{3}$$

$$\Delta' = \{|x_j - x_j^0| < \delta, j = 1, \ldots, m\}, \quad \Delta'' = \{|x_i - x_i^0| < \sigma, i = m+1, \ldots, n\}$$

and (uniquely determined) continuously differentiable functions $x_j = \mu_j(x_{m+1}, \ldots, x_n)$ $(j = 1, \ldots, m)$ defined on Δ'' and satisfying (1), i.e.

$$\varphi_j(\mu_1, \ldots, \mu_m, x_{m+1}, \ldots, x_n) = 0 \qquad (j = 1, \ldots, m)$$

where $|\mu_j - x_j^0| < \delta$ $(j = 1, \ldots, m)$.

Let us substitute the functions $\mu_1, \ldots, \mu_m$ for the corresponding variables $x_1, \ldots, x_m$ into the function f. The resultant expression (we retain the notation f for it) is a function dependent only on the (mutually) independent variables $x_{m+1}, \ldots, x_n$:

$$f(\mu_1, \ldots, \mu_m, x_{m+1}, \ldots, x_n) = \Phi(x_{m+1}, \ldots, x_n)$$

It is evident that if f attains its local conditional maximum (minimum) at the point $x^0 = (x_1^0, \ldots, x_n^0)$ then $\Phi(x_{m+1}, \ldots, x_n)$ attains at the point $(x_{m+1}^0, \ldots, x_n^0)$ its local unconditional maximum (minimum). As we know, for the latter case to take place it is necessary that the equalities

$$\left(\frac{\partial \Phi}{\partial x_l}\right)_0 = 0 \qquad (l = m+1, \ldots, n) \tag{4}$$

should be fulfilled simultaneously. Here the symbol $(\ )_0$ indicates that in the function in the parentheses we put $x_{m+1} = x_{m+1}^0, \ldots, x_n = x_n^0$. (We hope that this will not lead to a confusion.)

Now we state the definition of a stationary point of f: a point $x^0 \in E$ is called a *stationary point of the function f under constraints* (1) if equalities (4) hold for it or, which is the same, if

$$d\Phi = \sum_{l=m+1}^{n} \left(\frac{\partial \Phi}{\partial x_l}\right)_0 dx_l = 0$$

for any values of the independent differentials dx_l $(l = m+1, \ldots, n)$.

This definition is equivalent to the following:

A point $x^0 \in E$ is a *stationary point of f under constraints* (1) if the total differential df turns into zero, that is

$$df = \sum_{k=1}^{n} \left(\frac{\partial f}{\partial x_k}\right)_0 dx_k = 0 \tag{5}$$

for all dx_k satisfying the linear constraint equations

$$\sum_{k=1}^{n} \left(\frac{\partial \varphi_j}{\partial x_k}\right)_0 dx_k = 0 \qquad (j = 1, \ldots, m) \tag{6}$$

Indeed, by the invariance of the form of the first differential, we have $df = d\Phi$ where the (dependent) differentials $dx_1, \ldots, dx_m$ entering into df

are respectively equal to

$$dx_k = \sum_{j=m+1}^{n} \frac{\partial \mu_k}{\partial x_j} dx_j \qquad (k = 1, \ldots, m)$$

These expressions for $dx_1, \ldots, dx_m$ and the independent differentials $dx_{m+1}, \ldots, dx_n$ are connected by relations (6).

Our considerations incidentally provide the method for determining stationary points. It reduces to the solution of $n-m$ equations (4) with respect to $(x_{m+1}, \ldots, x_n)$. But this requires that equations (1) should be resolved in $x_1, \ldots, x_m$ beforehand. In most cases this method proves practically ineffective. A more effective and simpler method is the so-called *method of Lagrange's multipliers.*

The method of Lagrange's multipliers is based on the introduction of the auxiliary (Lagrangian) function

$$F(x) = f(x) - \sum_{1}^{n} \lambda_j \varphi_j(x) \tag{7}$$

dependent on the independent variables $x = (x_1, \ldots, x_n)$ *where* λ_j *are some constant numbers (Lagrange's multipliers) which should be determined simultaneously with the coordinates of the unknown stationary point* x^0. *The numbers* $\lambda_1, \ldots, \lambda_m$ *are first taken as undetermined parameters, and for the function* $F(x)$ *of the independent variables* $x_1, \ldots, x_n$ *(not connected by equations* (1)) *the problem of unconditional extremum is solved. More precisely, all its partial derivatives are equated to zero:*

$$\frac{\partial F}{\partial x_k} = \frac{\partial f}{\partial x_k} - \sum_{j=1}^{m} \lambda_j \frac{\partial \varphi_j}{\partial x_k} = 0 \qquad (k = 1, \ldots, n) \tag{8}$$

System (8) *of n equations is considered together with system* (1) *consisting of m constraint equations. The entire system of* $n+m$ *equations thus formed is then solved with respect to the unknowns* $x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_m$. *It turns out that to each solution* $x_1^0, \ldots, x_n^0, \lambda_1^0, \ldots, \lambda_m^0$ *there corresponds the stationary point* $\boldsymbol{x}^0 = (x_1^0, \ldots, x_n^0)$ *of the original problem and, conversely, if* $\boldsymbol{x}^0$ *is a stationary point then there are values of the multipliers* $\lambda_1^0, \ldots, \lambda_m^0$ *corresponding to it such that equations* (8) *with these values of* λ_j, $j = 1, \ldots, m$ *are satisfied by the coordinates of the point* $\boldsymbol{x} = \boldsymbol{x}^0$.

Let us consider the n-dimensional vectors

$$(\operatorname{grad} f)_0 = \left\{ \left(\frac{\partial f}{\partial x_1}\right)_0, \ldots, \left(\frac{\partial f}{\partial x_n}\right)_0 \right\} \tag{9}$$

and

$$(\operatorname{grad} \varphi_j)_0 = \left\{ \left(\frac{\partial \varphi_j}{\partial x_1}\right)_0, \ldots, \left(\frac{\partial \varphi_j}{\partial x_n}\right)_0 \right\} \qquad (j = 1, \ldots, m) \tag{10}$$

The fact that $\boldsymbol{x}^0$ is a stationary point can now be expressed as follows. From the orthogonality of the vector $(dx_1, \ldots, dx_n)$ to all the vectors $(\operatorname{grad} \varphi_i)_0$

($i = 1, \dots, m$), that is from the fact that system (6) is satisfied, it follows that equality (5) is satisfied, that is the vector $(dx_1, \dots, dx_n)$ is orthogonal to the vector $(\operatorname{grad} f)_0$ as well. Therefore, as is known from vector algebra (see the lemma below), there are m numbers $\lambda_1, \dots, \lambda_m$ such that

$$(\operatorname{grad} f)_0 = \lambda_1(\operatorname{grad} \varphi_1)_0 + \dots + \lambda_m(\operatorname{grad} \varphi_m)_0 \tag{11}$$

In other words, equalities (1) and (8) hold for $x = x^0$, and we have thus proved that to every stationary point x^0 there corresponds a set of numbers $\lambda_1, \dots, \lambda_m$ such that (1) and (8) hold simultaneously. Conversely, if for a point $x = x^0$ and for some numbers $\lambda_1, \dots, \lambda_m$ equalities (8) hold, that is if the vector equality

$$(\operatorname{grad} f)_0 = \sum_{i=1}^{m} \lambda_i (\operatorname{grad} \varphi_i)_0$$

is satisfied, then the fulfilment of equalities (6), that is the fact that the vector $(dx_1, \dots, dx_n)$ is orthogonal to vectors (10), obviously implies that this vector is orthogonal to vector (9), which means that equality (5) holds. We have thus proved the converse implication as well.

To find the answer to the question as to whether a given stationary point x^0 is a point of conditional extremum and what this extremum is (i.e. a maximum or a minimum) it is convenient to use the Lagrangian function F. Let us suppose that Jacobian (2) is different from zero at the point x^0. Then, by virtue of constraint equations (1), the variables $x_{m+1}, \dots, x_n$ can be regarded as independent in a neighbourhood of x^0 while the variables $x_1, \dots, x_m$ depend on them. For the sake of symmetry we can consider the whole set of the variables $x_1, \dots, x_n$ as being dependent on $x_{m+1}, \dots, x_n$. Proceeding from the theory of unconditional extremum, we can derive sufficient conditions for the existence of a maximum or a minimum by examining the second differential $d^2 f_0$ at the point x^0 on condition that the independent variables are $x_{m+1}, \dots, x_n$.

It follows from (1) that in a neighbourhood of the point x^0 there hold the relations $d\varphi_j = 0$ and $d^2\varphi_j = 0$ ($j = 1, \dots, m$). Therefore in this neighbourhood the equalities

$$d^2 f = d^2 F + d^2 \sum_{1}^{m} \lambda_j^0 \varphi_j = d^2 F = \sum_{k=1}^{n} \sum_{l=1}^{n} \frac{\partial^2 F}{\partial x_k \partial x_l} dx_k\, dx_l + \sum_{1}^{n} \frac{\partial F}{\partial x_k} d^2 x_k$$

hold for $\lambda_j = \lambda_j^0$. Since $\left(\frac{\partial F}{\partial x_k}\right)_0 = 0$, $k = 1, \dots, n$, it also follows that

$$d^2 f_0 = \sum_{k=1}^{n} \sum_{l=1}^{n} \left(\frac{\partial^2 F}{\partial x_k \partial x_l}\right)_0 dx_k\, dx_l$$

for $x = x^0$ and for all dx_k satisfying constraint relations (6).

Summing up what has been said we can state the following rule. Let it be necessary to compute the second differential $d^2 f_0$ of a function f at its stationary point x^0 under constraints $\varphi_j = 0$ ($j = 1, \dots, m$). To this end we form the Lagrangian function $F = f - \sum_{j=1}^{n} \lambda_j \varphi_j$ and compute its second differential $d^2F_0 = \sum_{k=1}^{n} \sum_{l=1}^{n} \left(\frac{\partial^2 F}{\partial x_k \, \partial x_l}\right)_0 dx_k \, dx_l$ regarding formally $dx_1, \ldots, dx_n$ as independent quantities. Then there holds the equality

$$d^2f_0 = d^2F_0 \tag{12}$$

which is fulfilled for any dx_k ($k = 1, \ldots, n$) connected by the linear constraint equations (6).

In this sense the investigation of d^2f_0 can be reduced to the investigation of d^2F_0.

Let us suppose that d^2F_0 is a positive definite quadratic form, that is $d^2F_0 > 0$ for any (mutually independent) dx_k not equal to zero simultaneously. If we arbitrarily choose certain values of $dx_{m+1}, \ldots, dx_n$ not equal to zero simultaneously then equations (6) determine some definite values of $dx_1, \ldots, dx_m$. We thus obtain a system of differentials $dx_1, \ldots, dx_n$ not vanishing simultaneously; for them we have $d^2F_0 > 0$, and therefore $d^2f_0 > 0$ as well, i.e. the quadratic form d^2f_0 in the variables $dx_{m+1}, \ldots, dx_n$ (which were not written explicitly) is positive definite. Therefore f as function of independent variables $x_{m+1}, \ldots, x_n$ attains at x^0 a local unconditional minimum or, which is the same, f as function of $x_1, \ldots, x_n$ attains at x^0 its local conditional minimum under constraints (1).

We can similarly conclude that if d^2F_0 is a negative definite form then f has at x^0 a local conditional maximum.

There are also more complicated cases when d^2F_0 is originally an indefinite quadratic form which becomes definite when the differentials $dx_1, \ldots, dx_n$ are subjected to constraint equations (6). In this case it is also convenient to use equality (12) when this method is applied.

The solution of the conditional extremum problem in a given domain Ω is carried out in accordance with the following scheme. In Ω we find the subset Ω_1 of points x at which the functions $f, \varphi_1, \ldots, \varphi_m$ have continuous partial derivatives and the matrix $\left\| \frac{\partial \varphi_j}{\partial x_k} \right\|$ has rank m. In Ω_1 we determine the stationary points using the method described above. Then each of these points is tested for extremum. If the continuous partial derivatives of the second order exist at such a point then the above method of investigating the second differential of the Lagrangian function F may prove effective. If this theory leads to an ambiguous case some special investigation is needed. The points $x \in \Omega - \Omega_1$ also require a special investigation.

Example 1. Let us determine the local conditional extrema of the function

$$f(x, y) = xy$$

on the circle Γ described by the equation

$$\varphi(x, y) = x^2 + y^2 - 1 = 0 \tag{13}$$

The functions f and φ are doubly continuously differentiable throughout the xy-plane. Besides, the rank of the matrix

$$\left\| \frac{\partial \varphi}{\partial x}, \frac{\partial \varphi}{\partial y} \right\| = \|2x, 2y\|$$

is equal to 1 (i.e. to the number of the constraints) throughout the xy-plane except at the point (0, 0) which does not belong to Γ. Consequently the points at which a local extremum is possible can only be among the stationary points of the problem.

Forming the Lagrangian function $F(x, y) = xy - \lambda(x^2+y^2-1)$ of the problem and equating to zero its partial derivatives we obtain the equations

$$\frac{\partial F}{\partial x} = y - 2\lambda x = 0, \quad \frac{\partial F}{\partial y} = x - 2\lambda y = 0 \tag{14}$$

On solving these equations together with equation (13) we obtain four stationary points (x, y) with coordinates $x = \pm 1/\sqrt{2}$ and $y = \pm 1/\sqrt{2}$ corresponding to all the possible combinations of the signs "+" and "−".

To the point (x_1, y_1) with $x_1 = y_1 = 1/\sqrt{2}$ there correspond the Lagrange multiplier $\lambda_1 = 1/2$ and the Lagrangian function

$$F(x, y) = xy - \frac{1}{2}(x^2+y^2-1)$$

The second differential of F at the point (x_1, y_1) is $d^2F = -dx^2 + 2dx\,dy - dy^2 = -(dx-dy)^2$. By (13), we have $2x\,dx + 2y\,dy = 0$ whence $dy = -dx$, and, finally,

$$d^2F = -(2dx)^2 = -4dx^2$$

where dx is an independent differential. Consequently there is a local conditional maximum at the point (x_1, y_1), the corresponding maximum value of f being $f(1/\sqrt{2}, 1/\sqrt{2}) = 1/2$. Using the symmetry properties of f we readily conclude that at the point $(-1/\sqrt{2}, -1/\sqrt{2})$ there is another local conditional maximum with maximum value equal to 1/2.

Since the circle Γ is a bounded closed set and since the function f is continuous on Γ, it attains on Γ its supremum; finally, since the supremum of f on Γ is necessarily its local maximum on Γ, we conclude that

$$\max_{\Gamma} f = f(1/\sqrt{2}, 1/\sqrt{2}) = f(-1/\sqrt{2}, -1/\sqrt{2}) = 1/2$$

and, similarly,

$$\min_{\Gamma} f = f(1/\sqrt{2}, -1/\sqrt{2}) = f(-1/\sqrt{2}, 1/\sqrt{2}) = -1/2$$

Lemma. *Let $\boldsymbol{a}, \boldsymbol{b}^1, \ldots, \boldsymbol{b}^m$ be vectors in the n-dimensional space $R_n (m < n)$. For the representation*

$$\boldsymbol{a} = \sum_{j=1}^{m} \lambda_j \boldsymbol{b}^j \tag{15}$$

where λ_j are some numbers to hold it is necessary and sufficient that every vector c orthogonal to all b^j, that is satisfying the conditions

$$(c, b^j) = 0 \qquad (j = 1, \ldots, m) \tag{16}$$

should be orthogonal to a:

$$(c, a) = 0 \tag{17}$$

Proof. If (15) holds then (16) implies

$$(c, a) = \left(c, \sum_{j=1}^{m} \lambda_j b^j\right) = \sum_{j=1}^{m} \lambda_j (c, b^j) = 0$$

and we have thus proved the necessity of the condition of the lemma.

Let us prove the sufficiency. Without loss of generality we can assume that the vectors

$$b^1, \ldots, b^m \tag{18}$$

are linearly independent. Then, as is proved in linear algebra, there exist mutually orthogonal vectors $b^{m+1}, \ldots, b^n$ each of which is also orthogonal to every vector (18) such that the vectors $b^1, \ldots, b^n$ form a basis in the n-dimensional space. Suppose that, for any vector c satisfying equalities (16), equality (17) also holds. Then, in particular, $(a, b^j) = 0$ $(j = m+1, \ldots, n)$.

The vector a is representable in the form of the sum

$$a = \sum_{j=1}^{m} \lambda_j b^j + \sum_{i=m+1}^{n} \mu_i b^i$$

whence $(a, b^i) = \mu_i = 0$ $(i = m+1, \ldots, n)$ and consequently $a = \sum_{1}^{m} \lambda_j b^j$, which is what we set out to prove.

§ 7.23. Singular Points of a Curve

As is known from the theory of implicit functions, if a function $F(x, y)$ turns into zero at a point (x_0, y_0), has continuous partial derivatives in a neighbourhood of this point and

$$\left(\frac{\partial F}{\partial y}\right)_0 \neq 0 \tag{1}$$

then there are $\delta_1 > 0$ and $\delta_2 > 0$ such that the set E of all points (x, y) belonging to the rectangle

$$|x - x_0| < \delta_1, \qquad |y - y_0| < \delta_2 \tag{2}$$

and satisfying the equality $F(x, y) = 0$ is described by a function $y = \varphi(x)$ $(y_0 = \varphi(x_0))$ having a continuous derivative in the interval $(x_0 - \delta_1, x_0 + \delta_1)$.

If we replace (1) in this formulation by the condition $\left(\frac{\partial F}{\partial x}\right)_0 \neq 0$ then there is rectangle (2) such that the corresponding set E is described by an equation $x = \psi(y)$ $(x_0 = \psi(y_0))$ where ψ is a continuously differentiable function.

In this section we shall be interested in the case when the function F satisfies all the conditions imposed above except one of them, namely, we shall suppose that both partial derivatives of F vanish at the point (x_0, y_0): $\left(\frac{\partial F}{\partial x}\right)_0 = \left(\frac{\partial F}{\partial y}\right)_0 = 0$.

When considering the set Γ of all (x, y) points for which $F = 0$ we shall call it a curve but it should be taken into account that, under the conditions imposed, Γ may not be a curve in the ordinary sense. For instance, if the function F is identically equal to zero the set Γ coincides with the whole xy-plane. We shall be particularly interested in the question as to what is the shape of Γ in a sufficiently small neighbourhood of the point (x_0, y_0) which will be referred to as a *singular point of the curve* Γ.

Without loosing generality we can assume that $x_0 = 0$ and $y_0 = 0$. We shall also suppose that the function F has continuous derivatives of the fourth order with the exception of the case $AC-B^2 > 0$ (A, B and C are defined below) when it is sufficient that only the second derivatives exist and are continuous.

Let us denote $A = \left(\frac{\partial^2 F}{\partial x^2}\right)_0$, $B = \left(\frac{\partial^2 F}{\partial x\, \partial y}\right)_0$ and $C = \left(\frac{\partial^2 F}{\partial y^2}\right)_0$.

We shall assume that the numbers A, B and C are not equal to zero simultaneously and shall consider separately the following possible cases.

(i) $AC-B^2 > 0$. In this case, according to the theory of unconditional extrema, the function F attains its (strict) local maximum or minimum at the point (0, 0) whence it follows that $F(x, y) \neq 0$ for the points (x, y) $(x \neq 0, y \neq 0)$ belonging to a circle σ of sufficiently small radius with centre at the origin.

Therefore within the circle σ the curve Γ described by the equation

$$F(x, y) = 0 \tag{3}$$

degenerates into the single point (0, 0).

In such a case we say that the point (0, 0) is an *isolated point* of Γ. The point (0, 0) itself belongs to Γ but there are no other points of Γ in its sufficiently small neighbourhood.

A simple example of this kind is the equation $x^2+y^2 = 0$.

(ii) $AC-B^2 < 0$. By Taylor's formula, we have

$$F(x, y) = Ax^2+2Bxy+Cy^2+\mu(x, y) \tag{4}$$

where μ possesses the second derivatives vanishing at the origin and

$$\mu = O(\varrho^3) \qquad (\varrho^2 = x^2+y^2) \tag{5}$$

In the first approximation it is natural to replace the equation $F(x, y) = 0$ by the equation $Ax^2+2Bxy+Cy^2 = 0$ which in this case $(AC-B^2 < 0)$ represents a pair of (nonimaginary) straight lines:

$$(\alpha_1 x+\beta_1 y)(\alpha_2 x+\beta_2 y) = 0 \tag{6}$$

where

$$\begin{vmatrix} \alpha_1 & \beta_1 \\ \alpha_2 & \beta_2 \end{vmatrix} \neq 0 \tag{7}$$

On making in (4) the substitution

$$\xi = \alpha_1 x + \beta_1 y, \quad \eta = \alpha_2 x + \beta_2 y \tag{8}$$

we obtain

$$F(x, y) = \Phi(\xi, \eta) = \xi\eta + \psi(\xi, \eta) \tag{9}$$

where

$$\psi = O(r^3) \qquad (r^2 = \xi^2 + \eta^2) \tag{10}$$

Here (10) follows from (5) because by (7) we have

$$c_1\varrho < r < c_2\varrho$$

where $c_1 > 0$ and $c_2 > 0$ are some constants.

By the uniqueness of the Taylor formula expansion with the remainder in Peano's form (see § 7.14), formula (10) implies that representation (9) is the Taylor expansion of the function $\Phi(\xi, \eta)$ at the point $\xi = \eta = 0$ with the remainder of order $l = 3$. Therefore

$$\psi(\xi, \eta) = \sum_{i=0}^{3} \xi^i \eta^{3-i} \psi_i(\xi, \eta)$$

where ψ_i are functions with continuous partial derivatives of the first order (see the end of § 7.13).

Thus,

$$\Phi(\xi, \eta) = \xi\eta + \sum_{i=0}^{3} \xi^i \cdot \eta^{3-i} \psi_i(\xi, \eta) \tag{11}$$

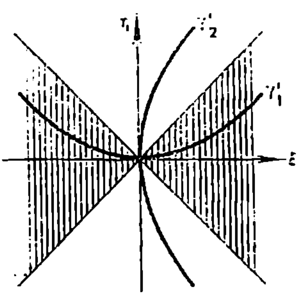

Fig. 7.10

Let us draw the two bisectors of the coordinate angles in the $\xi\eta$-plane (Fig. 7.10). To investigate the part γ_1' of the curve Γ' described by the equation $\Phi(\xi, \eta) = 0$ which lies (within a sufficiently small neighbourhood of the origin) in the shaded region in Fig. 7.10 let us perform the transformation of variables from (ξ, η) to (ξ, u) of the form

$$\eta = \xi u \qquad (|u| \leqslant 1) \tag{12}$$

To investigate the part γ_2' of the curve Γ' falling outside the shaded part we replace (ξ, η) by (u, η) according to the formula

$$\xi = u\eta \qquad (|u| \leqslant 1) \tag{13}$$

In the first case, by virtue of (11) and (12), we have

$$\Phi(\xi, \eta) = \xi^2 u + \xi^3 \varphi(\xi, u) = 0 \tag{14}$$

where $\varphi(\xi, u)$ has continuous partial derivatives.

On cancelling by ξ^2 we obtain

$$\lambda(\xi, u) = u+\xi\varphi(\xi, u) = 0 \tag{15}$$

To the point $\xi = 0$, $\eta = 0$ there may correspond points (ξ, u) with $\xi = 0$ and $|u| \leq 1$ specified by (12). But if $\xi = 0$ equation (15) is only satisfied when $u = 0$. The left-hand side of the equation has continuous partial derivatives in the vicinity of the point $\xi = 0$, $u = 0$ and $\left(\frac{\partial \lambda}{\partial u}\right)_0 = 1$. Hence, by the existence theorem for implicit functions, there is a uniquely determined function $u = \mu(\xi)$ $(\mu(0) = 0)$ defined in a sufficiently small interval $|\xi| < \delta$, having a continuous derivative and satisfying equation (15), and therefore the corresponding function

$$\eta = \xi\mu(\xi) \tag{16}$$

is also continuously differentiable in the interval $|\xi| < \delta$, possesses the derivative $(\eta')_0 = 0$ and satisfies the equation $\Phi(\xi, \eta) = 0$. This function describes the part γ_1' of the curve Γ'. The other part γ_2' of Γ' which is tangent to the η-axis at the origin is investigated with the aid of substitution (13).

Thus, in the case under consideration the curve Γ (which is the image of Γ' under the inverse transformation from ξ, η to x, y) splits into two portions γ_1 and γ_2 in the vicinity of the origin which intersect at a (nonzero) angle (see Fig. 7.11).

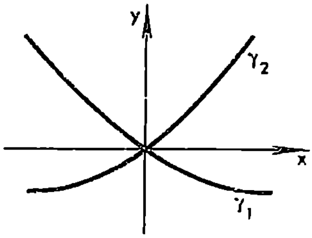

Fig. 7.11

(iii) $AC-B^2 = 0$. In this case we have

$$Ax^2+2Bxy+Cy^2 = (\alpha x+\beta y)^2$$

where α and β are not equal to zero simultaneously. Putting $\xi = \beta x-\alpha y$, $\eta = \alpha x+\beta y$ and arguing as above we conclude that in the new (rectangular) coordinates (ξ, η) the curve in question is described by the equation

$$\eta^2+\sum_{i=0}^{3} \xi^i\eta^{3-i}\psi_i(\xi, \eta) = 0 \tag{17}$$

where $\psi_i(x)$ are continuously differentiable functions. The substitution $\eta = u\xi$ leads, after the cancellation by ξ^2, to the equation

$$\varkappa(\xi, u) = u^2+\xi\varphi(\xi, u) = 0 \tag{18}$$

Now we again have $\varkappa(0, 0) = 0$ $(\varkappa(0, u) \neq 0$ for $u \neq 0)$. If $(\varphi)_0 = \varphi(0, 0) \neq$ $\neq 0$ (this case is more general while the case $(\varphi)_0 = 0$ is exceptional) then $\left(\frac{\partial \varkappa}{\partial \xi}\right)_0 = (\varphi)_0 \neq 0$. Consequently we can apply the theorem on implicit functions to equality (18); by this theorem, there is a uniquely determined function $\xi = \nu(u)$ $(\nu(0) = 0)$ possessing a continuous derivative and satisfying equation (18) in a sufficiently small neighbourhood of $u = 0$.

We have

$$\nu(u) = -u^2/\varphi(\xi, u),\ \nu'(u) = -\left\{2u\varphi - u^2 \frac{d}{du}\varphi(\nu(u), u)\right\}/\varphi^2$$

Let $(\varphi)_0 < 0$; then obviously there is $\delta > 0$ such that

$$\nu'(u) > 0 \text{ for } 0 < u \leqslant \delta,\ \nu'(u) < 0 \text{ for } -\delta \leqslant u < 0,\ \nu'(u) = 0 \text{ for } u = 0$$

and the function $\nu(u)$ is strictly decreasing to zero on $[-\delta, 0]$ and strictly increasing from zero on $[0, \delta]$. Therefore, the function $\xi = \nu(u)$ has inverse functions on each of these intervals. The inverse functions can be written as $u = \pm\sqrt{-\xi[(\varphi)_0 + \varepsilon(\xi)]}$ $(\varepsilon(\xi) \to 0,\ \xi \to 0)$ where $\xi > 0$.

We have thus obtained two smooth parts of the curve Γ':

$$\eta = \pm\xi\sqrt{-\xi[(\varphi)_0 + \varepsilon]} \approx \pm\xi^{3/2}\sqrt{-(\varphi)_0} \qquad (\xi \to 0,\ 0 \leqslant \xi \leqslant \delta)$$

The original curve Γ corresponding to Γ' is shown in Fig. 7.12.

In this case (0, 0) is spoken of as a *cuspidal point* of the curve Γ.

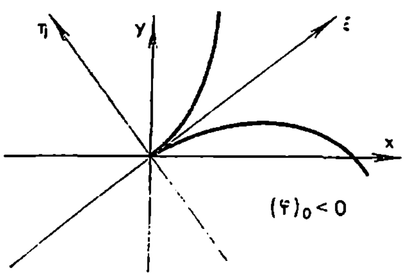

Fig. 7.12

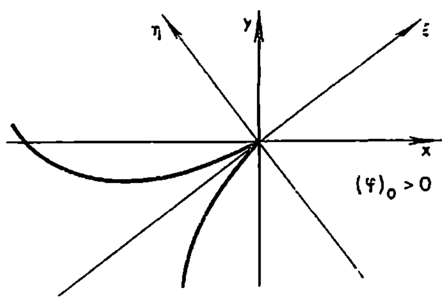

Fig. 7.13

The situation remains analogous in the case $(\varphi)_0 > 0$ but the branches of the curve lie in the domain $\xi < 0$ (where the radicand is positive) in the vicinity of the cusp (Fig. 7.13).

Of course the case $(\varphi)_0 = 0$ is more complicated and requires further investigation.

Note that the substitution $\xi = u\eta$ does not lead to a new branch of the curve because after the cancellation of (17) by η^2 we obtain the equation $1 + \eta\nu(\eta, u) = 0$ having no solution for $\eta = 0$ and finite u.

§ 7.24. Curves on a Surface

Let us consider a doubly continuously differentiable surface σ (or, as we also say, a surface *continuously differentiable of order* 2) specified by an equation

$$z = f(x, y) \tag{1}$$

and a doubly differentiable smooth curve Γ lying on σ which is described by an equation $\boldsymbol{r}(s) = \varphi(s)\boldsymbol{i}+\psi(s)\boldsymbol{j}+\xi(s)\boldsymbol{k}$ where s is the arc length of Γ (thus $f(x, y)$ and $\boldsymbol{r}(s)$ have continuous derivatives of the second order with respect to their arguments).

We shall use the notation

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x\, \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2} \tag{2}$$

(the letter s denotes the arc length of Γ and, simultaneously, the derivative $\frac{\partial^2 z}{\partial x\, \partial y}$ but we hope this will not lead to a confusion).

The unit normal to the surface σ at its point $A\,(x, y, z)$ forming an acute angle with the positive direction of the z-axis is

$$\boldsymbol{n} = \frac{-p}{\sqrt{1+p^2+q^2}}\,\boldsymbol{i}+\frac{-q}{\sqrt{1+p^2+q^2}}\,\boldsymbol{j}+\frac{1}{\sqrt{1+p^2+q^2}}\,\boldsymbol{k}$$

The unit vector of the principal normal of Γ at A is given by the formula (see § 6.11 (3))

$$\boldsymbol{\nu} = \frac{\ddot{\boldsymbol{r}}}{|\ddot{\boldsymbol{r}}|} = R\ddot{\boldsymbol{r}} = R\left(\frac{d^2x}{ds^2}\,\boldsymbol{i}+\frac{d^2y}{ds^2}\,\boldsymbol{j}+\frac{d^2z}{ds^2}\,\boldsymbol{k}\right) \qquad (\ddot{\boldsymbol{r}} \neq 0)$$

where R is the radius of curvature of Γ ($R > 0$) and x, y, z are the components of $\boldsymbol{r}$.

Let us denote by θ the angle between $\boldsymbol{n}$ and $\boldsymbol{\nu}$. Then

$$\frac{\cos\theta}{R} = \frac{-p\, d^2x-q\, d^2y+d^2z}{ds^2\sqrt{1+p^2+q^2}} \tag{3}$$

Since $\Gamma \subset \sigma$ the differentials of the components of $\boldsymbol{r}$ (corresponding to ds) satisfy the constraint relation $dz = p\, dx+q\, dy$ whence

$$d^2z = p\, d^2x+q\, d^2y+r\, dx^2+2s\, dx\, dy+t\, dy^2 \tag{4}$$

and therefore, by (3), we have

$$\frac{\cos\theta}{R} = \frac{r\, dx^2+2s\, dx dy+t\, dy^2}{ds^2\sqrt{1+p^2+q^2}} = \frac{r\alpha^2+2s\alpha\beta+t\beta^2}{\sqrt{1+p^2+q^2}} \tag{5}$$

where $\alpha = \frac{dx}{ds}$ and $\beta = \frac{dy}{ds}$ are the cosines of the angles formed by the tangent to Γ with the axes x and y respectively.

From formula (5) it follows immediately that *all the curves $\Gamma \subset \sigma$ having a common osculating plane L at the point $A \in \sigma$ not coinciding with the tangent plane to σ at A have one and the same curvature at A.* Indeed, the value of the right-hand side of (5) at the point A is one and the same number for all such curves and $\cos\theta$ has one and the same value for all of them, and their curvature at A is equal to the ratio of the right-hand side of (5) to $\cos\theta$. Hence, the curvature of any of these curves is equal to the curvature of one of them, for instance, to the curvature of the curve which is the section of the surface σ by the osculating plane L.

Generally speaking, this assertion fails when $\cos\theta = 0$, i.e. when the osculating plane L of Γ coincides with the tangent plane to σ. To show this it is sufficient to note that two curves in a plane tangent to each other do not necessarily have equal radii of curvature at the point of tangency.

Let Γ be a *normal section* of the surface σ at its point A, that is a curve along which σ intersects a plane passing through the normal to σ at A, and let Γ' be some other section of σ by a plane passing through the tangent line to Γ at the point A. Then the right-hand sides of (5) assume equal values at the point A for the curves Γ and Γ'. Besides, we have $\theta = 0$ or $\theta = \pi$ for Γ. Therefore, there holds the equality

$$\pm\frac{1}{R} = \frac{\cos\theta}{R'} \qquad (6)$$

where R and R' are, respectively, the radii of curvature of the normal section and of the section having a common tangent with the former.

Equality (6) is known as *the Meusnier* formula*. It can also be rewritten in the form $R' = \pm R\cos\theta = R\,|\cos\theta|$ (since $R > 0$ and $R' > 0$).

For the normal section formula (5) gives the expression of the radius of curvature R:

$$\frac{1}{R} = \frac{r\alpha^2+2s\alpha\beta+t\beta^2}{\sqrt{1+p^2+q^2}} \qquad (7)$$

More precisely, we should put the sign "+" or "−" in front of the left member of (7) depending on whether $\theta = 0$ or $\theta = \pi$. But we shall not do so because it is more convenient for our further aims to allow the radius of curvature of a normal section to assume positive or negative values ($R > 0$ or $R < 0$) depending on whether the concavity of the normal section is in the direction of the positive or negative half-axis z.

In all the considerations above including the deduction of formula (7) we assumed that $|\ddot{r}(s)| > 0$, i.e. that R is finite. Note that the notion of the principal normal to a curve does not make sense when $\ddot{r}(s) = 0$. But formula (7) continues to hold for the case $\ddot{r}(s) = 0$ as well. Indeed, in this case the left-hand side of (7) is equal to zero ($R = \infty$) and the right-hand side also turns into zero because the right-hand side of (3) is equal to zero and, by virtue of (4), the right-hand sides of (5) and of (7) turn into zero together with it.

Let us translate the origin of the rectangular coordinates to the point A (of the surface σ) under consideration and direct the z-axis along the normal to σ. Then $p = q = 0$ and $\alpha^2+\beta^2 = 1$, and we can put $\alpha = \cos\theta$ and $\beta = \sin\theta$. Formula (7) for the radius of curvature of a normal section is then brought to the form

$$\frac{1}{R} = r\cos^2\theta+2s\cos\theta\sin\theta+t\sin^2\theta = \frac{r+t}{2}+\frac{r-t}{2}\cos 2\theta+s\sin 2\theta \quad (8)$$

The curvature $1/R$ is a continuous function of θ on the closed interval

* J. B. M. Meusnier (1754-1799), a French mathematician.

$[0, 2\pi]$. Therefore it attains its maximum and minimum on that interval which can be found by equating its derivative to zero. This results in

$$\tan 2\theta = \frac{2s}{r-t} \qquad (r \neq t) \tag{9}$$

The function $\tan 2\theta$ is strictly increasing on the intervals $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ and $\left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$ and assumes all the values from $-\infty$ to $+\infty$ on it. Therefore on each of the intervals there is exactly one root of equation (9); we denote these roots as θ_1 and θ_2 respectively. Besides, $\theta_2-\theta_1 = \pi/2$. There are two more roots $\theta_1+\pi$ and $\theta_2+\pi$ on the period of the function but they determine the same sections.

Thus, if $r \neq t$ then there are two and only two mutually perpendicular directions along which the curvature of the normal section attains its maximum $1/R_1$ and minimum $1/R_2$. If the axes x and y go in these directions, equation (8) takes the form $1/R = r\cos^2\theta+t\sin^2\theta$ because in this case relation (9) must be fulfilled for $\theta = 0$.

We have $1/R_1 = r$ and $1/R_2 = t$ and thus obtain *Euler's** *formula*

$$\frac{1}{R} = \frac{\cos^2\theta}{R_1}+\frac{\sin^2\theta}{R_2} \tag{10}$$

The numbers R_1 and R_2 are termed the *principal radii of curvature of the surface (at the point A)*. They correspond to the so-called *principal normal sections* of the surface (which are mutually perpendicular).

By formula (10), to the section $\theta' = \theta+\pi/2$ there corresponds the curvature

$$\frac{1}{R'} = \frac{\sin^2\theta}{R_1}+\frac{\cos^2\theta}{R_2}$$

On adding together (10) and the last equality we arrive at the equality

$$\frac{1}{R}+\frac{1}{R'} = \frac{1}{R_1}+\frac{1}{R_2}$$

showing that *the sum of curvatures of any two mutually perpendicular normal sections is a constant quantity*. It is called the *mean curvature of the surface at the point in question.*

The points of a doubly continuously differentiable surface can be classified in the following way.

1. In the case $R_1R_2 > 0$ we have a so-called *elliptic point* of the surface. For an elliptic point the radii of curvature R_1 and R_2 are of one sign and therefore R is also of the same sign for any section, that is all the normal sections have the concavity in the direction of the positive or negative half of the z-axis depending on whether there is the sign "+" or "−". For instance, all the points of the surface of an ellipsoid possess this property.

---

* L. Euler (1707-1783), the great mathematician and physicist, a member of St. Petersburg Academy of Sciences.

2. A *hyperbolic point* corresponds to the case $R_1R_2 < 0$. Here the concavities of the principal sections (and of the sections adjoining them) are in different directions. The curvature $1/R$ being a continuous function of θ, there must exist at least two sections with zero curvature.

There are in fact exactly two such sections; the values of θ corresponding to them satisfy the relation $\tan\theta = \pm\sqrt{-\frac{R_2}{R_1}}$; the radii of curvature of these sections are infinite. A hyperboloid of one sheet is an example of such a surface.

3. In the case $\frac{1}{R_1R_2} = 0$ (where one of the numbers R_1 and R_2 is finite) we have a *parabolic point*. Hence, in this case either $R_1 > 0$ and $1/R_2 = 0$, and then

$$\frac{1}{R} = \frac{\cos^2\theta}{R_1}$$

or $1/R_1 = 0$ and $R_2 < 0$, and then $1/R = \sin^2\theta/R_2$.

Here there is only one section with zero curvature, all the other sections having curvature of one sign and concavity in one direction. An example of a surface whose all points are of this kind is a cylindrical surface.

To find to what class belongs a point of a surface $z = f(x, y)$ with tangent plane not necessarily parallel to the xy-plane we consider the sign of $rt-s^2$. Let $r > 0$ and $rt-s^2 > 0$, then the quadratic form

$$r\,dx^2+2s\,dx\,dy+t\,dy^2 \tag{11}$$

is positive definite and $\cos\theta$ in formula (5) retains constant sign for all the values of θ; consequently the point in question is elliptic.

If $rt-s^2 < 0$ then there are two different pairs (dx, dy) for which form (12) changes sign; this indicates that the point is hyperbolic. If $rt-s^2 = 0$ then form (11) retains sign for all (dx, dy) except for one direction, and hence the point is parabolic.

These considerations incidentally show that the sign of $rt-s^2$ is invariant with respect to all transformations of rectangular coordinates relative to which a small portion of the surface containing the point A can be written in the form $z = f(x, y)$. This can also be shown by means of rather lengthy calculations connected with change of variables in partial derivatives.

Now, let $r = t$; then $rt-s^2 = r^2-s^2$ and formula (8) assumes the form $1/R = r+s\sin 2\theta$. Here we have the following possibilities.

(i) If $r^2-s^2 > 0$ the point in question is *elliptic*. If $s \neq 0$ then, as in Case 1 on p. 300, there are two principal sections with maximum and minimum radii of curvature. If $s = 0$ then $1/R = r = \text{const}$, which means that all the sections have one and the same curvature; this is the so-called *umbilical* (or *circular*) *point*.

(ii) If $r^2-s^2 < 0$ we have a *hyperbolic point*.

(iii) If $r^2-s^2 = 0$ then the point is obviously *parabolic*.

Let us consider a smooth surface S represented parametrically as

$$\boldsymbol{r} = x\boldsymbol{i}+y\boldsymbol{j}+z\boldsymbol{k} = \varphi(u, v)\,\boldsymbol{i}+\psi(u, v)\boldsymbol{j}+\chi(u, v)\boldsymbol{k}\ (u, v \in \Omega) \tag{12}$$

where Ω is a domain. According to (12), to the differentials du and dv there correspond the differentials of the components of the vector $\boldsymbol{r}$ given by the expressions

$$dx = \frac{\partial x}{\partial u}du+\frac{\partial x}{\partial v}dv, \quad dy = \frac{\partial y}{\partial u}du+\frac{\partial y}{\partial v}dv,$$
$$dz = \frac{\partial z}{\partial u}du+\frac{\partial z}{\partial v}dv \tag{13}$$

to which there corresponds the following expression for the square of the differential of arc length of a curve on the surface S:

$$ds^2 = dx^2+dy^2+dz^2 = E\,du^2+2F\,du\,dv+G\,dv^2, \tag{14}$$
$$E = \left(\frac{\partial x}{\partial u}\right)^2+\left(\frac{\partial y}{\partial u}\right)^2+\left(\frac{\partial z}{\partial u}\right)^2,$$
$$F = \frac{\partial x}{\partial u}\frac{\partial x}{\partial v}+\frac{\partial y}{\partial u}\frac{\partial y}{\partial v}+\frac{\partial z}{\partial u}\frac{\partial z}{\partial v},$$
$$G = \left(\frac{\partial x}{\partial v}\right)^2+\left(\frac{\partial y}{\partial v}\right)^2+\left(\frac{\partial z}{\partial v}\right)^2$$

The right-hand member of (14) is a quadratic form in du and dv. For a smooth surface this form is positive definite since its discriminant is positive:

$$EG-F^2 = \left(\frac{D(y,z)}{D(u,v)}\right)^2+\left(\frac{D(z,x)}{D(u,v)}\right)^2+\left(\frac{D(x,y)}{D(u,v)}\right)^2 = |\dot{\boldsymbol{r}}_u\times\dot{\boldsymbol{r}}_v|^2 > 0 \tag{15}$$

A concrete smooth curve $\Gamma \subset S$ can be specified with the aid of continuously differentiable functions

$$u = \lambda(t),\ v = \mu(t), \qquad (\lambda'^2+\mu'^2 > 0) \tag{16}$$

If λ and μ are substituted into (12) for u and v respectively we obtain a vector equation of a smooth curve Γ lying on S for which

$$\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2+\left(\frac{dz}{dt}\right)^2 = E\left(\frac{du}{dt}\right)^2+2F\frac{du}{dt}\frac{dv}{dt}+G\left(\frac{dv}{dt}\right)^2 > 0 \tag{17}$$

for any t (this is implied by the condition written in parentheses in (16) and by the fact that the form is positive definite).

On multiplying the left-hand side of (17) by dt^2 we obtain (14), that is the square of the differential of arc length of the curve Γ in question (corresponding to the differential dt). For a given arc $\Gamma \subset S$ the differentials du and dv are mutually dependent but if we are interested in the various curves $\Gamma \subset S$ passing through $A \in S$ their specification determines the various pairs of the differentials du and dv.

Let us derive the parametric analogue of formula (5). We shall suppose that the surface S is not only smooth but also doubly continuously differentiable.

The left member of (5) can be rewritten as

$$\frac{\cos\theta}{R} = \frac{(\boldsymbol{\nu},\boldsymbol{n})}{R}(\ddot{\boldsymbol{r}}(s),\boldsymbol{n}) = \left(\ddot{\boldsymbol{r}}(s),\ \frac{\dot{\boldsymbol{r}}_u\times\dot{\boldsymbol{r}}_v}{|\dot{\boldsymbol{r}}_u\times\dot{\boldsymbol{r}}_v|}\right) \tag{18}$$

where $\boldsymbol{n}$ is the unit normal to the surface S, $\boldsymbol{r}(u, v)$ is the radius vector of the moving point of the surface S and $\boldsymbol{r}(s)$ is the radius vector of the moving point of Γ. We have

$$\dot{\boldsymbol{r}}(s) = \dot{\boldsymbol{r}}_u \frac{du}{ds} + \dot{\boldsymbol{r}}_v \frac{dv}{ds}$$

and

$$\ddot{\boldsymbol{r}}(s) = \ddot{\boldsymbol{r}}_{uu}\left(\frac{du}{ds}\right)^2 + 2\ddot{\boldsymbol{r}}_{uv}\frac{du}{ds}\,\frac{dv}{ds} + \ddot{\boldsymbol{r}}_{vv}\left(\frac{dv}{ds}\right)^2 + \dot{\boldsymbol{r}}_u \frac{d^2v}{ds^2} + \dot{\boldsymbol{r}}_v \frac{d^2v}{ds^2}$$

and, taking into account that the vectors $\dot{\boldsymbol{r}}_u$ and $\dot{\boldsymbol{r}}_v$ are orthogonal to $\boldsymbol{n}$, we receive

$$\frac{\cos\theta}{R} = \frac{(\ddot{\boldsymbol{r}}_{uu}\,du^2 + 2\ddot{\boldsymbol{r}}_{uv}\,du\,dv + \ddot{\boldsymbol{r}}_{vv}\,dv^2)\,(\dot{\boldsymbol{r}}_u \times \dot{\boldsymbol{r}}_v)}{ds^2\,|\dot{\boldsymbol{r}}_u \times \dot{\boldsymbol{r}}_v|}$$

where ds^2 can be replaced by expression (14).

§ 7.25. Curvilinear Coordinates in a Neighbourhood of Smooth Boundary of a Domain

Let Ω be a bounded spatial domain whose boundary is a (closed) smooth surface S. Let us issue a normal to S directed to the interior of Ω from each point $A \in S$ and mark a point A_λ on it at a distance of $\lambda > 0$ from A. The collection of all the points A_λ corresponding to a fixed $\lambda > 0$ forms a surface S_λ (see Fig. 7.14 where are depicted the sections of S and S_λ by a plane). If λ is large some points of S_λ may belong to different normals. But if λ is sufficiently small this is not the case. For small λ it appears natural to expect that in the layer between S and S_λ the normals do not intersect, and then each point of the layer is exactly on one normal issued from a point $A \in S$. This does in fact take place when S is a surface continuously differentiable of order 2 (doubly continuously differentiable), that is when the functions representing it locally have continuous second partial derivatives.

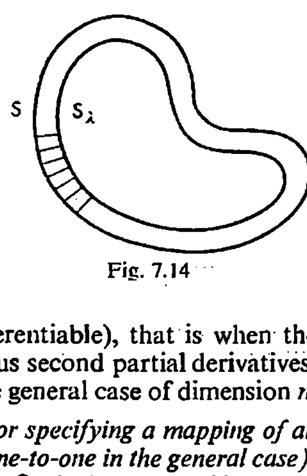

Fig. 7.14

In the lemma below this fact is proved for the general case of dimension n.

Lemma. *Let $y = A(x)$ be a continuous operator specifying a mapping of an open set $\Omega \subset R_n$ on $\Omega' \in R_n$ (which may not be one-to-one in the general case). Suppose that there is a bounded closed subset $F \subset \Omega$ which is mapped by A in a one-to-one fashion and that for each point $\boldsymbol{x} \in F$ there is a neighbourhood $\Omega_x \subset \subset \Omega$ which undergoes a one-to-one mapping.*

For any $\lambda > 0$ let us consider the set F^λ consisting of the points belonging to Ω such that each of them is at a distance not greater than λ from at least one point belonging to F.

Then there is $\lambda_0 > 0$ such that F^{λ_0} is mapped in a one-to-one manner under $A: \Omega \supset F^{\lambda_0} \rightleftharpoons (F^{\lambda_0})' \subset \Omega'$.

Proof. Suppose that the assertion of the lemma fails to hold. Let us take a decreasing sequence of numbers $\lambda_k \to 0$. Then for any k there are two different points $x_k, y_k \in F^{\lambda_k}$ such that $Ax_k = Ay_k$. The set of the points $x_k, y_k \in \in F^{\lambda_k} \subset F^{\lambda_1}$ being bounded, there is a subsequence of indices k (let us renumber them as $k = 1, 2, \ldots$) such that $x_k \to x^0 \in F$, $y_k \to y^0 \in F$, $Ax_k \to Ax^0$, $Ay_k \to Ay^0$ and $Ax^0 = Ay^0$; it follows that $x^0 = y^0$ since the mapping A of F on F' is one-to-one. But this is impossible because the point $x^0 = y^0$ belongs to the definite neighbourhood Ω_{x^0} and $x_k, y_k \in \Omega_{x^0}$ for sufficiently large k, and, since $x_k \neq y_k$, there must be $Ax_k \neq Ay_k$.

Let us consider an $(n-1)$-dimensional manifold S (see § 17.1) determined by the equations

$$x_i = \varphi_i(u), \qquad (u = (u_1, \ldots, u_{n-1}) \in \omega, \quad i = 1, \ldots, n) \tag{1}$$

Let us associate with each point $x \in S$ a unit vector

$$\nu = (\alpha_1, \ldots, \alpha_n) \tag{2}$$

specified by the formulas

$$\alpha_i = \frac{A_i}{\sqrt{\sum_1^n A_i^2}} \qquad (i = 1, \ldots, n) \tag{3}$$

where A_i are the signed minors of the elements of the first column of the determinant

$$\Delta = \begin{vmatrix} \alpha_1 & \dfrac{\partial\varphi_1}{\partial u_1} & \cdots & \dfrac{\partial\varphi_1}{\partial u_{n-1}} \\ \cdots & \cdots & \cdots & \cdots \\ \alpha_n & \dfrac{\partial\varphi_n}{\partial u_1} & \cdots & \dfrac{\partial\varphi_n}{\partial u_{n-1}} \end{vmatrix} = \sum_1^n \alpha_i A_i = \sqrt{\sum_1^n A_i^2} \tag{4}$$

The numbers A_i are not all equal to zero simultaneously since the rank of the matrix $\left\| \dfrac{\partial\varphi_i}{\partial u_j} \right\|$ is supposed to be equal to $n-1$.

The vector ν is called the *unit normal vector to S at the point $x \in S$*. The other unit normal vector to S at x differs from ν in sign.

It is important to stress that, given equations specifying S, the vector ν can be effectively determined. (This is a manifestation of the property that the manifold S is *orientable*.)

Let us choose a neighbourhood of S and consider the change of variables

$$x_i = t\alpha_i + \varphi_i(u_1, \ldots, u_{n-1}) \qquad (u \in \omega, i = 1, \ldots, n) \tag{5}$$

for its points $x = (x_1, \ldots, x_n)$ where t is a new real variable. Functions (5) themselves are defined for any $u \in \omega$ and any real t. They are continuously differentiable because φ_i are doubly continuously differentiable. It is also important that for any domain $\omega' \subset \bar{\omega}' \subset \omega$ there is $\delta > 0$ such that the set

$$S^\delta = \{|t| \leqslant \delta, u \in \bar{\omega}'\}$$

is mapped in a one-to-one way with the aid of equations (5) having a positive Jacobian, i.e. this mapping is continuously differentiable in both directions. This follows from the lemma proved above where the operator specified by formulas (5) can be taken as A because it determines a continuous mapping of the open set $\Omega = \{-\infty < t < \infty, u \in \omega\}$ of the space R_n into R_n, and under this mapping the points of the bounded closed set $F = \{t = 0, u \in \bar{\omega}'\} \subset \Omega$ are in a one-to-one correspondence with the points of the image of F; besides, for each point of F there is its neighbourhood undergoing a one-to-one mapping by the operator A (for at each such point the Jacobian of transformation (5) is positive: $\Delta > 0$).

Let us also consider a curve Γ, continuously differentiable of order 3 (i.e. triply continuously differentiable) and not intersecting itself, represented by an equation

$$r = \varphi(s)i + \psi(s)j + \chi(s)k \qquad (0 \leq s \leq s_0)$$

(we suppose that $\dot{r}(t) \times \ddot{r}(t) = 0$) where s is its arc length and φ, ψ and χ are functions possessing continuous derivatives of the third order with respect to s. In this case the unit vector β of the principal normal and the unit vector γ of the binormal are (once) continuously differentiable functions of s (see § 6.11). Let us put

$$\varrho = r + \lambda\beta + \mu\gamma \tag{6}$$

where λ and μ are arbitrary real numbers. This equality assigns to each triple of numbers (s, λ, μ) $(0 \leq s \leq s_0;\ -\infty < \lambda, \mu < \infty)$ the triple (ξ, η, ζ) of the Cartesian coordinates of vector ϱ. Let us denote by H_p $(p > 0)$ the set of points of the space (s, λ, μ) specified by the inequalities $0 \leq s \leq s_0$, $\lambda^2 + \mu^2 \leq p^2$. The coordinates ξ, η, ζ of the vector ϱ are continuously differentiable functions of s, λ, μ with a Jacobian $D(s, \lambda, \mu)$ equal to

$$D = \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix} = |\alpha(\beta \times \gamma)| = 1$$

for $\lambda = \mu = 0$ (α_i, β_i and γ_i are respectively the components of the unit vectors of the tangent, principal normal and binormal). The curve Γ not intersecting itself, equality (6) sets up a one-to-one correspondence between the points $(s, 0, 0)$ $(0 \leq s \leq s_0)$ forming a bounded closed set in the coordinate space (s, λ, μ) and the points (ξ, η, ζ) of the curve Γ. Therefore, by the lemma proved above, there is a sufficiently small p for which operator (6) maps the set H_p of points (s, λ, μ) onto the set Ω of points (ξ, η, ζ) in a one-to-one manner. By the theorem on implicit functions, the continuous differentiability of the corresponding inverse operator on the set Ω takes place for at least sufficiently small values of p for which $D(s, \lambda, \mu) > 0$.

The construction we have discussed makes it possible to introduce the curvilinear coordinates (s, λ, μ) in a sufficiently small neighbourhood Ω of the curve Γ such that in the coordinates (s, λ, μ) the equations determining Γ are written as $\lambda = 0$, $\mu = 0$; besides, there is a one-to-one correspondence continuously differentiable in both directions between the coordinates (s, λ, μ) and the Cartesian coordinates (ξ, η, ζ) of the points belonging to Ω:

$$\xi = f_1(s, \lambda, \mu), \quad \eta = f_2(s, \lambda, \mu), \quad \zeta = f_3(s, \lambda, \mu)$$

(on this question see also § 17.2).

§ 7.26. Change of Variables in Partial Derivatives

We shall limit ourselves to the case of dimension 2. In the n-dimensional case the calculations are quite analogous.

(i) Let us consider a function

$$z = f(x, y) \tag{1}$$

where

$$x = \varphi(u, v), \qquad y = \psi(u, v) \tag{2}$$

We shall find the expressions of the derivatives $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$ in terms of the derivatives of z with respect to u and v. To this end we differentiate (1) with respect to u and v:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v} \tag{3}$$

On solving these equations in p and q we obtain

$$p = \frac{\dfrac{D(z, y)}{D(u, v)}}{\dfrac{D(x, y)}{D(u, v)}}, \qquad q = \frac{\dfrac{D(x, z)}{D(u, v)}}{\dfrac{D(x, y)}{D(u, v)}} \tag{4}$$

In these considerations we of course suppose that φ and ψ possess continuous partial derivatives with respect to u and v, the corresponding Jacobian being different from zero. We shall always assume that the conditions guaranteeing the solvability of the corresponding equations hold and therefore this assumption will not be stipulated in what follows.

Equalities (4) can be rewritten as

$$\frac{\partial z}{\partial x} = A\frac{\partial z}{\partial u} + B\frac{\partial z}{\partial v}, \qquad \frac{\partial z}{\partial y} = C\frac{\partial z}{\partial u} + D\frac{\partial z}{\partial v} \tag{5}$$

where (this is important!) the coefficients A, B, C and D depend solely on u and v and do not depend on z. Therefore we have

$$\begin{aligned} r = \frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = A\frac{\partial}{\partial u}\left(\frac{\partial z}{\partial x}\right) + B\frac{\partial}{\partial v}\left(\frac{\partial z}{\partial x}\right) = A\frac{\partial}{\partial u}\left(A\frac{\partial z}{\partial u} + B\frac{\partial z}{\partial v}\right) + \\ &+ B\frac{\partial}{\partial v}\left(A\frac{\partial z}{\partial u} + B\frac{\partial z}{\partial v}\right) = A^2\frac{\partial^2 z}{\partial u^2} + 2AB\frac{\partial^2 z}{\partial u\,\partial v} + B^2\frac{\partial^2 z}{\partial v^2} + \\ &+ \left(A\frac{\partial A}{\partial u} + B\frac{\partial A}{\partial v}\right)\frac{\partial z}{\partial u} + \left(A\frac{\partial B}{\partial u} + B\frac{\partial B}{\partial v}\right)\frac{\partial z}{\partial v} \end{aligned} \tag{6}$$

Hence, we have found the expression of the partial derivative $\frac{\partial^2 z}{\partial x^2}$ in terms of the partial derivatives of z with respect to u and v.

To compute $s = \frac{\partial^2 z}{\partial x \, \partial y}$ and $t = \frac{\partial^2 z}{\partial y^2}$ we use the same procedure. Further derivatives are consecutively found in the same manner. For instance, to find $\frac{\partial^3 z}{\partial x^3}$ we substitute $A \frac{\partial z}{\partial u} + B \frac{\partial z}{\partial v}$ for z into (6) and then perform the necessary differentiations.

(ii) Let us consider a more general problem. Let there be given equations

$$F_j(x, y, z, u, v, w) = 0 \qquad (j = 1, 2, 3) \tag{7}$$

connecting x, y and z with the new variables u, v and w. It is required to express the derivatives p, q, r, s and t in terms of the derivatives of w with respect to u and v. Since z is a function of x and y the variables u, v and w are also functions of x and y.

Let us differentiate equalities (7) regarding x and y as independent variables:

$$\frac{\partial F_j}{\partial x} dx + \frac{\partial F_j}{\partial y} dy + \frac{\partial F_j}{\partial z} dz + \frac{\partial F_j}{\partial u} du + \frac{\partial F_j}{\partial v} dv + \frac{\partial F_j}{\partial w} dw = 0 \tag{8}$$
$$(j = 1, 2, 3)$$

On solving system (8) in du, dv and dw we obtain the expressions

$$\left.\begin{aligned} du &= A_1^1\, dx + A_2^1\, dy + A_3^1\, dz \\ dv &= A_1^2\, dx + A_2^2\, dy + A_3^2\, dz \\ dw &= A_1^3\, dx + A_2^3\, dy + A_3^3\, dz \end{aligned}\right\} \tag{9}$$

where A_i^j are coefficients dependent on x, y, z, u, v, w. Since $dw = \frac{\partial w}{\partial u} du + \frac{\partial w}{\partial v} dv$, the substitution of expressions (9) for the differentials du, dv, dw into this equality and the solution of the resulting equation with respect to dz yield the equality

$$dz = M\, dx + N\, dy \tag{10}$$

where M and N depend on x, y, z, u, v, w, $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$.

Since the differentials dx and dy are independent we must have

$$p = M, \qquad q = N \tag{11}$$

which solves the problem in question for the first-order derivatives.

To solve this problem for the derivatives of the second order we compute the second differentials of u, v and w corresponding to the independent differentials dx and dy. In the course of the computation there appear, together with the differentials dx, dy, the differentials dz, du, dv and dw which should be expressed in terms of dx, dy with the aid of formulas (9) and (10);

there also appears the differential d^2z, and we thus obtain

$$\left.\begin{aligned} d^2u &= B_1^1\,dx^2+B_2^1\,dx\,dy+B_3^1\,dy^2+B_4^1\,d^2z \\ d^2v &= B_1^2\,dx^2+B_2^2\,dx\,dy+B_3^2\,dy^2+B_4^2\,d^2z \\ d^2w &\doteq B_1^3\,dx^2+B_2^3\,dx\,dy+B_3^3\,dy^2+B_4^3\,d^2z \end{aligned}\right\} \tag{12}$$

On the other hand, we have

$$d^2w = \frac{\partial^2 w}{\partial u^2}\,du^2+2\frac{\partial^2 w}{\partial u\,\partial v}\,du\,dv+\frac{\partial^2 w}{\partial v^2}\,dv^2+\frac{\partial w}{\partial u}\,d^2u+\frac{\partial w}{\partial v}\,d^2v \tag{13}$$

Finally, let us replace the differentials entering into (13) by the corresponding expressions (9) and (12) and solve the resultant equation in d^2z:

$$d^2z = P\,dx^2+2Q\,dx\,dy+R\,dy^2 \tag{14}$$

It follows that $r = P$, $s = Q$ and $t = R$ where the right-hand members depend on x, y, z, u, v, w and on the derivatives of w with respect to u and v of orders not higher than 2. Of course, x, y and z can be eliminated from the right-hand side of (13) by means of (7).

The method we have used is based on the consecutive computation of differentials. It can also be applied to the first problem we dealt with (see (1), (2)), then we should put $z = w$.

Example 1. Let us find the expression of the (two-dimensional) *Laplacian* operator*

$$\Delta u = \frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2} \tag{15}$$

in polar coordinates. We shall solve this problem by means of the method of differentials used above (it can also be solved with the aid of the technique used in the solution of the first problem at the beginning of this section).

We have

$$x = \varrho\sin\theta, \qquad y = \varrho\cos\theta \tag{16}$$

On differentiating (16) under the assumption that the variables x and y are independent we obtain

$$dy = \cos\theta\,d\varrho-\varrho\sin\theta\,d\theta, \qquad dx = \sin\theta\,d\varrho+\varrho\cos\theta\,d\theta$$

whence

$$d\varrho = \sin\theta\,dx+\cos\theta\,dy, \qquad d\theta = (-\sin\theta\,dy+\cos\theta\,dy)\frac{1}{\varrho}$$

Further, we have

$$d^2\varrho = (-\sin\theta\,dy+\cos\theta\,dx)\,d\theta = \varrho\,d\theta^2$$

* P. S. Laplace (1749-1827), a French astronomer, mathematician and physicist.

and

$$d^2\theta = \frac{1}{\varrho}(-\cos\theta\, dy - \sin\theta\, dx)\, d\theta - \frac{d\varrho}{\varrho^2}(-\sin\theta\, dy + \cos\theta\, dx) =$$
$$= -\frac{d\varrho\, d\theta}{\varrho} - \frac{d\varrho\, d\theta}{\varrho} = -2\frac{d\varrho\, d\theta}{\varrho}$$

Substituting these expressions into the equality

$$d^2u = \frac{\partial^2 u}{\partial\varrho^2}\, d\varrho^2 + 2\frac{\partial^2 u}{\partial\varrho\,\partial\theta}\, d\varrho\, d\theta + \frac{\partial^2 u}{\partial\theta^2}\, d\theta^2 + \frac{\partial u}{\partial\varrho}\, d^2\varrho + \frac{\partial u}{\partial\theta}\, d^2\theta$$

and combining similar terms involving dx^2, $dx\, dy$ and dy^2 we find the second derivatives of u and, in particular, $\frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial^2 u}{\partial y^2}$, which leads to

$$\Delta u = \frac{\partial^2 u}{\partial\varrho^2} + \frac{1}{\varrho^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{1}{\varrho}\frac{\partial u}{\partial\varrho} \tag{17}$$

Example 2. Let us express the three-dimensional Laplacian operator

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

in spherical coordinates.

We have $x = \varrho\cos\theta\cos\varphi$, $y = \varrho\cos\theta\sin\varphi$ and $z = \varrho\sin\theta$. Let us introduce the auxiliary variable $r = \varrho\cos\theta$. Then $x = r\cos\varphi$, $y = r\sin\varphi$ ($z = z$), and, by virtue of formula (17), we find

$$\Delta u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 u}{\partial\varphi^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial z^2}$$

Now we make the substitution $r = \varrho\cos\theta$, $z = \varrho\sin\theta$ ($\varphi = \varphi$); by formula (17), this gives

$$\frac{\partial^2 u}{\partial r^2} + \frac{\partial^2 u}{\partial z^2} = \frac{\partial^2 u}{\partial\varrho^2} + \frac{1}{\varrho^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{1}{\varrho}\frac{\partial u}{\partial\varrho}$$

and, by formula (4),

$$\frac{\partial u}{\partial r} = -\frac{\dfrac{D(z, u)}{D(\varrho, \theta)}}{\dfrac{D(r, z)}{D(\varrho, \theta)}} = -\frac{1}{\varrho}\sin\theta\frac{\partial u}{\partial\theta} + \cos\theta\frac{\partial u}{\partial\varrho}$$

Finally,

$$\Delta u = \frac{\partial^2 u}{\partial\varrho^2} + \frac{1}{\varrho^2}\frac{\partial^2 u}{\partial\theta^2} + \frac{1}{\varrho^2\cos^2\theta}\frac{\partial^2 u}{\partial\varphi^2} + \frac{2}{\varrho}\frac{\partial u}{\partial\varrho} - \frac{\sin\theta}{\varrho^2\cos\theta}\frac{\partial u}{\partial\theta} \tag{18}$$

We have supposed that the angle θ (the latitude) is reckoned from the equator of the sphere ($-\pi/2 < \theta < \pi/2$). The substitution $\theta' = \pi/2 - \theta$ ($0 < \theta' < \pi$) introduces the latitude θ' reckoned from the north pole of the sphere. In this case we have $\partial^2 u/\partial\theta^2 = \partial^2 u/\partial\theta'^2$, $\cos\theta' = \sin\theta$, $\sin\theta' = \cos\theta$ and $\partial u/\partial\theta = -\partial u/\partial\theta'$, which yields

$$\Delta u = \frac{\partial^2 u}{\partial\varrho^2} + \frac{1}{\varrho^2}\frac{\partial^2 u}{\partial\theta'} + \frac{1}{\varrho^2\sin^2\theta'}\frac{\partial^2 u}{\partial\varphi^2} + \frac{2}{\varrho}\frac{\partial u}{\partial\varrho} - \frac{1}{\varrho^2}\frac{\cos\theta'}{\sin\theta'}\frac{\partial u}{\partial\theta'} \tag{18'}$$

Exercises

1. Show that the curvature of a plane curve $y = f(x)$ is expressed in polar coordinates ($x = r\cos\theta$, $y = r\sin\theta$) by the formula

$$\frac{\left|\frac{d^2y}{dx^2}\right|}{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}} = \frac{\left|r^2+2\left(\frac{dr}{d\theta}\right)^2-r\frac{d^2r}{d\theta^2}\right|}{\left[r^2+\left(\frac{dr}{d\theta}\right)^2\right]^{3/2}}$$

2. Show that the substitution $\xi = x+at$, $\eta = x-at$ brings the differential equation $\partial^2u/\partial\xi\partial\eta = 0$ to $\partial^2u/\partial t^2 = a^2(\partial^2u/\partial x^2)$.

§ 7.27. Dependent Functions

Let

$$y_j = f_j(x) = f_j(x_1, \ldots, x_n) \qquad (\boldsymbol{x} \in G; j = 1, \ldots, m) \tag{1}$$

be a given system of m ($m \leq n$) functions of n independent variables $x_1, \ldots, x_n$ defined and continuously differentiable in a domain G in the n-dimensional space R_n.

Definition. System (1) is said to be *dependent in the domain G* (or the functions f_j are said to be *dependent in G*) if at least one of the functions, say, y_m, is expressed in the domain G in terms of the other functions by means of an equality

$$y_m = \Phi(y_1, \ldots, y_{m-1}) \tag{2}$$

where Φ is a continuously differentiable function of $y_1, \ldots, y_{m-1}$, that is relation (2) becomes an identity with respect to $\boldsymbol{x} = (x_1, \ldots, x_n)$ in the domain G if we put $y_j = f_j(\boldsymbol{x})$ ($j = 1, \ldots, m$) in it. If case (2) holds we shall also say that *the function y_m is dependent on the functions $y_1, \ldots, y_{m-1}$ in the domain G.*

If none of the functions f_j is expressible in terms of the others (in G) the system of the given functions is said to be *independent in G* (or we say that the functions f_j are *independent* in G).

Theorem 1. *If system* (1) *is dependent in the domain G, all the minors of the mth order of the matrix*

$$\left\|\begin{matrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \cdots & \cdots & \cdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{matrix}\right\| \tag{3}$$

are identically equal to zero in G.

Indeed, let for instance y_m be dependent on $y_1, \ldots, y_{m-1}$ and expressed by equality (2). Then

$$\frac{\partial f_m}{\partial x_j} = \sum_{k=1}^{m-1} \frac{\partial \Phi}{\partial y_k}\frac{\partial f_k}{\partial x_j} \qquad (j = 1, \ldots, n; \text{in } G)$$

Therefore

$$\left\| \begin{matrix} \dfrac{\partial f_1}{\partial x_1} \cdots \dfrac{\partial f_1}{\partial x_m} \\ \cdots\cdots\cdots \\ \dfrac{\partial f_m}{\partial x_1} \cdots \dfrac{\partial f_m}{\partial x_m} \end{matrix} \right\| \equiv 0 \quad (\text{in } G)$$

because if we multiply the first $(m-1)$ rows of this determinant by, respectively, $\frac{\partial \Phi}{\partial y_k}$ $(k = 1, \ldots, m-1)$, subtract them from the mth row the resultant row will consist of zeros. Arguing similarly we conclude that any other minor of the mth order of matrix (3) is also identically equal to zero in G.

Theorem 2. *Let all the minors of the mth order of matrix* (3) *be identically equal to zero in G and let s* $(s < m)$ *be the greatest of the indices for which at least one of the minors of the sth order of matrix* (3) *is different from zero at a point* $x^0 \in G$. *Without loss of generality we can suppose that this is the minor*

$$\left| \begin{matrix} \dfrac{\partial f_1}{\partial x_1} \cdots \dfrac{\partial f_1}{\partial x_s} \\ \cdots\cdots\cdots \\ \dfrac{\partial f_s}{\partial x_1} \cdots \dfrac{\partial f_s}{\partial x_s} \end{matrix} \right| \tag{4}$$

Then there is a neighbourhood $\Omega \subset G$ *of the point* x^0 *within which the system of functions* $f_1, \ldots, f_s$ *is independent while the other functions* $f_{s+1}, \ldots, f_n$ *depend on* $f_1, \ldots, f_s$ *in the neighbourhood* Ω:

$$f_\lambda(x) = \Phi_\lambda(f_1(x), \ldots, f_s(x)) \qquad (\lambda = s+1, \ldots, m) \tag{5}$$

In other words, there exist continuously differentiable functions Φ_j *for which identity* (5) *holds in* Ω.

Proof. By the hypothesis, determinant (4) is different from zero at the point x^0 and therefore, since it is a continuous function of the variables $x_1, \ldots, x_n$, it is also different from zero in a neighbourhood Ω_1 of this point; consequently the system $f_1, \ldots, f_s$ is independent in the domain Ω_1 and in any neighbourhood $\omega \ni x^0$ of the point x^0 contained in Ω_1 $(\omega \subset \Omega_1)$ (see Theorem 1).

Further, since determinant (4) is different from zero in Ω_1, the system

$$y_j - f_j(x_1, \ldots, x_s, x_{s+1}, \ldots, x_n) = 0 \qquad (j = 1, \ldots, s) \tag{6}$$

is solvable with respect to $(x_1, \ldots, x_s)$; more precisely, there is a rectangle

$$\Delta = \{|x_1 - x_1^0| \leq \sigma, \ldots, |x_s - x_s^0| \leq \sigma, |x_{s+1} - x_{s+1}^0| \leq \delta, \ldots$$
$$\ldots, |x_n - x_n^0| \leq \delta, |y_1 - y_1^0| \leq \delta, \ldots, |y_s - x_s^0| \leq \delta\}$$

and there exist (uniquely determined) continuously differentiable functions

$$x_j = \mu_j(x_{s+1}, \ldots, x_n, y_1, \ldots, y_s) \qquad (j = 1, \ldots, s) \tag{7}$$

defined on the projection

$$\begin{aligned} \Delta' = \{|x_{s+1}-x_{s+1}^0| \leqslant \delta, \ldots, |x_n-x_n^0| \leqslant \delta, \\ |y_1-y_1^0| \leqslant \delta, \ \ldots, \ |y_s-y_s^0| \leqslant \delta\} \end{aligned} \tag{8}$$

of the rectangle Δ which turn equalities (6) into identities with respect to $(x_{s+1}, \ldots, x_n, y_1, \ldots, y_s)$, that is

$$y_j-f_j(\mu_1, \ldots, \mu_s, x_{s+1}, \ldots, x_n) = 0 \qquad (j = 1, \ldots, s) \tag{9}$$

so that $|\mu_j-x_j^0| \leqslant \sigma$ $(j = 1, \ldots, s)$ and the rectangle

$$\Delta'' = \{|x_1-x_1^0| \leqslant \sigma, \ldots, |x_s-x_s^0| \leqslant \sigma, |x_s-x_{s+1}| \leqslant \delta, \ldots, |x_n-x_n^0| \leqslant \delta\}$$

belongs to Ω_1. Moreover, if a point $(x_1, \ldots, x_n, y_1, \ldots, y_s) \in \Delta$ satisfies system (6) its coordinates must be connected by equalities (7) (the last property expresses the uniqueness).

For our aims it is convenient to introduce the vectors

$$(a)^{(l)} = \left(\frac{\partial f_l}{\partial x_1}, \ldots, \frac{\partial f_l}{\partial x_s}, \frac{\partial f_l}{\partial x_k}\right) \qquad (x \in \Delta''), \ b = \left(\frac{\partial \mu_1}{\partial x_k}, \ldots, \frac{\partial \mu_s}{\partial x_k}, 1\right)$$

where $k > s$ is a natural number.

On differentiating (9) with respect to x_k we get

$$a^{(j)}b = \sum_{i=1}^{s} \frac{\partial f_j}{\partial x_i} \frac{\partial \mu_i}{\partial x_k} + \frac{\partial f_j}{\partial x_k} \equiv 0 \qquad (j = 1, \ldots, s) \tag{10}$$

The substitution of μ_i into f_λ $(\lambda = s+1, \ldots, n)$ results in the functions

$$f_\lambda(\mu_1, \ldots, \mu_s, x_{s+1}, \ldots x_n) \tag{11}$$

which a priori depend on $x_{s+1}, \ldots, x_n, y_1, \ldots, y_s$. But in fact they only depend on $y_1, \ldots, y_s$. Let us prove this property. The total partial derivatives of these functions with respect to x_k are equal to the expressions

$$a^{(\lambda)}b = \sum_{i=1}^{s} \frac{\partial f_\lambda}{\partial x_i} \frac{\partial \mu_i}{\partial x_k} + \frac{\partial f_\lambda}{\partial x_k} \tag{12}$$

By the hypothesis, we have

$$\begin{vmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_s} & \dfrac{\partial f_1}{\partial x_k} \\ \cdots & \cdots & \cdots & \cdots \\ \dfrac{\partial f_s}{\partial x_1} & \cdots & \dfrac{\partial f_s}{\partial x_s} & \dfrac{\partial f_s}{\partial x_k} \\ \dfrac{\partial f_\lambda}{\partial x_1} & \cdots & \dfrac{\partial f_\lambda}{\partial x_s} & \dfrac{\partial f_\lambda}{\partial x_k} \end{vmatrix} \equiv 0$$

while the minor of the sth order of this determinant occupying the upper left corner is different from zero in $\Delta'' \subset \Omega_1$. Thus, the vector $a^{(\lambda)}$ is representable as a linear combination of the vectors $a^{(1)}, \ldots, a^{(s)}$. By (10), the latter vectors are orthogonal to b, and hence the vector $a^{(\lambda)}$ is also orthogonal to b. This means that expression (12) is identically equal to zero in Δ'. We have thus shown that the total derivatives of functions (11) with respect to x_k $(k = s+1, \ldots, n)$ are identically equal to zero in Δ' for $\lambda > s$. Therefore these functions do not in fact depend on $x_{s+1}, \ldots, x_n$. Consequently

$$f_\lambda(\mu_1, \ldots, \mu_s, x_{s+1}, \ldots, x_n) = \Phi_\lambda(y_1, \ldots, y_s) \qquad (\lambda = s+1, \ldots, n) \quad (13)$$

where Φ_λ are continuously differentiable functions of $y_1, \ldots, y_s$. Now, using the continuity of the functions $f_j(x)$ for the indicated values of δ and σ, we find $\delta_1 > 0$ $(\delta_1 < \delta, \delta_1 < \sigma)$ such that for the values of x_j satisfying the conditions

$$|x_j - x_j^0| < \delta_1 \qquad (\delta_1 < \delta, \delta_1 < \sigma) \quad (14)$$

there hold the inequalities

$$|f_j(x) - f_j(x^0)| < \delta \qquad (j = 1, \ldots, s) \quad (15)$$

Then for the points $x = (x_1, \ldots, x_n)$ belonging to cube (14), which we shall denote Ω, we have the relations (see the explanations below)

$$\begin{aligned} f_\lambda(x_1, \ldots, x_n) &= f_\lambda(\mu_1, \ldots, \mu_s, x_{s+1}, \ldots, x_n) = \\ &= \Phi_\lambda(y_1, \ldots, y_s) = \Phi_\lambda(f_1(x), \ldots, f_s(x)) \end{aligned} \quad (16)$$

which means that y_λ $(\lambda = s+1, \ldots, n)$ are dependent on $y_1, \ldots, y_s$ in Ω.

For $x = (x_1, \ldots, x_n) \in \Omega$ the points $(x_{s+1}, \ldots, x_n, y_1, \ldots, y_s)$ belong to Δ' $\big(y_j = f_j(x)$; see (8)$\big)$ and therefore

$$x_j = \mu_j(x_{s+1}, \ldots, x_n, y_1, \ldots, y_s) \qquad (j = 1, \ldots, s)$$

which proves the first equality (16); the second equality (16) follows from (13) and the derivation of the third one reduces to the reverse replacement of y_j by $f_j(x)$.

CHAPTER 8

Indefinite Integral. Properties of Polynomials

§ 8.1. Introduction. Integration by Change of Variables and Integration by Parts

In § 1.6 we introduced the notions of the antiderivative and of the indefinite integral. The reader should recall this material before proceeding to the study of the present chapter. The aim of this chapter is to provide practical means for the computation of indefinite integrals of some elementary functions.

In the theory of the definite integral (Chapter 9) we shall prove a theorem asserting that every function $f(x)$ continuous in an open interval (a, b) (or in a closed interval $[a, b]$) possesses an antiderivative $F(x)$ defined on that interval which is also continuous. Since by the indefinite integral of f on (a, b) is meant the general expression for the family of all the antiderivatives of f, and since any two antiderivatives of f only differ by a constant summand, the indefinite integral of f on (a, b) is written as $\int f(x)\,dx = F(x)+C$ where $F(x)$ is an arbitrary concrete antiderivative of f and C is an arbitrary constant which should be appropriately chosen to obtain a definite antiderivative. On the basis of the theorem we have mentioned we can say that every function f continuous on an interval has the indefinite integral on that interval. But it turns out that an antiderivative F of a given elementary function f (see § 1.3) is itself by far not always an elementary function. For some elementary functions their antiderivatives are also elementary functions but in the general case this is not so. Here lies an essential difference between the differential calculus and the integral calculus: the derivative of an elementary function is always an elementary function while the indefinite integral of an elementary function may not be an elementary function. In other words, there exist elementary functions which, as we say, are *nonintegrable in elementary functions.* Their indefinite integrals exist but cannot be represented in terms of elementary functions.

Fortunately, there are some rather wide classes of elementary functions important for practical applications of mathematics whose indefinite integrals are expressible in terms of elementary functions, that is their antiderivatives are elementary functions.

This chapter is devoted to the study of the integration methods for functions belonging to such classes.

To begin with we present the table of indefinite integrals implied by the basic table of the derivatives of the simplest elementary functions:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}+C \qquad (n+1 \neq 0)$$

$$\int \frac{1}{x}\,dx = \ln|x|+C$$

$$\int e^x\,dx = e^x+C$$

$$\int a^x\,dx = \frac{a^x}{\ln a}+C \qquad (a > 0)$$

$$\int \cos x\,dx = \sin x+C$$

$$\int \sin x\,dx = -\cos x+C$$

$$\int \sec^2 x\,dx = \tan x+C$$

$$\int \csc^2 x\,dx = -\cot x+C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \operatorname{arc\,sin} x+C = -\operatorname{arc\,cos} x+C' \qquad \left(C' = \frac{\pi}{2}+C\right)$$

$$\int \frac{dx}{1+x^2} = \operatorname{arc\,tan} x+C$$

$$\int \cosh x\,dx = \sinh x+C$$

$$\int \sinh x\,dx = \cosh x+C$$

On the left-hand side of each of these equalities stands an arbitrary (but definitely chosen) antiderivative of the corresponding integrand function while on the right-hand side there is a concrete antiderivative to which a constant C (we call it an *arbitrary constant* or a *constant of integration*) is added such that the equality between these antiderivatives holds.

The antiderivatives entering into these formulas are defined and continuous in the intervals where the corresponding integrands are defined and continuous. This is due to the fact mentioned above: every function continuous in an interval has a continuous antiderivative in that interval.

In § 1.6 we deduced the formula

$$\int (A_1u(x)+A_2v(x))\,dx = A_1\int u(x)\,dx+A_2\int v(x)\,dx+C \tag{1}$$

expressing the *linear property* of the indefinite integral.

One of the most important methods of the integral calculus is *integration by change of variable (by substitution)*:

$$\int f(x)\,dx = \int f(\varphi(t))\,\varphi'(t)\,dt+C = \int f(\varphi(t))\,d\varphi(t)+C \tag{2}$$

In this formula it is supposed that $x = \varphi(t)$ is a continuously differentiable function (i.e. having a continuous derivative) in an interval of variation of t and $f(x)$ is a continuous function in the corresponding interval of the x-axis. The first equality (2) asserts that its left-hand side is identically equal to the right-hand side in which, after the integration has been performed, the substitution $x = \varphi(t)$ is made and the constant C is appropriately chosen. Let us prove what has been said.

On the left-hand side of (2) stands a function which is an antiderivative of $f(x)$. If x is replaced by $x = \varphi(t)$ in this antiderivative, then its derivative with respect to t is

$$\frac{d}{dt}\int f(x)\,dx = \frac{d}{dx}\left(\int f(x)\,dx\right)\varphi'(t) = f(\varphi(t))\varphi'(t)$$

Consequently, if we make the substitution $x = \varphi(t)$ in the antiderivative of $f(x)$ the resultant expression is an antiderivative of $f(\varphi(t))\varphi'(t)$. As to the integral on the right-hand side of (2), it gives, by definition, the general expression for an arbitrary antiderivative of $f(\varphi(t))\varphi'(t)$. As we know, two antiderivatives of one and the same function only differ from each other by a constant C. This very fact is expressed by the first equality (2). The second equality (2) is of a purely formal character: we simply agree to write

$$\int F(t)\varphi'(t)\,dt \doteq \int F(t)\,d\varphi(t) \tag{3}$$

For instance,

$$\int e^{x^2}x\,dx = \frac{1}{2}\int e^{x^2}2x\,dx+C = \frac{1}{2}\int e^{x^2}\,d(x^2)+C =$$
$$= \frac{1}{2}\int e^{u}\,du+C_1 = \frac{1}{2}e^{u}+C_2 = \frac{1}{2}e^{x^2}+C_2 \qquad (u = x^2) \tag{4}$$

The first equality in (4) is written in accordance with (1), the second is written by virtue of (3), the third equality corresponds to (2) (the constant has been changed here) and the fourth one is written by virtue of the tabular formula (where the constant has been changed again). In practical computations we usually do not write the arbitrary constant C in the intermediate expressions involving indefinite integrals; for instance, the above calculations (4) can be written in the simplified form

$$\int e^{x^2}x\,dx = \frac{1}{2}\int e^{x^2}2x\,dx = \frac{1}{2}\int e^{x^2}\,dx^2 = \frac{1}{2}e^{x^2}+C$$

where the obvious third and fourth equalities are omitted.

Here are another two examples of this kind:

$$\int e^{3x}\,dx = \frac{1}{3}\int e^{3x}\,d(3x) = \frac{1}{3}e^{3x}+C$$

$$\int \sin kx\,dx = \frac{1}{k}\int \sin kx\,d(kx) = -\frac{1}{k}\cos kx+C$$

We also give some more examples which will be of use for the theory of integration of rational fractions:

$$\int \frac{dx}{(x-a)^m} = \int \frac{d(x-a)}{(x-a)^m} = \frac{1}{(x-a)^{m-1}(1-m)} + C \qquad (m \neq 1)$$

$$\int \frac{dx}{x-a} = \int \frac{d(x-a)}{x-a} = \ln|x-a| + C$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\int \frac{d(x/a)}{1+(x/a)^2} = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\int\left(\frac{1}{x-a}-\frac{1}{x+a}\right)dx = \frac{1}{2a}(\ln|x-a|-\ln|x+a|)+C =$$

$$= \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$$

$$\int \frac{dx}{x^2+px+q} = \int \frac{dx}{(x+(p/2))^2} = \int \frac{d(x+(p/2))}{(x+(p/2))^2} =$$

$$= -\frac{1}{x+(p/2)} + C \quad \left(q-\frac{p^2}{4}=0\right) \qquad (5)$$

$$\int \frac{dx}{x^2+px+q} = \int \frac{dx}{(x+(p/2))^2+(q-(p^2/4))} = \int \frac{d(x+(p/2))}{(x+(p/2))^2+a^2} =$$

$$= \frac{1}{a}\int \frac{d\left(\frac{x+(p/2)}{a}\right)}{1+\left(\frac{x+(p/2)}{a}\right)^2}\,dx = \frac{1}{a}\arctan\frac{x+(p/2)}{p} + C$$

$$\left(q-\frac{p^2}{4} = a^2 > 0,\ a > 0\right) \qquad (6)$$

$$\int \frac{dx}{x^2+px+q} = \int \frac{d(x+(p/2))}{(x+(p/2))^2-a^2} = \frac{1}{2a}\ln\left|\frac{x+(p/2)-a}{x+(p/2)+a}\right| + C$$

$$\left(q-\frac{p^2}{4} = -a^2,\ a > 0\right) \qquad (7)$$

$$\int \frac{(2x+p)\,dx}{x^2+px+q} = \int \frac{d(x^2+px+q)}{x^2+px+q} = \ln|x^2+px+q| + C$$

$$\int \frac{Ax+B}{x^2+px+q}dx = \frac{A}{2}\int \frac{2x+(2B/A)}{x^2+px+q}dx = \frac{A}{2}\int \frac{(2x+p)\,dx}{x^2+px+q} + \frac{A}{2}\int \frac{(2B/A)-p}{x^2+px+q}dx =$$

$$= \frac{A}{2}\ln|x^2+px+q| + D\int \frac{dx}{x^2+px+q} \quad \left(A \neq 0,\ D = \frac{A}{2}\left(\frac{2B}{A}-p\right)\right) \quad (8)$$

(further computations of the last integral reduce to (7)).

For the theory of integration of rational fractions it is important that the computation of integrals (5)-(8) in which a, A, B, p and q are constants leads to elementary functions (namely, to rational functions, logarithmic functions and to the arc tangent).

Now we proceed to the formula of *integration by parts*:

$$\int uv'\,dx = uv-\int vu'\,dx+C \tag{9}$$

or, equivalently, $\int u\,dv = uv-\int v\,du+C$.

In this formula u and v are continuously differentiable functions. The derivative of the left-hand side of the formula is equal to uv' and the derivative of the right-hand side is also equal to $uv' = (uv)'-vu'$; therefore they only differ by a constant, which is expressed by (9).

Consider the following examples:

$$\int \ln x\,dx = x\ln x-\int dx = x(\ln x-1)+C$$

$$\int xe^x\,dx = \int x\,de^x = e^x x-\int e^x\,dx = e^x(x-1)+C$$

$$\int x\sin x\,dx = \int x\,d(-\cos x) = -x\cos x+\int \cos x\,dx =$$
$$= -x\cos x+\sin x+C$$

$$\int e^{ax}\cos x\,dx = e^{ax}\sin x-a\int e^{ax}\sin x\,dx =$$
$$= e^{ax}\sin x-a\left[e^{ax}(-\cos x)+a\int e^{ax}\cos x\,dx\right]$$

whence

$$\int e^{ax}\cos x\,dx = \frac{e^{ax}(\sin x+a\cos x)}{1+a^2}+C$$

We also present one more example which will be of use in the theory of integration of rational fractions.

Let $k > 1$ be a natural number and $a > 0$ an arbitrary real number. Then

$$\int \frac{dx}{(x^2+a^2)^{k-1}} = a^2\int \frac{dx}{(x^2+a^2)^k}+\frac{1}{2}\int \frac{x\,2x\,dx}{(x^2+a^2)^k} =$$
$$= a^2\int \frac{dx}{(x^2+a^2)^k}+\frac{1}{2}\left\{\frac{x}{(1-k)(x^2+a^2)^{k-1}}-\frac{1}{1-k}\int \frac{dx}{(x^2+a^2)^{k-1}}\right\}$$

whence

$$a^2\int \frac{dx}{(x^2+a^2)^k} = \frac{x}{2(k-1)(x^2+a^2)^{k-1}}+\frac{3-2k}{2(1-k)}\int \frac{dx}{(x^2+a^2)^{k-1}}$$

Now, if $k > 2$, the integral on the right-hand side admits of the application of the same technique, which reduces by unity the exponent in the power function entering as the denominator in the integrand fraction. Proceeding in this way we ultimately arrive at an integral of $(x^2+a^2)^{-1}$ which is reducible to the arc tangent.

Thus, for $q-(p^2/4) = a^2 > 0$ and any natural k the integral

$$\int \frac{dx}{(x^2+px+q)^k} = \int \frac{du}{(u^2+a^2)^k}+C \qquad \left(u = x+\frac{p}{2}\right) \tag{10}$$

is expressible in terms of elementary functions.

Examples

Change of Variable (Substitution):

1. $\int \frac{dx}{1+x^2} = (\operatorname{arc\,tan} x \pm \operatorname{arc\,tan} 1)+C = \operatorname{arc\,tan} \frac{x\pm 1}{1\mp x}+C$

2. $\int \frac{dx}{\sqrt{a^2-x^2}} = \int \frac{d(x/a)}{\sqrt{1-(x/a)^2}} = \operatorname{arc\,sin} \frac{x}{a}+C \qquad (a > 0)$

3. $\int \frac{dx}{\sin x \cos x} = \int \frac{d(\tan x)}{\tan x} = \ln |\tan x|+C$

4. $\int \frac{x\,dx}{\sqrt{a^2-x^2}} = -\int \frac{1}{2} \frac{d(a^2-x^2)}{\sqrt{a^2-x^2}} = -\sqrt{a^2-x^2}+C$

5. $\int \tan x\,dx = -\int \frac{d(\cos x)}{\cos x} = -\ln |\cos x|+C$

6. $\int \frac{dx}{1+\cos x} = \int \frac{dx}{2\cos^2 x/2} = \int \sec^2 \frac{x}{2}\, d\left(\frac{x}{2}\right) = \tan \frac{x}{2}+C$

7. $\int \frac{dx}{1+\cos^2 x} = \int \frac{dx}{2\cos^2 x+\sin^2 x} = \int \frac{d(\tan x)}{2+\tan^2 x} =$

$$= \frac{1}{\sqrt{2}} \operatorname{arc\,tan} \left(\frac{1}{\sqrt{2}} \tan x\right)+C$$

Integration by Parts:

8. $\int \operatorname{arc\,sin} x\,dx = x \operatorname{arc\,sin} x - \int \frac{x}{\sqrt{1-x^2}}\,dx = x \operatorname{arc\,sin} x + \sqrt{1-x^2}+C$

9. $\int \operatorname{arc\,sin} x \frac{x\,dx}{\sqrt{1-x^2}} = -\sqrt{1-x^2} \operatorname{arc\,sin} x + \int dx =$

$$= x-\sqrt{1-x^2} \operatorname{arc\,sin} x+C$$

10. $\int \frac{x\,dx}{\cos^2 x} = x \tan x - \int \tan x\,dx = x \tan x - \ln |\cos x| +C$

Combined Methods:

11. $\int \tan^2 x\,dx = \int (\sec^2 x-1)\,dx = \tan x-x+C$

12. $\int \sqrt{\frac{1+x}{1-x}}\,dx = \int (1+x)\frac{dx}{\sqrt{1-x^2}} = (1+x) \operatorname{arc\,sin} x - \int \operatorname{arc\,sin} x\,dx =$

$$\operatorname{arc\,sin} x - \sqrt{1-x^2}+C$$

§ 8.2. Complex Numbers

Complex numbers can be defined formally as pairs (α, β) of real numbers (α and β) for which the equality relation and the arithmetical operations are defined according to the following rules:

(i) equality $(\alpha, \beta) = (\alpha_1, \beta_1)$ takes place if and only if $\alpha = \alpha_1$ and $\beta = \beta_1$;

(ii) $(\alpha, \beta) \pm (\alpha_1, \beta_1) = (\alpha \pm \alpha_1, \beta \pm \beta_1)$

$$(\alpha, \beta)(\alpha_1, \beta_1) = (\alpha\alpha_1 - \beta\beta_1, \alpha\beta_1 + \alpha_1\beta)$$

$$\frac{(\alpha, \beta)}{(\alpha_1, \beta_1)} = \left(\frac{\alpha\alpha_1 + \beta\beta_1}{\alpha_1^2 + \beta_1^2}, \frac{\alpha_1\beta - \alpha\beta_1}{\alpha_1^2 + \beta_1^2}\right)$$

For the pairs of the form $(\alpha, 0)$ *(and only for them!)* the notion expressed by the symbol " $<$ " is introduced:

$$(\alpha, 0) < (\beta, 0), \quad \text{if} \quad \alpha < \beta \tag{1}$$

There holds the obvious one-to-one correspondence $\alpha \sim (\alpha, 0)$ between the pairs $(\alpha, 0)$ and the real numbers α. It is important that this correspondence is isomorphic with respect to the arithmetical operations and with respect to the symbol " $<$ ":

$$\alpha \pm \beta \sim (\alpha \pm \beta, 0) = (\alpha, 0) \pm (\beta, 0)$$

$$\alpha\beta \sim (\alpha\beta, 0) = (\alpha, 0)(\beta, 0)$$

$$\frac{\alpha}{\beta} \sim \left(\frac{\alpha}{\beta}, 0\right) = \frac{(\alpha, 0)}{(\beta, 0)} \qquad (\beta \neq 0)$$

(also see (1)).

This shows that the pairs $(\alpha, 0)$ are real numbers (more precisely, they can be identified with real numbers). Hence, the symbols $(\alpha, 0)$ can be regarded as representing the real numbers and therefore we shall simply write $\alpha = (\alpha, 0)$.

Thus, the set of all pairs contains as its subset the set of all real numbers. As to the pairs (α, β) with $\beta \neq 0$, they are not real numbers and represent new objects which we call *complex numbers*.

It can readily be checked that the pairs satisfy the axioms of arithmetical operations similar to the axioms of real numbers (see § 2.4, Axioms II, III) with the exception of those involving the symbol " $<$ " which thus should be excluded. (The pairs $(\alpha, 0)$ satisfy, of course, all the axioms of real numbers.)

The pairs $(\alpha, 0)$ have already been denoted as α. For the pair $(0, 1)$ we introduce the symbol i: $i = (0, 1)$.

We can perform the arithmetical operations on the real numbers α and on the symbol i since they are pairs of real numbers for which these operations have already been defined. These operations result in new pairs which can be written as some arithmetical combinations of real numbers $\alpha, \beta, \ldots$ and the symbol i. The calculations are simplified if we take into account the relations $i^2 = ii = (0, 1)(0, 1) = (-1, 0) = -1$.

Every pair can be presented as an arithmetical combination of real numbers and the number i:

$$(\alpha, \beta) = (\alpha, 0) + (0, \beta) = \alpha + \beta(0, 1) = \alpha + i\beta \tag{2}$$

Thus, there is a one-to-one correspondence between the pairs (α, β) and the expressions $\alpha+i\beta$ specified by equality (2). This correspondence is an isomorphism relative to the arithmetical operations because by virtue of the arithmetical rules for real numbers and for the symbol i we have

$$(\alpha+i\beta)\pm(\alpha_1+i\beta_1) = (\alpha\pm\alpha_1)+i(\beta\pm\beta_1)$$
$$(\alpha+i\beta)(\alpha_1+i\beta_1) = (\alpha\alpha_1-\beta\beta_1)+i(\alpha\beta_1+\alpha_1\beta)$$
$$\frac{\alpha+i\beta}{\alpha+i\beta_1} = \frac{\alpha\alpha_1+\beta\beta_1}{\alpha_1^2+\beta_1^2}+i\frac{\alpha_1\beta-\alpha\beta_1}{\alpha_1^2+\beta_1^2} \qquad (\alpha_1^2+\beta_1^2 \neq 0)$$

Now we can discard the symbols (α, β) and operate only with the expressions (complex numbers) $\alpha+i\beta$ representing the pairs. A further step is to denote a complex number $\alpha+i\beta$, where α and β are real numbers, by a single letter, for instance, to write $a = \alpha+i\beta$; in this notation α is called the *real part* of the complex number a and β is called the *imaginary part* of a (the number β itself is of course real).

When saying that we are given a complex number $a = \alpha+i\beta$ we mean, without further stipulations, that α and β are real numbers.

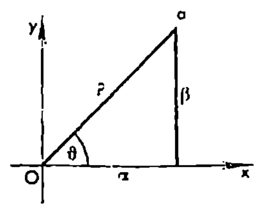

Fig. 8.1

The complex numbers can be represented geometrically as points of the so-called *complex plane* (see Fig. 8.1). In this construction with each complex number $a = \alpha+i\beta$ is associated a point (the point a) with rectangular coordinates α and β in the complex plane. The x-axis (Fig. 8.1) along which α is reckoned is called the *real axis* (or the *axis of reals*) and the y-axis along which β is reckoned is the *imaginary axis* (the *axis of imaginaries*). Denoting by ϱ the length of the radius vector of the point a and by θ (for $a \neq 0$) the angle (expressed in radians) formed by the radius vector with the positive half-axis x we clearly have $\alpha = \varrho\cos\theta$, $\beta = \varrho\sin\theta$ and $\varrho = \sqrt{\alpha^2+\beta^2} \geqslant 0$ and consequently

$$a = \alpha+i\beta = \varrho(\cos\theta+i\sin\theta) \tag{3}$$

If $a = 0$ then $\varrho = 0$ and equality (3) continues to hold for any θ.

Thus, we have shown that every complex number a is representable in form (3) where ϱ is a nonnegative number. In this *trigonometric form of complex numbers* ϱ is a uniquely determined (nonnegative) number for a given complex number a; generally speaking, θ is not determined uniquely for a given complex number but if we additionally impose the requirement that it should satisfy the inequalities

$$0 \leqslant \theta < 2\pi \tag{4}$$

then, for $a \neq 0$, it becomes unique. If, under condition (4), ϱ or θ (or both) is changed in (3), the original complex point goes to another point distinct from the former.

The number ϱ is called the *modulus* (also the *absolute value)* of a and is denoted $|a| = \varrho = \sqrt{\alpha^2+\beta^2}$. If a is a real number then its modulus coincides with its ordinary absolute value.

The number θ is called the *argument* of a and is denoted Arg a. If a given value of θ satisfies equation (3) then any other value differing from the former by $2k\pi$ $(k=0, \pm 1, \ldots)$ also satisfies it. For a given complex number a the corresponding (uniquely determined) value of θ satisfying (3) and the inequalities $0 \leqslant \theta < 2\pi$ is termed the *principal value of the argument of a* and is denoted arg a*. It is clear that

$$\theta = \operatorname{Arg} a = \arg a+2k\pi \qquad (k = 0, \pm 1, \pm 2, \ldots) \tag{5}$$

and that Arg a is an infinite-valued function of a, the value arg a specifying a single-valued branch of this function. Any solution θ of equation (3) can be written in form (5).

Let us put (at present purely formally!)

$$e^{i\theta} = \cos\theta + i\sin\theta \qquad (-\infty < \theta < \infty)$$

This relation can be regarded as a definition of the function e^z for the pure imaginary argument $z = i\theta$. For an arbitrary complex variable $z = x+iy$ the function e^z can now be defined by means of the equality

$$e^z = e^x e^{iy} = e^x(\cos y+i\sin y) \tag{6}$$

The expression $e^{i\theta}$ is a complex function (assuming complex values) of the real argument θ. When θ varies continuously in the half-interval $0 \leqslant \theta < 2\pi$ the point $e^{i\theta}$ in the complex plane describes continuously a circle of radius 1 with centre at 0. Thus, $|e^{i\theta}| = \sqrt{\cos^2\theta+\sin^2\theta} = 1$. It is evident that $e^{i\theta}$ is a periodic function with period 2π: $e^{i(\theta+2\pi)} = e^{i\theta}$. It possesses the properties

$$e^{i(\theta+\theta_1)} = e^{i\theta}e^{i\theta_1} \tag{7}$$

and

$$e^{-i\theta} = \frac{1}{e^{i\theta}}$$

which hold for any values θ and θ_1 since we have

$$e^{i\theta}e^{i\theta_1} = (\cos\theta+i\sin\theta)(\cos\theta_1+i\sin\theta_1) = (\cos\theta\cos\theta_1-\sin\theta\sin\theta_1)+$$
$$+i(\cos\theta\sin\theta_1+\sin\theta_1\cos\theta) = \cos(\theta+\theta_1)+ i\sin(\theta+\theta_1) = e^{i(\theta+\theta_1)}$$

and

$$\frac{1}{e^{i\theta}} = \frac{1}{\cos\theta+i\sin\theta} = \cos\theta-i\sin\theta = e^{-i\theta}$$

* It may sometimes be convenient to define the principal value of the argument of a as the number θ satisfying (3) and specified by an inequality of the form $\alpha_0 \leqslant \theta < \alpha_0+2\pi$ or $\alpha_0 < \theta \leqslant \alpha_0+2\pi$ where α_0 is an arbitrary number.

It also follows from (7) that $e^{i(\theta-\theta_1)} = e^{i\theta}e^{-i\theta_1} = \frac{e^{i\theta}}{e^{i\theta_1}}$. What was said above implies that every complex number a is representable in the *exponential form* $a = \varrho e^{i\theta}$ where $\varrho \geqslant 0$ is a uniquely determined number equal to $|a|$ and $\theta = \text{Arg } a$ is determined to within a summand of the form $2k\pi$ $(k = 0, \pm 1, \ldots)$.

For any complex numbers a and b there holds the inequality

$$|a+b| \leqslant |a|+|b| \tag{8}$$

and another inequality

$$||a|-|b|| \leqslant |a-b| \tag{9}$$

implied by the former. They express the geometrical fact that (see Fig. 8.2) the length of a side of a triangle does not exceed the sum of the lengths of the other two sides and is not less than their difference.

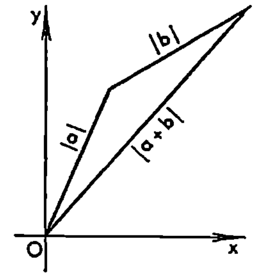

Fig. 8.2

Inequality (8) can be rewritten in terms of the real and imaginary parts of a and b, namely it reduces to the inequality (see § 6.2 (9))

$$\sqrt{(\alpha+\alpha_1)^2+(\beta+\beta_1)^2} \leqslant \sqrt{\alpha^2+\beta^2}+\sqrt{\alpha_1^2+\beta_1^2}$$

There also hold the relations

$$|ab| = |a|\,|b| \tag{10}$$

and

$$\text{Arg }(ab) = \text{Arg } a+\text{Arg } b+2k\pi \tag{11}$$

which imply

$$\left|\frac{a}{b}\right| = \left|\frac{a}{b}\right|, \qquad b \neq 0 \tag{12}$$

and

$$\text{Arg } \frac{a}{b} = \text{Arg } a-\text{Arg } b+2k\pi \qquad (k = 0, \pm 1, \ldots) \tag{13}$$

Equality (11) should be understood in the sense that if some admissible numbers θ_1, θ_2 and θ are chosen as arguments of a, b and ab, respectively, then θ differs from $\theta_1+\theta_2$ by a summand $2k\pi$ where k is an integer. The meaning of (13) is analogous.

Let us prove (10) and (11). Let $a = \varrho_1 e^{i\theta_1}$, $b = \varrho_2 e^{i\theta_2}$ and $ab = \varrho e^{i\theta}$ where θ_1, θ_2 and θ are some definite (but arbitrarily chosen) values of the arguments of a, b and ab respectively. Then

$$\varrho e^{i\theta} = \varrho_1\varrho_2 e^{i(\theta_1+\theta_2)}$$

and, by the obvious properties of the exponential representation of a complex number, there must be $\varrho = \varrho_1\varrho_2$ and $\theta = \theta_1+\theta_2+2k\pi$ where k is an integer.

Given a complex number $a = \alpha + i\beta$, the number $\bar{a} = \alpha - i\beta$ is called its *conjugate (complex)* number. If $a = \alpha$ is a real number then $\bar{a} = a$.

We obviously have

$$\overline{a \pm b} = \bar{a} \pm \bar{b}, \quad \overline{ab} = \bar{a}\bar{b}, \quad \overline{\left(\frac{a}{b}\right)} = \frac{\bar{a}}{\bar{b}} \qquad (b \neq 0) \tag{14}$$

because if $a = \alpha_1 + i\beta_1 = \varrho_1 e^{i\theta_1}$ and $b = \alpha_2 + i\beta_2 = \varrho_2 e^{i\theta_2}$ then

$$\overline{a \pm b} = \overline{(\alpha_1 \pm \alpha_2) + (\beta_1 \pm \beta_2) i} = (\alpha_1 \pm \alpha_2) - (\beta_1 \pm \beta_2) i =$$
$$= (\alpha_1 - \beta_1 i) \pm (\alpha_2 - \beta_2 i) = \bar{a} \pm \bar{b}$$

and

$$\overline{ab} = \overline{\varrho_1 \varrho_2 e^{i(\theta_1 + \theta_2)}} = \varrho_1 \varrho_2 e^{-i(\theta_1 + \theta_2)} = \varrho_1 e^{-i\theta_1} \varrho_2 e^{-i\theta_2} = \bar{a}\bar{b}$$

Let us consider the problem of computing the nth root of a complex number $a = \varrho e^{i\theta} (\varrho > 0)$. It is required to find all the numbers $b = re^{i\varphi}$ such that $b^n = a$. For such b's we have $r^n e^{in\varphi} = \varrho e^{i\theta} (r, \varrho \geqslant 0)$, and, by the properties of representation of a complex number in the exponential form, we obtain $\varrho = r^n$ and $n\varphi = \theta + 2k\pi$, $k = 0, \pm 1, \ldots$ From the first of these equalities it follows that $r = \sqrt[n]{\varrho} > 0$ (r is the arithmetic nth root of the positive number ϱ). The second of these equalities shows that $\varphi = \frac{\theta}{n} + \frac{2k\pi}{n}$ $(k = 0, \pm 1, \ldots)$.

The values of φ which yield essentially different values of the nth root of a only correspond to n successive values of k, for instance,

$$\varphi_k = \frac{\theta}{n} + \frac{2k\pi}{n} \qquad (k = 0, 1, \ldots, n-1) \tag{15}$$

To the other integers k there correspond the values of φ differing from one of the values (15) by an integer multiple of 2π which do not give new values of the root.

We have shown that there are exactly n different values of the nth root of a complex number $a \neq 0$ expressed by the formula

$$\sqrt[n]{a} = \sqrt[n]{\varrho e^{i\theta}} = \sqrt[n]{\varrho}\, e^{i\varphi_k} \qquad (k = 0, 1, \ldots, n-1)$$

where φ_k are given by equalities (15).

Examples

1. $$\frac{1}{2} + \cos x + \cos 2x + \ldots + \cos nx = \frac{\sin \frac{2n+1}{2} x}{2 \sin \frac{x}{2}} = D_n(x) \tag{16}$$

2. $$\sin x + \ldots + \sin nx = \frac{\cos \frac{x}{2} - \cos \left(n + \frac{1}{2}\right) x}{2 \sin \frac{x}{2}} \tag{17}$$

If we equate the real and imaginary parts on both sides of the equality

$$\sum_{k=1}^{n} e^{ikx} = \frac{e^{i(n+1)x}-e^{ix}}{e^{ix}-1} = \frac{e^{i(n+(1/2))x}-e^{ix/2}}{e^{ix/2}-e^{-ix/2}} =$$
$$= \frac{\sin(n+(1/2))x}{2\sin(x/2)} - \frac{1}{2} + i\,\frac{\cos(x/2)-\cos(n+(1/2))x}{2\sin(x/2)}$$

we directly obtain (16) and (17). Sums (16) and (17) play an important role in the theory of Fourier's series; functions (16) are termed *Dirichlet's sums* or *Dirichlet's kernels*.

§ 8.3. Limit of a Sequence of Complex Numbers. Function of a Complex Variable

The limit $\lim\limits_{n\to\infty} z_n$ *of a sequence of complex numbers* $z_n = \alpha_n + i\beta_n$ $(n = 1, 2, \ldots)$ is defined as a (complex) number $z_0 = \alpha_0 + i\beta_0$ for which

$$|z_n - z_0| = \sqrt{(\alpha_n-\alpha_0)^2+(\beta_n-\beta_0)^2} \to 0 \qquad (n\to\infty) \tag{1}$$

or, which is obviously the same, as a number $z_0 = \alpha_0 + i\beta_0$ for which $\alpha_n \to \alpha_0$ and $\beta_n \to \beta_0$ $(n\to\infty)$ simultaneously. If z_0 is the limit of z_n we write $\lim\limits_{n\to\infty} z_n = z_0$ or $z_n \to z_0$.

Since the existence of the limit of z_n reduces to the existence of the limits of its real and imaginary parts we readily conclude that, as in the case of real sequences, *Cauchy's criterion* holds for complex sequences: for a sequence $\{z_n\}$ to have a limit it is necessary and sufficient that, given any $\varepsilon > 0$, there should exist N such that for all $n, n' > N$ the inequality $|z_n - z_{n'}| < \varepsilon$ takes place.

It can also be easily shown that the existence of limit (1) implies that $|z_n| \to |z_0|$ $(n\to\infty)$.

There also hold the equalities

$$\left.\begin{aligned} \lim(z_n \pm z_n') &= \lim z_n \pm \lim z_n' \\ \lim(z_n z_n') &= \lim z_n \lim z_n' \\ \lim \frac{z_n}{z_n'} &= \frac{\lim z_n}{\lim z_n'} \qquad (\lim z_n' \neq 0) \end{aligned}\right\} \tag{2}$$

where, as usual, the existence of the limits on the left-hand sides of the equalities is asserted as a consequence of the existence of the limits of z_n and z_n'.

Let G be a set of complex numbers $z = x+iy$ (which can be equivalently interpreted as a set of points in the complex plane). If there is a law according to which to every $z \in G$ there corresponds a number w (which is complex in the general case) we say that *there is a function* $w = f(z)$ *of the complex variable* z *defined on* G.

By analogy with a function of a real variable, the notions of the limit, continuity and derivative are defined for such a complex function.

By a *neighbourhood of* z_0 is meant a set of points z (in the complex plane) for which the inequality

$$|z-z_0| < \delta \tag{3}$$

holds where δ is an arbitrary positive number.

A (complex) number A is said to be *the limit of the function* f *at the point* z_0 if f is defined in a neighbourhood of the point z_0 except possibly at that point itself and if for any $\varepsilon > 0$ there is $\delta > 0$ such that $|f(z)-A| < \varepsilon$ for all z satisfying the inequality $0 < |z-z_0| < \delta$.

If A is the limit of $f(z)$ at z_0 we write

$$\lim_{z\to z_0} f(z) = A \quad \text{or} \quad f(z) \to A,\ z \to z_0$$

This definition is equivalent to the following: for any sequence $\{z_n\}$, $z_n \neq z_0$, convergent to z_0 the relation $\lim f(z_n) = A$ holds.

The following properties of limits hold:

$$\left.\begin{aligned} \lim_{z\to z_0} \{u(z)\pm v(z)\} &= \lim_{z\to z_0} u(z) \pm \lim_{z\to z_0} v(z) \\ \lim_{z\to z_0} \{u(z)\,v(z)\} &= \lim_{z\to z_0} u(z) \lim_{z\to z_0} v(z) \\ \lim_{z\to z_0} \frac{u(z)}{v(z)} &= \frac{\lim\limits_{z\to z_0} u(z)}{\lim\limits_{z\to z_0} v(z)} \quad \left(\lim_{z\to z_0} v(z) \neq 0\right) \end{aligned}\right\} \tag{4}$$

Their proof is completely analogous to the case of real functions of a real variable (see § 4.1, Theorem 6).

We say that *a function* f *is continuous at a point* z^0 if it is defined in a neighbourhood of z^0 including the point z^0 itself and if $\lim\limits_{z\to z_0} f(z) = f(z^0)$.

It follows of course from (4) that the sum, the difference, the product and the quotient of functions continuous at z^0 are continuous functions at z^0 (in the case of the quotient the usual stipulation that the divisor is different from zero should be made).

The derivative $f'(z_0)$ *of a function* f *at a point* z_0 is defined as

$$\lim_{\Delta z\to 0} \frac{f(z_0+\Delta z)-f(z_0)}{\Delta z} = f'(z_0) \tag{5}$$

The kth derivative of f *at a point* z is defined by induction:

$$f^{(k)}(z) = \left(f^{(k-1)}(z)\right)' \qquad (k = 2, 3, \ldots) \tag{6}$$

The relations below are proved like in the case of real functions:

$$\left.\begin{aligned} [u(z)\pm v(z)]' &= u'(z)\pm v'(z) \\ [u(z)\,v(z)]' &= u(z)\,v'(z)+u'(z)\,v(z) \\ \left(\frac{u(z)}{v(z)}\right)' &= \frac{v(z)\,u'(z)-u(z)\,v'(z)}{v(z)^2} \qquad (v(z) \neq 0) \end{aligned}\right\} \tag{7}$$

$$((z-a)^n)' = n(z-a)^{n-1} \qquad (n = 0, \pm 1, \pm 2, \ldots) \tag{8}$$

In this chapter we shall also deal with complex (complex-valued) functions of a real variable

$$f(x) = \varphi(x)+i\psi(x) \tag{9}$$

(where φ and ψ are real functions defined on a set of real numbers x). The function $e^{i\theta} = \cos\theta+i\sin\theta$ $(-\infty<\theta<\infty)$ is an example of such a function. The notions of the derivative and of the indefinite integral are introduced for such functions in a natural manner:

$$f'(x) = \varphi'(x)+i\psi'(x) \tag{10}$$

and

$$\int f(x)\,dx = \int \varphi(x)\,dx+i\int \psi(x)\,dx+C \tag{11}$$

(where C is a complex arbitrary constant).

From (11) follows the equality

$$\int (Af_1(x)+Bf_2(x))\,dx = A\int f_1(x)\,dx+B\int f_2(x)\,dx+C \tag{12}$$

where A and B are complex constants and f_1 and f_2 continuous complex functions.

Remark. Let a function $f(z)$ be defined in an open set G of points z of the complex plane. Suppose that G intersects the real axis x along an interval (a, b) (Fig. 8.3). Then f can also be regarded as a (generally speaking, complex) function $f(x)$ of the real variable $x\in(a, b)$. If f has the derivative at a point $x\in(a, b)$ in the sense of the definition stated for *complex functions of a complex argument*, that is if there is the limit $\lim\limits_{\Delta z\to 0}\dfrac{f(x+\Delta z)-f(x)}{\Delta z} = f'(x)$ when Δz tends to zero running through any complex values then, moreover, the function f has the derivative $f'(x)$ (equal to the former) at the point x in the sense of the definition stated *for functions of a real variable* which requires that the limit $\lim\limits_{\Delta x\to 0}\dfrac{f(x+\Delta x)-f(x)}{\Delta x} = f'(x)$ should exist as Δx tends to zero only running through any real values. The converse does not hold; this can be confirmed by the example of the function $|z|$: its derivative in the sense of a real variable exists at any point $x>0$ and is equal to 1 while its derivative at the same point in the sense of a complex variable does not exist because the ratio $\dfrac{|x+\Delta z|-|x|}{\Delta z}$ (in which $\Delta z = \varrho e^{i\theta}$) has no definite limit as $\varrho = |\Delta z|\to 0$:

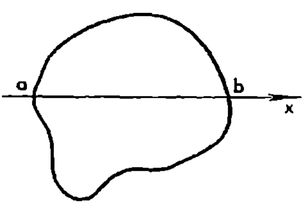

Fig. 8.3

$$\frac{|x+\Delta z|-|x|}{\Delta z} = \frac{\sqrt{(x+\varrho\cos\theta)^2+(\varrho\sin\theta)^2}-x}{\varrho e^{i\theta}} = \frac{\sqrt{x^2+2x\varrho\cos\theta+\varrho^2}-x}{\varrho e^{i\theta}} =$$

$$= \frac{(2x\varrho\cos\theta+\varrho^2)\dfrac{1}{2x}+o(\varrho)}{\varrho e^{i\theta}} = \frac{\cos\theta+o(1)}{e^{i\theta}} \to \frac{\cos\theta}{e^{i\theta}}, \qquad \varrho\to 0 \tag{13}$$

where the limit $\lim\limits_{\varrho \to 0} \frac{\cos\theta}{e^{i\theta}}$ has no definite value. The limit in (13) exists when $\Delta z \to 0$ so that the corresponding point in the complex plane moves toward the origin along a ray passing through the origin but, generally, the value of the limit is different for different rays. In the third equality (13) we have taken x outside the root and then used the equality $\sqrt{1+u} = 1+u/2+o(u)$, $u \to \to 0$.

§ 8.4. Polynomials

In § 5.9 we dealt with polynomials of the nth degree

$$Q_n(x) = a_0+a_1x+\ldots+a_nx^n \qquad (a_n \neq 0)$$

where the coefficients a_k and the variable x were considered real. In particular, we derived Taylor's formula

$$Q_n(x) = \sum_{k=0}^{n} \frac{Q_n^{(k)}(x_0)}{k!}(x-x_0)^k$$

in which the derivatives were interpreted in the sense of a real variable.

In the present section we shall consider more general polynomials of the nth degree

$$Q_n(z) = a_0+a_1z+\ldots+a_nz^n \qquad (a_n \neq 0) \tag{1}$$

where a_k are, generally, complex coefficients and $z = x+iy$ is a variable assuming any complex values.

If we substitute $(z-z_0)+z_0$ for z into the right-hand side of (1) and combine similar terms with equal powers of $(z-z_0)$ the polynomial $Q(z)$ is brought to the form of a sum involving powers of $(z-z_0)$:

$$Q(z) = b_0+b_1(z-z_0)+\ldots+b_n(z-z_0)^n \tag{2}$$

The derivative of the kth order of $Q(z)$ (understood in the sense of a complex variable, see the foregoing section) is equal to $Q^{(k)}(z) = k!\,b_k+(k+1)\ldots \ldots 2(z-z_0)+\ldots$. Therefore

$$b_k = \frac{Q^{(k)}(z_0)}{k!} \tag{3}$$

and

$$Q(z) = \sum_{0}^{n} \frac{Q^{(k)}(z_0)}{k!}(z-z_0)^k \tag{4}$$

We have thus derived Taylor's representation in powers of $(z-z_0)$ for the given complex polynomial $Q(z)$. It implies that $Q(z)$ has a unique expansion of type (2): if two polynomials are equal to each other identically (i.e. for all z) their coefficients in like powers of $(z-z_0)$ are also equal because they are defined by the same formulas (3). In particular, the coefficients of a polynomial of the nth degree identically equal to zero are all equal to zero.

If z_0 is a point such that

$$Q(z_0) = Q'(z_0) = \dots = Q^{(k-1)}(z_0) = 0, \qquad Q^{(k)}(z_0) \neq 0 \tag{5}$$

the polynomial $Q(z)$ can be represented in the form

$$Q(z) = (z-z_0)^k R(z) \qquad (R(z_0) \neq 0) \tag{6}$$

where R is a polynomial of degree $n-k$. Conversely, (6) implies (5). Indeed, by Taylor's formula (4), formula (6) follows from (5) and, on the other hand, if (6) holds then we can multiply all the terms of the expansion of $R(z)$ in powers of $(z-z_0)$ by $(z-z_0)^k$ and add them together, which results, by virtue of the uniqueness, in Taylor's expansion of Q in powers of $(z-z_0)$ satisfying conditions (5).

If (5) or, which is the same, (6) takes place we say that z_0 is a *root of multiplicity k* (or a *k-fold root*) *of the polynomial Q*. We can also say that z_0 is a k-fold root of Q if $Q(z)$ is divisible by $(z-z_0)^k$ and not divisible by $(z-z_0)^{k+1}$.

The well-known *fundamental theorem of algebra* asserts that *any polynomial Q of degree $n > 0$ has at least one complex root.*

As readily follows from this theorem, a polynomial of degree n has in fact exactly n roots counting their multiplicities. Indeed, let z_1 be a root of a polynomial Q of the nth degree. Let k_1 be its multiplicity. Then

$$Q(z) = (z-z_1)^{k_1} Q_1(z) \qquad (Q_1(z_1) \neq 0)$$

where Q_1 is a polynomial of degree $n-k_1$. If $n-k_1 > 0$ then, by virtue of the same theorem, Q_1 has a root z_2 of some multiplicity k_2, and therefore

$$Q_1(z) = (z-z_1)^{k_1}(z-z_2)^{k_2}Q_2(z) \qquad (Q_2(z_1),\ Q_2(z_2) \neq 0)$$

If the degree of Q_2 is positive the process can be continued. After a finite number of such steps we arrive at the conclusion that $Q(z)$ possesses (pairwise distinct) roots $z_1, \dots, z_m$ of multiplicities $k_1, \dots, k_m$, respectively, where $n = k_1 + \dots + k_m$, and can be factored as

$$Q(z) = a_n(z-z_1)^{k_1} \dots (z-z_m)^{k_m} \qquad (a_n \neq 0,\ n = k_1 + \dots + k_m) \tag{7}$$

There are no other roots of Q because, by virtue of (7), we obviously have $Q(z_0) \neq 0$ for any $z_0 \neq z_j$ $(j = 1, \dots, m)$. This shows that factorization of form (7) is unique for a given polynomial $Q(z)$.

Now we shall discuss an interesting relationship between the polynomial $Q(z)$ in question and its derivative $Q'(z)$. It turns out that *the greatest common divisor (factor) of $Q(z)$ and $Q'(z)$* (that is a polynomial of the highest possible degree by which both Q and Q' are divisible) *is equal, to within a constant nonzero factor, to the polynomial*

$$L(z) = (z-z_1)^{k_1-1} \dots (z-z_m)^{k_m-1} \tag{8}$$

for z_j is a k_j-fold root of the polynomial Q and hence there hold the conditions

$$Q(z_j) = Q'(z_j) = \dots = Q^{(k_j-1)}(z_j) = 0, \qquad Q^{(k_j)}(z_j) \neq 0$$

whence we arrive at the conditions

$$\Lambda(z_j) = \ldots = \Lambda^{(k_j-2)}(z_j) = 0, \qquad \Lambda^{(k_j-1)}(z_j) \neq 0$$

for $\Lambda(z) = Q'(z)$. They mean that $(z-z_j)^{k_j-1}$ is a root of multiplicity k_j-1 for the polynomial $Q'(z)$. Therefore $(z-z_j)^{k_j-1}$ is a common factor of $Q(z)$ and $Q'(z)$ while $Q'(z)$ is not divisible by $(z-z_j)^{k_j}$. Since this argument* can be carried out for any $j = 1,\ldots, m$ we see that polynomial L is the greatest common factor of Q and Q'. If $k_j = 1$ for some j then the corresponding factor in (8) is equal to 1.

It should be noted that the polynomial $L(z)$ can always be effectively found by means of the so-called *Euclidean algorithm* without computing the roots of L themselves.

We shall remind the reader of this method. Let there be given two polynomials M_0 and M_1 of degrees m and n respectively where $m \geqslant n$. Arranging them in descending powers of z and dividing M_0 by M_1 we obtain a quotient R_0 and a remainder M_2. The degree of the remainder is less than that of M_1. Next we similarly divide M_1 by M_2, the quotient being denoted R_1 and the remainder M_3, etc. Since at each of this stages the degree of the remainder is reduced at least by 1 the process ultimately stops since the remainder turns out to be identically zero. We thus obtain the chain of the relations

$$M_0 = M_1R_0+M_2, \qquad M_1 = M_2R_1+M_3, \ \ldots$$
$$M_{l-2} = M_{l-1}R_{l-2}+M_l, \qquad M_{l-1} = M_lR_{l-1}$$

where R_k and M_k are polynomials, the degrees of M_k decreasing with the increase of k. These equalities show that

$$\text{G.C.D } (M_0, M_1) = \text{G.C.D } (M_1, M_2) = \ \ldots = M_l$$

where M_l is G.C.D of M_0 and M_1 (G.C.D means the greatest common divisor).

The fundamental theorem of algebra only guarantees the existence of a root (which is complex in the general case) of a polynomial of the nth degree but does not provide any effective means for its computation. It should be mentioned that the proof of this theorem is carried out with the aid of mathematical analysis, not algebraically, and the only reason why this proof is not given in our course is that it does in fact have a closer relation to the theory of functions of a complex variable which we do not treat here.

There are formulas for solving general algebraic equations of the second, third and fourth degrees. As was proved by Abel**, there cannot exist such formulas for equations of degree $n > 4$. This should be understood in the sense that for $n > 4$ the roots of an equation $a_0x^n+\ldots+a_{n-1}x+a_n = 0$ $(a_0 \neq 0)$ cannot be expressed in terms of the coefficients a_k with the aid of

* In these considerations concerning the relationship between Q and Q' it is also allowable to suppose that the coefficients of Q and the variable x are real. This very case will be of use in § 8.7.

** N. H. Abel (1802-1829), a distinguished Norwegian mathematician.

functions of these coefficients reducing to a finite number of operations belonging to the following group: addition, subtraction, multiplication, division and extraction of a root.

A polynomial $Q(z)$ (see (1)) is called *real* if all its coefficients are real numbers. A real polynomial, when regarded for only real $z = x$, is a real function $Q(x)$ of the real variable x, that is a function assuming real values.

An important property of real polynomials is that for any complex z there holds the relation

$$\overline{Q(z)} = Q(\bar{z}) \tag{9}$$

It is derived on the basis of formulas (14) in § 8.2 (where it should be taken into account that, since a_k are real, we have $\bar{a}_k = a_k$) with the aid of the following calculations:

$$\overline{a(z)} = \overline{\sum_{k=0}^{n} a_k z^k} = \sum_{k=0}^{n} \overline{a_k z^k} = \sum_{k=0}^{n} \bar{a}_k \bar{z}^k = \sum_{0}^{n} a_k \bar{z}^k = Q(\bar{z})$$

Let us prove the following theorem.

Theorem. *If a real polynomial $Q(z)$ has a complex root $z_0 = \alpha + i\beta$ of multiplicity k then it also has the conjugate complex root $\bar{z}_0 = \alpha - i\beta$ of the same multiplicity.*

Proof. By hypothesis, $Q(z_0) = Q'(z_0) = \ldots = Q^{(k-1)}(z_0) = 0$, $Q^{(k)}(z_0) \neq$ $\neq 0$. It is easy to see that if $Q(z)$ is a real polynomial then its derivative of the lth order $Q^{(l)}(z)$ is also a real polynomial. Therefore, by (9), we have $\overline{Q^{(l)}(z_0)} = Q^{(l)}(\bar{z}_0)$ for any $l = 0, 1, 2, \ldots$, and consequently

$$Q(\bar{z}_0) = Q'(\bar{z}_0) = \ldots = Q^{(k-1)}(\bar{z}_0) = 0, \qquad Q^{(k)}(\bar{z}_0) \neq 0$$

Using this theorem and taking into account factorization (7) for a polynomial $Q(z)$ of the nth degree into linear factors we conclude that any real polynomial of degree n ($a_n \neq 0$) can be factored as

$$Q(z) = a_n(z-a_1)^{l_1} \ldots (z-a_r)^{l_r}(z^2+p_1z+q_1)^{m_1} \ldots (z^2+p_sz+q_s)^{m_s} \tag{10}$$

where $a_1, \ldots, a_r, p_1, \ldots, p_s, q_1, \ldots, q_s$ are real numbers, every polynomial $z^2+p_jz+q_j$ has a pair of (mutually) conjugate complex roots and $n = l_1+$ $+\ldots+l_r+2(m_1+\ldots+m_s)$.

Indeed, let us consider factorization (7). If among the roots z_k entering into it there are real numbers we renumber them denoting $a_1, \ldots, a_r$. The corresponding powers in which the binomials $(z-z_k)$ enter into (7) will be denoted as $l_1, \ldots, l_r$.

If factorization (7) involves a factor $(z-z_j)^{k_\mu}$ with a complex root z_j then, by the above theorem, it necessarily involves the factor of the form $(z-\bar{z}_j)_{k_\nu}$ as well where $k_\mu = k_\nu = k$. On putting $z_j = \alpha+\beta i$, ($\bar{z}_j = \alpha-\beta i$) we obtain

$$(z-z_j)^k(z-\bar{z}_j)^k = (z-\alpha-\beta i)^k(z-\alpha+\beta i)^k = [(z-\alpha)^2+\beta^2]^k =$$
$$= (z^2+pz+q)^k, \qquad p = -2\alpha,\ q = \alpha^2+\beta^2 \tag{11}$$

where p and q are real numbers.

Now to complete the proof it only remains to renumber the factors corresponding to different pairs of conjugate complex roots of Q and to substitute them for the corresponding factors in (7), which results in (10).

§ 8.5. Decomposition of a Rational Fraction into Partial Fractions

In this section we shall consider an arbitrary real rational function

$$f(x) = \frac{P(x)}{Q(x)} \tag{1}$$

which is supposed to be a proper fraction. This means that P and Q are real polynomials and that the degree of P is less than that of Q. Let us denote by m the degree of P and by n the degree of Q; thus, $m < n$. The variable x will be supposed real so that $f(x)$ is a real function. The degree of a polynomial $M(x)$ will be denoted s_M (so we have $s_P = m$ and $s_Q = n$).

Lemma 1. *Let a be a k-fold real root of the denominator $Q(x)$ of fraction* (1):

$$Q(x) = (x-a)^k N(x) \qquad (N(a) \neq 0) \tag{2}$$

Then fraction (1) *can be represented in a unique way as*

$$\frac{P(x)}{Q(x)} = \frac{A}{(x-a)^k} + \frac{M(x)}{(x-a)^{k-1}N(x)} \tag{3}$$

where A is a constant and the second term in (3) *is a proper rational fraction. The number A is real and $M(x)$ is a real polynomial.*

The uniqueness of decomposition (3) is understood in the sense that the numbers A and the polynomial $M(x)$ (i.e. its coefficients) are determined uniquely.

Proof. Suppose that equality (3) holds. On reducing the fractions on the right-hand side of (3) to a common denominator and equating the numerators on the right-hand and left-hand sides we obtain the identity

$$P(x) \equiv AN(x) + (x-a)M(x) \tag{4}$$

which holds for any x. Putting $x = a$ in it and taking into account that $N(a) \neq 0$ we receive

$$A = \frac{P(a)}{N(a)} \tag{5}$$

where the number A is real since P and N are real polynomials and a is a real number. The substitution of the value of A thus found into (4) leads to the (uniquely determined) expression

$$M(x) = \frac{P(x) - AN(x)}{x-a} \tag{6}$$

Since the numerator in (6) is a polynomial and A is chosen so that the polynomial turns into zero for $x = a$ this polynomial is divisible by $x-a$ and therefore $M(x)$ is a (real) polynomial. By the hypothesis, s_p, $s_N \leqslant n-1$.

Therefore, by virtue of (6), $s_M \leq n-2$, i.e. s_M is less than the degree $(n-1)$ of the denominator of the second fraction on the right-hand side of (3) and hence this fraction is proper.

Remark. The lemma continues to hold for an arbitrary (not necessarily real) proper fraction (1) and a complex a with the exception of its last assertion since, generally, in this case A is a complex number and $M(x)$ is not necessarily real either.

Lemma 2. *Let $Q(x)$ be factorizable in the form*

$$Q(x) = (x^2+px+q)^k N(x) \tag{7}$$

where p and q are real numbers, $q-p^2/4 > 0$, k is a natural number and $N(x)$ is a (real) polynomial with roots distinct from those of the trinomial x^2+px+q.

Then fraction (1) *is uniquely representable in the form*

$$\frac{P(x)}{Q(x)} = \frac{Ax+B}{(x^2+px+q)^k} + \frac{M(x)}{(x^2+px+q)^{k-1}N(x)} \tag{8}$$

where A and B are constants and the second fraction on the right-hand side of (8) *is proper. The numbers A and B and the polynomial $M(x)$ are real.*

Proof. Let us denote as $a = \alpha+i\beta$ and $\bar{a} = \alpha-i\beta$ the roots of the trinomial $x^2+px+q\,(\beta \neq 0)$. As follows from the conditions of the lemma, these roots are also k-fold roots of the real polynomial $Q(x)$. Suppose that representation (8) holds. On reducing the fractions on the right-hand side of (8) to a common denominator, clearing of fractions and equating the numerators on the right-hand and left-hand sides we arrive at the identity

$$P(x) \equiv (Ax+B)N(x)+(x^2+px+q)\,M(x) \tag{9}$$

Substituting the numbers a and $\bar{a}$ into it we obtain

$$P(a) = (Aa+B)N(a) \quad \text{and} \quad P(\bar{a}) = (A\bar{a}+B)N(\bar{a})$$

whence

$$Aa+B = \frac{P(a)}{N(a)} = \Lambda \quad \text{and} \quad A\bar{a}+B = \frac{P(\bar{a})}{N(\bar{a})} = \left(\overline{\frac{P(a)}{N(a)}}\right) = \bar{\Lambda} \tag{10}$$

(by the hypothesis, $N(a) \neq 0$ and $N(\bar{a}) \neq 0$). Equalities (10) is a system of equations with respect to A and B which should be solved in A and B. Its determinant is

$$\begin{vmatrix} a & 1 \\ \bar{a} & 1 \end{vmatrix} = a-\bar{a} = 2\beta i \neq 0$$

Hence this system is uniquely resolvable and its solution is real numbers A and B. The last assertion can be proved without computing the solution A and B. To this end we pass to the conjugate complex quantities in equations (10):

$$\bar{A}\bar{a}+\bar{B} = \bar{\Lambda} \quad \text{and} \quad \bar{A}a+\bar{B} = \Lambda \tag{10$'$}$$

The coefficients and the right members of systems (10) and (10′) completely coincide and therefore their solutions coincide as well: $A = \bar{A}$ and $B = \bar{B}$. Therefore the numbers A and B are real.

Now we substitute the numbers A and B found into (9) and receive

$$M(x) = \frac{P(x)-(Ax+B)N(x)}{x^2+px+q}$$

Since the roots of the trinomial x^2+px+q turn the numerator of the obtained fraction into zero (because of the choice of A and B) it is divisible by the denominator and hence $M(x)$ is an (obviously real) polynomial. It can also be easily shown that the second fraction on the right-hand side of (8) is proper. The lemma has been proved.

Lemmas 1 and 2 make it possible to *decompose the given real proper rational fraction into the so-called partial fractions*. Since $Q(x)$ is a real polynomial of the nth degree it can be factored as was indicated in § 8.4 (10). Therefore, using this factorization and applying Lemmas 1 and 2 we consecutively find the decomposition (see the explanations below)

$$\frac{P(x)}{Q(x)} = \frac{A_{l_1}^{(1)}}{(x-a_1)^{l_1}} + \frac{M_1(x)}{(x-a_1)^{l_1-1}N_1(x)} = \tag{11}$$

$$= \frac{A_{l_1}^{(1)}}{(x-a_1)^{l_1}} + \frac{A_{l_1-1}^{1}}{(x-a_1)^{l_1-1}} + \frac{M_2(x)}{(x-a_1)^{l_1-2}N_1(x)} = \ldots \tag{12}$$

$$\ldots = \frac{A_{l_1}^{(1)}}{(x-a_1)^{l_1}} + \ldots + \frac{A_1^{(1)}}{(x-a_1)} + \frac{M_{l_1}(x)}{(x-a_2)^{l_2}N_2(x)} = \ldots \tag{13}$$

$$\ldots = \frac{A_{l_1}^{(1)}}{(x-a_1)^{l_1}} + \ldots + \frac{A_1^{(1)}}{x-a_1} + \ldots + \frac{A_{l_r}^{(r)}}{(x-a_r)^{l_r}} + \ldots$$

$$\ldots + \frac{A_1^{(r)}}{x-a_r} + \frac{B_{m_1}^{(1)}x+C_{m_1}^{(1)}}{(x^2+p_1x+q_1)^{m_1}} + \ldots + \frac{B_1^{(1)}x+C_1^{(1)}}{x^2+p_1x+q} + \ldots$$

$$\ldots + \frac{B_{m_s}^{(s)}x+C_{m_s}^{(s)}}{(x^2+p_sx+q_s)^{m_s}} + \ldots + \frac{B_1^{(s)}x+C_1^{(s)}}{x^2+p_sx+q_s} \tag{14}$$

where the constants A, B and C (supplied with the corresponding indices) are *real* and are determined *uniquely*.

Relation (11) is based on Lemma 1; the polynomial $M_1(x)$ entering into it is real and can be found from the equality

$$Q(x) = (x-a_1)^{l_1} M_1(x) \qquad (M_1(a_1) \neq 0) \tag{15}$$

The passage from (11) to (12) is also based on the application of Lemma 1, which is legitimate since the second fraction on the right-hand side of (11) is real and proper and $M_1(a_1) \neq 0$. The dots after (12) symbolize the continuation of the process of finding the partial fractions corresponding to the real root a_1, this process being completed in (13). The dots after (13) symbolize the application of the same procedure for the other real roots. After (13) we consecutively write the partial fractions corresponding to the complex roots of Q using Lemma 2.

This procedure has been described for the general case; in particular cases some of its steps may not be present. For instance, if Q has no real roots the procedure begins with the application of Lemma 2, and Lemma 1 is not used.

It should be noted that the uniqueness of the numbers A, B and C (with various indices) in decomposition (14) has not yet been completely proved because their determination is connected with a definite process and, generally speaking, this does not exclude the possibility that some other process may lead to other values of A, B and C. Below we present a method for determining A, B and C based on the comparison of the coefficients, and in the justification of this method it will be seen that these numbers do in fact form a uniquely specified system.

Using Lemmas 1 and 2 we have already proved that for a given polynomial $Q(x)$ and any polynomial $P(x)$ with $s_P < s_Q$ there is a system of numbers $A, B, C, \dots$ for which identity (14) holds (for all real x distinct from the roots of Q). Let us reduce the fractions in (14) to a common denominator, combine similar terms in the numerator, clear of fractions and then equate coefficients on the right-hand side (they are linear combinations of A, B, $C, \dots$) to the corresponding coefficients of the polynomial $P(x)$ in like powers of x. This results in a system of n linear algebraic equations in the unknowns $A, B, C, \dots$ The number of the unknowns and the number of the equations coincide here. As was shown, this system is solvable for any polynomial $P(x)$, that is for any right-hand terms of the system in question, this was in fact proved with the aid of Lemmas 1 and 2. It follows that the determinant of the system is different from zero and consequently the system of the coefficients $A, B, C, \dots$ is in fact determined uniquely.

Example. What has been said shows that there holds the equality

$$\frac{ax^3+bx^2+cx+d}{(x^2+x+1)(x-2)^2} = \frac{Ax+B}{x^2+x+1}+\frac{C}{(x-2)^2}+\frac{D}{x-2}$$

In other words, for any polynomial ax^3+bx^2+cx+d of the third degree there are constants A, B, C and D such that the above equality holds for all $x \neq 2$. To find these constants we reduce the fractions on the right-hand side of the given equality to the least common denominator. The numerators on both sides of the resultant relation must be equal:

$$ax^3+bx^2+cx+d = (Ax+B)(x-2)^2+C(x^2+x+1)+D(x^2+x+1)(x-2)$$

for $x \neq 2$. By the continuity of the functions entering into this equality we can assert that it holds for $x = 2$ as well. On equating the coefficients in like powers of x we arrive at the linear algebraic system with the four unknowns A, B, C and D:

$$\left.\begin{aligned} a &= A+D, & c &= 4A-4B+C-D, \\ b &= B-4A+C-D, & d &= 4B+C-2D \end{aligned}\right\}$$

As is already known, this system is solvable in A, B, C and D for any a, b, c and d. Therefore, as is proved in algebra, the system of numbers A, B,

C and D serving as a solution to the linear equations in question is determined uniquely for given a, b, c and d.

We shall also discuss another type of the decomposition of P/Q in partial fractions which is based solely on the application of Lemma 1.

Let $\lambda_1, \ldots, \lambda_m$ be all the pairwise distinct (real and complex) roots of Q with multiplicities $k_1, \ldots, k_m$ respectively. Let us consecutively apply Lemma 1 alone and the remark to the lemma in order to obtain the partial fractions of the form $A/(x-\lambda)^j$ corresponding to these roots. This results in the decomposition

$$\frac{P(x)}{Q(x)} = \frac{A_1^{(1)}}{(x-\lambda_1)} + \ldots + \frac{A_{k_1}^{(1)}}{(x-\lambda_1)^{k_1}} + \ldots + \frac{A_1^{(m)}}{x-\lambda_m} + \ldots + \frac{A_{k_m}^{(m)}}{(x-\lambda_m)^{k_m}} \quad (16)$$

where the numbers A are no longer necessarily real and may assume complex values.

If, for instance, we want to find the coefficients corresponding to the root λ_1 we multiply both sides of (16) by $(x-\lambda_1)^{k_1}$, which gives

$$(x-\lambda_1)^{k_1}\frac{P(x)}{Q(x)} = A_{k_1}^{(1)} + A_{k_1-1}^{(1)}(x-\lambda_1) + \ldots$$
$$\ldots + A_1^{(1)}(x-\lambda_1)^{k_1-1} + (x-\lambda_1)^{k_1}R(x) \quad (17)$$

where $R(x)$ is a function having at the point λ_1 derivatives of any order. It is therefore evident that

$$A_{k_1-\mu}^{(1)} = \lim_{x\to\lambda_1} \frac{1}{\mu!}\frac{d\mu}{dx^\mu}\left\{(x-\lambda_1)^{k_1}\frac{P(x)}{Q(x)}\right\} \quad (18)$$

which shows, in particular, that the numbers A are determined uniquely. Formula (18) may be of use for practical computations.

It should be noted that decomposition (16) obviously holds (see the remark to Lemma 1) for any proper fraction P/Q, not necessarily real. But in the case of a real fraction there is a characteristic feature which does not apply to complex fractions. Namely, in decomposition (16) of a real fraction P/Q every term of the form $A/(x-\lambda)^j$ has a real coefficient A for any real root λ while for any complex root λ there necessarily appear both terms $A/(x-\lambda)^j$ and $\bar{A}/(x-\bar{\lambda})^j$. It is this fact which we have mentioned as a feature of decomposition (16) for *real* fractions.

Indeed, we can write

$$\frac{P(x)}{Q(x)} = \frac{A}{(x-\lambda)^k} + \frac{B}{(x-\bar{\lambda})^k} + \ldots \quad (19)$$

Passing to the complex conjugates in (19) and taking into account that the fraction P/Q and the variable x are real we obtain

$$\frac{P(x)}{Q(x)} = \frac{\bar{A}}{(x-\bar{\lambda})^k} + \frac{\bar{B}}{(x-\lambda)^k} + \ldots \quad (20)$$

By the uniqueness of the decomposition, it follows from (19) and (20) that

$\bar{A} = B$. Note that the sum $\frac{A}{(x-\lambda)^k} + \frac{\bar{A}}{(x-\bar{\lambda})^k}$ is a real function of x (the variable x is real) since it involves the pair of mutually conjugate complex summands.

§ 8.6. Integrating Rational Fractions

Let us consider the indefinite integral

$$\int \frac{P(x)}{Q(x)} dx \tag{1}$$

of a real rational fraction P/Q on an interval containing no real roots of $Q(x)$. The function $P(x)/Q(x)$ is continuous on such an interval and its antiderivative makes sense. If the degree s_P of the polynomial P is not less than the degree s_Q of Q ($s_P \geqslant s_Q$) we begin with the division of P by Q following the well-known algebraic rules:

$$\frac{P}{Q} = R + \frac{P_1}{Q}$$

The integration of the polynomial R is carried out quite simply; the expression P_1/Q is a proper rational fraction, and hence the whole question reduces to the integration of the proper fraction P_1/Q which we shall again denote as P/Q.

Let us suppose that Q is factorized as written in § 8.4 (formula (10)). Then P/Q can be written in the form of the decomposition into partial fractions in accordance with formula (14) of § 8.5. As is already known, each partial fraction can readily be integrated in elementary functions.

We have shown that, at least in principle, the integral of any rational fraction can be expressed in terms of elementary functions. The practical integration can be completed if all the roots of Q and their multiplicities are determined. As was mentioned in § 8.4, this is by far not always possible in the analytical sense. In this connection it becomes clear that various simplifications of integral (1) are extremely important for practice. One of such simplification techniques suggested by Ostrogradsky* will be discussed in § 8.7.

§ 8.7. Ostrogradsky's Method of Integrating Rational Fractions

Let it be necessary to find the indefinite integral

$$\int \frac{P(x)}{Q(x)} dx \tag{1}$$

where P/Q is a real proper rational function (i.e. a real proper rational fraction) and the polynomial Q is of degree n. Before proceeding to Ostrogradsky's method we shall discuss some purely theoretical facts needed for our further aims.

---

* Academician M. V. Ostrogradsky (1801-1862), a prominent Russian mathematician.

Let $\lambda_1, \ldots, \lambda_m$ be the pairwise distinct (real and complex) roots of Q of multiplicities $k_1, \ldots, k_m$ respectively. Let us decompose P/Q following the scheme of § 8.5 (16). This decomposition can be written as

$$\frac{P(x)}{Q(x)} = \sum \frac{A}{(x-\lambda)^k} \tag{2}$$

where A, λ and k must of course be supplied with the corresponding indices.

We can also rewrite (2) in the form

$$\frac{P}{Q} = \sum \frac{A}{x-\lambda} + \sum_{k>1} \frac{A}{(x-\lambda)^k} \tag{3}$$

where the first sum involves all the partial fractions with the first powers of $(x-\lambda)$ and the second sum contains all the other partial fractions. Each of these sums is a real function of x although, in general, some of their terms may be complex (see the end of § 8.5).

On reducing the fraction in the first sum to a common denominator we obtain

$$\frac{N(x)}{K(x)} = \sum \frac{A}{x-\lambda} \frac{b_0+b_1x+\ldots+b_{m-1}x^{m-1}}{(x-\lambda_1)\ldots(x-\lambda_m)} \tag{4}$$

(N/K is a real proper fraction). The integration of the second sum is readily performed, the integrals being expressed in terms of rational functions:

$$\int \sum_{k>1} \frac{A}{(x-\lambda)^k} dx = \sum_{k>1} \frac{A}{(1-k)(x-\lambda)^{k-1}} + C =$$

$$= \frac{a_0+a_1x+\ldots+a_{n-m-1}x^{n-m-1}}{(x-\lambda_1)^{k_1-1}\ldots(x-\lambda_m)^{k_m-1}} + C = \frac{M(x)}{L(x)} + C \tag{5}$$

We have performed here the termwise integration of functions which, in the general case, are complex-valued (see § 8.3 (12)). But the ultimate expression is a real function since each summand it contains is either real or enters into the sum together with its complex conjugate. The expression M/L is obviously a proper fraction; the arbitrary constant C is taken real here.

What has been said shows that the integration of equality (3) yields

$$\int \frac{P(x)}{Q(x)} dx = \frac{M(x)}{L(x)} + \int \frac{N(x)\,dx}{K(x)} \tag{6}$$

where

$$K(x) = (x-\lambda_1)\ldots(x-\lambda_m) \tag{7}$$

and

$$L(x) = (x-\lambda_1)^{k_1-1}\ldots(x-\lambda_m)^{k_m-1} \tag{8}$$

(the arbitrary constant on the right-hand side of (6) is included in the indefinite integral entering into it).

We have thus proved that for a given Q and any polynomial $P(x) = p_0+$ $+p_1x+\ldots+p_{n-1}x^{n-1}$ there are $a_0, a_1, \ldots, a_{n-m-1}, b_0, \ldots, b_{m-1}$ (their total number is n) such that for the polynomials $M(x)$ and $N(x)$ with these constants as coefficients there holds equality (6) in which M/L and N/K are proper fractions.

Let us show that M and N can be effectively found without computing the roots of Q. To this end we differentiate (6):

$$\frac{P}{Q} = \left(\frac{M}{L}\right)' + \frac{N}{K} = \frac{M'}{L} - \frac{1}{L}\,\frac{ML'}{L} + \frac{N}{K}$$

From this relation, on reducing the fractions on the right-hand side to the common denominator $Q = LK$ and equating the numerators of both sides, we derive

$$P = KM' - \frac{MKL'}{L} + LN \tag{9}$$

Since K and L are expressed by formulas (7) and (8), and L' involves the factor $(x-\lambda_1)^{k_1-2} \dots (x-\lambda_m)^{k_m-2}$ $(k_j > 1)$, the product KL' is divisible by L and hence KL'/L is a polynomial.

Now we add together the polynomials on the right-hand side of (9), combine similar terms with like powers of x and equate the coefficients of the resultant expression to the corresponding coefficients p_k of the polynomial P. This leads to a linear system of algebraic equations of the form

$$\alpha_0^k a_0 + \dots + \alpha_{n-m-1}^k a_{n-m-1} + \beta_0^k b_0 + \dots + \beta_{m-1}^k b_{m-1}^k = p_k \qquad (k = 0, \dots, n) \tag{10}$$

with the unknowns a_i and b_j. This system is in fact linear because the right-hand side of (9) is a linear combination of the polynomials M', L' and N which alone contain a_i and b_j.

The coefficients α_k and β_s of system (10) are known; they are determined by the given polynomial Q and are real. The determinant of system (10) is sure to be different from zero. This is implied by the fact that to the right-hand side of (10) there corresponds a definite polynomial $P(x)$ to which, in their turn, there correspond the numbers A entering into decomposition (2) from which, with the aid of (4) and (5), the numbers a_i and b_j are found, the latter, as was proved, satisfying equations (10). In other words, system (10) is solvable for any right-hand members whence follows that its determinant is nonzero. Since the numbers α_i, β_j and p_k are real the polynomials M and N are also real.

It should be noted that the factorization of Q in linear factors and the introduction of the numbers A were only involved in the considerations to justify the method. The polynomial L can be obtained practically with the aid of the Euclidean algorithm and the polynomial K can be found by dividing P by L. Therefore the determination of the roots of Q is not in fact needed. To obtain system (10) we simply write the sought-for polynomials M and N in the form

$$M(x) = a_0 + a_1 x + \dots + a_{n-m-1} x^{n-m-1}$$

and

$$N(x) = b_0+b_1x+\ldots+b_{m-1}x^{m-1}$$

(where the coefficients a_j and b_j are not yet known; this is the well-known *method of undetermined coefficients*), substitute these expressions into (6), differentiate the resultant relation and, finally, equate the coefficients of the numerators in like powers of x.

As to the expression N/K, its integration can only be completed with the aid of the techniques presented in this chapter when the roots of K (they are all simple, that is of multiplicity 1) are known. In this connection it should be mentioned that there are approximate methods of integration of functions which are applicable irrespective of whether or not the integral of the given integrand function is expressible in terms of elementary functions. If it is required to find an approximation to the integral of a rational fraction P/Q it is advisable first to apply Ostrogradsky's method in order to find its rational part and only then perform an approximate integration of the remaining part N/K.

We also note that the decomposition of N/K into partial fractions (see (4)) may involve some terms $A/(x-\lambda)$ with real λ. The integral of such a term has the form

$$\int \frac{A}{x-\lambda}\,dx = A\ln|x-\lambda|+C$$

In case $\lambda=\alpha+i\beta$ is a complex root, sum (4) involves both the term $A/(x-$ $-\lambda)$ and its complex conjugate $\bar{A}/(x-\bar{\lambda})$. The sum of the two terms is

$$\frac{A}{x-\lambda}+\frac{\bar{A}}{x-\bar{\lambda}} = \frac{Bx+D}{(x-\alpha)^2+\beta^2}$$

where B and D are some real numbers. The integration of such a fraction leads, as we know, to the logarithmic function and to the arc tangent. Hence, the integration of a rational fraction N/K leads, in the general case, to transcendental (not rational) functions (i.e. to the logarithm and the arc tangent).

Example. Let us find the integral $\int \frac{dx}{(x^3-1)^2}$. The denominator of the integrand in this integral has multiple roots and therefore it is advisable to use Ostrogradsky's method. We represent the integral in the form

$$\int \frac{dx}{(x^3-1)^2} = \frac{a_0+a_1x+a_2x^2}{x^3-1}+\int \frac{b_0+b_1x+b_2x^2}{x^3-1}\,dx$$

where a_i and b_j are the constants to be determined. On differentiating this equality and reducing the fraction to the common denominator $(x^3-1)^2$ we obtain

$$\frac{1}{(x^3-1)^2} = \left(\frac{a_0+a_1x+a_2x^2}{x^3-1}\right)'+\frac{b_0+b_1x+b_2x^2}{x^3-1}$$

Equating the numerators we find

$$1 = (x^3-1)(2a_2x+a_1)-3x^2(a_0+a_1x+a_2x^2)+(x^3-1)(b_2x^2+b_1x+b_0)$$

whence, comparing the coefficients in like powers of x we arrive at the linear system of equations

$$b_2 = 0,\ b_1-a_2 = 0,\ b_0-2a_1 = 0,\ b_2+3a_0 = 0,\ b_1+2a_2 = 0,\ b_0+a_1 = -1$$

The solution of the system which is easily performed results in the final expression

$$\int \frac{dx}{(x^3-1)^2} = -\frac{1}{3}\,\frac{x}{x^3-1}-\frac{2}{3}\int \frac{dx}{x^3-1}$$

Now let us find the same integral by using the decomposition of the integrand fraction in partial fractions:

$$\frac{1}{x^3-1} = \frac{A}{x-1}+\frac{Bx+C}{x^2+x+1} \tag{11}$$

On clearing of fraction we get the identity

$$1 = A(x^2+x+1)+(Bx+C)(x-1).$$

holding for any x. The substitution of $x = 1$ in it gives $A = 1/3$. Comparing the coefficients in the highest power of x and the terms not involving x we also find $0 = A+B$ and $1 = A-C$, whence $B = -1/3$ and $C = -2/3$. It now only remains to substitute the values of A, B and C thus found into (11) and to perform the integration, which, of course, leads to the same final result as above.

§ 8.8. Integrating Irrational Expressions

We shall use the symbol $R(x, u, v, \ldots, w)$ for a rational function of $x, u, v, \ldots, w$ (we take a finite number of letters denoting the corresponding variables). Such a function is obtained by applying to $x, u, v, \ldots, w$ a finite number of arithmetical operations (addition, subtraction, multiplication and division).

The integral

$$\int R\left(x, \left(\frac{ax+b}{cx+d}\right)^{\lambda}, \ldots, \left(\frac{ax+b}{cx+d}\right)^{\mu}\right) dx \tag{1}$$

where $\lambda, \ldots, \mu$ are *rational* numbers with the lowest common denominator m, is reduced with the aid of the substitution

$$t^m = \frac{ax+b}{cx+d} \qquad (ad-bc \neq 0) \tag{2}$$

to an integral of a rational fraction (as we say, it is *rationalized*). Indeed, it is obvious that x is expressible as a rational function $\mu(t)$ of t and, together with x, the function $\nu(t)$ entering into the differential $dx = \nu(t)dt$ is also rational. Therefore, denoting by $p, \ldots, q$, respectively, the numerators (which are integers) of the fractions $\lambda, \ldots, \mu$ (which are reduced to their common denominator m) we see that substitution (2) transforms the integral to an integral of the form

$$\int R(\mu(t), t^p, \ldots, t^q)\, \nu(t)\, dt = \int R_1(t)\, dt$$

where $R_1(t)$ is a rational function of t.

Examples

1. $\displaystyle\int \frac{x^3\, dx}{\sqrt{x-1}} = 2\int (1+t^2)^3\, dt \qquad \left(t = \sqrt{x-1},\ x = 1+t^2,\ dx = 2t\, dt\right)$

2. $\displaystyle\int \frac{dx}{x^{1/2}+x^{1/3}} = \int \frac{6t^5}{t^3+t^2}\, dt = 6\int \frac{t^3}{1+t}\, dt = 6\int (1-t+t^2)\, dt - 6\int \frac{dt}{1+t}$

$$(x = t^6,\ dx = 6t^5\, dt)$$

§ 8.9. Euler's Substitutions

Euler's substitutions are used to rationalize an integral of the form

$$\int R(x, y)\, dx \qquad \left(y = \sqrt{a+bx+cx^2}\right) \tag{1}$$

where $R(x, y)$ is a rational function of x and y.

The first substitution is used in the case when the roots α and β ($\alpha \neq \beta$) of the trinomial $a+bx+cx^2$ are real. It has the form

$$t = \frac{\sqrt{a+bx+cx^2}}{x-\alpha} = \frac{\sqrt{c(x-\alpha)(x-\beta)}}{x-\alpha} \tag{2}$$

which is equivalent to $t^2 = \dfrac{c(x-\beta)}{x-\alpha}$.

Here both the function $x = \varphi(t)$ and its derivative φ' are rational functions and hence

$$\int R(x, y)\, dx = \int R(\varphi(t), t[\varphi(t)-\alpha])\, \varphi'(t)\, dt = \int R_1(t)\, dt$$

where R_1 is a rational function. To return from t to x we use formula (2).

The second substitution. Let the roots of the trinomial $a+bx+cx^2$ be complex. In this case we must suppose that $c > 0$ since, if otherwise, the trinomial is negative for all x.

Let us put

$$y = t \mp x\sqrt{c} \tag{3}$$

On squaring this equality and replacing y^2 by its expression we receive

$$a+bx = t^2 \mp 2tx\sqrt{c}$$

whence

$$x = \frac{t^2-a}{b \pm 2t\sqrt{c}} = \varphi(t), \quad dx = \varphi'(t)\, dt \tag{4}$$

Therefore

$$\int R(x, y)\, dx = \int R\big(\varphi(t),\ t\mp\varphi(t)\sqrt{c}\big)\,\varphi'(t)\, dt = \int R_1(t)\, dt$$

where $R_1(t)$ is a rational function of t. To return from t to x we use the inverse transformation

$$t = \sqrt{a+bx+cx^2}\pm x\sqrt{c} \tag{5}$$

Remark. To the first substitution $y = t-x\sqrt{c}$ (see (3)) there corresponds the function (see § 5.8, Example 3)

$$x = \frac{t^2-a}{b+2t\sqrt{c}}$$

Its graph splits into two continuous branches corresponding to the variation of t over the intervals $(-\infty,\ -b/2\sqrt{c})$ and $(-b/2\sqrt{c},\ \infty)$ respectively. It is strictly increasing from $-\infty$ to $+\infty$ on either interval. To these branches there correspond the two one-valued inverse functions

$$t = x\sqrt{c}\pm\sqrt{a+bx+cx^2}$$

The first of them (involving the sign "+") has been used in substitution (5) rationalizing the given integral, the second one only differing from the former in sign.

We also note that the substitution $x = 1/z$ transforms the square root of the trinomial to the expression $\sqrt{az^2+bz+c}/(\pm z)$ (in case $a \neq 0$) and hence brings integral (1) to an integral of the same type. Here the sign "+" or "−" is taken depending on whether the interval of variation of z is a part of the ray $z > 0$ or $z < 0$. Generally speaking, this substitution performs a discontinuous transformation: under the transformation a neighbourhood of 0 goes into a neighbourhood of the point at infinity and vice versa. Therefore this substitution transforms a function continuous on the x-axis into a function of z which, in the general case, has a nonremovable discontinuity at $z = 0$. If the integrand function $R(x, y)$ in the original integral has a discontinuity at the point $x = 0$ the substitution $x = 1/z$ does not aggravate, in this sense, the situation.

Examples

1. Consider the integral $\displaystyle\int \frac{dx}{\sqrt{a+bx+x^2}}$.

Suppose that the trinomial in the denominator has complex roots. Then the computations are carried out as follows:

$$\int \frac{dx}{\sqrt{a+bx+x^2}} = \int \frac{dx}{t-x} = \int \frac{dt}{(b/2)+t} = \ln\left|\frac{b}{2}+x+\sqrt{a+bx+x^2}\right| +C$$

Here we have used the second substitution:

$$\sqrt{a+bx+x^2} = t-x,\ a+bx = t^2-2tx$$

$$b\,dx = 2t\,dt-2t\,dx-2x\,dt,\ \frac{dx}{t-x} = \frac{2dt}{b+2t}$$

In particular,

$$\int \frac{dx}{\sqrt{x^2+a}} = \ln\left|x+\sqrt{x^2+a}\right|+C$$

2. $$\int \frac{dx}{\sqrt{a+bx-x^2}} = \int \frac{dx}{\sqrt{\left(a+\frac{b^2}{4}\right)-\left(x-\frac{b}{2}\right)^2}} = \int \frac{\alpha\,dt}{\sqrt{\alpha^2-(\alpha t)^2}} = \int \frac{dt}{\sqrt{1-t^2}} =$$

$$= \arcsin \frac{2x-b}{\sqrt{b^2+4a}}+C \quad \left(a+\frac{b^2}{4} = \alpha^2 > 0, \quad x-\frac{b}{2} = \alpha t\right)$$

In the integrals below it is assumed that the roots of the trinomial in the denominator are complex (the upper sign corresponds to positive values of x and z)

3. $$\int \frac{dx}{x\sqrt{ax^2+bx+1}} = \mp\int \frac{dz}{\sqrt{a+bz+z^2}} =$$

$$= \mp\ln\left|\frac{b}{2}+z+\sqrt{a+bz+z^2}\right|+C = \mp\ln\left|\frac{b}{2}+\frac{1\pm\sqrt{ax^2+bx+1}}{x}\right|+C$$

4. $$\int \frac{dx}{x\sqrt{ax^2+bx-1}} = \mp\int \frac{dz}{\sqrt{a+bz-z^2}} = \mp\arcsin \frac{2z-b}{\sqrt{b^2+4a}}+C =$$

$$= \pm\arcsin \frac{bx-2}{x\sqrt{b^2+4a}}+C$$

In particular,

$$\int \frac{dx}{x\sqrt{x^2-1}} = \mp\arcsin\frac{1}{x}+C$$

5. The integrals $\int \frac{dx}{\sqrt{a+bx+cx^2}}$ and $\int \frac{dx}{(x-m)\sqrt{a+bx+cx^2}}$ can be reduced to the foregoing ones if we introduce, respectively, the new variables $z = \sqrt{|c|}\,x$ and $z = \sqrt{|c|}(x-m)$.

§ 8.10. Binomial Differentials. Chebyshev's* Theorem

Let us consider the integral

$$\int x^m(a+bx^n)^p\,dx \tag{1}$$

* Academician P. L. Chebyshev (1821-1894), the great Russian scientist who contributed many outstanding results to mathematics and mechanics.

where a and b are arbitrary nonzero numbers and m, n and p are rational numbers. The element of integration $x^m(a+bx^n)^p\,dx$ in (1) is termed a *binomial differential.*

The substitution $x^n = t$, $x = t^{1/n}$, $dx = (1/n)\,t^{(1/n)-1}\,dt$ brings (1) to the form

$$\frac{1}{n}\int t^{(m+1/n)-1}(a+bt)^p\,dt \tag{2}$$

On putting $(m+1/n)-1 = q$ we reduce the integration problem to an integral of the form

$$\int t^q(a+bt)^p\,dt \tag{3}$$

where p and q are rational numbers.

If at least one of the numbers p, q and $p+q$ is an integer (positive or negative or zero) integral (3) *can be expressed in elementary functions.*

Indeed, if p is an integer the integral has the form $\int R(t, t^q)\,dt$ where q is a rational number. If q is an integer then it has the form $\int R(t,(a+bt)^p)\,dt$ where p is a rational number and, finally, if $p+q$ is an integer then the integral can be written as $\int t^{p+q}\left(\frac{a+bt}{t}\right)^p dt$.

As we already know (see § 8.8), all the three integrals can be transformed with the aid of the corresponding substitutions to integrals of rational functions.

Chebyshev proved a remarkable theorem stating that *if for given rational numbers p and q none of the three indicated conditions is fulfilled integral* (3) *cannot be expressed in elementary functions.*

Exercises

Find the integrals:

1. $\int \frac{x^3\,dx}{\sqrt{x-1}}$; 2. $\int \frac{dx}{x\sqrt{a+bx}}$; 3. $\int \frac{1+x^{1/3}}{1+x^{1/4}}\,dx$;

4. $\int \frac{dx}{x^3\sqrt{x-1}}$; 5. $\int (a+x)^{2/3}\,x^3\,dx$;

6. $\int \frac{dx}{(1+x)^{1/3}-(1+x)^{1/2}}$

Show that

7. $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}} = \frac{1}{\sqrt{2}}\arctan\frac{x\sqrt{2}}{\sqrt{1-x^2}}+C$;

8. $\int \frac{dx}{(1-x^2)\sqrt{1+x^2}} = \frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{1+x^2}+x\sqrt{2}}{\sqrt{1+x^2}-x\sqrt{2}}\right|+C$

§ 8.11. Integrating Trigonometric Expressions

Let us consider the integral

$$\int R(\cos x, \sin x)\,dx \tag{1}$$

where

$$R(u, v) = \frac{P(u, v)}{Q(u, v)} \tag{2}$$

is a rational function of u and v (i.e. P and Q are polynomials in u and v).

1. If one of the polynomials P and Q is an even function of v while the other is an odd function of v we can represent R in the form $R(u, v) = \frac{M(u, v^2)\,v}{N(u, v^2)}$ (by multiplying, if necessary, the numerator and the denominator in (1) by v) where $M(\mu, v)$ and $N(\mu, v)$ are polynomials in μ and v. Therefore the substitution $t = \cos x$ reduces integral (1) to the form

$$\int R(\cos x, \sin x)\,dx = -\int \frac{M(t, 1-t^2)}{N(t, 1-t^2)}\,dt = \int R(t)\,dt$$

where $R(t)$ is a rational function of t.

2. If one of the polynomials P and Q is even as function of u while the other is an odd function of u the substitution $t = \sin x$ rationalizes the integral, which is proved in the same manner as above.

3. If (i) the polynomials P and Q do not change when u and v are replaced by $-u$ and $-v$ respectively or (ii) both polynomials change sign under this transformation of the arguments u and v, then integral (1) can be rationalized by the substitution

$$t = \tan x \tag{3}$$

(or $t = \cot x$).

We shall limit ourselves to the first case because the other case can be reduced to the former with the aid of the multiplication of the numerator and the denominator of the fraction in (2) by u or v.

Every term of the polynomial P has the form

$$Au^l v^m = A\left(\frac{u}{v}\right)^l v^{l+m} = Aw^l v^{l+m} \quad \left(w = \frac{u}{v}\right)$$

where A is a constant coefficient. Therefore

$$P(u, v) = M(w, v^2) = \sum \alpha(w)\,v^{2s} \tag{4}$$

is a polynomial in w and v^2 (see the end of § 5.9).

The sum on the right-hand side of (4) extends over a finite number of terms, $\alpha(w)$ are some polynomials dependent solely on w, and s are nonnegative integers. We similarly conclude that

$$Q(u, v) = N(w, v^2) = \sum \beta(w)\,v^{2s}$$

is a polynomial in w and v^2.

By substitution (3), we have $\sin^2 x = t^2/(1+t^2)$ and $dx = dt/(1+t^2)$; hence integral (1) is transformed as

$$\int \frac{M\left(t, \frac{t^2}{1+t^2}\right) dt}{N\left(t, \frac{t^2}{1+t^2}\right)(1+t^2)} = \int r(t)\, dt$$

where $r(t)$ is a rational function.

4. For any rational function $R(u, v)$ the substitution $t = \tan x/2$ rationalizes integral (1). Indeed, we have

$$\cos x = \frac{1-\tan^2 \frac{x}{2}}{1+\tan^2 \frac{x}{2}} = \frac{1-t^2}{1+t^2}$$

$$\sin x = \frac{2 \tan \frac{x}{2}}{1+\tan^2 \frac{x}{2}} = \frac{2t}{1+t^2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2}\, dx = \frac{1}{2}(1+t^2)\, dx \quad \text{and} \quad dx = \frac{2dt}{1+t^2}$$

5. An expression of the form

$$T_n(x) = \frac{a_0}{2} + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx) \tag{5}$$

where a_k and b_k are constant coefficients is called a *trigonometric polynomial of order n*. The integral of (5) is easily found:

$$\int T_n(x)\, dx = \frac{a_0}{2} x + \sum_{k=1}^{n} \frac{1}{k} (a_k \sin kx - b_k \cos kx) + C \tag{6}$$

We often deal with expressions of the form

$$\cos^m x \cos^l x, \quad \cos^m x \sin^l x, \quad \sin^m x \sin^l x \tag{7}$$

where m and l are nonnegative integers. These expressions are trigonometric polynomials of order $m+l$, that is they can be brought to form (5) with constant coefficients a_k and b_k. This can readily be proved by induction.

Indeed, we have (see the explanations below)

$$\cos^m x = \left(\frac{e^{ix}+e^{-ix}}{2}\right)^m = \frac{1}{2^m}\left(e^{imx} + C_m^1 e^{i(m-2)x} + \ldots + e^{-imx}\right) =$$

$$= \frac{1}{2^m}\left(\cos mx + C_m^1 \cos (m-2)x + \ldots + \cos(-mx)\right) \tag{8}$$

Here one should take into account that $\cos^m x$ is a real function and therefore the last term in this chain of equalities is obtained from the preceding term by taking its real part. The imaginary part of $\cos^m x$ is automatically equal to zero. On replacing x by $(x+\pi/2)$ in (8) we obtain, depending on whether m is even or odd, the expressions

$$\sin^m x = \frac{(-1)^{m/2}}{2^m}(\cos mx - C_m^1 \cos(m-2)x + C_m^2 \cos(m-4)x + \ldots$$
$$\ldots + (-1)^{m/2}\cos(-mx)) \quad (9)$$

or

$$\sin^m x = \frac{(-1)^{m+1/2}}{2^m}(\sin mx - C_m^1 \sin(m-2)x + \ldots) \quad (10)$$

The fact that expressions (7) are trigonometric polynomials of the indicated order follows now from (8)-(10) and from the relations

$$\left.\begin{aligned} \sin\lambda x\cos\mu x &= \frac{1}{2}[\sin(\lambda+\mu)x+\sin(\lambda-\mu)x] \\ \cos\lambda x\cos\mu x &= \frac{1}{2}[\cos(\lambda+\mu)x+\cos(\lambda-\mu)x] \end{aligned}\right\} \quad (11)$$

Examples

1. $\displaystyle\int \frac{dx}{\sin x\cos x} = \int \frac{dx}{\tan x\cos^2 x} = \int \frac{d(\tan x)}{\tan x} = \ln|\tan x| + C$

(substitution (3)).

2. $\displaystyle\int \frac{dx}{\sin x} = \frac{1}{2}\int \frac{dx}{\tan(x/2)\cos^2(x/2)} = \int \frac{d(\tan(x/2))}{\tan(x/2)} = \ln|\tan(x/2)| + C$

(substitution 4, § 8.11)).

3. $\displaystyle\int \frac{dx}{a+b\cos x} = \int \frac{dx}{a(\cos^2(x/2)+\sin^2(x/2))+b(\cos^2(x/2)-\sin^2(x/2))} =$

$$= \int \frac{dx}{[a(1+\tan^2(x/2))+b(1-\tan^2(x/2))]\cos^2(x/2)} = \int \frac{2\,dt}{a(1+t^2)+b(1-t^2)}$$

(substitution 4, § 8.11).

4. $\displaystyle\int \frac{dx}{a+b\cos x+c\sin x} = \int \frac{dx}{a+r\cos(x-\varphi)}$

where the constants r and φ are chosen so that $b = r\cos\varphi$ and $c = r\sin\varphi$. Hence this integral is reducible to the foregoing one.

5. $\displaystyle\int \sin^5 dx = -\int (1-\cos^2 x)^2\, d(\cos x) = -\int (1-t^2)^2\, dt.$

6. $\displaystyle\int \frac{dx}{\sin^6 x} = -\int (1+\cot^2 x)^2\, d(\cot x) = -\int (1+t^2)^2\, dt.$

7. $\displaystyle\int \frac{dx}{\sin^5 x} = \int \left(\frac{1+t^2}{2t}\right)^5 \frac{2dt}{1+t^2} = \frac{1}{16}\int \frac{(1+t^2)^4}{t^5}\, dt \quad \left(t = \tan\frac{x}{2}\right).$

8. $\displaystyle\int \frac{\sin^3 x}{\cos^4 x}\, dx = -\int \frac{1-t^2}{t^4}\, dt \qquad (t = \cos x).$

9. $\int \sin^4 x\,dx = \int \left(\frac{1-\cos 2x}{2}\right)^2 dx = \frac{1}{4}\int \left(1-2\cos 2x+\frac{1+\cos 4x}{2}\right) dx.$

10. $\int \sin^4 x \cos^2 x\,dx = \int \left(\frac{1-\cos 2x}{2}\right)^2 \left(\frac{1+\cos 2x}{2}\right) dx =$

$$= \frac{1}{8}\int (1-\cos^2 2x)(1-\cos 2x)\,dx = \frac{1}{8}\int \left[1-\left(\frac{1+\cos 4x}{2}\right)\right](1-\cos 2x)\,dx$$

(further computations are performed with the aid of formulas (11)).

11. $\int \frac{dx}{a^2\cos^2 x+b^2\sin^2 x} = \int \frac{dx}{(a^2+b^2\tan^2 x)\cos^2 x} = \int \frac{dt}{a^2+b^2t^2} =$

$$= \frac{1}{ab}\arctan\left(\frac{b}{a}\tan x\right)+C \qquad (t = \tan x).$$

12. $\int \frac{2-\sin x}{2+\cos x}\,dx = 2\int \frac{dx}{2+\cos^2(x/2)-\sin^2(x/2)}+\ln|2+\cos x| =$

$$= 2\int \frac{dx}{2(\cos^2(x/2)+\sin^2(x/2))+(\cos^2(x/2)-\sin^2(x/2))}+\ln|2+\cos x| =$$

$$= 2\int \frac{dx}{3\cos^2(x/2)+\sin^2(x/2)}+\ln|2+\cos x| = 4\int \frac{d\tan(x/2)}{3+\tan^2(x/2)}+\ln|2+\cos x|.$$

§ 8.12. Trigonometric Substitutions

Integrals of the form

(i) $\int R(x, \sqrt{a^2-x^2})\,dx$, (ii) $\int R(x, \sqrt{a^2+x^2})\,dx$,

(iii) $\int R(x, \sqrt{x^2-a^2})\,dx$

can be transformed to integrals involving rational expressions in $\sin t$ and in $\cos t$ by means of the following substitutions (called *trigonometric substitutions*):

(i) $x = a\sin t$, whence $dx = a\cos t\,dt$ and $\sqrt{a^2-x^2} = a\cos t$

(ii) $x = a\tan t$ whence $dx = a\cos^{-2} t\,dt$ and $\sqrt{a^2+x^2} = a\cos^{-1} t$

(iii) $x = a\sec t$ whence $dx = a\tan t\sec t\,dt$ and $\sqrt{x^2-a^2} = a\tan t$

Example

$$\int \sqrt{a^2-x^2}\,dx = \int a\cos t\, a\cos t\,dt = a^2\int \frac{1+\cos 2t}{2}\,dt =$$

$$= \frac{a^2}{2}\left(t+\frac{\sin 2t}{2}\right)+C = \frac{a^2}{2}\left(t+\sin t\sqrt{1-\sin^2 t}\right)+C =$$

$$= \frac{a^2}{2}\left(\arcsin\frac{x}{a}+\frac{x}{a}\sqrt{1-\frac{x^2}{a^2}}\right)+C = \frac{a^2}{2}\left(\arcsin\frac{x}{a}+\frac{x}{a^2}\sqrt{a^2-x^2}+C\right)$$

$$(x = a\sin t)$$

3.13. Some Important Integrals Inexpressible in Terms of Elementary Functions

that the indefinite integral of the function e^{-x^2} which plays ... role in the *theory of probability* cannot be expressed in terms of elementary functions. The same applies to the integral of the function $\frac{\sin x}{x}$ often occurring in mathematical analysis.

The following important integrals are also encountered in applications:

$$\int \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}}, \quad \int \frac{x^2\, dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \tag{1}$$

$$\int \frac{dx}{(1+hx^2)\sqrt{(1-x^2)(1-k^2x^2)}} \qquad (0 < k < 1)$$

They are known as *elliptic integrals of the first, the second and the third kind* respectively. The first two of them involve the parameter k and the third one depends on k and on the additional parameter h. It can be proved that none of the three integrals is expressible in terms of elementary functions.

The substitution $x = \sin\varphi$ $(0 \leqslant \varphi \leqslant \pi/2)$ transforms the first of the integrals to

$$\int \frac{d\varphi}{\sqrt{1-k^2\sin^2\varphi}} \tag{2}$$

the second to

$$\frac{1}{k^2}\int \frac{d\varphi}{\sqrt{1-k^2\sin^2\varphi}} - \frac{1}{k^2}\int \sqrt{1-k^2\sin^2\varphi}\, d\varphi \tag{3}$$

and the third one to

$$\int \frac{d\varphi}{(1+h\sin^2\varphi)\sqrt{1-k^2\sin^2\varphi}} \tag{4}$$

In (3) there appears the integral

$$\int \sqrt{1-k^2\sin^2\varphi}\, d\varphi \tag{5}$$

distinct from (2). Integrals (2), (5) and (4) are called *elliptic integrals of the first, the second and the third kind, respectively, in Legendre's* form***.

There is extensive literature devoted to elliptic integrals. It includes tables of values of some important definite integrals corresponding to them, in particular tables of values of the definite integral corresponding to (5) taken over the interval $(0, \pi/2)$.

---

* A. M. Legendre (1752-1833), a French mathematician.

** By the elliptic integrals of the first, second and third kinds in Legendre's (*normal* or *standard*) form are usually meant some concrete antiderivatives of $\frac{d\varphi}{\sqrt{1-k^2\sin^2\varphi}}$, $\sqrt{1-k^2\sin^2\varphi}$ and $\frac{1}{(1+h\sin^2\varphi)\sqrt{1-k^2\sin^2\varphi}}$, namely those equal to the definite integrals of these functions taken from 0 to the variable upper limit φ. — *Tr.*

CHAPTER 9

Definite Integral

§ 9.1. Introductory Remarks. Definition

The notion of the definite integral was already discussed in § 1.7. The reader is advised to recall the content of this section before proceeding to the study of the present chapter devoted to the theory of the Riemann* definite integral.

We begin with the formal definition of the Riemann integral. Then we investigate its properties and establish conditions on the integrand function which guarantee the integrability (the existence of the integrals); we also study some applications of the definite integral and present the theory of improper integrals. As will be seen, for a function to be integrable in the sense of the proper Riemann integral it is necessary for it to be defined in a finite interval and be bounded on it. The (proper) Riemann integral does not exist for unbounded functions. But it is possible to define the notion of an improper (Riemann) integral which involves a two-fold passage to the limit. The introduction of the notion of an improper integral makes it possible to state a correct definition of the area of a figure bounded by the graph of a function which tends to infinity at some points but not too fast. Another type of improper integral is introduced for a function defined throughout the real axis. With the aid of this type of integral it is possible to compute the work of a force acting upon a body along an infinite path of motion.

Let us consider a function f defined in a finite closed interval $[a, b]$ and divide the interval into n parts with the aid of points of division $a = x_0 < x_1 < \ldots < x_n = b$. Such a construction will be referred to as a *partition* R of the interval $[a, b]$. On choosing an arbitrary point ξ_i in each subinterval $[x_i, x_{i+1}]$ ($\xi_i \in [x_i, x_{i+1}$; $i = 0, 1, \ldots, n-1$) we form the sum

$$S = S_R = \sum_{i=0}^{n-1} f(\xi_i)\, \Delta x_i$$

called a *Riemann integral sum* (for the function f defined on the interval $[a, b]$) corresponding to the partition R. Such an integral sum is not uniquely

* G. F. B. Riemann (1826-1866), a distinguished German mathematician.

determined since it depends not only on the function f and the partition R but also on the choice of the intermediate points $\xi_i \in [x_i, x_{i+1}]$.

The Riemann integral of f on $[a, b]$ is defined as the limit

$$\lim_{\max \Delta x_i \to 0} \sum_{i=0}^{n-1} f(\xi_i)\, \Delta x_i = \int_a^b f(x)\, dx = I \tag{1}$$

In other words, by the (Riemann) definite integral is meant a number I such that, given any $\varepsilon > 0$, there is $\delta > 0$ such that the inequality $|S_R - I| < \varepsilon$ holds for all partitions R in which $\Delta x_i < \delta$, irrespective of the choice of the points $\xi_i \in [x_i, x_{i+1}]$.

Another equivalent definition of limit (1) is the following: given an arbitrary sequence of partitions $R^k = \{a = x_0^k < x_2^k < \ldots < x_{n_k}^k = b\}$, for any definite choice of arbitrary intermediate points $\xi_i^k \in [x_i^k, x_{i+1}^k]$ for every k, such that $\max \Delta x_i^k \to 0$ as $k \to \infty$, the corresponding sequence of integral sums has the limit

$$\lim_{k\to\infty} S_{R_k} = \lim_{k\to\infty} \sum_{i=0}^{n_k - 1} f(\xi_i^k)\, \Delta x_i^k = I$$

(independent of the choice of R^k and ξ_i^k).

The equivalence of the two definitions of limit (1) is proved by analogy with the proof of the equivalence of the definitions of the limit of a function stated in terms of ε, δ-formalism and in terms of sequences.

The existence of the integral can also be expressed in terms of Cauchy's criterion: for any $\varepsilon > 0$ there is $\delta > 0$ such that for any partitions R and R' with subintervals of length not exceeding δ the inequality $|S_R - S_{R'}| < \varepsilon$ holds.

A function f for which integral (1) exists is said to be *Riemann integrable* (in the proper sense).

§ 9.2. Boundedness of Integrable Function

Theorem 1. *If a function f is integrable on an interval* $[a, b]$ *it is bounded on that interval.*

Indeed, suppose that f is unbounded on $[a, b]$, and let $S_R = \sum_{j=0}^{n-1} f(\xi_j)\, \Delta x_{ij}$ be its integral sum corresponding to an arbitrary partition R. Since f is unbounded on $[a, b]$ it is unbounded on at least one of the subintervals $[x_j, x_{j+1}]$ of the partition, say on $[x_{j_0}, x_{j_0+1}]$. Then we have

$$S_R = f(\xi_{j_0})\Delta x_{j_0} + \sum{}' f(\xi_j)\, \Delta x_j = f(\xi_{j_0})\, \Delta x_{j_0} + A, \qquad A = \sum{}' f(\xi_j)\, \Delta x_j$$

where the sum $\sum'$ extends over all $j \neq j_0$. We consider all ξ_j entering into this sum as being arbitrary but fixed. It follows that $|S_R| \geqslant |f(\xi_{j_0})|\, \Delta x_{j_0} - |A|$. Let us choose an arbitrarily large number N and write the inequality

$$|f(\xi_{j_0})|\, \Delta x_{j_0} - |A| > N, \quad \text{i.e.} \quad |f(\xi_{j_0})| > \frac{|A| + N}{\Delta x_{j_0}}$$

The unboundedness of f on $[x_{j_0}, x_{j_0+1}]$ guarantees the existence of a point $\xi_{j_0} \in [x_{j_0}, x_{j_0+1}]$ such that this inequality is fulfilled.

We see that if f is unbounded on $[a, b]$ then, given any number $N > 0$ and any partition R, there is an integral sum corresponding to R such that, due to an appropriate choice of the points ξ_j, it exceeds N in its absolute value. This means that f is not integrable on $[a, b]$.

In what follows we shall only consider bounded functions (in the theory of the proper Riemann integral).

§ 9.3. Darboux* Sums

Let $f(x)$ be an arbitrary bounded function on $[a, b]$ and let $R = \{a = x_0 < < x_1 < \ldots < x_n = b\}$ be an arbitrary partition of $[a, b]$. Putting $m_j = = \inf\limits_{x \in [x_j, x_{j+1}]} f(x)$ and $M_j = \sup\limits_{x \in [x_j, x_{j+1}]} f(x)$ we form the sums

$$\underline{S}_R = \sum_{j=0}^{n-1} m_j \Delta x_j \quad \text{and} \quad \bar{S}_R = \sum_{j=0}^{n-1} M_j \Delta x_j$$

which are called *the lower* and *the upper Darboux sums*, respectively, corresponding to the partition R (a more precise notation is $\underline{S}_R(f)$ and $\bar{S}_R(f)$). These are definite numbers dependent solely on f and R.

It is evident that $\underline{S}_R \leqslant \bar{S}_R$.

Let R_1, R_2 and R_3 be some partitions of $[a, b]$. If all points of division entering into R_1 belong to the system of points of division with the aid of which R_2 is formed we shall say that R_2 is a *refinement* of R_1 and shall write $R_1 \subset R_2$. If the set of points of division of R_3 is the set-theoretic sum of the points of division entering into R_1 and R_2 we shall write $R_3 = R_1 + R_2$.

If $R \subset R'$ then

$$\underline{S}_R \leqslant \underline{S}_{R'} \leqslant \bar{S}_{R'} \leqslant \bar{S}_R \tag{1}$$

Indeed, let

$$R = \{x_0 < x_1 < \ldots < x_n\}$$

and

$$R' = \{x_0 = x_{0,0} < x_{0,1} < \ldots < x_{0,l_0} = x_1 = x_{1,0} < x_{1,1} < \ldots$$
$$\ldots < x_{n-2,l_{n-2}} = x_{n-1} = x_{n-1,0} < \ldots < x_{n-1,l_{n-1}} = x_n\}$$

Then, obviously,

$$M_{jk} = \sup_{x \in [x_{j,k}, x_{j,k+1}]} f \leqslant \sup_{x \in [x_j, x_{j+1}]} f = M_j$$

and

$$\bar{S}_{R'} = \sum_{j=0}^{n-1} \sum_{k=0}^{l_j-1} M_{jk} \Delta x_{jk} \leqslant \sum_{j=0}^{n-1} M_j \sum_{k=0}^{l_j-1} \Delta x_{jk} = \sum_{j=0}^{n-1} M_j \Delta x_j = \bar{S}_R$$

whence follows the last inequality (1). The first one is proved analogously (the middle inequality (1) has been already established).

Given any partitions R_1 and R_2, there holds $\underline{S}_{R_1} \leqslant \bar{S}_{R_2}$ because $\underline{S}_{R_1} \leqslant \leqslant \underline{S}_{R_1+R_2} \leqslant \bar{S}_{R_1+R_2} \leqslant \bar{S}_{R_2}$.

---

* J. G. Darboux (1842-1917), a French mathematician.

Now let us fix R_1, and let R be arbitrary; then

$$\underline{S}_{R_1} \leqslant \bar{S}_R \text{ and, consequently, } \underline{S}_{R_1} \leqslant \inf_R \bar{S}_R = \bar{I}$$

The number $\bar{I} = \inf_R \bar{S}_R$ is called *the upper (Darboux) integral of the function f on* $[a, b]$. We have proved the existence of the upper integral and the fact that for any R (we now replace R_1 by R) there holds

$$\underline{S}_R \leqslant \bar{I}$$

It follows that there exists the supremum

$$\underline{I} = \sup_R \underline{S}_R \leqslant \bar{I}$$

which is called *the lower (Darboux) integral of f on* $[a, b]$. Hence, we have established the existence of the lower integral $\underline{I}$ and the upper integral $\bar{I}$ of f on $[a, b]$ and the inequality $\underline{I} \leqslant \bar{I}$.

Lemma 1. *If E_1 and E_2 are point sets in the real line then*

$$\sup_{\substack{x \in E_1 \\ y \in E_2}} (x+y) = \sup_{x \in E_1} x + \sup_{y \in E_2} y$$

The proof is left to the reader.

Lemma 2. *If f is a bounded function defined on a closed interval* $[c, d]$ *then*

$$\sup_{\xi, \eta \in [c, d]} [f(\xi)-f(\eta)] = \sup_{\xi, \eta \in [c, d]} |f(\xi)-f(\eta)| = M-m \tag{2}$$

where $M = M_{[c, d]} = \sup_{x \in [c, d]} f(x)$ *and* $m = m_{[c, d]} = \inf_{x \in [c, d]} f(x)$.

Proof. For any $\xi, \eta \in [c, d]$ we have

$$f(\xi)-f(\eta) \leqslant |f(\xi)-f(\eta)| \leqslant M-m \tag{3}$$

On the other hand, there are $\xi, \eta \in [c, d]$ such that $f(\xi) > M-\varepsilon/2$ and $f(\eta) < m+\varepsilon/2$; for such ξ and η we have

$$f(\xi)-f(\eta) > (M-\varepsilon/2)-(m+\varepsilon/2) = M-m-\varepsilon$$

Hence, we have proved that the first and the third terms in (2) are equal. Then, moreover, (3) shows that they are equal to the second term in (2).

The number

$$M-m = \omega = \omega_{[c, d]}$$

is called *the oscillation of f on* $[c, d]$.

It follows from Lemmas 1 and 2 that

$$\sup_{\xi_j, \eta_j \in [x_j, x_{j+1}]} \sum_{j=0}^{n-1} |f(\xi_j)-f(\eta_j)| \, \Delta x_j = \sum_{j=0}^{n-1} \sup |f(\xi_j)-f(\eta_j)| \, \Delta x_j =$$

$$= \sum_{j=0}^{n-1} (M_j-m_j) \, \Delta x_j = \sum_{j=0}^{n-1} \omega_j \, \Delta x_j = \bar{S}_R-\underline{S}_R, \qquad \omega_j = \omega_{[x_j, x_{j+1}]} \tag{4}$$

(all these relations involve the sign of strict equality!).

§ 9.4. Key Theorem

Theorem 1 (Key Theorem). *Let f be a bounded function defined on a finite closed interval $[a, b]$.*

The following four properties are equivalent:

(i) $\underline{I} = \bar{I}$;

(ii) *for any $\varepsilon > 0$ there is a partition R such that*

$$\bar{S}_R - \underline{S}_R < \varepsilon \tag{1}$$

(iii) *for any $\varepsilon > 0$ there is $\delta > 0$ such that inequality* (1) *holds for all partitions R with subintervals $\Delta x_j < \delta$;*

(iv) *the integral*

$$\int_a^b f(x)\, dx = I \tag{2}$$

exists and $I = \underline{I} = \bar{I}$.

Here it is meant that $\underline{I}$ and $\bar{I}$ are the upper and the lower integral of f on $[a, b]$ respectively and that $\underline{S}_R$ and $\bar{S}_R$ are the upper and the lower Darboux sum of f corresponding to the partition R. It should be noted that for an arbitrary bounded function f on $[a, b]$ these properties may not hold but if at least one of them holds then the other three hold as well and if one of them fails to hold so do the others.

The key theorem can be restated as follows:

For the integral of f on $[a, b]$ to exist it is necessary and sufficient that (at least) one of the conditions (i)-(iii) *holds. Then the value of the integral I coincides with $\underline{I} = \bar{I}$.*

Proof: Implication (i) $\rightarrow$ (ii) (this means that Property (i) implies Property (ii)). If (i) holds then, putting $\underline{I} = \bar{I} = I$, we can assert that there are partitions R_1 and R_2 such that $\underline{I} - \varepsilon/2 < \underline{S}_{R_1}$ and $\bar{S}_{R_2} < \bar{I} + \varepsilon/2$. Then for $R = R_1 + R_2$ we have

$$\underline{I} - \varepsilon/2 < \underline{S}_{R_1} \leq \underline{S}_R \leq \bar{S}_R \leq \bar{S}_{R_2} \leq \bar{I} + \varepsilon/2$$

whence follows (ii).

Implication (ii) $\rightarrow$ (i). Let R be a partition for which (1) is fulfilled. Then, by virtue of the inequalities $\underline{S}_R \leq \underline{I} \leq \bar{I} \leq \bar{S}_R$ we have $\bar{I} - \underline{I} < \varepsilon$. Now, since $\varepsilon > 0$ can be arbitrarily small and $\underline{I}$ and $\bar{I}$ are definite numbers independent of ε, we conclude that $\underline{I} = \bar{I}$, i.e. (i) holds.

Implication (iv) $\rightarrow$ (iii). Let integral (2) exist. From the definition of the integral it follows that, given any $\varepsilon > 0$, there is $\delta > 0$ such that for any partition R with $\Delta x_j < \delta$, irrespective of the choice of the points $\xi_j \in [x_j, x_{j+1}]$, hold the inequalities

$$I - \varepsilon/2 < \sum_1^{n-1} f(\xi_j)\, \Delta x_j < I + \varepsilon/2$$

Now, taking the supremum and the infimum of the sum $\sum\limits_1^{n-1} f(\xi_j)\, \Delta x_j$ over all $\xi_j \in [x_j, x_{j+1}]$ entering into these inequalities we obtain

$$I-\varepsilon/2 \leq \underline{S}_R \leq \bar{S}_R \leq I+\varepsilon/2 \tag{3}$$

that is (3) takes place.

Implication (iii) → (ii). The proof is quite obvious.

Implication (ii) → (iii). This is the most intricate part of the theorem. We have to prove that if for any $\varepsilon > 0$ there is a partition $R_* = \{a = x_0^* < x_1^* < \ldots < x_n^* = b\}$ dependent on ε such that $\bar{S}_{R_*} - \underline{S}_{R_*} < \varepsilon$, then there is $\delta > 0$ such that (1) holds for all partitions R with $\Delta x_j < \delta$.

To prove this part of the theorem let us take as $\delta > 0$ the number satisfying the inequalities $2\delta < x_{j+1}^* - x_j^*$ $(j = 0, 1, \ldots, n-1)$ and $4n\,\delta K < \varepsilon$ where $K = \sup\limits_{x \in [a, b]} |f(x)|$. Then we have

$$\bar{S}_R - \underline{S}_R = \sum{}'(M-m)\,\Delta x + \sum{}''(M-m)\,\Delta x$$

(for brevity we omit the index j and write M, m and Δx instead of M_j, m_j and Δx_j) where the sum $\sum'$ extends over all (closed) subintervals of the partition R such that each of them contains one of the points entering in R_* and $\sum''$ extends over all the other subintervals of R.

The sum $\sum'$ involves not more than $2n$ summands: it includes one subinterval containing the point a and one subinterval containing the point b, and each of the points $x_1, \ldots, x_{n-1}$ is contained in one or two subintervals. Thus, we have

$$\sum{}'(M-m)\,\Delta x \leq 2K\,\delta\,2n < \varepsilon$$

Let us represent $\sum''$ as the two-fold sum $\sum'' = \sum\limits_i \sum^i$ where $\sum^i$ denotes the sum of the terms entering into $\sum''$ which correspond to those subintervals of R each of which is contained in exactly one subinterval $[x_i^*, x_{i+1}^*]$ of the former partition R_*.

We have

$$\sum{}''(M-m)\,\Delta x = \sum_i{}'' \sum{}^i(M-m)\,\Delta x \leq \sum_i (M_i^* - m_i^*) \sum{}^i \Delta x \leq$$
$$\leq \sum_i (M_i^* - m_i^*)\,\Delta x_i^* < \varepsilon$$

Therefore $\bar{S}_R - \underline{S}_R < 2\varepsilon$ for all partitions R with $\Delta x < \delta$, that is (iii) holds.

Implication (iii) → (iv). Let (iii) hold. Then, as has already been proved, (i) also holds. Let us choose an arbitrary $\varepsilon > 0$ and find $\delta > 0$ as was indicated in (iii). Then for the partition R mentioned in (iii) we obtain

$$\underline{S}_R \leq \sum f(\xi_j)\,\Delta x_j \leq \bar{S}_R, \quad \underline{S}_R \leq \underline{I} = \bar{I} \leq \bar{S}_R$$

whence, putting $I = \underline{I} = \bar{I}$, we derive

$$|I - \sum f(\xi_j)\,\Delta x_j| < \varepsilon$$

which means that I is the definite integral of f on [a, b]. Thus, we have proved (iv).

The proof of the theorem has been completed.

As a consequence of the key theorem we obtain

Theorem 2. *Let there be a sequence of partitions R_k of the form*

$$a = x_0^k < x_1^k < \ldots < x_{n_k}^k = b \qquad (k = 1, 2, \ldots)$$

for which $\max_j \Delta x_j^k = \delta_k \to 0$.

If the function f satisfies at least one of the conditions

$$\lim_{k\to\infty} (\bar{S}_{R_k}(f) - \underline{S}_{R_k}(f)) = 0 \tag{4}$$

$$\lim_{k\to\infty} \sum_j f(\xi_j^k)\, \Delta x_j^k = \lim_{k\to\infty} S_{R_k}(f) = I \tag{5}$$

and

$$\lim_{k\to\infty} \bar{S}_{R_k}(f) = \lim_{k\to\infty} \underline{S}_{R_k}(f) = I \tag{6}$$

then the integral $I = \int\limits_a^b f(x)\, dx$ *exists.*

Conversely, the existence of the integral of f on $[a, b]$ implies the fulfilment of all conditions (4)-(6).

From (4) (and also from (6)) obviously follows property (ii) mentioned in the formulation of the key theorem. Further (5) implies that for any $\varepsilon > 0$ there is k such that

$$I - \varepsilon/2 < S_{R_k} < I + \varepsilon/2 \tag{8}$$

for any $\xi_j^k \in [x_j^k, x_{j+1}^k]$. Taking the supremum and the infimum of S_{R_k} over the indicated ξ_j^k's we receive

$$I - \varepsilon/2 \leqslant \underline{S}_{R_k} \leqslant \bar{S}_{R_k} \leqslant I + \varepsilon/2 \tag{9}$$

whence follows property (ii) of the key theorem by virtue of which there exists the integral of f on $[a, b]$ which is obviously equal to I.

Conversely, if the integral exists and is equal to I then, by its definition, limit (5) exists for any sequence of partitions with $\delta_k \to 0$, and, in particular, for the sequence in question as well. Therefore, given any $\varepsilon > 0$, there is k_0 such that for $k > k_0$ there hold inequalities (8) and, consequently, inequalities (9) as well; hence (6) and, moreover, (4) take place.

The theorem has been proved.

Theorem 2 shows that *in order to check the existence of the integral it is sufficient to verify that limit* (5) *exists (for any choice of ξ_j^k) for at least one of the sequences of partitions R_k with $\delta_k \to 0$, for instance for the partition obtained by splitting $[a, b]$ into n equal parts ($n \to \infty$).*

Remark. Darboux proved a theorem (whose proof we do not present here) asserting that

$$\lim_{\delta\to 0} \bar{S}_R = \bar{I} \quad \text{and} \quad \lim_{\delta\to 0} \underline{S}_R = \underline{I} \tag{10}$$

for any bounded function f defined on $[a, b]$ although this function may not be integrable (i.e. it may happen that $\underline{I} < \bar{I}$).

Example. For the Dirichlet function f defined on $[0, 1]$ which is equal to 1 at the rational points of the interval $[0, 1]$ and to 0 at the irrational points we have the equality $\bar{S}_R = 1$ for the upper Darboux sum and $\underline{S}_R = 0$ for the lower sum for any partition R of the interval $[0, 1]$. Hence, $\underline{I} = 0 < 1 = \bar{I}$, which means that the Dirichlet function, although bounded, is not Riemann integrable.

§ 9.5. Existence Theorem for the Integral of a Continuous Function on an Interval $[a, b]$. Integrability of a Monotone Function

Theorem 1. *If a function f is continuous on a closed interval $[a, b]$ it is integrable on that interval.*

Proof. Let f be continuous on $[a, b]$; then for any partition R whose subintervals satisfy the inequality $\Delta x_j < \delta$ we have

$$\sum_{j=0}^{n-1} [f(\xi_j)-f(\eta_j)]\,\Delta x_j \leqslant \sum_{j=0}^{n-1} \omega(\delta)\,\Delta x_j = \omega(\delta)\,(b-a)$$

where $\omega(\delta) = \sup\limits_{\substack{x', x'' \in [a, b] \\ |x'-x''| < \delta}} |f(x')-f(x'')|$ is the modulus of continuity of f on $[a, b]$. Therefore

$$\bar{S}_R - \underline{S}_R = \sup_{\xi_j, \eta_j} \sum_{j=0}^{n-1} [f(\xi_j)-f(\eta_j)]\,\Delta x_j \leqslant \omega(\delta)\,(b-a)$$

As we know, the modulus of continuity $\omega(\delta)$ of the continuous function f defined on the closed interval $[a, b]$ tends to zero as $\delta \to 0$, and hence, given any $\varepsilon > 0$, there is $\delta > 0$ such that $\bar{S}_R - \underline{S}_R < \varepsilon$.

By the equivalence of properties (ii) and (iv) (see the key theorem), we conclude that the integral of f on $[a, b]$ exists.

Theorem 2. *A monotone bounded function f defined on a closed interval $[a, b]$ is integrable on it.*

For definiteness, let f be a nondecreasing function. Then, given an arbitrary partition R with $\Delta x_j \leqslant \delta$, we have

$$\sum_{j=0}^{n-1} [f(\xi_j)-f(\eta_j)]\,\Delta x_j \leqslant \sum_{j=0}^{n-1} [f(x_{j+1})-f(x_j)]\,\Delta x_j \leqslant$$

$$\leqslant \delta \sum_{j=0}^{n-1} [f(x_{j+1})-f(x_j)] = \delta[f(b)-f(a)] < \varepsilon \qquad (\xi_j, \eta_j \in [x_j, x_{j+1}])$$

provided that $\delta > 0$ is sufficiently small. Whence, taking the supremum over

ξ_j and η_j, we obtain

$$\bar{S}_R - \underline{S}_R = \sup \sum [f(\xi_j) - f(\eta_j)]\, \Delta x_j < \varepsilon$$

and therefore, by the equivalence of properties (ii) and (iv), we conclude that the function f is integrable on $[a, b]$.

It should be stressed that a monotone function f on $[a, b]$ can only have a finite or a countable number of points of discontinuity.

Indeed, let $a \leqslant x_1 < x_2 < \ldots < x_N \leqslant b$ be some (not necessarily all) points of discontinuity (of the first kind) of a monotone function $f(x)$ defined on $[a, b]$ (as above, we suppose that f is nondecreasing). Let us take $\varepsilon > 0$ so small that

$$x_1 + \varepsilon < x_2 - \varepsilon < x_2 + \varepsilon < x_3 - \varepsilon < \ldots$$

Then

$$\sum_{k=1}^{N} [f(x_k + \varepsilon) - f(x_k - \varepsilon)] \leqslant f(b) - f(a) = \varkappa$$

(if $x_1 = a$ then we put $f(a-\varepsilon) = f(a)$ and if $x_N = b$ then we put $f(b+\varepsilon) = f(b)$). On passing to the limit as $\varepsilon \to 0$ we obtain

$$\sum_{k=1}^{N} [f(x_k + 0) - f(x_k - 0)] \leqslant \varkappa$$

(if $x_1 = a$ then $x_1 - 0 = a$ and if $x_N = b$ then $x_N + 0 = b$). This inequality shows that the function f can have not more than one point of discontinuity with a jump exceeding $\varkappa/2$. If there is such a point we supply it with the index 1 and then check whether there are points of discontinuity with jump exceeding $\varkappa/3$. There can be not more than two points of this kind; if among the remaining points there is one or two points of the indicated type we supply them with the subsequent indices and so on. As the result of this procedure, all the points of discontinuity receive natural indices, which proves the assertion.

§ 9.6. Lebesgue's* Theorem

In the foregoing section we proved that any function continuous on a closed interval $[a, b]$ is Riemann integrable on that interval. We also proved that every monotone (bounded) function on $[a, b]$ is integrable on $[a, b]$. A monotone function may be discontinuous, and, as was shown, the number of its points of discontinuity is finite or countable. There arises the natural question as to how many points of discontinuity a function may have to remain Riemann integrable. The exhaustive answer to the question is given by the following theorem.

Theorem (Lebesgue). *For a function f to be (Riemann) integrable on a (finite) closed interval $[a, b]$ it is necessary and sufficient that it be bounded on*

---

* H. L. Lebesgue (1875-1941), a noted French mathematician, one of the creators of the modern theory of functions of a real variable.

$[a, b]$ and continuous on $[a, b]$ everywhere except possibly on a set of Lebesgue measure zero.

The proof of the theorem will be given in § 12.10 for the general case of an arbitrary dimension n.

By definition, *a set e has Lebesgue measure zero if, given any $\varepsilon > 0$, there is a finite or countable system of open intervals covering e such that the sum of the lengths of the intervals is less than ε.*

A finite or a countable set e is always of Lebesgue measure zero. Indeed, let us number the points of the set e as $x_1, x_2, \ldots$ Each of the points can be covered by an interval of length less than $\varepsilon/2^n$, the sum of the lengths of the intervals being less than ε.

Thus, it follows from Lebesgue's theorem that every bounded function on $[a, b]$ with a finite or countable number of points of discontinuity is Riemann integrable.

It should be mentioned that among the sets of Lebesgue measure zero there are also uncountable sets.

Example. Let us divide the closed interval $[0, 1]$ into three equal parts and delete from $[0, 1]$ the middle (open) interval $(1/3, 2/3)$ (which is an open set); each of the remaining closed intervals $[0, 1/3]$ and $[2/3, 1]$ we again divide into three equal parts and remove the corresponding open middle intervals. On the remaining four closed intervals we again perform the analogous operations and so on. Continuing this process indefinitely we arrive at a subset E of the original interval $[0, 1]$ obtained by deleting from $[0, 1]$ a countable number of open intervals the sum of whose lengths is equal to $1/3+2/9+4/27+ \ldots = 1$. The set E is closed.

It can be proved (e.g. see P. S. Alexandrov, A. N. Kolmogorov, *An Introduction to the Theory of Functions of a Real Variable*, GTTI, 3rd ed., Moscow, 1938) that the set E possesses the following properties: it is not only closed but also *perfect*, which means that every point of E is its limit point, it is *nowhere dense in* $[0, 1]$, i.e. any open interval belonging to $[0, 1]$ contains points not belonging to E, and finally, it is uncountable and at the same time its Lebesgue measure (see § 19.1, Vol. 2) is equal to zero. The set E is referred to as the *Cantor* set*.

Let us consider the function

$$f(x) = \begin{cases} 1 & \text{for} \quad x \in E \\ 0 & \text{for} \quad x \in [0, 1]-E \end{cases}$$

It is obviously continuous at all the points belonging to $[0, 1]-E$ and discontinuous at all the points of E. Thus, f can serve as an example of a Riemann integrable function with an uncountable set of points of discontinuity.

* After G. Cantor (1845-1918), a famous German mathematician, one of the creators of modern set theory.

§ 9.7. Additivity and Homogeneity of the Integral

Theorem 1. *If a function f is integrable on $[a, b]$ and if $a < c < b$ then f is also integrable on $[a, c]$ and on $[c, b]$ and vice versa. In this case*

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \tag{1}$$

Proof. Let us take an arbitrary sequence of partitions R_k, $[a, b]$ containing the point c as a point of division, with the maximum subinterval tending to zero. R_k generates the corresponding partitions R_k' and R_k'' of $[a, c]$ and $[c, b]$, respectively, and

$$\bar{S}_{R_k} - \underline{S}_{R_k} = (\bar{S}_{R_k'} - \underline{S}_k') + (\bar{S}_{R_k''} - \underline{S}_{R_k''}) \tag{2}$$

If the integral of f on $[a, b]$ exists then, by virtue of the theorem proved in § 9.4, the left-hand side of equality (2) tends to zero as $k \to \infty$ and consequently each of the (nonnegative) summands on its right-hand side also tends to zero; by the same theorem, this implies the existence of the integrals of f on $[a, c]$ and on $[c, b]$. Therefore, on passing to the limit as $k \to \infty$ in the obvious equality

$$S_{R_k} = S_{R_k'} + S_{R_k''}$$

we obtain equality (1).

Conversely, if the integrals of f on $[a, c]$ and on $[c, b]$ exist then, for arbitrary partitions R_k' and R_k'' corresponding to them whose maximum subintervals tend to zero, each of the differences on the right-hand side of (2) tends to zero as $k \to \infty$ and therefore the left-hand side of (2) tends to zero as well; this implies the existence of the integral of f on $[a, b]$.

The integral of f on $[a, b]$ was defined for the case $a < b$. It is convenient to extend this definition to the case $a > b$ by putting

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx \text{ for } a > b \text{ and } \int_a^a f(x)\,dx = 0 \text{ (for } a = b)$$

As can readily be checked, in this generalized interpretation of the symbol $\int_a^b$ equality (1) continues to hold for any a, b and c provided that the integral over the greatest of the intervals $[a, b]$, $[a, c]$ and $[b, c]$ exists. Here by an interval $[a, b]$ is meant the (directed) line segment connecting the point a to the point b, and $[a, a]$ is understood as the point a.

Theorem 2. *Let $f(x)$ and $\varphi(x)$ be two functions integrable on $[a, b]$ and C be a constant, then the functions*

(i) $f(x) \pm \varphi(x)$, (ii) $Cf(x)$, (iii) $|f(x)|$, (iv) $f(x)\varphi(x)$ *and* (v) $1/f(x)$

(*where in the last case it is supposed that* $|f(x)| > d > 0$ *on* $[a, b]$) *are inte-*

grable on $[a, b]$,

$$\int_a^b (f(x)\pm\varphi(x))\,dx = \int_a^b f(x)\,dx \pm \int_a^b \varphi(x)\,dx \tag{2}$$

and

$$\int_a^b Cf(x)\,dx = C\int_a^b f(x)\,dx \tag{3}$$

The integrability of the indicated functions is implied directly by Lebesgue's theorem because the Lebesgue measure of the sum of two sets of Lebesgue measure zero is again a set of measure zero, but it is also possible to prove the theorem without resorting to Lebesgue's theorem.

Let us take an arbitrary partition $a = x_0 < x_1 < \ldots < x_n = b$. Then

$$\lim_{\max \Delta x_j \to 0} \sum_j [f(\xi_j)\pm\varphi(\xi_j)]\,\Delta x_j = \lim \sum_j f(\xi_j)\,\Delta x_j \pm \lim \sum_j \varphi(\xi_j)\,\Delta x_j =$$

$$= \int_a^b f(x)\,dx \pm \int_a^b \varphi(x)\,dx$$

because, by the hypothesis, the integrals of $f(x)$ and $\varphi(x)$ exist. Consequently, the limit on the left-hand side of these relations exists and is equal to that on the right-hand side. This means that (2) holds. Similarly,

$$\int_a^b Cf\,dx = \lim_{\max \Delta x_j \to 0} \sum Cf(\xi_j)\,\Delta x_j = C\lim \sum f(\xi_j)\,\Delta x_j = C\int_a^b f\,dx$$

We have proved properties (i), (ii) and equalities (2) and (3).

Let us denote $M_f = \sup\limits_{x\in[x_j, x_{j+1}]} f$ and $m_f = \inf\limits_{x\in[x_j, x_{j+1}]} f$. The numbers M_f and m_f depend of course on j but we omit the index j for the sake of brevity. Let us also put $K_f = \sup\limits_{x\in[a,b]} |f|$. For arbitrary $\xi, \eta \in [x_j, x_{j+1}]$ we have

$$|f(\xi)|-|f(\eta)| \leqslant |f(\xi)-f(\eta)| \leqslant M_f - m_f$$

$$f(\xi)\varphi(\xi)-f(\eta)\varphi(\eta) \leqslant |f(\xi)|\,|\varphi(\xi)-\varphi(\eta)| +$$
$$+ |\varphi(\eta)|\,|f(\xi)-f(\eta)| \leqslant K_f(M_\varphi - m_\varphi)+K_\varphi(M_f - m_f)$$

and

$$\frac{1}{f(\xi)}-\frac{1}{f(\eta)} = \frac{f(\xi)-f(\eta)}{f(\xi)f(\eta)} \leqslant \frac{1}{d^2}(M_f - m_f)$$

On taking the suprema of the left-hand sides of these equalities over $\xi, \eta \in [x_j, x_{j+1}]$, multiplying them by Δx_j and summing with respect to j we receive

$$\sum (M_{|f|} - m_{|f|})\,\Delta x \leqslant \sum (M_f - m_f)\,\Delta x \tag{4}$$

$$\sum (M_{f\varphi} - m_{f\varphi})\,\Delta x \leqslant K_f \sum (M_\varphi - m_\varphi)\,\Delta x + K_\varphi \sum (M_f - m_f)\,\Delta x \tag{5}$$

and

$$\sum (M_{1/f}-m_{1/f})\, \Delta x \leqslant \frac{1}{d^2} \sum (M_f-m_f)\, \Delta x \tag{6}$$

(here we have omitted the index j in Δx_j). By the integrability of f, the right-hand sides of (4) and (6) can be made less than $\varepsilon > 0$ for an appropriate choice of the partition and then the left-hand sides also become less than ε. In the case of relation (5) the argument is slightly more complicated: for any given $\varepsilon > 0$ we find partitions R_1 and R_2 for which, respectively,

$$\bar{S}_{R_1}(f)-\underline{S}_{R_1}(f) < \varepsilon \quad \text{and} \quad \bar{S}_{R_2}(\varphi)-\underline{S}_{R_2}(\varphi) < \varepsilon$$

Therefore, moreover, both R_1 and R_2 can be replaced by $R = R_1+R_2$.

If we consider relation (5) for that very partition R it can be rewritten as

$$\bar{S}_R(f\varphi)-\underline{S}_R\, f\varphi) \leqslant (K_f+K_\varphi)\varepsilon \tag{7}$$

Thus, there is a partition (the partition R) for which the left-hand side of (7) becomes arbitrarily small, whence follows that $f\varphi$ is integrable on $[a, b]$.

Note that the integrability of $|f(x)|$ does not imply that of $f(x)$, which is readily confirmed by the example of the function equal to 1 at the rational points of $[a, b]$ and to -1 at the irrational points.

§ 9.8. Integrating Inequalities. Mean Value Theorem

Theorem 1. *If functions f and φ are integrable on a closed interval $[a, b]$ and satisfy the inequality $f(x) \leqslant \varphi(x)$ on that interval then*

$$\int_a^b f\, dx \leqslant \int_a^b \varphi\, dx \tag{1}$$

By the hypothesis, both integrals in (1) exist, and it only remains to prove the inequality itself. For any partition R we obviously have

$$\sum f(\xi_j)\, \Delta x_j \leqslant \sum \varphi(\xi_j)\, \Delta x_j$$

On passing to the limit as $\max \Delta x_j \to 0$ we arrive at (1).

Theorem 2. *If f is integrable on $[a, b]$ then*

$$\left| \int_a^b f\, dx \right| \leqslant \int_a^b |f|\, dx \leqslant K(b-a) \tag{2}$$

where $K = \sup\limits_{a \leqslant x \leqslant b} |f(x)|$.

We have $-|f(x)| \leqslant f(x) \leqslant |f(x)|$ and therefore the foregoing theorem shows that

$$-\int_a^b |f|\, dx = \int_a^b (-|f|)\, dx \leqslant \int_a^b f\, dx \leqslant \int_a^b |f|\, dx$$

whence follows the first inequality (2). Further, we have $|f| \leqslant K$ and therefore

$$\int_a^b |f|\,dx \leqslant \int_a^b K\,dx = K(b-a)$$

which gives the second inequality (2).

In Theorem 2 we meant that $a < b$. If $a > b$ then

$$\left|\int_a^b f\,dx\right| = \left|\int_b^a f\,dx\right| \leqslant \int_b^a |f|\,dx \leqslant (a-b)\,K = K\,|b-a|$$

Theorem 3 (Mean Value Theorem*). *If f and φ are integrable on $[a, b]$ and $\varphi(x) \geqslant 0$ then*

$$\int_a^b f\varphi\,dx = \Lambda \int_a^b \varphi\,dx \tag{3}$$

where $m \leqslant \Lambda \leqslant M$, $m = \inf\limits_{a \leqslant x \leqslant b} f(x)$ and $M = \sup\limits_{a \leqslant x \leqslant b} f(x)$.

Indeed, since $\varphi(x) \geqslant 0$ we have

$$m\varphi(x) \leqslant f(x)\,\varphi(x) \leqslant M\varphi(x) \tag{4}$$

On integrating these inequalities we obtain

$$m\int_a^b \varphi\,dx \leqslant \int_a^b f\varphi\,dx \leqslant M\int_a^b \varphi\,dx \tag{5}$$

If $\int_a^b \varphi\,dx = 0$ the second integral in these relations is also equal to zero, and equality (3) becomes obvious; if $\int_a^b \varphi\,dx > 0$ then (5) implies

$$m \leqslant \frac{\int_a^b f\varphi\,dx}{\int_a^b \varphi\,dx} \leqslant M$$

that is the second term in the above relations is equal to a number Λ satisfying the inequalities $m \leqslant \Lambda \leqslant M$, which is what we had to prove.

---

* This theorem is also called the *first mean value theorem* (on the *second mean value theorem* see § 9.10). — *Tr.*

Corollary. *If the function f in Theorem 3 is continuous on $[a, b]$ then there are points $x_1, x_2 \in [a, b]$ such that $f(x_2) = M$ and $f(x_1) = m$ and also a point $\xi \in [x_1, x_2]$ such that $f(\xi) = \Lambda$; consequently in the case of a continuous function f defined on $[a, b]$ equality* (3) *can be written in the form*

$$\int_a^b f\varphi\, dx = f(\xi) \int_a^b \varphi\, dx \qquad (a \leqslant \xi \leqslant b) \tag{6}$$

Theorem 4. *If f is an integrable nonnegative function on $[a, b]$, $x_0 \in [a, b]$ is a point of continuity of f and $f(x_0) > 0$ then*

$$\int_a^b f\, dx > 0$$

Indeed, the conditions of the theorem imply that there is a number $\lambda > 0$ and a closed interval $\sigma \subset [a, b]$ containing x_0 such that $f(x) \geqslant \lambda$ on σ. Let δ denote the closure of the set $[a, b]-\sigma$: $\delta = \overline{[a, b]-\sigma}$ (it consists of one or two closed intervals). Then

$$\int_a^b f\, dx = \int_\sigma f\, dx + \int_\delta f\, dx \geqslant \int_\sigma f\, dx \geqslant \lambda\, |\sigma| > 0$$

where $|\sigma|$ denotes the length of the interval σ.

§ 9.9. Integral as Function of Its Upper Limit. Newton-Leibniz Theorem

Let f be an integrable function defined on a closed interval $[a, b]$. It should be noted that

$$\int_a^b f(x)\, dx = \int_a^b f(u)\, du$$

This means that the value of the definite integral $\int_a^b f(x)\, dx$ does not depend on the notation of the variable of integration, that is its value does not depend on the letter (x, u and the like) written under the sign of f in the definite integral of f over the interval $[a, b]$. Such a variable (in this case the variable of integration x) is said to be *dummy*.

Let us take an arbitrary value $x \in [a, b]$ and take the new function $F(x) = \int_a^x f(t)\, dt$. It is defined for all the values of x belonging to $[a, b]$ because, as is known, if the integral of f on $[a, b]$ exists then the integral of f on $[a, x]$ also exists for any x satisfying the relation $a \leqslant x \leqslant b$. We remind the reader that, by definition,

$$F(a) = \int_a^a f\,dt = 0 \tag{1}$$

and also note that

$$F(b) = \int_a^b f(t)\,dt$$

Let us show that the function F is *continuous* on $[a, b]$. Indeed, if $x \in [a, b]$ and $x+h \in [a, b]$ then

$$F(x+h)-F(x) = \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt = \int_x^{x+h} f(t)\,dt$$

and, denoting $K = \sup |f(t)|$ $(a \leqslant t \leqslant b)$, we obtain

$$|F(x+h)-F(x)| \leqslant \left|\int_x^{x+h} f(t)\,dt\right| \leqslant K\,|h| \to 0 \qquad (h \to 0)$$

Hence, F is continuous on $[a, b]$, irrespective of whether or not the integrand f has discontinuities, provided that f is integrable on $[a, b]$.

Fig. 9.1 shows the graph of f. The area of the variable figure $aABx$ is equal to $F(x)$. Its increment $F(x+h)-F(x)$ is equal to the area of the figure $xBC(x+h)$ which, by the boundedness of f, obviously tends to zero as $h \to 0$ irrespective of whether x is a point of continuity or a point of discontinuity of f; for instance, this increment tends to zero for the point $x = d$ which is a point of discontinuity of f.

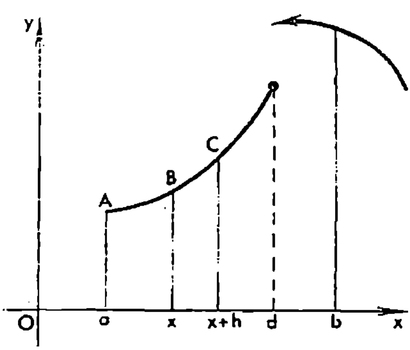

Fig. 9.1

Now let us suppose that *the function f is not only integrable on $[a, b]$ but also continuous at a point $x \in [a, b]$.* We shall show that *in this case F has a derivative at the point x equal to $f(x)$*:

$$F'(x) = f(x) \tag{2}$$

Indeed, for that point x we have (see the explanations below)

$$\frac{F(x+h)-F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)\,dt = \frac{1}{h}\int_x^{x+h} [f(x)+\eta(t)]\,dt =$$

$$= \frac{1}{h}\int_x^{x+h} f(x)\,dt + \frac{1}{h}\int_x^{x+h} \eta(t)\,dt = f(x)+o(1) \qquad (h \to 0) \quad (3)$$

Here we have put $f(t) = f(x)+\eta(t)$; since $f(x)$ as function of the variable t is a constant, we have $\int_x^{x+h} f(x)\,dt = f(x)h$. Further, by the continuity of f at the point x, for any $\varepsilon > 0$ there is $\delta > 0$ such that $|\eta(t)| < \varepsilon$ for all t satisfying the inequality $|x-t| < \delta$. Therefore

$$\left|\frac{1}{h}\int_x^{x+h} \eta(t)\,dt\right| \leqslant \frac{1}{|h|}|h|\,\varepsilon = \varepsilon \quad \text{for } |h| < \delta$$

which shows that the left-hand member of this inequality is $o(1)$ as $h \to 0$.

The passage to the limit in (3) as $h \to 0$ shows that the derivative of F at the point x does exist and that equality (2) takes place. In the cases $x = a$ and $x = b$ we must of course speak of the right-hand and the left-hand derivative of F respectively.

If f is a continuous function on $[a, b]$, then what has been proved shows that the corresponding function

$$F(x) = \int_a^x f(t)\,dt \tag{4}$$

has a derivative at each point $x \in [a, b]$ equal to $f(x)$: $F'(x) = f(x)$ for $a \leqslant x \leqslant b$. Consequently, in this case the function $F(x)$ is an antiderivative of f on $[a, b]$.

We have thus proved that *every function f defined and continuous on a closed interval $[a, b]$ has an antiderivative on that interval given by equality* (4). This shows that, as was mentioned (see § 8.1), *every function continuous on a closed interval possesses an antiderivative on that interval.*

Let $\Phi(x)$ be an arbitrary antiderivative of the function $f(x)$ on $[a, b]$. We know that $\Phi(x) = F(x)+C$ where C is a constant. On putting $x = a$ in this equality and taking into account that $F(a) = 0$ we get $\Phi(a) = C$. Consequently, $F(x) = \Phi(x)-\Phi(a)$. But we have

$$\int_a^b f(x)\,dx = F(b)$$

and therefore

$$\int_a^b f(x)\,dx = \Phi(b)-\Phi(a) = \Phi(x)\Big|_{x=a}^{x=b} \tag{5}$$

We have proved the following important theorem:

Theorem 1 (Newton-Leibniz). *If f is a continuous function on a closed interval $[a, b]$ and Φ is its arbitrary antiderivative on that interval then there holds equality* (5).

By Lagrange's theorem on finite increments, it follows from (5) that

$$\int_a^b f(x)\,dx = f(\xi)(b-a) \qquad (a<\xi<b)$$

where ξ is a point belonging to (a, b). We thus obtain a more precise relation than equality (6) in § 9.8 taken for $\varphi(x) \equiv 1$ which only asserts that $\xi \in [a, b]$.

Theorem 1 admits of the following generalization:

Theorem 2. *If $F(x)$ is a continuous piecewise smooth function on $[a, b]$ then*

$$F(b)-F(a) = \int_a^b F'(x)\,dx \tag{6}$$

Proof. Let us take the partition

$$a = x_0 < x_1 < \ldots < x_n = b$$

where $x_1, \ldots, x_{n-1}$ are the points of discontinuity (of the first kind!) of the derivative $F'(x)$ (see § 5.15). Then we have (see the explanations below)

$$F(b)-F(a) = \sum_0^{n-1} [F(x_{j+1})-F(x_j)] = \sum_{j=0}^{n-1} \int_{x_j}^{x_j+1} F'(x)\,dx = \int_a^b F'(x)\,dx \tag{7}$$

The second equality in (7) holds because we have

$$F(x_{j+1})-F(x_j) = \int_{x_j}^{x_{j+1}} F'(x)\,dx \tag{8}$$

for any j, for the derivative $F'(x)$ exists and is continuous in the open interval (x_j, x_{j+1}) and besides, the limits $F'(x_j+0)$ and $F'(x_{j+1}-0)$ are sure to exist and are equal to the right-hand derivative of F at the point $x = a$ and to the left-hand derivative of F at $x = b$ respectively.

The integrability of F' on each of the closed intervals $[x_j, x_{j+1}]$ implies its integrability on $[a, b]$ and the last equality (7).

Remark. The fact that the function F' is not defined at the points $x_1, \ldots$ $\ldots, x_{n-1} \in [a, b]$ does not violate its integrability on $[a, b]$ (this question is discussed in more detail in § 9.11).

Theorem 3 (on Integration by Parts). *For continuous piecewise smooth functions $u(x)$ and $v(x)$ defined on $[a, b]$ the formula of integration by parts takes place:*

$$\int_a^b u'(x)\,v(x)\,dx = u(x)\,v(x)\Big|_a^b - \int_a^b u(x)\,v'(x)\,dx \tag{9}$$

Indeed, the product $u(x)v(x)$ is also a continuous piecewise smooth function on $[a, b]$ and thus it has a derivative everywhere on $[a, b]$ except possibly at finite number of points, the derivative being given by the formula

$$(u(x)\, v(x))' = u(x)\, v'(x)+u'(x)\, v(x)$$

Taking into account that the functions $u'(x)\, v(x)$ and $u(x)\, v'(x)$ are also integrable on $[a, b]$ we see that, by the foregoing theorem,

$$u(x)\, v(x)\Big|_a^b = \int_a^b u'(x)\, v(x)\, dx+\int_a^b u(x)\, v'(x)\, dx$$

whence follows (9).

Theorem 4 (on Integration by Change of Variable). *If a function $x = \varphi(t)$ is continuously differentiable on $[c, d]$, $a = \varphi(c)$, $b = \varphi(d)$ and the values $\varphi(t)$ $(c \leqslant t \leqslant d)$ of the function φ belong to an interval $[A, B]$ of continuity of a function $f(x)$ defined on $[a, b]$* (i.e. $[a, b] \subset [A, B]$) *then*

$$\int_a^b f(x)\, dx = \int_c^d f[\varphi(t)]\, \varphi'(t)\, dt \tag{10}$$

Indeed, let $F(x)$ and $\Phi(t)$ be some antiderivatives of the functions $f(x)$ and $f[\varphi(t)]\, \varphi'(t)$ respectively. Then (see § 8.1 (2)) we have the identity $\Phi(t) = F[\varphi(t)]+C$, $c \leqslant t \leqslant d$, where C is a constant.

We now see that, by the Newton-Leibniz theorem, (10) follows from the obvious equality

$$\Phi(d)-\Phi(c) = F(b)-F(a)$$

Example 1. According to the Newton-Leibniz theorem, we have $\int_0^\pi \sin x\, dx = -\cos x \Big|_0^\pi = 2$ because the function $\sin x$ is continuous on $[0, \pi]$ and $-\cos x$ is its antiderivative.

Example 2. By Theorem 2, we have

$$\int_a^b \operatorname{sgn} t\, dt = |t|\Big|_a^b = |b|-|a| \tag{11}$$

since $|x|$ is a continuous piecewise smooth function on the closed interval $[a, b]$ (or even a smooth function if $ab \geqslant 0$) and $\operatorname{sgn} x$ is its derivative existing everywhere on $[a, b]$ except at the point $x = 0$ (provided the latter belongs to $[a, b]$).

In particular (11) implies

$$\int_0^x \operatorname{sgn} t\, dt = |x|$$

§ 9.10. Second Mean Value Theorem*

Theorem. *If a function φ is nonnegative and nondecreasing on a closed interval $[a, b]$ and a function f is integrable on $[a, b]$ there exists a point $\xi \in [a, b]$ such that*

$$\int_a^b \varphi(x)f(x)\,dx = \varphi(b)\int_\xi^b f(x)\,dx \tag{1}$$

Proof. Let us first consider the special case when φ has the continuous derivative φ' on $[a, b]$. Integrating by parts we receive

$$\int_a^b \varphi(x)f(x)\,dx = -\varphi(x)\int_x^b f(u)\,du\,\Big|_a^b + \int_a^b \varphi'(x)\int_x^b f(u)\,du\,dx =$$

$$= \varphi(a)\int_a^b f(u)\,du + \int_a^b \varphi'(x)\int_x^b f(u)\,du\,dx \tag{2}$$

Let

$$m = \min_{a\leq x\leq b}\int_x^b f(u)\,du \quad \text{and} \quad M = \max_{a\leq x\leq b}\int_x^b f(u)\,du$$

then the right-hand side of (2) is not greater than $\left(\varphi(a)+\int_a^b \varphi'(x)\,dx\right)M =$ $= \varphi(b)M$, since $\varphi(a)$, $\varphi'(x) \geqslant 0$, and not less than $\left(\varphi(a)+\int_a^b \varphi'(x)\,dx\right)m =$ $= \varphi(b)m$. Consequently there is $\xi \in [a, b]$ for which equality (1) holds.

If φ is a nondecreasing nonnegative function, generally speaking, discontinuous, then it is integrable on $[a, b]$, and there is a sequence of continuously differentiable nondecreasing nonnegative functions $\psi_n(x)$ for which (see § 18.2 and 18.5, Vol. 2)

$$\int_a^b |\varphi(x)-\psi_n(x)|\,dx \to 0, \qquad n \to \infty \tag{3}$$

By what has already been proved, for any n there is a point $\xi_n \in [a, b]$ such that

$$\int_a^b \psi_n(x)\,f(x)\,dx = \varphi(b)\int_{\xi_n}^b f(x)\,dx \tag{4}$$

---

* This theorem is also known as *Bonnet's theorem*. P. O. Bonnet (1819-1892), a French mathematician. — *Tr.*

The sequence $\{\xi_n\}$ contains a subsequence convergent to a point $\xi \in [a, b]$. Consequently, since the integral on the left-hand side as function of the lower limit ξ_n is continuous from the right and since

$$\left| \int_a^b \psi_n f\, dx - \int_a^b \varphi f\, dx \right| \leqslant K \int_a^b |\psi_n - \varphi|\, dx \to 0, \qquad n \to \infty, K \geqslant |f(x)|$$

we obtain (1) from (4) on passing to the limit as $n \to \infty$ in the latter.

§ 9.11. Modification of Integrand Function

Theorem. *If the values of an integrable function on* $[a, b]$ *are changed at a finite number of points belonging to that interval the new (modified) function is integrable and the value of the integral remains unchanged.*

Proof. It is clear that the new function is representable in the form

$$f_1(x) = f(x) + \varphi(x)$$

where φ is equal to zero everywhere on $[a, b]$ except at the points mentioned in the statement of the theorem. It is also clear that φ is integrable and that the integral of φ over $[a, b]$ is equal to zero. Therefore f_1 is integrable and

$$\int_a^b f_1\, dx = \int_a^b f\, dx + \int_a^b \varphi\, dx = \int_a^b f\, dx$$

Up till now, when speaking of an integrable function f on $[a, b]$, we supposed that f was defined throughout $[a, b]$. The theorem proved above shows that the integrability of f does not depend on the values which f assumes at a finite system of points of the interval $[a, b]$. Therefore it is allowable to suppose that f is not defined at those points at all. In this sense we can speak of the integrability of a bounded function f on $[a, b]$ which is in fact defined on the set obtained from $[a, b]$ by deleting a finite number of points; in this very sense we can speak of the integrability of $\sin \frac{1}{x}$ or of $\frac{\sin x}{x}$ on the interval $[0, 1]$. Both functions are only defined in the half-open interval $(0, 1]$ where they are bounded and continuous but we can say that they are integrable on $[0, 1]$.

If a function f defined and integrable on an interval $[a, b]$ is redefined on a countable set of points, the modified function f_1 may be nonintegrable on $[a, b]$. For instance, the function $f(x) \equiv 1$ is Riemann integrable on $[0, 1]$, its integral being equal to 1 (its any integral sum is equal to 1). But if we replace its values at the rational points by zero the new function f_1 is the Dirichlet function which is not Riemann integrable: its any upper Darboux sum is equal to 1 while any lower Darboux sum is equal to 0. But this is not always the case.

Let e be the set on which a given function f (which is supposed to be integrable on $[a, b]$) is redefined and let f_1 denote the new (modified) function. Then $f(x) = f_1(x)+\varphi(x)$ where $\varphi(x) = 0$ on $[a, b]-e$. If the set e is such that the integral over $[a, b]$ of any such function φ (i.e. of any function which is bounded on $[a, b]$ and assumes zero values on $[a, b]-e$) exists and is equal to zero $\left(\int_a^b \varphi\, dx = 0\right)$ then the function f can obviously be arbitrarily redefined on e without violating its integrability and changing the value of the integral. In such a case we can speak of the integral of the function f irrespective of whether or not it is in fact defined on e. We then say that the integral $\int_a^b f(x)\, dx$ taken over the interval $[a, b]$ exists although the function f itself may only be defined on $[a, b]-e$.

Example. The function $\psi(x) = \sin\left(\sin \frac{1}{x}\right)^{-1}$ is defined on $[0, 1]-e$ where e is the set consisting of the point 0 and of the points $x_k = \frac{1}{k\pi}$ $(k=1, 2, \ldots)$; this set is obviously countable. If we extend ψ to e by putting it equal to any numbers at these points on condition that these (new) values form a bounded set the resultant function $\psi_1(x)$ is defined throughout the interval $[0,1]$ and is integrable on it because it is bounded on $[0,1]$ and continuous everywhere on $[0, 1]$ except at the points of the set e of Lebesgue measure zero.

The integrability of ψ is also implied by the following theorem.

Theorem. *A function f bounded on $[a, b]$ and integrable on any closed interval $[a, x]$ where $a < x < b$ is integrable on $[a, b]$.*

The same assertion holds if $[a, x]$ is replaced by $[x, b]$.

Proof. Let us choose an arbitrary $\varepsilon > 0$. Let $|f(x)| \leqslant M$ on $[a, b]$ and $\delta = \frac{\varepsilon}{4M}$. Since f is integrable on the interval $[a, b-\delta]$ there exists a partition R' of this interval $(R' = \{a = x_0 < x_1 < \ldots < x_{n-1} = b-\delta\}$ such that

$$\bar{S}_{R'}-\underline{S}_{R'} = \sum_{j=0}^{n-2} (M_j-m_j)\, \Delta x_j < \frac{\varepsilon}{2}$$

Let us also take the partition $R = \{a = x_0 < x_1 < \ldots < x_{n-1} < x_n = b\}$ of the interval $[a, b]$. Then

$$\bar{S}_R-\underline{S}_R=(\bar{S}_{R'}-\underline{S}_{R'})+(M_{n-1}-m_{n-1})\delta < \frac{\varepsilon}{2}+2M\delta = \varepsilon$$

which shows (see § 9.4, the key theorem) that f is integrable on $[a, b]$.

In the above example the function ψ is bounded on $[0, 1]$ and integrable on any closed interval $[\delta, 1]$ with $\delta > 0$. For there are not more than a finite number of points of discontinuity of the function ψ on $[\delta, 1]$.

§ 9.12. Improper Integrals

Let a function f be defined in a half-open finite interval $[a, b)$. We shall suppose that it is integrable on any closed interval $[a, b']$ with $b' < b$ and is unbounded in a neighbourhood of the point b. Then the ordinary (Riemann) integral of f on $[a, b)$ or, which is the same, on $[a, b]$ cannot exist since a Riemann integrable (in the proper sense) function on $[a, b]$ must be bounded. But it may happen that the limit $\lim_{b'\to b} \int_a^{b'} f(x)\,dx$ exists. If it does exist we call it *the improper integral of f on the interval* $[a, b]$ and write

$$\int_a^b f(x)\,dx = \lim_{b'\to b} \int_a^{b'} f(x)\,dx \tag{1}$$

In such a case we say that *the (improper) integral* $\int_a^b f\,dx$ *exists* or that *it is convergent*. If otherwise, i.e. if this limit does not exist, we say that *the integral is divergent* or, synonymously, that it *does not exist* (in the sense of an improper Riemann integral).

Now let us suppose that a function f is defined in a semi-infinite interval $[a, \infty)$ and is integrable on every finite closed interval $[a, b']$ where $a < b' < \infty$. Then if the limit

$$\lim_{b'\to\infty} \int_a^{b'} f(x)\,dx$$

exists we call it *the improper integral of f on* $[a, \infty)$ and write

$$\int_a^{\infty} f(x)\,dx = \lim_{b'\to\infty} \int_a^{b'} f(x)\,dx$$

We shall use the following terminology. A (formal) expression

$$\int_a^b f(x)\,dx \tag{2}$$

will be referred to as *the integral (of f) with only one singularity at the point b* if the following conditions hold: if b is a finite point then the function f is integrable on $[a, b']$ for any b' satisfying the inequalities $a < b' < b$ and is unbounded in a neighbourhood of the point b; if $b = +\infty$ it is only required that the function f should be integrable on $[a, b']$ for any finite $b' > a$.

An integral of the form $\int_a^b f(x)\,dx$ with only one singularity at the point a is understood in the same sense. In this case b is a finite point, and if the point a ($a < b$) is also finite it is required that the function f should be unbounded in a neighbourhood of the point a and integrable on every closed interval $[a', b]$ where $a < a' < b$; if $a = -\infty$ the function f is only supposed to be integrable on $[a', b]$ for any $a' < b$.

For definiteness, we shall deal with integral (2) with only one singularity at the point b which can be finite or infinite. By analogy with this case, all the conclusions can readily be extended to the case of integral (2) with only one singularity at the point a.

Theorem. *Let there be given an integral of type* (2) *with only one singularity at the point b. For the existence of the integral it is necessary and sufficient that the conditions of Cauchy's criterion hold: given any $\varepsilon > 0$, there is $b_0 < b$ such that*

$$\left| \int_{b'}^{b''} f(t)\, dt \right| < \varepsilon \tag{3}$$

for any b' and b'' satisfying the inequalities $b_0 < b' < b'' < b$.

Proof. Let us consider the function

$$F(x) = \int_a^x f(t)\, dt \qquad (a < x < b)$$

The existence of integral (2) is equivalent to the existence of the limit $\lim\limits_{\substack{x\to b\\ x<b}} F(x)$, which, in its turn, is equivalent to the conditions of Cauchy's criterion for the existence of the limit of a function which reads: for any $\varepsilon > 0$ there is b_0, $a < b_0 < b$, such that for all b' and b'' satisfying the inequalities $b_0 < b' < b'' < b$ the inequality $|F(b'') - F(b')| < \varepsilon$ holds. But

$$F(b'') - F(b') = \int_{b'}^{b''} f(t)\, dt$$

and thus we see that the assertion of the theorem holds.

Example 1. The integral

$$\int_0^1 \frac{dx}{x^\alpha} \tag{4}$$

where $\alpha > 0$ is a constant number obviously has only one singularity at the point $x = 0$. To find whether the integral is convergent we must consider the corresponding limit:

$$\lim_{\substack{\varepsilon\to 0\\ \varepsilon>0}} \int_\varepsilon^1 \frac{dx}{x^\alpha} = \lim_{\varepsilon\to 0} \frac{1}{1-\alpha} \cdot \frac{1}{x^{\alpha-1}} \Bigg|_\varepsilon^1 = \lim_{\varepsilon\to 0} \frac{1}{1-\alpha} [1-\varepsilon^{1-\alpha}] = \begin{cases} \dfrac{1}{1-\alpha} & \text{for } \alpha < 1 \\ \infty & \text{for } \alpha > 1 \end{cases}$$

Thus, integral (4) is convergent for $\alpha < 1$ and is equal to $(1-\alpha)^{-1}$ in this case and is divergent for $\alpha > 1$.

For $\alpha = 1$ it is also divergent:

$$\lim_{\varepsilon\to 0} \int_{\varepsilon}^{1} \frac{dx}{x} = -\lim_{\varepsilon\to 0} \ln \varepsilon = +\infty$$

Example 2. For the integral $\int_1^\infty \frac{dx}{x^\alpha}$ we have

$$\int_1^\infty \frac{dx}{x^\alpha} = \lim_{N\to\infty} \int_1^N \frac{dx}{x^\alpha} = \frac{1}{1-\alpha} \lim_{N\to\infty} x^{1-\alpha}\Big|_1^N = \begin{cases} \frac{1}{\alpha-1} & \text{for } \alpha > 1 \\ +\infty & \text{for } \alpha < 1 \end{cases}$$

Hence, it is convergent for $\alpha > 1$ and divergent for $\alpha < 1$.

The integral $\int_1^\infty \frac{dx}{x}$ (the case $\alpha = 1$) is divergent:

$$\int_1^\infty \frac{dx}{x} = \lim_{N\to\infty} \int_1^N \frac{dx}{x} = \lim_{N\to\infty} \ln N = +\infty$$

Let us come back to the general expression of an integral

$$\int_a^b f(x)\,dx \tag{5}$$

having only one singularity at the point b. In this case the integral

$$\int_a^b f\,dx \tag{6}$$

where $a < c < b$ also has only one singularity at the point b. It is obvious that the conditions of Cauchy's criterion for the existence of the integral are stated in exactly the same manner for both integrals (5) and (6). Consequently, these integrals are simultaneously convergent or divergent. For $a < c < b$ we also have the obvious relation

$$\int_a^b f\,dx = \lim_{b'\to b} \int_a^{b'} f\,dx = \lim_{b'\to b} \left(\int_a^c f\,dx + \int_c^{b'} f\,dx \right) =$$

$$= \int_a^c f\,dx + \lim_{b'\to b} \int_c^{b'} f\,dx = \int_a^c f\,dx + \int_c^b f\,dx \tag{7}$$

where $\int\limits_a^c$ is an ordinary (proper) Riemann integral while the integrals $\int\limits_a^b$ and $\int\limits_c^b$ are improper.

We also note that for any constants A and B the equality

$$\int\limits_a^b (Af+B\varphi)\,dx = \lim_{b'\to b}\int\limits_a^{b'} (Af+B\varphi)\,dx = A\lim_{b'\to b}\int\limits_a^{b'} f\,dx+B\lim_{b'\to b}\int\limits_b^{b'} \varphi\,dx =$$

$$= A\int\limits_a^b f\,dx+B\int\limits_a^b \varphi\,dx \tag{8}$$

takes place. It should be understood in the sense that if the integrals on the right-hand side exist then the integral on the left-hand side also exists and equality (8) holds.

An integral of type (5) (with only one singularity at the point b) is said to be *absolutely convergent* if the integral

$$\int\limits_a^b |f(x)|\,dx < \infty \tag{9}$$

of the modulus of the function $f(x)$ is convergent.

Let us prove that *an absolutely convergent integral is convergent.* Suppose that integral (9) is convergent. Then, given any $\varepsilon > 0$, there is a point b_0 belonging to the open interval (a, b) such that the inequalities $b_0 < b' < < b'' < b$ imply

$$\varepsilon > \int\limits_{b'}^{b''} |f(x)|\,dx \geqslant \left|\int\limits_{b'}^{b''} f(x)\,dx\right|$$

that is integral (5) satisfies the conditions of Cauchy's criterion.

Since

$$\left|\int\limits_a^{b'} f(x)\,dx\right| \leqslant \int\limits_a^{b'} |f(x)|\,dx$$

we conclude, on passing to the limit as $b' \to b$, that for an absolutely convergent integral of type (5) the inequality

$$\left|\int\limits_a^b f(x)\,dx\right| \leqslant \int\limits_a^b |f(x)|\,dx \tag{10}$$

is always fulfilled.

A convergent improper integral may not be absolutely convergent* (see

---

* A convergent improper integral which is not absolutely convergent is said to be *conditionally convergent*. — *Tr.*

the examples in §§ 9.14 and 9.15). But, of course, a convergent improper integral of a nonnegative function is always absolutely convergent.

The following theorem can be proved quite easily.

Theorem. *If a function $F(x)$ is continuous on a closed interval $[a, b]$ and has the continuous derivative $F'(x)$ on $[a, b)$ then*

$$F(b)-F(a) = \lim_{\substack{b'\to b\\ b'<b}} [F(b')-F(a)] = \lim_{\substack{b'\to b\\ b'<b}} \int_a^{b'} F'(x)\,dx = \int_a^b F'(x)\,dx$$

where the integral on the right-hand side may be proper or improper.

For instance, we have

$$\sqrt{x}\Big|_0^1 = \frac{1}{2}\int_0^1 \frac{dx}{\sqrt{x}}$$

Here the singularity is at the left end point of the interval $[0, 1]$.

§ 9.13. Improper Integrals of Nonnegative Functions

Let us consider the integral

$$\int_a^b f(x)\,dx \tag{1}$$

with only one singularity at the point b, and let $f(x) \geqslant 0$ in the interval of integration $[a, b)$.

In this case the function

$$F(b') = \int_a^{b'} f(x)\,dx \qquad (a < b' < b)$$

dependent on b' is obviously monotone nondecreasing. Therefore, if it is bounded, i.e. if $F(b') \leqslant M$ $(a < b' < b)$ for some $M > 0$, integral (1) is convergent (exists):

$$\int_a^b f(x)\,dx = \lim_{b'\to b} \int_a^{b'} f(x)\,dx \leqslant M$$

If F is unbounded integral (1) is divergent (does not exist):

$$\int_a^b f(x)\,dx = \lim_{b'\to b} \int_b^{b'} f(x)\,dx = +\infty$$

(in such a case it is sometimes said that integral (1) is *divergent to* $+\infty$).

If $f(x) \geqslant 0$ on $[a, b)$ we write

$$\int_a^b f(x)\,dx < \infty \quad \text{or} \quad \int_a^b f\,dx = \infty$$

depending on whether the integral is convergent or divergent (to $+\infty$).

Theorem 1. *Let each of the integrals*

$$\int_a^b f(x)\,dx \tag{1}$$

and

$$\int_a^b \varphi(x)\,dx \tag{2}$$

have only one singularity at the point b and let the inequalities

$$0 \leqslant f(x) \leqslant \varphi(x) \tag{3}$$

be fulfilled in the interval $[a, b)$.

Then the convergence of integral (2) *implies the convergence of integral* (1) *and the inequality*

$$\int_a^b f\,dx \leqslant \int_a^b \varphi\,dx$$

while the divergence of integral (1) *implies the divergence of integral* (2).

Proof. Inequalities (3) imply that

$$\int_a^{b'} f\,dx \leqslant \int_a^{b'} \varphi\,dx \tag{4}$$

for $a < b' < b$. If integral (2) is convergent the right member of (4) does not exceed integral (2) and therefore the left member is also bounded above. Since the left member is monotone nondecreasing as b' increases, it tends to a limit which is equal to the integral $\int_a^b f\,dx$:

$$\int_a^b f\,dx = \lim_{b'\to b} \int_a^{b'} f\,dx \leqslant \int_a^b \varphi\,dx$$

Now, if integral (1) is divergent the limit of the left member of (4) as $b' \to \infty$ is equal to ∞, and consequently the limit of the right member is also equal to ∞.

Theorem 2. *Let each of the integrals* (1) *and* (2) *have only one singularity at the point b, let the integrands be positive and the limit*

$$\lim_{x\to b}\frac{f(x)}{\varphi(x)} = A > 0 \tag{5}$$

exist.

Then the integrals are simultaneously convergent or divergent.

Proof. It follows from (5) that for any positive ε such that $\varepsilon < A$ there is $c \in [a, b)$ such that

$$A-\varepsilon < \frac{f(x)}{\varphi(x)} < A+\varepsilon \qquad (c < x < b)$$

and, since $\varphi(x) > 0$, we have

$$(A-\varepsilon)\,\varphi(x) < f(x) < (A+\varepsilon)\,\varphi(x) \qquad (c < x < b) \tag{6}$$

The convergence of the integral $\int_a^b \varphi\, dx$ implies the convergence of the integral $\int_c^b \varphi\, dx$ and also the convergence of the integral $\int_c^b (A+\varepsilon)\varphi\, dx$. Therefore, by virtue of the foregoing theorem, the integral $\int_c^b f\, dx$ also converges and, together with it, so does the integral $\int_a^b f dx$. Conversely, if $\int_a^b f\, dx$ is convergent then $\int_c^b \varphi\, dx$ is also convergent because, together with (5), there holds the equality

$$\lim_{x\to b}\frac{\varphi(x)}{f(x)} = \frac{1}{A} > 0$$

Remark 1. Theorem 2 admits of the following generalization. Let, as before, the functions f and φ satisfy the conditions of Theorem 2 and let ψ be a continuous nonnegative function defined on (a, b). Then the integrals $\int_a^b f\psi\, dx$ and $\int_a^b \varphi\psi\, dx$ are simultaneously convergent or divergent. To prove this proposition one should make use of the inequalities

$$(A-\varepsilon)\,\varphi(x)\,\psi(x) \leqslant f(x)\,\psi(x) \leqslant (A+\varepsilon)\,\varphi(x)\,\psi(x)$$

implied by (6).

Remark 2. If $A = 0$ in Theorem 2, the convergence of the integral $\int_a^b \varphi\, dx$ implies that of the integral $\int_a^b f\, dx$, which follows from the second inequality (6).

Remark 3. It is allowable to assume in the statement of Theorem 2 that only one of the functions f and φ is positive on $[a, b)$ because (6) shows that under this assumption the other function is positive on $[c, b)$ for some c.

Examples. Let us agree that the symbol $\sim$ placed between two integrals means that they are simultaneously convergent or divergent. In the examples below, when writing $\sim$, we resort to Theorem 2.

$$\text{(i)} \quad \int_0^1 \frac{dx}{\ln(1+\sqrt{x})} = \int_0^1 \frac{dx}{\sqrt{x}+o(\sqrt{x})} \sim \int_0^1 \frac{dx}{\sqrt{x}} < \infty$$

$$\text{(ii)} \quad \int_0^1 \frac{dx}{x-\ln(1+x)} = \int_0^1 \frac{dx}{\frac{x^2}{2}+o(x^2)} \sim \int_0^1 \frac{dx}{x^2} = \infty$$

The singularities in integrals (i) and (ii) are only at the point $x = 0$. In the denominators in the integrands of these integrals we have written the principal power terms (see §§ 4.10 and 5.11) and then applied Theorem 2. Integral (i) is convergent while integral (ii) is divergent.

$$\text{(iii)} \quad \int_1^\infty x^\alpha e^{-x^\beta}\, dx = \int_1^\infty (x^{\alpha+2}e^{-x^\beta}) \frac{1}{x^2}\, dx \leqslant K \int_1^\infty \frac{dx}{x^2} < \infty \qquad (\beta > 0)$$

The function in the parentheses is continuous on $[1, \infty)$ and tends to zero for $x \to \infty$; therefore it is bounded above on $[1, \infty)$ by a constant K. Consequently, this integral having only one singularity (at $x = \infty$) is convergent.

§ 9.14. Integration by Parts

Let two continuous functions $\varphi(x)$ and $\psi(x)$ be defined in an interval $[a, \infty)$ and, besides, let ψ be continuously differentiable. Denoting by $\Phi(x)$ an arbitrary antiderivative of $\varphi(x)$ we can write

$$\int_a^N \varphi(x)\,\psi(x)\, dx = \psi(N)\,\Phi(N) - \psi(a)\,\Phi(a) - \int_a^N \psi'(x)\,\Phi(x)\, dx,$$

$$a < N < \infty \tag{1}$$

If the improper integral

$$\int_a^\infty \psi'(x)\,\Phi(x)\, dx = A \tag{2}$$

exists and if the limit

$$\lim_{x\to+\infty} \psi(x)\,\Phi(x) = B \tag{3}$$

exists then the improper integral

$$\int_a^b \varphi(x)\,\psi(x)\,dx = B-\psi(a)\,\Phi(a)-A \tag{4}$$

also exists.

Below we give two special sufficient tests for the existence of integral (2) and limit (3) and thus for the existence of integral (4).

(i) Let the function $\Phi(x)$ be bounded, i.e.

$$|\Phi(x)| \leqslant M \tag{5}$$

and let the function $\psi(x)$ tend to zero as $x \to \infty$:

$$\psi(x) \to 0 \qquad (x \to \infty) \tag{6}$$

Also, let

$$\int_a^\infty |\psi'(x)|\,dx < \infty \tag{7}$$

then integral (2) and limit (3) exist.

Indeed, under these conditions, integral (2) is convergent and even absolutely convergent because

$$\int_a^\infty |\psi'(x)\,\Phi(x)|\,dx \leqslant M \int_a^\infty |\psi'(x)|\,dx < \infty$$

and we also have

$$|\psi(x)\,\Phi(x)| \leqslant M\,|\psi(x)| \to 0 \qquad (x \to \infty)$$

in this case. Consequently integral (4) converges and $B = 0$.

(ii) *Dirichlet's test.* This test requires that the function Φ should satisfy inequality (5) and the function ψ should be decreasing on $[a, \infty)$ and tend to zero as $x \to \infty$. As before, the function ψ is supposed to be continuously differentiable, and therefore the above requirement implies that its derivative is nonpositive. Condition (6) holds in this case, and condition (7) also holds because the limit

$$\lim_{N\to\infty} \int_a^N |\psi'(x)|\,dx = -\lim_{N\to\infty} \int_a^N \psi'(x)\,dx = \lim_{N\to\infty} [\psi(a)-\psi(N)] = \psi(a)$$

exists. Consequently, Dirichlet's test is a particular case of Test (i).

Example. The only singularity of the integral

$$\int_0^\infty \frac{\sin x}{x}\,dx \tag{8}$$

is at the point $x = \infty$. One should take into account that the function $\frac{\sin x}{x}$

has a removable discontinuity at the point $x = 0$. If we put the function equal to 1 at that point the (new) function will be continuous. Integral (8) converges because, by Dirichlet's test, the integral

$$\int_1^{\infty} \frac{\sin x}{x}\, dx$$

converges since the function $1/x$ is monotone decreasing, tends to zero as $x \to \infty$ and has the continuous derivative $-(x)^{-2}$ while the function $\sin x$ is continuous and its antiderivative $-\cos x$ is bounded. It should be stressed that although integral (8) is convergent it is not absolutely convergent (see § 9.15, Example 1); hence, it is conditionally convergent.

§ 9.15. Relationship Between Improper Integrals and Series*

Let us consider the integral

$$\int_a^b f(x)\, dx \tag{1}$$

with only one singularity at the point b. Let us take a sequence

$$a = b_0 < b_1 < b_2 < \ldots < b, \qquad b_k \to b$$

Together with integral (1) we shall consider the series

$$\int_{b_0}^{b_1} f\, dx + \int_{b_1}^{b_2} f\, dx + \ldots = \sum_{k=0}^{\infty} \int_{b_k}^{b_{k+1}} f\, dx \tag{2}$$

whose kth term is

$$u_k = \int_{b_k}^{b_{k+1}} f\, dx$$

Theorem 1. *If integral* (1) *converges so does series* (2) *and there holds the equality*

$$\int_a^b f\, dx = \sum_0^{\infty} \int_{b_k}^{b_{k+1}} f\, dx \tag{3}$$

For we have

$$\lim_{n\to\infty} \sum_0^n \int_{b_k}^{b_{k+1}} f\, dx = \lim_{n\to\infty} \int_{b_0}^{b_{n+1}} f\, dx = \int_a^b f\, dx$$

* In this section we use some elementary notions of the theory of series (see § 11.1).

If f is nonnegative on $[a, \infty)$ then, conversely, the convergence of series (2) implies the convergence of integral (1). Indeed, let the series be convergent and its sum be equal to S. Then for any b' such that $a < b' < b$ there is $n = n(b')$ such that $b' < b_n$. Therefore, taking into account that $f(x) \geqslant 0$, we can write

$$\int_a^{b'} f\,dx \leqslant \int_a^{b_n} f\,dx = \sum_{k=0}^{n-1} \int_{b_k}^{b_{k+1}} f\,dx \leqslant S$$

which means that the integral of the nonnegative function f on the left-hand side is bounded above and, consequently, improper integral (1) exists. Therefore, as was proved above, equality (3) is fulfilled.

In the case when the function f does not retain sign on $[a, b)$ the convergence of series (2) does not, generally, imply the convergence of the integral. For instance, the series

$$\sum_{k=0}^{\infty} \int_{2k\pi}^{2(k+1)\pi} \sin x\,dx = \sum_{0}^{\infty} 0 = 0$$

is convergent while the integral $\int_0^\infty \sin t\,dt$ is divergent because the integral

$$\int_0^x \sin t\,dt = 1 - \cos x$$

as function of x has no limit as $x \to \infty$.

Theorem 2. *If f is a continuous nonincreasing function on $[0, \infty)$ the integral*

$$\int_0^\infty f(x)\,dx \tag{3}$$

and the series

$$\sum_{k=0}^{\infty} f(k) = f(0)+f(1)+f(2)+\ \dots \tag{4}$$

are simultaneously convergent or divergent.

Proof. We have the obvious inequalities

$$f(k+1) \leqslant \int_k^{k+1} f(x)\,dx \leqslant f(k), \qquad (k = 0, 1, \dots)$$

On summing them with respect to k we obtain

$$\sum_{1}^{n+1} f(k) = \sum_{0}^{n} f(k+1) \leqslant \int_0^{n+1} f(x)\,dx \leqslant \sum_{0}^{n} f(k)$$

Since all the terms in these relations are monotone nondecreasing as n increases, we arrive at the assertion of the theorem.

From this theorem it follows that the series

$$1+1/2^{\alpha}+1/3^{\alpha}+\ \dots \tag{5}$$

is convergent for $\alpha > 1$ and divergent for $\alpha \leqslant 1$ because the function $1/(1+x)^{\alpha}$ is monotone decreasing on $[0, \infty)$ for $\alpha > 0$ and

$$\int_0^{\infty} \frac{dx}{(1+x)^{\alpha}} < \infty \quad \text{if} \quad \alpha > 1 \quad \text{and} \quad \int_0^{\infty} \frac{dx}{(1+x)^{\alpha}} = \infty \quad \text{if} \quad \alpha \leqslant 1$$

In the case $\alpha \leqslant 0$ it is readily seen that series (5) diverges.

Theorem 3. *Let the functions $\varphi(x)$ and $\psi(x)$ entering into the integral*

$$\int_0^{\infty} \frac{\varphi(x)}{\psi(x)}\, dx \tag{6}$$

be continuous and let $\psi(x)$ be positive and increasing (nondecreasing). Also let $\varphi(x)$ be nonnegative and periodic with period l and let $\int_0^{l} \varphi(x)\, dx > 0$. Then the integral

$$\int_0^{\infty} \frac{dx}{\psi(x)} \tag{7}$$

and integral (6) *are simultaneously convergent or divergent.*

Proof. Let us represent formally integral (6) in the form of the series

$$\int_0^{\infty} \frac{\varphi(x)}{\psi(x)}\, dx = \sum_{k=0}^{\infty} \lambda_k \quad \text{where} \quad \lambda_k = \int_{kl}^{(k+1)l} \frac{\varphi(x)}{\psi(x)}\, dx \tag{8}$$

By the periodicity of φ, we have

$$\int_{kl}^{(k+1)l} \varphi(x)\, dx = \int_0^{l} \varphi(kl+u)\, du = \int_0^{l} \varphi(u)\, du = \mu > 0$$

and, since ψ increases, we obviously have

$$\frac{\mu}{\psi((k+1)l)} \leqslant \lambda_k < \frac{\mu}{\psi(kl)} \qquad (k = 0, 1, \dots) \tag{9}$$

The integral on the left-hand side of (8) converges or diverges simultaneously with the series on the right-hand side of (8) which, in its turn, converges or diverges, by virtue of (9), simultaneously with the series $\sum_{k=0}^{\infty} \frac{1}{\psi(kl)}$. Finally, by the foregoing theorem, the last series converges or diverges simultaneously with integral (7) and thus the theorem has been proved.

Example 1. By Theorem 3, we have

$$\int_{\pi}^{\infty} \frac{|\sin x|}{x}\, dx = \infty$$

Here when using this theorem we put $l = \pi$, $\varphi(x) = |\sin x|$ and $\psi(x) = x$.

Example 2. Let us consider the integral

$$\int_{1}^{\infty} \frac{\sin x}{x^{\alpha}+\sin x}\, dx \qquad (\alpha > 0) \tag{10}$$

Its only singularity is at the point $x = \infty$. It is absolutely convergent for $\alpha > 1$:

$$\int_{2}^{\infty} \frac{|\sin x|}{|x^{\alpha}+\sin x|}\, dx \leqslant \int_{2}^{\infty} \frac{dx}{x^{\alpha}-1} < \infty$$

because $x^{\alpha} \sim x^{\alpha}-1\ (x \to \infty)$ and $\int_{2}^{\infty} \frac{dx}{x^{\alpha}} < \infty$.

For $\alpha \leqslant 1$ integral (10) is not absolutely convergent because, by virtue of the last theorem,

$$\int_{\pi}^{\infty} \frac{|\sin x|}{|x^{\alpha}+\sin x|}\, dx \geqslant \int_{\pi}^{\infty} \frac{|\sin x|}{x^{\alpha}+1}\, dx = \infty$$

Indeed, the function $|\sin x|$ is continuous and periodic with period π and $\int_{0}^{\pi} |\sin x|\, dx = \int_{0}^{\pi} \sin x\, dx = 2 > 0$ while the function $x^{\alpha}+1$ is continuous and increasing, and $\int_{\pi}^{\infty} (x^{\alpha}+1)^{-1}\, dx = \infty$.

But nevertheless integral (10) is convergent (not absolutely, i.e. conditionally) for $1/2 < \alpha \leqslant 1$. Indeed, the integration by parts gives us

$$\int_{1}^{N} \frac{\sin x}{x^{\alpha}+\sin x}\, dx = -\cos x \left.\frac{1}{x^{\alpha}+\sin x}\right|_{1}^{N} - \int_{1}^{N} \frac{\cos x\,(\alpha x^{\alpha-1}+\cos x)}{(x^{\alpha}+\sin x)^2}\, dx$$

where the first term on the right-hand side has a finite limit as $N \to \infty$ for $\alpha > 0$ and the second term is the sum of the two integrals

$$I'_N = -\int_{1}^{N} \frac{\cos^2 x}{(x^{\alpha}+\sin x)^2}\, dx \quad \text{and} \quad I''_N = \alpha \int_{1}^{N} \frac{\cos x \cdot x^{\alpha-1}}{(x^{\alpha}+\sin x)^2}\, dx$$

We have

$$\int_2^\infty \frac{|\cos x|\, x^{\alpha-1}}{(x^\alpha+\sin x)^2}\, dx \le \int_2^\infty \frac{x^{\alpha-1}}{(x^\alpha-1)^2}\, dx < C\int_2^\infty \frac{dx}{x^{1+\alpha}} < \infty$$

and therefore I''_N tends to a finite limit as $N \to \infty$ for any $\alpha > 0$, and the problem thus reduces to the investigation of I'_N.

For $\alpha > 0$ the integral $\int\limits_\pi^\infty \frac{\cos^2 x}{(x^\alpha+\sin x)^2}\, dx$ converges or diverges simultaneously with the integral

$$\int_\pi^\infty \frac{\cos^2 x}{x^{2\alpha}}\, dx \tag{11}$$

(see Remark 1 at the end of § 9.13) because $x^\alpha + \sin x \sim x^\alpha$ $(x \to \infty)$. As to integral (11), it is convergent or divergent simultaneously with the integral $\int\limits_\pi^\infty \frac{dx}{x^{2\alpha}}$ for which we have

$$\int_\pi^\infty \frac{dx}{x^{2\alpha}} < \infty \quad \text{if} \quad \frac{1}{2} < \alpha \quad \text{and} \quad \int_\pi^\infty \frac{dx}{x^{2\alpha}} = +\infty \quad \text{if} \quad \frac{1}{2} \geqslant \alpha$$

(see the foregoing theorem).

Hence, the limit of I'_N as $N \to \infty$ only exists when $\alpha > 1/2$ and therefore integral (10) is only convergent for $\alpha > 1/2$.

§ 9.16. Improper Integrals with Singularities at Several Points

Let (a, b) be an open interval (finite or infinite) on which a function f is defined such that the integral

$$\int_a^b f(x)\, dx \tag{1}$$

has singularities only at the points a and b. This means that either $a = -\infty$ or a is a finite point and the function f is unbounded in its neighbourhood and, similarly, either $b = +\infty$ or b is a finite point and the function f is unbounded in its neighbourhood; besides, the function f is integrable on any closed interval $[a', b']$ such that $a < a' < b' < b$.

An arbitrary point c belonging to the interval (a, b) divides it into two subintervals (a, c) and (c, b).

The integral

$$\int_a^c f(x)\,dx \tag{2}$$

has a singularity only at the point a; the integral

$$\int_c^b f(x)\,dx \tag{3}$$

also has only one singularity (at the point b). We already know in what sense the existence (convergence) of integrals of types (2) and (3) is understood.

Now we state the definition of the existence (convergence) of integral (1): this integral exists (converges) if and only if each of the integrals (2) and (3) exists and, by definition,

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

provided that both (2) and (3) are convergent integrals.

The convergence or divergence of the integral understood in the sense of this definition does not in fact depend on the choice of c. For if $a < c < c' < b$ then

$$\int_c^b = \int_c^{c'} + \int_{c'}^b \tag{4}$$

where $\int_c^{c'}$ is a proper integral, and, similarly,

$$\int_a^c + \int_c^{c'} = \int_a^{c'} \tag{5}$$

On adding together (4) and (5) and cancelling $\int_c^{c'}$ we obtain

$$\int_a^c + \int_c^b = \int_a^{c'} + \int_{c'}^b$$

Let us proceed to a more complicated case. Let us take (by now purely formally) an integral

$$\int_a^b f(x)\,dx \tag{6}$$

where the interval (a, b) is finite or infinite. Suppose that this interval can be split with the aid of points of division $a = c_0 < c_1 < c_2 < \ldots < c_{n-1} < < c_n = b$ into a finite number n of subintervals (c_k, c_{k+1}) such that each of the integrals

$$\int_{c_k}^{c_{k+1}} f(x)\,dx \qquad (k = 0, 1, \ldots, n-1) \tag{7}$$

has a singularity at only one of the end points c_k and c_{k+1}.

Then, if all improper integrals (7) exist (converge) we regard (by definition) integral (6) as existing (convergent) and put

$$\int_a^b f(x)\,dx = \sum_{k=0}^{n-1} \int_{c_k}^{c_{k+1}} f\,dx$$

If at least one of integrals (7) is divergent, integral (6) is regarded as divergent (nonexisting).

Similarly, we say that integral (6) is absolutely convergent if and only if all integrals (7) are absolutely convergent.

In connection with the question under consideration the following remark should be taken into account. Let us suppose that, for instance, the point a is finite, b is a finite number exceeding a $(a < b < \infty)$ and the integral

$$\int_a^\infty f\,dx \tag{8}$$

has, in addition to the point $x = \infty$, one more singularity at the point b.

Let $b < c < \infty$. Then, in accordance with the above, improper integral (8) (we suppose that it exists) can also be defined in the following way:

$$\int_a^\infty f\,dx = \int_a^b f\,dx + \int_b^c f\,dx + \int_c^\infty f\,dx =$$

$$= \lim_{\varepsilon_1 \to 0} \int_a^{b-\varepsilon_1} f\,dx + \lim_{\varepsilon_2 \to 0} \int_{b+\varepsilon_2}^{c} f\,dx + \lim_{\varepsilon_3 \to 0} \int_c^{1/\varepsilon_3} f\,dx =$$

$$= \lim_{\varepsilon_1, \varepsilon_2, \varepsilon_3 \to 0} \left\{ \int_a^{b-\varepsilon_1} f\,dx + \int_{b+\varepsilon_2}^{c} f\,dx + \int_c^{1/\varepsilon_3} f\,dx \right\} \qquad (\varepsilon_1, \varepsilon_2, \varepsilon_3 > 0) \tag{9}$$

This means that if improper integral (8) exists then the limit of the expression in the braces on the right-hand side of (9) also exists when the positive variables ε_1, ε_2 and ε_3 tend to zero independently of each other*. It is im-

* Here we deal with the limit of a function of three variables (ε_1, ε_2 and ε_3) defined on the set of points with positive coordinates as the variable point (belonging to that set) tends in an arbitrary way to the point with zero coordinates (to the origin).

portant to stress that the converse assertion also holds, that is if the limit of the right-hand side of (9) exists when ε_1, ε_2 and ε_3 independently tend to zero retaining positive values, each of the three limits entering into the middle member of (9) also exists, that is improper integral (8) is convergent (exists). To prove what has been stated, let us denote

$$\varphi_1(\varepsilon_1) = \int\limits_a^{b-\varepsilon_1} f\,dx, \quad \varphi_2(\varepsilon_2) = \int\limits_{b+\varepsilon_2}^{c} f\,dx, \quad \varphi_3(\varepsilon_3) = \int\limits_c^{1/\varepsilon_3} f\,dx$$

Suppose that the limit

$$\lim_{\varepsilon_1, \varepsilon_2, \varepsilon_3 \to 0} \{\varphi_1(\varepsilon_1)+\varphi_2(\varepsilon_2)+\varphi_3(\varepsilon_3)\}$$

is known to exist. Then the conditions of Cauchy's criterion must hold: given any $\eta > 0$, there is $\delta > 0$ such that if $0 < \varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_1', \varepsilon_2', \varepsilon_3' < \delta$ then

$$|\varphi_1(\varepsilon_1)+\varphi_2(\varepsilon_2)+\varphi_3(\varepsilon_3)-\varphi_1(\varepsilon_1')-\varphi_2(\varepsilon_2')-\varphi_3(\varepsilon_3')| < \eta$$

In particular, here it is allowable to put $\varepsilon_2 = \varepsilon_2'$ and $\varepsilon_3 = \varepsilon_3'$; then we see that for any $\eta > 0$ there is $\delta > 0$ such that $|\varphi_1(\varepsilon_1)-\varphi_1(\varepsilon_1')| < \eta$ for $0 < \varepsilon_1, \varepsilon_1' < \delta$, which shows that the limit

$$\lim_{\varepsilon_1 \to 0} \varphi_1(\varepsilon_1) \tag{10}$$

exists. The existence of the other two limits is proved analogously, that is the limits

$$\lim_{\varepsilon_2 \to 0} \varphi_2(\varepsilon_2) \quad \text{and} \quad \lim_{\varepsilon_3 \to 0} \varphi_3(\varepsilon_3) \tag{11}$$

also exist.

Thus, we have proved that for integral (8) to exist it is necessary and sufficient that the limit on the right-hand side of (9) exist when ε_1, ε_2 and ε_3 tend to zero independently of each other retaining positive sign.

The assumption that the variables ε_1, ε_2 and ε_3 are independent is essential for this proposition. If, for instance, it is only known that the limit on the right-hand side of (9) exists when $\varepsilon_1 = \varepsilon_2 = \varepsilon_3 \to 0$ this does not, generally speaking, imply the existence of each of the limits (10) and (11) taken separately.

These considerations can be extended in a natural way to all the other possible types of location of the singularities of the integral.

As an instance, let us consider the integral

$$\int\limits_{-1}^{1} \frac{dx}{x} \tag{12}$$

It has only one singularity (at the point $x = 0$). It does not exist (is divergent) since neither of the integrals $\int\limits_{-1}^{0} \frac{dx}{x}$ and $\int\limits_{0}^{1} \frac{dx}{x}$ exists separately.

In accordance with the discussion we can also say that integral (12) does not exist because the limit

$$\int_{-1}^{-\varepsilon_1} \frac{dx}{x} + \int_{\varepsilon_2}^{1} \frac{dx}{x} = \ln|x|\Big|_{-1}^{-\varepsilon_1} + \ln|x|\Big|_{\varepsilon_2}^{1} = \ln\frac{\varepsilon_1}{\varepsilon_2}$$

does not exist when ε_1 and ε_2 tend to zero independently of each other retaining positive sign.

We thus conclude that the improper (Riemann) integral of the function $1/x$ on the closed interval $[-1, 1]$ does not exist.

Here we also mention an important generalization of the improper integral in the sense of *Cauchy's principal value*; in this sense the integral in question is understood as the limit

$$\text{v.p.} \int_{-1}^{1} \frac{dx}{x} = \lim_{\varepsilon\to 0}\left\{\int_{-1}^{\varepsilon} \frac{dx}{x} + \int_{\varepsilon}^{1} \frac{dx}{x}\right\} = 0$$

(here take we $\varepsilon = \varepsilon_1 = \varepsilon_2$!; the notation "v.p." originates from French *valeur principale* — principal value).

§ 9.17. Taylor's Formula with Integral Form of the Remainder

Let f be a function defined in an open interval containing a point a and let the function f have continuous piecewise smooth derivative of the $(r-1)$th order in this interval. Then the derivatives $f^0 \equiv f, f', f'', \ldots, f^{r-1}$ exist on that interval and are continuous and the derivative $f^{(r)}(x)$ exists throughout the indicated interval except at a finite number of points and is a piecewise continuous function (see § 5.15). For any point x belonging to this interval there holds Taylor's formula

$$f(x) = \sum_{k=0}^{r-1} \frac{f^{(k)}(a)}{k!}(x-a)^k + R(x) \tag{1}$$

with the remainder in the integral form

$$R(x) = \frac{1}{(r-1)!}\int_a^x (x-t)^{r-1} f^{(r)}(t)\,dt \qquad (0! = 1) \tag{2}$$

Indeed, successive integrations by parts of $R(x)$ yield

$$R(x) = -\frac{f^{(r-1)}(a)}{(r-1)!}(x-a)^{r-1} + \frac{1}{(r-2)!}\int_a^x (x-t)^{r-2} f^{(r-1)}(t)\,dt =$$

$$= -\frac{f^{(r-1)}(a)}{(r-1)!}(x-a)^{r-1} - \frac{f^{(r-2)}(a)}{(r-2)!}(x-a)^{r-2} +$$

$$+\frac{1}{(r-3)!}\int\limits_a^x (x-t)^{r-3}f^{(r-2)}(t)\,dt = \ldots$$

$$\ldots = -\sum_0^{r-1}\frac{f^{(k)}(a)}{k!}(x-a)^k+f(x)$$

The substitution $t = a+(x-a)u$, $dt = (x-a)\,du$ in integral (2) leads to the following expression for the remainder:

$$R(x) = (x-a)^r\,\psi(x)$$

where

$$\psi(x) = \frac{1}{(r-1)!}\int\limits_0^1 (1-u)^{r-1}f^{(r)}\big(a+(x-a)\,u\big)\,du \tag{3}$$

If $f^{(r)}(x)$ is continuous then $\psi(x)$ is also a continuous function of x because integral (3) satisfies the conditions of the theorem on the continuity of an integral dependent on a parameter proved in § 12.13, Vol. 2. If the function f possesses continuous derivatives of higher order $f^{(r+s)}(x)$ $(s > 0)$ then it is legitimate to differentiate $\psi(x)$ s times under the integral sign (see § 13.12). Therefore, in this case $\psi(x)$ is s times continuously differentiable. This conclusion cannot be drawn from Lagrange's form of the remainder in Taylor's formula because it involves the function θ whose differentiability properties are unknown *a priori*.

§ 9.18. Wallis'* and Stirling's** Formulas

Wallis' formula is written as

$$\sqrt{\pi} = \lim_{m\to\infty}\frac{(m!)^2 2^{2m}}{(2m)!\sqrt{m}} \tag{1}$$

To derive it we take the integral $\int\limits_0^{\pi/2}\sin^n x\,dx$ and integrate by parts:

$$\int\limits_0^{\pi/2}\sin^n x\,dx = -\sin^{n-1}x\cos x\Big|_0^{\pi/2}+(n-1)\int\limits_0^{\pi/2}\sin^{n-2}x\cos^2 x\,dx =$$

$$= (n-1)\int\limits_0^{\pi/2}\sin^{n-2}x\,dx-(n-1)\int\limits_0^{\pi/2}\sin^n x\,dx$$

* J.. Wallis (1616-1703), an English mathematician.
** J Stirling (1692-1770), a Scotch mathematician.

On transposing the second integral on the right-hand side to the left and dividing by n we receive

$$\int_0^{\pi/2} \sin^n x\,dx = \frac{n-1}{n}\int_0^{\pi/2} \sin^{n-2} x\,dx \tag{2}$$

Proceeding from an even n and an odd n and applying in succession equality (2) (which reduces the power of $\sin x$ by 2) we obtain respectively

$$\int_0^{\pi/2} \sin^{2m} x\,dx = \frac{2m-1}{2m}\,\frac{2m-3}{2m-2}\cdots\frac{1}{2}\int_0^{\pi/2} dx = \frac{(2m-1)!!}{2m!!}\,\frac{\pi}{2}$$

and

$$\int_0^{\pi/2} \sin^{2m+1} x\,dx = \frac{2m}{2m+1}\,\frac{2m-2}{2m-1}\cdots\frac{2}{3}\int_0^{\pi/2} \sin x\,dx = \frac{(2m)!!}{(2m+1)!!}$$

Now, dividing these equalities by each other and putting

$$\mu_m = \frac{\int_0^{\pi/2} \sin^{2m} x\,dx}{\int_0^{\pi/2} \sin^{2m+1} x\,dx} \qquad (m = 1, 2, \ldots)$$

we arrive at the equality

$$\pi = 2\frac{[(2m)!!]^2}{(2m-1)!!\,(2m+1)!!}\,\mu_m \tag{3}$$

Note that for $x \in [0, \pi/2]$ we have

$$\sin^{2m+1} x \leqslant \sin^{2m} x \leqslant \sin^{2m-1} x$$

whence

$$\int_0^{\pi/2} \sin^{2m+1} x\,dx \leqslant \int_0^{\pi/2} \sin^{2m} x\,dx \leqslant \int_0^{\pi/2} \sin^{2m-1} x\,dx$$

On dividing the members of these inequalities by $\int_0^{\pi} \sin^{2m+1} x\,dx$ and applying equality (2) we obtain

$$1 \leqslant \mu_m \leqslant \frac{\int_0^{\pi/2} \sin^{2m-1} x\,dx}{\int_0^{\pi/2} \sin^{2m+1} x\,dx} = \frac{2m+1}{2m} = 1 + \frac{1}{2m}$$

whence it follows that

$$\mu_m \to 1 \qquad (m \to \infty) \tag{4}$$

From (3) and (4) we derive the asymptotic relation

$$\pi \approx 2\frac{[(2m)!!]^2}{(2m-1)!!\,(2m+1)!!} \approx 2\left(\frac{(2m-2)!!}{(2m-1)!!}\right)^2 2m \qquad (m \to \infty)$$

obtained by discarding in succession the factors $\mu_m \to 1$ and $\frac{2m}{2m+1} \to 1$, whence

$$\sqrt{\pi} \approx 2\sqrt{m}\,\frac{(2m-2)!!}{(2m-1)!!} = 2\sqrt{m}\,\frac{[(2m-2)!!]^2}{(2m-1)!} =$$

$$= 2\sqrt{m}\,\frac{[(2m)!!]^2}{(2m)!}\cdot\frac{1}{2m} = \frac{(m!)^2 2^{2m}}{(2m)!\,\sqrt{m}} \qquad (m \to \infty)$$

that is (1) takes place.

Stirling's formula is written as the limiting relation

$$\lim_{n\to\infty}\frac{n!}{\sqrt{2\pi}\,n^{n+1/2}\,e^{-n}} = 1 \tag{5}$$

or, which is the same, as the asymptotic equality

$$n! \approx \sqrt{2\pi}\,n^{n+1/2}\,e^{-n} \qquad (n \to \infty) \tag{6}$$

We shall derive the inequalities

$$\sqrt{2\pi}\,n^{n+(1/2)}\,e^{-n} < n! < \sqrt{2\pi}\,n^{n+(1/2)}\,e^{-n+(1/4n)} \tag{7}$$

which directly imply (5) and (6).

Let us put

$$a_n = \frac{n!}{n^{n+1/2}\,e^{-n}} \qquad (n = 1, 2, \ldots) \tag{8}$$

Then

$$\frac{a_n}{a_{n+1}} = \frac{1}{e}\left(1+\frac{1}{n}\right)^{n+(1/2)}$$

and $$\ln\frac{a_n}{a_{n+1}} = \left(n+\frac{1}{2}\right)\ln\left(1+\frac{1}{n}\right)-1$$

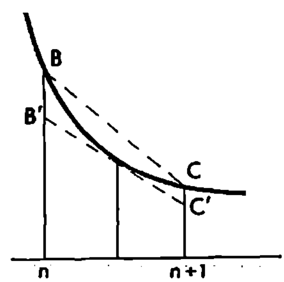

Fig. 9.2

The function $1/x$ is monotone decreasing and its graph is convex downward for $x > 0$; therefore the area of the figure bounded by the graph, by the x-axis and by the straight lines $x = n$ and $x = n+1$ (see Fig. 9.2) is less than the area of the trapezoid $nBC(n+1)$ and greater than the area of the trapezoid $nB'C'(n+1)$ where $B'C'$ is the segment of the tangent to the graph at its point with abscissa $x = n+1/2$:

$$\frac{1}{n+\frac{1}{2}} < \int_n^{n+1}\frac{dx}{x} = \ln\left(1+\frac{1}{n}\right) < \frac{1}{2}\left(\frac{1}{n}+\frac{1}{n+1}\right)$$

On multiplying the members of these inequalities by $n+1/2$ and subtracting 1 from all of them we obtain

$$0 < \left(n+\frac{1}{2}\right)\ln\left(1+\frac{1}{n}\right)-1 < \frac{\left(n+\frac{1}{2}\right)^2}{n(n+1)}-1 = \frac{1}{4n(n+1)} = \frac{1}{4}\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

It follows that $0 < \ln\frac{a_n}{a_{n+1}} < \frac{1}{4}\left(\frac{1}{n}-\frac{1}{n+1}\right)$ whence

$$1 < \frac{a_n}{a_{n+1}} < e^{\frac{1}{4}\left(\frac{1}{n}-\frac{1}{n+1}\right)} \tag{9}$$

Now we substitute $n+j$ $(j = 0, 1, \ldots, k-1)$ for n into these inequalities and multiply them all, which results in

$$1 < \frac{a_n}{a_{n+k}} < e^{\frac{1}{4}\left(\frac{1}{n}-\frac{1}{n+k}\right)} \tag{10}$$

The first inequality (9) shows that the sequence of the positive numbers a_j $(j = 1, 2, \ldots)$ is monotone decreasing and, consequently, tends to a nonnegative limit which we denote by α. From inequalities (10), on passing to the limit as $k \to \infty$, we obtain

$$1 < \frac{a_n}{\alpha} \leqslant e^{1/4n} \tag{11}$$

whence it follows that $\alpha > 0$.

Wallis' formula can obviously be written in terms of the numbers a_n as

$$\sqrt{\pi} = \lim_{n\to\infty}\frac{a_n^2}{a_{2n}\sqrt{2}} = \frac{\alpha}{\sqrt{2}}$$

and consequently

$$\alpha = \sqrt{2\pi} \tag{12}$$

Finally, (8), (11) and (12) imply (7).

CHAPTER 10

Some Applications of Integrals. Approximate Methods

§ 10.1. Area in Polar Coordinates

The area S of the figure bounded by two rays $\theta = \theta_0$ and $\theta = \theta_*$ issued from the pole O and by a curve Γ represented in polar coordinates by a continuous function $\varrho = f(\theta)$ (see Fig. 10.1) can be computed in the following way.

We partition the interval $[\theta_0, \theta_*]$ of variation of θ with the aid of points of division

$$\theta_0 < \theta_1 < \ldots < \theta_n = \theta_*$$

and take the expression $1/2\varrho_k^2 \Delta\theta_k$ $(\Delta\theta_k = \theta_{k+1} - \theta_k)$ as an approximation to the area of the element of the figure bounded by the curve Γ and by the two rays $\theta = \theta_k$ and $\theta = \theta_{k+1}$; this means that we replace the area of the element by the area $k+1$ of the circular sector of radius $\varrho_k = f(\theta_k)$ bounded by the same rays. Now it appears natural to define the area of the figure in question by putting

$$S = \lim_{\max \Delta\theta_k \to 0} \sum_0^{n-1} \varrho_k^2 \Delta\theta_k = \frac{1}{2} \int_{\theta_0}^{\theta_*} \varrho^2 \, d\theta = \frac{1}{2} \int_{\theta_0}^{\theta_*} f^2(\theta) \, d(\theta) \tag{1}$$

Formula (1) expresses the area S of the figure in polar coordinates. As is already known, integral (1) is sure to exist if the function $f(\theta)$ is continuous.

In connection with formula (1) there arises the question as to whether the area S would retain the same numerical value if we computed it in Cartesian coordinates. The general theory of the Jordan measure developed in § 12.4, Vol. 2 gives a positive answer to this question.

Example. The circumference of the circle shown in Fig. 10.2 is described in polar coordinates by the equation $\varrho = 2R \cos \theta$. According to (1), its area is found by computing the integral

$$S = 2R^2 \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta = 4R^2 \int_0^{\pi/2} \frac{1 - \cos 2\theta}{2} \, d\theta = \pi R^2$$

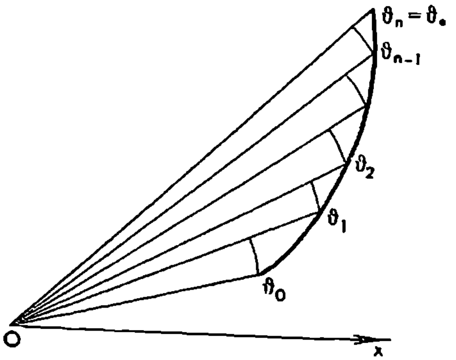

Fig. 10.1

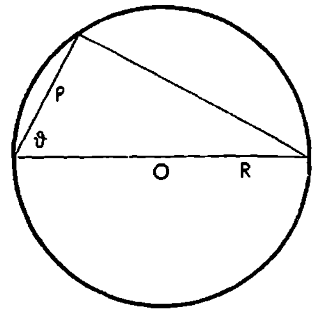

Fig. 10.2

§ 10.2. Volume of a Solid of Revolution

Let Γ be a curve described in rectangular coordinates x, y by a continuous positive function $y = f(x)$ $(a \leqslant x \leqslant b)$. We shall compute the volume V of the solid of revolution bounded by the planes $x = a$ and $x = b$ and by the surface of revolution generated by the rotation of the curve Γ about the x-axis.

Let us partition the interval $[a, b]$ with the aid of points of division $a = x_0 < < x_1 < \ldots < x_n = b$ and assume that the volume of the element ΔV of the solid bounded by the planes $x = x_k$ and $x = x_{k+1}$ is approximately equal to the volume of the circular cylinder with radius $y_k = f(x_k)$ and altitude $\Delta x_k = x_{k+1} - x_k$:

$$\Delta V_k \sim \pi y_k^2 \cdot \Delta x_k = \pi f(x_k)^2 \Delta x_k$$

The sum $V_n = \pi \sum_0^{n-1} f(x_k)^2 \Delta x_k$ gives us an approximation to the sought-for volume V which is ultimately obtained by passing to the limit:

$$V = \lim_{\max \Delta x_k \to 0} \sum_0^{n-1} f(x_k)^2 \Delta x_k = \pi \int_a^b f^2(x)\, dx \tag{1}$$

Formula (1) expresses the volume of the solid of revolution.

We shall also present an alternative derivation of formula (1) based on the notion of the differential of volume. Let $V(x)$ denote the volume of the part of the solid in question lying between the planes passing through the points a and x of the x-axis perpendicularly to the latter (Fig. 10.3). The increment $\Delta V(x)$ of the volume $V(x)$ corresponding to the increment $\Delta x > 0$ of the coordinate x is equal to the volume of the part of the body lying between the planes perpendicular to the x-axis and passing through the points x and $x+\Delta x$.

Let us prove the equality

$$\Delta V = \pi f^2(x)\,\Delta x + o(\Delta x) \qquad (\Delta x \to 0) \qquad (2)$$

To this end we put

$$m = \min_{x \le \xi \le x+\Delta x} f(\xi) \quad \text{and} \quad M = \max_{x \le \xi \le x+\Delta x} f(\xi)$$

Then, obviously,

$$\pi m^2 \Delta x \le \Delta V(x) \le \pi M^2 \Delta x,$$
$$\pi m^2 \Delta x \le \pi f^2(x)\,\Delta x \le \pi M^2 \Delta x \qquad (3)$$

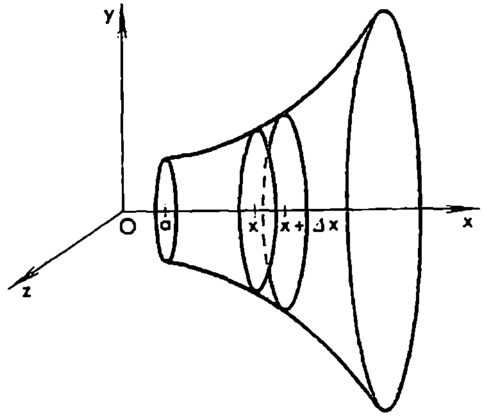

Fig. 10.3

and, since the function $f(x)$ is continuous, $M-m \to 0$ $(\Delta x \to 0)$. This means that

$$\pi(M^2-m^2)\,\Delta x = o(\Delta x) \qquad (\Delta x \to 0) \qquad (4)$$

From (3) and (4) now follows (2).

Equality (2) shows that the first term on its right-hand side is the differential of V:

$$dV = \pi f^2(x)\,\Delta x = \pi f^2(x)\,\Delta x$$

Finally, by the Newton-Leibniz theorem, the sought-for volume V is

$$V = V(b) = V(b) - V(a) = \pi \int_a^b f^2(x)\,dx$$

Example. The ellipsoid of revolution (about the x-axis) described by the equation

$$\frac{x^2}{a^2} + \frac{y^2+z^2}{b^2} = 1$$

is a body bounded by the surface of revolution generated by the rotation about the x-axis of the curve

$$y = b\sqrt{1-\frac{x^2}{a^2}} \qquad (-a \le x \le a)$$

Formula (1) gives the following expression for its volume:

$$V = \pi b^2 \int_{-a}^{a} \left(1-\frac{x^2}{a^2}\right) dx = \pi b^2 \left(x - \frac{x^3}{3a^2}\right)_{-a}^{a} = \frac{4}{3}\pi a b^2$$

§ 10.3. Arc Length of a Smooth Curve

Let Γ be a smooth curve described by functions

$$x = \varphi(t), \quad y = \psi(t), \quad z = \chi(t), \quad a \leq t \leq b \tag{1}$$

(see § 6.5). The smoothness of Γ means that the functions $\varphi(t)$, $\psi(t)$ and $\chi(t)$ have continuous derivatives on $[a, b]$. Let us take a partition $a = t_0 < t_1 < \ldots < t_n = b$ of $[a, b]$ and form the sum (see § 6.8)

$$S_n = \sum_0^{n-1} \sqrt{\Delta x_k^2 + \Delta y_k^2 + \Delta z_k^2}$$

where

$$\Delta x_k = \varphi(t_{k+1}) - \varphi(t_k), \quad \Delta y_k = \psi(t_{k+1}) - \psi(t_k)$$
$$\Delta z_k = \chi(t_{k+1}) - \chi(t_k) \quad \text{and} \quad \delta = \max_k \Delta t_k, \;\; \Delta t_k = t_{k+1} - t_k$$

This sum is obviously equal to the length of the polygonal line inscribed in Γ with vertices at the points corresponding to the values $t = t_k$.

Now we can write

$$S_n = \sum_0^{n-1} \sqrt{\varphi'(\mu_k)^2 + \psi'(\nu_k)^2 + \chi'(\lambda_k)^2}\, \Delta t_k = \sum_0^{n-1} \sqrt{\varphi'(t_k)^2 + \psi'(t_k)^2 + \chi'(t_k)^2}\, \Delta t_k +$$
$$+ \sum_0^{n-1} \varepsilon_k \Delta t_k \to \int_a^b \sqrt{\varphi'(t)^2 + \psi'(t)^2 + \chi'(t)^2}\, dt$$

where μ_k, ν_k and λ_k are some points belonging to the interval (t_k, t_{k+1}): $t_k < \mu_k, \nu_k, \lambda_k < t_{k+1}$ $(k = 0, 1, \ldots, n-1)$.

Here in the first equality we have used Lagrange's formula of finite increments.

To justify the fact that $\sum \varepsilon_k \Delta t_k \to 0$ as $\delta \to 0$ let us consider the auxiliary function

$$\alpha(u, v, w) = \sqrt{\varphi'(u)^2 + \psi'(v)^2 + \chi'(w)^2}$$

which is obviously continuous in the cube $\Delta = \{a \leq u, v, w \leq b\}$. Its modulus of continuity on Δ will be denoted by $\omega(\delta)$. Since the distance between the points (t_k, t_k, t_k) and $(\mu_k, \nu_k, \lambda_k)$ belonging to the cube does not exceed $\delta\sqrt{3}$ we have

$$|\varepsilon_k| = |\alpha(t_k, t_k, t_k) - \alpha(\mu_k, \nu_k, \lambda_k)| \leq \omega(\delta\sqrt{3})$$

and therefore

$$\left| \sum_0^{n-1} \varepsilon_k \Delta t \right| \leq \omega(\delta\sqrt{3}) \sum_0^{n-1} \Delta t_k = (b-a)\,\omega(\delta\sqrt{3}) \to 0, \qquad \delta \to 0$$

We have proved that the arc length of smooth curve (1) exists and is given by the formula

$$S = \int_a^b \sqrt{\varphi'(t)^2+\psi'(t)^2+\chi'(t)^2}\, dt \tag{2}$$

Making the change of variable $t = \lambda(\tau)$ ($\lambda'(\tau) > 0$, $c \leqslant \tau \leqslant d$) in formula (2) where $\lambda(\tau)$ is a continuously differentiable function we obviously obtain

$$S = \int_c^d \sqrt{\varphi_1'(\tau)^2+\psi_1'(\tau)^2+\chi_1'(\tau)^2}\, d\tau \tag{3}$$

where $\varphi_1(\tau) = \varphi(\lambda(\tau))$ etc., which shows that formula (2) expressing the arc length is invariant with respect to change of variable.

If a plane curve is represented by an equation

$$y = f(x), \qquad (a \leqslant x \leqslant b)$$

where f possesses a continuous derivative on $[a, b]$ its arc length is expressed by the formula

$$S = \int_a^b \sqrt{1+f'(x)^2}\, dx$$

which readily follows from (2) if we put $t = x$, $y = f(x)$ and $z = 0$.

Example. Using formula (2) we obtain for arc length of the screw line

$$x = a\cos\theta, \quad y = a\sin\theta, \quad z = b\theta, \qquad (0 \leqslant \theta \leqslant \theta_0)$$

the expression

$$S = \int_0^{\theta_0} \sqrt{a^2+b^2}\, d\theta = \theta_0\sqrt{a^2+b^2}$$

§ 10.4. Area of a Surface of Revolution

Let Γ be a plane curve described by a positive function $y = f(x)$ ($a \leqslant x \leqslant b$) in rectangular coordinates and let f be continuously differentiable on $[a, b]$.

We shall compute the area S of the surface of revolution generated by the rotation of Γ about the x-axis. To this end we take a partition

$$a = x_0 < x_1 < \ldots < x_n = b \tag{1}$$

of the interval $[a, b]$ and inscribe in the curve Γ the polygonal line Γ_n with vertices at the points $(x_k, f(x_k))$. The area of the surface of revolution generated by the rotation of the polygonal line about the x-axis is

$$S_n = \pi\sum_0^{n-1} [f(x_k)+f(x_{k+1})]\sqrt{\Delta x_k^2+\Delta y_k^2}, \quad \Delta y_k = f(x_{k+1})-f(x_k)$$

To find the sought-for area S we pass to the limit as $\max \Delta x_k \to 0$ and receive

$$S = 2\pi \int_a^b f(x)\sqrt{1+f'(x)^2}\, dx = 2\pi \int_a^b y\sqrt{1+y'^2}\, dx \tag{2}$$

Indeed, on taking Δx_k ($\Delta x_k > 0$) outside the root and applying to Δy_k the formula of finite increments we obtain (see the explanations below)

$$S_n = \pi \sum_0^{n-1} [f(x_k)+f(x_{k+1})]\sqrt{1+f'(\xi_k)^2}\,\Delta x_k =$$

$$= 2\pi \sum_0^{n-1} f(x_k)\sqrt{1+f'(x_k)^2}\,\Delta x_k+\alpha \to 2\pi \int_a^b f(x)\sqrt{1+f'(x)^2}\, dx,$$

$$(\max \Delta x_k \to 0,\ x_k < \xi_k < x_{k+1})$$

To prove that

$$\alpha \to 0 \text{ as } \max \Delta x_k \to 0 \tag{3}$$

let us consider the auxiliary function $\Phi(\xi, \eta, \zeta) = [f(\xi)+f(\eta)]\sqrt{1+f'(\zeta)^2}$, which is defined and continuous in the cube $\Delta = \{a \leqslant \xi, \eta, \zeta \leqslant b]$; since the cube Δ is a closed set the function $\Phi(\xi, \eta, \zeta)$ is uniformly continuous in Δ.

It follows that, given any $\varepsilon > 0$, there is $\delta > 0$ such that

$$|\Phi(\xi, \eta, \zeta)-\Phi(\xi, \eta', \zeta')| < \frac{\varepsilon}{(b-a)\pi}$$

provided that $|\eta-\eta'|<\delta$, $|\zeta-\zeta'| < \delta$ and $(\xi, \eta, \zeta), (\xi, \eta', \zeta') \in \Delta$. Therefore, if partition (1) is such that $\max \Delta x_k < \delta$ we have

$$|\Phi(x_k, x_{k+1}, \xi_k)-\Phi(x_k, x_k, x_k)| < \frac{\varepsilon}{(b-a)\pi}$$

and consequently

$$|\alpha| = \left|\pi \sum_0^{n-1} \Phi(x_k, x_{k+1}, \xi_k)\,\Delta x_k-\pi \sum_0^{n-1} \Phi(x_k, x_k, x_k)\,\Delta x_k\right| \leqslant \frac{\varepsilon}{b-a} \sum_0^{n-1} \Delta x_k = \varepsilon$$

which proves (3).

The general definition of area of an arbitrary smooth surface will be stated in § 12.23, Vol. 2.

Example. Applying formula (2) we readily find the area of the surface of revolution generated by the rotation of the segment of the parabola $y = x^2$ $(0 \leqslant x \leqslant 1)$ about the x-axis:

$$S = 2\pi \int_0^1 x^2\sqrt{1+4x^2}\, dx$$

§ 10.5. Lagrange's Interpolation Formula

Consider a function f defined on a closed interval $[a, b]$ and a system of points

$$x_0, x_1, \ldots, x_n \tag{1}$$

belonging to $[a, b]$. Let us consider the following problem: it is required to find a polynomial $P(x) = a_0+a_1x+\ldots+a_nx^n$ of degree n whose values at the indicated points coincide with the values of the function $f(x)$, that is

$$f(x_k) = P(x_k) \qquad (k = 0, 1, \ldots, n) \tag{2}$$

To solve the problem we construct the polynomials

$$Q_k(x) = \frac{(x-x_0)\ldots(x-x_{k-1})(x-x_{k+1})\ldots(x-x_n)}{(x_k-x_0)\ldots(x_k-x_{k-1})(x_k-x_{k+1})\ldots(x_k-x_n)} \qquad (k = 0, 1, \ldots, n) \tag{3}$$

It is obvious that, for every $k = 0, 1, \ldots, n$, the function Q_k is a polynomial of the nth degree equal to 1 at the point x_k and to 0 at the other points of system (1):

$$Q_k(x_j) = \delta_{kj} \qquad (k, j = 0, 1, \ldots, n)$$

where the symbol δ_{kj}* is specified by the relations

$$\delta_{kj} = \begin{cases} 1 & \text{for} \quad k = j \\ 0 & \text{for} \quad k \neq j \end{cases}$$

Let us put

$$P(x) = \sum_0^n Q_k(x) f(x_k) \tag{4}$$

The function $P(x)$ is a polynomial of degree n possessing the properties

$$P(x_j) = \sum_0^n Q_k(x_j) f(x_k) = Q_j(x_j) f(x_j) = f(x_j) \qquad (j = 0, 1, \ldots, n)$$

Thus, it is a solution to the problem in question; besides, it is determined uniquely because if we supposed that there is another polynomial $P_1(x)$ of the nth degree serving as a solution to the same problem, the difference $P(x)-$ $-P_1(x)$ would be a polynomial of the nth degree having $(n+1)$ distinct roots whence it would follow that $P(x)-P_1(x) \equiv 0$.

It should be noted that if the original function f is itself a polynomial of degree n then we have the identity $f(x) \equiv P(x)$ because two polynomials whose values coincide at $(n+1)$ distinct points are identically equal to each other. Relations (3), (4) express the so-called *Lagrange interpolation formula*.

---

* The symbol δ_{kj} is called the *Kroneker delta* after L. Kroneker (1823-1891), a German mathematician. — *Tr.*

§ 10.6. Rectangle and Trapezoid Formulas

Let it be required to compute the definite integral of a continuous function f on an interval $[a, b]$. If its antiderivative is known we can naturally apply the Newton-Leibniz theorem. But, as is known, an antiderivative of a continuous function by far not always can be found analytically, and therefore there arises the problem of an approximate computation of the integral.

One of the simplest methods for approximate computation of a definite integral is directly implied by its definition. Let us divide the interval of integration $[a, b]$ into equal parts with the aid of the points

$$x_k = a+k\frac{b-a}{N} \qquad (k = 0, 1, \ldots, N) \tag{1}$$

and put

$$\int_a^b f(x)\,dx \approx \frac{b-a}{N}\sum_0^{N-1} f\left(\frac{x_k+x_{k+1}}{2}\right) \tag{2}$$

where the sign "$\approx$" indicates an approximate equality.

Relation (2) is referred to as the *rectangle formula* (or the *rectangle rule*; formulas of this kind are termed *quadrature formulas*). Fig. 10.4 shows that in the case of a positive function $f(x)$, for which the integral in question is

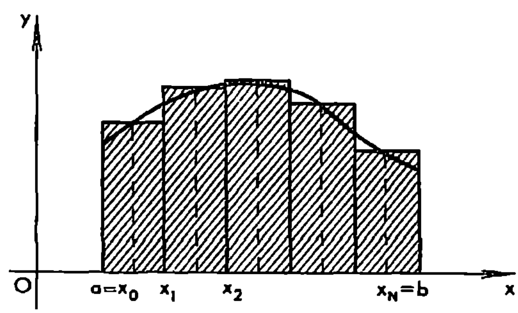

Fig. 10.4

equal to the area of the figure bounded by the curve $y = f(x)$, by the x-axis and by the straight lines $x = a$ and $x = b$, formula (2) gives an approximation to this area equal to the sum of the areas of the rectangles seen in the figure.

As is known, in the case of a continuous function f on $[a, b]$ the limit, as $N \to \infty$, of the right-hand member of formula (2) is exactly equal to its left-hand member; therefore it is natural to expect that the error of the approximate formula (2), that is the absolute value of the difference between its right-hand and left-hand members, is small for large N.

In this connection there arises the question as to how the error of such an approximation can be estimated. As will be shown, such an estimation can readily be found if we require additionally that the function f (which is supposed to be continuous) satisfy some smoothness conditions.

It is important to stress that *if f is a linear function of the form $f(x) = Ax+$ $+B$, formula* (2) *gives an exact equality*: in this case the right-hand side of (2) exactly coincides with its left-hand side. Since a linear function is a polynomial of the first degree we can say that *the rectangle formula provides an exact result for all polynomials of degree not higher than* 1.

Another natural method for computing approximately a definite integral leads to the so-called *trapezoid (quadrature) formula (trapezoid rule)*. According to this method, the interval of integration $[a, b]$ is split into equal parts with the aid of a system of points of type (1) and the approximate expression for the sought-for integral is written as

$$\int_a^b f(x)\,dx \approx \frac{b-a}{N}\left(\frac{f(x_0)+f(x_1)}{2}+\frac{f(x_1)+f(x_2)}{2}+\ldots+\frac{f(x_{N-1})+f(x_N)}{2}\right)=$$

$$=\frac{b-a}{2N}\left(f(x_0)+2f(x_1)+2f(x_2)+\ldots+2f(x_{N-1})+f(x_N)\right) \quad (3)$$

As is seen from Fig. 10.5, the trapezoid formula expresses approximately the sought-for area as the sum of the areas of the trapezoids shown in the figure. It is important to note that *the trapezoid formula provides an exact*

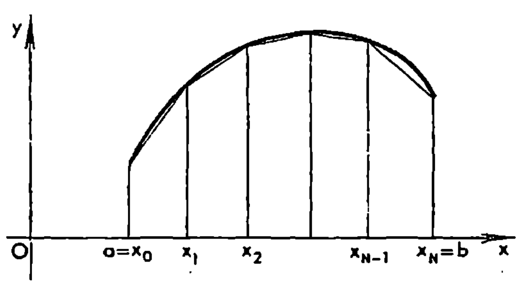

Fig. 10.5

result for any linear function $Ax+B$ (where A and B are constants), that is for any polynomial of degree not higher than 1: if an expression $Ax+B$ is substituted for $f(x)$ into (3) this relation becomes exact. In this sense the trapezoid formula has no advantages over the rectangle formula since they are both exact for linear functions.

§ 10.7. General Quadrature Formula. The Notion of a Functional

In this section we shall generalize the notion of a quadrature formula. Let us take a system of points

$$a \leqslant x_0 < x_1 < \ldots < x_N \leqslant b \quad (1)$$

splitting a given interval $[a, b]$ and a system of numbers

$$p_0, p_1, \ldots, p_N \tag{2}$$

Taking an arbitrary continuous function f defined on $[a, b]$ we put

$$L(f) = \sum_0^N p_k f(x_k) \tag{3}$$

$L(f)$ will be regarded (at present purely formally) as an approximation to the integral of f taken over $[a, b]$

$$\int_a^b f(x)\,dx \approx L(f) \tag{4}$$

An approximate relation of form (4) will be referred to as the *quadrature formula with nodes* (1) *and weights* (2).

Let $\mathfrak{M}$ denote a set of functions f and let there be a law according to which with each function $f \in \mathfrak{M}$ is associated a number $F(f)$. Then we say that F is a *functional* defined on $\mathfrak{M}$. If $\mathfrak{M}$ is a linear space (see § 6.1) and if F possesses the property

$$F(\alpha f_1+\beta f_2) = \alpha F(f_1)+\beta F(f_2)$$

for any numbers α and β and any functions $f_1, f_2 \in \mathfrak{M}$ the functional F (defined on $\mathfrak{M}$) is said to be *linear*.

The set of all continuous functions defined on $[a, b]$ is usually denoted as $C = C(a, b)$. It is a linear space because if α and β are arbitrary numbers and $f_1, f_2 \in C$ then $\alpha f_1+\beta f_2 \in C$. The integral $\int_a^b f\,dx$ can obviously be regarded as a linear functional on C. It is readily seen that the expression $L(f)$ (see (3)) is also a linear functional defined on C. It follows that if equality (4) is exact for a finite number of continuous functions $f_1, \ldots, f_l$ it is automatically exact for the functions *(linear combinations)* $\sum_1^l \alpha_k f_k(x)$ where α_k are arbitrary numbers.

§ 10.8. Simpson's* Formula

In this section we shall derive an important quadrature formula used in approximate calculations known as *Simpson's formula* (or *Simpson's rule*). It is quite simple and at the same time possesses a remarkable property: it is exact for all polynomials of the third degree.

* T. Simpson (1710-1761), an English mathematician.

We shall begin with the auxiliary problem: it is required to determine three numbers A, B and C such that the quadrature formula

$$\int_{-1}^{1} f(x)\,dx \approx Af(-1)+Bf(0)+Cf(1) \tag{1}$$

is exact for the functions 1, x and x^2. On substituting these functions for f into (1) and writing the exact equalities we arrive at the system of three equations

$$\left.\begin{aligned} A+B+C &= 2\\ -A+C &= 0\\ A+C &= \frac{2}{3} \end{aligned}\right\}$$

whence $A = C = 1/3$ and $B = 4/3$. Now, since $A = C$ it can readily be verified that formula (1) with the values of A, B and C found is exact for the function x^3 as well; since the functionals entering into formula (1) are linear (see the foregoing section) the formula is also exact for all polynomials of degree not higher than the third.

For an arbitrary interval $[a, b]$ the generalized formula of type (1) is written as

$$\int_{a}^{b} f(x)\,dx \approx \frac{b-a}{6}\left(f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right) \tag{2}$$

Let us show that formula (2) is exact for all polynomials of the third degree. Indeed, on putting

$$x = \frac{a+b}{2}+t\,\frac{b-a}{2} \quad \text{and} \quad F(t) = f\left(\frac{a+b}{2}+t\,\frac{b-a}{2}\right)$$

in (2) and cancelling by $\frac{b-a}{2}$ we obtain

$$\int_{-1}^{1} F(t)\,dt \approx \frac{1}{3}\,(F(-1)+4F(0)+F(1)) \tag{3}$$

Since formula (3) is exact for the polynomials in t of degree not higher than the third, formula (2) is exact for the polynomials in x of degree not higher than the third because the substitution $x = \frac{a+b}{2}+t\,\frac{b-a}{2}$ transforms the polynomials of degree not higher than the third dependent on x into the polynomials of degree not higher than the third dependent on t.

If we split a given closed interval $[a, b]$ into $2N$ equal parts with the aid of the points of division

$$x_k = a+\frac{b-a}{2N}\,k \qquad (k = 0, 1, \ldots, 2N)$$

and apply formula (2) to each of the subintervals $[x_0, x_2]$, $[x_2, x_4]$, ..., the addition of the results obtained leads to *Simpson's formula*

$$\int_a^b f(x)\,dx \approx \frac{b-a}{6N}\left(f(x_0)+4f(x_1)+2f(x_2)+4f(x_3)+\ldots+f(x_{2N})\right) \quad (4)$$

From the point of view of practical applications the amounts of calculations involved in Simpson's formula and in the rectangle formula are essentially the same. But if the integrand function f is sufficiently smooth then, for large N, the error of the approximation computed by Simpson's formula is considerably smaller than the corresponding error introduced by the rectangle formula (see § 10.9).

§ 10.9. General Method for Estimating Errors of Quadrature Formulas

The formula

$$\int_0^1 f(t)\,dt \approx \sum_0^n p_k f(t_k) = L(f) \quad (1)$$

written for the interval $[0, 1]$ will be referred to as the *standard quadrature formula with nodes*

$$0 \leqslant t_0 < t_1 < \ldots < t_n \leqslant 1 \quad (2)$$

and weights

$$p_0, p_1, \ldots, p_n \quad (3)$$

To standard formula (1) there corresponds the quadrature formula

$$\int_c^d f(x)\,dx \approx (d-c)\sum_0^n p_k f(c+(d-c)\,t_k) = (d-c)L(f(c+(d-c)t)) \quad (4)$$

for the interval $[c, d]$. Its nodes $c+(d-c)\,t_k$ divide the interval $[c, d]$ in the same ratio in which the nodes t_k of the standard formula divide the interval $[0, 1]$, the corresponding weights being $(d-c)p_k$ $(k = 0, 1, \ldots, n)$.

If an interval $[a, b]$ is split into equal parts by the points

$$x_k = a+\frac{b-a}{N}k \qquad (k = 0, 1, \ldots, N)$$

we can apply to each subinterval $[x_k, x_{k+1}]$ the formula

$$\int_{x_k}^{x_{k+1}} f(x)\,dx \approx \frac{b-a}{N} L\left(f\left(x_k+\frac{b-a}{N}\,t\right)\right)$$

corresponding to the standard formula and then perform the summation of its left and right members with respect to k. This results in the formula

$$\int_a^b f(x)\,dx \approx \frac{b-a}{N}\sum_{k=0}^{N-1} L\left(f\left(x_k+\frac{b-a}{N}\,t\right)\right) \tag{5}$$

which will be referred to as the *modified quadrature formula corresponding to standard formula* (1).

To estimate the error introduced by formula (5) let us suppose that the original standard formula is exact for all polynomials of degree not higher than $r-1$.

Let $W^r(a, b)$ denote the class of functions f defined on the closed interval $[a, b]$ and having the continuous piecewise smooth derivative $f^{(r-1)}$ on $[a, b]$ (see the beginning of § 9.17).

If f is a function belonging to $W^r(0, 1)$ ($f \in W^r(0, 1)$) we can write its expansion by Taylor's formula with remainder in integral form (see § 9.17): $f(x) = P(x)+R(x)$ where

$$P(x) = \sum_0^{r-1} a_k x^k$$

and

$$R(x) = \frac{1}{(r-1)!}\int_0^x (x-t)^{r-1} f^{(r)}(t)\,dt = \int_0^1 K(x-t) f^{(r)}(t)\,dt$$

Here we have introduced the function

$$K(u) = \begin{cases} 0 & \text{for} \quad u \leqslant 0 \\ \dfrac{u^{r-1}}{(r-1)!} & \text{for} \quad u > 0 \end{cases}$$

Let us substitute f into standard formula (1) and transpose all the terms to the left-hand side. Since formula (1) is exact for the polynomial $P(x)$ we obtain

$$\int_0^1 f(x)\,dx - L(f) =$$

$$= \int_0^1 \left(\int_0^1 K(x-t) f^{(r)}(t)\,dt\right)dx - \int_0^1 \left(\sum_0^n p_k K(x_k-t)\right) f^{(r)}(t)\,dt =$$

$$= \int_0^1 \left(\int_0^1 K(x-t)\,dx\right) f^{(r)}(t)\,dt - \int_0^1 \left(\sum_0^n p_k K(x_k-t)\right) f^{(r)}(t)\,dt =$$

$$= \int_0^1 \Lambda(t) f^{(r)}(t)\,dt\,^*$$

* Here we have changed the order of integration. The justification of such a procedure is presented in the theory of multiple integrals (see Chapter 12, Vol. 2).

where

$$\Lambda(t) = \int_0^1 K(x-t)\,dx - \sum_0^n p_k K(x_k - t) \tag{6}$$

Now, denoting

$$\varkappa = \int_0^1 |\Lambda(x)|\,dt \tag{7}$$

and

$$\|f^{(r)}\| = \sup_{0\le t\le 1} |f^{(r)}(t)| \tag{8}$$

we receive the inequality

$$\left|\int_0^1 f(x)\,dx - L(f)\right| \le \varkappa \|f^{(r)}\| \tag{9}$$

The function $\Lambda(t)$ and, together with it, the constant $\varkappa$ are dependent on the weights and on the location of the nodes in formula (1) but are independent of the function f. The constant $\varkappa$ can be computed for a given standard formula and then automatically used in calculations. Its value is determined precisely since for the function $f \in W^{(r)}(0, 1)$ having the derivative $f^{(r)}(x) = \operatorname{sgn} \Lambda(x)$ relation (9) turns into an exact equality.

Now let us derive an estimation for the error of formula (4) corresponding to a closed interval $[c, d]$ on condition that $f \in W^r(c, d)$. On transposing the second term in (4) to the left-hand side and making the substitution

$$x = c+(d-c)t, \qquad F(t) = f(c+(d-c)t)$$

we obtain, taking into account that $F^{(r)}(t) = (d-c)^r f^{(r)}(x)$, the equality

$$\left|\int_c^d f(x)\,dx - (d-c)L\big(f(c+(d-c)t)\big)\right| =$$

$$= (d-c)\left|\int_2^1 F(t)\,dt - L(F)\right| \le (d-c)\varkappa \|F^{(r)}\|_{0,1} =$$

$$= (d-c)^{r+1}\varkappa \|f^{(r)}\|_{(c,d)} \tag{10}$$

where $\|\varphi\|_{(c,d)} = \sup\limits_{c\le x\le d} |\varphi(x)|$.

Inequality (10) involves as a factor the above constant $\varkappa$ and, besides, contains the new factor $(d-c)^{r+1}$ dependent on the length $d-c$ of the interval $[c, d]$. This factor tends to zero together with $d-c$, the rate of its decrease increasing together with r.

Finally, let us estimate the error for modified formula (5) on condition that $f \in W^r(a, b)$. Since in this case we have $f \in W^r(x_k, x_{k+1})$ for any k, it follows, by virtue of (10), that

$$\left|\int_a^b f(x)\,dx-\frac{b-a}{N}\sum_{k=0}^{N-1} L\left(f\left(x_k+\frac{b-a}{N}\,t\right)\right)\right| \leqslant$$

$$\leqslant \sum_0^{N-1}\left|\int_{x_k}^{x_{k+1}} f(x)\,dx-\frac{b-a}{N}\,L\left(f\left(x_k+\frac{b-a}{N}\,t\right)\right)\right| \leqslant$$

$$\leqslant \sum_{k=0}^{N-1}\left(\frac{b-a}{N}\right)^{r+1}\varkappa\,\|f^{(r)}\|_{(x_k,\,x_{k+1})} \leqslant \frac{(b-a)^{r+1}\varkappa}{N^r}\,\|f^{(r)}\|_{(a,\,b)} \qquad (11)$$

We shall say that a given modified quadrature formula possesses property T^{r-1} if it is exact for the polynomials of the $(r-1)$th degree. As has been proved, if a function f belongs to $W^r(a,b)$ ($f\in W^r(a,b)=W^r$) and if the integral of that function is approximated on $[a,b]$ by a modified quadrature formula (5) possessing property T^{r-1}, the error of the approximation is of the order of N^{-r}.

Here we also note (without proof) that if $f\in W^k$ and if formula (5) has property T^{r-1} the errors of the approximations for $k<r$ and for $k\geqslant r$ are of the orders of N^{-k} and N^{-r} respectively.

As an example let us take Simpson's formula; it possesses property T^3 and therefore the error of the approximation of the integral of a function $f\in W^k$ with the aid of this formula is of the order of N^{-4} for $k\geqslant 4$ and of the order of N^{-k} for $k<4$.

In conclusion we make the following remark. Let us consider a system of nodes $x_0<x_1<\ldots<x_r$. For this system an arbitrary polynomial $P(x)$ of the rth degree can be represented exactly (identically) with the aid of Lagrange's interpolation formula (3), (4) derived in § 10.5. Putting

$$\lambda_k=\int_a^b Q_k(x)\,dx \qquad (k=0,1,\ldots,n)$$

we obtain the quadrature formula

$$\int_a^b f(x)\,dx\approx\sum_0^r \lambda_k f(x_k)$$

which is exact for any polynomial of degree r.

§ 10.10. More on Arc Length

In this section we suppose without further stipulations that Γ is a continuous curve not intersecting itself, determined by equations

$$x=\varphi(t),\quad y=\psi(t),\quad z=\chi(t),\quad a\leqslant t\leqslant b \qquad (1)$$

where the functions $\varphi(t)$, $\psi(t)$ and $\chi(t)$ are continuous.

Let us introduce the following notation: if ϱ is a partition

$$a = t_0 < t_1 < t_2 < \ldots < t_N = b \tag{2}$$

of the interval $[a, b]$ then

$$\delta_\varrho = \max_k \Delta t_k, \qquad \sigma_\varrho = \max_k \sqrt{\Delta x_k^2 + \Delta y_k^2 + \Delta z_k^2}$$

Γ_ϱ is the polygonal line inscribed in Γ with vertices corresponding to the values $t_j \in \varrho$ and, finally, $|\Gamma|$ and $|\Gamma_\varrho|$ are the lengths of Γ and Γ_ϱ respectively. If the curve Γ is not rectifiable (has no length) we put formally $|\Gamma| = \infty$.

There hold the following properties:

(i) *The relations*

$$\delta_\varrho \to 0 \tag{3}$$

and

$$\sigma_\varrho \to 0 \tag{4}$$

imply each other and, consequently, the arc length of a continuous curve Γ can be defined by means of one of the two equivalent equalities

$$|\Gamma| = \lim_{\delta_\varrho \to 0} |\Gamma_\varrho| = \lim_{\sigma_\varrho \to 0} |\Gamma_\varrho| \tag{5}$$

Indeed, (3) implies (4) because the functions φ, ψ and χ are uniformly continuous on $[a, b]$. Further, equations (1) specify a mapping under which the interval $[a, b]$ of variation of t goes in the set of points (x, y, z) belonging to Γ. This mapping is continuous and therefore the point set Γ is bounded and closed (see Theorem 1 in § 12.20). Since Γ does not intersect itself this mapping establishes a one-to-one correspondence $[a, b] \rightleftharpoons \Gamma$, and consequently, by the same theorem, the inverse mapping is also continuous and thus is described by a continuous function $t = \Phi(x, y, z)$ defined on the bounded closed set Γ which is therefore uniformly continuous on Γ.

(ii) *If $\varrho \prec \varrho'$, that is if all the points of division t_j entering into ϱ ($t_j \in \varrho$) also belong to ϱ', then*

$$|\Gamma_\varrho| \leqslant |\Gamma_{\varrho'}| \tag{6}$$

This is quite obvious. The addition to ϱ of one more point t_j results in the replacement of a segment of the polygonal line Γ_ϱ by two segments forming together with the former segment a triangle (which may be degenerated, i.e. lying in a straight line).

(iii) *There hold the relations*

$$\sup_\varrho |\Gamma_\varrho| = \lim_{\delta_\varrho \to 0} |\Gamma_\varrho| = |\Gamma| \tag{7}$$

The number $|\Gamma|$ specified by them can be finite and then the curve Γ is rectifiable and $|\Gamma|$ is its length (see § 6.8). It is also possible that $|\Gamma| = +\infty$; then the curve Γ is not rectifiable.

To prove (7) let us put $\Lambda = \sup_\varrho |\Gamma_\varrho|$ and choose arbitrary numbers Λ' and Λ'' such that $\Lambda' < \Lambda'' < \Lambda$. According to the property of the supremum, there is a partition $\varrho^* = \{a = t_0^* < t_1^* < \ldots < t_N^* = b\}$ such that $\Lambda'' < |\Gamma_{\varrho^*}|$. Let us take an arbitrary $\varepsilon < \Lambda'' - \Lambda'$ and choose, using the uniform continuity of the functions $\varphi(t)$, $\psi(t)$ and $\chi(t)$ on $[a, b]$, a number $\delta > 0$ such that the inequality

$$\lambda = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} < \frac{\varepsilon}{2N}$$

(where λ is the length of the chord joining the points of Γ corresponding to t and $t+\Delta t$) is fulfilled for all $t, t+\Delta t \in [a, b]$, $|\Delta t| < \delta$.

For any arbitrary partition ϱ of the interval $[a, b]$ with $\delta_\varrho < \delta$ there hold the inequalities (see the explanations below)

$$\Lambda'' < |\Gamma_{\varrho^*}| < |\Gamma_{\varrho+\varrho^*}| < |\Gamma_\varrho| + \varepsilon \tag{8}$$

which imply

$$\Lambda' < |\Gamma_\varrho|$$

The second inequality (8) holds by virtue of Property (ii). The third one is explained as follows. Let AA' be the chord connecting two neighbouring vertices of Γ_ϱ and let ν be the number of vertices of Γ_{ϱ^*} contained inside the arc $\overset{\frown}{AA'} \subset \Gamma$. Let us inscribe in $\overset{\frown}{AA'}$ the polygonal line $\gamma_{AA'}$ with vertices coinciding with the indicated vertices of Γ_{ϱ^*}. The number of segments of this polygonal line is $\nu+1$ and the length of its each segment does not exceed $\frac{\varepsilon}{2N}$; therefore the length of $\gamma_{AA'}$ is not greater than $\frac{\varepsilon(\nu+1)}{2N}$.

If all the chords $\overline{AA'} \subset \Gamma_\delta$ are replaced by the corresponding polygonal lines $\gamma_{AA'}$ as described above for all the arcs $\overset{\frown}{AA'}$ inside which there are points of Γ_{ϱ^*}, this results in the transformation of Γ_ϱ into $\Gamma_{\varrho+\varrho^*}$ and, since the number of vertices of Γ_{ϱ^*} which may fall in an arc $\gamma_{AA'}$ is not greater than N, we have

$$|\Gamma_{\varrho+\varrho^*}| < |\Gamma_\varrho| + \varepsilon$$

that is the third inequality (8) holds. We have proved that for any number $\Lambda' < \Lambda$ there is $\delta > 0$ such that the inequality $\Lambda' < |\Gamma_\varrho| \leqslant \Lambda$ is fulfilled for all Γ_ϱ, with $\delta_\varrho < \delta$. This proves (7) where $|\Gamma| = \Lambda$.

(iv). *If Γ is rectifiable on $[a, b]$ then it is also rectifiable on $[a, c]$ and $[c, b]$, $a < c < b$; conversely, if Γ is rectifiable on $[a, c]$ and on $[c, b]$ it is rectifiable on $[a, b]$ as well. In this case we have*

$$|\Gamma| = |\Gamma_{ac}| + |\Gamma_{cb}| \tag{9}$$

Proof. Let us choose a sequence of partitions ϱ^k of the interval $[a, b]$ with $\delta_{\varrho^k} \to 0$ $(k \to \infty)$. Suppose that Γ_{ϱ^k} contains a point c for any k. Then Γ_{ϱ^k} generates the corresponding partitions ϱ_1^k and ϱ_2^k of $[a, c]$ and $[c, b]$ respectively with $\delta_{\varrho_1^k}, \delta_{\varrho_2^k} \to 0$ $(k \to \infty)$ and

$$|\Gamma_{\varrho^k}| = |\Gamma_{\varrho_1^k}| + |\Gamma_{\varrho_2^k}| \qquad (k = 1, 2, \ldots) \tag{10}$$

If Γ is rectifiable, i.e. $|\Gamma| < \infty$, then $|\Gamma_{\varrho^k}| \to |\Gamma|$. Therefore $|\Gamma_{\varrho_1^k}|$ and $|\Gamma_{\varrho_2^k}|$ are bounded and, since $\delta_{\varrho_1^k}, \delta_{\varrho_2^k} \to 0$, they converge to $|\Gamma_{ac}|$ and $|\Gamma_{cb}|$. Thus Γ_{ac} and Γ_{cb} are rectifiable. On passing to the limit in (10) as $k \to \infty$ we obtain (9). Conversely, if Γ_{ac} and Γ_{cb} are rectifiable then $|\Gamma_{\varrho_1^k}| \to |\Gamma_{ac}|$, $|\Gamma_{\varrho_2^k}| \to |\Gamma_{cb}|$ and therefore, by (10), the limit of $|\Gamma_\varrho{}^k|$ exits. Since $\delta_\varrho{}^k \to 0$ it follows, by (iii), that Γ is rectifiable.

(v). *Let* $\Gamma_\varepsilon\left(0 < \varepsilon < \frac{b-a}{2}\right)$ *denote the arc of Γ corresponding to the interval* $[a+\varepsilon, b-\varepsilon]$. *If Γ_ε is rectifiable for any such ε then*

$$\lim_{\varepsilon\to 0} |\Gamma_\varepsilon| = |\Gamma| \tag{11}$$

Thus, for Γ to be rectifiable it is necessary and sufficient that the limit in (11) *be finite.*

Proof. Let us inscribe in Γ_ε a polygonal line Γ'_ε with segments whose lengths do not exceed ε such that

$$|\Gamma_\varepsilon| - \varepsilon < |\Gamma'_\varepsilon| \leqslant |\Gamma_\varepsilon| \tag{12}$$

and let Γ''_ε be the polygonal line inscribed in Γ and obtained by adding two segments to Γ'_ε. By the continuity of Γ, we have

$$|\Gamma''_\varepsilon| - |\Gamma'_\varepsilon| \to 0, \qquad \varepsilon \to 0 \tag{13}$$

Evidently, $|\Gamma_\varepsilon|$ does not decrease as $\varepsilon \to 0$. Therefore limit (11) (finite or infinite) exists and, by (7), (12) and (13),

$$\lim_{\varepsilon\to 0} |\Gamma_\varepsilon| = \lim_{\varepsilon\to 0} |\Gamma'_\varepsilon| = \lim_{\varepsilon\to 0} |\Gamma''_\varepsilon| = |\Gamma|$$

§ 10.11. The Number π. Trigonometric Functions

Let us consider the circle $x^2+y^2 = 1$. Its upper half (semicircle) Γ is described by the continuous function $f(x) = \sqrt{1-x^2}$, $-1 \leqslant x \leqslant 1$. As to the derivative $f'(x) = -x/\sqrt{1-x^2}$, it is only continuous in the open interval $(-1, 1)$. Therefore at present formula (4) in § 10.3 expressing arc length can only be applied to a closed interval $[-1+\varepsilon, 1-\varepsilon]$ $(0 < \varepsilon < 1)$ on which f is continuous together with its derivative:

$$|\Gamma_\varepsilon| = \int_{-1+\varepsilon}^{1-\varepsilon} \sqrt{1+f'(x)^2}\,dx = \int_{-1+\varepsilon}^{1-\varepsilon} \frac{dx}{\sqrt{1-x^2}}$$

But the function $(1-x^2)^{-1}$ is integrable on the closed interval $[-1, +1]$ if the integral is interpreted in the improper sense, and therefore, by Property (10) in § 10.10, we have

$$|\Gamma| = \lim_{\varepsilon\to 0} |\Gamma_\varepsilon| = \int_{-1}^{+1} \frac{dx}{\sqrt{1-x^2}} < \infty \tag{1}$$

which proves that the semicircle Γ is rectifiable and that its length is the number equal to the integral on the right-hand side of (1). It is this number that is called the number π:

$$\pi = \int_{-1}^{+1} \frac{dx}{\sqrt{1-x^2}} \tag{2}$$

These considerations provide a rigorous proof of the existence of this important limit which meets the requirements of modern mathematical analysis. In elementary geometry the definition of the arc length of a circle is stated quite correctly but its existence is justified by means of an intuitive argument accompanied by calculations.

The function arc cos x can be defined with the aid of the equality

$$\theta = \text{arc cos } x = \int_{x}^{1} \frac{dt}{\sqrt{1-t^2}}, \quad -1 \leqslant x \leqslant 1 \tag{3}$$

where θ is the length of the arc $\overset{\frown}{AB}$ (see Fig. 10.6) and x is the abscissa of the point B belonging to the upper semicircle.

We see that, according to the property of the definite integral as function of its lower limit, the function arc cos x is continuous; also it is strictly decreasing on the closed interval $[-1, +1]$ and possesses the derivative

$$(\text{arc cos } x)' = -\frac{1}{\sqrt{1-x^2}}, \quad -1 < x < 1$$

It is clear that arc cos $1=0$, arc cos $0 = \pi/2$ and arc cos $(-1) = \pi$.

What has been established guarantees the existence of the inverse function of arc cos x defined on the interval $[0, \pi]$ which is monotone decreasing on this interval; this function is denoted $x = \cos\theta$ and is called the cosine of the arc θ (the latter is supposed to be expressed in radians).

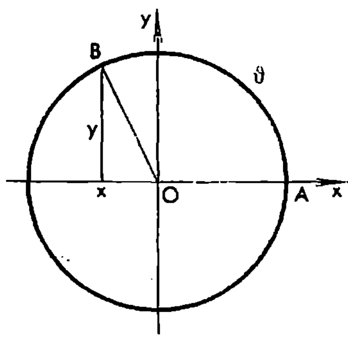

Fig. 10.6

The notion of the arc length θ is extended in an ordinary way to the whole real axis $-\infty < \theta < \infty$. The function $\cos\theta$ is extended accordingly. Namely, we define $\cos\theta$ $(-\infty < \theta < \infty)$ as an even function of period 2π which coincides with the function $\cos\theta$ defined above for the interval $[0, \pi]$. This definition of the cosine corresponds to the usual definition in which $\cos\theta$ is understood as the abscissa of the point B of the unit circle with coordinate θ (equal to arc length). From the definition of the extension of the cosine and from the properties of $\cos\theta$ on the interval $[0, \pi]$ it readily follows that $\cos\theta$ is continuously differentiable throughout the real axis $-\infty < \theta < \infty$. Besides, from formula (3) it obviously follows that the function $\cos(\pi/2+u)$ is odd (as function of u): $\cos(\pi/2+u) = -\cos(\pi/2-u)$.

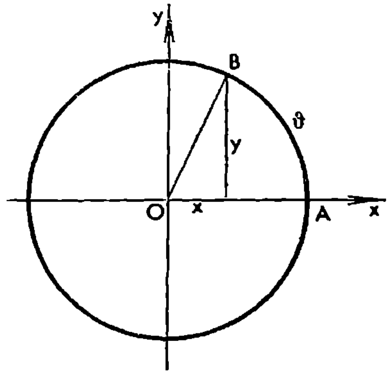

Fig. 10.7

Now let us suppose that the variable point B is in the right semicircle (Fig. 10.7):

$$x = \sqrt{1-y^2}, \qquad -1 \leqslant y \leqslant 1$$

The function

$$\theta = \arcsin y = \int_0^y \frac{dt}{\sqrt{1-t^2}},$$

$$-1 \leqslant y \leqslant 1 \tag{4}$$

expresses the length of the arc $\overset{\frown}{AB}$ taken with the corresponding sign. This function is obviously continuous, odd and strictly increasing on $[-1, +1]$ and satisfies the conditions $\theta(-1) = -\pi/2$, $\theta(0) = 0$ and $\theta(+1) = \pi/2$. Besides, it is continuously differentiable on $(-1, +1)$. Its inverse function

$$y = \sin\theta$$

is strictly increasing and continuous on $[-\pi/2, \pi/2]$. Its extension to the whole real axis $-\infty < \theta < \infty$ is obtained by assuming that the extended function $\sin\theta$ is periodic with period 2π and that the function $\sin(\pi/2+u)$ $(-\infty < < u < \infty)$ is odd (as function of u). It can easily be verified that $\sin\theta$ is a continuous function on $(-\infty, \infty)$ equal to the ordinate of the point B of the unit circle corresponding to the arc θ.

It is clear that $\cos^2\theta+\sin^2\theta = 1$, $-\infty < \theta < \infty$, because $\cos\theta$ and $\sin\theta$ are, respectively, the abscissa and the ordinate of one and the same point belonging to the unit circle.

There holds the equality

$$\arcsin x+\operatorname{arc}\cos x = \int_0^x \frac{dt}{\sqrt{1-t^2}}+\int_x^1 \frac{dt}{\sqrt{1-t^2}} = \int_0^1 \frac{dt}{\sqrt{1-t^2}} = \frac{\pi}{2}, \quad -1 \leqslant x \leqslant 1 \tag{5}$$

expressing geometrically the fact that, for any $x \in [-1, +1]$, the sum of the arc belonging to $[-\pi/2, \pi/2]$, whose sine is equal to x, and the arc belonging to $[0, \pi]$, whose cosine is equal to x, is equal to the number $\pi/2$.

If $\theta \in [0, \pi]$ and $x = \cos\theta$ then $\theta = \operatorname{arc}\cos x$, and, by virtue of (5), we have $\arcsin x = \pi/2-\theta$; consequently

$$\cos\theta = \sin\left(\frac{\pi}{2}-\theta\right) \tag{6}$$

Similarly, if $\theta \in [-\pi/2, \pi/2]$ and $x = \sin\theta$ then $\theta = \arcsin x$ and $\pi/2- -\theta = \operatorname{arc}\cos x$, and, consequently,

$$\sin\theta = \cos\left(\frac{\pi}{2}-\theta\right) \tag{7}$$

Using the symmetry and periodicity properties of $\cos\theta$ and $\sin\theta$ we can readily extend equalities (6) and (7) to the whole real axis $-\infty<\theta<\infty$. We also have the equalities

$$(\sin\theta)' = \cos\theta,\quad (\cos\theta)' = -\sin\theta,\quad -\infty<\theta<\infty \tag{8}$$

For instance, from (4) follows

$$\frac{d\theta}{dy} = \frac{1}{\sqrt{1-y^2}} \qquad (-1\leq y\leq 1)$$

whence we obtain the first equality (8):

$$(\sin\theta)' = \frac{dy}{d\theta} = \sqrt{1-\sin^2\theta} = \cos\theta$$

for $-\pi/2\leq\theta\leq\pi/2$, and, by virtue of the symmetry and periodicity of the functions $\sin\theta$ and $\cos\theta$, for all the values of θ as well.

Using formulas (8) and the fact that $\cos 0 = 1$ and $\sin 0 = 0$ we can compute derivatives of any order of the functions $\cos\theta$ and $\sin\theta$ at the point $\theta = 0$ and then represent these functions by Taylor's formula with the remainer corresponding to an arbitrarily large n as was already done in § 5.10. Moreover, taking into account the boundedness of the higher-order derivatives of the functions in question $\left(\left|\frac{d^n\cos\theta}{d\theta^n}\right|\leq 1 \text{ and } \left|\frac{d^n\sin\theta}{d\theta^n}\right|\leq 1\right)$ we conclude, like in § 5.10, that the remainder terms in the corresponding Taylor expansions tend to zero as $n\to\infty$ for any θ. We thus arrive at the expansions of the functions in question in the power series

$$\sin\theta = \theta-\frac{\theta^3}{3!}+\ldots \quad\text{and}\quad \cos\theta = 1-\frac{\theta^2}{2!}+\ldots \tag{9}$$

by means of which it is possible to derive (see § 11.13) the basic trigonometric formulas

$$\left.\begin{aligned}\cos(\alpha+\beta) &= \cos\alpha\cos\beta-\sin\alpha\sin\beta\\ \sin(\alpha+\beta) &= \sin\alpha\cos\beta+\cos\alpha\sin\beta\end{aligned}\right\} \tag{10}$$

Of course, it is also possible to obtain formulas (10) directly from (3) and (4). For instance, let $0<\alpha,\beta$ and $\alpha+\beta\leq\pi$. Then (see the explanations below) we have

$$\beta = \int_{\cos\beta}^{1}\frac{dx}{\sqrt{1-x^2}} = \int_{\varkappa}^{\cos\alpha}\frac{dx'}{\sqrt{1-x'^2}} = \int_{\varkappa}^{1}\frac{dx'}{\sqrt{1-x'^2}} - \int_{\cos\alpha}^{1}\frac{dx'}{\sqrt{1-x'^2}} \tag{11}$$

and $\alpha+\beta = \int_{\varkappa}^{1}\frac{dx'}{\sqrt{1-x'^2}}$ where $\varkappa = \cos\alpha\cos\beta-\sin\alpha\sin\beta$, whence, using (3), we obtain the first formula (10) for the indicated α and β.

The second equality in (11) is obtained by means of the change of variable $x' = x\cos\alpha - \sqrt{1-x^2}\sin\alpha$. The equality

$$\frac{dx'}{\sqrt{1-x'^2}} = \frac{dx}{\sqrt{1-x^2}}$$

follows from

$$dx' = \frac{1}{\sqrt{1-x^2}}\left(\cos\alpha\sqrt{1-x^2}+x\sin\alpha\right)dx$$

and $1-x'^2 = \left(\cos\alpha\sqrt{1-x^2}+x\sin\alpha\right)^2$. It should also be taken into account that the expression in the parentheses in this equality is nonnegative. Putting $x = \cos u$, $0 \leqslant u \leqslant \beta$, we can rewrite this expression as $\cos\alpha\sin u+\cos u\sin\alpha$. It is clearly nonnegative if $\alpha, u \leqslant \pi/2$. In the case $\alpha < \pi/2$, $u > \pi/2$, taking into account that $\sin t$ and $\cos t$ are decreasing on $[-u, \pi-\alpha]$, we obtain

$$\cos\alpha\sin u+\cos u\sin\alpha \geqslant \cos\alpha\sin(\pi-\alpha)+\cos(\pi-\alpha)\sin\alpha =$$
$$= \cos\alpha\sin\alpha-\cos\alpha\sin\alpha = 0$$

The case $u < \pi/2, \alpha > \pi/2$ is treated similarly by interchanging u and α.

In conclusion we note that the inequality

$$|\sin\theta| \leqslant |\theta| \tag{12}$$

(see § 4.2, Example 5) can be proved on the basis of the material of this section in the following way. We have

$$|\theta| = \left|\int_0^{\sin\theta}\frac{dt}{\sqrt{1-t^2}}\right| \geqslant \int_0^{|\sin\theta|} 1\,dt = |\sin\theta|, \qquad |\theta| \leqslant \frac{\pi}{2}$$

and since

$$\pi = \int_{-1}^{1}\frac{dt}{\sqrt{1-t^2}} \geqslant \int_{-1}^{1} dt = 2$$

and $|\sin\theta| \leqslant 1$, inequality (12) holds for all θ. Further, the inequality $\theta \leqslant \tan\theta$, $0 \leqslant \theta \leqslant \pi/2$ (see § 4.9) can be obtained by using Lagrange's theorem on finite increments:

$$\tan\theta-\theta = \theta(\sec^2\theta_1-1) \geqslant 0, \qquad 0 \leqslant \theta_1 \leqslant \theta \leqslant \frac{\pi}{2}$$

CHAPTER 11

Series

§ 11.1. The Notion of a Series

An expression

$$u_0+u_1+u_2, \ \ldots \tag{1}$$

where the numbers u_k (*the terms of the series* which are complex in the general case) depend on the index $k = 0, 1, 2, \ldots$ is called a *(number) series*. At present we consider this expression purely formally and do not assign any number (sum) to it because the ordinary addition of an infinite number of summands does not make sense. Series (1) can be written in the abbreviated form as

$$\sum_{k=0}^{\infty} u_k = \sum_{0}^{\infty} u_k \tag{2}$$

The latter expression is also purely formal; in some cases it proves more convenient than (1).

The number

$$S_n = u_0+u_1+ \ldots +u_n \qquad (n = 0, 1, \ldots)$$

is called the *nth partial sum of series* (1).

We say that series (1) is *convergent* if the limit

$$\lim_{n\to\infty} S_n = S$$

exists. In this case we write

$$S = u_0+u_1+u_2+ \ldots = \sum_{k=0}^{\infty} u_k \tag{3}$$

and call S the *sum* of the series; in this very sense we assign to expression (1) or (2) the number S. We also say that series (3) *converges to S*.

By Cauchy's criterion (which holds for complex numbers as well), *for series* (1) *to be convergent it is necessary and sufficient that, given any* $\varepsilon > 0$, *there should be N such that for all natural* $n > N$ *and any natural p the inequality*

$$|u_{n+1}+ \ldots +n_{n+p}| = |S_{n+p}-S_n| < \varepsilon$$

holds.

In particular, putting $p = 1$ we see that if series (1) is convergent its *general term* u_k tends to zero:

$$\lim_{n\to\infty} u_n = 0 \tag{4}$$

Condition (4) is necessary for series (1) to be convergent but, as will be seen from examples, it is not sufficient.

Let us consider the series called the *remainder series* of series (1)

$$u_{n+1}+u_{n+2}+\ldots = \sum_{k=1}^{\infty} u_{n+k} \tag{5}$$

Since conditions of Cauchy's criterion are stated in exactly the same manner for both series (1) and (5) they are simultaneously convergent or divergent. If they are convergent the sum of series (5) is

$$\lim_{m\to\infty} \sum_{k=1}^{m} u_{n+k} = \lim_{m\to\infty} (S_{n+m}-S_n) = S-S_n$$

If the terms of series (1) are real and nonnegative its partial sums form a nondecreasing sequence $S_1 \leqslant S_2 \leqslant S_3 \leqslant \ldots$ and therefore if this sequence is bounded, that is

$$S_n \leqslant M \qquad (n = 1, 2, \ldots)$$

then the series is convergent and its sum satisfies the inequality

$$\lim_{n\to\infty} S_n = S \leqslant M$$

If this sequence is unbounded the series is divergent:

$$\lim_{n\to\infty} S_n = \infty$$

In this case we write

$$\sum_{k=0}^{\infty} u_k = \infty$$

and say that series (1) with nonnegative terms is *divergent to* ∞ or is *properly divergent.*

Example. The nth partial sum of the series

$$1+z+z^2+\ldots \tag{6}$$

is $S_n(z) = \dfrac{1-z^{n+1}}{1-z}$ (for $z \neq 1$). If $|z| < 1$ then $|z^{n+1}| = |z|^{n+1} \to 0$, that is $z^{n+1} \to 0$ $(n \to \infty)$; if $|z| > 1$ then $|z^{n+1}| \to \infty$, and, finally, if $|z| = 1$ then $z^{n+1} = \cos(n+1)\theta + i\sin(n+1)\theta$ where θ is the argument of z, and we see that the variable z^{n+1} has no limit as $n \to \infty$ because its real or imaginary part (or both) has no limit as $n \to \infty$. For $z = 1$ the divergence of series (6) is quite obvious.

We see that series (6) is convergent and has a sum equal to $(1-z)^{-1}$ in the open circle $|z| < 1$ of the complex plane and is divergent at all the other points z.

§ 11.2. Operations on Series

If $\sum_0^\infty u_k$ *and* $\sum_0^\infty v_k$ *are convergent series and* α *is a number then the series* $\sum_0^\infty \alpha u_k$ *and* $\sum_0^\infty (u_k \pm v_k)$ *are also convergent and we have*

$$\sum_0^\infty \alpha u_k = \alpha \sum_0^\infty u_k \tag{1}$$

and

$$\sum_0^\infty (u_k \pm v_k) = \sum_0^\infty u_k \pm \sum_0^\infty v_k \tag{2}$$

Indeed,

$$\sum_0^\infty \alpha u_k = \lim_{n\to\infty} \sum_0^n \alpha u_k = \alpha \lim_{n\to\infty} \sum_0^n u_k = \alpha \sum_0^\infty u_k$$

and

$$\sum_0^\infty (u_k \pm v_k) = \lim_{n\to\infty} \sum_0^n (u_k \pm v_k) = \lim_{n\to\infty} \sum_0^n u_k \pm \lim \sum_0^n v_k = \sum_0^\infty u_k \pm \sum_0^\infty v_k$$

It should be stressed that, generally speaking, the convergence of series on the left-hand side of (2) does not imply the convergence of each of the series on the right-hand side, which is readily confirmed by the example of the series

$$(1-1)+(1-1)+\ \dots \tag{3}$$

This series is convergent since all its terms are equal to 0 but the expression $\sum_0^\infty 1 - \sum_0^\infty 1$ does not make sense because the series entering into it are divergent (to infinity).

If a series

$$u_0+u_1+u_2+\ \dots \tag{4}$$

is convergent and has a sum S its terms can be arbitrarily grouped in brackets (without rearranging them!), for instance,

$$u_0+(u_1+u_2)+(u_3+u_4+u_5)+\ \dots$$

The new series thus formed whose terms are the sums of the numbers in brackets is also convergent and its sum is again the number S, for the partial sums of the new series form a subsequence of the sequence of partial sums of the original series (4).

On the contrary, if we are given a series whose terms are certain sums enclosed in brackets then, generally speaking, it is illegitimate to remove the brackets because this may destroy the convergence of the original series and may change its sum. For instance, series (3) is convergent but if the brackets are removed we obtain the divergent series $1-1+1-\ldots$. If the terms in the brackets are only nonnegative or nonpositive numbers the removal of the brackets in such a series does not of course affect its convergence and its sum.

§ 11.3. Series with Nonnegative Terms

In this section we shall deal with series whose terms are nonnegative or positive numbers. A series with positive terms is referred to as a *positive series.*

Theorem (Comparison Tests for Positive Series). *Let there be given two series*

$$\text{(i)}\ \sum_0^\infty u_k \quad \text{and} \quad \text{(ii)}\ \sum_0^\infty v_k$$

with nonnegative terms.

(a) *If* $u_k \leqslant v_k$ $(k = 0, 1, 2, \ldots)$, *the convergence of series* (ii) *implies the convergence of series* (i) *and the divergence of series* (i) *implies the divergence of series* (ii).

(b) *If*

$$\lim_{k\to\infty} \frac{u_k}{v_k} = A > 0 \tag{1}$$

then series (i) *and* (ii) *are simultaneously convergent or divergent.*

Proof. Let series (ii) be convergent and let S be its sum. Then

$$\sum_0^n u_k \leqslant \sum_0^n v_k \leqslant S \qquad (n = 0, 1, \ldots)$$

which means that the partial sums of series (i) are bounded and hence it is convergent. Its sum S' obviously satisfies the inequality $S' \leqslant S$.

Now suppose that series (i) is divergent; then (see § 11.1) its nth partial sum increases indefinitely together with n, whence, by virtue of the inequality

$$\sum_0^n u_k \leqslant \sum_0^n v_k \qquad (n = 0, 1, \ldots)$$

it follows that the nth partial sum of series (ii) also increases indefinitely, that is the latter series is divergent.

Now we suppose that equality (1) is fulfilled. Then the terms v_k are in fact positive $(v_k > 0)$, and for any positive $\varepsilon < A$ there is N such that $A-\varepsilon < \frac{u_k}{v_k} < A+\varepsilon$ $(k > N)$ whence

$$v_k(A-\varepsilon) < u_k < (A+\varepsilon)v_k \tag{2}$$

If series (ii) is convergent the series $\sum_{N+1}^{\infty}(A+\varepsilon)v_k$ is also convergent and, by virtue of the second inequality (2), the series $\sum_{N+1}^{\infty} u_k$ is convergent together with the former; therefore series (i) is convergent. If series (ii) is divergent the series $\sum_{N+1}^{\infty} v_k(A-\varepsilon)$ is also divergent and, together with it, so is the series $\sum_{N+1}^{\infty} u_k$. Hence, series (i) is also divergent in this case.

The theorem has been proved.

Theorem 2 (D'Alembert's* Test). *Let there be a positive series*

$$\sum_{0}^{\infty} u_k \tag{3}$$

$$(u_k > 0 \textit{ for all } k = 0, 1, 2, \ldots)$$

(a) *If*

$$\frac{u_{k+1}}{u_k} \leqslant q < 1 \qquad (k = 0, 1, 2, \ldots) \tag{4}$$

series (3) *is convergent; if*

$$\frac{u_{k+1}}{u_k} \geqslant 1 \qquad (k = 0, 1, 2, \ldots) \tag{5}$$

series (3) *is divergent.*

(b) *If*

$$\lim_{k\to\infty} \frac{u_{k+1}}{u_k} = q \tag{6}$$

then series (3) *is convergent for $q < 1$ and divergent for $q > 1$***.

Proof. We have

$$u_n = u_0 \frac{u_1}{u_0} \frac{u_2}{u_1} \cdots \frac{u_n}{u_{n-1}} \qquad (n = 0, 1, 2, \ldots)$$

and therefore it follows from (4) that

$$u_n \leqslant u_0 q^n, \ q < 1 \qquad (n = 0, 1, 2, \ldots)$$

This relation shows that, since series $\sum_{1}^{\infty} u_0 q^n$ is convergent, series (3) is also convergent.

---

* D'Alembert, Jean le Rond (1717-1783), a French philosopher and mathematician.

** The following more general tests are proved analogously: series (3) is convergent if $\overline{\lim} \frac{u_{k+1}}{u_k} < 1$ and divergent if $\underline{\lim} \frac{u_{k+1}}{u_k} > 1$.

From (5) it follows that $u_n \geqslant u_0$ $(n = 0, 1, 2, \ldots)$ and, since the series $u_0+u_0+\ldots$ is divergent, series (3) is divergent as well.

Now, if property (6) holds and $q < 1$ then for any positive ε satisfying the condition $q+\varepsilon < 1$ we have $u_{k+1}/u_k < q+\varepsilon < 1$ $(k \geqslant N)$ where N is sufficiently large. By test (4), in this case the series $\sum_N^\infty u_k$ is convergent and, together with it, series (3) is also convergent.

If property (6) is fulfilled for $q > 1$ it follows that $u_{k+1}/u_k > 1$ $(k \geqslant N)$ for a sufficiently large N, and therefore, by Test (5), the series $\sum_N^\infty u_k$ is divergent and, together with it, so is series (3).

Theorem (Cauchy's Test). *Let* (3) *be a series with positive terms.*

(a) *If*

$$\sqrt[k]{u_k} < q < 1 \qquad (k = 0, 1, \ldots) \tag{7}$$

series (3) *is convergent; if*

$$\sqrt[k]{u_k} \geqslant 1 \qquad (k = 0, 1, \ldots) \tag{8}$$

it is divergent.

(b) *If*

$$\lim_{k\to\infty} \sqrt[k]{u_k} = q \tag{9}$$

then series (3) *is convergent for* $q < 1$ *and divergent for* $q > 1$.*

Proof. Inequality (7) implies that $u_k < q^k$ $(q < 1;\ k = 0, 1, \ldots)$ and, since in this case the series $\sum_0^\infty q^k$ is convergent, so is series (3). If inequality (8) is fulfilled then $u_k \geqslant 1$ $(k = 0, 1, \ldots)$, and, since the series $1+1+\ldots$ is divergent, series (3) is also divergent.

Further, from property (9) with $q < 1$ it follows that

$$\sqrt[k]{u_k} < q+\varepsilon < 1 \qquad (k \geqslant N)$$

for sufficiently large N, whence

$$u_k < (q+\varepsilon)^k \qquad (k \geqslant N)$$

Since the series $\sum_N^\infty (q+\varepsilon)^k$ is convergent the last inequality implies that $\sum_N^\infty u_k$ is convergent and hence series (3) is also convergent. If property (9) is fulfilled for $q > 1$ it follows that $u_k > 1$ $(k \geqslant N)$ for sufficiently large N, whence obviously follows the divergence of series (3).

---

* The following more general tests are proved similarly: if $\overline{\lim}\sqrt[k]{u_k} < 1$ series (3) is convergent and if $\underline{\lim}\sqrt[k]{u_k} > 1$ it is divergent.

Remark 1. The series with general term $u_n = n^{-\alpha}$ ($\alpha > 0$) is convergent when $\alpha > 1$ and divergent when $\alpha \leq 1$ (see § 9.15 (5)*). In both cases we have

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = 1 \tag{10}$$

and also

$$\lim_{n\to\infty} \sqrt[n]{u_n} = 1 \tag{11}$$

This shows that there are both convergent and divergent series for which conditions (10) or (11) are fulfilled.

The series $1+1/2+1/3+\ldots$ is called the *harmonic series*. It is divergent.

Remark 2. If for a sequence of positive numbers u_n there exists the limit

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = q > 0 \tag{12}$$

it follows automatically that

$$\lim_{n\to\infty} \sqrt[n]{u_n} = q \tag{13}$$

Therefore, theoretically, the limiting form of D'Alembert's test (i.e. (6)) does not provide any new information in comparison with the limiting form of Cauchy's test (i.e. (9)) but in practice D'Alembert's test proves extremely convenient.

Let us justify what has been said. Suppose that (12) takes place. Then for any positive $\varepsilon < q$ there is n such that $q-\varepsilon < \frac{u_{n+k}}{u_{n+k-1}} < q+\varepsilon$ ($k = 0, 1, \ldots$). It follows that

$$(q-\varepsilon)^p < \frac{u_{n+1}}{u_n}\,\frac{u_{n+2}}{u_{n+1}}\cdots\frac{u_{n+p}}{u_{n+p-1}} < (q+\varepsilon)^p$$

i.e. $(q-\varepsilon)^p < \frac{u_{n+p}}{u_n} < (q+\varepsilon)^p$, whence

$$u_n^{1/(n+p)}(q-\varepsilon)^{p/(n+p)} < \sqrt[n+p]{u_{n+p}} < u_n^{1/(n+p)}(q+\varepsilon)^{p/(n+p)}$$

On passing to the limit in the last relation as $p \to \infty$ we obtain

$$q-\varepsilon \leq \underline{\lim}\, \sqrt[n]{u_n} \leq \overline{\lim}\, \sqrt[n]{u_n} \leq q+\varepsilon$$

Since $\varepsilon > 0$ can be made arbitrarily small we obtain

$$q = \underline{\lim}\, \sqrt[n]{u_n} = \overline{\lim}\, \sqrt[n]{u_n} = \lim_{n\to\infty} \sqrt[n]{u_n}$$

* In § 9.15 we only used some properties of series presented in § 11.1.

The converse is not true: limit (13) may exist when limit (12) does not exist. For example, let $u_n = q^{n\pm\sqrt{n}}$ where "+" is written for even n and "−" for odd n. Then we have $\sqrt[n]{u_n} = q^{1\pm n^{-1/2}} \to q$ $(n \to \infty)$; on the other hand,

$$\frac{u_{n+1}}{u_n} = \begin{cases} q^{1+\sqrt{n+1}+\sqrt{n}} & \text{(for odd } n\text{)} \\ q^{1-\sqrt{n+1}-\sqrt{n}} & \text{(for even } n\text{)} \end{cases}$$

If $q \neq 1$ the ratio u_{n+1}/u_n is unbounded and thus has no finite limit.

Examples

(i) $\sum\limits_0^\infty \frac{x^k}{k!}$; (ii) $\sum\limits_1^\infty \frac{x^k}{k^\alpha}$ $(\alpha > 0)$; (iii) $\sum\limits_1^\infty (e^{1/k}-1)$;

(iv) $\sum\limits_1^\infty \ln\left(1+\frac{1}{k}\right)$; (v) $\sum\limits_1^\infty q^{k+\sqrt{k}}$ $(q > 0)$

Series (i) and (ii) are obviously convergent for $x = 0$. Series (i) is also convergent for any $x > 0$ because $\frac{u_{k+1}}{u_k} = \frac{x}{k} + 1 \to 0$, $k \to \infty$. As to series (ii), it is convergent for $0 < x < 1$ and divergent for $x > 1$ because for this series we have $\frac{u_{k+1}}{u_k} = x\left(\frac{k}{k+1}\right)^\alpha \to x, k \to \infty$; for the case $x = 1$ see Remark 1. Series (iii) and (iv) are divergent because $e^{1/k}-1 \approx \frac{1}{k}$, $k \to \infty$, and $\ln\left(1+\frac{1}{k}\right) \approx \frac{1}{k}, k \to \infty$ (here "$\approx$" is the sign of asymptotic equality; see § 4.10), and the series $\sum\limits_0^\infty \frac{1}{k}$ is divergent. Series (v) is convergent for $0 \leqslant q < 1$ and divergent for $q > 1$ because for this series we have $\sqrt[k]{u_k} = q^{1+k^{-1/2}} \to q$ $(k \to \infty)$. It is also divergent for $q = 1$ since in this case its general term is equal to 1.

Theorem. *Let a series*

$$u_0+u_1+u_2+ \dots \tag{14}$$

with nonnegative terms be convergent to a sum S. Then the series

$$u'_0+u'_1+u'_2+u'_3+ \dots \tag{15}$$

obtained from the former by rearranging and renumbering its terms in an arbitrary way is also convergent and has the same sum S.

Proof. Let

$$S'_n = u'_0+u'_1+ \dots +u'_n$$

be the nth partial sum of series (15). Its terms $u'_0, u'_1, \dots, u'_n$ have some indices $k_0, k_1, \dots, k_n$ as terms of the original series (14). Let N be the greatest of the indices $k_0, k_1, \dots, k_n$ and S_N be the Nth partial sum of series (14).

We obviously have $S'_n \leqslant S_N \leqslant S$, and, since n is arbitrary, series (15) is convergent and has a sum $S' \leqslant S$. Now, interchanging the roles of series (14) and (15), we can repeat the same argument whence $S \leqslant S'$. Consequently $S = S'$.

§ 11.4. Alternating Series

A series of the form

$$a_0 - a_1 + a_2 - a_3 + \dots \tag{1}$$

where the numbers a_k are positive ($a_k > 0$) is called an *alternating series*. If the numbers a_k form a monotone decreasing sequence convergent to zero ($a_k \geqslant a_{k+1}$ and $a_k \to 0$ for $k \to \infty$) alternating series (1) is called a *Leibniz series*.

We shall show that a Leibniz series is convergent and that its sum satisfies the inequality $S \leqslant a_0$.

Indeed, its partial sum S_{2n+1} with an odd index $2n+1$ can be written in the form

$$S_{2n+1} = a_0 - (a_1 - a_2) - (a_3 - a_4) - \dots - (a_{2n-1} - a_{2n}) - a_{2n+1}$$

whence it obviously follows that it is bounded above by the number a_0:

$$S_{2n+1} \leqslant a_0$$

On the other hand, this sum can be rewritten as

$$S_{2n+1} = (a_0 - a_1) + (a_2 - a_3) + \dots + (a_{2n} - a_{2n+1})$$

whence we see that it is monotone nondecreasing. Therefore there exists the limit

$$\lim_{n \to \infty} S_{2n+1} = S \leqslant a_0$$

It is also obvious that

$$\lim_{n \to \infty} S_{2n} = \lim_{n \to \infty} (S_{2n+1} - a_{2n+1}) = S - 0 = S$$

The theorem has been proved.

Example. The series $1 - 1/2 + 1/3 - \dots$ is obviously a Leibniz series; it is convergent and its sum S does not exceed 1 (as follows from § 5.11 (5), its sum S is in fact equal to $\ln 2$).

§ 11.5. Absolutely Convergent Series

A series

$$u_0 + u_1 + u_2 + \dots \tag{1}$$

with complex terms is said to be *absolutely convergent* if the series

$$|u_0| + |u_1| + |u_2| + \dots \tag{2}$$

of the moduli of its terms is convergent.

An absolutely convergent series is always convergent. Indeed, suppose that series (1) is absolutely convergent; then series (2) is convergent and, by Cauchy's criterion, given any $\varepsilon > 0$, there is N such that $\varepsilon > |u_{n+1}| + \ldots \ldots + |u_{n+p}|$ for all p and $n > N$. Then moreover, $\varepsilon > |u_{n+1} + \ldots + u_{n+p}|$, and, by Cauchy's criterion, series (1) is convergent.

It is quite obvious that a convergent series with (real) nonnegative terms is absolutely convergent. But there are series (with arbitrary terms) which are convergent but not absolutely convergent. This can be demonstrated by the example of the series $1 - 1/2^\alpha + 1/3^\alpha - \ldots (\alpha > 0)$ which is convergent for all $\alpha > 0$ since it is a Leibniz series but is absolutely convergent only for $\alpha > 1$.

Theorem. *If a series is absolutely convergent, any rearrangement of its terms does not violate its absolute convergence, and the sum of the new series remains the same.*

Proof. We shall begin with the special case when the terms u_k of the series are real numbers.

Let us put (on condition that u_k are real)

$$u_k^+ = \begin{cases} u_k & \text{for} \quad u_k \geq 0 \\ 0 & \text{for} \quad u_k < 0 \end{cases} \qquad u_k^- = \begin{cases} -u_k & \text{for} \quad u_k \leq 0 \\ 0 & \text{for} \quad u_k > 0 \end{cases} \tag{3}$$

The numbers u_k^+ and u_k^- are obviously nonnegative and

$$u_k = u_k^+ - u_k^- \tag{4}$$

Together with series (1) we shall consider the two series

$$\sum_0^\infty u_k^+ \quad \text{and} \quad \sum_0^\infty u_k^- \tag{5}$$

(with nonnegative terms).

As was agreed, we suppose that series (1) is absolutely convergent and that its terms u_k are real numbers. Then series (5) are also convergent because we obviously have $u_k^+ \leq |u_k|$ and $u_k^- \leq |u_k|$.

Let the series obtained by rearranging the terms of the original series (1) have the form $v_1 + v_2 + v_3 + \ldots$. For its terms we can construct the corresponding numbers v_k^+ and v_k^- as was done above for series (1). Then (see the explanations below)

$$\sum_0^\infty u_k = \sum_0^\infty (u_k^+ - u_k^-) = \sum_0^\infty u_k^+ - \sum_0^\infty u_k^- = \sum_0^\infty v_k^+ - \sum_0^\infty v_k^- = \sum_0^\infty (v_k^+ - v_k^-) = \sum_0^\infty v_k$$

The first equality in these relations follows from (4), the second follows from § 11.2 (2) if we take into account that series (5) is convergent; the third equality is implied by the fact that any rearrangement of the terms of a convergent series with nonnegative terms does not violate its convergence and does not alter its sum, the fourth equality is implied by § 11.2 (2), and, finally, the fifth equality is fulfilled because $v_k = v_k^+ - v_k^-$. Thus, for real u_k the assertion of the theorem has been proved.

Now we pass to the general case when $u_k = \alpha_k + i\beta_k$ are complex numbers and, as before, v_k denote the terms of the rearranged series. Since $|\alpha_k| \leqslant |u_k|$ and $|\beta_k| \leqslant |u_k|$ the series $\sum\limits_0^\infty \alpha_k$ and $\sum\limits_0^\infty \beta_k$ (with real terms) are absolutely convergent and their terms, as was proved above, can be rearranged arbitrarily; therefore, putting $v_k = \gamma_k + i\delta_k$ we obtain

$$\sum_0^\infty u_k = \sum_0^\infty (\alpha_k + i\beta_k) = \sum_0^\infty \alpha_k + i\sum_0^\infty \beta_k =$$

$$= \sum_0^\infty \gamma_k + i\sum_0^\infty \delta_k = \sum_0^\infty (\gamma_k + i\delta_k) = \sum_0^\infty v_k$$

The proof of the theorem has been completed.

§ 11.6. Conditional and Unconditional Convergence of Series with Real Terms

Let us consider a series

$$u_0 + u_1 + u_2 + \ \ldots \tag{1}$$

with real terms. As was done in the foregoing section, let us construct the two series

$$\sum_0^\infty u_k^+ \quad \text{and} \quad \sum_0^\infty u_k^- \tag{2}$$

(with nonnegative terms) corresponding to it.

If series (1) is absolutely convergent then, as we know, series (2) are also convergent. The converse is also obviously true: the convergence of both series (2) implies the absolute convergence of series (1) because

$$|u_k| = u_k^+ + u_k^-$$

Thus, for series (1) to be absolutely convergent it is necessary and sufficient that series (2) generated by it be convergent.

Now let us suppose that series (1) is convergent but not absolutely. Then at least one of series (2), for definiteness, the first one, is divergent, that is $\sum\limits_0^\infty u_k^+ = \infty$ (because $u_k^+ \geqslant 0$). Let us write down the equality

$$\sum_0^n u_k^- = \sum_0^n u_k^+ - \sum_0^n u_k \tag{3}$$

The first sum on the right-hand side of (3) increases indefinitely together with n while the second one tends to a finite limit since series (1) is convergent; consequently the left member of (3) increases indefinitely as $n \to \infty$. Hence, series (2) is divergent.

We also note that the convergence of series (1) implies that $u_k \to 0$ and then, obviously, we also have $u_k^+ \to 0$ and $u_k^- \to 0$.

We have shown that if series (1) is convergent but not absolutely both series (2) generated by it are divergent but at the same time $u_k^+ \to 0$ and $u_k^- \to 0$.

This proposition can be restated as follows:

(i) *For a series to be absolutely convergent it is necessary and sufficient that the series composed of its positive terms and negative terms be convergent.*

It may however happen that one of these series is a finite sum or does not exist at all.

(ii) *If a convergent series is not absolutely convergent then the series composed of its positive and of its negative terms are divergent while their general terms tend to zero.*

The following terminology is often used. A series is said to be *unconditionally convergent* if it is convergent and if any rearrangement of its terms does not violate its convergence; it is said to be *conditionally convergent* if it is convergent and if its terms can be rearranged so that the convergence violated, i.e. so that the rearranged series is divergent.

The theorem on the commutativity of an absolutely convergent series proved in the foregoing section implies that (a) *an absolutely convergent series is unconditionally convergent.* From proposition (ii) and from the theorem whose proof is given below it follows that (b) *a series which converges not absolutely is conditionally convergent.*

Combining propositions (a) and (b) we can say that *for a series to be unconditionally convergent it is necessary and sufficient that it be absolutely convergent.*

Our discussion shows that we can restate equivalently the definition of unconditional convergence in the following way: *a convergent series is said to be unconditionally convergent if the series obtained from it by an arbitrary rearrangement of its terms continues to be convergent and has the same sum as the original series.*

Now we proceed to the theorem mentioned in connection with proposition (b).

Theorem. *Let $\sum\limits_0^\infty \alpha_k$ and $\sum\limits_0^\infty \beta_k$ be two divergent series with positive terms tending to zero as $k \to \infty$ ($\alpha_k \to 0$ and $\beta_k \to 0$ as $k \to \infty$).*

Then for any number S, finite or infinite ($-\infty \leqslant S \leqslant \infty$), it is possible to construct a series of the form

$$\begin{aligned}\alpha_0+\alpha_1+\ldots+\alpha_{k_1}-\beta_0-\ldots-\beta_{k_1'}+\alpha_{k_1+1}+\ldots\\ \ldots+\alpha_{k_2}-\beta_{k_1'+1}-\ldots-\beta_{k_2'}+\alpha_{k_2+1}+\ldots\end{aligned} \tag{4}$$

whose sum is equal to S.

In the cases $S=+\infty$ and $S=-\infty$ series (4) is divergent. This series includes all α_k's and β_k's and each of these numbers enters into the series only once.

Proof. For definiteness, let S be a finite positive number. The indices $k_1 < k_2 < \ldots$ and $k_1' < k_2' < \ldots$ can be chosen in this case as the smallest natural numbers for which the corresponding inequalities written below are fulfilled:

$$\text{(i)}\quad A_1 = \sum_0^{k_1} \alpha_j > S, \text{ (ii)} \qquad A_2 = A_1 - \sum_0^{k_1'} \beta_j < S,$$

$$\text{(iii)}\quad A_3 = A_2 + \sum_{k_1+1}^{k_2} \alpha_j > S, \qquad \text{(iv)}\quad A_4 = A_3 - \sum_{k_1'+1}^{k_2'} \beta_j < S$$

At the lth stage ($l = 1, 2, \ldots$) of this construction it is possible to choose the numbers k_l and k_l' satisfying the lth inequality because the series $\sum_0^\infty \alpha_j$ and $\sum_0^\infty \beta_j$ are divergent. The fact that the series thus constructed is convergent to S follows from the above inequalities and from the assumption that $\alpha_k \to 0$ and $\beta_k \to 0$ as $k \to \infty$.

To prove the theorem for the case $S = \infty$ we can, for instance, replace S on the right-hand sides of inequalities (i), (ii), ... by the numbers 2, 1, 4, 3, 6, 5, ... respectively.

§ 11.7. Sequences and Series of Functions. Uniform Convergence

Let us consider a sequence of functions $\{f_k(x)\}$ defined on a set of points $x = (x_1, \ldots, x_n)$ of the n-dimensional space R_n. They are allowed to take on complex values $(f_k(x) = \alpha_k(x) + i\beta_k(x))$. We can also assume that x is a (variable) complex point ($x = x = \xi + i\eta$) running through a set E in the complex plane; then $f_k(x)$ are functions of the complex variable x.

Let the (number) sequence $\{f_k(x)\}$ tend to a number $f(x)$ for every fixed $x \in E$. Since x can vary within the set E the limit $f(x)$ of this sequence is a function defined on the set E. Let

$$\varrho_n = \sup_{x \in E} |f(x) - f_n(x)| \tag{1}$$

denote the supremum of the absolute values of the deviations of $f_n(x)$ from $f(x)$ taken over the set E. We shall suppose that ϱ_n is finite for every n ($\varrho_n < < \infty$).

A sequence of functions $\{f_n(x)\}$ is said to be uniformly convergent on E to $f(x)$ if $\varrho_n \to 0$ as $n \to \infty$.

Another equivalent definition of uniform convergence reads: a sequence $\{f_n(x)\}$ is uniformly convergent to $f(x)$ on E if for any $\varepsilon > 0$ there is N such that the inequality

$$|f(x) - f_n(x)| > \varepsilon \quad \text{for all } x \in E \tag{2}$$

holds for all $n > N$.

If the conditions of the former definition are fulfilled then for any $\varepsilon > 0$ there is N such that $\varrho_n < \varepsilon$ for all $n > N$ whence

$$|f(x)-f_n(x)| \leq \varrho_n < \varepsilon \tag{3}$$

for all $x \in E$ and $n > N$, that is the conditions of the latter definition hold as well. Conversely, if the conditions of the latter definition hold then, given any $\varepsilon > 0$, there exists N such that inequality (2) is fulfilled for all $n > N$. Taking the supremum of the left-hand side of (2) over all $x \in E$ we obtain $\varrho_n \leq \varepsilon\ (n > N)$ whence $\varrho_n \to 0$, that is the conditions of the former definition are fulfilled.

We can also state a third definition of uniform convergence (in terms of Cauchy's criterion): a sequence $\{f_n(x)\}$ is uniformly convergent on E if, given any $\varepsilon > 0$, there is N such that the inequality

$$|f_{n+p}(x)-f_n(x)| < \varepsilon \tag{4}$$

is fulfilled for all $n > N$ and $p > 0$ and for all $x \in E$.

If a sequence $\{f_n(x)\}$ is uniformly convergent in the sense of the second definition it follows that for any $\varepsilon > 0$ there is N such that for $n > N$ and any p there holds the inequality

$$|f_{n+p}(x)-f_n(x)| \leq |f_{n+p}(x)-f(x)|+|f(x)-f_n(x)| < 2\varepsilon \qquad \text{for all } x \in E$$

that is the conditions of the third definition are fulfilled. On the other hand, if the conditions of the third definition hold then for every fixed $x \in E$ the ordinary Cauchy's criterion for number sequences holds, and therefore the sequence is convergent on E to a function $f(x)$. Now, taking an arbitrary $\varepsilon > 0$, we can find the number N as indicated in the third definition and pass to the limit as $p \to \infty$ in inequality (4) where $n > N$ is fixed; this results in

$$|f(x)-f_n(x)| \leq \varepsilon \qquad (x \in E)$$

whence

$$\varrho_n = \sup_{x \in E} |f(x)-f_n(x)| \leq \varepsilon$$

and, since $n > N$ can be taken arbitrarily, we have $\varrho_n \to 0\ (n \to \infty)$, that is the conditions of the first definition are fulfilled.

It is easy to see that if α is a number and $\{f_k(x)\}$ and $\{\varphi_k(x)\}$ are two sequences of functions uniformly convergent on E then the sequences $\{\alpha f_k(x)\}$ and $\{f_k(x) \pm \varphi_k(x)\}$ are also uniformly convergent on E. It can also be readily shown that if a sequence of functions is uniformly convergent on E it is also uniformly convergent on every subset $E' \subset E$. The converse is not true in the general case.

To every sequence of functions $\{f_n(x)\}$ there corresponds the series

$$f_0(x)+\big(f_1(x)-f_0(x)\big)+\big(f_2(x)-f_1(x)\big)+\ \dots$$

whose nth partial sums are equal to $f_n(x)$ $(n = 0, 1, \dots)$ respectively.

Now let us consider a series

$$u_0(x)+u_1(x)+u_2(x)+\ \dots \tag{5}$$

whose terms are functions of $x \in E$ (generally speaking, assuming complex values) where, as before, E is a point set in the n-dimensional space R_n or in the complex plane.

Series (5) *is said to be uniformly convergent on the set E to a function $S(x)$ if the sequence of partial sums $\{S_k(x)\}$ is uniformly convergent to $S(x)$ on E.*

In particular, the definition of a uniformly convergent series can be stated as follows: series (5) is uniformly convergent on the set E if for any $\varepsilon > 0$ there is N such that for $n > N$ and $p > 0$ and for any $x \in E$ there holds the inequality $|u_{n+1}(x)+ \dots +u_{n+p}(x)| < \varepsilon$.

The theorem below provides an important test for uniform convergence of a series.

Theorem 1 (Weierstrass). *If the terms of series* (5) *satisfy the inequalities*

$$|u_k(x)| \leqslant \alpha_k \qquad (k = 0, 1, \dots) \tag{6}$$

where $x \in E$ and α_k are numbers (independent of x) such that the series with the terms α_k is convergent, then series (5) *is uniformly and absolutely convergent on E.**

Indeed, if the series with the terms α_k is convergent, inequalities (6) imply that for any $\varepsilon > 0$ there is N such that for any $n > N$ and $p > 0$ and for arbitrary $x \in E$ the inequality

$$\varepsilon > \alpha_{n+1}+ \dots +\alpha_{n+p} \geqslant |u_{n+1}(x)|+ \dots +|u_{n+p}(x)| \geqslant$$
$$\geqslant |u_{n+1}(x)+ \dots + u_{n+p}(x)|$$

is fulfilled. This means that series (5) is uniformly convergent on E. Its absolute convergence is evident.

The lemma below will be of use for our further aims.

Lemma 1. *If a function $F(x)$ defined on a set E in the n-dimensional space R_n is such that for any $\varepsilon > 0$ there exist $\delta > 0$ and a representation of the form*

$$F(x) = F_1(x)+F_2(x) \tag{7}$$

such that F_2 is a continuous function (relative to E) at a point $x^0 \in E$ and $|F_1(x)| < \varepsilon$ for all $x \in E$ satisfying the inequality $|x-x^0| < \delta$ then F is continuous (relative to E) at the point x^0.

Proof. We have

$$|F(x)-F(x^0)| \leqslant |F_1(x)-F_1(x^0)|+|F_2(x)-F_2(x^0)| \leqslant$$
$$\leqslant |F_1(x)|+|F_1(x^0)|+|F_2(x)-F_2(x^0)| < \varepsilon+\varepsilon+\varepsilon = 3\varepsilon$$

provided that $|x-x^0| < \delta_1$, $x \in E$ where δ_1 is sufficiently small. For any

---

* The number series $\sum\limits_{k=0}^{\infty} \alpha_k$ is called a *dominant series* of the functional series $\sum\limits_{k=0}^{\infty} u_k(x)$, and the series $\sum\limits_{k=0}^{\infty} u_k(x)$ is spoken of as a *dominated series*. — *Tr.*

given $\varepsilon > 0$ we can first find some $\delta > 0$ and a representation of form (7) mentioned in the conditions of the lemma and, taking into account the continuity of F_2 at the point x^0, we can choose $\delta_1 < \delta$ so that $|F_2(x)-F_2(x^0)| < \varepsilon$ for $|x-x^0| < \delta_1$ and $x \in E$, which proves the lemma.

Now we proceed to another important theorem.

Theorem 2. *If a sequence of functions $\{f_n\}$ is uniformly convergent on a set E to a function f, and f_n are continuous (relative to E) at a point x^0 then the function f is also continuous at x^0.*

This theorem can be restated in terms of series: *the sum of a uniformly convergent series of functions continuous at a point $x^0 \in E$ is a continuous function at that point.*

Proof. Taking an arbitrary $\varepsilon > 0$ we can represent $f(x)$ as the sum

$$f(x) = f_N(x)+\psi_N(x)$$

and choose N such that $|\psi_N(x)| < \varepsilon$ on E. This is possible since the sequence of functions f_n is uniformly convergent to f on E. By the hypothesis, the function f_N is continuous at x^0. Therefore, by Lemma 1, the function f is also continuous at the point x^0.

We shall also prove more subtle tests for uniform convergence based on the so-called *Abel identity* (which is an analogue of the formula for integration by parts).

Let us consider a series of the form

$$\alpha_0\beta_0+\alpha_1\beta_1+\alpha_2\beta_2+ \dots \tag{8}$$

where α_k and β_k are functions of $x \in E$ (or constant numbers).

On putting $B_k = \beta_{n+1}+\beta_{n+2}+ \dots +\beta_{n+k}$ (here n is fixed temporarily) we write for the sum $\alpha_{n+1}\beta_{n+1} + \dots +\alpha_{n+p}\beta_{n+p}$ Abel's identity

$$\begin{aligned}\alpha_{n+1}\beta_{n+1}+ \dots +\alpha_{n+p}\beta_{n+p} &= \alpha_{n+1}\beta_1+\alpha_{n+2}(B_2-B_1)+ \dots \\ &\quad \dots +\alpha_{n+p}(B_p-B_{p-1}) = (\alpha_{n+1}-\alpha_{n+2})B_1+ \\ &\quad +(\alpha_{n+2}-\alpha_{n+3})B_2+ \dots +(\alpha_{n+p-1}-\alpha_{n+p})B_{p-1}+\alpha_{n+p}B_p = \\ &= \sum_{k=1}^{p-1}(\alpha_{n+k}-\alpha_{n+k+1})B_k+\alpha_{n+p}B_p \end{aligned} \tag{9}$$

whose validity can be easily checked.

Now we can establish the following two tests for uniform convergence of a series of the indicated form (in the case of constant α_k and β_k we obtain a new test for convergence of a number series).

Theorem 3 (Dirichlet's Test for Uniform Convergence of a Series). *If the partial sums of the series*

$$\beta_0+\beta_1+\beta_2+ \dots \tag{10}$$

are uniformly bounded and the sequence of the real functions $\alpha_k(x)$ is uniformly *(relative to x) convergent to zero (as k increases) on the set E and is decreasing for every fixed x then series* (8) *is uniformly convergent on E.*

Indeed, let M be a constant such that $|\sigma_n| < M$ where σ_n are the partial sums of series (10). Then for any n and k we have

$$|B_k| = |\sigma_{n+k}-\sigma_n| \leqslant |\sigma_{n+k}|+|\sigma_n| \leqslant 2M$$

Therefore, by (9), by virtue of the fact that α_s is decreasing and uniformly tends to zero, we conclude that the inequality

$$\left|\sum_1^p \alpha_{n+k}\beta_{n+k}\right| \leqslant 2M\sum_{k=1}^{p-1}(\alpha_{n+k}-\alpha_{n+k+1})+\alpha_{n+p}2M = 2M\alpha_{n+1} < \varepsilon$$

is fulfilled for any $n > N$ and $p > 0$ and any $x \in E$ provided that N is sufficiently large. Consequently, series (8) is uniformly convergent. The last inequality in the above relations holds for all $x \in E$ because $\alpha_{n+1}(x)$ tends to zero uniformly as $n \to \infty$.

Theorem 4 (Abel's Test for Uniform Convergence of a Series). *If the real functions α_k form a monotone decreasing sequence (as k increases) for every fixed $x \in E$ and are uniformly bounded on E, and if series* (10) *is uniformly convergent on E then series* (8) *is also uniformly convergent.*

Indeed, Let $M \geqslant |\alpha_k|$ ($k = 0, 1, \dots$; the functions α_k can be positive or negative!). By the uniform convergence of series (10), for any $\varepsilon > 0$ there is N such that $|B_k| < \varepsilon$ for any $n > N$ and k. Therefore, by virtue of (9) and by the monotonicity of α_k, for any $n > N$ and p there holds

$$\left|\sum_1^p \alpha_{n+k}\beta_{n+k}\right| \leqslant \varepsilon\sum_{k=1}^{p-1}(\alpha_{n+k}-\alpha_{n+k+1})+\varepsilon|\alpha_{n+p}| =$$
$$= \varepsilon(\alpha_{n+1}-\alpha_{n+p})+\varepsilon|\alpha_{n+p}| \leqslant 3\varepsilon M$$

which means that series (8) is uniformly convergent.

Example 1. The series

$$1+(x-1)+(x^2-x)+(x^3-x^2)+\dots \qquad (0 \leqslant x \leqslant 1) \tag{11}$$

is convergent on the closed interval [0, 1] but not uniformly. Indeed, its nth partial sum is $S_n(x) = x^n$ and

$$\lim_{n\to\infty} S_n(x) = S(x) = \begin{cases} 1 & \text{for} \quad x = 1 \\ 0 & \text{for} \quad 0 \leqslant x < 1 \end{cases}$$

Therefore $\varrho_n = \sup\limits_{x \in [0,1]} |S(x)-S_n(x)| = \sup\limits_{x \in [0,1]} |x^n| = 1$, and hence ϱ_n does not tend to zero as $n \to \infty$.

On the other hand, series (11) is uniformly convergent on any closed interval $[0, q]$ where $0 < q < 1$ because in this case

$$\varrho_n = \sup_{x \in [0,q]} |S(x)-S_n(x)| = \sup_{x \in [0,q]} |x^n| = q^n \to 0 \qquad (n \to \infty)$$

The sum of series (11) is a discontinuous function at the point $x = 1$ although the terms of the series are continuous on [0, 1]. This indicates that the sum of a series of continuous functions which is convergent not uniformly is not necessarily a continuous function. In the example below there also exist series (and sequences) of continuous functions which are convergent to continuous functions although their convergence is not uniform.

Example 2. Let

$$f_n(x) = \begin{cases} 0 & \text{for } x = 0 \quad \text{and} \quad \text{for } 1/n \leq x \leq 1 \\ n & \text{for } x = 1/2n \end{cases} \tag{12}$$

and let $f_n(x)$ be extended to the whole interval [0, 1] so that it is linear and continuous on each of the subintervals [0, 1/2n] and [1/2n, 1/n] (see Fig. 11.1). Obviously, the function $f_n(x)$ thus defined is continuous throughout the closed interval [0, 1]. It is evident that $\lim\limits_{n\to\infty} f_n(x) = 0\ (0 \leq x \leq 1)$.

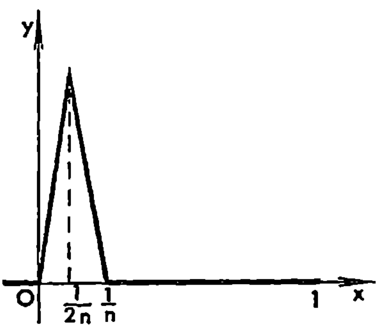

Fig. 11.1

On the other hand, the convergence of the sequence $f_n(x)$ is not uniform on the closed interval [0, 1] since $\varrho_n = \sup\limits_{x\in[0,1]} |f_n(x) - 0| = n \to \infty$. But on any closed interval of the form $[\varepsilon, 1]$ $(0 < \varepsilon < 1)$ the convergence is uniform because $f_n(x) \equiv 0$ on $[\varepsilon, 1]$ for $n > 1/\varepsilon$.

Example 3. The series

$$\sum_1^\infty \frac{\cos kx}{k^\alpha} \quad \text{and} \quad \sum_1^\infty \frac{\sin kx}{k^\alpha} \qquad (\alpha > 0) \tag{13}$$

are uniformly and absolutely convergent for $\alpha > 1$ throughout the real axis $-\infty < x < \infty$. Indeed, the absolute values of their kth terms do not exceed $k^{-\alpha}$, and the series $\sum k^{-\alpha}$ is convergent for $\alpha > 1$, and hence, by Weierstrass' test, the given series are uniformly convergent (for $\alpha > 1$). Weierstrass' test is no longer applicable when $\alpha \leq 1$ since in this case the series $\sum k^{-\alpha}$ is divergent. But if we take a narrower interval $[\varepsilon, 2\pi - \varepsilon]$ where ε is an arbitrary number such that $0 < \varepsilon < 2\pi - \varepsilon$, both series turn out to be uniformly convergent on $[\varepsilon, 2\pi - \varepsilon]$ for $0 < \alpha \leq 1$. For the partial sums of the series

$$\frac{1}{2} + \cos x + \cos 2x + \cos 3x + \dots \text{ and } \sin x + \sin 2x + \dots$$

are respectively equal to

$$D_n(x) = \frac{\sin\left(n+\frac{1}{2}\right)x}{2\sin(x/2)} \quad \text{and} \quad K_n(x) = \frac{\cos\frac{x}{2} - \cos\left(n+\frac{1}{2}\right)x}{2\sin(x/2)}$$

$$(n = 1, 2, \dots)$$

(see the example at the end of § 8.2). They are uniformly bounded on $[\varepsilon, 2\pi-\varepsilon]$:

$$|D_n(x)| \leqslant \frac{1}{2\sin(\varepsilon/2)} \quad \text{and} \quad |K_n(x)| \leqslant \frac{1}{\sin(\varepsilon/2)} \quad (n = 1, 2, \ldots)$$

Besides, $n^{-\alpha} \geqslant (n+1)^{-\alpha}$ and $n^{-\alpha} \to 0$, and therefore, by Dirichlet's test, series (13) are uniformly convergent on $[\varepsilon, 2\pi-\varepsilon]$.

§ 11.8. Integration and Differentiation of Uniformly Convergent Series on a Closed Interval

Theorem 1. *Let there be given a sequence $\{f_n\}$ of (real or complex) continuous functions on an interval $[a, b]$ convergent to a function f. If the convergence is uniform then*

$$\lim_{n\to\infty} \int_a^x f_n(t)\,dt = \int_a^x f(t)\,dt \tag{1}$$

and the sequence $\left\{\int_a^x f_n(t)\,dt\right\}$ also converges to its limit function $\int_a^x f(t)\,dt$ uniformly on $[a, b]$. In particular, for $x = b$ we have

$$\lim_{n\to\infty} \int_a^b f_n(t)\,dt = \int_a^b f(t)\,dt \tag{2}$$

Proof. From the conditions of the theorem it follows (see § 11.7, Theorem 2) that the function f is continuous on $[a, b]$ and

$$\max_{a\leqslant t\leqslant b} |f_n(t)-f(t)| = r_n \to 0 \qquad (n \to \infty)$$

Therefore

$$\left|\int_a^x f_n(t)\,dt - \int_a^x f(t)\,dt\right| \leqslant \int_a^x |f_n(t)-f(t)|\,dt \leqslant \int_a^b r_n\,dt = (b-a)r_n$$

where the right-hand side is independent of x and tends to zero as $n \to \infty$ which proves the theorem.

Theorem 2. *A series*

$$S(x) = u_0(x)+u_1(x)+u_2(x)+\ldots \tag{3}$$

of (real or complex) continuous functions uniformly convergent on a closed interval $[a, b]$ can be integrated termwise:

$$\int_{x_0}^x S(t)\,dt = \int_{x_0}^x u_0(t)\,dt + \int_{x_0}^x u_1(t)\,dt + \ldots \qquad (a \leqslant x_0 \leqslant b) \tag{4}$$

The integrated series (4) *is uniformly convergent on* $[a, b]$.

In particular,

$$\int_a^b S(t)\,dt = \int_a^b u_0(t)\,dt + \int_a^b u_1(t)\,dt + \ldots \tag{5}$$

Proof. According to the hypothesis, the nth partial sum

$$S_n(x) = u_0(x) + \ldots + u_n(x)$$

of series (3) is a continuous function on $[a, b]$, and the integral of this function

$$\int_{x_0}^{x} S_n(t)\,dt = \int_{x_0}^{x} u_0(t)\,dt + \ldots + \int_{x_0}^{x} u_n(t)\,dt$$

is the nth partial sum of series (4). Since, by the hypothesis, series (3) is uniformly convergent on the closed interval $[a, b]$, the sequence of its partial sums $\{S_n(x)\}$ converges uniformly on $[a, b]$ and, moreover, on $[x_0, x]$ to a continuous function $S(x)$ which is the sum of series (3). Consequently, by virtue of Theorem 1, we have the equality

$$\lim_{n\to\infty} \int_{x_0}^{x} S_n(t)\,dt = \int_{x_0}^{x} S(t)\,dt$$

where the convergence is uniform on $[a, b]$. This equality shows that its right member is the sum of series (4) which is convergent to it uniformly on $[a, b]$.

Theorem 3. *Let*

$$u_0(x) + u_1(x) + u_2(x) + \ldots \tag{6}$$

be a series of (real or complex) continuously differentiable functions defined on a closed interval $[a, b]$.

If series (6) *is convergent at a point* $x_0 \in [a, b]$ *and the series*

$$u_1'(x) + u_2'(x) + u_3'(x) + \ldots \tag{7}$$

obtained from (6) *by the formal term-by-term differentiation is uniformly convergent on* $[a, b]$, *then series* (6) *is uniformly convergent on* $[a, b]$, *its sum* $S(x)$ *is a differentiable function, and the derivative of* $S(x)$ *is equal to the sum of series* (7).

This theorem can be restated in terms of sequences:

Theorem 3′. *Let a sequence* $\{S_n(x)\}$ *of continuously differentiable functions on a closed interval* $[a, b]$ *be convergent at a point* $x_0 \in [a, b]$ *and let the sequence of the derivatives* $\{S_n'(x)\}$ *be uniformly convergent on that interval to a function* $\varphi(x)$. *Then* $S_n(x)$ *is convergent throughout the interval* $[a, b]$ *to a continuously differentiable function* $S(x)$, *the convergence is uniform and* $S'(x) = \varphi(x)$.

Theorem 3 follows from Theorem 3' if we assume that in the latter the function $S_n(x)$ ($n = 1, 2, \ldots$) is the nth partial sum of series (6); then it follows automatically that $S_n'(x)$ is the nth partial sum of series (7).

Proof of Theorem 3'. By the hypothesis, the function $S_n(x)$ is continuously differentiable for any natural n. Therefore, by the Newton-Leibniz theorem,

$$S_n(x) = S_n(x_0) + \int_{x_0}^{x} S_n'(t)\,dt, \quad x \in [a, b], \quad n = 1, 2, \ldots \tag{10}$$

By the conditions of the theorem, the limit

$$\lim_{n\to\infty} S_n(x_0) = A$$

of the number sequence $S_n(x_0)$ exists. This sequence can be regarded as a uniformly convergent sequence of constant functions. The function $\varphi(x)$ is continuous on the closed interval $[a, b]$ because it is the limit of the sequence of continuous functions $\{S_n'(x)\}$ uniformly convergent on that interval. We also have the equality

$$\lim_{n\to\infty} \int_{x_0}^{x} S_n'(t)\,dt = \int_{x_0}^{x} \varphi(t)\,dt$$

where the convergence is uniform on $[a, b]$ (see Theorem 1).

What has been said shows that the right-hand side of (10) and, together with it, the left-hand side are uniformly convergent, as $n \to \infty$, on $[a, b]$ to the function

$$A + \int_{x_0}^{x} \varphi(t)\,dt$$

which we shall denote by $S(x)$.

The function φ being continuous, the equality obtained can be differentiated with respect to x. This results in the equality $\varphi(x) = S'(x)$ which holds for any $x \in [a, b]$.

The theorem has been proved.

Remark. Theorem 3' is widely applied in mathematical analysis. It obviously continues to hold when the index n on which $S_n(x)$ depends runs through any number sequence n_k tending to a (finite or infinite) number n_0 or when n tends to n_0 continuously.

Example 1. Let us consider the series

$$A_\alpha(x) = \sum_{k=1}^{\infty} \frac{\cos\left(kx + \frac{\alpha\pi}{2}\right)}{k^\alpha} \qquad (k = 1, 2, \ldots) \tag{11}$$

For even α it has the form

$$\text{(a)} \quad \pm\sum_{1}^{\infty} \frac{\cos kx}{k^\alpha}$$

and for odd α the form

$$\text{(b)} \quad \pm\sum_{1}^{\infty} \frac{\sin kx}{k^{\alpha}}$$

Since $\frac{d}{dx}\cos\left(kx+\frac{\alpha\pi}{2}\right) = -k\cos\left(kx+(\alpha-1)\frac{\pi}{2}\right)$ we can write, formally,

$$\frac{d}{dx}A_{\alpha}(x) = -A_{\alpha-1}(x) \tag{12}$$

But this equality does in fact hold for any real x when $\alpha = 3, 4, \ldots$; it also holds for $\alpha = 2$ when x is any real number satisfying the conditions

$$x \neq 2k\pi \qquad (k = 0, \pm 1, \pm 2, \ldots) \tag{13}$$

This follows from Theorem 3 and from Example 3 in § 11.7 where some properties of series (a) and (b) were discussed. To prove equality (12) for $\alpha = 2$ and for a fixed x satisfying inequalities (13) we take a closed interval $[a, b]$ containing x strictly inside and not containing the points of the form $2k\pi$ $(k = 0, \pm 1, \ldots)$. Both series (11) are uniformly convergent on $[a, b]$ for $\alpha = 1$ and $\alpha = 2$, which makes it possible to apply Theorem 3.

Example 2. Let us take the sequence of the functions

$$f_n(x) = \begin{cases} 0 & \text{for } x = 0 \text{ and for } 1/n \leqslant x \leqslant 1 \\ \alpha_n & \text{for } x = 1/2n \end{cases} \tag{14}$$

which are extended, for every n, to the whole interval $[0, 1]$ so that they are linear and continuous on each of the intervals $[0, 1/2n]$ and $[1/2n, 1/n]$, the numbers α_n forming an arbitrary sequence. Then we obviously have $\lim\limits_{n\to\infty} f_n(x) = 0$ for all $x \in [0, 1]$, and

$$\int_0^1 f_n(x)\,dx = \int_0^{1/2n} 2n\alpha_n x\,dx - \int_{1/2n}^{1/n} 2n\alpha_n\left(x-\frac{1}{n}\right)dx = \frac{\alpha_n}{2n}$$

Further, it is evident that

$$r_n = \sup_{0\leqslant x\leqslant 1} |f_n(x)-0| = \alpha_n$$

The sequence $\{f_n(x)\}$ is uniformly convergent on $[0, 1]$ if and only if $\alpha_n \to 0$. The relation

$$\int_0^1 f_n(x)\,dx \to \int_0^1 f(x)\,dx \qquad (f(x) \equiv 0) \tag{15}$$

is fulfilled if and only if $\frac{\alpha_n}{2n} \to 0$ $(n \to \infty)$.

We see that the uniform convergence of f_n to $f \equiv 0$ on $[0, 1]$ implies the convergence of integrals (15), which agrees with Theorem 2. But the sequence $\{f_n\}$ can also be convergent nonuniformly and in this case property (15) may continue to hold, for instance, when $\alpha_n = 1$. But if we put $\alpha_n = n$ then not only the sequence $\{f_n\}$ converges to zero nonuniformly but property (15) fails to hold either.

Example 3. The equality $(1-z)^{-1} = 1+z+z^2+\ldots$ $(z = \varrho e^{i\theta},\ \varrho < 1)$ implies that

$$\frac{1+\varrho e^{i\theta}}{2(1-\varrho e^{i\theta})} = \frac{1}{2}+\varrho e^{i\theta}+\varrho^2 e^{i2\theta}+\ldots$$

On separating the real and imaginary parts we obtain

$$P_\varrho(\theta) = \frac{1}{2}\,\frac{1-\varrho^2}{1-2\varrho\cos\theta+\varrho^2} = \frac{1}{2}+\varrho\cos\theta+\varrho^2\cos 2\theta+\ldots \tag{16}$$

and

$$Q_\varrho(\theta) = \frac{\varrho\sin\theta}{1-2\varrho\cos\theta+\varrho^2} = \varrho\sin\theta+\varrho^2\sin 2\theta+\ldots \tag{17}$$

The function $P_\varrho(\theta)$ is known as the *Poisson* kernel* and $Q_\varrho(\theta)$ is its *conjugate kernel.*

Exercise. Show that $P_\varrho(\theta)$ and $Q_\varrho(\theta)$ (as functions of ϱ and θ) are harmonic functions (for $\varrho < 1$), that is they satisfy the Laplace equation $\Delta u = 0$ (see § 7.26 (15) and (17)). *Hint:* show that $\varrho^n\cos n\theta$ and $\varrho^n\sin n\theta$ are harmonic functions for any n and $\varrho \geq 0$ and then apply the theorem on term-by-term differentiation of uniformly convergent series.

§ 11.9. Multiple Series. Multiplication of Absolutely Convergent Series

An expression

$$\sum_{k=0}^{\infty}\sum_{l=0}^{\infty} a_{kl} \tag{1}$$

where a_{kl} are (real or complex) numbers dependent on the pair of indices $k, l = 0, 1, 2, \ldots$ is called a *double series.* The numbers a_{kl} are its *terms* and the numbers

$$S_{mn} = \sum_{k=0}^{m}\sum_{l=0}^{n} a_{kl} \quad (m, n = 0, 1, 2\ldots) \tag{2}$$

are the *partial sums* of the series.

By definition, series (1) is said to be *convergent to a number S* called the *sum* of series (1) if the limit

$$\lim_{m,\,n\to\infty} S_{mn} = S \tag{3}$$

exists, that is if, given any $\varepsilon > 0$, there is N such that

$$|S-S_{mn}| < \varepsilon$$

* S. D. Poisson (1781-1840), a French mathematician.

for all $m, n > N$. Then we write

$$S = \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} a_{kl}$$

Now let us dwell on the special case when the terms of series (1) are nonnegative ($a_{kl} \geqslant 0$). Let us put

$$A = \sup_{m, n} S_{mn} \tag{4}$$

If $A < \infty$ is a finite number then for any $\varepsilon > 0$ there is a pair of numbers m_0, n_0 such that $A - \varepsilon < S_{m_0 n_0} \leqslant A$, and, since a_{kl} are nonnegative, we have

$$S_{m_0 n_0} \leqslant S_{mn}, \quad m, n > N = \max(m_0, n_0)$$

Therefore $A - \varepsilon < S_{mn} < A + \varepsilon$ $(m, n > N)$ and the limit $\lim\limits_{m, n \to \infty} S_{mn} = S = A$ exists.

If $A = \infty$ then (on condition that $a_{kl} \geqslant 0$!) we obviously have $\lim\limits_{m, n \to \infty} S_{mn} = S = \infty$. In this case we write

$$\sum_{k=0}^{\infty} \sum_{l=0}^{\infty} a_{kl} = \infty$$

Series (1) (with arbitrary terms) is said to be *absolutely convergent* if the series $\sum\limits_{k=0}^{\infty} \sum\limits_{l=0}^{\infty} |a_{kl}|$ is convergent. As in the case of ordinary series, it can be proved (using Cauchy's criterion) that an absolutely convergent series is convergent.

Together with series (1) we can consider the expression

$$\sum_{k=0}^{\infty} \left(\sum_{l=0}^{\infty} a_{kl} \right)$$

to which a certain number (sum) A can be attributed in the following natural way: if the series in brackets is convergent for every $k = 0, 1, \ldots$ and has a sum A_k, and the series $\sum\limits_{0}^{\infty} A_k$ converges to a number A we put

$$A = \sum_{k=0}^{\infty} A_k = \sum_{k=0}^{\infty} \left(\sum_{l=0}^{\infty} a_{kl} \right) \tag{5}$$

Theorem 1. *If series* (1) *is absolutely convergent then*

$$\sum_{k=0}^{\infty} \sum_{l=0}^{\infty} a_{kl} = \sum_{k=0}^{\infty} \left(\sum_{l=0}^{\infty} a_{kl} \right) \tag{6}$$

Proof. Let us first suppose that the numbers a_{kl} are nonnegative. Let the left-hand side of (6) (which makes sense!) be denoted as S.

For any nonnegative s and n such that $s \leqslant m$ we have

$$\sum_{l=0}^{n} a_{sl} \leqslant \sum_{k=0}^{m} \sum_{l=0}^{n} a_{kl} \leqslant S \tag{7}$$

whence follows that the series $\sum_{l=0}^{\infty} a_{sl} (s = 0, 1, 2, \ldots)$ are convergent; consequently, if we fix m in the second inequality and pass to the limit as $n \to \infty$, the resultant inequality

$$\sum_{k=0}^{m} \left(\sum_{l=0}^{\infty} a_{kl} \right) \leqslant S$$

holds for any m, which implies the existence of the number A (see (5)) and the fact that $A \leqslant S$.

On the other hand, if the number A is finite then, for any m and n, we have

$$S_{mn} = \sum_{k=0}^{m} \sum_{l=0}^{n} a_{kl} \leqslant \sum_{k=0}^{m} \left(\sum_{l=0}^{\infty} a_{kl} \right) \leqslant A$$

and therefore

$$S = \sup_{m,n} \ S_{mn} \leqslant A$$

We have proved equality (6) for $a_{kl} \geqslant 0$.

Now let a_{kl} be arbitrary real numbers. Let us put

$$a_{kl}^{+} = \begin{cases} a_{kl} & \text{for} \quad a_{kl} \geqslant 0 \\ 0 & \text{for} \quad a_{kl} < 0 \end{cases} \quad \text{and} \quad a_{kl}^{-} = \begin{cases} -a_{kl} & \text{for} \quad a_{kl} \leqslant 0 \\ 0 & \text{for} \quad a_{kl} > 0 \end{cases}$$

Then

$$a_{kl} = a_{kl}^{+} - a_{kl}^{-} \quad \text{and} \quad a_{kl}^{+} + a_{kl}^{-} = |a_{kl}|$$

Consequently, the convergence of the series $\sum\sum |a_{kl}|$ implies that the series $\sum\sum a_{kl}^{+}$ and $\sum\sum a_{kl}^{-}$ with nonnegative terms are also convergent, and therefore

$$\sum\sum a_{kl} = \sum\sum a_{kl}^{+} - \sum\sum a_{kl}^{-} = \sum_{k} \left(\sum_{l} a_{kl}^{+} \right) - \sum_{k} \left(\sum_{l} a_{kl}^{-} \right) = \sum_{k} \left(\sum_{l} a_{kl} \right)$$

Finally, if $a_{kl} = \alpha_{kl} + i\beta_{kl}$ are complex numbers and the series $\sum\sum |a_{kl}|$ is convergent, the series $\sum\sum |\alpha_{kl}|$ and $\sum\sum |\beta_{kl}|$ where α_{kl} and β_{kl} are real numbers are also convergent, and consequently

$$\sum\sum a_{kl} = \sum\sum \alpha_{kl} + i \sum\sum \beta_{kl} = \sum_{k} \left(\sum_{l} \alpha_{kl} \right) + i \sum_{k} \left(\sum_{l} \beta_{kl} \right) = \sum_{k} \left(\sum_{l} a_{kl} \right)$$

The proof of the theorem has been completed.

Let us proceed to the following question. Let there be given an absolutely convergent series of form (1). Its sum S and also the sum S' of the absolute values of its terms can be represented as the limits of the sequences

$$S = \lim_{n\to\infty} \sum_0^n \sum_0^n a_{kl} = \lim_{n\to\infty} S_{nn} \quad \text{and}$$

$$S' = \lim_{n\to\infty} \sum_0^n \sum_0^n |a_{kl}| = \lim_{n\to\infty} S'_{nn}$$

respectively, whose terms depend only on one index n. To the sequences $\{S_{nn}\}$ and $\{S'_{nn}\}$ there correspond the series

$$S = a_{00}+(a_{10}+a_{11}+a_{01})+ \\ +(a_{20}+a_{21}+a_{22}+a_{12}+a_{02})+(a_{30}+\dots)+\dots \tag{8}$$

and

$$|a_{00}|+(|a_{10}|+|a_{11}|+|a_{01}|)+(|a_{20}|+|a_{21}|+|a_{22}|+|a_{12}|+|a_{02}|)+\dots \tag{9}$$

with terms equal to the sums of numbers in brackets. In the brackets of the second series there are nonnegative numbers and therefore its convergence is not altered if the brackets are removed:

$$|a_{00}|+|a_{10}|+|a_{01}|+|a_{20}|+\dots \tag{10}$$

It follows that the series

$$a_{00}+a_{10}+a_{01}+a_{20}+\dots \tag{11}$$

obtained from (8) by removing all the brackets is absolutely convergent and hence it obviously converges to S.

We have shown that if a double series of type (1) is absolutely convergent to a number S then the ordinary (one-fold) series (11) obtained from it is also absolutely convergent to S. Now, since the terms of an absolutely convergent series can be rearranged in an arbitrary way without altering its sum, we arrive at the following theorem.

Theorem 2. *If the terms of a double series of type* (1) *which is absolutely convergent to a number S are renumbered in an arbitrary manner as v_0, v_1, v_2, ... with the aid of one index the series $v_0+v_1+v_2+\dots$ thus obtained is (absolutely) convergent to the same number S.*

In conclusion we shall prove the following theorem.

Theorem 3. *Let $\sum_0^\infty u_k$ and $\sum_0^\infty v_l$ be two absolutely convergent (one-fold) series and let all the possible products $u_k v_l$ $(k, l = 0, 1, 2, \dots)$ be numbered with the aid of one index as $w_1, w_2, w_3, \dots$ Then there holds the equality*

$$\sum_0^\infty u_k \times \sum_0^\infty v_l = \sum_0^\infty w_k$$

where the series on the right-hand side is absolutely convergent.

Proof. Indeed, let $\sum\limits_0^\infty |u_k| = M$ and $\sum\limits_0^\infty |v_l| = N$. The double series $\sum\limits_{k=0}^\infty \sum\limits_{l=0}^\infty u_k v_l$ is absolutely convergent because we have

$$\sum_0^m \sum_0^n |u_k v_l| = \sum_0^m |u_k| \times \sum_0^n |v_l| \leq MN$$

for any m and n. Therefore

$$\sum_{k=0}^\infty u_k \times \sum_{l=0}^\infty v_l = \lim_{n\to\infty} \sum_{k=0}^n u_k \times \lim_{n\to\infty} \sum_{l=0}^n v_l = \lim_{n\to\infty} \sum_{k=0}^n \sum_{l=0}^n u_k v_l =$$

$$= \sum_{k=0}^\infty \sum_{l=0}^n u_k v_l = \sum_{j=0}^\infty w_j$$

where the last equality is fulfilled by virtue of the foregoing theorem.

Example. The series $\psi(z) = \sum\limits_0^\infty \frac{z^k}{k!}$ is absolutely convergent for any complex z. Consequently, by Theorem 2, we have

$$\psi(z)\,\psi(u) = 1 + \frac{z}{1} + \frac{u}{1} + \frac{z^2}{2!} + \frac{z}{1}\cdot\frac{u}{1} + \frac{u^2}{2!} + \frac{z^3}{3!} + \frac{z^2}{2!}\cdot\frac{u}{1} + \frac{z}{1} + \frac{u^2}{2!} + \frac{u^3}{3!} + \ldots$$

for any complex z and u where the series on the right-hand side is absolutely convergent.

Taking into account that

$$\frac{z^n}{n!} + \frac{z^{n-1}}{(n-1)!}\,\frac{u}{1} + \ldots + \frac{u^n}{n!} = \frac{(z+u)^n}{n!}$$

we see that

$$\psi(z)\,\psi(u) = \sum_0^\infty \frac{(z+u)^n}{n!} = \psi(z+u)$$

When two absolutely convergent series $\sum\limits_0^\infty u_k$ and $\sum\limits_0^\infty v_l$ are multiplied by each other their product is often represented in the form of the one-fold series

$$\sum_0^\infty u_k \times \sum_0^\infty v_l = u_0v_0 + u_1v_0 + u_0v_1 + u_2v_0 + u_1v_1 + u_0v_2 + u_3v_0 + \ldots$$

(especially in the theory of power series; see § 11.11). This very operation has been performed in the above example.

In conclusion we note that a similar theory can be developed for *multiple series* of form

$$\sum_{k=0}^\infty \sum_{l=0}^\infty \sum_{m=0}^\infty a_{klm}, \ldots$$

(this is a *triple* series) and, generally,

$$\sum_{k_1=0}^{\infty} \cdots \sum_{k_n=0}^{\infty} a_{k_1, \ldots, k_n}$$

For such series it is possible to prove, by analogy, the theorems similar to Theorems 1-3.

§ 11.10. Method of Arithmetic Means for Summation of Series and Sequences

Let us consider a number series

$$u_0+u_1+u_2+\ldots \tag{1}$$

and denote

$$S_n = u_0+u_1+u_2+\ldots+u_n \tag{2}$$

and

$$\sigma_n = \frac{S_0+S_1+\ldots+S_n}{n+1} \qquad (n = 0, 1, 2, \ldots) \tag{3}$$

If there exists the limit

$$\lim_{n\to\infty} \sigma_n = \sigma \tag{4}$$

the method of arithmetic means attributes the "sum" σ to series (1) (or the "limit" σ to the sequence $\{S_n\}$).

Theorem. *If series* (1) *converges to a number S then its summation by the method of arithmetic means leads to the same number S.*

Proof. Let series (1) be convergent, then there is $M > 0$ such that

$$|S_j| \leqslant M \qquad (j = 0, 1, \ldots) \tag{5}$$

and there is also a sufficiently large natural n (which will be regarded as fixed while the indices k and p below will be understood as variable) such that

$$|S_{n+k}-S| < \varepsilon \qquad (k = 1, 2, \ldots) \tag{6}$$

Further, we have

$$S-\sigma_{n+p} = \left(S-\frac{1}{p}\sum_{k=1}^{p} S_{n+k}\right)+\left(\frac{1}{p}-\frac{1}{n+p+1}\right)\sum_{k=1}^{p} S_{n+k}-\frac{1}{n+p+1}\sum_{k=0}^{n} S_k$$

whence, taking into account that $\frac{1}{p}-\frac{1}{n+p+1} = \frac{n+1}{p(n+p+1)}$, we obtain

$$|S-\sigma_{n+p}| < \varepsilon+\frac{n+1}{n+p+1}M+\frac{n+1}{n+p+1}M < \varepsilon+\varepsilon+\varepsilon = 3\varepsilon \qquad (p > p_0)$$

if p is sufficiently large. Consequently $\sigma_{n+p} \to S$ $(p\to\infty)$ or, which is the same, $\sigma_j \to S\,(j\to\infty)$, which completes the proof of the theorem.

Example. The series $1-1+1-\ldots$ is divergent while the summation by the method of arithmetic means gives the value $\sigma = 1/2$.

§ 11.11. Power Series

A series of the form

$$a_0+a_1z+a_2z^2+\ldots \tag{1}$$

where a_k $(k = 0, 1, 2, \ldots)$ are constant numbers (which may be complex in the general case) and z is a complex variable is called a *power series*, the numbers a_k being referred to as its *coefficients*.

Theorem (Abel). *If series* (1) *is convergent at a point $z_0 \neq 0$ of the complex plane then it is convergent absolutely and uniformly in the circle*

$$|z| \leqslant q \tag{2}$$

where q is any (fixed) number satisfying the inequalities

$$0 < q < |z_0| \tag{3}$$

Proof. The convergence of series (1) at the point z_0 implies the existence of a constant $M > 0$ such that

$$|a_k z_0^k| \leqslant M \qquad (k = 0, 1, \ldots) \tag{4}$$

Consequently, for the points z satisfying inequality (2) we have

$$|a_k z^k| = |a_k z_0^k| \left|\frac{z}{z_0}\right|^k \leqslant M\alpha^k \qquad (k = 0, 1, \ldots) \qquad \alpha = \frac{q}{|z_0|} < 1$$

Thus, the moduli of the terms of series (1) do not exceed the corresponding terms of the convergent dominant series

$$M+M\alpha+M\alpha^2 \quad (M\alpha^n > 0;\ n = 0, 1, 2, \ldots)$$

for all z belonging to circle (2). Hence, by Weierstrass' test, series (1) is absolutely and uniformly convergent for all such z's. The theorem has been proved.

Note that, according to Weierstrass' test, series (1) is convergent for every z belonging to the open circle $|z| < |z_0|$ since q in Abel's theorem is an arbitrary number satisfying the inequality $q < |z_0|$. At the same time, as is seen from examples, the convergence of the series in that open circle is by far not always uniform.

Let

$$R = \sup_{z \in \mathfrak{M}} |z| \tag{5}$$

be the supremum taken over the set $\mathfrak{M}$ of points z for which series (1) is convergent. In the general case $0 \leqslant R \leqslant \infty$. The (finite) number R is called *the radius of convergence of power series* (1).

(i) If $0 < R \leq \infty$ then, for any z such that $|z| < R$, series (1) is convergent because, according to the definition of supremum, there is $z_1 \in \mathfrak{M}$ such that $|z| < |z_1| < R$ and series (1) is convergent for $z = z_1$. Therefore, by Abel's theorem, it is also convergent for z.

(ii) On the other hand, in the case of finite R, series (1) is sure to be divergent for any z such that $|z| > R$; this is a direct consequence of the definition of the number R.

There can obviously exist only one number R possessing properties (i), (ii). What has been said allows us to assert that the radius of convergence of a power series convergent for at least one value $z \neq 0$ can be defined as the number $R > 0$ (finite or infinite) for which properties (i) and (ii) hold. If a given power series is only convergent at one point $z = 0$ then, by the definition stated above, its radius of convergence is equal to zero ($R = 0$).

If $R > 0$, the (open) circle $|z| < R$ in the complex plane is called *the circle of convergence of power series* (1).

Finally, we can also give a third definition of R by putting it equal to the reciprocal of the number

$$\overline{\lim_{k\to\infty}} \sqrt[k]{|a_k|} = A$$

that is

$$R = 1/A \tag{6}$$

on condition that $1/\infty = 0$ and $1/0 = \infty$.

Indeed, let us begin with the case $R > 0$ (it can be, in particular, that $R = \infty$) where R is understood in the sense of the first or second definition. If R_1 is an arbitrary number satisfying the inequalities $0 < R_1 < R$, series (1) converges at the point $z = R_1$, and the absolute value of its general term is bounded: $|a_k R_1^k| \leq M$ $(k = 1, 2, \ldots)$. Therefore $R_1 \sqrt[k]{|a_k|} \leq \sqrt[k]{M}$, whence

$$R_1 \overline{\lim} \sqrt[k]{|a_k|} = \overline{\lim} R_1 \sqrt[k]{a_k} \leq \lim_{k\to\infty} \sqrt[k]{M} = 1$$

that is

$$A \leq 1/R_1 \tag{7}$$

By the arbitrariness of $R_1 < R$, relation (7) implies the inequality

$$A \leq 1/R \tag{8}$$

which obviously continues to hold for $R = 0$.

Now, if R is finite, series (1) is divergent for $z = R_1 > R$ and, moreover, the series $\sum |a_k| R_1^k$ is also divergent, this shows that there must be $\overline{\lim} \sqrt[k]{|a_k| R_1^k} \geq 1$ (if otherwise, the series would satisfy the conditions of Cauchy's first test, see footnote on p. 421). Consequently, $R_1 A = R_1 \overline{\lim} \times \times \sqrt[k]{|a_k|} \geq 1$ for any $R_1 > R$, whence $RA \geq 1$, that is

$$A \geq 1/R \tag{9}$$

Obviously inequality (9) continues to hold for $R = \infty$.

Finally, (8) and (9) imply (6).

Examples

$$1+z+z^2+\dots \tag{10}$$

$$1+\frac{z}{1^\alpha}+\frac{z^2}{2^\alpha}+\dots \qquad (\alpha > 0) \tag{11}$$

$$1+z+2!z^2+3!z^3+\dots \tag{12}$$

$$1+\frac{z}{1}+\frac{z^2}{2!}+\frac{z^3}{3!}+\dots \tag{13}$$

On the basis of formula (6) we conclude that the radii of convergence of series (10) and (11) are equal to 1, for series (12) the radius of convergence is equal to 0 and for series (13) to ∞.

The sum of series (10) (called *the geometric series*) is equal to $(1-z)^{-1}$ in the open circle $|z| < 1$; its remainder is

$$r_n(z) = \frac{z^{n+1}}{1-z} \to 0 \qquad (n \to \infty)$$

But the convergence of the series in the indicated open circle is not uniform. For instance, this can be demonstrated by the fact that even for positive $z = x$ satisfying the condition $0 < x < 1$ there is no number N such that inequality $\varepsilon > x^{n+1}/(1-x)$ is satisfied for all $n > N$ and all $x \in [0, 1]$.

For $\alpha > 1$ series (11) is uniformly convergent throughout the closure of its circle of convergence since $|z^a|/k^\alpha \leq k^{-\alpha}$ and $\sum k^{-\alpha} < \infty$ ($|z| \leq 1$).

Series (12) is convergent at only one point $z = 0$ while series (13) is convergent throughout the complex plane.

§ 11.12. Differentiation and Integration of Power Series

We shall prove that *if a power series*

$$f(z) = a_0+a_1z+a_2z^2+\dots \tag{1}$$

having a radius of convergence $R > 0$ (in particular, there can be $R = \infty$) is formally differentiated term-by-term, the resultant (differentiated) series

$$\varphi(z) = a_1+2a_2z+3a_3z^2+\dots \tag{2}$$

has the same radius of convergence R, and

$$f'(z) = \varphi(z), \qquad |z| < R \tag{3}$$

Indeed, let R' be the radius of convergence of series (2). It is evident that the series

$$a_1z+2a_2z^2+3a_3z^2+\dots \tag{4}$$

obtained from series (2) by multiplying its terms by z has the same radius of convergence R'. If $|z| < R'$, then series (4) is absolutely convergent, and, by virtue of the inequality $|a_nz^n| \leq n|a_nz^n|$, series (1) is also absolutely convergent. This means that $R' \leq R$. On the other hand, if z is an arbitrary complex number such that $|z| < R$ then there is another number z_1 such that $|z| < |z_1| < R$, and therefore there is a number $M > 0$ for which $|a_n z_1^n| \leqslant M$. Then we have

$$n|a_n z^n| = n\left|\frac{z}{z_1}\right|^n |a_n z_1^n| \leqslant Mn\left|\frac{z}{z_1}\right|^n \qquad (n = 0, 1, \ldots) \tag{5}$$

and, since, by Cauchy's test, the series formed of the right members of (5) is convergent, series (2) is also convergent, whence $R \leqslant R'$.

Hence we have shown that $R = R'$.

Now we proceed to prove (3). Let z be an arbitrary complex point belonging to the circle $|z| < R$ and let R_1 be a (finite!) number such that $|z| < R_1 < R$. Below we consider only those complex numbers h for which $|h| \leqslant R_1 - |z| = \delta$ ($\delta > 0$). Then $|z+h| < R_1$, and we have

$$\frac{f(z+h)-f(z)}{h} = \alpha_1(h) + \alpha_2(h) + \ldots \tag{6}$$

where

$$\alpha_n(h) = a_n \frac{(z+h)^n - z^n}{h} = a_n\big((z+h)^{n-1} + (z+h)^{n-2}z + \ldots + z^{n-1}\big)$$

$$(|h| > 0, \ n = 1, 2, \ldots)$$

Let us also put

$$\alpha_n(0) = a_n \lim_{n\to 0} \frac{(z+h)^n - z^n}{h} = na_n z^{n-1} \tag{7}$$

By this convention, the terms of series (6) are defined not only for $|h| > 0$ but also for $h = 0$, they are continuous functions of h in the closed circle $|h| \leqslant \delta$ and obviously satisfy the inequalities

$$|\alpha_n(h)| \leqslant |a_n|\,(R_1^{n-1} + R_1^{n-2}R_1 + \ldots + R_1^{n-1}) \leqslant n|a_n|\,R_1^{n-1} \qquad (|h| \leqslant \delta)$$

The positive numbers on the right-hand sides of these inequalities are independent of h, and the series formed of them is convergent. Indeed, R is not only the radius of convergence of series (1) but of series (2) as well. Therefore, by Weierstrass' theorem, series (6) of functions continuous in the closed circle $|h| \leqslant \delta$ is uniformly convergent on that circle. It follows that the sum of series (6) is also a continuous function of h in that circle, and, in particular, at the point $h = 0$. Consequently, the limit

$$\lim_{h\to 0} \frac{f(z+h)-f(z)}{h} = \alpha_1(0) + \alpha_2(0) + \ldots$$

exists when the complex variable h tends to zero in an arbitrary way. Thus, the derivative $f'(z)$ (understood in the sense of a complex variable) exists for any z satisfying the condition $|z| < R$ and is equal to

$$f'(z) = a_1 + 2a_2 z + 3a_3 z^2 + \ldots$$

whence follows (3).

A thorough investigation of term-by-term integration of power series involves the notion of a line integral of a complex function which will not be treated here. Therefore we shall confine ourselves to considering the integration problem for power series

$$f(x) = a_0+a_1x+a_2x^2+\dots \tag{8}$$

in the real variable x ($z = x$).

Let us take a series of type (8) convergent in an interval $-R < x < R$ where $0 < R \leq \infty$. The numbers a_k can be real or complex. Choosing a point $x_0 \in (-R, R)$ which will be regarded as fixed and denoting by x the variable point belonging to $(-R, R)$ we can find $q > 0$ such that $-R < q < x_0$, $x < q < R$. Power series (8) is uniformly convergent in the closed interval $[-q, q]$ lying strictly inside the interval of convergence of the series. Therefore it can be integrated termwise (see § 11.8, Theorem 2) over the closed interval with end points x_0 and x:

$$\int_{x_0}^{x} f(t)\,dt = a_0(x-x_0)+\frac{a_1}{2}(x^2-x_0^2)+\frac{a_2}{3}(x^3-x_0^3)+\dots$$

$$(-R < x,\ x_0 < R) \tag{9}$$

In particular, putting $x_0 = 0$ we obtain

$$\int_{0}^{x} f(t)\,dt = a_0x+\frac{a_1}{2}x^2+\frac{a_2}{3}x^3+\dots \qquad (-R < x < R) \tag{10}$$

Examples

$$\arctan x = x-\frac{x^3}{3}+\frac{x^5}{5}-\dots, \qquad (-1 \leq x \leq 1) \tag{11}$$

$$\arcsin x = x+\frac{x^3}{2\cdot 3}+\frac{1\cdot 3}{2!}\frac{x^5}{2^25}+\frac{1\cdot 3\cdot 5}{3!}\frac{x^7}{2^37}+\dots \qquad (-1 \leq x \leq 1) \tag{12}$$

These equalities are obtained for $x \in (-1, 1)$ by means of term-by-term integration of the well-known equalities

$$\frac{1}{1+x^2} = 1-x^2+x^4-\dots$$

and

$$\frac{1}{(1-x^2)^{1/2}} = 1+\frac{x^2}{2}+\frac{1\cdot 3}{2!}\frac{x^4}{2^2}+\frac{1\cdot 3\cdot 5}{3!}\frac{x^6}{2^3}+\dots \tag{13}$$

respectively over the closed interval with end points 0 and x. By Leibniz' test, series (11) is convergent at the points $x = 1$ and $x = -1$. Equality (11) itself holds by virtue of Abel's second theorem proved below.

Series (13) is not convergent for $x = 1$ and $x = -1$ because, if otherwise, by Abel's second theorem, its sum would be a continuous function on $[-1, +1]$. At the same time series (12) is convergent at the points $x = 1$ and $x = -1$ because for these points the absolute value of its general term satisfies (see the explanations bellow) the asymptotic relation

$$|u_n| = \frac{(2n-1)!!}{(2n)!!\,(2n+1)} = \frac{(2n-1)!!\,(2n)!!}{((2n)!!)^2\,(2n+1)} = \frac{(2n)!}{2^{2n}(n!)^2\,(2n+1)} \approx$$

$$\approx \frac{\sqrt{2\pi}\,(2n)^{2n+(1/2)}e^{-2n}}{2^{2n}2\pi n^{2n+1}e^{-2n}(2n+1)} = \sqrt{\frac{1}{\pi}}\,\frac{1}{\sqrt{n}\,(2n+1)} \qquad (n\to\infty)$$

Here we have written the symbol $\approx$ which was already used in § 9.18. In the fourth of the above equalities (with $\approx$) we have applied Stirling's formula (§ 9.18(6)). The series whose general term is equal to the rightmost member of the last relations is convergent and therefore series $\sum u_n$ is also convergent (see § 11.3 (1)).

According to Abel's second theorem, the convergence of series (12) at the points $x = \pm 1$ implies the continuity of its sum $S(x)$ on the interval $[-1, +1]$. For the open interval $(-1, +1)$ we have the equality $S(x) = \arcsin x$; since the function $\arcsin x$ is continuous on $[-1, +1]$ this equality also holds for the entire closed interval $[-1, +1]$.

Abel's Second Theorem. *If a power series*

$$f(x) = a_0+a_1x+a_2x^2+\ldots \tag{14}$$

has a radius of convergence $R < \infty$ and is convergent at the point $x = R$ then the function $f(x)$ is continuous not only in the open interval $(-R, R)$ but also in the half-open interval $(-R, R]$.

Indeed, the general term of series (14) can be written in the form

$$a_nx^n = a_nR^n(x/R)^n$$

where the (constant) numbers a_nR^n can be regarded as the terms of a convergent number series while the expressions $(x/R)^n$ form a sequence of functions bounded and nonincreasing on $[0, R]$ $(1 \geqslant (x/R)^n \geqslant (x/R)^{n+1}$; $n = 0, 1, \ldots)$. Therefore, by Abel's test (see § 11.7, Theorem 4), series (14) of functions continuous on the closed interval $[0, R]$ is uniformly convergent on that interval and, consequently, its sum $f(x)$ is a continuous function on $[0, R]$.

§ 11.13. Power Series Expansions for the Functions e^z, cos z and sin z of Complex Variable z

The functions e^z, $\cos z$ and $\sin z$ of the complex variable z are defined as the sums of the series

$$\exp z = e^z = 1+\frac{z}{1}+\frac{z^2}{2!}+\frac{z^3}{3!}+\ldots \tag{1}$$

$$\cos z = 1-\frac{z^2}{2!}+\frac{z^4}{4!}-\ldots \tag{2}$$

and

$$\sin z = z-\frac{z^3}{3!}+\frac{z^5}{5!}-\ldots \tag{3}$$

respectively.

These series are convergent for any complex z because the radius of convergence of each series is equal to ∞. Thus, the functions e^z, $\cos z$ and $\sin z$ are defined throughout the complex plane. For real $z = x$ these definitions lead to the well-known real functions e^x, $\cos x$ and $\sin x$ (see § 5.11).

The function e^z possesses a remarkable property expressed by the identity

$$e^{z+u} = e^z e^u \tag{4}$$

holding for any complex z and u (see the example in § 11.9).

It is evident that for any complex z we have

$$e^{iz} = \cos z + i \sin z \tag{5}$$

$$\cos z = \frac{1}{2}(e^{iz}+e^{-iz}) \quad \text{and} \quad \sin z = \frac{1}{2i}(e^{iz}-e^{-iz}) \tag{6}$$

Equalities (6) are known as *Euler's formulas*. From (6) and (4) follow the formulas

$$\sin (z+u) = \sin z \cos u + \cos z \sin u$$

and

$$\cos (z+u) = \cos z \cos u - \sin z \sin u$$

which hold not only for real but also for any complex z and u and are the generalization of the well-known trigonometric formulas.

Finally, (4) implies that for $z = x+iy$ we have

$$e^z = e^x e^{iy} = e^x(\cos y + i \sin y) \tag{7}$$

The function $z = \ln w$ of the complex variable w is defined as the inverse function of the function

$$w = e^z \tag{8}$$

We can write (for $w \neq 0$) in the exponential form $w = \varrho e^{i\theta}$ ($\varrho = |w| > 0$), and equality (8) is then written as

$$\varrho e^{i\theta} = e^x e^{iy} \qquad (z = x+iy)$$

Consequently

$$z = \ln w = \ln |w| + i \operatorname{Arg} w = \ln |w| + i \arg w + i2k\pi \tag{9}$$

$$(k = 0, \pm 1, \pm 2, \ldots)$$

where $\ln |w|$ ($|w| > 0$) is understood in the ordinary sense. It is seen from (9) that $\ln w$ ($w \neq 0$), together with $\operatorname{Arg} w$, are many-valued (more precisely infinite-valued) functions of w, irrespective of whether w is real or complex. For instance, from the point of view of the theory of functions of a complex variable the natural logarithm of 1 ($\ln 1$) is equal to any of the numbers $2k\pi i$ ($k = 0, \pm 1, \pm 2, \ldots$). In real analysis $\ln 1$ is understood as the number 0 which is one of the above values.

Here we shall not consider further topics of the theory of functions of a complex variable and shall only confine ourselves to the following remark concerning the formula

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots \qquad (-1 < x \leqslant 1) \tag{10}$$

which was derived in § 5.11 for real x. If x is replaced in the right-hand side of (10) by a complex number z with $|z| < 1$ the series continues to be convergent. We can say that its sum is equal to $\ln(1+z)$ where the logarithm is understood in the sense of the above general definition, more precisely, its sum is equal to one of the branches of the infinite-valued function $\ln(1+z)$.

Functions of a complex variable which can be expanded into convergent power series (Taylor's series) are called *analytic functions*. They are studied in a special division of higher mathematics called *the theory of analytic functions* or *the theory of functions of a complex variable.*

NAME INDEX

SUBJECT INDEX

TO THE READER

Mir Publishers would be grateful for your comments on the content, translation and design of this book. We would also be pleased to receive any other suggestions you may wish to make.

Our address is: USSR, 129820, Moscow I-110, GSP
Pervy Rizhsky Pereulok, 2
MIR PUBLISHERS

Printed in the Union of Soviet Socialist Republics

A textbook for university students (physicists and mathematicians) with special supplementary material on mathematical physics. Based on the course read by the author at the Moscow Engineering Physics Institute.

Volume 1 contains Fundamentals of Mathematical Analysis, Real Numbers Limit of Sequence, Limit of Function, Functions of One Variable, Functions of Several Variables, Indefinite Integral, Definite Integral, Some Applications of Integrals, Series.